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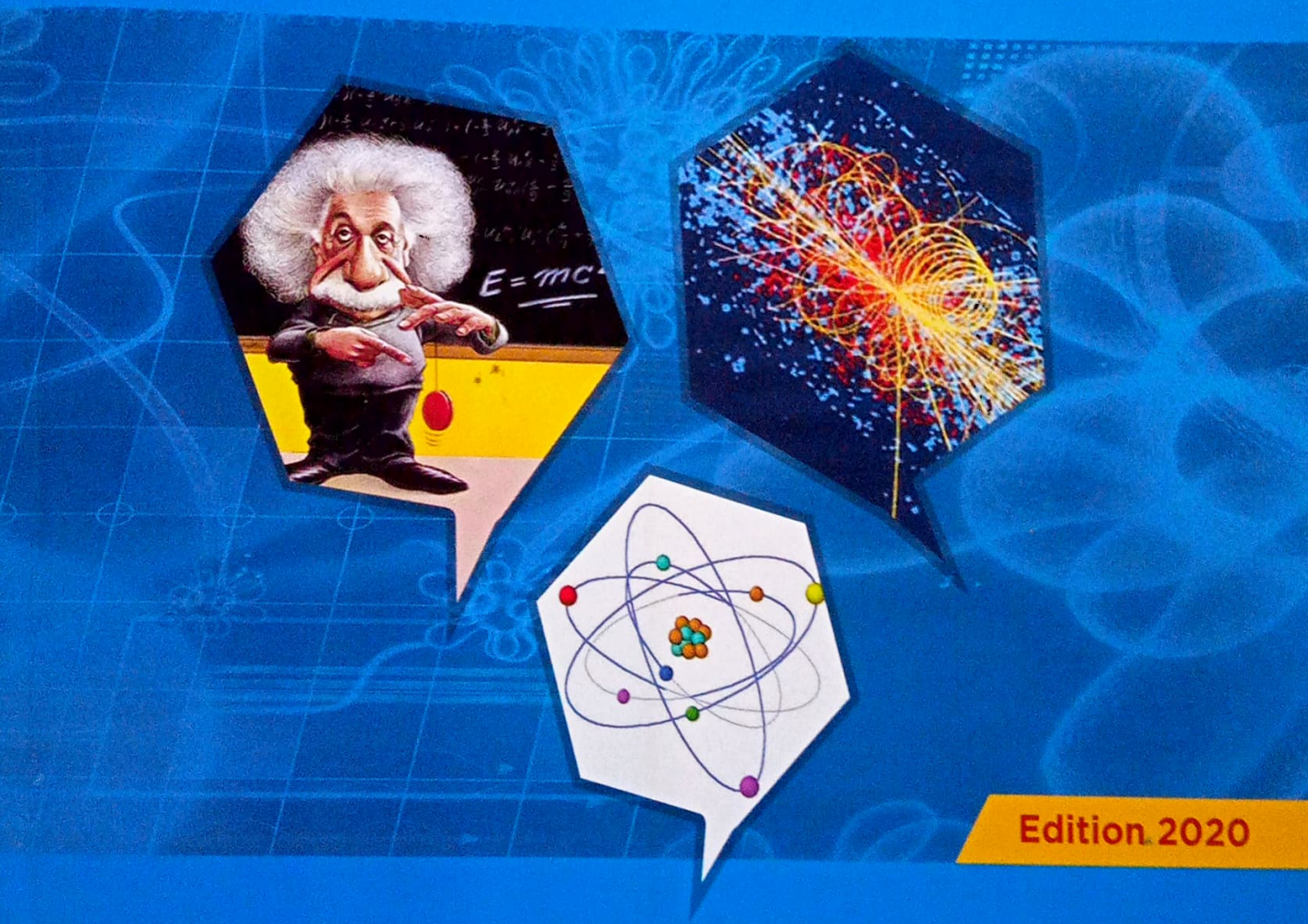

Edition 2020

PHYSICS

PHYSICS

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KINEMATICS 1

REST AND MOTION

Rest or Motion of a body must always be specified w.r.t. a particular reference frame.

Distance (Δs) = length of path actually traversed by the body/particle.

Displacement $(\Delta \vec{r})$ = change in position vector of a body.

The basic equations of kinematics are:

(a) $v_{av} = \frac{\Delta s}{\Delta t}$; (b) $\vec{v}_{av} = \frac{\Delta \vec{r}}{\Delta t}$;

(c) $v_{instantaneous} = \frac{ds}{dt}$; $\vec{v}_{instantaneous} = \frac{d\vec{r}}{dt}$;

(d) $\vec{a}_{av} = \frac{\Delta \vec{v}}{\Delta t}$; (e) $\vec{a}_{instantaneous} = \frac{d\vec{v}}{dt}$;

(f) $v = |\vec{v}|$; (g) $s = \int_0^t |\vec{v}|\,dt = \int_0^t v\,dt$;

(h) $\vec{r} = \int_0^t \vec{v}\,dt + \vec{r}_0$

The position vector, velocity and acceleration of a particle can be written as follows in Cartesian co-ordinates:

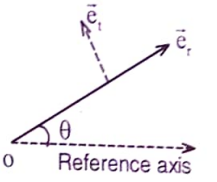
$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$

$\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$

We confine our analysis to two-dimensional motion only, as three-dimensional motion is out of the scope of present discussion.

Also, $\vec{r} = r\vec{e}_r$, $\vec{e}_r$ = radial unit vector and $\vec{e}_t$ = tangential unit vector.



$$\frac{d\vec{r}}{dt} = \frac{dr}{dt}\vec{e}_r + r\omega\vec{e}_t, \text{ where } \omega = \frac{d\theta}{dt}$$

$$\frac{d^2\vec{r}}{dt^2} = \left[\frac{d^2r}{dt^2} - r\omega^2\right]\vec{e}_r + \left[r\frac{d\omega}{dt} + 2\omega\frac{dr}{dt}\right]\vec{e}_t$$

In case of circular motion, r = constant

Therefore,

$$\frac{d\vec{r}}{dt} = r\omega\vec{e}_t \text{ and } \frac{d^2\vec{r}}{dt^2} = -r\omega^2\vec{e}_n + r\frac{d\omega}{dt}\vec{e}_t$$

where a_n = normal acceleration = $\omega^2 r$ and $a_t = r\frac{d\omega}{dt} = \frac{dv}{dt}$ = tangential acceleration

Note: The component of acceleration of a particle in the direction of velocity of the particle is responsible for change in speed, whereas the component of the acceleration of a particle normal to the direction of velocity is responsible for change in direction only.

When the acceleration of a body is constant, the following equations also hold true.

$$\vec{v} = \vec{v}_0 + \vec{a}t$$

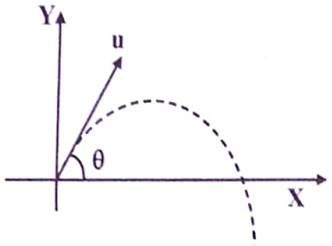
$$\vec{r} = \vec{r}_0 + \vec{u}_0 t + \frac{1}{2}\vec{a}_0 t^2$$

$$v^2 = v_0^2 + 2\vec{a}.(\vec{r} - \vec{r}_0)$$

$$\vec{s}_n = \vec{v}_0 + \frac{\vec{a}}{2}(2n-1)$$

When the acceleration of a body is in the direction of its velocity, it is termed as straight line motion. Sometimes, the acceleration of the body is not in the direction of its velocity, e.g. projectile motion we deal such situations by resolving the velocity and acceleration into perpendicular components.

PROJECTILE MOTION



$u_x = u\cos\theta$

$u_y = u\sin\theta$

$a_x = 0$ [as there is no acceleration along the X-axis].

$a_y = -g$ [the acceleration is downward and equal to g].

$\Rightarrow x = u\cos\theta\, t$ ...(1)

$\Rightarrow y = u\sin\theta\, t - \frac{1}{2}gt^2$ ...(2)

From Eqs. (1) and (2), we get

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$$ [equation of trajectory] ... (3)

Time of flight, $T = \frac{2u\sin\theta}{g}$,

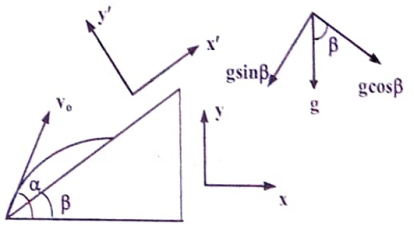
Range, $R = \frac{u^2\sin 2\theta}{g}$

This implies that we get the maximum range for $\theta = 45°$. Also, maximum range is same for complimentary angle of projection for a given speed.

Maximum height reached, $H_{max} = \frac{u^2\sin^2\theta}{2g}$.

Projectile Motion on an Inclined Plane

It is often advantageous to resolve the velocity into two components, one parallel to the direction of the plane and the other perpendicular to the direction of the plane.



$a_x = -g\sin\beta$ and $a_y = -g\cos\beta$

$v_x = v_0\cos(\alpha-\beta) - g\sin\beta t$

and $v_y = v_0\sin(\alpha-\beta) - g\cos\beta t$

$x = v_0\cos(\alpha-\beta)\,t - \frac{1}{2}g\sin\beta t^2$

and $y = v_0\sin(\alpha-\beta)\,t - \frac{1}{2}g\cos\beta t^2$

$$T = \frac{2v_0\sin(\alpha-\beta)}{g\cos\beta}$$

$$\Rightarrow R = \frac{v_0^2\left[\sin(2\alpha-\beta) - \sin\beta\right]}{g\cos^2\beta}$$

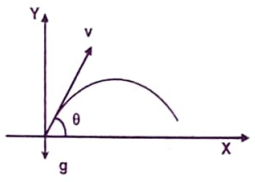
$$\Rightarrow R_{max} = \frac{v_0^2[1-\sin\beta]}{g(1-\sin^2\beta)} = \frac{v_0^2}{g(1+\sin\beta)}$$ up the plane

and $R_{max} = \frac{v_0^2}{g(1-\sin\beta)}$ down the plane.

For two values of α_1 and α_2, we get the same value of range, where α_1 and α_2 are related by

$$\alpha_1 + \alpha_2 = \frac{\pi}{2} + \beta$$

Example: A projectile is launched at t = 0, with a velocity v and at an angle α with the horizontal. Let us find out the rate of change of speed and radius of curvature at the point of projection and at the highest point.



(a) At the point of projection, $\left|\frac{dv}{dt}\right| = g\sin\theta$

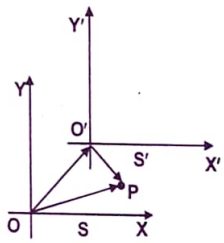
$$\Rightarrow R = \frac{v^2}{a_N} = \frac{v^2}{g\cos\theta}$$

(b) At the highest position, $\left|\frac{dv}{dt}\right|$ = zero

$$\Rightarrow \frac{v^2\cos^2\theta}{a_N} = \frac{v^2\cos^2\theta}{g} = R \text{ ; Radius of curvature}$$

RELATIVE VELOCITY

Rest and Motion are related to each other.
The position, velocity and acceleration of a particle depend on the reference frame from which the observations are made.



Consider a particle P which is moving and is being observed from two different reference frames S and S′. The frame S is stationary and the frame S′ is in motion. Suppose that at any time, the position vector of the particle P with respect to S be denoted by

$$\overrightarrow{OP} = \vec{r}_{p,s}$$

and with respect to S′ by

$$\overrightarrow{O'P} = \vec{r}_{p,s'}$$

The position vector of the origin of S′ with respect to S is, by geometry,

$$\overrightarrow{OO'} = \vec{r}_{s',s}$$

From the vector triangle OO′P, we get

$$\overrightarrow{O'P} = \overrightarrow{OP} - \overrightarrow{OO'}$$

i.e. $\vec{r}_{p,s'} = \vec{r}_{p,s} - \vec{r}_{s',s}$

Differentiating,

$$\Rightarrow \frac{d}{dt}(\vec{r}_{p,s'}) = \frac{d}{dt}(\vec{r}_{p,s}) - \frac{d}{dt}(\vec{r}_{s',s})$$

Differentiating again,

or $\vec{v}_{p,s'} = \vec{v}_{p,s} - \vec{v}_{s',s}$ (or $\vec{v}_{ps'} = \vec{v}_{ps} + \vec{v}_{ss'}$)

or $\vec{v}_{p,s'} = \vec{v}_p - \vec{v}_{s'}$

$$\vec{a}_{p,s'} = \vec{a}_p - \vec{a}_{s'}$$

ILLUSTRATIVE PROBLEMS

1. A fighter plane is flying horizontally at a height of 250 m from ground with constant velocity of 500 m/s. It passes exactly over a cannon which can fire a shell at any time in any direction with a speed of 100 m/s. Find the duration of time for which the plane is in danger of being hit by a cannon shell.

Sol. We first find the area in which the cannon shell can reach. The equation of trajectory for cannon shell is

$$y = x\tan\theta - \frac{\frac{1}{2}gx^2}{u^2}\sec^2\theta \qquad \ldots(1)$$

For maximum y for a given value of x –

$$\frac{dy}{d(\tan\theta)} = x - \frac{\frac{1}{2}gx^2}{u^2}(2\tan\theta) = 0 \qquad \Rightarrow \qquad \tan\theta = \frac{u^2}{gx}$$

Putting in equation (1)

$$y_{max} = x\frac{u^2}{gx} - \frac{\frac{1}{2}gx^2}{u^2}\left[1+\frac{u^4}{g^2x^2}\right] = \frac{u^2}{2g} - \frac{\frac{1}{2}gx^2}{u^2}$$

$\therefore$ The cannon shell can hit an area given by

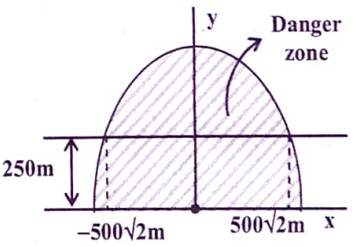
$$y \le \frac{u^2}{2g} - \frac{\frac{1}{2}gx^2}{u^2}$$

Given in the problem y = 250m, u = 100m/s, g = 10 m/s.

Putting these value we get,

$$\frac{x^2}{2000} \le 250 \qquad \Rightarrow -500\sqrt{2} \le x \le 500\sqrt{2}$$

$\therefore$ Plane is in danger for a period of $\frac{1000\sqrt{2}}{500} = 2\sqrt{2}$ sec



2. A balloon moves up vertically such that if a stone is thrown from it with a horizontal velocity v_0 relative to it the stone always hits the ground at a fixed point $2v_0^2/g$ horizontally away from it. Find the height of the balloon as a function of time.

Sol: Range = v_0 × time of flight (t)

$$\Rightarrow t = \frac{2v_0^2}{gv_0} = \frac{2v_0}{g}$$

If y is the height of balloon at any instant t and $\frac{dy}{dt}$ its velocity then

$$y = -\left(\frac{dy}{dt}\right)\left(\frac{2v_0}{g}\right) + \frac{1}{2}g\left(\frac{2v_0}{g}\right)^2$$

$$\Rightarrow \frac{dy}{dt} + \frac{g}{2v_0}y - v_0 = 0$$

$$\frac{dy}{v_0 - \frac{g}{2v_0}y} = dt$$

$$\Rightarrow -\frac{2v_0}{g}\ell n\left(v_0 - \frac{g}{2v_0}y\right) = t + c$$

At $t = 0,\ y = 0 \Rightarrow c = -\frac{2v_0}{g}\ell n\, v_0$

Simplifying, we get

$$y = \frac{2v_0^2}{g}\left[1 - e^{-gt/2v_0}\right]$$

3. A sailor in a boat, which is going due east with a speed of 8 m/s, observes that a submarine is heading towards north at a speed of 12m/s and sinking at a rate of 2 m/s. The commander of submarine observes a helicopter ascending at a rate of 5 m/s and heading towards west with 4m/s. Find the actual speed of the helicopter and its speed w.r.t. boat.

Sol: Taking east, north and vertically upward directions along +x, +y and +z axes respectively.

For boat $\vec{v}_B = 8\hat{i}$

Relative velocity of submarine w.r.t boat $\vec{v}_{S/B} = 12\hat{j} - 2\hat{k}$

$\therefore\ \vec{v}_S - \vec{v}_B = 12\hat{j} - 2\hat{k} \quad \Rightarrow \quad \vec{v}_S = 8\hat{i} + 12\hat{j} - 2\hat{k}$

Relative velocity of helicopter w.r.t. submarine

$\vec{v}_{H/S} = \vec{v}_H - \vec{v}_S$

$5\hat{k} - 4\hat{i} = \vec{v}_H - \left(8\hat{i} + 12\hat{j} - 2\hat{k}\right)$

$\vec{v}_H = +4\hat{i} + 12\hat{j} + 3\hat{k}$

$|\vec{v}_H| = \sqrt{4^2 + 12^2 + 3^2} = 13 m/s$

$\vec{v}_{H/B} = \vec{v}_H - \vec{v}_B = \left(4\hat{i} + 12\hat{j} + 3\hat{k}\right) - 8\hat{i} = -4\hat{i} + 12\hat{j} + 3\hat{k}$

$|\vec{v}_{H/B}| = 13 m/s$

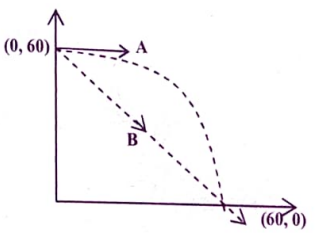
4. Two particles A and B are situated at point (60, 0) at t = 0. The particle A move on a circle given by $X^2 + Y^2 = 3600$ and with a speed 6π m/s, while B moves with a constant velocity.

(a) Find the velocity of B so that the collision takes places between A and B at point (60, 0).

(b) Calculate average acceleration of particle A during the motion.

Sol: (a) Time for A to move from (0, 60) to (60, 0)

$$= \frac{2\pi \times 60}{4} \times \frac{1}{6\pi} = 5 \text{ second.}$$

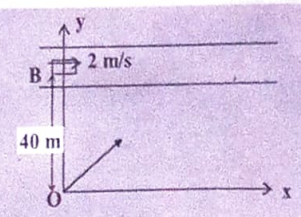


Displacement of $B = 60\hat{i} - 60\hat{j}$

$\therefore$ Velocity of $B = (60\hat{i} - 60\hat{j})/5 = 12(\hat{i} - \hat{j})$ m/s

(b) $\vec{a}_{\text{avg of A}} = \frac{\Delta\vec{v}}{t} = \frac{-6\pi\hat{j} - 6\pi\hat{i}}{5} = -\frac{6\pi}{5}\left[\hat{i} + \hat{j}\right]$ m/s².

5. A block B is floating in the river. The position of the block is as shown in figure at time t = 0. Flow speed of river is 2 m/s. A stone is thrown from point O at time t = 0 with a velocity $\vec{v} = (v_1\hat{i} + v_2\hat{j} + 20)\hat{k}$ m/s.



(a) Find v_1, v_2 such that the stone hits the block.

(b) If the speed of throw is kept fixed, what is the maximum value of time t at which it is possible to throw the stone to hit the block.

Sol: (a) Position vectors of the block and the stone are

$$\vec{r}_B = (2t)\hat{i} + (40)\hat{j}$$

$$\vec{r}_s = v_1 t\hat{i} + v_2 t\hat{j} + \left(20t - \frac{1}{2}gt^2\right)\hat{k}$$

When the stone hits the block

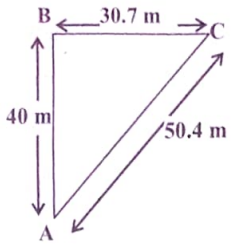
$$2t = v_1 t;\quad 40\text{ m} = v_2 t;\quad 20\,t - \frac{1}{2}gt^2 = 0$$

$$v_1 = 2\text{ m/s};\quad v_2 = \frac{40}{t}\text{ m/s};\quad t = 4\text{ s}$$

$\Rightarrow$ $v_2 = 10$ m/s.

(b) For the same speed of throw, the maximum range

$$R_{max} = \frac{v^2}{g} = \frac{v_1^2 + v_2^2 + 20^2}{g} = \frac{504}{10} = 50.4\text{ m}$$



$\Rightarrow$ BC = 30.7 m.

Time taken by the stone to reach C = $\frac{30.7}{2} = 15.35$ s

Time of flight of the stone = $\frac{2(v/\sqrt{2})}{g} = \frac{\sqrt{2}\sqrt{504}}{10} = 3.17$ s

Maximum value of t = 15.35 – 3.17 = 12.18 s.

SOLVED PROBLEMS

1. A driver applies the brakes on seeing traffic signal 400 m ahead. At the time of applying the brakes, vehicle was moving with 15 m/s and afterwards it starts retarding with 0.3 m/s^2. The distance of vehicle after 1 min, from the traffic light is

(A) 25 m **(B) 375 m**

(C) 360 m **(D) 40 m**

Sol: The maximum distance covered by the vehicle before coming to rest = $\frac{v^2}{2a} = \frac{(15)^2}{2(0.3)} = 375$ m

The corresponding time (t) = $\frac{v}{a} = \frac{15}{0.3} = 50$ s. Therefore after 50 seconds, the distance covered by the vehicle = 375 m from the instant of application of the brakes.

$\Rightarrow$ The distance of the vehicle from the traffic light after one minute = (400 – 375) m = 25 m.

$\therefore$ (A)

2. **A particle is moving along the path given by $y = \frac{C}{6}t^6$ (where C is a positive constant). The relation between the acceleration (a) and the velocity (v) of the particle at t=5sec is**

(A) 5a = v

(B) a = 5v

(C) $a = \sqrt{v}$

(D) a = v

Sol: $y = \frac{C}{6}t^6 \Rightarrow v = \frac{dy}{dt} = Ct^5$

$a = \frac{dv}{dt} = 5Ct^4$

$\frac{a}{v} = \frac{5}{t}$

So, at t = 5 sec, $\frac{a}{v} = 1 \Rightarrow a = v$

$\therefore$ (D)

3. **Which of the following statements can be possible cases in one/two-dimensional motion?**

(A) A body has zero velocity and is still accelerating.

(B) The velocity of an object reverses direction when acceleration is constant.

(C) An object's speed goes on increasing even magnitudes of its acceleration decreases.

(D) None of these.

Sol: (A) When the body is projected vertically up, at the highest point its speed becomes zero whereas it is accelerating downwards with $g = 9.8\ m/s^2$.

(B) When a body is projected up, the velocity at the highest point reverses its direction whereas the acceleration of the body remains constant, that is $\vec{g}$.

(C) Acceleration means, increasing speed (magnitude of velocity). Therefore, the speed of a particle increases with an acceleration. When the acceleration is decreased the speed of the particle goes on increasing till the acceleration reduces to zero.

$\therefore$ (A), (B) and (C)

4. **A motor boat is to reach at a point 30° upstream on other side of a river flowing with velocity 5 m/s. Velocity of motor boat with respect to water is $5\sqrt{3}$ m/sec. The driver should steer the boat at an angle of**

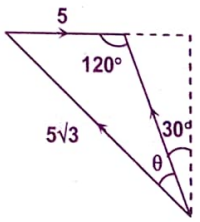
(A) 30° up w.r.t. the line of destination from the starting point

(B) 60° up w.r.t.. normal to the bank

(C) 120° w.r.t. stream direction

(D) None of these

Sol: The velocity of motor boat is given as $\vec{v}_m = \vec{v}_{mw} + \vec{v}_w$



$\Rightarrow \frac{5}{\sin\theta} = \frac{5\sqrt{3}}{\sin 120°}$

$\Rightarrow \sin\theta = ½ \Rightarrow \theta = 30°$

$\therefore$ (A), (B) and (C)

5. **Choose the wrong statement from the following.**

(A) Zero velocity of a particle does not necessarily mean that its acceleration is zero.

(B) Zero acceleration of a particle does not necessarily mean that its velocity is zero.

(C) If speed of a particle is constant, its acceleration must be zero.

(D) None of these.

Sol: If speed, that is the magnitude of velocity, is constant, whereas the direction of velocity changes, we cannot say that velocity is constant. Therefore, the particle has non-zero acceleration.

$\therefore$ (C)

6. **The raindrops are hitting the back of a man walking at a speed of 5 km/hr. If he now starts running in the same direction with a constant acceleration, the magnitude of the velocity of the rain with respect to him will**

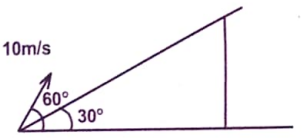
(A) gradually increase **(B) gradually decrease**

(C) first decrease then increase **(D) first increase then decrease**

Sol: The magnitude will decrease till the direction of the velocity with respect to man becomes vertical. It will increase thereafter.

$\therefore$ (C)

7. **A particle is thrown at time t = 0 with a velocity of 10 m/s at an angle of 60° with the horizontal from a point on an incline plane, making an angle of 30° with the horizontal. The time when the velocity of the projectile becomes parallel to the incline is**



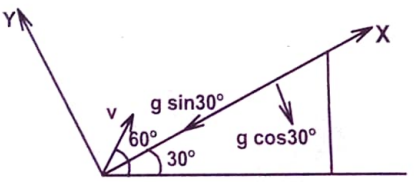
(A) $\frac{2}{\sqrt{3}}$ **sec** **(B)** $\frac{1}{\sqrt{3}}$ **sec**

(C) $\sqrt{3}$ **sec** **(D)** $\frac{1}{2\sqrt{3}}$ **sec**

Sol: $v_y = v \sin 30° - g \cos 30° t$

$$t = \frac{v \sin 30°}{g \cos 30°} = \frac{v}{g}\frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}} \text{ sec}$$

$\therefore$ (B)



8. **When a body is thrown up in a lift with a velocity u relative to the lift, the time of flight is found to be t. The acceleration with which the lift is moving up is**

(A) $\frac{u-gt}{t}$ **(B)** $\frac{u+gt}{t}$

(C) $\frac{2u-gt}{t}$ **(D)** $\frac{u}{t} - 2g$

Sol: $\frac{2u}{g+a} = t \Rightarrow a = \frac{2u}{t} - g$

$\therefore$ (C)

9. **A particle of mass 3m is dropped from certain height at t = 0 at time t = t_0. The particle explodes and breaks into two parts m and 2m. At time t = $3t_0$ velocity of lighter mass is $\vec{v} = v_1\hat{i} + v_2\hat{j}$. The velocity of heavier mass at t = t_0 (just after explosion) is (if y represents vertical and x represents horizontal)**

(A) $-v_1\hat{i} - v_2\hat{j}$

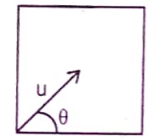
(B) $-\frac{v_1}{2}\hat{i} + \frac{(-v_2 + 5gt_0)}{2}\hat{j}$

(C) $-\frac{v_1}{2}\hat{i} + \frac{(-v_2 + 5gt_0)\hat{j}}{2}$

(D) none of the above.

Sol: (C)

10. **A box is falling freely. Inside the box, a particle is projected with some velocity v with respect to the box at an angle θ as shown in the figure.**



Assertion (A): The path of the particle with respect to an observer sitting in the box will be a straight line.

Reasoning (R): The acceleration of particle with respect to the box is zero. Then,

(A) Both (A) and (R) are correct and R is correct explanation of A.

(B) Both (A) and (R) are correct but R is not correct explanation of A.

(C) (A) is correct but (R) is wrong.

(D) Both (A) and (R) are wrong.

Sol: (A)

COMPREHENSION

The trajectory of a projectile is given by $y = x\tan\theta - \frac{1}{2}\frac{gx^2}{u^2\cos^2\theta}$. This equation can be used for calculating various phenomena such as finding the minimum velocity required to make a stone reach a certain point, maximum range for a given projection velocity and the angle of projection required for maximum range. The range of a particle thrown from a tower is defined as the distance between the foot of the tower and the point of landing.

11. **From a certain tower of unknown height it is found that the maximum range at a certain projection velocity is obtained for a projection angle of 30^0 and this range is $10\sqrt{3}$ m. The projection velocity must be**

(A) 10 m/s

(B) $10\sqrt{3}$ **m/s**

(C) $\frac{10}{\sqrt{3}}$ **m/s**

(D) 5 m/s.

Sol: $\tan\theta = \frac{u^2}{Rg} \Rightarrow u = \sqrt{Rg\tan\theta} = 10$ m/s.

12. **A tower is at a distance of 5 m from a man who can throw a stone with a maximum speed of 10 m/s. What is the maximum height that the man can hit on this tower.**

(A) 4.5 m

(B) 4m

(C) 5 m

(D) 3.75 m.

Sol: $\tan\theta = \frac{u^2}{Rg}$

$$H = R\tan\theta - \frac{1}{2}g\frac{R^2}{u^2}(1+\tan^2\theta)$$

$$= \frac{u^2}{g} - \frac{1}{2}g\frac{R^2}{u^2} - \frac{1}{2}\frac{u^2}{g} = 10 - 1.25 - 5 = 3.75\text{m}$$

13. In the previous problem, what should be the corresponding projection angle.

(A) $\tan^{-1}(1/2)$ **(B)** $\tan^{-1}(1/3)$

(C) $\tan^{-1} 2$ **(D)** $\tan^{-1} 3$

Sol: $\tan\theta = \frac{u^2}{Rg} = \frac{100}{5\times10} = 2$

$\Rightarrow \theta = \tan^{-1} 2.$

Matrix – Match Type

14. Match the following

	Column – I		Column – II
(A)	A particle is moving in a circle	(p)	The acceleration may be perpendicular to its velocity.
(B)	A particle is moving in a straight line	(q)	The acceleration may be in the direction of velocity.
(C)	A particle is undergoing projectile motion with angle of projection α to the horizontal, $0 < \alpha \le \pi/2$	(r)	The acceleration may make some angle, θ $(0 < \theta < \pi/2)$ with velocity
(D)	A particle is moving in space	(s)	The acceleration may be opposite to its velocity
		(t)	The acceleration may make some angle, θ $(\pi/2 < \theta < \pi)$ with velocity

Sol. (A) → (p), (r), (t), (B) → (q), (s), (C) → (p), (q), (r), (s), (t), (D) → (p), (q), (r), (s),(t)

Answer questions 15, 16 and 17 by appropriately matching the information given in the three columns of the following table.

Column –I lists the velocities at t = 0 and t = 5 s for particles moving along a straight line with constant acceleration column –II lists the acceleration and column –III lists 'distance' covered by particle from t = 0 to t = 5.

Column 1			Column 2		Column 3
(I)	u = 0, v = 10 m/s	(i)	$a = 3\ m/s^2$	(P)	30 m
(II)	u = 10 m/s, v = –10 m/s	(ii)	$a = 2\ m/s^2$	(Q)	52.5 m
(III)	u = –3 m/s, v = 12 m/s	(iii)	$a = -4\ m/s^2$	(R)	25 m
(IV)	u = 3 m/s, v = 18 m/s	(iv)	$a = -2\ m/s^2$	(S)	22.5 m

15. Pick correct option.

(A) (I) (iv) (R) **(B) (III) (i) (P)**

(C) (II) (i) (P) **(D) (IV) (iv) (S)**

Sol. B

16. **Pick correct option for minimum magnitude of acceleration.**
(A) (III) (ii) (P)
(B) (II)(iii) (P)
(C) (I) (ii) (R)
(D) (IV) (iv) (S)

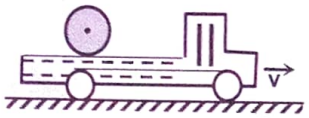
Sol. C

17. **Pick option for minimum displacement.**
(A) (II) (iii) (R)
(B) (I) (ii) (P)
(C) (II) (iv) (S)
(D) (I) (iii) (Q)

Sol. A

Numerical Type

18. **A solid cylinder rolls from the back of a large truck travelling at 10 m/s to the right. The cylinder is travelling horizontally at 8 m/s to the left relative to an observer in the truck. The ball lands on the roadway 1.25 m below its starting level. How far behind the truck does it land (in m)?**



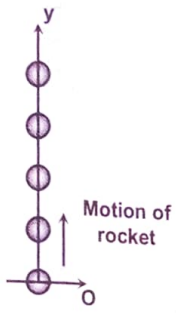
Sol. 4

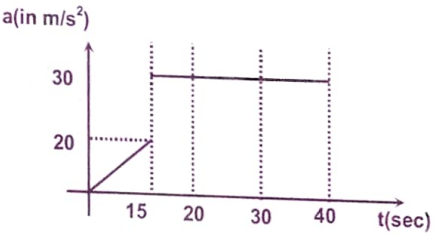
$1.25 = \frac{1}{2}(10)t^2$

$\therefore t = 0.5$ sec

Distance of the cylinder from the truck $= 8 \times 0.5 = 4$ m

19. **A two stage rocket is fired vertically upward from rest with acceleration as shown in a-t graphs. After 15 sec, the first stage burns out and second stage ignites. Choose the INCORRECT statement regarding motion of rocket in the time interval $0 \le t \le 40$ s. If In time interval t = 0 sec to t = 20 sec, the distance travelled by rocket is $\frac{3000}{n}$ m. Find n.**





Sol. 2

$$v = \begin{cases} \frac{2t^2}{9} & \text{if } 0 \le t \le 15 \text{ sec} \\ 150 + 30(t-15) & \text{if } 15 \text{ s} \le t \le 40 \text{ s} \end{cases}$$

at t = 15 sec

$v = \frac{3 \times 15 \times 15}{3} = 150$ m/s

$$\vec{r} = \begin{cases} \frac{2t^3}{3}\hat{j} & \text{if } 0 \le t \le 15 \text{ sec} \\ 150 + 150(t-15) + 15(t-15)^2\hat{j} & \text{if } 15 \text{ s} \le t \le 40 \text{ s} \end{cases}$$

at t = 15

$r = \frac{2 \times 15 \times 15 \times 15}{3 \times 3} = 750$ m

Numerical Decimal Type

20. A particle whose velocity is given as $\vec{v} = \hat{i} + 6\hat{t}\hat{j}$ m/s is moving in x-y plane. At t = 0, particle is at origin. Find the radius of curvature of path at point $\left(\frac{\sqrt{2}}{3}\text{m}, \frac{2}{3}\text{m}\right)$

Sol. 00004.50

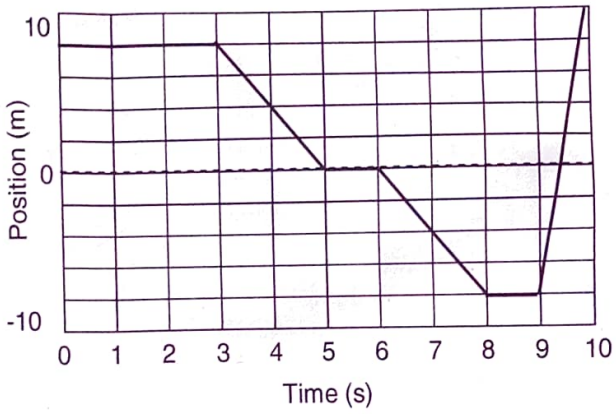
$v_x = 1 \Rightarrow x = t$ and $v_y = 6t \Rightarrow y = 3t^2 \Rightarrow y = 3x^2$

$$\Rightarrow \frac{dy}{dx} = 6x\,,\ \frac{d^2y}{dx^2} = 6 \Rightarrow \left.\frac{dy}{dx}\right|_{x=\frac{\sqrt{2}}{3}} = 2\sqrt{2}$$

As we know that

$$R = \frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}} = \frac{(1+8)^{3/2}}{6} = 4.5\,\text{m}$$

21. The position versus time graph of a particle moving along a straight line is shown. Find the is the total distance (in m) travelled by the particle from t = 0 s to t = 10s?



Sol. **00034.00**

Use the concept of graph

LAWS OF MOTION 2

NEWTON'S LAWS OF MOTION

A force is a push or a pull. It has a magnitude and direction and is therefore a vector.

Newton's First Law

A body remains in its state of rest or continues to move with a uniform velocity unless acted upon by a net external force.

In other words, a body remains unaccelerated only if the net force acting on the body is zero.

This law is, however, not valid in all frames of reference. Only in certain classes of frames, known as inertial frames of reference, the first law is valid.

Other frames, in which Newton's first law is not valid are called non-inertial frames.

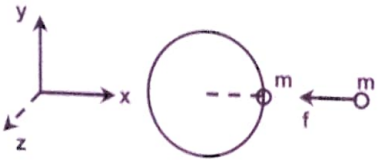
Any frame of reference which is moving with a uniform velocity w.r.t. an inertial frame is also inertial.

Any frame accelerating with some acceleration $\vec{a}$ w.r.t. an inertial frame is non-inertial.

(Newton's first law of motion may be extended to particles in a non-inertial frame of reference as well.)

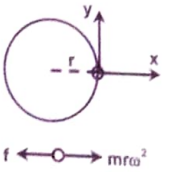
To every particle of mass m in the system, apply an imaginary force equal to $-m\vec{a}$ on the particle. Having done this, you can apply Newton's laws of motion on the system.

A particle is moving on a circular path of radius r with angular velocity ω. The centripetal acceleration of the particle is $r\omega^2$ in the frame of the earth. This centripetal acceleration is provided by the centripetal force f.



$\therefore \quad f = mr\omega^2$

Next, consider a frame attached to this particle. In this frame, the particle is at rest. But, the frame itself is having an acceleration directed radially inward. In this non-inertial frame, an imaginary force equal to $mr\omega^2$, called centrifugal force, is supposed to be acting on the particle in radially outward direction.



i.e. $f = mr\omega^2$

Newton's Second Law

The rate of change of linear momentum of a body is equal to the net external force acting on the body

$$\vec{F} = \frac{d}{dt}(m\vec{v}) = m\frac{d\vec{v}}{dt} + \vec{v}\frac{dm}{dt}$$

In constant mass system, $\frac{dm}{dt} = 0$

$\therefore \quad \vec{F} = m\frac{d\vec{v}}{dt}$

Newton's Third Law

To every action, there is an equal and opposite reaction.

If a system A applies a force $\vec{F}_{AB}$ on a system B, then the system B reacts on A by means of a force given by

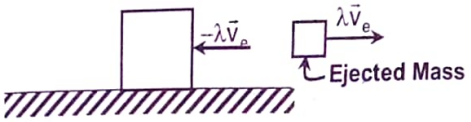
$$\vec{F}_{BA} = -\vec{F}_{AB}$$

For two particles A and B, the force which A exerts on B is along the straight line joining A and B.

Newton's third law does not apply to pseudo forces.

VARIABLE MASS SYSTEM

Consider a system from which mass is being ejected at a constant rate of λ kg/s at a relative velocity $\vec{v}_e$ w.r.t. the system.



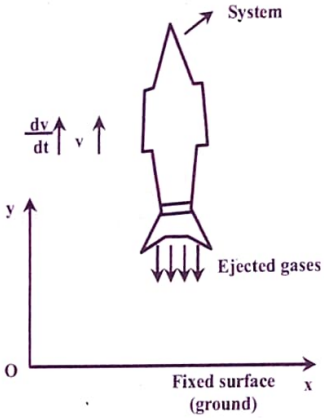
The force on the ejected mass due to system is equal to $\lambda\vec{v}_e$.

$\therefore$ The force on the system due to ejected mass is equal to $-\lambda\vec{v}_e$.

So while dealing with variable mass system we need to consider this thrust on the system.

ROCKET MOTION

At any instant of time let the total mass of the rocket be m and its velocity be velocity v. The gases are ejected at a constant rate of λ kg/sec and velocity v_e relative to the rocket.



After a small time interval dt, a small mass (λdt) of the gas gets ejected.

Initial momentum of mass λ dt just before ejection $= \lambda\, dt\, v\,\hat{j}$

Momentum after ejection $= (\lambda dt)(v_{eg})\,\hat{j}$,

where v_{eg} = velocity of ejected gas w.r.t. the ground

$= (v - v_e)\,\hat{j} = (\lambda dt)(v - v_e)\,\hat{j}$,

Change in the momentum of the ejected mass

$$= (\lambda\, dt)\,(v - v_e\,\hat{j}) - \lambda\, dt\, v\,\hat{j} = -\lambda\, dt\, v_e\,\hat{j}$$

Force exerted on ejected mass $= \dfrac{-\lambda\, dt\, v_e\,\hat{j}}{dt} = -\lambda v_e\,\hat{j}$

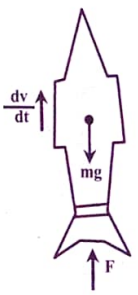
Force exerted on the rocket, $\vec{F} = \lambda v_e\,\hat{j}$

By writing the equation of motion of the rocket, we get

$$F - mg = m\frac{dv}{dt}$$

$$\lambda v_e - mg = m\frac{dv}{dt}$$

$$dv = \frac{\lambda v_e\, dt}{m} - g\, dt \qquad \ldots (1)$$



F.B.D. of rocket

$\therefore$ Let the initial mass of the rocket be m_0, then the mass of the rocket at any instant t is

$$m = m_0 - \lambda t$$

Putting the value of m in Eq. (1) and integrating both the sides, we get

$$\int_0^v dv = \lambda v_e \int_0^t \frac{dt}{(m_0 - \lambda t)} - \int_0^t g\,dt$$

Assuming g to be uniform

$$v = v_e \ln\left(\frac{m_0}{m_0 - \lambda t}\right) - gt = v_e \ln\left(\frac{m_0}{m}\right) - gt$$

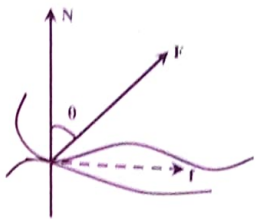
If we ignore the effect of gravity, then

$$\therefore \quad v = v_e \ln\left(\frac{m_0}{m}\right) \qquad \ldots (2)$$

$$\frac{dv}{dt} = \frac{\lambda v_e}{m} \qquad \ldots (3)$$

FRICTION

Whenever two surfaces are in contact, they interact with each other. The magnitude and direction of the force of interaction depend on the situation. If θ is the angle between the force of interaction and normal to the contact surface, as shown in the figure, then θ can have any value between zero to $\frac{\pi}{2}$.



The component of this force F normal to the surface of contact is called the normal reaction, and is denoted by N. The component of this force parallel to the surface of contact is called friction force and is denoted by f. These forces are represented as

$$N = R\cos\theta \qquad \ldots (1)$$

$$f = R\sin\theta \qquad \ldots (2)$$

Dividing Eq. (2) by Eq. (1), we get $\frac{f}{N} = \tan\theta$

or $f = N\tan\theta$

i.e. $f \propto N$

TYPES OF FRICTION

1. ***Kinetic friction*** acts between the two surfaces having relative motion w.r.t. each other.

   $f_k = \mu_k N$, where μ_k = coefficient of kinetic friction.

2. ***Static friction*** is the friction when contact surfaces have a tendency to perform relative motion, but there is no actual relative motion between the surfaces.

   Static friction is self adjusting in nature.

   The maximum value of static friction,

   $f_{s(max)} = \mu_s N$, where N = normal reaction and μ_s = coefficient of static friction.

How to choose the direction of friction force?

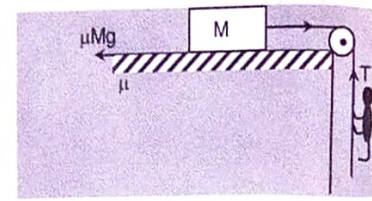
Remember the following simple rule:

"Force of friction between two surfaces acts in a direction so as to oppose relative motion either potential or actual, between two surfaces".

MEGACOSM
COGNITIONS PRIVATE LIMITED

ILLUSTRATIVE PROBLEMS

1. A monkey of mass 'm' climbs up to a rope hung over a fixed pulley with an acceleration g/4. The opposite end of the rope is tied to a block of mass M lying on a rough horizontal plane. The coefficient of friction between the block and horizontal plane is μ. Find the tension in the rope.



Sol. $a_{monkey/rope} = g/4$

Let acceleration of M = a_0

So, acceleration of rope = a_0

$\vec{a}_{monkey} = \vec{a}_{monkey/rope} + \vec{a}_{rope} = (a_0 - g/4)$

Now for mass M, $T - \mu Mg = Ma_0$ ...(1)

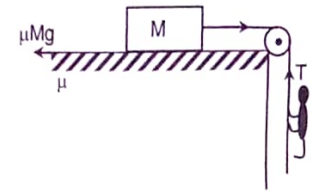
For monkey $mg - T = m(a_0 - g/4)$ ...(2)

From equation (1) and (2)

$$mg - \mu Mg = a_0(M + m) - \frac{mg}{4}$$

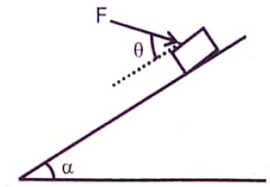
$$\Rightarrow a_0 = \left(\frac{5m}{4} - \mu M\right) g/(M+m) = \frac{(5m - 4\mu M)g}{4(M+m)}$$

$$\therefore T = \frac{M(5m - 4\mu M)g}{4(M+m)} + \mu Mg$$



2. Find the minimum force required to keep a block of mass m in equilibrium on a rough inclined plane with inclination α and coefficient of friction μ (<tanα).

Sol. In order to achieve a minimum of F, it should be directed as shown in the figure. The value of θ can be found using the FBD of the block.



Normal to the incline,

$N = mg\cos\alpha + F\sin\theta$ ... (i)

Along the incline

$\mu N + F\cos\theta = mg\sin\alpha$ ... (ii)

From (i) and (ii), we get

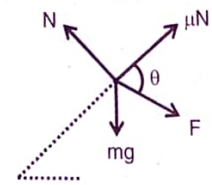
$$F = \frac{mg(\sin\alpha - \mu\cos\alpha)}{\cos\theta + \mu\sin\theta}$$

For F to be minimum

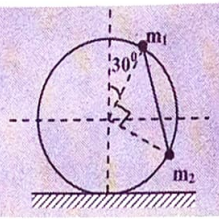
$$\frac{d}{d\theta}(\cos\theta + \mu\sin\theta) = 0$$

$\Rightarrow \mu = \tan\theta$

$$\Rightarrow F_{min} = \frac{mg(\sin\alpha - \mu\cos\alpha)\sqrt{1+\mu^2}}{1+\mu}$$



3. Two beads of masses m_1 and m_2 are connected to each other by massless string. Both beads are free to move over fixed smooth circular loop in vertical plane. At any instant they are in the position shown and subtends angle 90° at centre. Find tangential accelerations of both beads and tension in the string if string remains taut.



Sol. Let a_1 and a_2 be the tangential accelerations of two beads respectively. Then for bead A

$m_1 g \sin 30° + T \sin 45° = m_1 a_1$ ... (i)

For bead B

$m_2 g \sin 60° - T \sin 45° = m_2 a_2$ ... (ii)

since string remains taut

$\therefore a_1 \cos 45° = a_2 \cos 45°$

(Acceleration along string remains same)

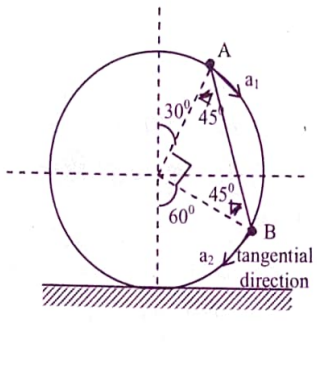
$\Rightarrow a_1 = a_2 = a$

$\therefore$ from (i) and (ii)

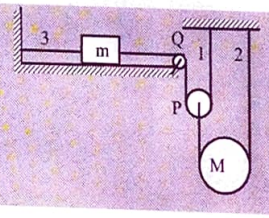
$$a = \frac{g(m_1 + \sqrt{3}m_2)}{2(m_1 + m_2)}$$

$$\frac{T}{\sqrt{2}} = \frac{m_1 g(m_1 + \sqrt{3}m_2)}{2(m_1 + m_2)} - \frac{m_1 g}{2}$$

$$T = \frac{(\sqrt{3} - 1)(m_1 m_2 g)}{\sqrt{2}(m_1 + m_2)}.$$



4. Shown in the diagram is a system of two bodies, a block of mass m and a disc of mass M, held in equilibrium. If the string 3 is burnt, find the acceleration of the disc. Neglect the masses of the pulleys P and Q. (The co-efficient of friction between the block and horizontal surface is μ.)



Sol. Referring to the free-body diagram we write the equation of motion.

For the block m:

$T_1 - \mu N = ma_1$ ... (1)

and $N - mg = 0$ ... (2)

For pulley P:

$m_p g + T_2 - 2T_1 = m_p a_p$

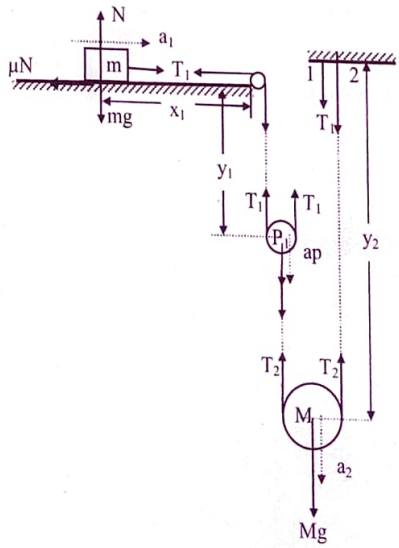
Since the pulley is light,

$\therefore m_p \approx 0$

$\Rightarrow T_2 = 2T_1$ ... (3)

For disc M

$Mg - 2T_2 = Ma_2$ ... (4)



From the geometry of the figure the positions of the parts m, P and M are related as $x_1 + 2y_1 = \ell_1$ and $(y_2 - y_1) + y_2 = \ell_2$, where ℓ_1 and ℓ_2 are the lengths of the strings 1 and 2, respectively. Eliminating y_1 from these two equations, we obtain,

$x_1 + 4y_2 = \ell_1 + 2\ell_2$;

Differentiating both sides twice w.r.t. time, we obtain $\frac{d^2x_1}{dt^2} + 4\frac{d^2y_2}{dt^2} = 0$

$\Rightarrow a_1 = 4a_2$ ... (5)

From Eqs. (1) and (2), we get

$T_1 - \mu mg = ma_1$ ... (6)

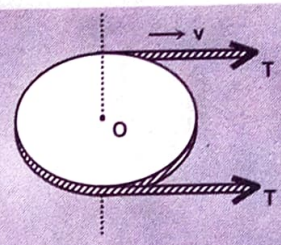
From Eqs. (3) and (4), we get

$Mg - 4T_1 = Ma_2$ ... (7)

Eliminating T_1 from Eqs. (6) and (7) and putting $a_1 = 4a_2$, we obtain

$a_2 = \frac{M - 4\mu m}{M + 16m} g.$

5. A flexible drive belt runs over a frictionless pulley as shown in figure. The pulley is rotating freely about the vertical axis passing through the centre O of the pulley. The vertical axis is fixed on the horizontal smooth surface. The mass per unit length of the drive belt is 1 kg/m and the tension in the drive belt 8 N. The speed of the drive belt is 2 m/s. Find the net normal force applied by the belt on the pulley in newton.

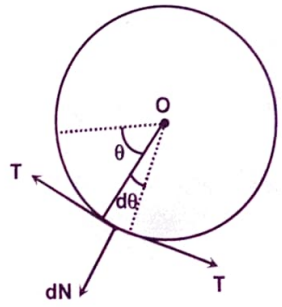


Sol. $2T\sin\frac{d\theta}{2} - dN = \lambda R d\theta \frac{v^2}{R}$ (λ is linear mass density of belt)

$\therefore T d\theta - dN = \lambda v^2 d\theta$

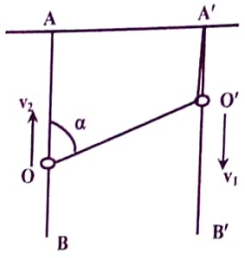
$\therefore$ so total normal $= \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} dN\cos\theta = \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} T\cos\theta d\theta - \lambda v^2 \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} \cos\theta d\theta$

$\therefore N = 8$ newton



SOLVED PROBLEMS

1. **Two small rings O and O′ are put on two vertical stationary rods AB and A′B′, respectively. One end of an inextensible thread is tied at point A′. The thread passes through ring O′ and its other end is tied to ring O.**



Assuming that ring O′ moves downwards at a constant velocity v_1, determine the velocity v_2 of the ring O, when $\angle AOO' = \alpha$.

(A) $v_1\left[\frac{2\sin^2\alpha/2}{\cos\alpha}\right]$ **(B)** $v_1\left[\frac{2\cos^2\alpha/2}{\sin\alpha}\right]$

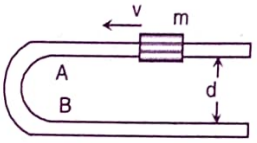
(C) $v_1\left[\frac{3\cos^2\alpha/2}{\sin\alpha}\right]$ **(D) None of these**

Sol. $v_2\cos\alpha + v_1\cos\alpha = v_1$

$\Rightarrow v_2 = v_1\left[\frac{2\sin^2(\alpha/2)}{\cos\alpha}\right]$

$\therefore$ (A)

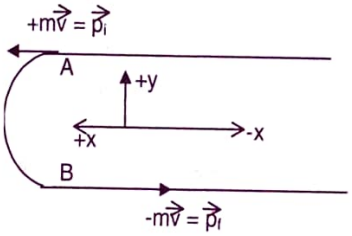
2. **A fixed U-shaped smooth wire has a semi-circular bending between A and B as shown in the figure. A bead of mass 'm' moving with uniform speed v through the wire enters the semicircular bend at A and leaves at B. The magnitude of average force exerted by the bead on the part AB of the wire is**



(A) 0 **(B)** $\frac{4mv^2}{\pi d}$

(C) $\frac{2mv^2}{\pi d}$ **(D) none of these**

Sol. Choosing the positive X–Y axis as shown in the figure, the momentum of the bead at A is $\vec{p}_i = +m\vec{v}$. The momentum of the bead at B is $\vec{p}_f = -m\vec{v}$.



Therefore, the magnitude of the change in momentum between A and B is

$\Delta\vec{p} = \vec{p}_f - \vec{p}_i = -2m\vec{v}$

The time interval taken by the bead to reach from A to B is

$\Delta t = \frac{\pi \cdot d/2}{v} = \frac{\pi d}{2v}$.

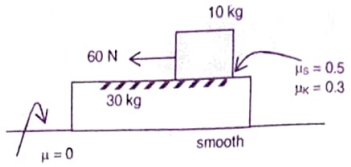
Therefore, the average force exerted by the bead on the wire is

$F_{av} = \frac{\Delta p}{\Delta t}$

$= \left(2mv \Big/ \frac{\pi d}{2v}\right) = \frac{4mv^2}{\pi d}$

$\therefore$ (B).

3. **Two blocks of mass 10 kg and 30 kg are kept as shown in the figure. The surface between the block is frictional while the ground is smooth. A horizontal force of magnitude 60 N is applied horizontally on the upper block. Find the frictional force on block of mass 30 kg?**



(A) 20 N **(B) 30 N**

(C) 40 N **(D) 50 N**

Sol. Let the both blocks are moving together, then

$a = \frac{60}{40} = 1.5 m/s^2.$

thus, force on lower block has to be $F = Ma = 30 \times 1.5 = 45$ N

Next, $f_L = \mu N = 0.5 \times 10 \times 10 = 50$ N

Since $F < f_L$; both the blocks will move together, thus force of friction = 45 N

$\therefore$ (B).

4. **Two identical particles A and B, each of mass m, are interconnected by a spring of stiffness k. If the particle B experiences an external force F along the line of spring and outward (as shown) and the elongation of the spring at this instant is x, the acceleration of particle B relative to particle A is equal to**

(A) $\frac{F}{2m}$ **(B)** $\frac{F-kx}{m}$

(C) $\frac{F-2kx}{m}$ **(D)** $\frac{kx}{m}$

Sol. Equation of motion for A:

$kx = ma \Rightarrow a = \frac{kx}{m}$

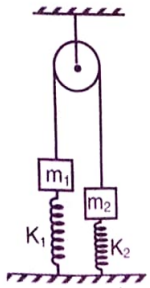
For B: $F - T = ma'$

$\Rightarrow a' = \frac{F-kx}{m}$

$\Rightarrow$ The relative acceleration $= a_r = |a' - a| = \frac{F-2kx}{m}.$

$\therefore$ (C)

5. **The system shown in the figure is in equilibrium. Masses m_1 & m_2 are 2 kg and 8 kg respectively. Spring constants k_1 & k_2 are 50 N/m and 70 N/m respectively. If the compression in second spring is 0.5 m. What is the compression in first spring? (Both springs have the same natural length)**



(A) 1.3 m **(B) –0.5 m**

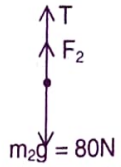
(C) 0.5 m **(D) 0.9 m**

Sol. As the springs are fixed to the horizontal and have the same natural length, hence if one spring is compressed, the other must be expanded. Hence, the compression will be negative.

F.B.D of m_2 $\quad T + F_2 = 80$ N

and $F_2 = 70 \times 0.5 = 35$ N

$\therefore T = 80 - 35 = 45$ N



F.B.D of m_2

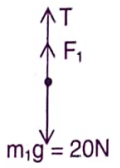
F.B.D of m_1

$T + F_1 = mg$

or $45 + F_1 = mg$

$\Rightarrow F_1 = -45 + 20 = -25$ N

$\therefore x_1 = \dfrac{-25}{50} = -0.5$m

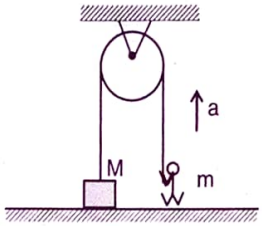


F.B.D of m_1

$\therefore$ Compression in first spring = –0.5 m

$\therefore$ (B).

6. In the figure the block of mass M is at rest on the floor. The acceleration with which a boy of mass m should climb along the rope of negligible mass so as to lift the block from the floor is,



(A) $\left(\dfrac{M}{m}-1\right)g$ **(B)** $>\left(\dfrac{M}{m}-1\right)g$

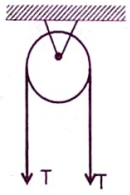
(C) $\dfrac{M}{m}g$ **(D)** $>\dfrac{M}{m}g$

Sol. Equation of motion for M:

Since M is stationary,

$T - Mg = 0$

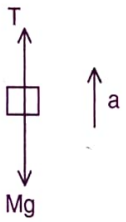
$\Rightarrow T = Mg$ ...(1)



Since the boy moves up with an acceleration 'a',

$T - mg = ma$

$\Rightarrow T = m(g + a)$ ...(2)



Equating Eqs. (1) and (2), we obtain

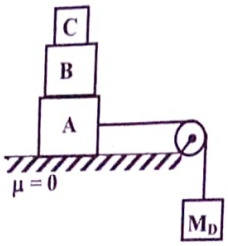
$Mg = m(g + a)$

$\Rightarrow a = \left(\dfrac{M}{m}-1\right) g$

That means, if $a > \left(\dfrac{M}{m}-1\right) g$, the block M can be lifted.

$\therefore$ **(B)**.

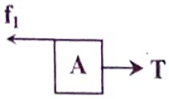
7. **Three blocks A, B and C of equal mass m are placed one over other on a frictionless surface (table) as shown in the figure. Coefficient of friction between any blocks A, B and C is μ. The maximum value of mass of block M_D so that the block A, B and C move without slipping over each other is**

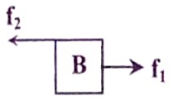


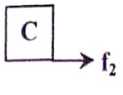
(A) $\frac{3m\mu}{\mu+1}$ **(B)** $\frac{3m(1-\mu)}{\mu}$

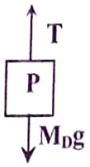
(C) $\frac{3m(1+\mu)}{\mu}$ **(D)** $\frac{3m\mu}{(1-\mu)}$

Sol.









$T - f_1 = ma$ $f_1 - f_2 = ma$ $f_2 = ma$ $M_Dg - T = M_Da$

Solving we get,

$M_Dg = (M_D + 3m)a$

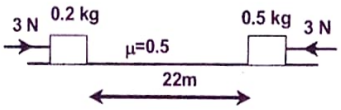
The maximum value of acceleration = μg

$\Rightarrow M_Dg = (M_D + 3m)\mu g$

$M_D = \frac{3m\mu}{1-\mu}$

∴ **(D)**

8. **Two blocks of masses 0.2 kg and 0.5 kg, which are placed 22 m apart on a rough horizontal surface (μ = 0.5), are acted upon by two forces of magnitude 3 N each as shown in figure at time t = 0. Then, the time t at which they collide each other is**



(A)1 sec **(B)** $\sqrt{2}$ **sec**

(C)2 sec **(D) none of these.**

Sol. $a_1 = \frac{F - f_1}{m_1} = \frac{F - \mu m_1 g}{m_1} = 10 m/s^2$

$a_2 = \frac{-F + \mu m_2 g}{m_2} = -1\ m/s^2$

$\therefore \quad s = \frac{1}{2} a_{real} t^2$

$22 = \frac{1}{2}[10 - (-1)]t^2 \Rightarrow t = 2\,sec$

∴ (C)

9. **A worker wishes to pile up sand on to a circular area of radius R. No sand is to spill on to the surrounding area. If μ_s is the coefficient of static friction between sand particles, then choose the correct statement(s).**

(A) The greatest height of sand pile that can be created is $\mu_s R$.

(B) The greatest height of sand pile that can be created is R/μ_s.

(C) Minimum work required to create greatest height sand pile is $\frac{\pi}{12}\mu_s^2 R^4 \rho g$. Where ρ is the density of sand volume.

(D) Sand pile will be in stable equilibrium.

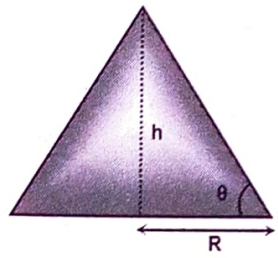
Sol. A, C

From free body diagram of sand particle

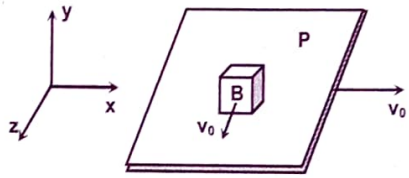
$\tan\theta = \mu_S = h/R$

work required $= \left(\frac{1}{3}\pi R^2 h\rho g\right)\frac{h}{4}$.

The system is in unstable equilibrium.



10. **A very large plank P of some unknown mass is being moved with velocity $v_0\hat{i}$ under application of an external force (not shown in figure). Simultaneously a block B of mass m placed on the plank is also moving with velocity $v_0\hat{k}$. All these velocities are with respect to ground frame and at t = 0. Coefficient of friction between the plank and the block is μ. Choose the correct option(s).**



(A) Kinetic friction force acting on the block, at t = 0 is $-\mu mg\hat{k}$.

(B) At t = 0, power developed (with respect to ground frame) by kinetic friction force on the block B is $-\frac{\mu mgv_0}{\sqrt{2}}$

(C) At t = 0, power developed (with respect to ground frame) by kinetic friction force on the block B is $-\mu mg\sqrt{2}v_0$

(D) At t = 0, heat dissipation per sec in the system is $2\sqrt{2}\,\mu mgv_0$.

Sol. B

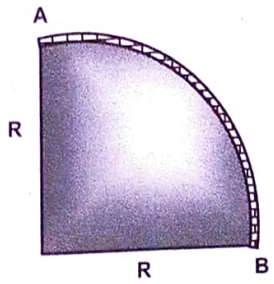
$\vec{f}_k$ on the block B $= \mu mg\left(\frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{k}\right)$

$$P_{f_k} = \mu mg\left[\frac{\hat{i}}{\sqrt{2}} - \frac{\hat{k}}{\sqrt{2}}\right]\cdot(v_0\hat{k})$$

$$= -\frac{\mu mgv_0}{\sqrt{2}}$$

$$\frac{dQ}{dt} = \mu mgv_0\frac{2}{\sqrt{2}} = \sqrt{2}\mu mgv_0$$

11. **A rope AB of linear mass density λ is placed on a quarter vertical fixed disc of radius R as shown in the figure. The surface between the disc and rope is rough such that the rope is just in equilibrium. Gravitational acceleration is g. Choose the correct option(s).**



(A) Coefficient of static friction between rope and disc is $\mu = 1$.

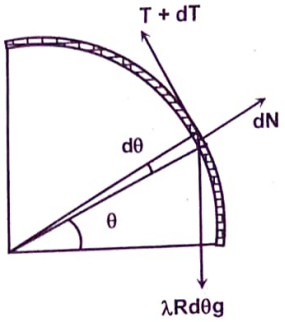
(B) Coefficient of static friction between rope and disc is $\mu = \frac{1}{\sqrt{2}}$.

(C) Maximum tension in the rope is at the top most point A of the rope.

(D) Maximum tension in the rope is $\lambda Rg(\sqrt{2}-1)$.

Sol. A, D

For equilibrium



$$\lambda Rg\int_0^{\frac{\pi}{2}}\cos\theta d\theta = \mu\lambda Rg\int_0^{\frac{\pi}{2}}\sin\theta d\theta$$

$\therefore \mu = 1$

At the position of maximum tension in the rope

$\lambda Rd\theta g\cos\theta = \mu(\lambda Rd\theta g\sin\theta)$

$\therefore \theta = 45°$

At any θ,

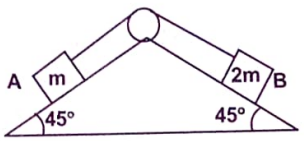
$dT = \lambda Rd\theta g\cos\theta - \mu\lambda Rd\theta g\sin\theta$

$$\int_0^{T_{max}} dT = \lambda Rg\int_0^{\frac{\pi}{4}}(\cos\theta - \sin\theta)d\theta$$

$$T_{max} = \lambda Rg\left[\sin\theta + \cos\theta\right]_0^{\frac{\pi}{4}} = \lambda Rg\left[\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1\right] = \lambda Rg\left(\sqrt{2}-1\right)$$

Matrix – Match Type

12. **Two blocks A and B of mass m and 2m are placed on a fixed triangular wedge by massless inextensible string as shown. The pulley is massless and frictionless. The coefficient of friction between block A and the wedge is 2/3 and that between B and the wedge is 1/3. (Take m = 1 kg, g = 10 m/s^2)**

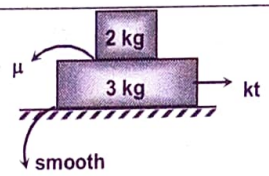


	Column – I		Column – II
(A)	Friction force between block A and wedge	(p)	$\frac{5}{\sqrt{2}}$ units
(B)	Friction force between block B and wedge	(q)	$\frac{10}{3\sqrt{2}}$ units
(C)	Tension in the string	(r)	$\frac{10\sqrt{2}}{3}$ units
(D)	Maximum friction force between Block B and wedge.	(s)	$30\sqrt{2}$ units
		(t)	$\frac{20\sqrt{2}}{3}$ units

Sol. (A) → (q) (B) → (r) (C) → (t) (D) → (r)

Answer questions 13, 14 and 15 by appropriately matching the information given in the three columns of the following table.

Figure shows two blocks placed on a smooth surface. A force F = kt starts acting on 3 kg block starting from t = 0. Column-I lists values of friction coefficient μ and force constant k, column –II lists the time instant when slipping starts between 2 kg and 3 kg block where as column –III lists the velocity of 3 kg block at that instant.



Column 1			Column 2		Column 3
(I)	μ = 0.1, k = 1	(i)	2.5 sec	(P)	10 m/s
(II)	μ = 0.2, k = 2	(ii)	10 sec	(Q)	2.5 m/s
(III)	μ = 0.2, k = 1	(iii)	5 sec	(R)	5 m/s
(IV)	μ = 0.1, k = 2	(iv)	7.5 sec	(S)	1.25 m/s

13. Pick correct option.

(A) (I) (iii) (Q) **(B) (II)(ii) (P)**
(C) (III) (iii) (Q) **(D) (IV) (iv) (P)**

Sol. A

14. Pick the option corresponding to maximum value of instant when slipping starts.

(A) (I) (i) (Q) **(B) (IV)(ii) (S)**
(C) (II) (iv) (R) **(D) (III) (ii) (P)**

Sol. D

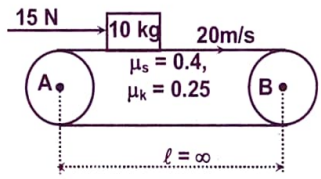
15. Pick the option corresponding to minimum speed of 3 kg block at the instant that slipping starts.

(A) (IV) (i) (S) **(B) (III) (ii) (R)**
(C) (II) (iii) (P) **(D) (I) (iv) (Q)**

Sol. A

COMPREHENSION

A block of mass 10 kg is put gently on a belt-conveyor system of infinite length at t = 0 sec, which is moving with constant speed 20 m/sec rightward at all time, irrespectively of any situation by means of a motor-system as shown in the figure.
A constant force of magnitude 15 N is applied on the block continuously during its motion.



16. The nature of friction force acting on the block of mass 10 kg during its motion is

(A) kinetic only **(B) Static only**
(C) some time kinetic some time static **(D) can't be predicted**

Sol: C

17. Work done by the kinetic friction on the block of mass 10 kg is

(A) 1250 Joule **(B) 2500 Joule**
(C) –1250 Joule **(D) none**

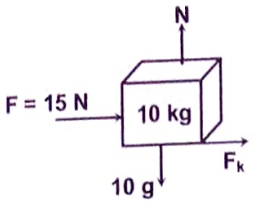
Sol: A

18. The magnitude of acceleration of the block of mass 10 kg at t = 6 sec is

(A) 4 m/s^2 **(B) 3 m/s^2**
(C) 2 m/s^2 **(D) none**

Sol: D

16-18. In the time interval t = 0 sec to t = t_0 sec, till the time. The relative velocity is not zero the nature of friction will be kinetic.



$\vec{u}_{AB} = 0\hat{i} - (20\hat{i}) = 20\ m/sec\,(\hat{i})$

$N = 100\ N \Rightarrow F_k = 0.25 \times 100 = 25\ N$

$\vec{a}_A = 4m/sec^2\,(\hat{i}),\ \vec{a}_B = 0m/sec^2\,(\hat{i})$

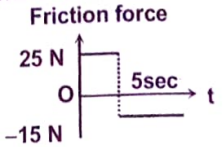
$\vec{a}_{AB} = \vec{a}_A - \vec{a}_B = 4m/sec^2\,(\hat{i})$

$\vec{V}_{AB} = \vec{u}_{AB} + \vec{a}_{AB}t = \vec{0}$

$\Rightarrow t = 5$ sec

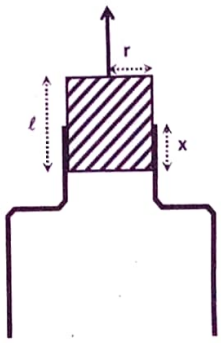
When the relative motion between block A and belt conveyor will be zero, the nature of friction will be static and its magnitude will equal to magnitude of unbalanced external force acting on the block A and its direction will be I the opposite direction of unbalanced external force.

$\Rightarrow F_r$ = Frictional force = $\begin{cases} 25N(\hat{i}) \Rightarrow \text{kinetic in nature} \\ 15N(-\hat{i}) \Rightarrow \text{Static in nature} \end{cases}$



Numerical Type

19. **A cylindrical massless cork of length ℓ = 10 cm and radius r = 3 cm is slowly extracted from the neck of a bottom opened bottle. If the normal pressure between the bottle and un-extracted part of the cork at any instant is constant and equal to $P = \frac{10^5}{\pi}$ N/m², find the work done (in Joule) in extracting it completely. The coefficient of friction between the cork and bottle is μ = 0.3.**



Sol: 9

$W = \int_0^{\ell} \mu 2\pi r x dx P$

$= \pi\mu r P\ell^2$

$= \pi(0.3)(3 \times 10^{-2})\frac{10^5}{\pi} \times 10^{-2} = 9\ J$

20. **A long plank begins to move at t = 0 and accelerates along a straight track with a speed given by $v = 2t^2$ for $0 \le t \le 2$ (where v is in m/s and t is in second). After 2 sec the plank continues to move at the constant speed acquired. A small block initially at rest on the plank begins to slip at t = 1 sec and stops sliding at t = 3 sec. If the coefficient of static friction and kinetic friction between the plank and the block is 0.s and 0.k (where s and k are digits) respectively, find s + k. (take g = 10m/s²)**

Sol: 7

$\mu_s g = \frac{dv}{dt}$ at t = 1 sec

$\therefore\ \mu_s = 0.4$

At t = 3 sec velocities of plank and block are equal.

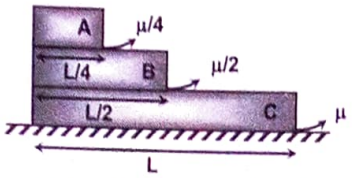
$\therefore\ 2 + \mu_k g(2) = v$ at t = 3 sec = 8

$\therefore\ \mu_k = 0.3$

$\therefore\ s + k = 4 + 3 = 7$

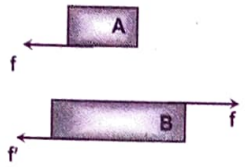
Numerical Decimal Type

21. Three blocks A, B and C of mass m, $\frac{m}{2}$ and m of different densities and dimensions are placed over each other as shown in the figure. The coefficients of friction are shown. Blocks placed in a vertical line are made to move towards right with same velocity at the same instant. Find the time (in sec) taken by the upper block A to topple from the middle block B. Assume that blocks B and C don't stop sliding before A topples from B. (given L = 36 m, $\mu = 0.4$ and $g = 10\ m/s^2$)



Sol: 00003.00

Since $\frac{\mu}{4} < \frac{\mu}{2} < \mu$, the upper block will move faster than the middle block and hence force of friction on upper block is towards left



$\therefore f = \frac{\mu}{4}mg$

$\therefore a_A = \frac{\mu g}{4}$ leftward

$f' = \frac{\mu}{2}\left(m + \frac{m}{2}\right)g = \frac{3}{4}\mu mg$

$\therefore a_B = \mu g$ (leftward)

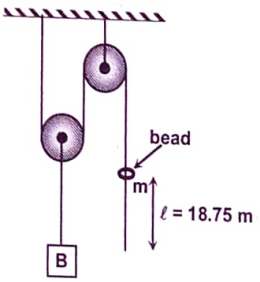
$\therefore a_{A/B} = \frac{3}{4}\mu g$ (rightward)

For sliding down the block B, A has to move a distance $\frac{3L}{8}$ relative to block B.

$\frac{3L}{8} = \frac{1}{2}\frac{3}{4}\mu g t^2$

$\therefore t = \sqrt{\frac{L}{\mu g}} = 3$ sec.

22. In the system shown in the figure, a bead of mass m can slide on the string. There is friction between the bead and the string. Block B has mass equal to twice that of the bead. The system is released from rest with length $\ell = 18.75$ m of the string hanging below the bead. Assuming the pulley and string to be massless. Find the distance (in meter) moved by the block B before the bead slips out of the thread.



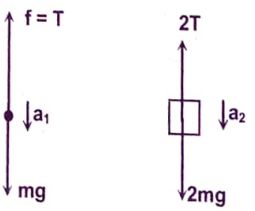
Sol: 00006.25

$mg - T = ma_1$

$2mg - 2T = 2ma_2$

So, $a_1 = a_2$

Relative acceleration of bead with respect to end = 3a



$\therefore$ displacement of block $x = \frac{1}{2}at^2 = \frac{\ell}{3} = 6.25$ m

3

WORK, ENERGY & POWER

WORK

Work is said to be done by a force when the point of application of force is displaced under the influence of the force. Work is a scalar quantity and it is measured by the product of the magnitude of force and the component of displacement along the direction of force. In fact, work is the scalar product (dot product) of the force vector and the displacement vector. Thus, $W = \vec{F}.\vec{S} = FS\cos\theta$

where F and S are the magnitude of force and displacement vectors and θ is the angle between them.

The equation $W = \vec{F}.\vec{S} = FS\cos\theta$ is applicable when $\vec{F}$ remains constant but when the force is variable, work is obtained by integrating $\vec{F}.d\vec{s}$.

Thus, $W = \int \vec{F}.d\vec{S}$

If the work done by a force field along a closed path is zero, the force is said to be conservative otherwise it is called non-conservative. Under such forces, the work done depends only upon the initial and final positions and is path independent.

ENERGY

Energy (or stored work) is that physical quantity which enables a system to do work. In this section, we restrict ourselves to mechanical energy which comprises of two forms:

(i) kinetic energy

(ii) potential energy

Kinetic Energy

The kinetic energy of a body is the energy possessed by the body by virtue of its motion. Mathematically, it is equal to $\frac{1}{2}mv^2$, where m is the mass of the body and v is the speed of the body.

Work–Energy Theorem

The change in kinetic energy of a body is equal to the sum of the work done by all the forces acting on the body.

Potential Energy

It is a function defined such that the negative of its derivative with respect to the position of a particle (system) gives the internal conservative force acting on the particle (system).

$$F_{int,const.} = -\frac{dU}{dr}$$

Alternatively, $U_f - U_i = -\int_{\vec{r}_i}^{\vec{r}_f} \vec{F}_{int,const.} \cdot d\vec{r}$

The restoring force and the potential energy of a spring either compressed or elongated by a distance x is equal to $-kx$ and $\frac{1}{2}kx^2$, respectively, where k is the spring constant of the spring.

Conservation of Mechanical Energy

The mechanical energy of a system remains constant if no external forces act on the system and all the internal forces are conservative. This is called law of conservation of mechanical energy. When external forces are also present, the work done by them is equal to the change in mechanical energy of the system.

POWER

Power is defined as the rate of work done.

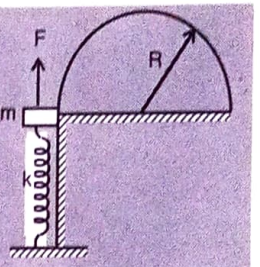
$$P = \vec{F}.\vec{v} \qquad \ldots(3)$$

If the force is variable, we calculate the average power as

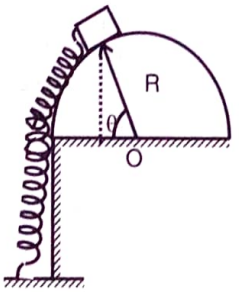
$$P_{av} = \frac{\Delta W}{\Delta t} = \frac{\int_0^t P\ dt}{\int_0^t dt}$$

ILLUSTRATIVE PROBLEMS

1. In the figure shown a semi-cylinder of radius R is rigidly fixed on a horizontal table of height ℓ_0 from the ground such that there is a common edge of the table and the semi-cylinder. A block of mass m is attached rigidly to a light spring of stiffness k, whose other end is rigidly attached to the ground. The natural length of the spring is ℓ_0. A force of constant magnitude F is applied on the block to move it on the frictionless surface of the semi-cylinder in a vertical plane such that the path followed by the block is a semicircle of radius R.



The force F is always tangential to the cylinder. Find the contact force between the cylinder and the block as a function of θ. Initial speed of the block is zero and initial length of the spring is ℓ_0. θ is the angular displacement of the block w.r.t. the centre of the circular path as shown in the figure.



Sol. W_N = work done by normal reaction = 0

$W_g = -mgR\sin\theta$

$W_{spring} = -\frac{1}{2}k(R\theta)^2$

$W_F = FR\theta$

$\Sigma W = \Delta KE$

$\Rightarrow -mgR\sin\theta - \frac{1}{2}kR^2\theta^2 + FR\theta = \frac{1}{2}mv^2$

$\Rightarrow mv^2 = -2mgR\sin\theta - kR^2\theta^2 + 2FR\theta \qquad \ldots$ (i)

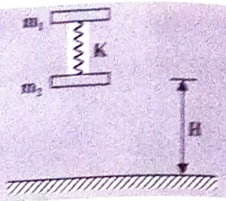
Also, $mg\sin\theta - N = \frac{mv^2}{R}$

$\Rightarrow N = mg\sin\theta - \frac{mv^2}{R}$

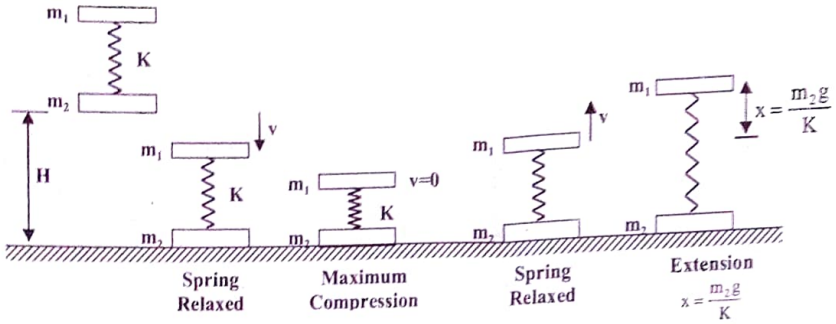
$= mg\sin\theta + 2mg\sin\theta + kR\theta^2 - 2F\theta$ using equation (i)

$= 3mg\sin\theta + kR\theta^2 - 2F\theta$

2. A spring–mass system is held at rest at height H from the ground with the spring in the relaxed state. Find the minimum value of H so that the system has a tendency to rebound after hitting the ground. The coefficient of restitution between m_2 and ground is zero.



Sol.



Using energy conservation

$$\frac{1}{2}m_1v^2 = m_1gx + \frac{1}{2}kx^2$$

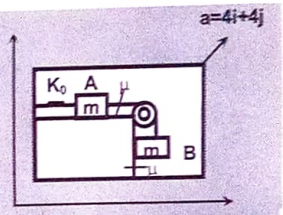
Since $v^2 = 2gH$ and

for the block M_2 to just lift off,

$$x = \frac{m_2g}{K}, \qquad \therefore \quad m_1gH = m_1g\left(\frac{m_2g}{K}\right) + \frac{1}{2}K\left(\frac{m_2g}{K}\right)^2$$

$$\text{or, } H = \frac{m_2g}{K}\left[\frac{2m_1 + m_2}{2m_1}\right].$$

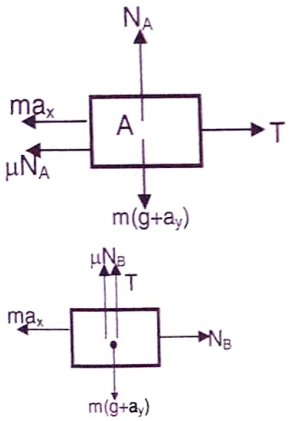
3. The arrangement shown in the diagram is moving in space with an acceleration $a = 4(i + j)$ m/s^2. An ideal spring of natural length ℓ_0 having spring constant k = 50 N/m, is connected to block A. Blocks A and B are connected by an ideal string passing through a frictionless pulley. Mass of each block A and B is equal to m = 2 kg.



(a) Calculate the minimum value of coefficient of friction so that spring remains in its natural length. (b) If μ is reduced to (9/35)th of the original value, calculate the maximum extension of the spring.

Sol. Case I: $a = (4i+4j)$ m/s^2, $a_x = 4$m/s^2, $a_y = 4$m/s^2

FBD of blocks A and B



$N_A = m(g + a_y) = 28$ N

$T = ma_x + \mu N_A = 8 + 28\mu$ ... (1)

$N_B = ma_x = 8$ N

$T + \mu N_B = m(g+a_y)$

$T + 8\mu = 28$ ... (2)

From Eqs. (1) and (2), $\mu_{min} = 5/9$

Case: II: Let extension of spring = x, $\mu' = (5/9) \times (9/35) = 1/7$

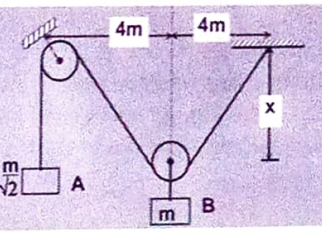
From work energy theorem:

$mgx - \mu' N_A x - \mu' N_B x - ma_x x + ma_y x = \frac{1}{2} kx^2$

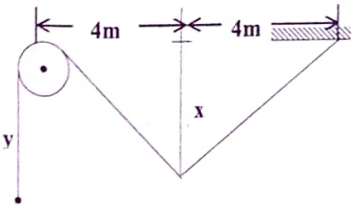
$20x - (1/7)x \times (28x) - (1/7)x \times (8x) = 25x^2$

$x = 0.6$ m.

4. The system of mass A and B shown in the figure is released from rest with x = 0, determine the velocity of mass B when x=3m. Also find the maximum displacement of mass B.



Sol. Constraint relation gives



$2\sqrt{x^2+16}+y=c$

Differentiating w.r.t. time, we get

$$\frac{dy}{dt}=-\frac{x}{\sqrt{x^2+dt}}\frac{dx}{dt}$$

Let $\frac{dx}{dt}$ = v = velocity of B

$$\therefore \quad \frac{dy}{dt} = \text{velocity of A} = -\frac{x}{\sqrt{x^2+16}}.v$$

Minus sign indicates that it in upward direction.

∴ Using energy conservation

$$mg.3-\frac{mg}{\sqrt{2}}.2=\frac{1}{2}mv^2+\frac{1}{2}\frac{m}{\sqrt{2}}\left(v.\frac{3}{5}\right)^2$$

$$\Rightarrow \left(3-\sqrt{2}\right)g=\frac{v^2\left(25\sqrt{2}+9\right)}{50\sqrt{2}}$$

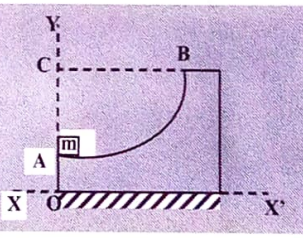
$$\Rightarrow v=\sqrt{\frac{\left(3-\sqrt{2}\right)\left(50\sqrt{2}\right)}{25\sqrt{2}+9}}\,g = 5 \text{ m/s}$$

Let x_1 be the maximum displacement of B, then

$$mgx_1-\frac{mg}{\sqrt{2}}2\left[\sqrt{x_1^2+16}-4\right]=0$$

$\Rightarrow x_1 = 8\sqrt{2}$ m.

5. A wedge with a rough circular track AB, is fixed on the plane xx'. The radius of the track is R and the coefficient of friction of the track varies as $\mu = \mu_0 x$. Find the work done on mass m in moving from A to B. Here C is the centre of the circular track AB and x is the distance along +ve x axis from origin O.

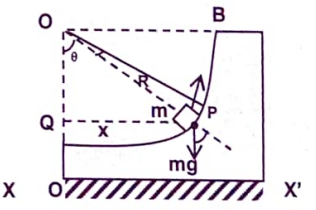


Sol. $F = \mu mg\cos\theta + mg\sin\theta$

$\mu = \mu_0 x = \mu_0 R\sin\theta$

Work done in moving block a small distance ds along track

$\int dW = \int Fds$

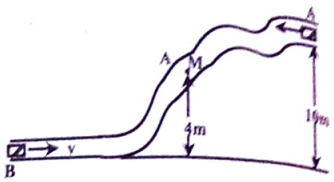


$$W=\int_0^{\pi/2} FR\,d\theta = \mu_0 mgR\int_0^{\pi/2}\sin\theta\cos\theta\,d\theta+\int_0^{\pi/2} mgR\sin\theta\,d\theta$$

$$W = mgR\left[\frac{\mu_0}{2}+1\right]$$

SOLVED PROBLEMS

1. **Two bodies A and B of each of mass 100 gm are allowed to move along a frictionless typical path as shown below. In order to have the same kinetic energy for both the bodies at M, the initial velocity that should be given to B, if A starts from rest is**



(A)10m/s **(B)** $\sqrt{2}$ **m/s**

(C)11 m/s **(D)** $10\sqrt{2}$ **m/s**

Sol. (KE_A) at $M = mg(10-4) = 6\ J$

(KE_B) at $M = 6\ J$ and (PE_B) at $M = \frac{1}{10}\times 10\times 4 = 4J$

$$\Rightarrow \left(\frac{1}{2}mv_B^2\right)_{\text{on ground}} = (T_{EB})_M = (6+4)J$$

$$\Rightarrow v = \sqrt{200} = 10\sqrt{2}\ m/s$$

$\therefore$ (D)

2. **A particle is moving on a circle of radius R such that at every instant the tangential and radial accelerations are equal in magnitude. If the velocity of the particle be v_0 at t = 0, the time for the completion of the half of the first revolution will be**

(A)R/v_0 **(B)** $(R/v_0)(1-e^{-\pi})$

(C) $(R/v_0)\,e^{-\pi}$ **(D)** $(R/v_0)(1-e^{-2\pi})$

Sol. $\omega^2 R = R\frac{d\omega}{dt}$, $\omega = \frac{d\theta}{dt} = \frac{\omega_0}{1-\omega_0 t}$

$$\Rightarrow T = \frac{R}{v_0}(1-e^{-\pi})$$

$\therefore$ **(B)**

3. **A force given by the relation F = 8t, acts on a body of mass 2 kg, initially at rest. Find the work done by this force on the body during first 2 seconds of its motion.**

(A) 64 J **(B) zero**

(C) –64 J **(D) none of these**

Sol. $F = 8t$

$$m.\frac{dv}{dt} = 8t$$

$$m\int_0^v dv = 8\int_0^t t\ dt$$

$$mv = 4t^2$$

$$\text{or } m\frac{dx}{dt} = 4t^2 \quad \Rightarrow \quad dx = \frac{4t^2}{m}dt = 2t^2 dt$$

$$W = \int F dx = \int_0^2 (8t\times 2t^2)dt = 64\ J$$

$\therefore$ (A)

4. A particle of mass m is moving horizontally with a constant velocity v towards a rigid wall that is moving in opposite direction with a constant speed u. Assuming elastic impact between the particle and wall, the work done by the wall in reflecting the particle is equal to

(A) (1/2) m(u+v)² **(B) (1/2) m(u+v)**

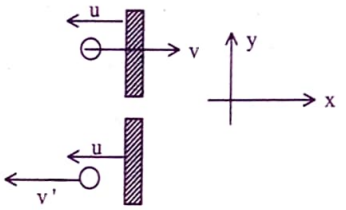
(C) (1/2) muv **(D) 2mu (u + v).**

Sol. Velocity of separation = – e (velocity of approach)

$\Rightarrow (-v'\hat{i}) - (-ui) = -e[(v_i) - (-u_i]$

$\Rightarrow u - v' = -1(v + u)$

$\Rightarrow v' = 2u + v$



$\therefore W = \Delta K.E. = \frac{1}{2}mv'^2 - \frac{1}{2}mv^2$

$= \frac{1}{2}m\left[(2u+v)^2 - v^2\right] = 2mu\,(u + v).$

$\therefore$ (D)

5. A body is acted upon by a force which is inversely proportional to the distance covered. The work done will be proportional to

(A) s **(B) s^2**

(C) $\sqrt{s}$ **(D) $k \ln(s/s_1)$**

Sol. $W = \int \vec{F}.d\vec{s} = \int_{s_1}^{s} \frac{k}{s} ds = k \ln(s/s_1)$

$\therefore$ (D)

6. A machine delivers power to a body which is proportional to instantaneous velocity 'v' of the body. If the body starts with a velocity which is almost negligible, then distance covered by the body is proportional to

(A) $\sqrt{v}$ **(B) $\sqrt[3]{\frac{v}{2}}$**

(C) $v^{5/3}$ **(D) v^2**

Sol. Power = $Fv = m\frac{dv}{dt}(v) \propto v$

$\Rightarrow mv\frac{dv}{dt} = k_0 v,$ where k_0 = constant

$\Rightarrow m\frac{dv}{dt} = k_0 \quad \Rightarrow \quad mv\frac{dv}{dx} = k_0$

$\Rightarrow v\,dv = \frac{k_0}{m}dx$

$\Rightarrow$ integrating both sides,

$\int_0^v v\,dv = \frac{k_0}{m}\int_0^x dx$

$\frac{v^2}{2} = \frac{k_0}{m}x$

Hence, displacement is proportional to square of instantaneous velocity.

$\therefore$ (D)

7. **The kinetic energy acquired by a mass m in travelling a certain distance d, starting from rest, under the action of a force F such that the force F is**
(A) directly proportional to t^2 **(B) independent of t**
(C) directly proportional to t^4 **(D) directly proportional to t**

Sol. $f = kt$

$$m\int_0^v dv = k\int_0^t tdt$$

$$mv = kt^2$$

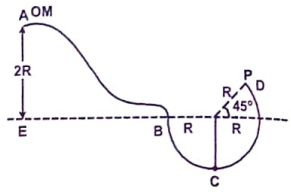
$$v = \frac{k}{m}t^2$$

$$K.E. = \frac{1}{2}mv^2 = \frac{1}{2}m\frac{k^2}{m^2}t^4 = \left(\frac{k^2}{2m}\right)t^4$$

$KE \propto t^4$

$\therefore$(C)

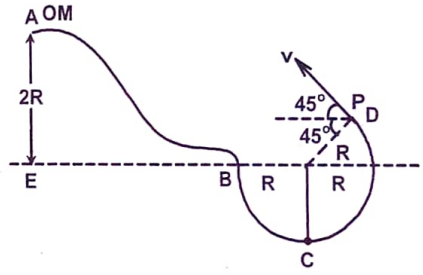
8. **A particle of mass m slides on a frictionless surface ABCD, starting from rest as shown in the figure. The part BCD is a circular arc. If it looses contact at point P, find the maximum height attained by the particle from point C.**



(A) $R\left[2+\frac{1}{2\sqrt{2}}\right]$ **(B)** $R\left[2-\frac{1}{2\sqrt{2}}\right]$

(C)3R **(D) none of these.**

Sol.. Apply conservation of energy



$$2mgR - mg\frac{R}{\sqrt{2}} = \frac{1}{2}mu^2$$

$$\Rightarrow u = \sqrt{4gR - \sqrt{2}gR}$$

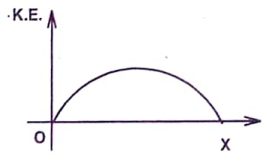
$$H_{max} = R + \frac{R}{\sqrt{2}} + \frac{u^2\sin^2 45°}{2g} = R + \frac{R}{\sqrt{2}} + \frac{4gR - \sqrt{2}gR}{2g\times 2}$$

$$= 2R + \frac{R}{\sqrt{2}} - \frac{R}{2\sqrt{2}} = R\left[2+\frac{1}{2\sqrt{2}}\right]$$

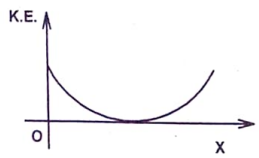
$\therefore$ (A)

9. **A projectile is fired with some velocity making certain angle with the horizontal. Which of the following graphs is the best representation for the kinetic energy of a projectile (K.E.) versus its horizontal displacement (x)?**

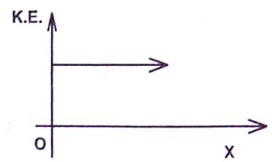
(A)



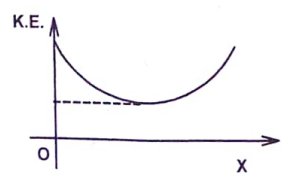
(B)



(C)



(D)



Sol. Velocity of a projectile at any instant of time (t) is

$$v^2 = v_x^2 + v_y^2 = (u\cos\theta)^2 + \left(u\sin\theta - g\frac{x}{u\cos\theta}\right)^2$$

$$\therefore KE = \frac{1}{2}mu^2 - mgx\tan\theta + \frac{gx^2}{u^2\cos^2\theta}$$

The given equation represents equation of a parabola

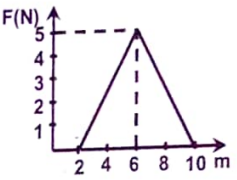
$\therefore$ (D)

Note: Answer can't be (B) as K.E. of a projectile never becomes zero.

10. A force shown in the F-x graph is applied to a 2 kg block horizontal as shown in the figure. The change in kinetic energy is

(A) 15 J(B) 20 J

(C) 25 J(D) 30 J



Sol. Change in kinetic energy $\Delta k = \frac{1}{2}\times(10-2)\times 5 = 20J$

$\therefore$(B)

11. The potential energy of a particle of mass 0.1 kg, moving along the x-axis, is given by U = 5x(x–4)J, where x is in meters. It can be concluded that

(A) The particle is acted upon by a variable force.

(B) The minimum potential energy during motion is –20 J

(C) The speed of the particle is maximum at x = 2m.

(D) The period of oscillation of the particle is (π/5) sec.

Sol. A, B, C, D

$$F = -\frac{dU}{dx} = -5(2x-4)$$

At mean position $F = 0 \Rightarrow x = 2m$

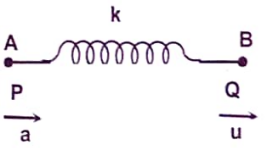
$U_{min} = -20$ J

$a = -50 \times 2(x-2)$

$\omega = 10$ rad/sec

$T = \pi/5$ sec

12. A massless spring of spring constant k has two ends P and Q as shown in the figure. End Q is being pulled by person B with constant speed u towards right. End P is being pushed by person A with constant acceleration a towards right starting from rest. Both starts simultaneously. Choose the correct option at the instant when B appears to be at rest with respect to A



(A) work done by B = $ku^4/3a^2$

(B) work done by A = $-5ku^4/24a^2$

(C) work done by A = $+5ku^4/24a^2$

(D) elastic potential energy stored in spring = $ku^4/8a^2$

Sol. A, B, D

$$\text{Work done by B} = \int_0^{u/a} k\left(ut - \frac{1}{2}at^2\right)udt = \frac{ku^4}{3a^2}$$

$$\text{Work done by A} = -\int_0^{u/a} k\left(ut - \frac{1}{2}at^2\right)atdt = -\frac{5ku^4}{24a^2}$$

$$\text{Energy stored in spring} = = \frac{1}{2}k\left(\frac{u^2}{2a}\right)^2 = \frac{ku^4}{8a^2}$$

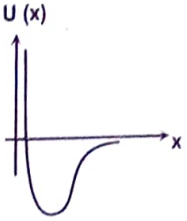
COMPREHENSION

The potential energy function for the force between two in a diatomic molecule can approximately be expressed as $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$, where a and b are positive constants, and x is the distance between the atoms. Answer the following question by selecting most appropriate alternative.

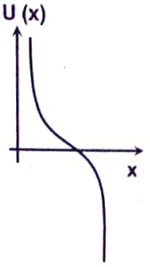
13. The graph between potential energy vs x will be

(A)

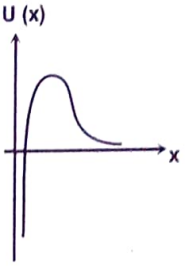
(B)



(C)



(D)



Sol. B

14. The dissociation energy of the molecule is (initially molecule is at rest at equilibrium)

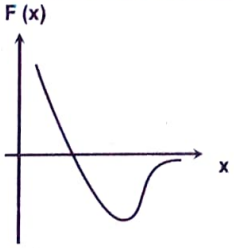
(A) $-\frac{b^2}{4a}$ **(B)** $-\frac{b^2}{2a}$

(C) $+\frac{b^2}{4a}$ **(D)** $+\frac{b^2}{2a}$

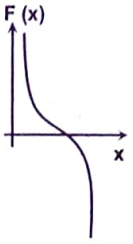
Sol. C

15. The graph between force between the atoms [F(x)] vs x will be

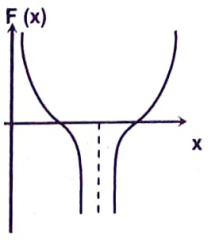
(A)



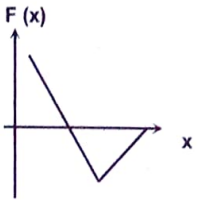
(B)



(C)



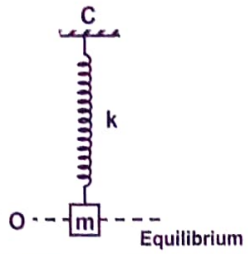
(D)



Sol. A

Matrix – Match Type

16. **An ideal spring-block system is suspended from point C as shown in the figure. The block is in equilibrium at t = 0 at position O. Match the column I to column II [Assume that acceleration due to gravity is g = 10 m/s^2]**

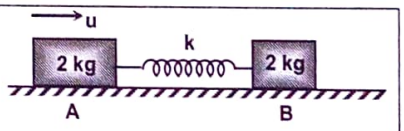


	Column – I		Column – II
(A)	Spring is pushed up by an external force F by a distance mg/3k to point A from point O	(p)	Magnitude of acceleration is g/3 just after removal of force.
(B)	Spring is pushed down by an external force F by a distance mg/3k to point B from point O.	(q)	The energy stored in spring is $\frac{8m^2g^2}{9k}$ just after removal of force.
(C)	Spring is pushed up by an external force F by a distance 4mg/3k to point C from point O	(r)	Magnitude of acceleration is 4g/3 just after removal of force.
(D)	Spring is pushed up by an external force F by a distance 7mg/3k to point D from point O	(s)	The energy stored in the spring is $\frac{2m^2g^2}{9k}$ just after removal of force
		(t)	The energy stored in the spring is $\frac{m^2g^2}{18k}$ just after the removal of force

Sol. (A) → (p, s) (B) → (p, q) (C) → (r, t) (D) → (q)

Answer questions 17, 18 and 19 by appropriately matching the information given in the three columns of the following table.

Figure shows a system with two identical masses connected by a spring which is initially in natural length. Now, block A is pushed towards block B with a speed 'u'. Column –I lists the value of u and spring constant k. Column –II lists the maximum compression of spring and column –III lists the (common) speed of blocks at maximum compression.



Column 1			Column 2		Column 3
(I)	u = 10 m/s, k = 400 N	(i)	20 cm	(P)	3 m/s
(II)	u = 10 m/s, k = 100 N	(ii)	50 cm	(Q)	5 m/s
(III)	u = 5 m/s, k = 2500 N	(iii)	100 cm	(R)	2.5 m/s
(IV)	u = 6 m/s, k = 900 N	(iv)	10 cm	(S)	7.5 m/s

17. **Pick correct option.**

(A) (IV) (ii) (Q) **(B) (IV)(i) (P)**
(C) (III) (ii) (P) **(D) (III) (iv) (Q)**

Sol. B

18. **Pick correct option corresponding to height value of spring's maximum compression.**

(A) (II) (iii) (Q) **(B) (I)(iii) (P)**
(C) (IV) (i) (R) **(D) (III) (iv) (S)**

Sol. A

19. **Pick correct option corresponding to lowest value of spring's maximum compression.**

(A) (I) (i) (S) **(B) (II) (iv) (S)**
(C) (III) (iv) (R) **(D) (IV) (i) (Q)**

Sol. C

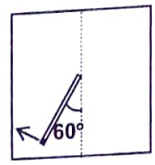
Numerical Type

20. **A light rod of length L, is hanging from the vertical smooth wall of a vehicle moving with acceleration $\sqrt{3}$ g having a small mass attached at it's one end is free to rotate about an axis passing through the other end. The minimum velocity given to the mass at it's equilibrium position so that it can complete vertical circular motion is $\sqrt{KgL}$. Find the value of K.**

Sol. 8

Conservation of energy $\frac{1}{2}mv^2 - mg\frac{\ell}{2} = mg\frac{\ell}{2} + m\sqrt{3}\left(2\ell\frac{\sqrt{3}}{2}\right)g$

$v = \sqrt{8g\ell}$



21. **A rod of length ℓ = 2m is maintained to rotate with a constant angular velocity ω = 1 rad/s about vertical axis passing through one end (fig). There is a spring of spring constant k = 1 N/m which just encloses rod inside it in natural length. One end of the spring is attached to axis of rotation. S is sleeve of mass m = 1kg which can just fit on rod. All surfaces are smooth. With what minimum kinetic energy (in J) sleeve should be projected so that it enters on the rod without impulse and completely compresses the spring.**

Sol. 6

For entering without jerk $v_2 = \ell\omega_0 = 2$rad/s

Using work energy theorem on sleeve after entering in the frame of rod.

$W_{spring} + W_{centrifugal} = \Delta K$

$$-\frac{1}{2}k\ell^2 - \frac{1}{2}m\omega^2\ell^2 = 0 - \frac{1}{2}mv_1^2 \qquad \Rightarrow v_1^2 = 8$$

Now $K = \frac{1}{2}m\left(v_1^2 + v_2^2\right) = 6$

Numerical Decimal Type

22. **A power output from a certain experimental car design to be shaped like a cube is proportional to the mass m of the car. The force of air friction on the car is proportional to Av^2, where v is the speed of the car and A is the cross-sectional area. On a level surface the car has a maximum speed v_{max}. Assume that all versions of this design have the same density. Then v_{max} is proportional to $m^{1/C}$. Find C.**

Sol. 00009.00

At v_{max}

$$Av_{max}^2 = F = \frac{km}{v_{max}}$$ (where k is a proportionalty constant)

$$m^{2/3}v_{max}^2 \propto \frac{m}{v_{max}}$$

$\therefore\ v_{max} \propto m^{1/9}$

23. **A car can pull a trailer of twice its mass up a certain slope at a maximum speed v = 1 m/s. Without the trailer the maximum speed of the car up the same slope is 2v. The resistance to the motion is proportional to mass and square of speed. If the car (without trailer) starts to move down the same slope, with its engine shut off, eventually it will acquire a constant speed v_t, then find v_t^2 (in m^2/s^2).**

Sol. 00005.00

For car + trailer system

$P_{max} = (3mg \sin\theta + 3kmv^2)v$

For car only

$P_{max} = (mg \sin\theta + 4kmv^2)2v$

$\therefore (3mg \sin\theta + 3kmv^2)v = (mg \sin\theta + 4kmv^2)2v$

$\therefore k = \dfrac{g \sin\theta}{5v^2}$

For down the incline $kmv_t^2 = mg \sin\theta$

$\therefore v_t^2 = 5v^2$

CONSERVATION OF MOMENTUM 4

CENTRE OF MASS OF A SYSTEM OF PARTICLES

$$\vec{r}_{cm} = \left[\sum_{i=1}^{n} m_i \vec{r}_i \Big/ \sum_{i=1}^{n} m_i\right]$$

$$\vec{v}_{cm} = \left[\sum_{i=1}^{n} m_i \vec{v}_i \Big/ \sum_{i=1}^{n} m_i\right]$$

$$\vec{a}_{cm} = \left[\sum_{i=1}^{n} m_i \vec{a}_i \Big/ \sum_{i=1}^{n} m_i\right] = \frac{\vec{F}_{ext}}{M}$$

CENTRE OF MASS OF CONTINUOUS BODIES

The co-ordinates x, y, z can be calculated as

$$X = \frac{1}{M}\int x\,dm\ ;\ Y = \frac{1}{M}\int y\,dm\ ;\ Z = \frac{1}{M}\int z\,dm$$

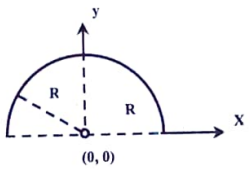
We will give the location of centre of mass for the following simple cases:

(a) **Uniform straight rod**

Located at mid point of the rod

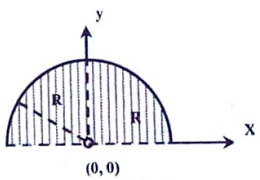
(b) **Uniform semi-circular wire**

Located at : $\left[0, \frac{2R}{\pi}\right]$



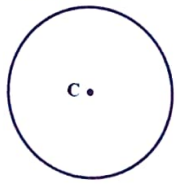
(c) **Uniform semi-circular plate**

Located at : $\left[0, \frac{4R}{3\pi}\right]$



(d) **Centre of mass of a uniform solid sphere or shell**

Located at the centre C of the sphere



CONSERVATION OF MOMENTUM

Impulse

If a force $\vec{F}$ acts on a body for time t, the impulse due to the force is defined as the change in linear momentum of the body produced by the force: $\vec{p}_f - \vec{p}_i = \int_0^t \frac{d\vec{p}}{dt}dt = \int_0^t \vec{F}\,dt$

Impulse is a vector quantity.

Law of Conservation of Linear Momentum

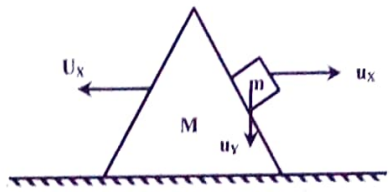
The net momentum of a system remains constant in the absence of net external force on the system.

If the net external force is zero along any one direction, the total momentum remains conserved in that particular direction. This is called the law of conservation of linear momentum for the system.

Applications of the law

(a) Suppose a small block of mass m slides down a wedge of mass M on frictionless ground. If there is no friction between the wedge and the ground we can apply conservation of linear momentum for the system

$m u_x = M U_x$



(b) **Collision:** When two smooth bodies collide with each other, impulsive forces act along the normal to the surfaces of contact and the momentum of individual bodies remains conserved along the tangential direction. If the surfaces are rough, frictional impulse is also to be taken into account, and momentum along the surface is not conserved for individual bodies.

Newton's Experimental Law

Coefficient of restitution,

$$e = \frac{\text{Magnitude of relative velocity of points of contact along the common normal after impact}}{\text{Magnitude of relative velocity of points of contact along the common normal before impact}}$$

where e is a constant for a particular collision and is known as the co-efficient of restitution. On the basis of conservation of kinetic energy of the system, collisions can be categorised into two types:

For elastic collisions:

$e = 1$ (kinetic energy of the system is conserved)

For inelastic collisions:

$e = 0$ (kinetic energy of the system is not conserved).

$e < 1$ (for partially inelastic collisions).

Simple Direct Collisions

Conserving momentum of the colliding bodies before and after the collision

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 \qquad \ldots (1)$$

Applying Newton's experimental law, we have $\frac{v_2 - v_1}{u_2 - u_1} = -e$

$$v_2 = v_1 - e(u_2 - u_1) \qquad \ldots . (2)$$

Putting Eq. (2) in Eq. (1), we obtain

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2\{v_1 - e(u_2 - u_1)\}$$

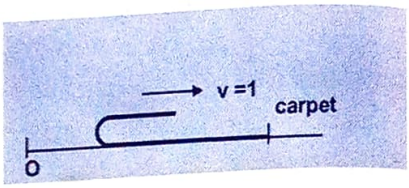
$$v_1 = u_1\frac{(m_1 - em_2)}{m_1 + m_2} + u_2\frac{m_2(1+e)}{m_1 + m_2} \qquad \ldots (3)$$

From Eq. (2), $v_2 = v_1 - e(u_2 - u_1)$

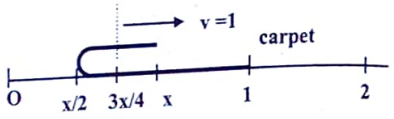
$$v_2 = u_1\frac{m_1(1+e)}{m_1 + m_2} + u_2\frac{(m_2 - m_1e)}{m_1 + m_2}$$

ILLUSTRATIVE PROBLEMS

1. A carpet of unit mass an unit length is laid on the floor. One end of the carpet is bent back and then pulled backwards with constant. Unit velocity, just above the part of the carpet which is still at rest on the floor. Find the speed of the centre of mass of the moving part. What is the minimum force needed to pull the moving part?



Sol. If the end of the folded part of the carpet is at a distance of x from the origin, the centre of mass of the folded part (which is moving) is at 3x/4 and hence has a speed of 0.75 m/s.

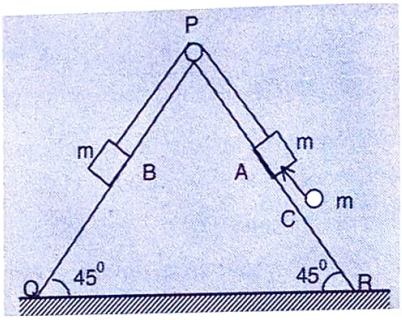


Now, force $= \frac{dp}{dt} = \frac{d(mv)}{dt} = m\left(\frac{dv}{dt}\right) = 0 + v\left(\frac{dm}{dt}\right) = 0.5$ N

Note that the moving part of the carpet starts from the origin and the whole carpet will be moving when it reaches x = 2 after two units of time giving $\left(\frac{dm}{dt}\right) = 0.5$

Note that the moving part of the carpet sets the next part into motion some energy is dissipated as heat.

2. PQR is a wedge fixed to the floor. Two blocks A & B of equal mass m = 1 kg is placed on either side of wedge connected through a string passing over the pulley P. Initially PB = PA. Another block C of mass m = 1 kg moving with a velocity 10 m/s collides inelasitcally with A. All the surfaces are smooth and pulley & string are massless.
(a) What should be the minimum length of the string AB so that block A does not hit the pulley?
(b) What is the loss in mechanical energy from the instant C hits A to the instant when B starts moving upward? (g = 10 m/s^2)



Sol. (a) Conservation of momentum,

$m_c v_c = (m_A + m_C) \times v_1$; v_1 = common velocity of (A + C) mass.

$\therefore \quad v_1 = \frac{m_c v_c}{m_A + m_C} = \frac{1 \times 10}{2} = 5$ m/s.

String slacks and B starts moving downward under acc = g sin45°.

$\therefore$ (A + C) moves under retardation of g sin45° and come to rest after time

$5 = 10 \times \frac{1}{\sqrt{2}} t \quad \Rightarrow \quad t = 0.707$ sec.

Distance moved in this time

$$s_1 = \frac{v_1^2}{2g\sin 45^0} = \frac{5 \times 5 \times \sqrt{2}}{2 \times 10} = \frac{5}{4}\sqrt{2} = 1.76 \text{ m.}$$

Now, string will again become taut and (A + C) will experience a jerk.

Velocity of B at this time = 5 m/s

$\therefore$ Conservation of momentum

for B : $1 \times 5 - J = 1 \times v_2$; J = impulse generated

for (A + C) : $J = 2 \times v_2$; v_2 = common velocity of (A+C) after string has become taut

$\therefore \quad v_2 = \frac{5}{3}$ m/s

Now (A + C) will start moving upward with v_2 velocity under retardation

$$a = \left(\frac{m_{(A+C)} - m_B}{m_{A+C} + m_B}\right) g \sin 45° = \left(\frac{2-1}{3}\right) \times 10 \times \frac{1}{\sqrt{2}} = \frac{10}{3\sqrt{2}} m/s^2$$

$\therefore$ (A + C) will come to rest after moving distance

$$s_2 = \frac{(5/3)\times(5/3)}{2\times(10/3\sqrt{2})} = \frac{5}{6\sqrt{2}} m = 0.59 m$$

$\therefore$ Net length risen by (A +C) = $S_1 + S_2 = 2.35$ m.

$\therefore$ Length of the string must be at least $\ell = 2 \times 2.35 = 4.7$ m

(b) Energy loss in collision $= \frac{1}{2} m_C v_c^2 - \frac{1}{2}(m_A + m_C) v_1^2$

$$= \frac{1}{2} \times 1 \times 100 - \frac{1}{2} \times 2 \times 25 = 25 J$$

Loss in impulsive jerk

$$= \frac{1}{2} m_B v_1^2 - \frac{1}{2}(m_B + m_A + m_C) v_2^2 = \frac{1}{2} \times 1 \times 25 - \frac{1}{2} \times 3 \times \frac{25}{9}$$

$$= \frac{25}{2}\left[1 - \frac{1}{3}\right] = \frac{25}{3} J$$

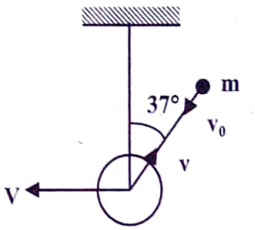
$\therefore$ Net loss $= 25 + \frac{25}{3} = \frac{4\times25}{3} = \frac{100}{3} = 33.33$ J.

3. An elastic ball of mass 'm' is suspended from a fixed point by an inextensible string. A small particle of same mass m , moving downwards at an angle of 37° with the vertical hits the ball directly with the velocity v_0. If the coefficient of restitution is 4/5,

(i) find the velocity of the ball just after the impact.

(ii) determine the impulsive tension (i.e. impulse) in the string at the instant of collision.

Sol. The particle after striking this ball will rebound with the velocity v exactly in the opposite direction.



Say V is the velocity attained by the ball after collision

Here the linear momentum remains conserved only in the horizontal direction.

$mv_0 \sin 37° + 0 = mV - mv \sin 37°$

Along common normal, $e(v_0 - 0) = (V \sin 37° + v)$

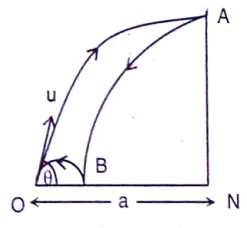
$V = (27/34)v_0$ and $v = (11/34)v_0$

Impulsive tension produced $= mv_0 \cos 37° + mv \cos 37°$

$$= mv_0 \frac{4}{5} + m\frac{11}{34} v_0 \frac{4}{5} = mv_0 \frac{18}{17}$$

4. A ball is projected from a point in a horizontal plane so as to strike a vertical wall at right angle A. after rebounding from the wall it strikes the horizontal plane once and returns to the point of projection just before second collision at horizontal surface. Find co-efficient of restitution for the two collisions, assuming it to be same for both the collision.

Sol. Let the distance of the point of projection be 'a' from the wall and angle of projection and velocity of projection be θ and u respectively.



As the ball strikes the wall at right angle, therefore a = half the horizontal range

$$\Rightarrow a = \frac{u^2 \sin 2\theta}{2g} \quad \ldots . (i)$$

and time of flight is $\frac{u \sin \theta}{g}$

After striking the wall, velocity of the ball becomes eu cosθ along horizontal.

Now $NB = eu\cos\theta \times \frac{u\sin\theta}{g} = \frac{eu\sin 2\theta}{2g} = ae.$

$OB = a - ae$ .... (ii)

After striking at B the vertical component becomes eu sin θ and the horizontal and the horizontal component remains unchanged.

Now the time of flight from B to O is given by

$$T = \frac{2eu\sin\theta}{g}$$

Range of the projectile is $eu\cos\theta \cdot \frac{2eu\sin\theta}{g} = \frac{e^2u^2\sin 2\theta}{g}$

Now $\frac{e^2u^2\sin 2\theta}{g} = a - ae = a(1-e) = \frac{u^2\sin 2\theta}{2g}(1-e)$ (from (i))

$2e^2 = 1 - e$

$2e^2 + 2e - e - 1 = 0$

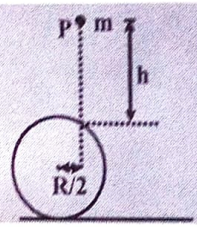
$2e(e+1) - 1(1+e) = 0$

$\Rightarrow (e+1)(2e-1) = 0$

$e = \frac{1}{2}$.

As $e = -1$ is not admissible.

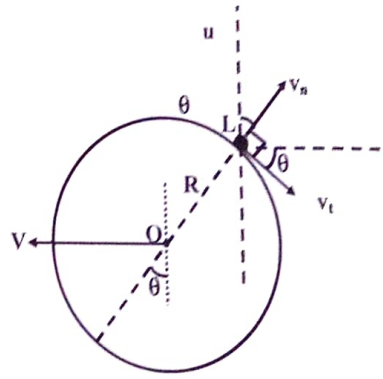
5. A tiny ball of mass m is released from the state of rest over a large smooth sphere of mass M and radius R, which is at rest on a smooth horizontal surface. If the ball strikes the sphere perfectly inelastically, find their velocities after collision.



Sol. Velocity attained by ball just before it strikes the surface

$u = \sqrt{2gh}$

$\therefore \sin\theta = \frac{R/2}{R} \Rightarrow \theta = 30^\circ$



After collision let the velocity of the sphere be V and the components of velocity of ball along common normal OL and along common tangent be v_n and v_t respectively.

In the absence of external force in the horizontal direction, the linear momentum conserved.

$0 + 0 = -MV + mv_n\sin\theta + mv_t\cos\theta$ (1)

Along the common normal at,

$V\sin\theta - (-v_n) = e[u\cos\theta - 0]$ (2)

$e = 0$ for inelastic collision.

$\therefore V\sin\theta = -v_n$

$\therefore v_n = -v/2$ (3)

Along the common tangent velocity components before and after collision remain same.

$\therefore u\sin\theta = v_t \Rightarrow v_t = \frac{u}{2}$ (4)

Using (3) and (4) in (1),

$$MV = m\left(-\frac{V}{2}\right)\frac{1}{2} + m\left(\frac{u}{2}\right)\frac{\sqrt{3}}{2}$$

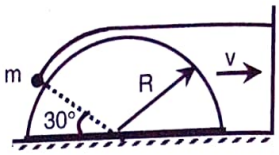
$$\left(M + \frac{m}{2}\right)V = m\frac{u\sqrt{3}}{4}$$

Velocity of sphere $V = \dfrac{m\sqrt{6gh}}{2(2M+m)}$ $\quad \because\ u = \sqrt{2gh}$

Velocity of ball $v = \sqrt{v_n^2 + v_t^2} = \sqrt{\dfrac{6ghm^2}{4(2M+m)^2} + \dfrac{gh}{2}}$

SOLVED PROBLEMS

1. **A point object of mass m is slipping down on a smooth hemispherical body of mass M and radius R. The point object is tied to a wall with an ideal string as shown. At a certain instant, speed of the hemisphere is v and its acceleration is a. Then speed v_P and acceleration a_P of a particle has value (Assume all the surfaces in contact are frictionless)**



(A) $v_P = v \sin 60°$ **(B)** $v_P = v$

(C) $a_P = a$ **(D)** $a_P = \sqrt{\left[\left(\frac{v^2}{R}\right)+\left(a\frac{\sqrt{3}}{2}\right)\right]^2+\left(\frac{a}{2}\right)^2}$

Sol. $x + R\theta = \text{constant}$

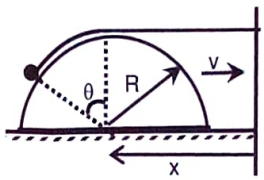
$$\frac{dx}{dt}+\frac{Rd\theta}{dt}=0$$

$$\therefore\ \omega=\frac{d\theta}{dt}=\frac{v}{R}$$

$\alpha = d\omega/dt = a/R$

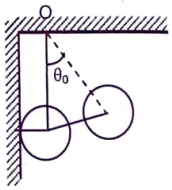
At $\theta = 60°$,

$$\therefore\ v_P = v \text{ and } a_p = \sqrt{\left(\frac{a\sqrt{3}}{2}+\frac{v^2}{R}\right)^2+\left(\frac{a}{2}\right)^2}$$



Hence (B) & (D) are correct

2. **An iron ball of mass m, suspended by a light inextensible string of length ℓ from a fixed point O, is shifted by θ_0 as shown so as to strike the vertical wall perpendicularly. The maximum angle made by the string with vertical after the first collision, if e is the coefficient of restitution, is**



(A) $\sin^{-1}\left\{1-e^2(1-\cos\theta_0)\right\}$ **(B)** $\cos^{-1}\left\{1-e^2(1-\cos\theta_0)\right\}$

(C) $\tan^{-1}\left\{1-e^2(1-\cos\theta_0)\right\}$ **(D) zero**

Sol. Speed just before collision $= \sqrt{2g\ell(1-\cos\theta_0)}$

Speed just after collision $= e\sqrt{2g\ell(1-\cos\theta_0)}$

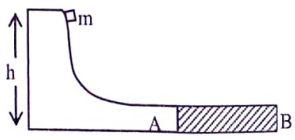
From COE,

$$\frac{1}{2}m e^2\,[2g\ell(1-\cos\theta_0)] = mg\ell\,(1-\cos\theta)$$

$$\therefore\ \theta=\cos^{-1}\left\{1-e^2(1-\cos\theta_0)\right\}$$

$\therefore$ (B)

3. **A particle of mass m is released from height 'h' on smooth quarter circular fixed wedge. The horizontal surface AB following the circular path at bottom of wedge is rough with coefficient of friction μ between surface and m. Find the distance from bottom of wedge where the mass will stop.**



(A) $\frac{2h}{\mu}$ (B) $\frac{h}{\mu}$

(C) $\frac{h}{\mu g}$ (D) none of these

Sol. By work energy theorem,

$mgh - \mu mgs = 0$

$\Rightarrow s = \frac{h}{\mu}$

$\therefore$ (B)

4. A ping-pong ball of mass m is floating in air by a jet of water emerging out of a nozzle. If the water strikes the ping-pong ball with a speed v and just after collision water falls dead, the mass flow rate of water in the nozzle is equal to:

(A) $\frac{2mg}{v}$ (B) $\frac{mv}{g}$

(C) $\frac{mg}{v}$ (D) None of these

Sol. The impact force $F = (\Delta p/\Delta t)$, where Δp = change of momentum of water of mass Δm striking the ball with a speed v during time Δt. Since water falls dead after collision with the ping-pong ball

$\Delta p = \Delta m\, v \qquad \Rightarrow F = v\frac{\Delta m}{\Delta t}$

where $\frac{\Delta m}{\Delta t}$ = rate of flow of water in the nozzle.

Since the ball is in equilibrium

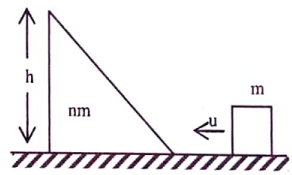
$F - mg = 0 \quad \Rightarrow F = mg$

$\Rightarrow \frac{v\Delta m}{\Delta t} = mg$

$\Rightarrow \frac{\Delta m}{\Delta t} = \frac{mg}{v}$.

$\therefore$(C)

5. A block of mass m is pushed towards a movable wedge of mass nm and height h, with a velocity u. All surfaces are smooth. The minimum value of u for which the block will reach the top of the wedge is



(A) $\sqrt{2gh}$ (B) n2gh

(C) $\sqrt{2gh\left(1+\frac{1}{n}\right)}$ (D) $\sqrt{2gh\left(1-\frac{1}{n}\right)}$

Sol. The centre of mass of the (block + wedge) must move with speed $= v_{cm} = \frac{mu}{m+nm} = \frac{u}{1+n}$

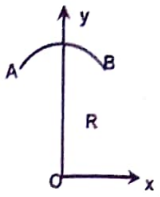
$\therefore \frac{1}{2}(m+nm)\, v_{cm}^2 = \frac{1}{2}mu^2 - mgh$

$\Rightarrow \frac{1}{2}m(1+n)\frac{u^2}{(1+n)^2} = \frac{1}{2}mu^2 - mgh$

$\Rightarrow u = \sqrt{2gh(1+1/n)}$

$\therefore$ (C)

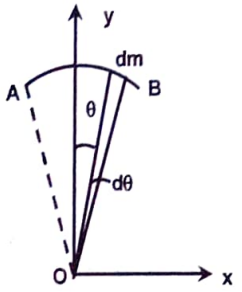
6. A circular arc (AB) of thin wire frame of radius R and mass M makes an angle of 90° at the origin. The centre of mass of the arc lies at



(A) $[0, (2/\pi)R]$ (B) $[0, (\sqrt{2}/\pi)R]$

(C) $[0, (2\sqrt{2}/\pi)R]$ (D) $[0, (4/\pi)R]$

Sol. By symmetry



$x_{cm} = 0$

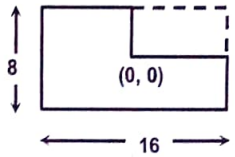
$dm = \lambda R\, d\theta$

$$y_{cm} = \frac{\int y\, dm}{\int dm} = \frac{1}{M}\int R\cos\theta\,\lambda R\, d\theta = \frac{R^2\lambda}{M}\int_{-\pi/4}^{+\pi/4}\cos\theta\, d\theta$$

$$= \frac{R^2\lambda}{R\times\lambda\times(\pi/2)}\left[2\sin\frac{\pi}{4}\right] = \frac{2R}{\pi}2\frac{1}{\sqrt{2}} = \frac{2\sqrt{2}R}{\pi}$$

$\therefore$ (C)

7. From a 16 cm × 8 cm rectangular uniform plane sheet, exactly one quarter of the sheet is removed as shown in the figure. If the origin be taken at the centre of the sheet considered as shown in the figure the coordinates of the centre of mass of the remaining sheet is



(A) (4/3, 2/3) (B) (0, 2/3)

(C) (4/3, 0) (D) (–4/3, –2/3)

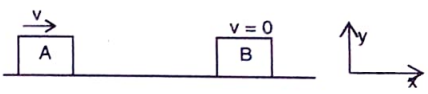
Sol. Let mass of the full sheet is 4M

$$X_{cm} = \frac{Mx - mx'}{M-m} = \frac{4M(0)-M(4)}{4M-M} = -\frac{4}{3}$$

$$y_{cm} = \frac{My - my'}{M-m} = \frac{4M(0)-M(2)}{4M-M} = -\frac{2}{3}$$

$\therefore$ (D)

8. A block B is at rest on the rough horizontal surface as shown. Another block A collides with B. Then, which of the following statement is correct?



(i) Impulsive friction acts on block B.

(ii) Conservation of momentum can not be applied on a system (A + B) along x-axis for the situation just before and just after the collision.

(iii) Conservation of momentum can be applied on system (A + B) along x-axis for the situation just before and just after the collision.

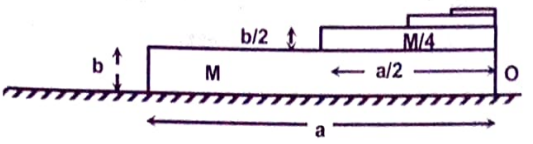
(iv) Block A always moves along positive x-axis after collision.

(A) (i) and (ii) (B) (i), (ii) and (iv)

(C) (i) (iv) (D) (iii)

Sol. $\therefore$ (D)

9. **Infinite number of bricks are placed one over the other as shown in the figure. Each succeeding brick having half the length and breadth of its preceding brick and the mass of each succeeding bricks being $(1/4)^{th}$ of the preceding one. Taking 'O' as the origin, the x coordinate of centre of mass of the system of bricks is at**



(A) –a/7
(B) 3a/7
(C) –3a/7
(D) –2a/7

Sol. $X_{com} = -\dfrac{\left[M\times\dfrac{a}{2}+\dfrac{M}{2}\times\dfrac{a}{4}+\dfrac{M}{16}\times\dfrac{a}{8}+\cdots\infty\right]}{M+\dfrac{M}{4}+\dfrac{M}{16}+\cdots\infty} = -\dfrac{3a}{7}$

$\therefore$ (C)

10. **A particle strikes a smooth horizontal surface at an angle of 45° with a velocity of 100 m/s and rebounds. If the coefficient of restitution between the floor and the particle is 0.57 then the angle which the velocity of the particle after it rebounds will make with the floor is**

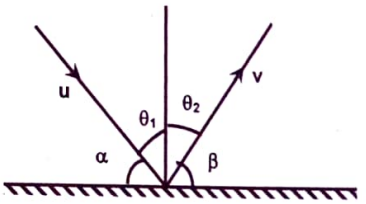
(A) 30°
(B) 45°
(C) 60°
(D) 90°

Sol. Along horizontal direction

$mu \sin\theta_1 = mv \sin\theta_2$ ... (1)

Along common normal

$eu\cos\theta_1 = v\cos\theta_2$ ... (2)

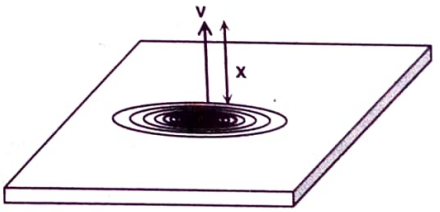


Solving Eqs. (1) and (2), $\theta_2 = \tan^{-1}\left(\dfrac{1}{0.57}\tan 45°\right) = 60°$

$\Rightarrow \beta = 30°$

$\therefore$ (A)

11. **A uniform rope of liner mass density λ and length ℓ is coiled on a smooth horizontal surface. One end is pulled up by an external agent with constant vertical velocity v. Choose the correct option(s).**



(A) Power developed by external agent as a function of x is $P = \lambda xgv$
(B) Power developed by external agent as a function of x is $P = (\lambda v^2 + \lambda xg)v$
(C) Energy lost during the complete lift of the rope is zero.
(D) Energy lost during the complete lift of the rope is $\dfrac{\lambda\ell v^2}{2}$.

Sol. B, D

F applied by external agent = Weight + thrust force = $\lambda xg + \lambda v^2$

Energy lost in the complete lift = $W_F - \Delta K.E. - \Delta U = \dfrac{\lambda\ell v^2}{2}$

12. Velocity of a particle of mass 2 kg changes from $\vec{v}_1 = -2\hat{i} - 2\hat{j}$ m/s to $\vec{v}_2 = (\hat{i} - \hat{j})$ m/s after colliding with a smooth plane surface.

(A) The angle made by the plane surface with the positive x-axis is $90° + \tan^{-1}\left(\frac{1}{3}\right)$

(B) The angle made by the plane surface with the positive x-axis is $\tan^{-1}\left(\frac{1}{3}\right)$

(C) The direction of change in momentum makes an angle $\tan^{-1}\left(\frac{1}{3}\right)$ with the positive x-axis.

(D) The direction of charge in momentum makes an angel $90° + \tan^{-1}\left(\frac{1}{3}\right)$ with the plane surface.

Sol. A, C

$\Delta\vec{P} = 2(\hat{i} - \hat{j}) - 2(-2\hat{i} - 2\hat{j})$

$= 6\hat{i} + 2\hat{j}$ kg-m/s

So, $\Delta\vec{P}$ makes angle $\tan^{-1}\left(\frac{1}{3}\right)$ with positive x-axis which is also the direction of normal force exerted on the ball. So the plane surface makes an angle $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{3}\right)$ with positive x-axis.

13. Identical constant forces push two identical cars A and B continuously from a starting line to a finish line. The cars move on a frictionless horizontal surface. If car A is initially at rest and car-B is initially moving right with speed v_0. Choose the INCORRECT statement.

(A) Car-A has the larger change in momentum.

(B) Car-B has the larger change in momentum.

(C) Both cars have the same change in momentum.

(D) Not enough information is given to decide.

Sol. B, C, D

Time taken by Car-A to reach the finish line is greater than that of Car-B, So

$$\int_0^{t_A} F\,dt > \int_0^{t_B} F\,dt \Rightarrow \Delta p_A > \Delta p_B$$

COMPREHENSION

Graphical solution of a two body head on collision

A block A of mass m moving with a uniform velocity v_0 strikes another identical block B kept at rest on a horizontal smooth surface as shown in the figure-(i). We can conserve linear momentum.

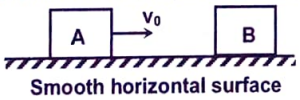


figure-(i)

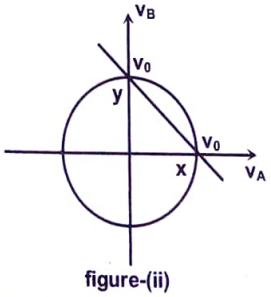


figure-(ii)

So, $mv_0 = mv_A + mv_B$ (v_A and v_B are the velocities of the blocks after collision)

$\therefore \quad v_0 = v_A + v_B \quad \ldots(i)$

If the collision is perfectly elastic

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2$$

$$\Rightarrow \quad v_0^2 = v_A^2 + v_B^2 \qquad \ldots(ii)$$

Both the above equations (i) and (ii) are plotted on $v_A - v_B$ plane as shown in figure (ii). This plot can be used to find the unknowns v_A and v_B.

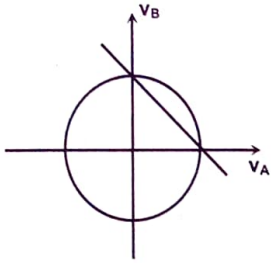
For example, the solution of the situation in figure (i) is $v_A = 0$, $v_B = v_0$ (point y in the plot)

Because $v_A = v_0$, $v_B = 0$ (point x in the plot) is not physically possible.

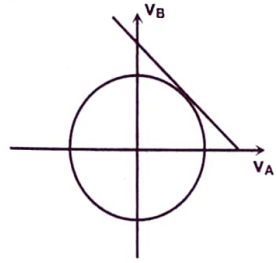
Now answer the following questions based on the above comprehension.

14. If the collision is perfectly inelastic, then the $v_A - v_B$ plot is

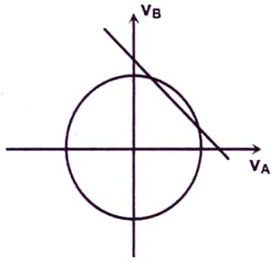
(A)



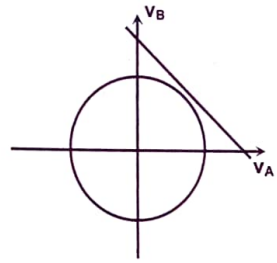
(B)



(C)



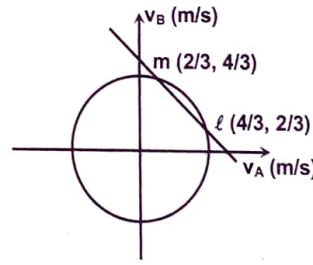
(D)



Sol. **B**

For perfectly in elastic collision both blocks have same velocity after collision.

15. In a situation, block A is moving with velocity 2 m/s and strikes another identical block B kept at rest. The $v_A - v_B$ plot for the situation is shown. m and ℓ are the intersection points whose v_A, v_B coordinates are given in the figure. The coefficient of restitution of the collision is



(A) 1/2

(B) 1/3

(C) 1

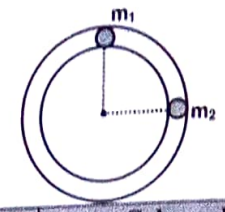
(D) Collision not possible

Sol. **B**

$$e = \frac{\frac{4}{3} - \frac{2}{3}}{2} = \frac{1}{3}$$

Matrix – Match Type

16. In a smooth horizontal groove two particles m_1 and m_2 collide first collision takes place at t = 0. Radius of the circular grove is R

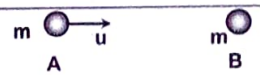


	Column I		Column II
(A)	If $m_1 = m_2$ and both are moving in opposite direction with speed v then time of 2nd collision (e =1)	(p)	$\frac{2\pi R}{V}$
(B)	If m_2 is at rest and $m_2 = 2m1$ and e = 1 then time of 2nd collision, m1 collides with velocity v.	(q)	$\frac{\pi R}{V}$
(C)	If m_2 is at rest, m_1 collides with velocity v, $m_1 = m_2$ and e = 1/2 then time of 2^{nd} collision.	(r)	$\frac{4\pi R}{V}$
(D)	If m_2 is at rest, m_1 collides with velocity v, $m_1 = m_2$ and e = 1 then time of 3^{rd} collision.	(s)	$\frac{3\pi R}{V}$
		(t)	$\frac{\pi R}{2V}$

Sol. (A) → (q) (B) → (p) (C) → (r) (D) → (r)

Answer questions 17, 18 and 19 by appropriately matching the information given in the three columns of the following table.

Figure shows a particle moving with speed u striking an identical particle kept at rest. Column –I lists four different values of e, the coefficient of restitution. Column –II lists the ratio of velocity of A and velocity of B just after collision and column –III lists the fraction of kinetic energy lost by A during collision.



Column 1			Column 2		Column 3
(I)	e = 1/4	(i)	0	(P)	48/64
(II)	e = 1/2	(ii)	3/5	(Q)	55/64
(III)	e = 0	(iii)	1/3	(R)	60/64
(IV)	e = 1	(iv)	1	(S)	64/64

17. Pick correct option.

(A) (I) (ii) (Q) **(B) (II)(i) (P)**
(C) (III) (iii) (R) **(D) (IV) (iv) (S)**

Sol. A

18. Pick correct option corresponding maximum value of the velocity ratio.

(A) (III) (iv) (P) **(B) (III)(ii) (P)**
(C) (IV) (iv) (P) **(D) (II) (ii) (S)**

Sol. A

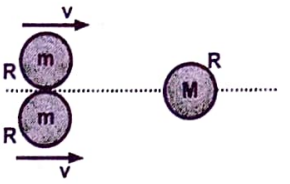
19. Pick correct option corresponding to maximum value of the fractional loss of K.E.

(A) (IV) (i) (Q) **(B) (IV) (ii) (P)**
(C) (IV) (i) (S) **(D) (IV) (iii) (P)**

Sol. C

Numerical Type

20. **Two identical discs of mass m and of radius R touch each other and move with the same velocity perpendicularly to the line segment which joins their centres of mass, along the surface of a horizontal smooth tabletop. There is a third disc of mass M and of radius R at rest, at a point on the perpendicular bisector of the line segment joining the centres of mass of the two moving discs as shown in the figure. The two moving discs collide elastically with the third one, which is at rest. There is no friction between the rims of the discs. What should the ratio of M/m be in order that after the collision the two discs of mass m move perpendicularly to their initial velocity?**



Sol. **3**

Conservation of momentum for (M +m)

$2mv_0 = Mv_1$ ...(i)

Newton's second law :

$v_0 \cos 30° = v_1 \cos 30° + v_2 \cos 60°$

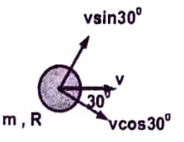
$\sqrt{3}v_0 = \sqrt{3}v_1 + v_2$ ...(ii)

For 'm': $v_0 \sin 30° = v_2 \sin 60°$ ...(iii)

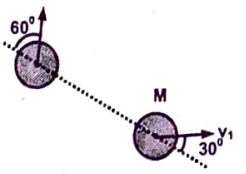
$\Rightarrow v_2 = \frac{v_0}{\sqrt{3}}, \; v_1 = \frac{2}{3}v_0$

from (i) $\frac{2mv_0}{M} = \frac{2v_0}{3}$

$\Rightarrow \left(\frac{m}{M}\right) = \frac{1}{3} \Rightarrow \left(\frac{M}{m}\right) = 3$

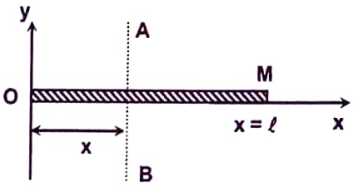


Before collision



After collision

21. **A non uniform rod OM (of length ℓ m) is kept along x-axis and rotating about an axis AB, which is perpendicular to rod as shown in the figure. The rod has linear mass density that varies with the distance x from left end of the rod according to** $\lambda = \lambda_0\left(\frac{x^3}{L^3}\right)$



Where unit of λ_0 is kg/m. $\frac{4\ell}{n}$ is the value of x so that moment of inertia of rod about axis AB (I_{AB}) is minimum. Find the value of n.

Sol. **5**

First Method:

$$I_{AB} = \int y^2 dm = \frac{\lambda_0}{\ell^3}\int_{-x}^{\ell-x}(y+x)^3 y^2 dy$$

$$\int (y+x)^3 y^2 dy = y^2\frac{(y+x)^4}{4} - \frac{1}{2}y(y+x)^4 dy$$

$$= \frac{y^2(y+x)^2}{4} - \frac{y}{10}(y+x)^5 + \frac{1}{10}(y+x)^5 dy$$

$$= \frac{y^2(y+x)^2}{4} - \frac{y(x+y)^5}{10} + \frac{(y+x)^2}{60}$$

$$\int_{-x_0}^{(\ell-x)} y^2(y+x)^3 dx = \frac{(L-x)^2(\ell)^4}{4} - \frac{(\ell-x)\ell^5}{10} + \frac{\ell^6}{60}$$

$$I_{AB} = \lambda_0\left[\frac{(\ell-x)^2}{4}\ell - \frac{1}{10}(\ell-x)\ell^2 + \frac{1}{60}\ell^3\right] = \lambda_0\ell\left[\frac{(\ell-x)^2}{4} - \frac{1}{10}(\ell-x) + \frac{1}{60}\ell^2\right]$$

$$\frac{dI_{AB}}{dx} = 0 \Rightarrow -\frac{(\ell-x)}{2} + \frac{\ell}{10} = 0 \Rightarrow 5\ell - 5x = \ell \Rightarrow 5x = 4\ell \Rightarrow x = \frac{4\ell}{5}$$

Second Method:

As we know that moment of inertia is minimum among all set of parallel axis, if axis is passing through centre of mass, so x must be distance of centre of mass.

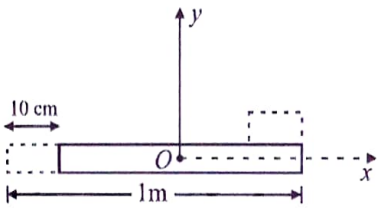
$$x_{CM} = \frac{1}{M}\int_0^{\ell} x\,dm \qquad \ldots.(i)$$

$$M = \int dm = \frac{\lambda_0}{\ell^3}\int_0^{\ell} x^3 dx = \frac{\lambda_0\ell}{4}, \text{ and } \int_0^{\ell} x\,dx = \frac{\lambda_0}{\ell^3}\int_0^{\ell} x^4 dx = \frac{\lambda_0\ell^2}{5}$$

Putting these values in equation (i) we have $x_{cm} = \frac{4\ell}{5}$

Numerical Decimal Type

22. **First 10 cm piece of a uniform meter strick is cut and kept on the other end as shown in figure. Find the shift in the x-coordinate of center of mass in cm.**

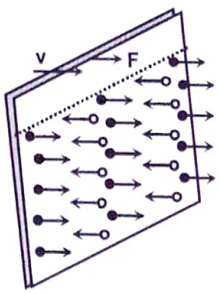


Sol. **00009.00**

Let initial location of center of mass is 0 then find location of cm

$$X_{cm} = \frac{\lambda(90\text{cm})(5\text{cm}) + \lambda(10\text{cm})(45)}{\lambda(90) + \lambda(10)} = 9 \text{ cm}$$

23. **A thin heavy metal plate is being bombarded by a perpendicular beam of gas particles from both sides as shown in the figure. The solid dots are representing the molecules hitting from left side and the faint dots are the molecules hitting from right side. The mass of these gas particles is $m = 10^{-26}$ kg and velocity before hitting is $v_0 = 5$ m/s. Volume density of the gas particles on both sides is $n = 10^{25}$ per m^3. Each beam has an area $A = 1\ m^2$ and the collisions are perfectly elastic. What is the external force F (in newton) required to move the plate with a constant velocity v = 2 m/s.**



Sol. **00008.00**

$$F = 2mn\eta A\left[(v_0+v)^2 - (v_0-v)^2\right]$$

$$= 2(10^{-26})(10^{25})(1)\left[(5+2)^2 - (5-2)^2\right]$$

$$= 0.2 \times 40 = 8 \text{ N}$$

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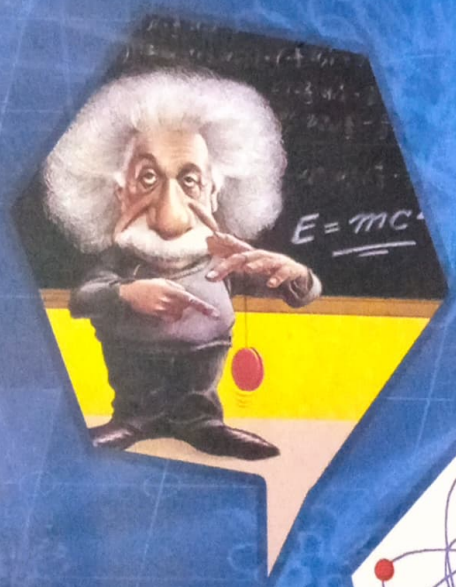
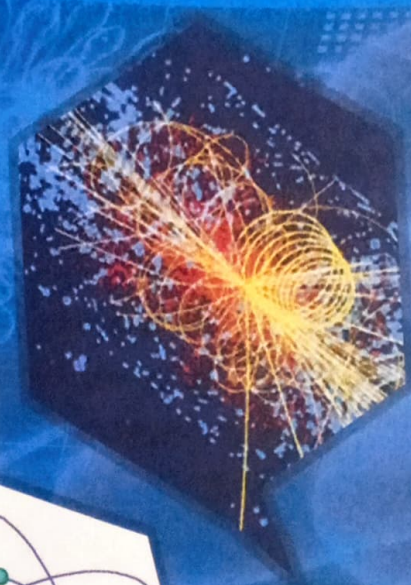
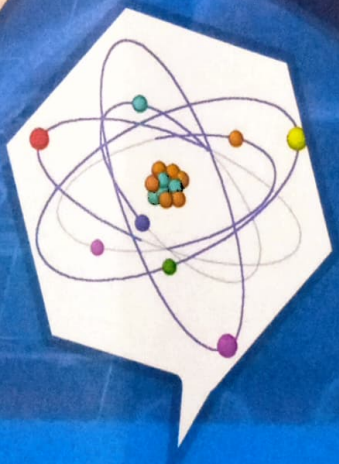






Edition 2020

PHYSICS

ROTATIONAL MECHANICS 5

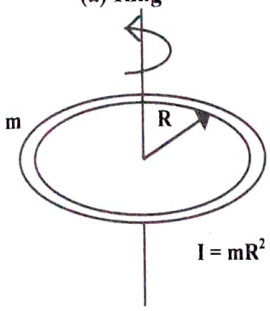
MOMENT OF INERTIA

Moment of inertia of a system of particles about an axis is defined by

$$I = \sum_{i=1}^{n} m_i r_i^2$$

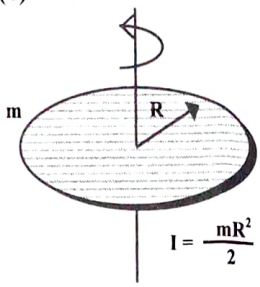
We shall give the moment of inertia for the following simple cases:

(a) Ring



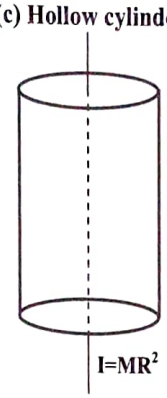
$I = mR^2$

(b) Disc



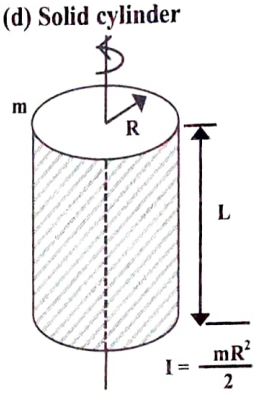
$I = \frac{mR^2}{2}$

(c) Hollow cylinder



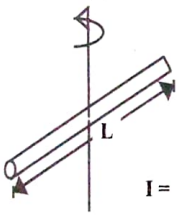
$I = MR^2$

(d) Solid cylinder



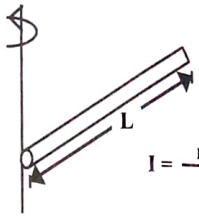
$I = \frac{mR^2}{2}$

(e) Rod



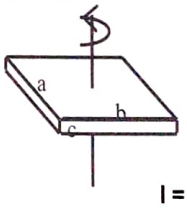
$I = \frac{mL^2}{12}$

(f) Rod



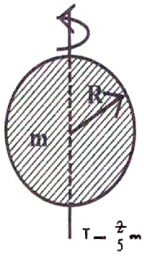
$I = \frac{mL^2}{3}$

(g) Solid box



$I = \frac{M(a^2+b^2)}{12}$

(h) Solid sphere



$I = \frac{2}{5} m D^2$

(i) Hollow sphere

$I = \frac{2}{3} m D^2$

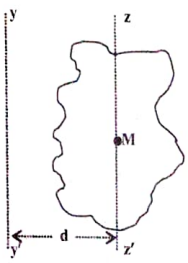
Parallel Axis Theorem

$$I = I_{cm} + md^2$$

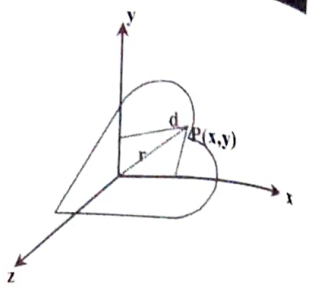
where m = total mass of the body

d = perpendicular distance between the parallel axes

I_{cm} = M.I. about a parallel axis passing through C.M.



Perpendicular Axis Theorem



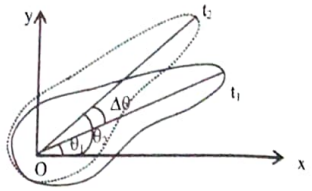
$I_z = I_x + I_y$

where x, y and z are three mutually perpendicular axes such that x and y lie in the plane of the laminar body.

Note: This theorem is applicable only for laminar bodies, e.g. ring, disc, rectangular, etc.

Moment of inertia of a combination about an axis can be found by simply adding the moment of inertia of different components about that axis

Angular Kinematics



Angular velocity, $\vec{\omega}_{\text{instantaneous}} = \frac{d\vec{\theta}}{dt}$

Angular acceleration, $\vec{\alpha}_{\text{instantaneous}} = \frac{d\vec{\omega}}{dt}$

The kinematics equations for constant angular acceleration:

$$\vec{\omega} = \vec{\omega}_0 + \vec{\alpha}t$$

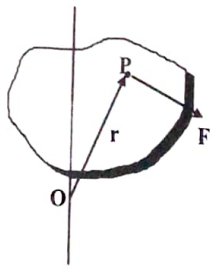
$$\vec{\theta} = \vec{\theta}_0 + \vec{\omega}_0 t + \frac{1}{2}\vec{\alpha}t^2$$

$$\omega^2 = \omega_0^2 + 2\vec{\alpha}\cdot\vec{\theta}$$

Torque about a Point

Torque, $\vec{\tau} = \vec{r} \times \vec{F}$, where r is the position vector of the point of application of the force.

Torque about an Axis

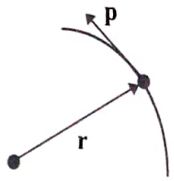


Torque, $\vec{\tau} = (\vec{r} \times \vec{F}) \cdot \hat{n}$, where $\hat{n}$ is a unit vector along the axis.

The angular acceleration α of any body about any inertial axis is equal to the net external torque acting on the body divided by the moment of inertia of the body about that axis.

$$\alpha = \frac{\vec{\tau}_{ext}}{I}$$

Angular Momentum



Angular momentum of a particle about any point is defined as

$\vec{L} = \vec{r}_i \times \vec{p}_i$, where $\vec{p} = m_i \vec{v}_i$

If a body is moving in pure rotation about an axis with angular velocity ω, then if the moment of inertia of the body about that axis is I the rotational kinetic energy of the body is equal to $\frac{1}{2}I\omega^2$ and the angular momentum of the body about that axis equal to $I\omega$.

Angular Impulse

If a torque τ acts about an axis on a body for time t, then the angular impulse is defined as the change in angular momentum produced by this torque.

$$J = \int_{t_1}^{t_2} \tau \, dt, \quad \text{where } J = \text{angular impulse.}$$

$$\Rightarrow I(\omega_2 - \omega_1) = L_2 - L_1 = \Delta L$$

Work Done by a Torque

If under the influence of a torque τ a body rotates through an angle $d\theta$, work done by the torque is equal to

$$dW = \tau\, d\theta$$

$$W = \int_{\theta_1}^{\theta_2} \tau d\theta$$

Also the power developed by a torque τ is given by

$P = \vec{\tau}.\vec{\omega}$, where ω is the instantaneous angular velocity of the body

Law of Conservation of Angular Momentum

If the net torque on a system about an axis is zero, the total angular momentum of the system about that axis remains unchanged.

While choosing conservation of angular momentum, you must be sure of the reference frame and the axis about which the angular momentum is being conserved. Remember that conservation of angular momentum is possible only when net torque about the chosen axis is zero.

Combined Translational and Rotational Motion

Suppose a body is undergoing rotation about an axis passing through its centre of mass and its moment of inertia being denoted by I_{cm} and angular velocity by ω_{cm}.

Let the centre of mass be moving with a velocity $\vec{v}_{cm}$ in the frame of earth and let the mass of the body be M.

Let the position vector of the centre of mass be $\vec{r}_{cm}$ in the frame of earth.

Then the following statements hold true

(a) Total kinetic energy = translational kinetic energy + rotational kinetic energy

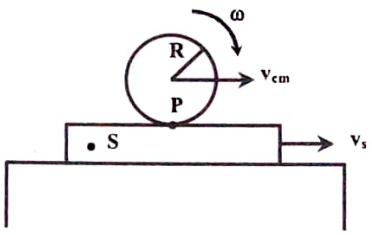
$$= \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$$

(b) Total angular momentum of the body

$$\vec{L}_P = I_{cm}\vec{\omega} + M\vec{r}_{cm} \times \vec{v}_{cm}$$

ROLLING

A body is said to be rolling when the velocity of its point of contact w.r.t. the surface on which it is placed is zero.



Consider a body rotating about its centre of mass with angular velocity ω_0 and the centre of mass moving with a velocity v_{cm} w.r.t. the earth, whereas the surface also moves with a velocity v_s w.r.t. earth.

For pure rolling,

$$v_{PS} = 0 = v_{PE} - v_{SE}$$

$$= v_{cm} - R\omega - v_s = 0$$

$$\therefore \quad v_{cm} = R\omega + v_s$$

If $v_s = 0$, it is reduced to the commonly known equation

$$v_{cm} = R\omega$$

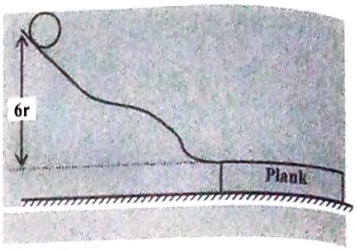
Conditions of Equilibrium for any Body

(a) The net external force acting on the body must be zero.

(b) The net torque acting on the body about any axis must be zero.

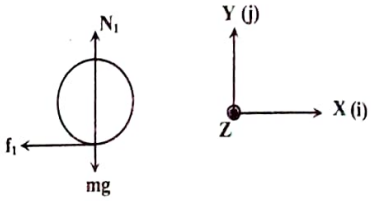
ILLUSTRATIVE PROBLEMS

1. **In the figure shown, a solid sphere of mass 'm' and radius r is released from a height 6r to slide down a smooth surface. A plank of same mass 'm' touches the horizontal portion of the surface at the ground. The co-efficient of friction between the plank and the sphere is μ and that between the plank and the ground is μ/4. Find the work done by the friction force between the plank and the ground till the sphere starts pure rolling on the plank. Neglect the height of the plank.**



Sol. The sphere will reach the plank with speed $v_0 = \sqrt{10gr}$ (in horizontal direction)

FBD of the sphere when it moves on the plank.



$N_1 = mg$ ... (i)

$\vec{a}_{sph} = -\mu g\,\hat{i}$

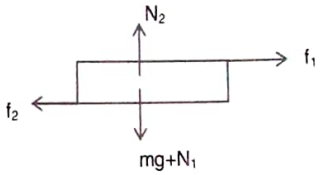
(as $f_1 = \mu N_1$ kinetic friction)

$\Rightarrow \vec{v}_{sph(t)} = (v_0 - \mu g t)\hat{i}$ ... (ii)

$\vec{\alpha}_{sph} = \mu mgr(-\hat{k})$

$\Rightarrow \vec{\omega}_{sph(t)} = -\dfrac{\mu mgr}{I_{cm}}t\hat{k} = \dfrac{-5\mu g}{2r}t\hat{i}$ ... (iii)

FBD of plank,



$N_2 = mg + N_1 = 2mg$ ... (iv)

$f_2 = \dfrac{\mu}{4}N_2 = \dfrac{\mu}{4}\times 2mg = \dfrac{\mu mg}{2}$ ... (v)

$\vec{a}_{p/k} = \dfrac{(f_1 - f_2)\hat{i}}{m} = \dfrac{\mu g}{2}\hat{i}$

$\Rightarrow \vec{v}_{P/k(t)} = \dfrac{\mu g}{2}t\hat{i}$

For pure cooling,

$\vec{v}_{sph} + \vec{\omega}_{sph} \times \vec{r} = \vec{v}_{p/k}$

(Here $\vec{r}$ is the position vector of the lower most point of the sphere with respect to the centre of the sphere).

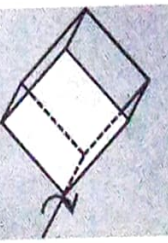
$\Rightarrow (v_0 - \mu g t)\hat{i} - \dfrac{5\mu g t}{2}\hat{i} = \dfrac{\mu g}{2}t\hat{i}$

$\Rightarrow t = \dfrac{v_0}{4\mu g}$

Distance moved by the plank during this time is $s = \dfrac{1}{2}\dfrac{\mu g}{2}\times\dfrac{v_0^2}{16(\mu g)^2} = \dfrac{v_0^2}{64\mu g}$

$\therefore\ W_{f2} = -\dfrac{\mu mg}{2}\times\dfrac{v_0^2}{64\mu g} = -\dfrac{mv_0^2}{128}$

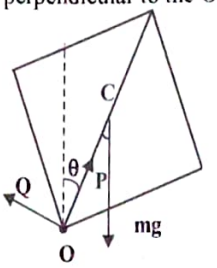
2. A cube of side a and mass M is pivoted on a horizontal axis such that it can rotate about one of its edges. Initially it is held in the highest position and then released. Find horizontal and vertical components of reaction of pivot on the cube at the moment when the reaction toward the centre of the cube becomes zero. Given that M.I of the cube about its edge is equal to $\frac{2}{3}Ma^2$.



Sol. Let P and Q be the components of reaction of pivot on the cube along and perpendicular to the OC.

M.I. of the cube about O is $\frac{2}{3}Ma^2$

$$\alpha = \frac{\tau}{I} = Mg\sin\theta \cdot \frac{a}{\sqrt{2}} \Big/ \frac{2}{3}Ma^2 = \frac{3}{2\sqrt{2}}\frac{g\sin\theta}{a}$$



Now along OC, $Mg\cos\theta - P = M\frac{a}{\sqrt{2}}\omega^2$

Perpendicular to OC, $Mg\sin\theta - Q = M\frac{a}{\sqrt{2}}\alpha$

$Q = (Mg\sin\theta)/4$

Integrating α, we get $\omega^2 = \frac{3g(1-\cos\theta)}{a\sqrt{2}}$ (1)

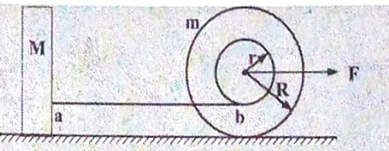
$$P = Mg\cos\theta - Ma\frac{3g(1-\cos\theta)}{2a} = \frac{-3}{2}Mg + \frac{5}{2}Mg\cos\theta$$

When $P = 0$ $\cos\theta = 3/5$, $\sin\theta = 4/5$

Now horizontal component of reaction is $P\sin\theta - Q\cos\theta = (-3/25)Mg$

Vertical component of reaction is $P\cos\theta + Q\sin\theta = (4/25)Mg$

3. A spool of thread of mass m and moment of inertia I is attached to the block of mass M. Along a horizontal line passing through the centre of the spool, a constant force F is applied as shown. The surface under the block is smooth while beneath the spool is rough.



Show that if $M = \frac{I}{r(R-r)}$, no frictional force will act on the spool. Now suppose $M = \frac{2I}{r(R-r)}$, find in which direction will the frictional force act? Also find in this case the rate of increase of length 'ab' of the thread at time t after the application of force. (No slipping occurs.)

Sol. For translational motion

$$F - T - f = ma_{c.m.} \quad \ldots(1)$$

For rotational motion

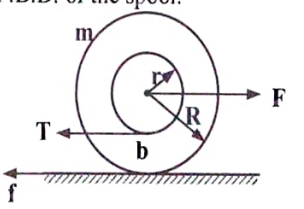
$$(Tr + fR) = I\alpha \quad \ldots(2)$$

As per given condition if frictional force is zero then,

$$\Rightarrow \alpha = \frac{a_{c.m.}}{R}$$

Hence, $Tr + fR = I\frac{a_{c.m.}}{R}$

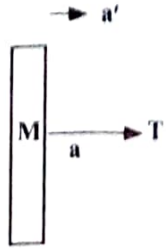
F.B.D. of the spool.



$$\Rightarrow T = \frac{Ia_{cm}}{Rr} - f.\frac{R}{r} \quad \ldots (3)$$

$\Rightarrow T = Ma'$

F.B.D. of M



Since string ab remains tangent, hence acceleration of point a and that of point b is same.

$$\Rightarrow a' = \frac{a_{cm}}{R}(R - r)$$

$$\Rightarrow T = M\frac{a_{cm}}{R}(R - r) \quad \ldots (4)$$

From Eqs. (3) and (4), we get

$$M\frac{a_{cm}}{R}(R - r) = I\frac{a_{cm}}{R.r} - f.\frac{R}{r} \quad \ldots (5)$$

Putting $f = 0$ in Eq. (5)

$$M = \frac{I}{r(R - r)}$$

Putting the value of $M = \frac{2I}{r(R - r)}$ in Eq. (5), we get

$$f = \frac{-I.a_{cm}}{R^2} \quad \ldots (6)$$

Negative sign indicates that friction is acting in the opposite direction of our assumption i.e. in forward direction.

From Eqs. (1), (3) and (6), we get

$$F - \frac{2I.a_{cm}}{Rr} + \frac{I.a_{cm}}{R^2} = m\,a_{cm}$$

$$\Rightarrow a_{cm} = \frac{F}{\left(m + \frac{2I}{Rr} - \frac{I}{R^2}\right)} \quad \Rightarrow \alpha = \frac{F}{R\left(m + \frac{2I}{Rr} + \frac{I}{R^2}\right)}$$

$\therefore T = ma'$

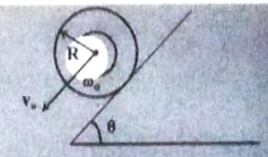
$$\frac{d^2(ab)}{dt^2} = \alpha(R - r) - a' = \frac{F(R - r)}{R\left(m + \frac{2I}{Rr} - \frac{I}{R^2}\right)} - \frac{2Ia_{cm}}{mRr}$$

$$= \frac{F(R - r)}{R\left(m + \frac{2I}{Rr} - \frac{I}{R^2}\right)} - \frac{2I}{mRr}\frac{F}{\left(m + \frac{2I}{Rr} - \frac{I}{R^2}\right)}$$

$$= \frac{F}{R\left(m + \frac{2I}{Rr} - \frac{I}{R^2}\right)}\left[R - r - \frac{2I}{mr}\right]$$

$\therefore$ Rate of increase in length ab $(= v_{ab}) = \frac{Ft}{R\left(m + \frac{2I}{Rr} - \frac{I}{R^2}\right)}\left[R - r - \frac{2I}{mr}\right]$.

4. A sphere of radius R is projected with a reverse spin ω_0 down a rough inclined plane with a speed v_0 for which coefficient of friction is $\mu > \tan\theta$, where θ is angle of the inclination. Show that it will turn back if



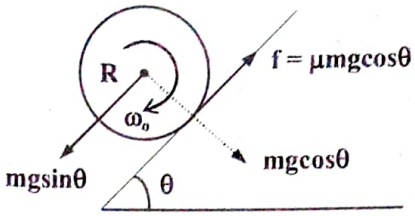
$$\omega_0 > \frac{5v_0\mu}{2R(\mu - \tan\theta)}$$

Sol. $\because \mu > \tan\theta$

$\Rightarrow \mu mg\cos\theta \geq mg\cos\theta\tan\theta$

$\Rightarrow f \geq mg\sin\theta$

Hence, the sphere will turn back if at the moment its centre of mass comes to stop it must have angular speed in same direction as ω_o has.



Let at time t, $v_{c.m.} = 0$

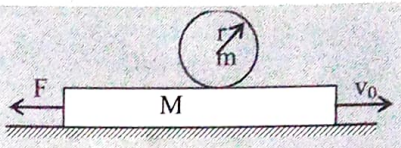
$\Rightarrow v_o + (g\sin\theta - \mu g\cos\theta)t = 0$

$$\Rightarrow t = \frac{v_o}{g(\mu\cos\theta - \sin\theta)} \qquad (1)$$

further, $\omega_o - \frac{\mu mg\cos\theta}{I_{c.m.}}t > 0 \qquad (2)$

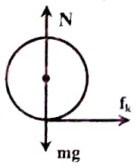
From Eq. (1) and (2), we get $\omega_o > \frac{5v_o\mu}{2R(\mu - \tan\theta)}$.

5. A plank of mass M is moving on a smooth horizontal surface with speed 'v_0'. At t = 0 a sphere of mass 'm' and radius 'r' is gently placed on it and simultaneously a constant horizontal force 'F' is applied on the plank in the opposite direction of v_0. Find the time at which sphere starts rolling on the plank. The coefficient of friction between the plank and the sphere is μ.



Sol. Till rolling starts kinetic friction acts on the bodies.

F.B.D. of sphere for no vertical acceleration N = mg ... (1)



$\Rightarrow f_k = \mu mg$ (as $f_k = \mu N$)

$\Rightarrow a = \mu g$

$v = \mu gt$

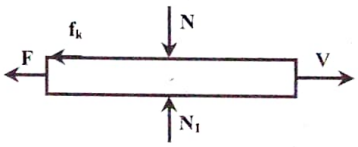
Also $\alpha = \frac{\mu mgr}{I} \Rightarrow \omega = \frac{\mu mgrt}{I}$

The velocity of the lower most point of the sphere at time t is

$v_s = v + \omega r$

$v_s = \mu gt + \frac{\mu mgr^2}{I}t$

The F.B.D. of the plank



$$\Rightarrow a_P = -\left(\frac{F + f_k}{M}\right) = -\left(\frac{F + \mu mg}{M}\right)$$

$$v_P = v_0 - \left(\frac{F + \mu mg}{M}\right)t$$

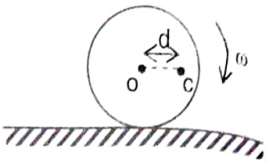
For no slipping, $v_s = v_P$

$$\Rightarrow \mu gt + \frac{\mu mgr^2}{I}t = v_0 - \left(\frac{F + \mu mg}{M}\right)t$$

$$\Rightarrow t = \frac{v_0}{\left[\frac{7}{2}\mu g + \left(\frac{F + \mu mg}{M}\right)\right]}.$$

SOLVED PROBLEMS

1. A solid sphere is rolling on a rough surface, whose centre of mass is at C at a certain instant. Find at that instant it has angular velocity ω. Its radius is R. Find the angular acceleration at that instant mass of sphere is m



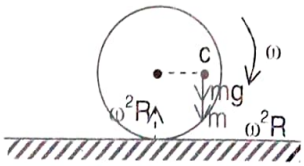
(A) $\frac{5}{2}\frac{gd}{R^2}$ **(B)** $\frac{5}{7}\frac{gd}{R^2}$

(C) $\frac{5}{2}\frac{d\left(g+\omega^2R\right)}{R^2}$ **(D)** $\frac{5}{7}\frac{d\left(g+\omega^2R\right)}{R^2}$

Sol. [$m\omega^2R$ is pseudo force]

If we will take torque about P then we have to apply pseudo force $m\omega^2R$ on centre of mass because this axis has

Acceleration = ω^2R towards 0.



Then $T_P\,\alpha = (mg + m\omega^2R)d$

$$\alpha\ \frac{5}{7}\frac{d\left(g+\omega^2R\right)}{R^2}$$

Hence (D) is correct

2. When a body rolls without sliding up an inclined plane, the frictional force is

(A) directed up the plane **(B) directed down the plane**

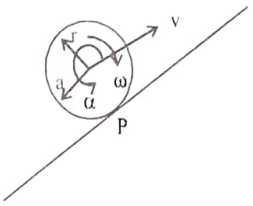
(C) zero **(D) dependent on its velocity**

Sol. Since the body rolls without sliding,

$a_p = 0$ and $v_p = 0$

$\Rightarrow v - r\omega = 0$ and $a - r\alpha = 0$

$\Rightarrow v = r\omega$ and $a = r\alpha$



Since the body acceleration 'a' is downwards parallel to the plane, therefore angular acceleration of the body must be in anticlockwise sense. That means the friction must act up the plane to produce an anticlockwise torque to produce anticlockwise angular acceleration

∴(A)

3. A particle m starts with zero velocity along a line y = 4d. The position of particle m varies as x = A sinωt. At ωt = π/2, its angular momentum with respect to the origin

(A) mAωd **(B) mωd/A**

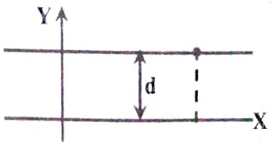
(C) mAd/ω **(D) zero**

Sol. At $\omega t = \pi/2$

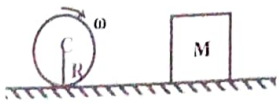
$u = A\omega\cos t = 0$

So, angular momentum of particle m with respect to the origin should be zero.

∴(D)



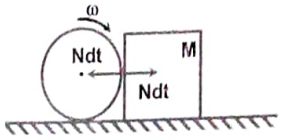
4. **A sphere of radius R and mass M collides elastically with a cubical block of mass M and side 2R. The entire system is on a smooth horizontal ground. Given that the sphere was rolling without slipping with an angular velocity ω at the time of collision. The velocities of the sphere and the block after the collision are**



(A) $\omega_{sphere} = 0,\ v_{sphere} = 0,\ v_{block} = v$ **(B)** $\omega_{sphere} = \omega,\ v_{sphere} = 0,\ v_{block} = v$

(C) $\omega_{sphere} = 0,\ v_{sphere} = 0,\ v_{block} = 0$ **(D)** $\omega_{sphere} = \frac{\omega}{2},\ v_{sphere} = \frac{v}{2},\ v_{block} = \frac{v}{2}$

Sol. When the sphere collides with the block due to elastic collision, the translational kinetic energy of the sphere gets totally transferred to the block, but rotational kinetic energy of the sphere does not change.



Hence, $\omega_{sphere} = \omega,\ v_{sphere} = 0,\ v_{block} = v$

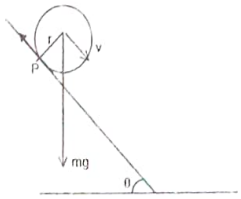
$\therefore$ (B)

5. **A body of mass m and radius r is released from rest along a smooth inclined plane of angle of inclination θ. The angular momentum of the body about the instantaneous point of contact after a time t from the instant of release is equal to**

(A) mgrt cos θ **(B) mgrt sin θ**

(C) (3/2) mgrt sin θ **(D) None of these**

Sol. Since the surface is frictionless, the body does not rotate about its centre of mass. Only it slides with certain velocity v parallel to the surface



$\therefore\ \omega = 0$ and $v = (g \sin \theta)\, t$

$\Rightarrow$ The angular momentum after a time t is given as

$L = mvr \quad \Rightarrow \quad L = mvr$

$\Rightarrow L = mg\, rt \sin \theta$

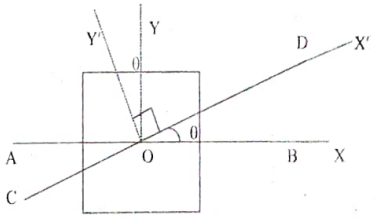
$\therefore$(B)

6. **Let I be the moment of inertia of a uniform square plate about an axis AB that passes though its centre and is parallel to two of its sides. CD is a line in the plane of the plate that passes through the centre of the plate and makes an angle θ with AB. The moment of inertia of the plate about the axis CD is then equal to**

(A) I **(B) I $\cos^2 \theta$**

(C) I $\sin^2 \theta$ **(D) I $\cos^2(\theta/2)$**

Sol. According to the perpendicular axis theorem



$I_Z = I_x + I_y$

Since the plate is quite symmetrical about x′ and y′, $I_{x'} = I_{y'}$

$\Rightarrow I_z = I'_x + I'_y = 2I'_x = 2I'_y$

$\Rightarrow I'_x = I'_y = I_z/2$

Similarly, $I_x = I_y = I_z/2$

$\therefore$ The required M.I. = I

where $\frac{I_z}{2} = I$

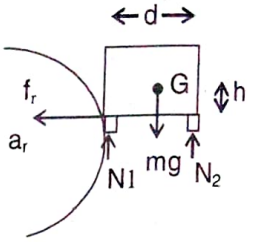
$\therefore$(A)

7. **When a car negotiates a curve, the normal force exerted on the inner and outer wheels are N_1 and N_2, respectively . Then N_1 / N_2 is**

(A) = 1 **(B). < 1**

(C) >1 **(D) zero**

Sol. Referring to the free body diagram we obtain,



$$\sum F_y = N_1 + N_2 - mg = 0$$

$$\Rightarrow N_1 + N_2 = mg \quad \ldots (1)$$

Along radial, $\sum F_r = ma_r$

$$\Rightarrow f_r = \frac{mv^2}{r} \quad \ldots (2)$$

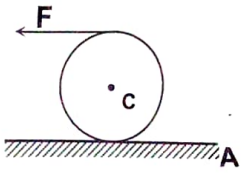
$$\sum \tau_G = 0$$

$$\Rightarrow N_1 \frac{d}{2} + f_r h - N_2 \frac{d}{2} = 0 \quad \ldots (3)$$

Using $N_2 - N_1 = \frac{2h}{d}\left(\frac{mv^2}{r}\right)$ and $N_2 = mg/2 + \frac{h\ mv^2}{dr} \Rightarrow \frac{N_1}{N_2} < 1.$

$\therefore$(B)

8. **A massless string is wrapped around a hollow cylinder having mass m and radius r. The cylinder is kept on a rough horizontal surface (coefficient of friction is μ). A constant force F is applied as shown in the figure. In case of pure rolling, the friction force acting on the bottom most point of the cylinder is**

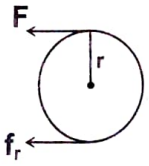


(A)zero **(B) μmg**

(C)μmg/2 **(D) 3μmg/2**

Sol. $F + f_r = ma_c \quad \ldots (1)$

Anticlockwise is taken as positive.

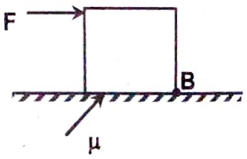


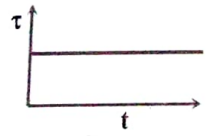
$$R(F - f_r) = I\frac{a_c}{R}, \text{ where } I = mR^2 \quad \ldots (2)$$

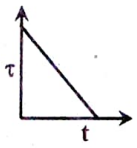
So, $f_r = 0$

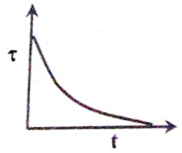
$\therefore$(A)

9. **A cubical box of side length L rests on a rough horizontal surface having coefficient of friction μ. A variable horizontal force $F = \alpha t$ is applied on the top of the block as shown in the figure, where α is a constant and t is time. The coefficient of friction is sufficient so that the block does not slide before toppling. The graph between torque due to normal reaction about end B and time before toppling start is**



(A) 

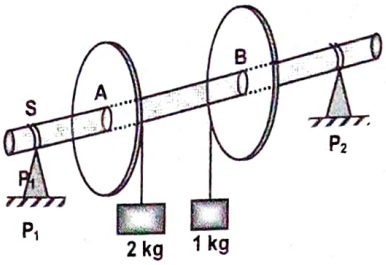
(B) 

(C) 

(D) None of these

Sol. (B)

10. Two discs A and B each of mass 2 kg and radius 0.5 m are rigidly fixed to the ends of a thin shaft S, which is supported in horizontal position with the help of two smooth bearings at points P_1 and P_2. Two blocks of masses 2 kg and 1 kg are connected to the light cords wrapped around the discs as shown in the figure. There is no slipping between the cord and the discs. If the system is released from rest from the position shown, then mark the correct statement(s). (Take $g = 10\ m/s^2$)



(A) Power developed by tension in cord connected to 2 kg block is, negative.
(B) Acceleration of each block is $2\ m/s^2$.
(C) Total work done by tension on the discs is zero during the motion.
(D) Acceleration of each block is $1\ m/s^2$.

Sol. A, B

$20 - T_1 = 2a$ (for 2 kg block) ....(i)

$T_2 - 10 = 1a$ (for 1 kg block) ....(ii)

$$T_1\frac{1}{2} - T_2\frac{1}{2} = (2)\left(\frac{1}{2}\right)^2\frac{1}{2}(2)\frac{a}{1/2} \text{ (for the shaft + disc system)}$$

So, $T_1 - T_2 = 2a$ ....(iii)

After solving these three equation

$a = 2\ m/s^2$

11. A uniform ball of radius R rolls without slipping between the rails such that the horizontal distance is $\sqrt{3}$ R between the two contact points of the rails of the ball. Figure (a) shows the front view of the ball and figure (b) shows the side view of the ball. v_{CM} is the velocity of centre of mass of the ball and ω is the angular velocity of the ball after rolling down a distance 2h along the incline then

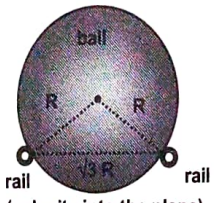


figure (a)

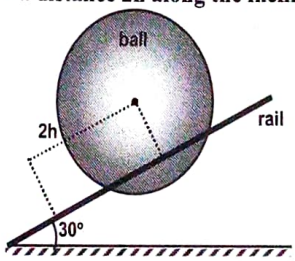


figure (b)

(A) $v_{CM} = \omega R$

(B) $v_{CM} = \omega\frac{R}{2}$

(C) $v_{CM} = \sqrt{\frac{10gh}{13}}$

(D) $v_{CM} = \sqrt{\frac{10gh}{7}}$

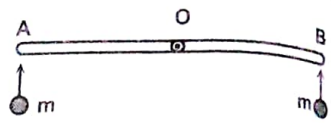
Sol. B, C

$$v_{CM} = \omega\sqrt{R^2 - \frac{3R^2}{4}} = \omega\frac{R}{2} \text{ (from pure rolling)}$$

$$mgh = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}\frac{2}{5}mR^2\omega^2 \text{ (from conservation of energy)}$$

$$\therefore v_{CM} = \sqrt{\frac{10gh}{13}}$$

COMPREHENSION



Two identical uniform rods OA and OB each of length ℓ and mass m are connected to each other by a massless pin connection (both the rods can rotate about O which is free to move), that allows free rotation. The assembly is kept on a frictionless horizontal plane. Now two point masses each of mass m moving with speed u perpendicular to the AB and hit the assembly inelastically at points A and B as shown in the figure.

12. The speed of the centre of mass of the whole assembly just after the collision will be

(A) u/4 **(B) u/2**

(C) u **(D) none of these**

Sol. B

13. Find the angular speed of rods just after the collision

(A) $\frac{3}{4}\frac{u}{\ell}$ **(B)** $\frac{6}{5}\frac{u}{\ell}$

(C) $\frac{3}{2}\frac{u}{\ell}$ **(D) zero.**

Sol. B

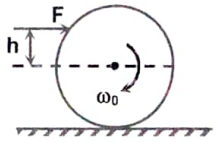
14. Find the speed of centre of rod AO just after the collision.

(A) $\frac{8}{10}u$ **(B)** $\frac{u}{5}$

(C) $\frac{7}{10}u$ **(D)** $\frac{u}{2}$

Sol. B

15. A cue stick applies a horizontal force F continuously on an object kept on a rough horizontal surface horizontally at a distance h above the centre as shown in the figure. Match the following.



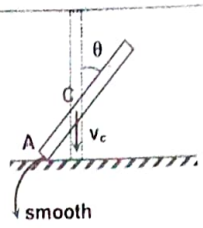
Column I	Column II
(A) If h = R and object is ring	(p) the object may roll with slipping.
(B) If h = R/2 object is sphere	(q) object will roll without slipping.
(C) If h = 2R/5 and object is sphere.	(r) friction will be in forward direction.
(D) If h = R/2 and object is cylinder	(s) friction will be in backward direction
	(t) After a certain value of F the body will be rolling with slipping.

Sol. (A) → (q) (B) → (p, r, t) (C) → (q) (D) → (q)

Answer questions 16, 17 and 18 by appropriately matching the information given in the three columns of the following table.

A thin rod of mass m, length ℓ was initially kept vertical on a smooth floor. It is tipped slightly so that it starts falling as shown. At any instant, the angle of rod with vertical is θ.

Column –I lists some values for the combination (θ, ℓ), column –II lists the corresponding values of the rod's angular speed ω, column –III lists the corresponding values for the speed of point A(shown in the diagram)



Column 1			Column 2		Column 3
(I)	θ = 60°, ℓ = 2m	(i)	$\sqrt{\frac{60}{13}}$ rad/s	(P)	Zero
(II)	θ = 60°, ℓ = 4m	(ii)	$\sqrt{15}$ rad/s	(Q)	$\sqrt{\frac{30}{13}}$ m/s
(III)	θ = 90°, ℓ = 2m	(iii)	$\sqrt{\frac{15}{2}}$ rad/s	(R)	$\sqrt{\frac{60}{13}}$ m/s
(IV)	θ = 90°, ℓ = 4m	(iv)	$\sqrt{\frac{120}{13}}$ rad/s	(S)	$\sqrt{\frac{120}{13}}$ m/s

16. Pick correct option.

(A) (I) (ii) (R) **(B) (II)(iii) (S)**
(C) (III) (i) (Q) **(D) (IV) (iii) (P)**

Sol. D

17. Pick the option for minimum value of V_A.

(A) (IV) (iii) (Q) **(B) (III)(ii) (P)**
(C) (IV) (i) (P) **(D) (III) (iv) (Q)**

Sol. B

18. Pick the option for maximum value of V_A.

(A) (II) (i) (R) **(B) (II) (ii) (R)**
(C) (I) (i) (R) **(D) (III) (ii) (R)**

Sol. A

Numerical Type

19. A uniform, solid cylinder having mass M and radius R is pulled by a horizontal force F acting through the center as shown. The cylinder rolls to the right without slipping. What is the magnitude of the force of friction between the cylinder and the ground?

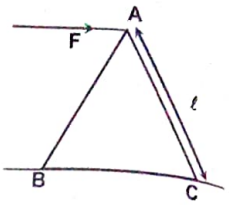
(A) F/4 **(B) F/3**
(C) F/2 **(D) 2F/3**

Sol. B

$F - f = Ma,\ fR = \frac{MaR}{2}$

$\Rightarrow\ f = F/3$

20. A triangular frame consists of three identical rods, each of mass m and length ℓ. It rests upright on a horizontal smooth surface with its lower right corner against a stop about which the frame could pivot. A horizontal force of magnitude F is applied to the upper corner of the frame as shown in the figure. What is the largest value of F may have without causing the frame to pivot upward about the stop?

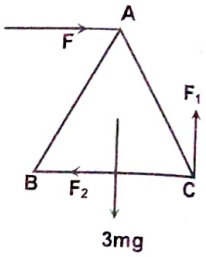


(A) $\sqrt{3}$ mg (B) 3 mg

(C) $\frac{mg}{\sqrt{3}}$ (D) $\frac{3}{2}mg$

Sol. A

Since system is in rotational equilibrium so net torque on the system about C is zero's

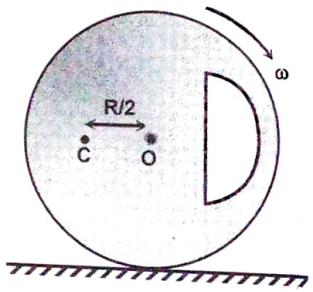


$$F_{max} \times \frac{\ell\sqrt{3}}{2} - 3mg\frac{\ell}{2} = 0$$

$$\Rightarrow F_{max} = \sqrt{3}mg$$

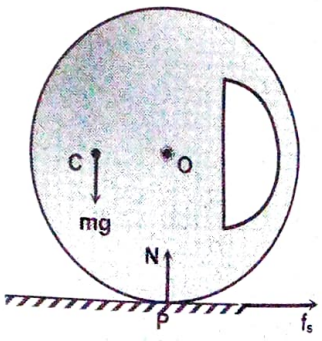
Numerical Decimal Type

21. A disc shaped body (having a hole as shown in the figure) of mass m = 10 kg and radius R = $\frac{10}{9}$m is performing pure rolling motion on a rough horizontal surface. In the figure point O is geometrical center of the disc and at this instant the centre of mass C of the disc is at same horizontal level with O. The radius of gyration of the disc about an axis passing through C and perpendicular to the plane of the disc is $\frac{R}{2}$ and at the instant shown the angular velocity of the disc is $\omega = \sqrt{\frac{g}{R}}$ rad/sec in clockwise sense. g is gravitation acceleration = 10 m/s^2. Find angular acceleration α(in rad/s^2) of the disc at this instant.



Sol. **00006.00**

considering torque of real and pseudo force in the frame of P,



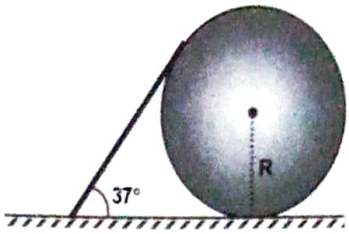
$$\tau = mg\frac{R}{2} + m\omega^2 R\frac{R}{2}$$

$$= mg\frac{R}{2} + m\frac{g}{R}R\frac{R}{2} = mgR$$

$$I_p = m\left(\frac{R}{2}\right)^2 + m\left(R^2 + \frac{R^2}{4}\right) = \frac{3}{2}mR^2$$

$$\therefore \alpha = \frac{2mgR}{3mR^2} = \frac{2g}{3R} = 6 \text{ rad/s}^2$$

22. A stick of mass density $\lambda = 8$ kg/m rests on a disc of radius $R = 20$ cm as shown in the figure. The stick makes an angle $\theta = 37°$ with the horizontal and is tangent to the disc at its upper end. Friction exists at all points of contact and assume that it is large enough to keep the system at rest. Find the friction force (in Newton) between the ground and the disc. (take $g = 10\ m/s^2$)



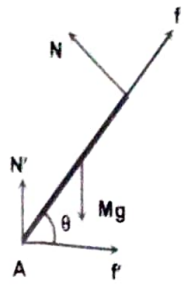
Sol. **00006.40**

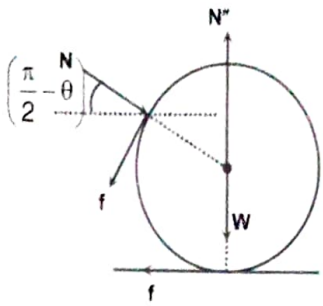
$\tau_A = 0$

$$N = \frac{Mg}{2}\cos\theta$$

For disc, $N \sin\theta = f + f\cos\theta$

$$\therefore \quad f = \frac{N\sin\theta}{1+\cos\theta} = \frac{Mg\sin\theta\cos\theta}{2(1+\cos\theta)}$$





$M = \lambda L$ and $L = \dfrac{R}{\tan\dfrac{\theta}{2}}$

$$f = \frac{1}{2}\lambda g R\cos\theta$$

$$= \frac{1}{2}\times 8\times 10\times\frac{2}{10}\times\frac{4}{5}$$

$$= \frac{32}{5} = 6.40\ N$$

GRAVITATION

6

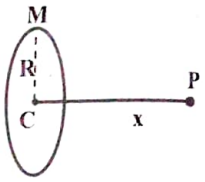
GRAVITATIONAL POTENTIAL

Potential due to a point mass m at a distance r is given by

$$V = -GM/r$$

Potential Due to a Thin Ring

$$V = -\frac{GM}{\sqrt{R^2 + x^2}}$$

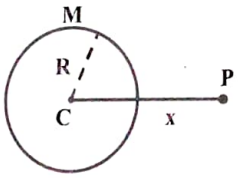


Gravitational Potential due to a Shell

(i) Outside the shell: $V = -GM/r \quad (r > R)$

(ii) On the surface of the shell: $-GM/R$

(iii) Inside the shell: $V = -GM/R$



Potential Due to the Solid Sphere

(i) Outside the sphere

$$V = -\frac{GM}{r}$$

(ii) Inside the shell,

$$V = -\frac{3GM}{R^3}\left(\frac{R^2}{2} - \frac{r^2}{6}\right)$$

(iii) On the surface, $V = -\frac{GM}{R}$

Gravitational potential energy of a system of two particles of masses m_1 and m_2 separated by a distance r

$$U = -\frac{Gm_1m_2}{r}$$

GRAVITATIONAL FIELD (E)

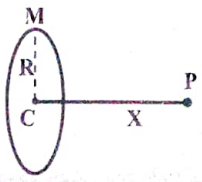
Field created by a point mass M at a distance r from the particle is given by

$$|E| = \frac{GM}{r^2}$$

A particle of mass m in a gravitation field E_G experiences a force equal to $\vec{F} = m\vec{E}_G$

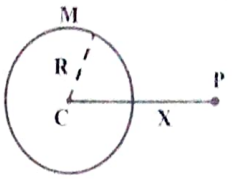
Field due to a circular ring

$$|E_G| = \frac{GM}{\sqrt{(R^2 + X^2)^3}}$$



Field Due to a Sphere

In a spherical shell:



For $X > R$, $\vec{E}_G = \frac{GM}{X^2}(-\hat{x})$

For $X \le R$, $\vec{E}_G$ = zero

In a solid sphere:

For $X > R$, $\vec{E}_G = \frac{GM}{X^2}(-\hat{x})$

For $X \le R$, $\vec{E}_G = \frac{GMX}{R^3}(-\hat{x})$

The gravitational field intensity due to earth is called acceleration due to gravity.

GRAVITY

If we consider the earth to be a perfect sphere of radius R_e and mass M_e, then the force acting on a particle of mass m, situated at a distance X from the centre of the earth is given by

$$\vec{F} = \frac{GM_e mX}{R_e^3}(-\hat{x}) \quad (X < R)$$

$$\vec{F} = \frac{GM_e m}{X^2}(-\hat{x}) \quad (X \ge R)$$

$$\vec{F} = \frac{GM_e m}{R^2}(-\hat{x}) \quad (X = R) \text{ (on the surface of the earth)}$$

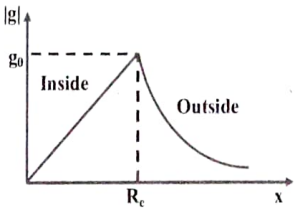
$$|F| = \frac{GM_e m}{R^2} = mg_0$$

$\therefore$ g_0 = acceleration due to gravity on the surface of the earth = $\frac{GM_e}{R^2}$

Variation in the Acceleration due to Gravity

(a) Inside the earth: $g = g_0 \frac{X}{R_e}$ (X is distance from the centre of the earth)

(b) Outside the earth: $g = g_0\left(\frac{R_e}{X}\right)^2$



(c) Variation of acceleration due to gravity close to the earth's surface,

$g = g_0\left(1 - \frac{h}{R}\right)$ [inside the earth]

If h is the small distance measured from the earth's surface.

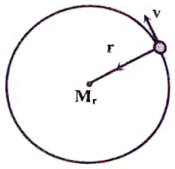
$g = g_0\left(1 - \frac{2h}{R}\right)$ [outside the earth]

(d) Variation of gravity with latitude α,

$g' = g_0 - R\omega^2\cos^2\alpha$, where g_0 is the acceleration due to gravity at that place in absence of spinning and

$\omega = \frac{2\pi}{T}$; T = 24 hrs

MOTION OF PLANETS AND SATELLITES IN CIRCULAR ORBITS



Planets are considered to revolve round the earth in nearly circular orbits.

The gravitational force of attraction provides the necessary centripetal acceleration.

$$F = \frac{GMm}{r^2} = m\left(\frac{v^2}{r}\right)$$

$$\Rightarrow \quad v^2 = \frac{GM}{r}$$

Time period of revolution,

$$T = \frac{2\pi r}{v} = \frac{2\pi r}{\sqrt{GM_e/r}} = \frac{2\pi}{\sqrt{GM_e}} r^{3/2}$$

Squaring, $T^2 = \frac{4\pi^2 r^3}{GM_e}$

$\therefore \quad T^2 \propto r^3$ [Kepler's third law]

K.E. of the satellite is given by, $K = +\ GM_e m/2r$

P.E. of the satellite is given by, $U = -GM_e m/r$

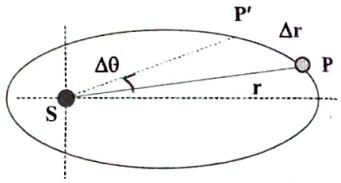
Escape velocity: $\frac{1}{2}mv_e^2 - \frac{GM_e m}{R} \geq 0$

$$\therefore \quad v_e \geq \sqrt{\frac{2GM_e}{R}}$$

Note: Direction of projection does not matter.

In case of problems related to elliptical orbits, conservation of angular momentum and conservation of mechanical energy should be used.

KEPLER'S LAWS OF MOTION



1. All planets move round the sun in elliptical orbits such that the sun remains at one of the foci.
2. An imaginary line joining the planet and the sun sweeps out equal areas in equal interval of time.
3. The square of time period of revolution of the planet is directly proportional to the cube of its mean distance from the sun.

Kepler's Second Law

$\Delta r = r\Delta\theta$

$A = \frac{1}{2}r^2\Delta\theta$;

The areal velocity

$$\frac{\Delta A}{\Delta t} = \frac{1}{2}r^2\frac{\Delta\theta}{\Delta t} = \text{constant}$$

$$\frac{dA}{dt} = \frac{1}{2}r^2\frac{\Delta\theta}{dt} = \frac{1}{2}r^2\omega = \frac{1}{2m}mr^2\omega = \frac{L}{2m}$$

where L = instantaneous angular momentum = $mr^2\omega$

$$\therefore \quad \frac{dA}{dt} = \frac{L}{2m}$$

Since, the line of action of the gravitational force passes through the axis

$\therefore \quad \tau = 0$ or L = constant

$$\therefore \quad \frac{dA}{dt} = \text{constant}$$

Note: Kepler's law of areal velocities is identical with the law of conservation of angular momentum.

ILLUSTRATIVE PROBLEMS

1. A satellite close to earth is in orbit above the equator with a period of rotation of 1.5 hrs. If it is above a point P on the equator at some time, after what time it will be above P again?

Sol. Let ω_0 = the angular velocity of earth above its axis $= (2\pi/24)$ rad/hr
Let ω be the angular velocity of the satellite.
$\Rightarrow \omega = 2\pi/1.5$
For a satellite rotating from west to east (same as earth), the relative angular velocity $\omega_1 = \omega - \omega_0$
$\therefore$ Time period of rotation relative to earth $= 2\pi/\omega_1 = 1.6$ h
Now, for a satellite rotating from east to west (opposite to earth) the relative angular velocity $\omega_2 = \omega + \omega_0$.
Time period of rotation relative to earth $= 2\pi/\omega_2 = (24/17)$ hr

2. Two satellites of the Earth move in a common plane along circular orbits, the radii being r and r - Δr (Δr ≪ r). What is the time interval between their periodic approaches to each other over the minimum distance Take M_e to be the mass of the Earth. ($M_e = 6 \times 10^{24}$ kg, r = 7000 km, Δr = 70 km).

Sol. $T = 2\pi\sqrt{\dfrac{r^3}{GM}} = 1.62$ hrs; $dT = \dfrac{2\pi}{\sqrt{GM}} \times \dfrac{3}{2} r^{1/2}\, dr$

$$\frac{dT}{T} = \frac{3}{2}\frac{dr}{r}$$

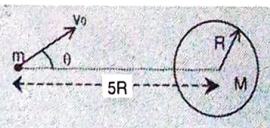
$$dT = \frac{3}{2} \times \frac{70}{7000} \times T = 0.015\ T.$$

Let first satellite catch up with the second after n revolution, then
$nT = (n + 1)(T - dT)$

$$\frac{n}{n+1} = 0.985 \Rightarrow n = 65.67$$

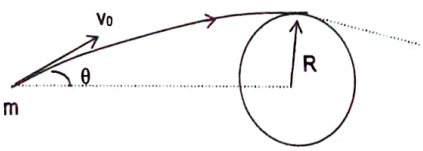
$\therefore$ Periodic time of approach $= nT = 106.38$ hrs.

3. A spaceship is sent to investigate a planet of mass M and radius R. While hanging motionless in space at a distance 5R from the center of the planet, the spaceship fires an instrument package with speed v_0 as shown in the figure. The packages has mass m, which is much smaller than the mass of the spaceship. For what angle θ will the package just graze the surface of the planet?



Sol. Let the speed of the instrument package is v when it grazes the surface of the planet.
Conserving angular momentum of the package about the centre of the planet

$$m v_0 \times 5R \sin(\pi - \theta) = mvR \sin 90^\circ$$



Conserving mechanical energy,

$$-\frac{GMm}{5R} + \frac{1}{2}mv_0^2 = -\frac{GMm}{R} + \frac{1}{2}mv^2$$

$$\Rightarrow \frac{1}{2}m(v^2 - v_0^2) = \frac{4GMm}{5R}$$

$$v^2 - v_0^2 = \frac{8GM}{5R}$$

Substituting the value of v from Eq. (1) in Eq. (2)

$$25\, v_0^2 \sin^2\theta - v_0^2 = \frac{8GM}{5R}$$

$$\Rightarrow \sin\theta = \frac{1}{5}\sqrt{1+\frac{8GM}{5v_0^2R}}$$

$$\text{or} \quad \theta = \sin^{-1}\left[\frac{1}{5}\sqrt{\left(1+\frac{8GM}{5v_0^2R}\right)}\right].$$

4. A body is projected vertically upwards from the surface of earth with a velocity sufficient to carry it to infinity. Calculate the time taken by it to reach a height 'h'.

Sol. If at a distance 'r' from the centre of the earth the body has velocity v, by conservation of mechanical energy

$$\frac{1}{2}mv^2+\left(-\frac{GMm}{r}\right)=\frac{1}{2}mv_e^2+\left(-\frac{GMm}{R}\right)$$

$$\Rightarrow v^2 = v_e^2 + \frac{2GMm}{R}\left(\frac{R}{r}-1\right)$$

Now, $v_e = \sqrt{2gR}$ and $g = \frac{GM}{R^2}$

$$\Rightarrow v^2 = 2gR + 2gR\,(R/r - 1)$$

$$v = \sqrt{\frac{2gR^2}{r}}$$

$$\Rightarrow \frac{dr}{dt} = R\frac{\sqrt{2g}}{\sqrt{r}}$$

$$\Rightarrow \int_0^t dt = \frac{1}{R\sqrt{2g}}\int_R^{R+h}\sqrt{r}\,dr$$

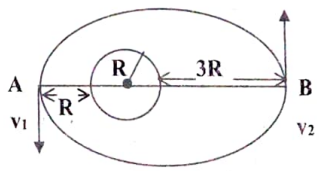
$$\Rightarrow t = \frac{1}{3}\sqrt{\frac{2R}{g}}\left[\left(1+\frac{h}{R}\right)^{3/2}-1\right]$$

5. The minimum and maximum distances of a satellite from the centre of the earth are 2 R and 4 R, respectively, where R is the radius of earth and M is the mass of the earth. Find
(a) its minimum and maximum speeds.
(b) radius of curvature at the point of maximum distance.

Sol. (a) Applying conservation of angular momentum

$$mv_1(2R) = mv_2(4R)$$

$$v_1 = 2v_2 \qquad \ldots(1)$$



From C.O.E.

$$\frac{1}{2}mv_1^2 - \frac{GMm}{2R} = \frac{1}{2}mv_2^2 - \frac{GMm}{4R} \qquad \ldots(2)$$

Solving Eqs. (1) and (2),

$$v_2 = \sqrt{\frac{GM}{6R}}, \quad v_1 = \sqrt{\frac{2GM}{3R}}$$

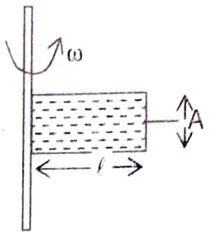
(b) If r is the radius of curvature at point B

$$\frac{mv_1^2}{r} = \frac{GMm}{(4R)^2}$$

$$r = \frac{16v_2^2R^2}{GM} = \frac{8R}{3} \text{ (putting value of } v_2)$$

SOLVED PROBLEMS

1. **A closed cylinder of length 'ℓ', containing a liquid of variable density $\rho(x) = \rho_0(1+\alpha x)$ is rotating with constant angular velocity ω. Find the net force exerted by the liquid on the axis of rotation. (Take the cylinder to be massless and A = cross sectional area of cylinder)**



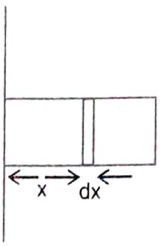
(A) $\rho_0 A\omega^2 \ell^2\left[\frac{1}{2}+\frac{1}{3}\alpha\ell\right]$ (B) $\rho_0 A\omega^2\ell^2\left[\frac{1}{2}+\frac{2}{3}\alpha\ell\right]$

(C) $\rho_0 A\omega^2 \ell^2\left[\frac{1}{2}+\alpha\ell\right]$ (D) $\rho_0 A\omega^2\ell^2\left[\frac{1}{2}+\frac{4}{3}\alpha\ell\right]$

Sol. $\int df = \int dm\omega^2 x$

$$\Rightarrow F = \int_0^{\ell}\rho_0(1+\alpha x)A\,dx.\omega^2 x = \rho_0 A\omega^2\int_0^{\ell}(1+\alpha x)\,x.dx = \rho_0 A\omega^2\left[\frac{x^2}{2}+\alpha\frac{x^3}{3}\right]_0^{\ell}$$

$$F = \rho_0 A\omega^2\ell^2\left[\frac{1}{2}+\alpha\frac{\ell}{3}\right]$$



$\therefore$ (A)

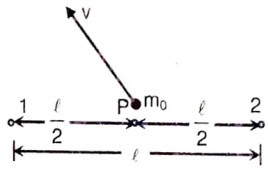
2. **A particle is projected from the mid-point of the line joining two fixed particles each of mass m. If the separation between the fixed particles is ℓ, the minimum velocity of projection of the particle so as to escape is equal to**

(A) $\sqrt{\frac{Gm}{\ell}}$ (B) $\sqrt{\frac{Gm}{2\ell}}$

(C) $\sqrt{\frac{2Gm}{\ell}}$ (D) $2\sqrt{\frac{2Gm}{\ell}}$

Sol. The gravitational potential at the mid-point P = V = $V_1 + V_2$

$$= \frac{-Gm}{(\ell/2)} - \frac{Gm}{(\ell/2)} = -\frac{4Gm}{\ell}$$



The gravitational potential energy $U = -4Gmm_0/\ell$, m_0 = mass of particle

When it is projected with a speed v, it just escapes to infinity, and the potential and kinetic energy will become zero.

$\Rightarrow \Delta K.E. + \Delta P.E. = 0$

$$\Rightarrow \left(0-\frac{1}{2}m_0v^2\right)+\left\{0-\left(-\frac{4Gmm_0}{\ell}\right)\right\} = 0$$

$$\Rightarrow v = 2\sqrt{\frac{2Gm}{\ell}}.$$

$\therefore$(D)

3. **A particle hanging from a spring stretches it by 1 cm at earth's surface. Radius of earth is 6400 km. At a place 800 km above the earth's surface, the same particle will stretch the spring by**

(A)1 cm (B) 8 cm

(C)0.1 cm (D) 0.79 cm

Sol. $\frac{kx_1}{kx_2} = \frac{F_{g_1}}{F_{g_2}} = \left(\frac{r_2}{r_1}\right)^2$

$\Rightarrow \frac{x_1}{x_2} = \left(\frac{800+6400}{6400}\right)^2 = \left(\frac{72}{64}\right)^2 = \left(\frac{9}{8}\right)^2$

$\Rightarrow \frac{x_1}{x_2} = \frac{81}{64}$

$\Rightarrow x_2 = \frac{64}{81}x_1 = \frac{64}{81}$ cm = 0.79 cm

$\therefore$(D)

4. A circular ring of mass M and radius 'a' is placed in a gravity free space. A small particle of mass m placed on the axis of the ring at a distance $\sqrt{3}a$ from the centre of the ring, is released from rest. The velocity with which the particle passes through the centre of the ring is

(A) $\sqrt{\frac{GM}{a}}$ **(B)** $\sqrt{\frac{GM}{2a}\left(\frac{m}{M}\right)}$

(C) $\sqrt{\frac{GM}{2a}\left(\frac{M}{m}\right)}$ **(D)** $\sqrt{\frac{GM}{a}\left(\frac{M}{M+m}\right)}$

Sol. From COM, $mv_1 - Mv_2 = 0$ ... (1)

From COE, $-\frac{GMm}{2a} = -\frac{GMm}{a} + \frac{1}{2}mv_1^2 + \frac{1}{2}Mv_2^2$ ... (2)

From Eqs. (1) and (2),

$v_1 = \sqrt{\frac{GM}{a}\left(\frac{M}{m+M}\right)}$

$\therefore$(D)

5. A satellite is moving round the earth in a circular orbit. The following statements are given.

(i) It is moving with a constant velocity.

(ii) It suffers no acceleration.

(iii) its angular momentum w.r.t. the earth remains conserved.

(iv) Its distance from centre must be equal to $\sqrt{2}$ times of earth's radius.

The correct option is

(A)(i) and (ii) are true **(B) (i), (iii) and (iv) are true**

(C)only (iii) is true **(D) (i) and (iv) are true**

Sol. When an earth satellite is moving round the earth in a circular orbit, it is moving with a constant speed. It has centripetal acceleration. Its angular momentum remains constant as no external torque is present. Its radius need not be equal to the $\sqrt{2}R_e$

$\therefore$(C)

6. The angular speed of rotation of earth about its axis when the weight of a man standing on equator will become half of its weight at the north pole (assume earth to be a perfect sphere) R-6400 km.

(A) 275×10^{-4} **rad sec^{-1}** **(B)** 9.75×10^{-4} **rad sec^{-1}**

(C) 8.75×10^{-4} **rad sec^{-1}** **(D)** 8.0×10^{-4} **rad sec^{-1}**

Sol. C

$$\frac{g}{2} = g - R\omega^2$$

$$\Rightarrow \omega = \sqrt{\frac{g}{2R}}$$

7. The escape velocity at the poles of a planet is v_e. If the effect of rotation of a planet makes the weight of a body at the equator 10% less than the weight at pole, the peripheral velocity of a point on the equator of the planet is (assume the planet to be a uniform solid sphere)

(A) $\frac{v_e}{\sqrt{5}}$ **(B)** $\frac{v_e}{2\sqrt{5}}$

(C) $\frac{\sqrt{5}v_e}{2}$ **(D)** $\frac{2v_e}{5}$

Sol. B

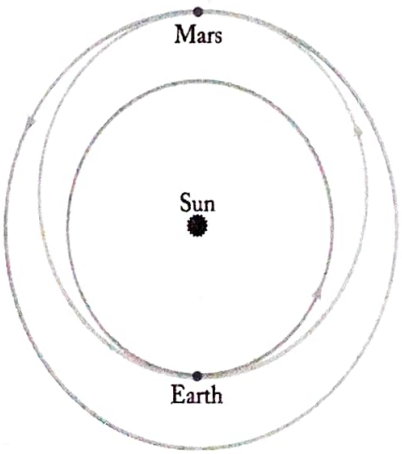
$$g_{equator} = (0.9)g_{poles}$$

$$\Rightarrow g_{pole} - \omega^2 R = (0.9)g_{pole}$$

$$\Rightarrow \omega^2 R = \frac{g}{10}$$

Peripheral velocity on equator $v = \omega R = \sqrt{\frac{g}{10R}} \times R = \sqrt{\frac{2gR}{20}} = \frac{v_e}{2\sqrt{5}}$

8. The Earth has a circular orbit of radius r and period t around the Sun; Mars has a circular orbit of radius R and period T. In order to send a spacecraft from the Earth to Mars, it is convenient to launch the spacecraft into an elliptical orbit whose perihelion coincides with the orbit of the Earth and whose aphelion coincides with the orbit of Mars; this orbit requires the least amount of energy for a trip to Mars. The time t' taken by a spacecraft to reach Mars from the Earth satisfies:



(A) $t' = (t + T)/2$

(B) $t'^2 = (t^2 + T^2)/2$

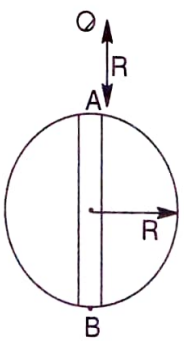
(C) $t'^{2/3} = (t^{2/3} + T^{2/3})/2$

(D) $(2t')^{2/3} = (t^{2/3} + T^{2/3})/2$

Sol. D

Use Kepler's Law of Periods.

9. A small particle is dropped from a height R in front of a narrow tunnel dug inside the earth (along a diameter). Let M be the mass of earth, R be radius of earth. Let v_0, T be speed of particle when it reaches A and time taken by particle to go from A to B respectively. Assuming mass of particle to be negligible as compare to mass of earth, pick the correct option(s)



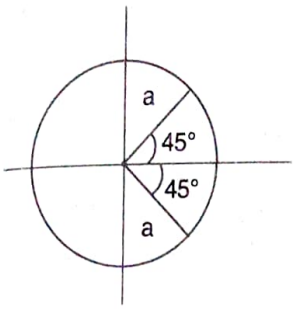
(A) $V_0 = \sqrt{\frac{GM}{2R}}$ **(B)** $V_0 = \sqrt{\frac{GM}{R}}$

(C) $R = \frac{\pi}{2}\sqrt{\frac{R^3}{GM}}$ **(D)** $T = \pi\sqrt{\frac{R^3}{GM}}$

Sol. B, C

$$\frac{-GMm}{2R} = \frac{-GMm}{R} + \frac{1}{2}mv_0^2$$

$$\Rightarrow \quad v_0 = \sqrt{\frac{GM}{R}}$$



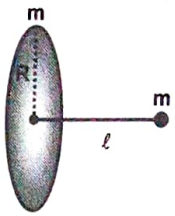
While in the tunnel;

$$\frac{d^2x}{dt^2} = -\left(\frac{GM}{R^3}\right)x$$

$$v_0^2 = \frac{GM}{R} = \omega^2(a^2 - R^2) \Rightarrow a = \sqrt{2}R$$

$$\Rightarrow \quad T_{A\to B} = \frac{1}{4} \times 2\pi\sqrt{\frac{R^3}{GM}}$$

10. A thin homogeneous disc of radius R and mass m and a small sphere of same mass is placed at a distance ℓ from the disc on its axis of symmetry. Initially both are in rest and then released from rest. They collide because of gravitation attraction. Assume $\ell >> R$, Find the magnitude of relative velocity of approach just before the collision.



(A) $\sqrt{4Gm\left(\frac{2}{R} - \frac{1}{\ell}\right)}$

(B) $\sqrt{2Gm\left(\frac{2}{R} - \frac{1}{\ell}\right)}$

(C) $\sqrt{Gm\left(\frac{2}{R} - \frac{1}{\ell}\right)}$

(D) $\sqrt{\frac{Gm}{2}\left(\frac{2}{R} - \frac{1}{\ell}\right)}$

Sol. A

Gravitation potential due to disc is

$$V = -\frac{2Gm}{R^2}(\sqrt{R^2 + \ell^2} - \ell)$$

$$\text{So, } U_i = -\frac{Gmm}{\ell} \quad (R << \ell)$$

$$U_f = -\frac{2Gmm}{R} \quad (\ell \to 0)$$

$$\Delta U = U_i - U_f = Gmm\left(\frac{2}{R} - \frac{1}{\ell}\right)$$

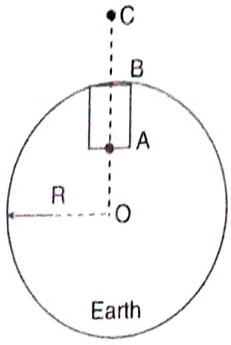
$$\frac{1}{2}mv^2 + \frac{1}{2}mv^2 = Gmm\left(\frac{2}{R} - \frac{1}{\ell}\right)$$

$$v = \sqrt{Gm\left(\frac{2}{R} - \frac{1}{\ell}\right)}$$

So, relative velocity

$$v_R = 2v = \sqrt{4Gm\left(\frac{2}{R} - \frac{1}{\ell}\right)}$$

11. First an object is slowly lifted from the bottom (point – A) of a shaft of depth $h_1 = \frac{R}{2}$ to earth's surface (point-B) and then it is slowly lifted still higher to attain an altitude $h_2 = \frac{R}{2}$ above the earth's surface (Point C). W_1 and W_2 are the work performed in two cases respectively. Choose the correct option(s)



(A) $W_1 > W_2$

(B) $W_1 < W_2$

(C) $\left|\frac{W_1 + W_2}{W_1 - W_2}\right| = 17$

(D) $\left|\frac{W_1 + W_2}{W_1 - W_2}\right| = 9$

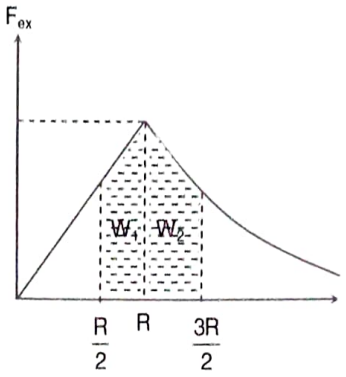
Sol. A, C

$$F_{ex} = \begin{cases} \frac{GMm}{R^3}\vec{r} & \text{if } 0 \le r \le R \\ \frac{GMm}{r^3}\vec{r} & \text{if } R \le r < \infty \end{cases} \Rightarrow W_1 > W_2$$

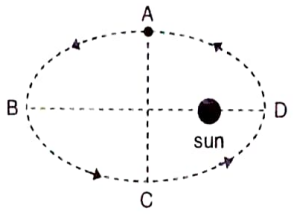
$$W_1 = \int_{R/2}^{R} \frac{4}{3}\pi G\rho m x dx = \frac{3GMm}{8R}$$

$$W_2 = \int_{R}^{3R/2} \frac{GMm}{x^2}dx = \frac{GMm}{3R}$$

$$\Rightarrow \frac{W_1}{W_2} = \frac{9}{8} \Rightarrow \left|\frac{W_1 + W_2}{W_1 - W_2}\right| = 17$$



12. A satellite is moving around the sun in an elliptical orbit. 'M' is the mass of sun and 'a' is semi major axis of elliptical orbit. The velocity of satellite when it reaches at A is



(A) $\sqrt{\frac{4GM}{a}}$

(B) $\sqrt{\frac{2GM}{a}}$

(C) $\sqrt{\frac{GM}{2a}}$

(D) $\sqrt{\frac{GM}{a}}$

Ans. D

Sol. $r^2 = b^2 + a^2e^2$ ...(1)

$b^2 = a^2(1 - e^2)$ ...(2)

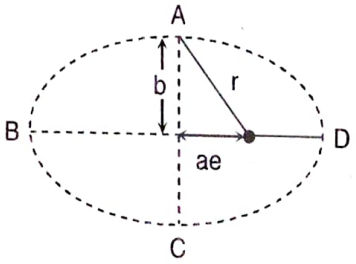
From equation (1) & (2)

$r = a$

Now from conservation of energy

$$-\frac{GMm}{2a} = \frac{1}{2}mv^2 - \frac{GMm}{a}$$

$$v = \sqrt{\frac{GM}{a}}$$



13. An artificial satellite of mass m revolves in a circular orbit of radius 4R around the centre of the moon (where R is the radius of the moon). Suddenly it starts experiencing a slight resistance due to cosmic dust. The resistance force depends on the speed of satellite as F = kv where k is a constant. If the radius of the moon is R and acceleration due to gravity on the moon's surface is g, then (assume that at every moment the satellite follows a circular path)

(A) The satellite will hit the moon's surface after time $t = \frac{m}{k}\ln 2$

(B) The satellite will hit the moon's surface after time $t = \frac{2m}{k}\ln 2$

(C) The work done by the resistance force on the satellite till it hits the moon surface $= -\frac{3}{8}mgR$.

(D) Work done by gravitational force till the satellite hits the moon's surface $= \frac{3}{4}mgR$.

Sol. A, C, D

for circular orbit of satellite

$$E = \frac{U}{2} = -K \Rightarrow -dE = dK$$

$$\Rightarrow (kv)vdt = mvdv$$

$$\frac{dv}{dt} = \frac{kv}{m} \Rightarrow \int dt = -\frac{m}{k}\int_{v_i}^{v_f}\frac{dv}{v}$$

$$\Rightarrow t = \frac{m}{k}\ln\left(\frac{v_f}{v_i}\right) = \frac{m}{k}\ln\sqrt{\frac{GM/R}{GM/4R}} = \frac{m}{k}\ln 2$$

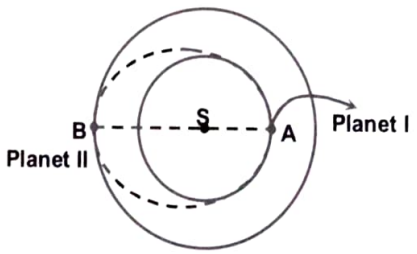
$$W_g = U_i - U_f = -\frac{GMm}{4R} - \left(-\frac{GmM}{R}\right) = \frac{3}{4}\frac{gR^2m}{R} = \frac{3}{4}mgR$$

$$W_g + W_r = \frac{1}{2}m\left(v_f^2 - v_i^2\right) = \frac{1}{2}m\left(\frac{GM}{R} - \frac{GM}{4R}\right)$$

$$\Rightarrow \frac{3}{4}mgR + W_r = \frac{3}{8}mgR \Rightarrow W_r = \frac{-3}{8}mgR$$

COMPREHENSION

One of the method to transfer a planetary body like the rocket, from one planet to another is not sending directly along the radial line. But to let it go in an elliptical orbit [with sun at one of its focus], which joins smoothly to the orbits of the both planets. Let the path of the planetary body starts tangentially to the orbit of one planet at A and joins the orbit of other planet tangentially at B, as shown in the figure.



SA = r_{min} ⇒ The minimum distance at point on the orbit of planetary body is called perigee = radius of orbit of first planet.

SB = r_{max} ⇒ The maximum distance at point on the orbit of planetary body is called apogee = radius of orbit of second planet.

$$e = \text{eccentricity of orbit} = \frac{r_{max} - r_{min}}{r_{max} + r_{min}}$$

$$p \Rightarrow \text{orbital parameter} = \frac{2r_{max}r_{min}}{r_{max} + r_{min}}$$

The total energy of the planetary body at any point of path (which is remains conserved) is given by $E = \frac{-GM_S m}{2a}$, where M_S mass of sun, m is mass of planetary body.

$a \Rightarrow$ semi-major axis of the elliptical orbit $= \frac{r_{max} + r_{min}}{2}$

If the rocket moves on the elliptical orbit of semi-major axis 'a', its time period of the motion will be directly proportional to "$a^{3/2}$".

A planetary body begins its journey in space, it is at perihelion of orbit of first planet at A and finally reaches near the second planet, it is at the apehelion of second orbit at B. So using conservation of energy

$$E_A = E_B = E$$

$$\frac{1}{2}mv_A^2 - \frac{GM_S m}{r_{min}} = \frac{1}{2}mv_B^2 - \frac{GM_S m}{r_{max}} = -\frac{GM_S m}{2a}$$

If v_1 and v_2 are the orbital velocity of first planet and second planet around sun then $v_1 < v_A$ and $v_2 > v_B$ always. Hence once, taken from the gravity of first planet, the planetary body is launched in the direction of orbital motion of first planet, the planetary body needs a small impulse to get a speed about $(v_A - v_1)$ relative to the first planet.

Again in order to tangential approach to second planet, the planetary body requires a final impulse which should increase its speed v_B to v_2 i.e. to get a speed relative to the second planet about $(v_2 - v_B)$. Once it catches up with the second planet, it would land on it, guided by the gravity of second planet. A hypothetical rocket is transfered from first planet to second planet for which following data is given.

$r_{min} = 2c$ unit, $r_{max} = 16$ c unit, $\sqrt{\frac{GM_S}{c}} = 30\sqrt{2}$ km/hr

Read above passage carefully and answer the following questions.

14. The speed of the rocket after first impulse (v_A) at planet A is

(A) 10 km/hr **(B) 5 km/hr**

(C) 12 km/hr **(D) None of these**

Sol. D

15. During the journey, how many times, it is required to provided impulse to the rocket

(A) Once **(B) twice**

(C) thrice **(D) none of these**

Sol. B

16. If the time period of the revolution of planet A about sun is $2\sqrt{2}/3$ years then the time required to transfer the rocket from planet A to planet B is

(A) 1.5 years **(B) 3 years**

(C) 4.5 years **(D) 9 years**

Sol. C

14-16. $\frac{1}{2}v_A^2 - \frac{GM_S}{2c} = -\frac{GM_S}{18c} \Rightarrow v_A = \frac{2\sqrt{2}}{3}\sqrt{\frac{GM_S}{c}} = 40$ km/hr

Using Keplers IIIrd law

$$\frac{T}{T_1} = \left(\frac{a}{2c}\right)^{3/2} = 3 \times 3 \times 3 \Rightarrow T = 9 \text{ years}$$

so time required to transfer planet = T/2 = 4.5 years

Matrix – Match Type

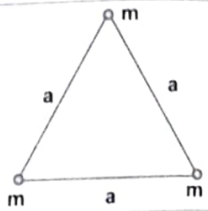
17. **For planet orbiting about sun in elliptical orbit, some incomplete statements regarding physical quantities are given in column – I. Which can be completed by using entries of column – II.**

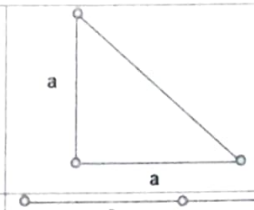
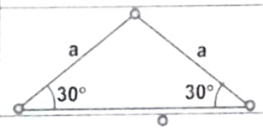
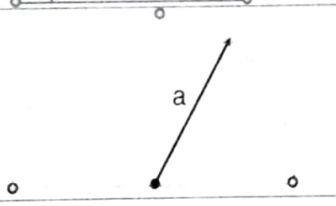
Column – I	Column – II
(A) Maximum potential energy of sun planet system	(p) Is dependent on semi-major axis of orbit.
(B) Maximum speed of planet	(q) Is at aphelion
(C) Minimum potential energy of sun planet system	(r) Is dependent of mass of planet.
(D) Minimum KE of planet	(s) Is at perihelion

Sol. (A) → (p, q) (B) → (p, r, s) (C) → (p, s) (D) → (q, r)

Answer questions 18, 19 and 20 by appropriately matching the information given in the three columns of the following table.

Figure shows three point masses placed at the vertices of an equilateral triangle. An external agent slowly moves the particles into final configurations. Column –I lists the final configurations, column –II lists the work done by the external agent in changing the configuration and column –III lists gravitational potential energy of the system in final configuration.



Column 1		Column 2		Column 3	
(I)		(i)	$\frac{Gm^2}{2a}$	(P)	$-\frac{Gm^2}{a}\left(\frac{1}{2}+\sqrt{2}\right)$
(II)		(ii)	$\frac{Gm^2}{a}\left(1-\frac{1}{\sqrt{2}}\right)$	(Q)	$-\frac{Gm^2}{a}\left(2+\frac{1}{\sqrt{2}}\right)$
(III)		(iii)	$\frac{Gm^2}{a}\left(\frac{5}{2}-\sqrt{2}\right)$	(R)	$-\frac{5}{2}\frac{Gm^2}{a}$
(IV)		(iv)	$\frac{Gm^2}{a}\left(1-\frac{1}{\sqrt{3}}\right)$	(S)	$\frac{Gm^2}{a}\left(2+\frac{1}{\sqrt{3}}\right)$

18. **Pick the correct option.**

(A) (IV) (ii) (Q) **(B) (III) (i) (S)**
(C) (II) (i) (R) **(D) (I) (iv) (Q)**

Sol. C

19. **Pick the option for maximum work done by the external agent.**

(A) (II) (iii) (P) **(B) (II) (iv) (S)**
(C) (IV) (iii) (P) **(D) (IV) (ii) (Q)**

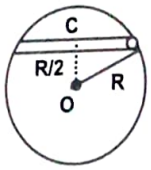
Sol. C

20. **Pick the option for minimum potential energy in the final configuration.**
(A) (IV) (ii) (Q) **(B) (III) (i) (S)**
(C) (II) (i) (P) **(D) (I) (ii) (Q)**

Sol. D

Numerical Type

21. **A tunnel is dug across the earth of mass 'M' and radius 'R' at a distance R/2 from the centre 'O' of the Earth as shown. A body is released from rest from one end of the smooth tunnel. The velocity acquired by the body when it reaches the centre 'C' of the tunnel is $\sqrt{\frac{3GM}{nR}}$. Find n.**



Sol. 4

Using conservation of energy

$$\frac{1}{2}mv^2 = -\frac{GMm}{R} - \left(\frac{-11GMm}{8R}\right)$$

$$\frac{1}{2}mv^2 = \frac{3GMm}{8R}$$

$$\therefore\ v = \sqrt{\frac{3GM}{4R}}$$

22. **An artificial satellite is moving in a circular orbit around the earth with a speed equal to 3/8 times of the magnitude of escape velocity from the earth. If the satellite is stopped suddenly in its orbit and allowed to fall freely onto the earth, then the speed with which it hits the surface of the earth is $\sqrt{\frac{23}{n}}$ Km/s. Find n. (take g = 10 m/s and R_e = 6400 Km)**

Sol. 2

$$V_e = \sqrt{2gR_e}$$

$$V_s = \frac{3}{8}\sqrt{2gR_e} = \sqrt{\frac{9gR_e}{32}} = \sqrt{\frac{R_e^2 g}{R_e + h}}$$

$$9R_e + 9h = 32\,R_e$$

$$h = \frac{23}{9}R_e$$

Now, total energy at height 'h' = total energy at the surface of the earth

$$0 - \frac{GM_em}{R_e} = \frac{1}{2}mv^2 - \frac{GM_em}{R_e}$$

$$\frac{1}{2}mv^2 = \frac{GM_em}{R_e} - \frac{GM_em}{R_e + \frac{23}{9}R_e}$$

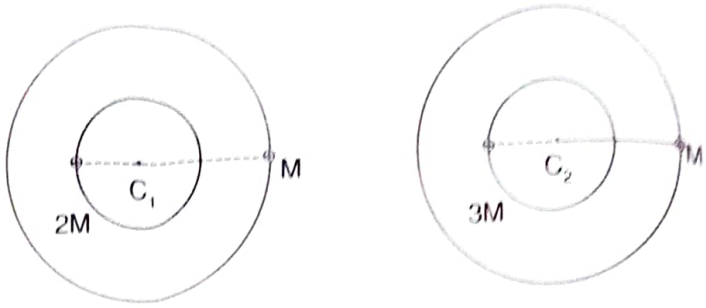
$$\frac{1}{2}mv^2 = \frac{23}{32}\frac{GM_em}{R_e}$$

$$v = \sqrt{\frac{23}{16}\frac{GM_e}{R_e}},\ \sqrt{gR_e} = 8\ \text{Km/sec}$$

$$v = \sqrt{\frac{23}{16}gR_e} = \sqrt{\frac{23}{2}}\ \text{Km/s}$$

Numerical Decimal Type

23. **Figure shows two binary star systems such that the distance of lighter star from the centre of rotation is same in both cases. If the ratio of time periods of rotation is $\frac{T_1}{T_2} = \frac{n}{8}\sqrt{\frac{3}{2}}$, where n is an integer, find n.**



Ans. **00009.00**

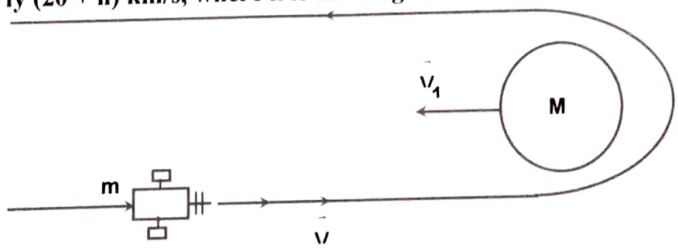
Sol. $\frac{2}{3}\ell_1 = \frac{3}{4}\ell_2 \Rightarrow \frac{\ell_1}{\ell_2} = \frac{9}{8}$.

$M_1 = 3M$, $M_2 = 4M$.

$$\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{\ell_1}{\ell_2}\right)^3 \times \frac{M_2}{M_1} = \left(\frac{9}{8}\right)^3 \times \frac{4}{3}$$

$$\Rightarrow \quad \frac{T_1}{T_2} = \frac{9}{8}\sqrt{\frac{3}{2}}$$

24. **An unmanned space probe (of mass m) and speed v = 2 km/s relative to the Sun, approaches the planet Jupiter (of mass M >>m) and speed V_J relative to the Sun as shown in the figure. The spacecraft rounds the planet and departs in the opposite direction. The mass of Jupiter is very much greater than the mass of the spacecraft (M >>m), and it orbits the sun at a distance of 5.29 AU, where 1 AU is the orbital radius of the earth around the sun: 1 AU = 1.5×10^8 km, 1 earth year = $\pi \times 10^7$ s, nearly. The speed of the space probe after it leaves the solar system, relative to the sun is nearly (20 + n) km/s, where n is an integer. Find the value of n.**



Sol. **00001.00**

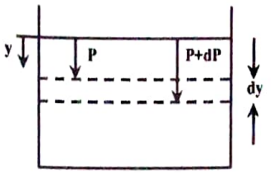
Use momentum conservation principle.

FLUID MECHANICS 7

(A) FLUID STATICS

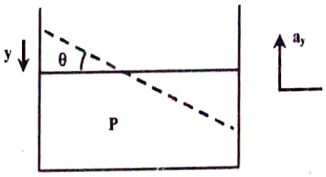
(a) Law of Hydrostatics

$$\frac{dP}{dy} = \rho g ,$$



(b) Acceleration of the Containers

$$\tan\theta = \frac{a_x}{a_y + g}$$



where a_x and a_y are horizontal and vertical components of acceleration of the containers and θ is the angle of inclination of the free surface of the liquid with the horizontal.

(c) $p_{total} = p_{gauge} + p_{atm}$

(d) Archimedes' Principle

Buoyant force = weight of the displaced liquid.

(e) Law of Flotation

The weight of a floating body is equal to the weight of volume of liquid displaced by submerge volume of the liquid.

(f) Pascal's Law

The pressure at any point in a liquid is same in all directions

(B) FLUID DYNAMICS

(i) $\rho_1 v_1 A_1 = \rho_2 v_2 A_2 = \text{constant}$

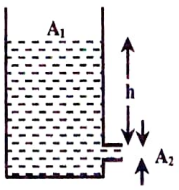
(ii) Bernoulli's theorem

$$\therefore \quad P + \rho gh + \frac{\rho v^2}{2} = \text{constant}$$

For an ideal fluid flow under steady state conditions, the sum total of pressure energy, kinetic energy and potential energy is constant along a stream line.

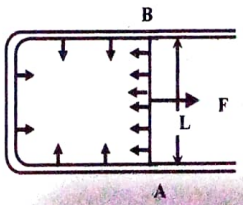
(iii) Velocity of efflux,

$v = \sqrt{2gh}$, assuming $A_1 \gg A_2$



Surface Tension of a Liquid

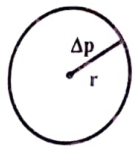
It is the property by virtue of which the free surface of a liquid at rest behaves like a stretched elastic membrane tending to contract so that it possesses minimum surface area. If F be the force acting in equilibrium and ℓ the length of the imaginary line AB, the force F is perpendicular to AB, and the surface tension is given by $\sigma = F/\ell$.



The direction of force due to surface tension is perpendicular to the line and tangential to the liquid surface.

SURFACE ENERGY

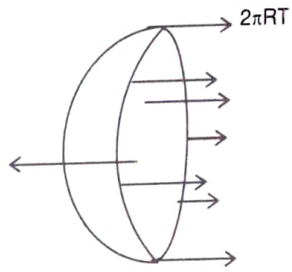
The work done in increasing the surface area of a liquid by one unit area is equal to the surface tension under isothermal conditions. Excess pressure inside an air bubble in a liquid is given by



$$2\pi RT = \Delta P(\pi R^2)$$

For a bubble in air, there are two surfaces.

$$\therefore \quad \Delta p = \frac{4T}{r}$$



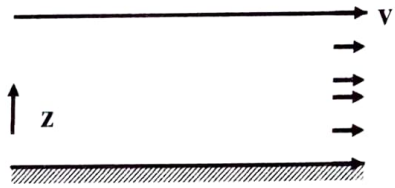
Note: A bubble inside a liquid has only one free surface area.

Rise of liquid in a capillary tube

$$h = \frac{2T\cos\theta}{r\rho g}$$

VISCOSITY

When a fluid flows such that a velocity gradient is set up within it, forces act within the fluid so as to prevent the velocity gradient from existing. This force is due to a property called viscosity.



Suppose that a glass plate in contact with a water column of height h is moved with a constant velocity v. Forces of viscosity appear between the solid surface and the layer in contact.

$F = -\eta A\frac{dv}{dz}$, where η is a constant called coefficient of viscosity, its C.G.S. unit is poise. The S.I. unit of viscosity equals 10 poise.

Stokes' Law and Terminal Velocity

When a sphere of radius r moves with a velocity v through a fluid of viscosity η, the viscous force opposing the motion of the sphere is

$$F = 6\pi\eta rv$$

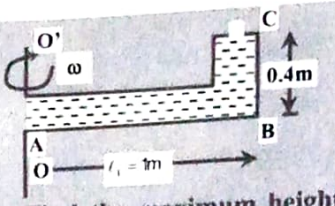
If, for a sphere, viscous force becomes equal to the net weight acting downward, the velocity of the body become constant and is known as the terminal velocity.

$$6\pi\eta r v_T = \frac{4}{3}\pi r^3(\rho - \sigma)g$$

$$\therefore \quad v_T = \frac{2}{9}r^2\left\{\frac{\rho - \sigma}{\eta}\right\}g.$$

ILLUSTRATIVE PROBLEMS

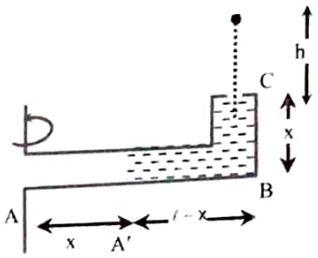
1. A horizontally oriented bent thin tube ABC of length 1.4 m rotates with a constant angular velocity ω about a stationary vertical axis OO' passing through the end A as shown in figure. The portion BC of tube is parallel to OO' and its length is equal 0.4 m. At t = 0, AB part of tube is filled with an ideal fluid.



The end A of tube is open, the closed end C has a very small orifice. Find the maximum height reached by fluid relative to end C. (take g = 10/s² and ω = 10 rad/s)

Sol. Let any time t fluid just fills the portion BC and comes out with velocity v.

$\ell = 1\text{m}, x = 0.4\text{m}$



In rotating frame, centrifugal force on portion A'B,

$$F = \rho A\int_{x}^{\ell}\omega^2 r\,dr = \frac{\rho A\omega^2(\ell^2 - x^2)}{2}$$

Due to pressure difference between C and A'

$$\Delta P = \frac{1}{2}\rho\omega^2\left(\ell^2 - x^2 - \rho g x\right)$$

$$\text{Now } \frac{1}{2}\rho V^2 = \Delta P = \frac{\rho\omega^2}{2}\left(\ell^2 - x^2\right) - \rho g x \qquad \ldots(1)$$

From Bernoulli theorem,

Let it reaches upto a height h, then,

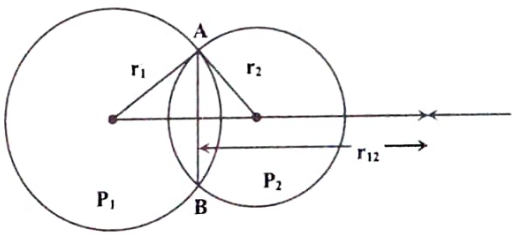
$$\rho g h = \frac{1}{2}\rho v^2 = \frac{\rho\omega^2(\ell^2 - x^2)}{2} - \rho g x$$

$$\rho \times 10 \times h = \rho \times \frac{(10)^2}{2}\left(1^2 - (.4)^2\right) - .4 \times 10 \times \rho \quad \Rightarrow \quad h = 3.8\text{ m.}$$

2. Two soap bubbles are stuck together with an intermediate film separating them. Compute the radius of curvature of this film given that the radii of the bubbles in this arrangement are r_1 and r_2 respectively. If $r_1 > r_2$ state clearly which way the intermediate film will bulge.

For the case when $r_1 = r_2 = 2$ cm, calculate the radius of the bubble formed by bursting the intermediate film. The volume of a spherical dome of radius R and height h is $\pi b^2(3R - b)/3$.

Sol. Excess pressure inside a soap bubble is 4T/r



If P is the atmospheric pressure

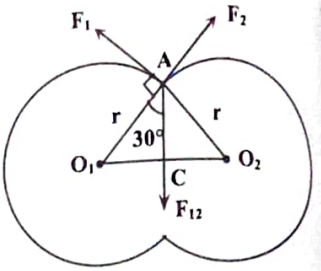
$$P_1 = \frac{4T}{r_1} - P, \quad P_2 = \frac{4T}{r_2} - P$$

$$\therefore \quad P_1 - P_2 = 4T\left(\frac{1}{r_1} - \frac{1}{r_2}\right) = \frac{4T}{r_{12}}$$

$$\therefore \quad \frac{1}{r_{12}} = \frac{1}{r_1} - \frac{1}{r_2} \Rightarrow r_{12} = \frac{r_1 r_2}{|r_1 - r_2|}$$

If $r_1 > r_2$, $P_2 > P_1$ and the intermediate film bulges towards the centre of circle of radius r_1.

If $r_1 = r_2 = 2$ cm, the intermediate film is a plane symmetrical positioned. The 3 films due to the two bubbles and the intermediate film meet at a common circle with diameter AB. These 3 films produce 3 forces at every point of this circle.



Since the surface tension T is the same for all the films,

$|F_1| = |F_2| = |F_{12}|$

Equilibrium at any point such as A is possible only if the three forces are at 120° to one another.

$\therefore\ O_1C = \frac{1}{2}r$

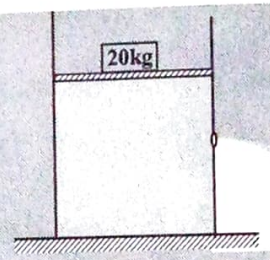
$b = r + \frac{1}{2}r = \frac{3}{2}r$

Volume of each bubble before bursting $= \frac{\pi}{3}\left(\frac{3}{2}r\right)^2\left(3r - \frac{3}{2}r\right) = \frac{9\pi r^3}{8}$

If R is the radius of the combined bubble after bursting the intermediate film,

$\frac{4\pi R^3}{3} = 2 \times \frac{9\pi r^3}{8} = \frac{9\pi r^3}{4}$

3. A long cylindrical tank of cross-section area $0.5m^2$ is filled with water. It has an opening at a height 50 cm from the bottom, having area of cross-section $1 \times 10^{-4}\ m^2$. A movable piston of cross-section area almost equal to $0.5\ m^2$ is fitted on the top of the tank such that it can slide in the tank freely. A load of 20 kg is applied on the top of the water by piston , as shown in the figure. Find the speed of the water jet with which it hits the surface when piston is 1m above the bottom. (Ignore the mass of the piston)



Sol. With respect to the opening the height of the piston is 0.5 m .

Pressure at the top is $P_1 = P_0 + \frac{20 \times 10N}{0.5m^2}$

where P_0 is the atmospheric pressure.

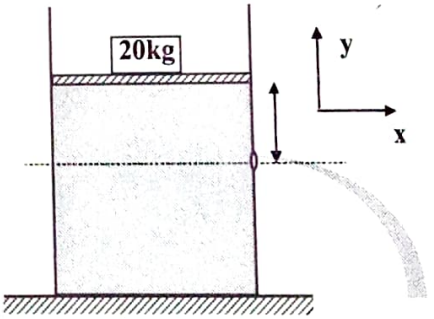
Pressure at the opeing is $P_2 = P_0$, From Bernoulli's equation, we have

$$P_1 + \rho gh + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

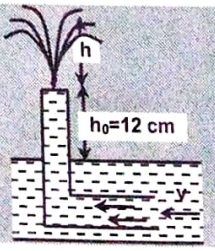
On solving, we get $v_2 = 3.3$ m/s $= v_x$ (say)

$v_y = \sqrt{2gh}$

$v = \sqrt{v_x^2 + v_y^2} = 4.56$ m/s



4. A bent tube is lowered into water stream as shown in the figure. The velocity of the stream relative to the tube is equal to v = 2.5 m/s. The closed upper end of the tube located at a height of 12 cm has a small orifice. To what height h will the water jet spurt?



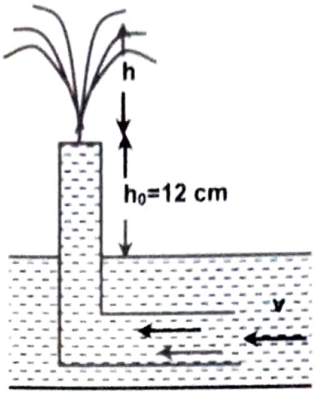
Sol. At maximum height total energy will be potential,

Apply Bernoulli's theorem, we have

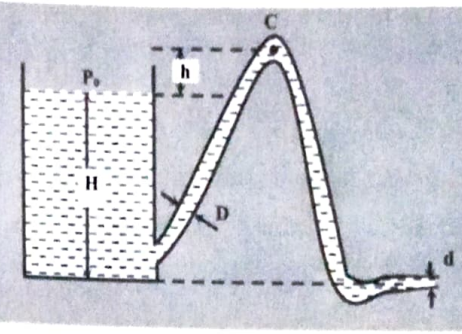
$$\frac{1}{2}\rho v^2 = (h + h_0)\rho g$$

$$\Rightarrow h = \frac{v^2}{2g} - h_0 = \frac{(2.5)^2}{2\times 10} - 0.12 \qquad [g = 9.8\ m/s^2]$$

$\approx$ 20 cm (approximately)



5. A pipe of diameter D is connected to a water tank of large cross-sectional area in which water is maintained at a height H as shown in the figure. A nozzle of diameter 'd' fitted at the end of the pipe discharges water into the atmosphere. Find the flow rate, given that the density of water is ρ. Also find pressure at point C. (P_0 is the atmospheric pressure.)



Sol. Taking the nozzle axis as the datum and applying Bernoulli's equation

$$\frac{P_0}{\rho g} + H + 0 = \frac{P_0}{\rho g} + \frac{v^2}{2g} + 0$$

$$v = \sqrt{2gH}$$

Flow rate, $Q = \frac{\pi}{4} d^2 \sqrt{2gH}$

Now, $\frac{\pi}{4} d^2 \sqrt{2gH} = \frac{\pi}{4} D^2 v_1$

$$v_1 = \frac{d^2}{D^2}\sqrt{2gH}$$

Applying Bernoulli's equation at the water surface and at point C

$$\frac{P_0}{\rho g} + H + 0 = \frac{P_c}{\rho g} + H + h + \frac{v_1^2}{2g}$$

$$P_C = P_0 - \left(h + \frac{Hd^4}{D^4}\right)\rho g$$

SOLVED PROBLEMS

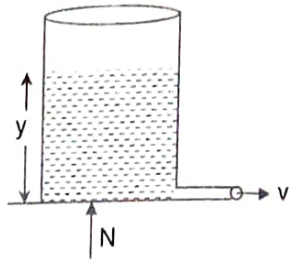
1. A light cylindrical vessel is kept on a horizontal surface. Its base area is A. A hole of cross sectional area a is made just at its bottom side. The minimum coefficient of friction necessary for sliding of the vessel due to the impact force of the emerging liquid is (a<< A)

(A) varying **(B) a/A**

(C) 2a/A **(D) None of these**

Sol. The velocity of efflux of the liquid is given as

$v = \sqrt{2gy}$



$\therefore$ The impact force of the emerging liquid on the vessel + liquid content is equal to

$$F = v\frac{dm}{dt} = va\rho v = a\rho v^2$$

$$\Rightarrow F = a\rho\,(\sqrt{2gy})^2 = 2a\rho gy$$

The force of friction $= f = F = 2a\rho gy$

$\Rightarrow \mu N = 2a\rho gy \quad \Rightarrow \mu\,(A\rho gy) = 2a\,\rho gy$

$$\Rightarrow \mu = \frac{2a}{A}.$$

$\therefore$ (C)

2. The density of ice is x gm/cc. and that of water is y gm/cc. What is the change in volume in cc., when m gm of ice melts ?

(A) m(y – x) **(B) (y – x)/m**

(C) mxy (x – y) **(D) m (1/y – 1/x)**

Sol. m = volume × density

Mass remains constant.

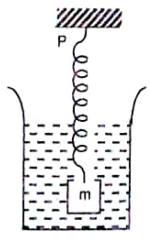
$$V_{ice} \times \rho_{ice} = V_{water} \times \rho_{water}$$

$$V_{water} = \frac{V_{ice} \times \rho_{ice}}{\rho_{water}} = \frac{m}{y}$$

$$\Delta V = V_{water} - V_{ice} = \frac{m}{y} - \frac{m}{x} = m\left(\frac{1}{y} - \frac{1}{x}\right).$$

$\therefore$(D)

3. A cube of mass m and density D is suspended from a point P by a spring of stiffness k. The system is kept inside a beaker filled with a liquid of density d. The elongation in the spring, assuming D > d, is:



(A) $\frac{mg}{k}\left(1 - \frac{d}{D}\right)$ **(B)** $\frac{mg}{k}\left(1 - \frac{D}{d}\right)$

(C) $\frac{mg}{k}\left(1 + \frac{d}{D}\right)$ **(D) None of these.**

Sol. The cube is in equilibrium under the following three forces,

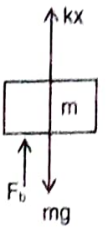
(A)spring force kx, where x = elongation of the spring,

(B)gravitational force w, weight of the cube = mg

(C) buoyant force F_b (or upward thrust) imparted by the liquid on the cube given as

$F_b = Vdg$ where

V = volume of the immersed portion of the cube. For complete immersion, V = volume of the cube.



For equilibrium of the cube, $kx + F_b = mg$

$$\Rightarrow \quad x = \frac{mg - F_b}{k} = \frac{mg - Vdg}{k}, \qquad \text{where } V = (m/D)$$

$$\Rightarrow x = \frac{mg}{k}\left(1 - \frac{d}{D}\right)$$

$\therefore$ (A)

4. Eight identical droplets each falling under gravity in the earth's atmosphere with terminal velocity v combine together to form a single drop. The terminal velocity of the resulting drop will be

(A) v **(B) 2 v**

(C) 3 v **(D) 4 v**

Sol. Use Stoke's law

$\therefore$ (D)

5. Water rises to a height of 10cm in a capillary tube and mercury falls to a depth of 3.42 cm in the same capillary tube. If the density of mercury is 13.6g/cc and the angle of contact of mercury and water are 135° and 0°, respectively, the ratio of surface tension of water and mercury is

(A) 1 : 0.15 **(B) 1 : 3**

(C) 1 : 6.5 **(D) 1.5 : 1**

Sol. $h = \dfrac{2T\cos\theta}{r\rho g}$

or $s = \dfrac{hr\rho g}{2\cos\theta}$

or $S \propto \dfrac{h\rho}{\cos\theta}$

$$\frac{S_W}{S_{Hg}} = \frac{h_1}{h_2} \times \frac{\cos\theta_2}{\cos\theta_1} \times \frac{\rho_1}{\rho_2}$$

Putting the values, we obtain 1 : 6.5

$\therefore$ (C)

6. An air bubble of radius 5mm rises through a vat of syrup at a steady speed of 2mm/s. If the syrup has a density of 1.4×10^3 kg/m^3, what is its viscosity?

(A) 1.15 kg/m-s **(B) 15kg/m-s**

(C) 2.25 kg/m-s **(D) 19.05 kg/m-s**

Sol. In this case the terminal velocity is upward. Since $\sigma > \rho$, the density of the air inside the bubble may be neglected completely.

From equation

$$v = \frac{2a^2(\rho - \sigma)g}{9\eta} = \frac{2a^2\rho g}{9\eta}$$

$$\Rightarrow \eta = \frac{2a^2\rho g}{9v}, \text{ where } a = 0\,0.005\text{mm}, \rho = 1.4\times10^3\text{kg/m}^3, v = 0.004\text{m/s}$$

$$\therefore \eta = \frac{2\times 25\times10^{-6}\times1.4\times10^3\times9.8}{9\times4\times10^{-3}}\text{ kg/m-s} = 19.05\text{ kg/m-s}$$

$\therefore$ (D).

7. n number of water droplets, each of radius r, coalesce, to form a single drop of radius R. The rise in temperature dθ is

(A) $\frac{2T}{rJ}$ **(B)** $\frac{3T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$

(C) $-\frac{3T}{J}$ **(D)** $\frac{3T}{J}\left(\frac{1}{r}+\frac{1}{R}\right)$

Sol. Let n drops coalesce to form a big drop. Then

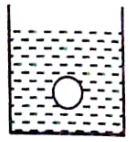
$$n\times\frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3 \Rightarrow n = \frac{R^3}{r^3}$$

$$\text{Heat produced} = \frac{\text{work done}}{J} = S.T\times\frac{\text{decrease in area}}{J} = mc\,d\theta$$

$$\Rightarrow d\theta = \frac{3T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$$

$\therefore$ (B).

8. A smooth spherical ball of radius 1 cm and density 4×10^3 kg/m^3 is dropped gently in a large container containing viscous liquid of density 2×10^3 kg/m^3, and $\eta = 0.1$ N-s/m^2. The distance moved by the ball in t = 0.1 sec after it attains terminal velocity is



(A) $\frac{4}{5}$ **m up** **(B)** $\frac{4}{9}$ **m up.**

(C) $\frac{2}{3}$ **m down** **(D)** $\frac{4}{9}$ **m down**

Sol. Once the body acquire the terminal velocity it will have uniform motion.

$$v_t = \frac{2}{9}\frac{r^2}{\eta}\left[(\rho-\sigma)g\right], \text{ down}$$

$$v_t = \frac{40}{9}\text{ m/s down}$$

$$\text{Distance travelled} = \frac{40}{9}\times0.1 = \frac{4}{9}\text{m}$$

$\therefore$ (D)

9. For a stream line flow of water following statements are given below

(a) Two streamlines do not intersect each other.

(b) streamlines must be straight.

(c) streamlines flow is more likely for liquids with low density and height viscosity.

(d) streamlines flow is more likely for liquids with high density and low viscosity.

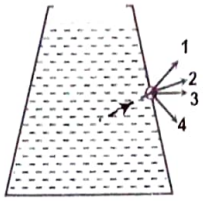
(A)(a) and (b) are true. (B) (a), (b) and (c) are true.

(C)(a) and (c) are true. (D) All are true.

Sol. Reason: Streamlines may be straight or curved.

∴(C)

10. Water has been filled in a conical vessel fixed on the horizontal surface at its base. When a very small hole is made in its curved surface as shown in the figure, water will come out in which of the direction shown

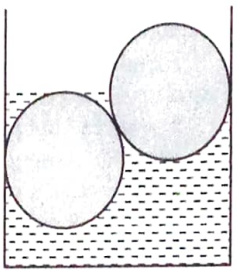


(A)1 (B) 2

(C)3 (D) 4

Sol. (B)

11. There are two cylinder-shaped wooden billets, each having a mass of mass 45 kg, in a vertical wall sewage, which contains water in it. The two billets have the same size and the same material; they touch each other and the walls of the sewage. One of them is totally under the water, whilst only half of the other one is immersed into the water. Friction is negligible everywhere. (density of water is 1000 kg/m^3 and g = 10 m/s^2). Choose the correct option(s)



(A) Density of the wood is 750 kg/m^3

(B) Density of the wood is 800 kg/m^3

(C) The forces exerted by the billets on the vertical walls is 260 N approximately

(D) The forces exerted by the billets on the vertical walls is 320 N approximately

Sol. A, C

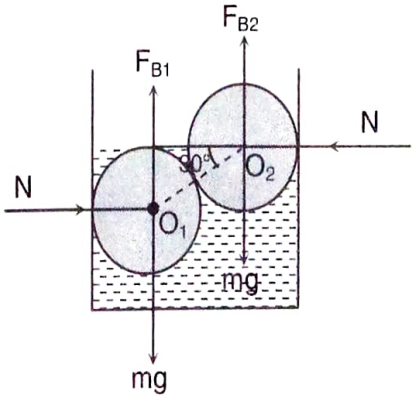
$2mg = F_{B1} + F_{B2}$

$\Rightarrow \rho_{wood} = \frac{3}{4}\rho_{water} = 750\,kg/m^3$

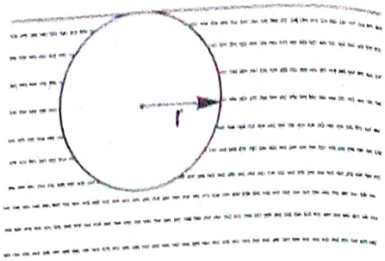
Taking torque about O_1

$N \times R = (mg - F_{B2}) \times 2R\cos 30°$

$\Rightarrow = 45 \times 10 \times \frac{1}{3} \times 1.732 = 259.8 \approx 260\,N$



12. **Figure shows a solid sphere positioned just below the surface of a fluid and held there. Let the density of the fluid be σ, force applied by fluid on the upper half of the sphere be F_1 and the force on lower half be F_2 (excluding the contribution due to atmospheric pressure). Pick correct option(s):-**



(A) $F_1 = \frac{1}{3}\pi r^3 \sigma g$

(B) $F_2 = \frac{5}{3}\pi r^3 \sigma g$

(C) $F_1 = \frac{2}{3}\pi r^3 \sigma g$

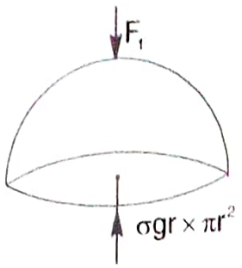
(D) $F_2 = \frac{6}{3}\pi r^3 \sigma g$

Sol. **A, B**

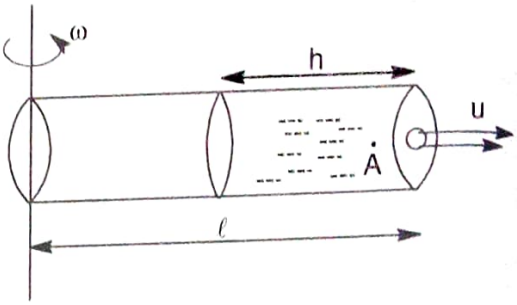
$F_B = \pi r^3 \sigma g - F_1 = \frac{2}{3}\pi r^3 \sigma g$

$\Rightarrow \quad F_1 = \frac{1}{3}\pi r^3 \sigma g$

Thus; $F_2 = \frac{5}{3}\pi r^3 \sigma g$



13. **Figure shows a thin tube open at one end rotating with constant angular speed ω. The other end of the tube has a small opening. Tube contains a fluid of density ρ which effuses with a speed u relative to the tube. Let p_A be the pressure at the point in the tube just behind the opening. Assume the atmospheric pressure to be p_0 and pick correct option(s) for the instant when $h = \frac{\ell}{2}$.**



(A) $p_A = p_0 + \frac{\rho\omega^2\ell^2}{4}$

(B) $p_A = p_0 + \frac{3}{8}\rho\omega^2\ell^2$

(C) $u = \frac{\omega\ell}{2}$

(D) $u = \frac{\sqrt{3}}{2}\omega\ell$

Sol. **B, D**

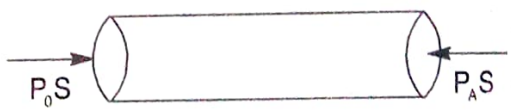
By Newton's second law;

$(p_A - p_0)s = \rho sh\omega^2\left(\ell - \frac{h}{2}\right)$

$\Rightarrow \quad p_A = p_0 + \rho\omega^2 h\left(\ell - \frac{h}{2}\right).$

By Bernoulli's equation;

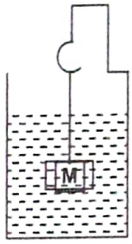
$u = \omega\sqrt{2h\left(\ell - \frac{h}{2}\right)}$



COMPREHENSION

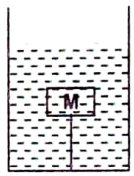
Buoyant force on an object completely immersed in a fluid does not depend on the density of the object. It would only depend on the weight of liquid displaced. If the vessel in which an object is immersed is accelerated then the pseudo force acting on the object in the frame of the vessel depends on the mass of the object. The direction of motion of an object relative to a vessel can be found by finding the resultant of buoyant force, the pseudo force, the gravitational force and any other force acting on the body.

14. **Consider a block immersed in a beaker containing water as shown:**
If the beaker now starts accelerating upwards then the tension in the string will:
(A) decrease
(B) increase
(C) remain unchanged
(D) data insufficient



Sol. (B)

15. **Consider a wooden block immersed in a beaker containing water. If the beaker now starts accelerating upwards then:**
(A) Tension in the string will increase
(B) Tension the string will decrease
(C) Tension in the string will remain unchanged
(D) data insufficient



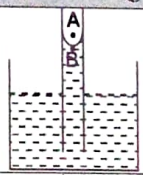
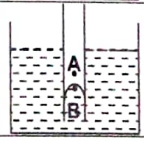
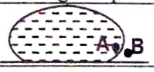
Sol. (A)

16. **If in the previous problem the beaker was given a rightward acceleration then relative to the beaker the block will have a tendency to move:**
(A) Leftwards **(B) Rightwards**
(C) Upwards **(D) Downwards**

Sol. (B)

Matrix – Match Type

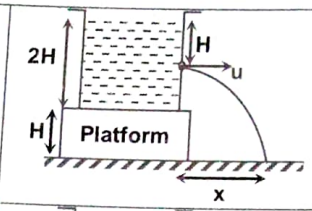
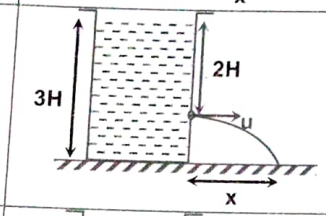
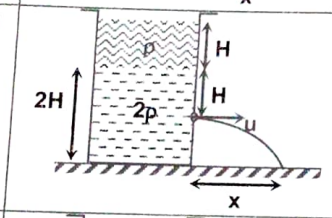
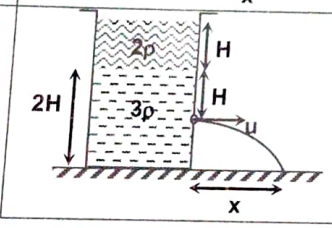
17. **Capillary rise and shape of droplets on a plate due to surface tension are shown in column-II. Match the following.**

Column – I		Column – II	
(A)	Adhesive forces is greater than cohesive force.	(p)	
(B)	Cohesive forces is greater than adhesive forces.	(q)	
(C)	Pressure at A > pressure at B	(r)	A mercury drop is pressed between two parallel glass plates.
(D)	Pressure at B > Pressure at A	(s)	

Sol. (A) → (p) (B)→ (q, r, s) (C)→ (p, r, s) (D)→ (q)

Answer questions 18, 19 and 20 by appropriately matching the information given in the three columns of the following table.

Column –I lists some situation with a fluid container open to atmosphere with a small opening. Column –II lists the efflux speed u for the fluid and column –III lists the horizontal range x of the fluid stream.

Column 1		Column 2		Column 3	
(I)		(i)	$2\sqrt{gH}$	(P)	$\sqrt{\frac{20}{3}}H$
(II)		(ii)	$\sqrt{\frac{10}{3}gH}$	(Q)	$\sqrt{6}H$
(III)		(iii)	$\sqrt{2gH}$	(R)	$2\sqrt{2}H$
(IV)		(iv)	$\sqrt{3gH}$	(S)	4H

18. **Pick correct option.**

(A) (IV) (ii) (P) **(B) (III)(i) (S)**
(C) (II) (iv) (Q) **(D) (I) (iii) (P)**

Sol. A

19. **Pick correct option for minimum efflux speed.**

(A) (II) (i) (R) **(B) (I) (iii) (R)**
(C) (I) (iii) (S) **(D) (II) (ii) (P)**

Sol. B

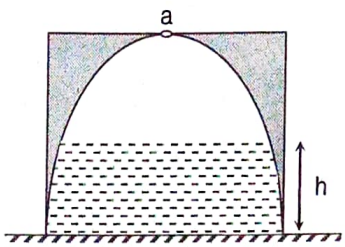
20. **Pick the correct option for minimum horizontal range.**

(A) (III) (iv) (Q) **(B) (II) (iii) (Q)**
(C) (III) (ii) (P) **(D) (IV) (i) (P)**

Sol. A

Numerical Type

21. **Consider a cuboidal vessel (2R × 2R × R) with a hemispherical cavity of radius R, kept at a horizontal smooth surface as shown in the figure. The vessel has very small hole of cross-sectional area "a". Now water of density ρ is poured into space developed due horizontal surface and vessel through hole very slowly. When the height of water level is h, the vessel lifts off the surface and liquid leaks through space generated. (R = 2m, $\rho = 10^3$ kg/m^3, m = 9π kg). If the value of h is 5k cm, find the valve of k. (Atmospheric pressure = 10^5 N/m^2).**

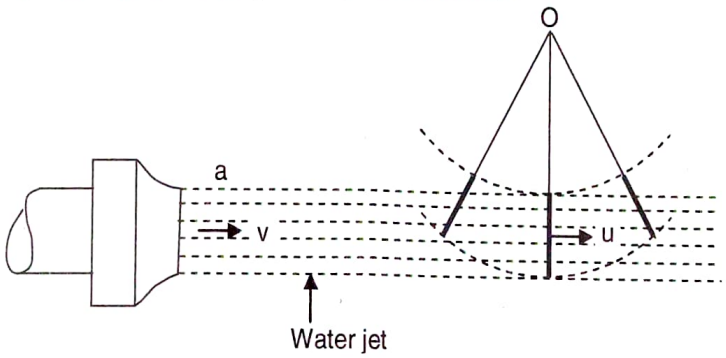


Sol. 6

$$mg = F_B = \left[\pi R^2 h - \int_0^h \pi\left(R^2 - z^2\right)dz\right]\rho g \quad \Rightarrow m = \frac{\pi h^3 \rho}{3}$$

$$\Rightarrow h^3 = \frac{3m}{\pi\rho} = \frac{3\times\pi\times 9}{\pi\times 10^3} \Rightarrow h = \frac{3}{10} = 0.3m = 30cm$$

22. **In an undershot water wheel, the cross-sectional area $a = 0.1\,m^2$ of the stream is striking the series of radial flat vanes of the wheel. The velocity of stream is 6 m/s. The velocity of vanes is 3 m/s. If the power supplied by jet (in watts) is 2700K, find K.**



Sol. 2

Mass of water striking vanes/sec = ρav

Initial velocity = v

Final velocity of water = u

Change in velocity = v – u

F = force on the vane = ρav(v – u)

Work done = f × u = ρavu(v – u) = 5400 watts

⇒ K = 2

Numerical Decimal Type

23. **The mean pressure is $\frac{6K}{5}$, which rain renders to vertical windshield of automobile, moving with constant velocity of magnitude v = 12 m/s. Consider that raindrops fall vertically with speed u = 5 m/s. The intensity of rainfall deposits h = 2 cm of sediments in time τ = 1 minute. [ρ = 10^3 kg/m^3 is the density of liquid] (Assume collisions are inelastic).**

Sol. 8

Let in time t, m mass of liquid collides with windshield. So

$$F = \left|\frac{p_f - p_i}{t}\right| = \left|\frac{0 - mv}{t}\right| = \frac{mv}{t}$$

$\lambda \Rightarrow$ density of liquid molecules in air so

$m = Svt \times \lambda = Sv\lambda t$

Where $S \Rightarrow$ surface area of the windshield

$$F = Sv^2\lambda \Rightarrow \text{Mean pressure} = \frac{F}{S} = \lambda v^2 \qquad \ldots(1)$$

Now as we assumed that λ is the density of liquid molecules of drops in air so in τ time $u\tau S_0$ volume of liquid will strikes on the surface of earth so mass of liquid strikes on the earth in time interval τ will be

$m_0 = S_0 u\tau\lambda$

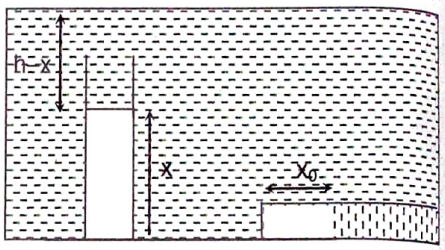
$\Rightarrow \quad S_0\rho h = S_0 u\tau\lambda$

$$\Rightarrow \quad \lambda = \frac{\rho h}{u\tau}$$

Putting this value in equation (1), we have

$$\text{Mean pressure} = \frac{\rho h v^2}{u\tau} = \frac{48}{5} \Rightarrow K = 8$$

24. **On the bottom of lake at a depth h = 100m, a horizontal pipe of length 80m with piston is kept as shown in the figure. The piston is light and movable. Between the piston and pipe some air is captured (x_0 = 9m). The pipe is slowly raised to vertical position and open end upward. The length of air column (x) in SI units is 5K. Find value of K. (Disregard the atmospheric pressure & piston and pipe are air tight. Temperature of gas remains constant).**



Sol. 2

$P_2 = \rho g(h - x), \; P_1 = \rho gh$

$V_2 = Ax \quad V_1 = Ax_0$

Using Boyle's low we can write

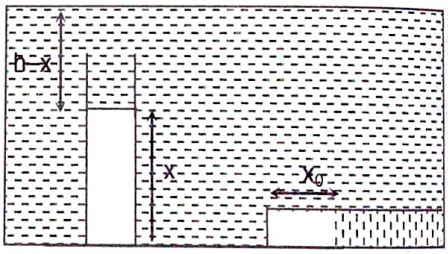
$\rho g(h - x)Ax = \rho gh(Ax_0)$

$$\Rightarrow \quad x^2 - hx + hx_0 = 0 \Rightarrow x = \frac{h}{2} \pm \sqrt{\frac{h^2}{4} - hx_0}$$

$\Rightarrow \quad x = 10\text{m}, 90\text{m}$

$x < \ell$ (length of pipe)

So answer will be x = 10 m



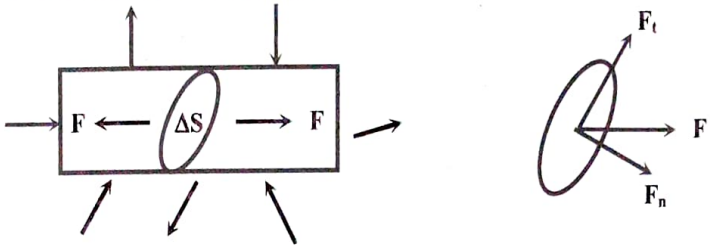
ELASTICITY AND WAVES 8

Elasticity

Solid bodies under the action of external forces get deformed. However, when these forces are removed it is observed that the bodies return to its original size of shape.

This property of a body whereby its natural shape is restored after external stress is removed is known as elasticity.

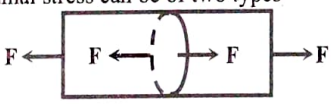
Longitudinal and Shearing Stress



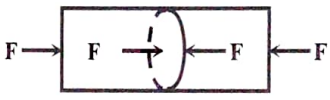
The internal restoring force F can be resolved into a component parallel to ΔS represented by F_t and a component normal to ΔS and is called F_n. Normal stress or longitudinal stress τ_n is defined as $\tau_n = \frac{F_n}{\Delta S}$.

Tangential or shearing stress is defined as $\tau_t = \frac{F_t}{\Delta S}$

Longitudinal stress can be of two types



Tensile Stress

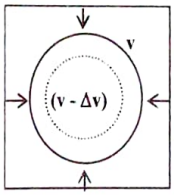


Compressive Stress

Volume Stress

Volume stress is equal to change in pressure acting on a given volume

ΔP = volume stress.



Strain

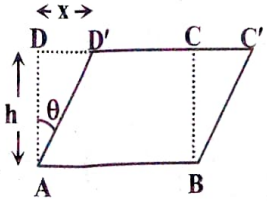
(a) Longitudinal strain = $\Delta L/L$

It can be tensile strain, if $\Delta L > 0$

It is compressive strain, if $\Delta L < 0$.

(b) Shearing strain

Shearing strain = $\frac{DD'}{DA} = \frac{x}{h}$



HOOKE'S LAW

If the deformation is small the stress in a body is proportional to the corresponding strain

By Hooke's law, $\frac{\text{Tensile Stress}}{\text{Tensile Strain}} = Y$ (Young's modulus)

Also $Y = \frac{FL}{\pi r^2 \Delta L}$, for a stressed wire

The ratio of volume stress over volume strain is called Bulk Modulus.

$$B = -\frac{\Delta p}{\Delta v / v} = -v\frac{dp}{dv} = \rho\frac{dp}{d\rho}$$ (Δp – change in pressure, ρ = density)

The isothermal bulk modulus of a gas is equal to p.
The adiabatic bulk modulus of a gas is equal to γp.

The ratio of shear stress to shear strain is called shear modulus.

Compressibility k is defined as the reciprocal of bulk modulus.

Stress–Strain graph for a metal wire

P – proportional limit

E – elastic limit

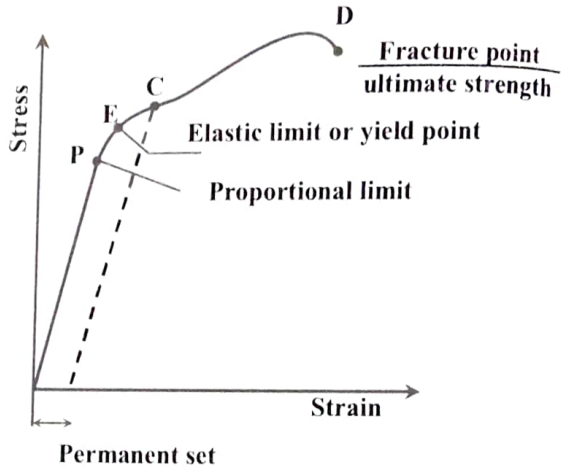
D – fracture point

$= \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$

= potential energy stored

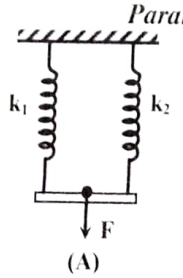
$F = kx$, $U = \frac{1}{2}kx^2$, k = spring constant

x = extension in spring.

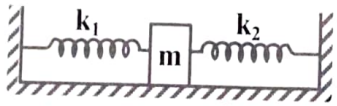


COMBINATIONS OF SPRINGS

Parallel Combination

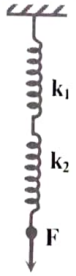


(A)



$k_{eff} = k_1 + k_2$

Series Combination



$$k_{eff} = \frac{k_1 k_2}{k_1 + k_2}$$

(B)

WAVES

Equation of Travelling Wave

(i) $y = f(x \pm vt)$ (General wave). For the function 'f' to be a wave function it must be defined for all real values of x and t and it should be bounded.

For example, (i) $y = A \log(kx \pm vt)$ is not a proper wave function since it is not defined for all real x and t.

(ii) $y = Ae^{k(x-vt)}$ is not a wave function as it is not bounded.

(iii) A harmonic wave can be represented as

$$y = A\sin\frac{2\pi}{\lambda}(x \pm vt) \qquad \text{(Sinusoidal wave)}$$

or, $y = A\sin(kx \pm \omega t)$

The negative sign refers to the wave travelling along the positive x - axis, and vice-versa.

$k = \frac{2\pi}{\lambda}$ = angular wave number

$\omega = \frac{2\pi}{T} = 2\pi f$ = angular frequency

$v = \lambda f = \frac{\omega}{k}$ = wave velocity

SOUND WAVES

In these longitudinal waves there is a phase gap of $\pi/2$ between the displacement and pressure waves, i.e at displacement minima there is pressure maxima and vice versa.

If $Y = A\sin(\omega t - kx)$, then $p = BkA\cos(\omega t - kx)$, where p = pressure difference from normal pressure and B = bulk modulus of the gas.

Velocity of Wave

(i) Transverse wave in a tight string

$$v = \sqrt{\frac{T}{\mu}}$$ [T = tension, μ = mass per unit length.]

(ii) Longitudinal wave in a solid

$$v = \sqrt{\frac{Y}{\rho}}$$ [Y = Young's Modulus, ρ= density]

(iii) Longitudinal wave in a fluid

$$v = \sqrt{\frac{K}{\rho}}$$ [K = Bulk Modulus]

For sound propagation in air, $K = \gamma p$,
where γ = adiabatic exponent, p = pressure of the air.

Energy of a Plane Progressive Wave

$$Y = A\sin(\omega t - kx)$$

Total energy $dE = \frac{1}{2}(\rho Sdx)(A\omega)^2$

Energy density $= \frac{1}{2}(\rho Sdx)(A\omega)^2 \Big/ S.dx = \frac{\rho}{2}(A\omega)^2$

$$= 2\pi^2 f^2 A^2 \rho\left[J/m^3\right]$$

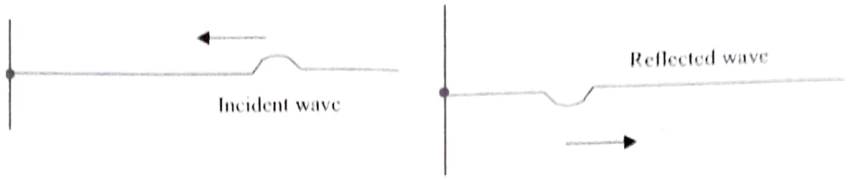
Energy per unit length $= PS(2\pi^2 f^2 A^2 \rho)$

Power transmitted $= PS(2\pi^2 f^2 A^2 \rho)V$ (watt = J/s)

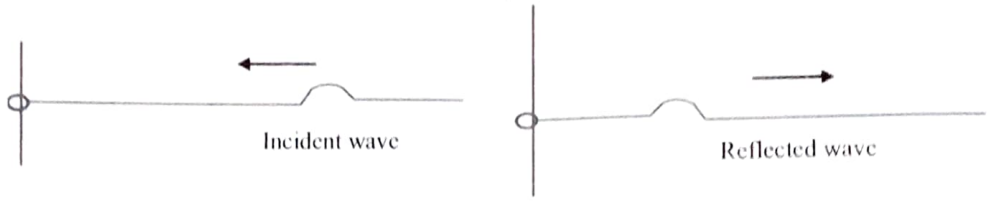
$$\text{Intensity} = \frac{\text{Power}}{\text{unit area}} = 2\pi^2 f^2 A^2 \rho V$$

Reflection of Waves

(a) Waves on reflection from a fixed end undergoes a phase change of 180°.



(b) On reflection from a free end is there is no change in phase.



STANDING WAVE

A standing wave is produced by the superposition of two identical waves travelling in opposite directions viz., $y_1 = a \sin [kx + \omega t]$ and $y_2 = a \sin (\omega t - kx)$, gives the standing wave, $y = 2a \sin \omega t \cos kx$

Note:

(a) In this equation, it is seen that a particle at any particular point x executes simple harmonic motion and all particles vibrate with the same frequency.

(b) The amplitude is not the same for different particles but varies with the location x of the particle.

(c). The points having the maximum amplitude are those where 2a cos kx has a maximum value of 2a, these are at the position,

$kx = 0, \pi, 2\pi, \ldots$

i.e. $x = 0, \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \ldots$

These points are called antinodes.

(d) The amplitudes reaches a minimum value of zero at the positions

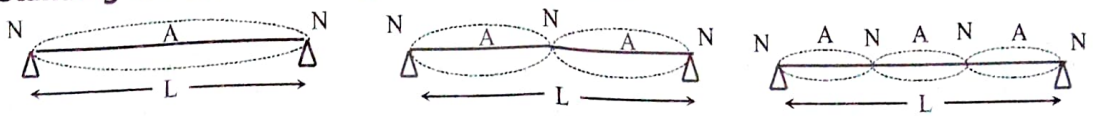
where $kx = \pi/2, 3\pi/2, 5\pi/2, \ldots$.

or, $x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \ldots$

These points are called nodes.

(e) Energy is not transported along the string to the right or to the left, because energy cannot flow parts from nodal points in the string which are permanently at rest.

Standing Waves on a String Fixed at both Ends



Fundamental First overtone 2nd overtone

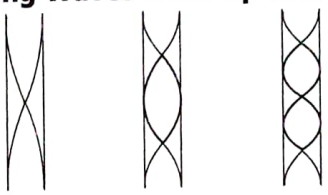
The allowed frequencies of standing waves in a string fixed at both ends are given by

$$f = \frac{n}{2\ell}\sqrt{\frac{T}{\mu}}, \text{ where } n = 1, 2, 3, \ldots.$$

L = distance between two fixed ends

T = tension is the string, μ = mass per unit length.

Standing Waves in an Open Pipe

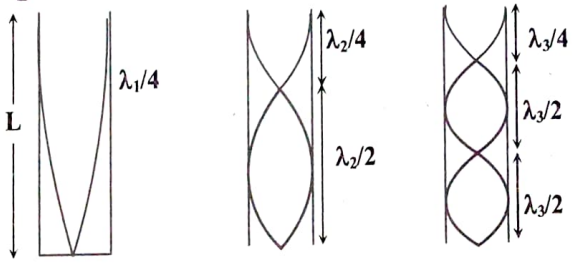


The allowed frequencies of standing waves in an open pipe

$$f = \frac{n}{2L}\sqrt{\frac{k}{\rho}}, \; n = 1, 2, 3, \ldots.$$

L = length of the pipe and k= bulk modulus

Standing Waves in a Closed Pipe



The allowed frequencies of standing waves in an open pipe

$$f = \frac{n}{4L}\sqrt{\frac{k}{\rho}}, \; n = 1, 3, 5, \ldots.$$

L = length of the pipe and k = bulk modulus

Note: n is the numbers of harmonics in case of closed pipe, only odd harmonics are present.

BEATS

If two or more waves of slightly different frequencies are superimposed, the intensity of the resulting wave has alternate maxima and minima. The number of maximas in one second is called beat frequency.

$y_1 = A \sin(2\pi f_1 t - \phi_1)$

$y_2 = A \sin(2\pi f_2 t - \phi_2)$

$y = y_1 + y_2 = A \sin(2\pi f_1 t - \phi_1) + A \sin(2\pi f_2 t - \phi_2)$

$$y = 2A \sin\left\{2\pi\left(\frac{f_1 + f_2}{2}\right)t - \left(\frac{\phi_1 - \phi_2}{2}\right)\right\} \cos 2\pi\left(\frac{f_1 - f_2}{2}\right)t$$

$$y = R \sin\left\{2\pi\left(\frac{f_1 + f_2}{2}\right)t - \left(\frac{\phi_1 - \phi_2}{2}\right)\right\}$$

where $R = 2A \cos \pi(f_1 - f_2)t$

The time for one beat is the time between consecutive maxima or minima.

First maxima would occur when

$$\cos \pi(f_1 - f_2)t = +1$$

$\therefore$ $t = 0$

Then $2\pi\left(\frac{f_1 - f_2}{2}\right)t = 0$

Second maxima would occur when

$$\cos \pi(f_1 - f_2)t = -1$$

Then $2\pi(f_1 - f_2)t = \pi$

or, $t = \frac{1}{f_1 - f_2}$

The time for one beat $= \left(\frac{1}{f_1 - f_2} - 0\right) = \frac{1}{f_1 - f_2}$

$\therefore$ Time for one beat $= \frac{1}{f_1 - f_2}$

Beat frequency $= |f_1 - f_2|$

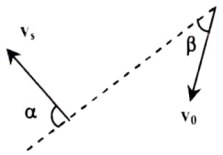
DOPPLER EFFECT

The apparent shift in frequency of the wave motion when the source of sound moves with respect to the observer, is called Doppler Effect.

The formula for calculating apparent frequency

$$n' = \left(\frac{c + v_0}{c - v_s}\right) \times n,$$ where c = velocity of sound, v_0 = velocity of observer,

v_s = velocity of source, n = frequency of sound both considered positive when they are moving towards each other.



For example in the case shown $n' = \left(\frac{c - v_s \cos\alpha}{c - v_0 \cos\beta}\right) \times n$

If the wind is blowing with a velocity ω then the effective value of c is taken as

(a) $c + \omega$, if wind is blowing in the same direction as that of sound travel.

(b) $c - \omega$, if wind in blowing in the opposite direction as that of sound travel.

ILLUSTRATIVE PROBLEMS

1. A person sitting at a distance of 10 m from a point source of sound of power W_0 writes down the following equation for the waves received by him.

$$y = A\sin(20\pi x + 7000\pi t)$$

(a) Determine the density of the air, if the Bulk modules of air $= 1.40 \times 10^5\ Nm^{-2}$

(b) If the largest pressure amplitude tolerable to a human ear is 28 Pa then determine the maximum power W_0 of the sound source and the displacement amplitude A of the waves received by him.

Sol. Since, $y = A\sin(2\pi x + 7000\pi t) = A\sin(kx + \omega t)$

$\therefore\ k = 20\pi,\ \omega = 7000\pi$

$\therefore$ Velocity of sound $v = \omega/k = 350$ m/s

Given that $B = 1.48 \times 10^5$.

$$\therefore\ v = \sqrt{\frac{B}{\rho}}$$

$$\rho = B/v^2 = \frac{1.40\times10^5}{(350)^2} = 1.143 kg/m^3$$

Pressure amplitude is defined as $P_0 = BkA$

Given that max. $P_0 = 28$ Pascal

$k = (20)\pi$

$$\therefore\ A = \frac{P_0}{BK} = \frac{28}{20\pi\times1.40\times10^5} = 3.183\mu m$$

Intensity received by the person $I = W_0/(4\pi(10)^2)$

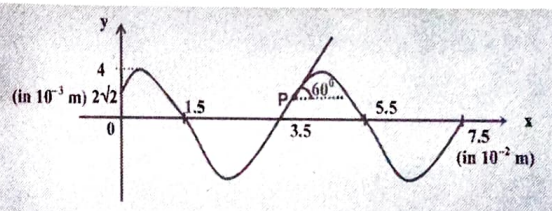
Also $I = \dfrac{vP_0^2}{2B}$

$$W_0 = \frac{4\pi(10)^2 vP_0^2}{2B} = \frac{200\pi\times(350)(28)^2}{2\times(1.4\times10^5)} = 615.75 \text{ Watts.}$$

2. The figure shows a snap photograph of a vibrating string at t = 0. The particle P is observed moving up with velocity $20\sqrt{3}$ cm/s. The tangent at P makes an angle 60° with x-axis. Find

(a) the direction in which the wave is moving

(b) the equation of the wave



Sol. (a) P is moving up i.e. has positive velocity, particle velocity,

$$v_p = -v\left(\frac{dy}{dx}\right) \quad \ldots(1)\ v = \text{wave velocity}$$

$$\frac{dy}{dx} = \text{slope at point P} = \tan 60^0 = \sqrt{3}$$

As v_p and $\dfrac{dy}{dx}$ are both positive hence v must be negative, hence the wave is moving in negative x-axis.

(b) The equation of the wave

$y = a \sin(\omega t - kx + \delta)$ ... (2)

$\omega = 2\pi n$ and $k = \frac{\omega}{v} = \frac{2\pi}{\lambda} = \frac{\pi}{2}$ cm^{-1}

$a = 4 \times 10^{-3}$ m $= 0.4$ cm At $t = 0$, $x = 0$ and $y = 2 \times 10^{-3}$ m $= 0.2$ cm

Substituting in (2)

$2\sqrt{2} \times 10^{-3} = 4 \times 10^{-3} \sin\delta$, which gives $\delta = \frac{\pi}{4}$ or $\frac{3\pi}{4}$

Also at $t = 0$, $x = 1.5$ cm. And $y = 0$

$0 = a \sin(0 - \frac{3\pi}{4} + \delta)$ which gives $\delta = \frac{3\pi}{4}$

hence, $\delta = \frac{3\pi}{4}$

$20\sqrt{3} = -v \times \tan 60^0$;

$v = -20$ cm /s (-sign because wave is moving in negative direction)

Hence, $n = v/\lambda = 5$ Hz.

Hence, $\omega = 2\pi \times 5 = 10\pi$

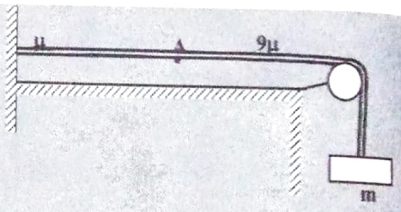
Substituting the values of a, ω, k and δ

We get,

$y = 0.4 \sin(10\pi t - \frac{\pi}{2}x + \frac{\pi}{6})$

or $y = 0.4 \sin 2\pi(5t - \frac{x}{4} + \frac{1}{12})$.

3. A light string is tied at one end to a fixed support and to a heavy string of equal length L at the other end as shown in figure. A block of mass m is tied to the free end of heavy string. Mass per unit length of the strings are μ and 9μ and the tension is T. Find the possible values of frequencies such that junction of two wire point A is a node.



Sol. For A_1 be a node

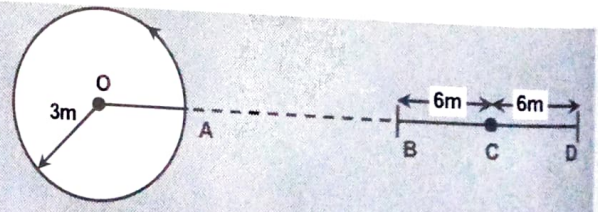
Suppose n_1 and n_2 are the complete loops formed on left and right side of point A.

$n_1 \frac{\lambda_1}{2} = L \quad \Rightarrow \quad f = \frac{v_1 n_1}{2L}$

$\frac{n_2 \lambda_2}{2} = L \quad \Rightarrow \quad f = \frac{v_2 n_2}{2L}$

$\Rightarrow v_1 n_1 = v_2 n_2 \quad \Rightarrow \quad n_1 : n_2 = 1 : 3$ (n_1 being any positive integer)

4. A source of sound is moving along a circular path of radius 3 m with an angular velocity of 10 rad/s. A sound detector located far away from the source is executing linear S.H.M. along the line BD shown in the figure with an amplitude BC = DC = 6m.



The frequency of oscillation of the detector is $5/\pi$ Hz. The source is at point A when the detector is at point B. If the source emits a continues sound wave of frequency 340 Hz, find the maximum and minimum frequencies recorded by the detector.

Sol. The speed of sound in air = 340 m/s.

r = 3 m, ω = 10 rad/s. The linear speed of the source along the circular orbit is

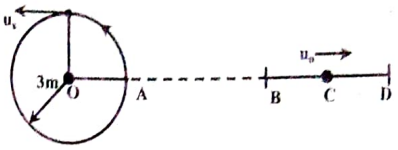
$v_s = r\omega = 3 \times 10$ m/s

The detector is executing SHM of amplitude a = 6 m and frequency n = 5/π Hz. The angular velocity of the detector is

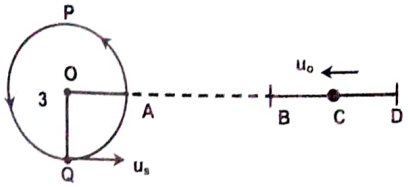
$\omega' = 2\pi n = 2\pi \times 5/\pi = 10$ rad/s

and the linear velocity of the detector is

$u_o = a\omega' = 6 \times 10 = 60$ m/s



Since the source and the detector have the same angular velocity (i.e. ω = ω'), they have the same time period T. At time t = T/4, the source will be at point P and the detector will be at point C.



Since they are receding from each other at the maximum relative velocity, the recorded frequency will be minimum which is given by

$$n_{min} = n\left(\frac{v - u_o}{v + u_s}\right) = 340 \times \left(\frac{340 - 60}{340 + 30}\right) = 257 \text{ Hz}$$

At a time t = 3T/4, the source will be at point Q and the detector will be at point C approaching the source with the maximum relative velocity. Hence the recorded frequency will be maximum which is given by

$$n_{max} = n\left(\frac{v + u_o}{v - u_s}\right) = 340 \times \left(\frac{340 + 60}{340 - 30}\right) = 485.7 \text{ Hz}$$

5. The displacement of the medium in a sound wave is given by the equation:
y = A cos (ax + bt) where A, a, b are +ve constants.
The wave is partially reflected by a denser medium at x = 0. The intensity of the reflected wave is 0.49 times that of the incident wave.
(i) Write the equation for the reflected wave.
(ii) Express the resultant wave as a superposition of a standing wave and a travelling wave.
(iii) What are the positions of the antinodes of the standing wave?
(iv) In the resultant wave formed after reflection find the maximum value of the particle speed in the medium.

Sol. (i) Amplitude of the reflected wave = $\sqrt{0.49}A = 0.7A$

As the reflecting surface is a denser medium there will be a phase change of π. Also, the direction of propagation will be reversed. Hence the equation of the reflected wave will be

$y = 0.7A \cos(-ax + bt + \pi)$

or $y = -0.7A \cos(bt - ax)$

(ii) Resultant wave

$y = A \cos(ax + bt) - 0.7A \cos(bt - ax)$

$y = 0.3A \cos(ax + bt) - 1.4A \sin ax \sin bt$

First term represents a progressive wave where as the second one is a standing wave.

(iii) For antinodes we have

$\sin ax = \pm 1$

$\Rightarrow ax = (2n - 1)\frac{\pi}{2}$; $x = (2n - 1)\pi/2a$

(iv) Maximum speed of a particle of medium will occur at antinode. Equation of motion of a particle at antinode.

$$y = 0.3A\cos\left[(2n-1)\frac{\pi}{2} + bt\right] - 1.4A \sin(2n-1)\,\pi/2 \sin bt$$

$$\frac{dy}{dt} = -0.3bA \sin[(2n-1)\pi/2 + bt] - 1.4\,bA \sin[(2n-1)\,\pi/2]\cos bt$$

$$\frac{dy}{dt} = 1.7A\,b\cos bt \quad \text{(if n is even)}$$

$$\frac{dy}{dt} = -1.7A\,b\cos bt \quad \text{(if n is odd)}$$

$\therefore$ Maximum speed of the particle = 1.7Ab

SOLVED PROBLEMS

1. When the load on a wire is increasing slowly from 2 kg to 4 kg, the elongation increases from 0.6 mm to 1 mm. The work done during this extension of the wire is ($g = 10\ m/s^2$)

(A) 14×10^{-3} J **(B) 0.4×10^{-3} J**

(C) 8×10^{-2} J **(D) 10^{-3} J**

Sol. $W = \left(\frac{1}{2}F_2 \times \Delta\ell_2\right) - \left(\frac{1}{2}F_1 \times \Delta\ell_1\right) = 20 \times 10^{-3} - 6 \times 10^{-3} = 14 \times 10^{-3}$ J

$\therefore$ (A)

2. In the interference of waves from two sources of intensities I_0 and $4I_0$, the intensity at a point where the phase difference is π is

(A) I_0 **(B) $2I_0$**

(C) $3I_0$ **(D) $4I_0$**

Sol. $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\theta = I_0 + 4I_0 + 2\sqrt{(I_0 \times 4I_0)}\cos\pi = I_0$.

$\therefore$ (A)

3. An air column in a pipe, which is closed at one end, will be in resonance with vibrating tuning fork of frequency 264 Hz, if the length of the column is

(A) 31.25 **(B) 62.50**

(C) 93.75 **(D) 125**

Sol. $f = \frac{V}{4L}$

$$264 = \frac{330}{4 \times L} \Rightarrow L = \frac{330}{4 \times 264} = 31.25 \text{ cm}$$

$\therefore$(A)

4. A rigid bar of mass M is supported symmetrically by three wires each of length ℓ. Those at each end are of copper and the middle one is of iron. The ratio of their diameters, if each is to have the same tension, is equal to

(A) $\frac{Y_{copper}}{Y_{iron}}$ **(B)** $\sqrt{\frac{Y_{iron}}{Y_{copper}}}$

(C) $\frac{Y_{iron}^2}{2Y_{copper}^2}$ **(D)** $\frac{2}{3}\frac{Y_{iron}}{Y_{copper}}$

Sol. B

$$\frac{T}{A} = \frac{Y\Delta\ell}{\ell}$$

5. When two simple harmonic motions of same periods, same amplitude, having phase difference of $3\pi/2$, and at right angles to each other are super imposed, the resultant wave form is a

(A) circle **(B) parabola**

(C) ellipse **(D) figure of eight**

Sol. The SHM's are $x = a \sin \omega t$ and $y = a \sin(\omega t + 3\pi/2) = -a \cos \omega t$

$\therefore\ x^2 + y^2 = a^2$.

$\therefore$ (A)

6. A transverse wave is described by the equation $y = y_0 \sin 2\pi (ft - x/a)$. The maximum particle velocity is equal to four times the wave velocity if a is equal to

(A) $\pi y_0 / 4$ **(B) $\pi y_0/2$**

(C) πy_0 **(D) $2\pi y_0$**

Sol. The maximum particle velocity of a SHM of amplitude y_0 and frequency f is $2\pi f y_0$. The wave velocity is $f\lambda$. For $2\pi f y_0$ to be equal to $4f\lambda$, λ has to be $\pi y_0 / 2$

(Here $\lambda = a$).

$\therefore$ (B)

7. The amplitude of a wave disturbance propagating in the positive x-direction is given by $y = 1/(1 + x^2)$ at time $t = 0$ and by $y = 1/[1 + (x - 1)^2$ at $t = 2$ seconds, where x and y are in metres. The shape of the wave disturbance does not change during the propagation. The velocity of the wave is

(A) 1 ms^{-1} **(B) 0.5 ms^{-1}**

(C) 1.5 ms^{-1} **(D) 2 ms^{-1}**

Sol. Writing the general expression for y in terms of x as

$y = \frac{1}{1+(x - vt)^2}$. At $t = 0$, $y = 1/(1 + x^2)$. At $t = 2$ s, $y = \frac{1}{1+[x - v(2)]^2}$

Comparing with the given equation we get $2v = 1$ and $v = 0.5$ m/s.

$\therefore$ (B)

8. The displacement y of a particle executing periodic motion is given by $y = 4 \cos^2(1/2\ t) \sin(1000\ t)$. This expression may be considered as a result of the superposition of

(A) two waves **(B) three waves**

(C) four waves **(D) five waves**

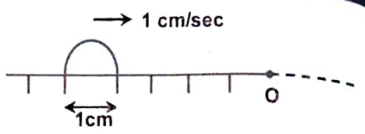
Sol. If we have a term cos kx sin ωt, it is the superposition of two wave motions. If we have an equation of the form $\cos^2$ at sin bt it can be shown to be superposition of three sine waves.

Here, $y = 4 \cos^2 (\frac{1}{2} t \sin(1000\ t)$

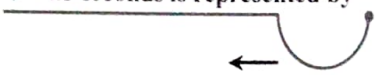
$= 2 [1 + \cos t] \sin 1000\ t = 2 [\sin 1000\ t + \frac{1}{2}(\sin 100\ t + \sin 999\ t)]$ = this comprises three waves.

$\therefore$ (B)

9. **A wave pulse on a string has the shape at a particular instant as shown in the figure. The wave speed is V = 1 cm/s. The point O is a fixed end. The total wave on the string at t = 4.5 seconds is represented by**



(A)



(B)

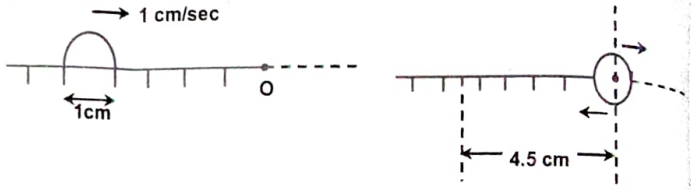
(C)

(D)

Sol. The formation of the reflected pulse from the fixed support is similar to the overlap of two in vertical pulse traveling in opposite direction as shown below

$\therefore$ At t = 4.5 net displacement of all particles of string will be zero, i.e. the string will be straight as shown

$\therefore$ (C)



10. **If the amplitude of waves at a distance r from line source is A. The amplitude at a distance 4r will be**

(A) 2A **(B) A**

(C) A/2 **(D) A/4**

Sol: For point source intensity (I) $\alpha \frac{1}{r^2}$

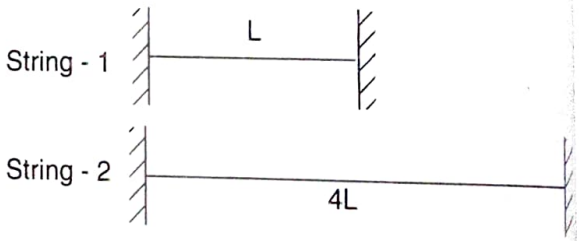
$I \propto \text{amplitude } (A)^2 \Rightarrow \text{Amplitude} \propto \frac{1}{r}$

$A_1 r_1 = A_2 r_2$

$A_2 = A/4$

$\therefore$ (D)

11. **Two strings made of same material, same cross-section but different lengths are fixed at both ends and excited in their respective fundamental modes such that total oscillation energy is the same in both cases. let f_1, f_2 be the oscillation frequencies and A_1, A_2 be the amplitudes of oscillation of the antinodes in the two cases. Assuming both strings to be under same tension, pick the correct option(s).**



(A) $\frac{f_1}{f_2} = \frac{1}{4}$ **(B)** $\frac{A_1}{A_2} = \frac{1}{2}$

(C) $\frac{f_1}{f_2} = \frac{4}{1}$ **(D)** $\frac{A_1}{A_2} = \frac{2}{1}$

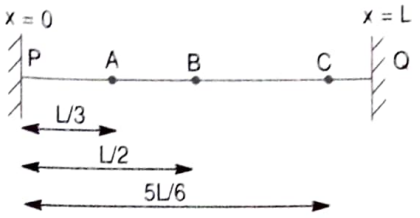
Sol. B, C

$E = m_1\omega_1^2A_1^2 = m_2\omega_2^2A_2^2$

$f = \frac{V}{2L} \Rightarrow \frac{f_1}{f_2} = \frac{\omega_1}{\omega_2} = \frac{L_2}{L_1} = \frac{4}{1}$

$\frac{A_1}{A_2} = \sqrt{\frac{m_2}{m_1} \times \left(\frac{\omega_2}{\omega_1}\right)^2} = \sqrt{\frac{L_2}{L_1} \times \left(\frac{L_1}{L_2}\right)^2} = \frac{1}{2}$

12. **Standing waves are established on a string of length L such that A is a node and B is an immediate anti node. Oscillation amplitude for point B is a_0. Let a′ be the oscillation amplitude for point C. Pick the suitable option(s) for correct value of a′ and the possible equation(s) for standing waves on the string.**



(A) $a' = \frac{a_0}{2}$

(B) $y = a_0 \sin\left(\frac{6\pi}{L}x\right)\cos(\omega t)$

(C) $a' = a_0$

(D) $y = a_0 \sin\left(\frac{3\pi x}{L}\right)\sin(\omega t)$

Sol. C, D

$y = a_0 \sin(kx) \sin(\omega t + \phi)$

$\frac{kL}{3} = n\pi$ and $\frac{kL}{2} = \left(n + \frac{1}{2}\right)\pi$

$\Rightarrow n = 1,\ k = \frac{3\pi}{L}$

Also; $a' = \left|a_0 \sin\left(\frac{3\pi}{L} \times \frac{5L}{6}\right)\right| = a_0$

COMPREHENSION

A large cylindrical chamber has a massless piston fitted at the top. An acoustic source is situated at the centre of the bottom of the cylinder, while an observer is situated at the bottom circumference (inside the chamber). The observer is moving directly towards the source with constant velocity and hears a frequency f

13. **If the chamber were to be isobarically heated then the observed frequency will be**
(A) same as f **(B) higher than f**
(C) lower than f **(D) data insufficient**

Sol. **C**

14. **If the chamber were to be isochorically cooled then the observed frequency will be:**
(A) same as f **(B) higher than f**
(C) lower than f **(D) Date insufficient**

Sol. **B**

15. **If the chamber were to be isothermally heated then, the observed frequency will be:**
(A) same as f **(B) higher than f**
(C) lower than f **(D) Data insufficient**

Sol. A
Use Doppler's effect, the speed of sound increases with increase in temperature.

Matrix – Match Type

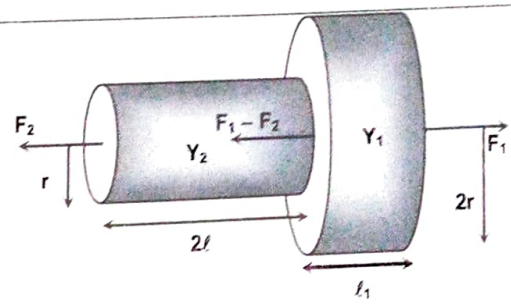
16. **Match the following:**
A source is emitting a sound of frequency f_s and moving with a velocity v_s. An observer at some distance from the source has velocity v_0 and heard sound of frequency f_0. Velocity of medium is v_m.

Column – I		Column- II
(A) $v_s = v_0 = 0$ and v_m has direction from source to observer.	(p)	$f_s > f_0$
(B) v_m has direction from source to observer and v_s is towards observer: $v_0 = 0$	(q)	$f_s < f_0$
(C) direction of v_m is as above and v_s is perpendicular to the line joining source and observer $v_0 = 0$	(r)	$f_s = f_0$
(D) $v_m = 0$ and v_0 & v_s is parallel to each other and have none zero component along the line joining source and observer.	(s)	Whether f_0 is greater than f_s or less than f_s will depend on magnitude of v_m, v_s and v_0.

Sol. (A) → (r) (B) → (q) (C) → (r) (D) → (s)

Answer questions 17, 18 and 19 by appropriately matching the information given in the three columns of the following table.

Figure shows a composite object made by rigidly welding two cylinders of different materials. The right part has length ℓ, cross section 4A and young's modulus $Y_1 = Y$ whereas the left part has length 2ℓ, cross section A and young's modulus $Y_2 = 4Y$. F_1, F_2, $F_1 - F_2$ are three external forces acting on the composite system and it is at rest under the action of these forces. Column –I lists different combinations of forces F_1, F_2. Column –II lists the corresponding total extension and column –III lists the total elastic energy stored for the composite system.



Column 1		Column 2		Column 3	
(I)	$F_1 = 2F, F_2 = F$	(i)	$\frac{5F\ell}{4AY}$	(P)	$\frac{3F^2\ell}{8AY}$
(II)	$F_1 = 3F, F_2 = 2F$	(ii)	$\frac{F\ell}{AY}$	(Q)	$\frac{11F^2\ell}{8AY}$
(III)	$F_1 = 3F, F_2 = F$	(iii)	$\frac{3F\ell}{4AY}$	(R)	$\frac{17}{8}\frac{F^2\ell}{AY}$
(IV)	$F_1 = F_2 = F$	(iv)	$\frac{7F\ell}{4AY}$	(S)	$\frac{3}{4}\frac{F^2\ell}{AY}$

17. **Pick the correct option.**
(A) (I) (i) (P)
(B) (II) (iii) (S)
(C) (III) (i) (Q)
(D) (IV) (ii) (R)

Sol. C

18. **Pick the correct option for maximum value of extension.**
(A) (II) (i) (R) (B) (IV)(i) (S)
(C) (III) (iv) (P) (D) (II) (iv) (R)

Sol. D

19. **Pick the correct option for minimum value of stored elastic energy.**
(A) (I) (iv) (P) (B) (I) (iii) (Q)
(C) (IV) (iii) (P) (D) (IV) (ii) (S)

Sol. C

Numerical Type

20. **A close organ pipe of diameter 10 cm has length 42 cm. The air column in pipe vibrates in its second overtone with the maximum amplitude ΔP_o. The pressure amplitude at middle of pipe is $\frac{\sqrt{n}\Delta P_o}{2}$. Find n.**

Sol. 3

Here

$5\frac{\lambda}{4} = 42 + 0.3\times 10 \Rightarrow \lambda = 36\text{cm}$

$\therefore$ pressure amplitude at middle of pipe is

$$\Delta P = \Delta P_o |\sin Kx| = \Delta P_o \left|\sin\left(\frac{2\pi}{36}\times 24\right)\right| = \frac{\sqrt{3}\Delta P_o}{2}$$

21. **Two cars A and B are moving towards each other with same speed 25 m/s. Wind is blowing with speed 5 m/s in the direction of motion of car A. The car A blows horn of frequency 300 Hz and the sound is reflected from car B. The wavelength of the reflected sound received by the driver of car A is $\frac{31}{4n}$ m. Find n. (velocity of sound in air = 330 m/s)**

Sol. 9

Frequency received by car B is $f_1 = \left(\frac{v+w+u}{v+w-u}\right)f$

Now the car B will be treated as a source of frequency f_1

$\therefore$ wavelength of reflected sound received by the driver of car A is

$$\lambda' = \frac{(v-w-u)}{f_1} = \frac{(v-w-u)(v+w-u)}{(v+w+u)f}$$

$$\frac{(330-5-25)(330+5-25)}{(330+5+25)300} = \frac{31}{36}\text{m}$$

Numerical Decimal Type

22. **A policeman on duty detects a drop of 10% in the pitch of the horn of a moving car as it crosses him. If the velocity of sound is 330 m/s. Find the speed (in m/s) of the car.**

Sol. **00017.30**

$$f_1 / f_2 = \left[\frac{c}{c-v} \Big/ \frac{c}{c+v}\right]$$

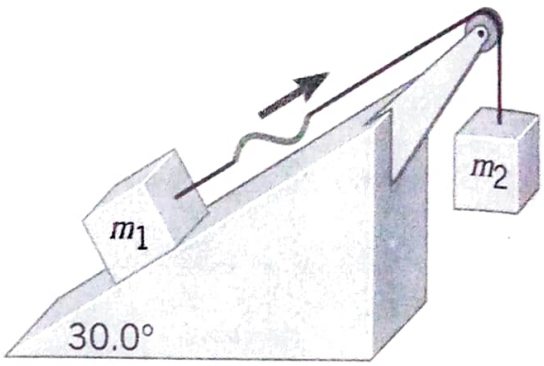
$$\Rightarrow \frac{f_1}{f_2} = \frac{c+v}{c-v} \Rightarrow \frac{100}{90} = \frac{c+v}{c-v}$$

$\Rightarrow 100\,c - 100\,v = 90\,C + 90\,v$

$\Rightarrow 10\,c = 190\,v$

$\Rightarrow 1/19 \times 330 = 17.3$ m/s.

23. **The drawing shows a frictionless incline and pulley. The two blocks are connected by a wire (mass per unit length, μ = 25 g/m) and remain stationary. A transverse wave on the wire has a speed of 60 m/s relative to it. Neglect the weight of the wire relative to the tension in the wire. If the mass m_2 be increased by 1%, Find the speed (in m/s) of the transverse wave. will be**



Sol. **00060.20**

Tension, $T = \frac{3m_1m_2}{2(m_1+m_2)}g$; $v = \sqrt{\frac{T}{\mu}}$; $\frac{\delta v}{v} = \frac{1}{2}\frac{\delta T}{T} = \frac{1}{3}\frac{\delta m_2}{m_2}$ (where, $m_1 \approx 2m_2$)

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PHYSICS

SIMPLE HARMONIC MOTION 9

LINEAR SHM

$$F = -kx = -m\omega^2 x$$

$$\therefore \quad \omega = \sqrt{\frac{k}{m}}$$

Further, $x = A \sin(\omega t + \phi)$

$$v = \frac{dx}{dt} = \omega A \cos(\omega t + \phi) = \omega\sqrt{A^2 - x^2}$$

$$a = \frac{dv}{dt} = -\omega^2 A \sin(\omega t + \phi) = -\omega^2 x$$

$$KE = \frac{1}{2} m\omega^2 A^2 \cos^2(\omega t + \phi)$$

$$PE = \frac{1}{2} m\omega^2 A^2 \sin^2(\omega t + \phi)$$

Total energy $(E) = \frac{1}{2} m\omega^2 A^2 = \text{constant.}$

ANGULAR SHM

Torque, $\tau = -k\theta$,

Angular acceleration, $-\alpha = \frac{-k\theta}{I} = -\omega^2\theta$

$$\therefore \quad \omega = \sqrt{\frac{k}{I}}$$

Also, $\theta = \theta_0 \sin(\omega t + \delta)$

$\omega = \theta_0\omega \cos(\omega t + \delta) = \omega\sqrt{\theta_0^2 - \theta^2}$ (ω is the angular velocity at any instant)

$\alpha = \theta_0\omega^2 \sin(\omega t + \delta) = -\omega^2\theta$

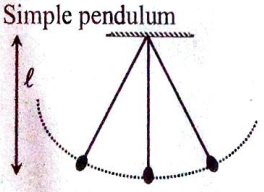
$$KE = \frac{1}{2} I\theta_0^2\omega^2 \cos^2(\omega t + \delta)$$

$$PE = \frac{1}{2} I\theta_0^2\omega^2 \sin^2(\omega t + \delta)$$

Total energy, $E = \frac{1}{2} I\theta_0^2\omega^2$

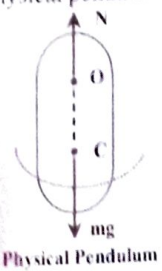
Time Period in Simple Cases involving SHM

(A) Time period, $T = 2\pi\sqrt{\frac{\ell}{g}}$



Simple pendulum

(B) Physical pendulum



Physical Pendulum

$$T = 2\pi\sqrt{\frac{I}{mgd}}$$

where OC = distance between the point of suspension and the centre of gravity and I = moment of inertia about the point of suspension.

Combination of Two SHMs in Same Direction

$$X_1 = A_1 \sin \omega t$$

$$X_2 = A_2 \sin(\omega t + \phi)$$

The resultant given by

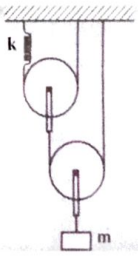
$$X = X_1 + X_2 = A \sin(\omega t + \delta)$$

where $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos\phi}$

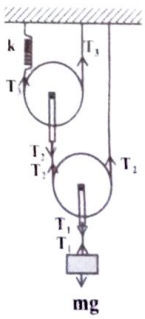
and $\tan\delta = \dfrac{A_2 \sin\phi}{A_1 + A_2 \cos\phi}$

ILLUSTRATIVE PROBLEMS

1. Find the natural frequency of the system as shown in the figure. The pulleys are massless and frictionless. (Take $k = 9\pi^2$ N/m and m = 4 kg)



Sol. To find frequency we have to first calculate the acceleration of the block as a function of displacement. In such type of complicated systems we can find this by using the equation of energy conservation and then differentiating it. One important point is to be noted is that the spring is already stretched by some amount and beyond this extension we can not take energy to be equal to (1/2) kx^2 so we have to first find out the initial extension.



$mg = T_1$, $T_1 = 2T_2$

$T_2 = 2T_3$, $T_3 = kx_0$

$$\Rightarrow x_0 = \frac{mg}{4k}$$

If suppose the block is further pulled down by x, spring stretches by 4x

$$U \text{ (Total energy of system)} = \frac{1}{2}k(x_0 + 4x)^2 - mgx + \frac{1}{2}mv^2$$

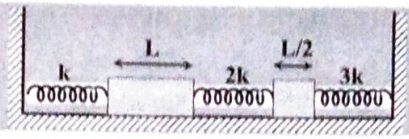
$$\frac{dU}{dt} = 0$$

$$k(x_0 + 4x)\,4\frac{dx}{dt} - mg\frac{dx}{dt} + mv\frac{dv}{dt} = 0$$

$$\frac{dv}{dt} = \frac{d^2x}{dt^2} = -16\frac{k}{m}x$$

$$\Rightarrow f = \frac{1}{2\pi}\sqrt{\frac{16k}{m}} \quad \Rightarrow f = 3.$$

2. The system shown consists of 3 springs and two rods. If the temperature of the rods is increased by ΔT, calculate the energy stored in each of the springs.



The springs are initially relaxed. There is no friction. Take the coefficient of linear expansion of the material of rods to be equal to α.

Sol. Let x_1, x_2 and x_3 be the compressions in the three springs, then

$$x_1 + x_2 + x_3 = L\alpha\Delta T + \frac{L}{2}\alpha\Delta T = \frac{3}{2}L\alpha\Delta T \qquad \ldots(1)$$

Considering the F.B.D.s of the two rods

Equilibrium of the rod requires that

$$x_1 = 2x_2 = =3x_3 \qquad \ldots(2)$$

$$\therefore\ x_1 + \frac{x_1}{2} + \frac{x_3}{3} = \frac{3}{2}L\alpha\Delta T$$

$$\Rightarrow x_1 = \frac{9}{11}L\alpha\Delta T$$

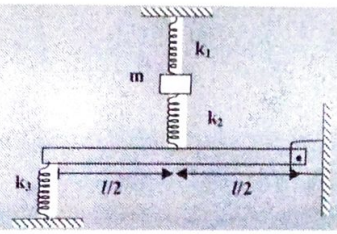
$\therefore$ Energy stored in spring of spring constant k

$$E_1 = \frac{1}{2}kx_1^2 = \frac{81}{242}kL^2\alpha^2\Delta T^2$$

Similarly

$$E_2 = \frac{81}{484}kL^2\alpha^2\Delta T^2 \quad \text{and} \quad E_3 = \frac{27}{242}kL^2\alpha^2\Delta T^2$$

3. Find the angular frequency of small oscillation of block m in the arrangement shown. Rod is massless. [Assume gravity to be absent]



Sol. Suppose the mass m is displaced a distance x downward and let θ be the angular displacement of the rod

It is clear that the spring of force constant k_2 will have a compression $-\left(x - \frac{l\theta}{2}\right)$ for small x and θ

$\therefore$ Balancing the torques on (massless) rod

$$\Rightarrow k_2\left(x - \frac{l\theta}{2}\right)\frac{l}{2}\cos\theta = k_1 l\theta l\cos\theta$$

$$\Rightarrow k_2\left(x - \frac{l\theta}{2}\right) = 2k_1 l\theta \qquad \Rightarrow \qquad l\theta = \frac{2k_2 x}{4k_1 + k_2}$$

Now for mass m

$$F_{rest} = -k_1 x - k_2\left(x - \frac{l\theta}{2}\right) = -k_1 x - k_2 x + \frac{k_2^2 x}{4k_1 + k_2} \quad \text{(putting the value of } l\theta\text{)}$$

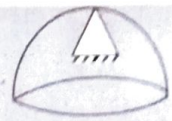
$$\Rightarrow m\frac{d^2x}{dt^2} = -\frac{4k_1k_3 + k_1k_2 + 4k_2k_3}{4k_1 + k_2}x$$

$$\therefore m\frac{d^2x}{dt^2} + \frac{4k_1k_3 + k_1k_2 + 4k_2k_3}{4k_1 + k_2}x = 0$$

$\therefore$ Motion is simple harmonic and

$$\omega = \sqrt{\frac{4k_1k_3 + k_1k_2 + 4k_2k_3}{(4k_1 + k_2)m}}.$$

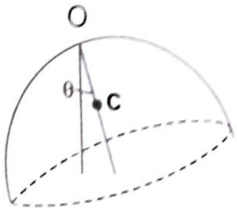
4. A hemispherical shell of mass M and radius R made to vibrate as shown in figure. Find its frequency of oscillation.



Sol. Centre of mass of the hemispherical shell is at a distance R/2 from the centre of hemisphere.

$\therefore$ The torque in displaced position is $Mg\frac{R}{2}\sin\theta$

M.I. of the hemispherical shell about the axis passing through O is $\frac{2MR^2}{3}$. (same as the M.I. about a diameter of the base as they are equidistant from the centre of mass.)



$$\Rightarrow I\alpha = -Mg\frac{R}{2}\sin\theta \qquad \text{(Torque and } \theta \text{ oppositely directed)}$$

$$\Rightarrow \frac{2}{3}MR^2.\alpha = -Mg\frac{R}{2}\theta \qquad (\theta \text{ is small})$$

$$\alpha = \frac{3}{4}\frac{g}{R}\theta$$

$$\omega^2 = \frac{3}{4}\frac{g}{R}$$

$$T = 2\pi\sqrt{\frac{4R}{3g}} = 4\pi\sqrt{\frac{R}{3g}}$$

5. The period of a particle executing SHM is T. There is a point P, at a distance 'x' from the mean position 'O'. When the particle passes P towards OP, it has speed v. Find the time in which it returns to P again.

Sol. Let the particle is at P at an instant t starting from mean position and it returns to p again after interval t'.

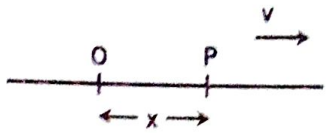
Then $x = a \sin \omega t$ ... (1)

Also $x = a \sin \omega(t + t')$ ... (2)

Form equation (1)

$$\frac{dx}{dt} = a\omega \cos \omega t = v$$

$\Rightarrow \cos\omega t = v/a\omega$ ... (3)



from equation (2)

$x = a [\sin\omega t' \cos\omega t + \cos\omega t' . \sin t]$

$= a[\sin\omega t' \frac{v}{a\omega} + \frac{x}{a} \cos\omega t']$ using equation (1) and (3)

$$t' = \frac{2}{\omega} \tan^{-1} \frac{v}{\omega x}$$

SOLVED PROBLEMS

1. A pendulum has time period T in air. When it is made to oscillate in water, it acquired a time period $T' = \sqrt{2}T$. The specific gravity of the pendulum bob is equal to

(A) $\sqrt{2}$ **(B) 2**

(C) $2\sqrt{2}$ **(D) None of these**

Sol. The effective acceleration of a bob in water $= g' = g\left(1 - \frac{d}{D}\right)$, where d and D are the density of water and the bob respectively.

$\frac{D}{d}$ = specific gravity of the bob.

Since the period of oscillation of the bob in air and water are given as

$$T = 2\pi \sqrt{\frac{\ell}{g}} \text{ and } T' = 2\pi \sqrt{\frac{\ell}{g'}}$$

$$\therefore T/T' = \sqrt{\frac{g'}{g}} = \sqrt{\frac{g(1-d/D)}{g}} = \sqrt{1 - \frac{d}{D}} = \sqrt{1 - \frac{1}{s}}$$

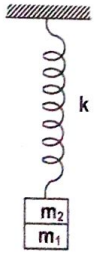
Putting $T/T' = 1/\sqrt{2}$, we obtain

$1/2 = 1 - 1/s$

$$\Rightarrow \frac{1}{s} = \frac{1}{2} \Rightarrow s = 2$$

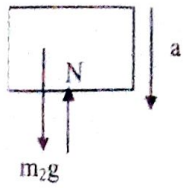
$\therefore$(B)

2. A spring is loaded with two blocks m_1 and m_2, where m_1 is rigidly fixed with the spring and m_2 is just kept on the block m_1 as shown in the figure. The maximum energy of oscillation that is possible for the system having the block m_2 in contact with m_1 is



(A) $\frac{m_1^2 g^2}{2k}$ **(B) $\frac{m_2^2 g^2}{2k}$**

(C) $\frac{(m_1 + m_2)^2 g^2}{2k}$ **(D) None of these**



Sol. Let the required amplitude be A. For this amplitude of oscillation the normal contact force between the blocks can be given as

$m_2 g - N = m_2 a$, where $a = \omega^2 A$

$\Rightarrow m_2 g - N = m_2 \omega^2 A$

Putting N = 0 for just losing contact for maximum amplitude, we obtain

$A = \frac{g}{\omega^2}$, where ω = angular frequency of oscillation of $(m_1 + m_2)$ and the spring.

Putting $\omega^2 = \frac{k}{m_1 + m_2}$, we obtain

$A = \frac{(m_1 + m_2)g}{k}$

$\therefore U_{max} = \frac{1}{2}kA^2 = \frac{(m_1 + m_2)^2 g^2}{2k}$.

$\therefore$ (C)

3. A particle of mass m is executing oscillations about the origin on the x-axis. Its potential energy is $V(x) = k|x|^3$, where k is a positive constant. If the amplitude of oscillation is a, then its time period T is

(A) proportional to $1/\sqrt{a}$ **(B) independent of a**

(C) proportional to $\sqrt{a}$ **(D) proportional to $a^{3/2}$**

Sol. $V = k|x|^3$

$F = -\frac{dv}{dx} = -3k|x|^2$ ... (1)

The equation of simple harmonic motion is given as

$x = a \sin \omega t$

$\Rightarrow m\frac{d^2x}{dt^2} = m(-a\omega^2 \sin \omega t) = -m\omega^2 x$ ... (2)

Using (I) and (ii), we obtain

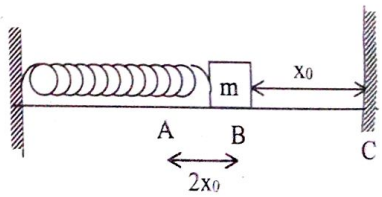
$3k|x|^2 = m\omega^2 x \Rightarrow \omega = \sqrt{3kx/m}$

$\Rightarrow T = 2\pi\sqrt{\frac{m}{3kx}} \Rightarrow T = 2\pi\sqrt{\frac{m}{3ka\sin\omega t}}$

$\Rightarrow T \propto \frac{1}{\sqrt{a}}$.

$\therefore$ (A)

4. One end of a spring of force constant k is fixed to a vertical wall and the other to a block of mass m resting on a smooth horizontal surface. There is another wall at a distance x_0 from the block. The spring is then compressed by $2x_0$ and released. The time taken to strike the wall is



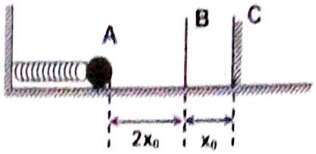
(A) $\frac{1}{6}\pi\sqrt{\frac{k}{m}}$ **(B) $\sqrt{\frac{k}{m}}$**

(C) $\frac{2\pi}{3}\sqrt{\frac{m}{k}}$ **(D) $\frac{\pi}{4}\sqrt{\frac{k}{m}}$**

Sol. The total time from A to C

$= t_{AC} = t_{AB} + t_{BC} = (T/4) + t_{BC}$

where T = time period of oscillation of spring mass system



t_{BC} can be obtained from, $BC = AB \sin(2\pi/T)\, t_{BC}$

Putting $\frac{BC}{AB} = \frac{1}{2}$, we obtain $t_{BC} = \frac{T}{12}$

$\Rightarrow t_{AC} = \frac{T}{4} + \frac{T}{12} = \frac{2\pi}{3}\sqrt{\frac{m}{k}}$.

$\therefore$(C)

5. A particle moves on x-axis according to the equation $x = x_0 \sin^2 \omega t$, the motion is simple harmonic

(A) with amplitude x_0 **(B) with amplitude $2x_0$**

(C) with time period $(2\pi/\omega)$ **(D) with time period (π/ω)**

Sol. $x = x_0 \sin^2 \omega t = \frac{x_0}{2}(1 - \cos 2\omega t) = \frac{x_0}{2} - \frac{x_0}{2}\cos 2\omega t$

Frequency $\omega' = 2\omega$

$\Rightarrow 2\pi/T' = 2\omega$

$\Rightarrow T' = (\pi/\omega)$

Amplitude $= x_0/2$

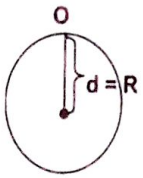
$\therefore$(D)

6. A disc of radius R and mass M is pivoted at the rim and is set for small oscillations. If simple pendulum has to have the same period as that of the disc, the length of the simple pendulum should be

(A) (5/4) R **(B) (2/3) R**

(C) (3/4)R **(D) (3/2) R**

Sol. Time period of a physical pendulum :



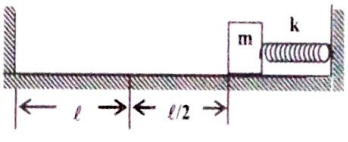
$$T = 2\pi\sqrt{\frac{I_0}{mgd}} = 2\pi\sqrt{\frac{\left(\frac{1}{2}mR^2 + mR^2\right)}{mgR}} = 2\pi\sqrt{\frac{3R}{2g}} \quad \ldots(1)$$

$$T_{\text{simple pendulum}} = 2\pi\sqrt{\frac{\ell}{g}} \quad \ldots(2)$$

Equating Eqs. (1) and (1), $\ell = \frac{3}{2}R$.

$\therefore$(D)

7. A block of mass m compresses a spring of stiffness k through a distance $\ell/2$ as shown in the figure. If the block is not fixed to the spring, the period of motion of the block is



(A) $2\pi\sqrt{\frac{m}{k}}$ **(B) $(\pi + 4)\sqrt{\frac{m}{k}}$**

(C) $(1 + \pi)\sqrt{\frac{m}{k}}$ **(D) none of these.**

Sol. The period of oscillation $= 2\pi\sqrt{\frac{m}{k}}$

$\Rightarrow$ The period of motion till the block is in contact with the spring is

$t = \frac{T}{2} = \pi\sqrt{\frac{m}{k}}$ then it leaves the spring with a speed $v = \omega A$

$v = \left(\sqrt{\frac{k}{m}}\right)\left(\frac{\ell}{2}\right)$

Then it moves with constant velocity v for a distance $D = \ell + \ell = 2\ell$

$\Rightarrow$ The corresponding time of motion $= t_2 = 2\ell/v$

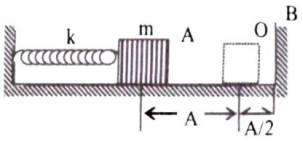
$\Rightarrow t_2 = \frac{2\ell}{\frac{\ell}{2}\sqrt{\frac{k}{m}}} = 4\sqrt{\frac{m}{k}}$

$\therefore$ The time period of motion $= t = t_1 + t_2 = \pi\sqrt{\frac{m}{k}} + 4\sqrt{\frac{m}{k}}$

$= \sqrt{\frac{m}{k}}[\pi + 4].$

$\therefore$ (B)

8. A block of mass m rigidly attached with a spring k is compressed through a distance A. If the block is released, the period of oscillation of the block for a complete cycle is equal to



(A) $\frac{4\pi}{3}\sqrt{\frac{m}{k}}$ **(B)** $\frac{\pi}{\sqrt{2}}\sqrt{\frac{m}{k}}$

(C) $\frac{2\pi}{3}\sqrt{\frac{m}{k}}$ **(D) None of these.**

Sol. The period of motion from A to O is equal to quarter of the time period T of oscillation of mass spring system.

$\Rightarrow t_{AO} = \frac{T}{4} = \frac{1}{4}\left[2\pi\sqrt{\frac{m}{k}}\right] = \frac{\pi}{2}\sqrt{\frac{m}{k}}.$

Since the motion is simple harmonics

$OB = OA \sin\frac{2\pi}{T}t_{OB}$, where t_{OB} is the time of motion from O to B.

$\Rightarrow t_{OB} = \frac{T}{2\pi}\sin^{-1}\frac{A/2}{A} = \frac{T}{2\pi}\left(\frac{\pi}{6}\right)$

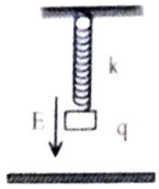
$= \frac{T}{12} = \frac{2\pi}{12}\sqrt{\frac{m}{k}} = \frac{\pi}{6}\sqrt{\frac{m}{k}}$

$\therefore$ The total time of motion for a complete cycle $= t = 2(t_{AO} + t_{OB})$

$\Rightarrow t = 2\left[\frac{\pi}{2}\sqrt{\frac{m}{k}} + \frac{\pi}{6}\sqrt{\frac{m}{k}}\right] = \frac{4\pi}{3}\sqrt{\frac{m}{k}}.$

$\therefore$(A)

9. **Time period of a block when suspended from the upper plate of a parallel plate capacitor by a spring of stiffness k is T. When block is uncharged. If a charge q is given to the block then, the new time period of oscillation will be**



(A) T **(B) > T**

(C) < T **(D) ≥ T**

Sol. The forces that act on the block are qE and mg. Since qE and mg are constant forces, the only variable elastic force changes by kx, where x is the elongation in the spring

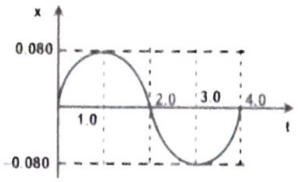
$\Rightarrow$ Unbalanced (restoring) force = F = – kx

$\Rightarrow -m\omega^2 x = -kx$

$\Rightarrow \omega = \sqrt{\frac{k}{M}} = T.$

$\therefore$ (A)

10. **An object of mass 0.8 kg is attached to one end of a spring and the system is set into simple harmonic motion. The displacement x of the object as a function of time is shown in the figure. With the aid of the data, the magnitude of the object's acceleration at t = 1.0 is**



(A) zero. **(B) 1.57 m/s²**

(C) 0.197 m/s² **(D) 0.157 m/s²**

Sol. From the figure

T = 4 sec

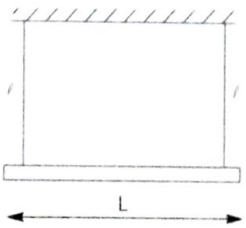
$\omega = \frac{2\pi}{T} = \frac{\pi}{2} = 1.57 \text{ rad / sec}$

$\omega = \sqrt{\frac{k}{m}} \Rightarrow k = (0.8)\left(\frac{\pi}{2}\right)^2 = 1.97 \text{ N/m}$

$|a| = \omega^2 x = 0.197 \text{ m/s}^2$

$\therefore$ (C)

11. **A rod of lengths L is supported by two ideal strings of length ℓ such that the system hangs in a vertical plane.**
Case 1: Rod is kept horizontal, displaced slightly perpendicular to the plane and allowed by oscillate.
Case 2: Rod is given a small twist about central axis and then allowed to oscillate.
Let T_1, T_2 be the periods of oscillations in the two respective cases.

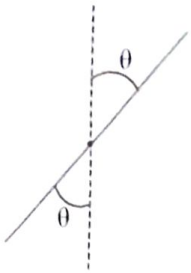


(A) $T_1 = 2\pi\sqrt{\frac{\ell}{g}}$ **(B)** $T_1 = 2\pi\sqrt{\frac{\ell}{2g}}$

(C) $T_2 = 2\pi\sqrt{\frac{L}{6g}}$ **(D)** $T_2 = 2\pi\sqrt{\frac{\ell}{3g}}$

Sol. A, D

Let θ be the angle that the rod rotates about its central axis and ϕ be the angle made by string with the vertical.



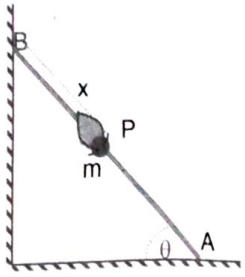
$$-(2T\phi)\ \frac{L}{2} = \frac{mL^2}{12}\frac{d^2\theta}{dt^2}$$

$$\Rightarrow \quad \frac{d^2\theta}{dt^2} = -\left(\frac{6g}{L}\right)\phi$$

$$\text{or} \quad \frac{d^2\theta}{dt^2} = -\left(\frac{3g}{\ell}\right)\theta$$

$$\Rightarrow \quad T_2 = 2\pi\sqrt{\frac{\ell}{3g}}$$

12. A light stick of length ℓ rests with its one end against the smooth wall and other end against the smooth horizontal floor as shown in the figure. The bug starts at rest from point B and moves such that the stick always remains at rest. a_P is the magnitude of acceleration of bug of mass m, which depends upon its distance of x from the top end of the stick. Choose the correct option(s)



(A) $a_P = \dfrac{g}{\sin\theta}\left(1-\dfrac{x}{\ell}\right)$

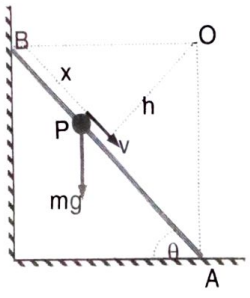
(B) $a_P = \dfrac{g}{\cos\theta}\left(1-\dfrac{x}{\ell}\right)$

(C) The time taken by the bug to reach the bottom of the stick having started at the top end from rest is $\dfrac{\pi}{2}\sqrt{\dfrac{\ell\sin\theta}{3g}}$

(D) The time taken by the bug to reach the bottom of the stick having started at the top end from rest is $\dfrac{\pi}{2}\sqrt{\dfrac{\ell\sin\theta}{g}}$

Sol. A, D

Consider the bug at point P at any time t, moving with speed v along the stick, so $L = mvh = mv\ell\sin\theta\cos\theta \Rightarrow$ Angular momentum of bug about point O



$$\Rightarrow \tau = \frac{dL}{dt} = (m\ell\sin\theta\cos\theta)\frac{dv}{dt}$$

$$\Rightarrow mg(\ell - x)\cos\theta = (m\ell\sin\theta\cos\theta)\frac{dv}{dt}$$

$$\Rightarrow \frac{dv}{dt} = \frac{g}{\sin\theta}\left(1-\frac{x}{\ell}\right)$$

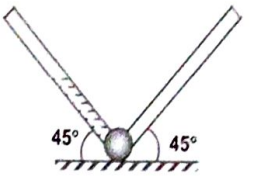
$$\Rightarrow \frac{dv}{dt} + \frac{g}{\ell\sin\theta}x = \frac{g}{\sin\theta} \Rightarrow \text{Equation of SHM with mean position at point A}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{\ell\sin\theta}{g}}$$

$$\text{Time required} = \frac{T}{4} = \frac{\pi}{2}\sqrt{\frac{\ell\sin\theta}{g}}$$

COMPREHENSION

A thin v-shaped glass tube is bent through right angle and is fixed in the vertical plane as shown. Initially the left tube contains a column of water of length d. A valve at the bottom of the tube prevents water from moving. At some time the valve is quickly opened. Neglect friction (given ρ = density of liquid and A = area of tube)



13. Find time period of oscillation of liquid column.

(A) $\pi\sqrt{\frac{d}{\sqrt{2}g}}$ **(B)** $\pi\sqrt{\frac{d}{g}}$

(C) $\pi\sqrt{\frac{2d}{g}}$ **(D)** $\pi\sqrt{\frac{d}{2g}}$

Sol. C

Time period of oscillation will be $\pi\sqrt{\frac{2d}{g}}$.

14. Minimum time after which all water moves to right tube is

(A) $\pi\sqrt{\frac{d}{2g}}$ **(B)** $\frac{\pi}{2}\sqrt{\frac{2d}{g}}$

(C) $\pi\sqrt{\frac{2d}{g}}$ **(D)** $\pi\sqrt{\frac{d}{\sqrt{2}g}}$

Sol. A

During half time period all liquid will move to the right side

15. Minimum time after which speed of liquid will be maximum.

(A) $\frac{\pi}{4}\sqrt{\frac{2d}{g}}$ **(B)** $\frac{\pi}{4}\sqrt{\frac{d}{g}}$

(C) $\pi\sqrt{\frac{2d}{g}}$ **(D)** $\pi\sqrt{\frac{d}{2g}}$

Sol. A

Speed of liquid will be maximum after one-fourth time period

Matrix – Match Type

16. A particle of mass 2 kg is moving on a straight line under the action of force F = (8 – 2x) newton along the x-axis, where x is the coordinates on x-axis. If a particle is released from x = 7 m, then for the subsequent motion match the following (all values in Column II are in S.I. units)

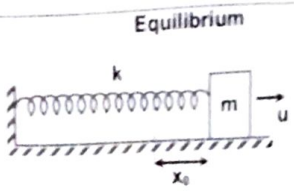
	Column I		Column II
(A)	Amplitude of SHM is	(p)	0.52
(B)	Time taken to move directly from x = 2.5 m to x = 4 m (approx)	(q)	3
(C)	Maximum kinetic energy of SHM system	(r)	6
(D)	Velocity of particle at x = 4m	(s)	9
		(t)	4.5

Sol. (A) → (q), (B) → (p), (C) → (s), (D) → (q)

Answer questions 17, 18 and 19 by appropriately matching the information given in the three columns of the following table.

Figure shows the situation at t = 0 for a block of mass 2 kg attached to spring of spring constant 2 N m and oscillating. Displacements and velocities to the right of mean position are considered as positive. The equation of SHM is written as

$x = A \sin(\omega t + \phi)$



Column –I lists different combinations of initial values x_0 and u. Column –II lists the corresponding amplitude and column –III lists the phase constant ϕ (given that $-\pi \le \phi \le \pi$).

Column 1		Column 2		Column 3	
(I)	$x_0 = 1$ m, u = 1 m/s	(i)	2	(P)	120°
(II)	$x_0 = -1$ m, u = $\sqrt{3}$ m/s	(ii)	4	(Q)	–143°
(III)	$x_0 = \sqrt{3}$ m, u = –1 m/s	(iii)	5	(R)	45°
(IV)	$x_0 = -3$ m, u = –4 m/s	(iv)	$\sqrt{2}$	(S)	–30°

17. **Pick the correct option.**
(A) (I) (ii) (P) **(B) (II)(i) (S)**
(C) (III) (iii) (R) **(D) (IV) (iv) (Q)**

Sol. B

18. **Pick the option for minimum amplitude.**
(A) (I) (iv) (R) **(B) (II)(i) (P)**
(C) (III) (iii) (S) **(D) (IV) (ii) (Q)**

Sol. A

19. **Pick the option for maximum magnitude of phase constant.**
(A) (I) (ii) (P) **(B) (II) (iii) (P)**
(C) (III) (iii) (S) **(D) (IV) (iii) (Q)**

Sol. D

Numerical Type

20. **A 2 kg block moving with 10 m/sec strikes a spring of constant π^2 N/m which is attached to 2 kg block at rest kept on a smooth horizontal floor. Find the time (in sec) for which rear moving block remain in contact with spring.**

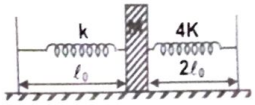
Sol. 1

$$T = 2\pi\sqrt{\frac{\mu}{k}}$$

$$T = \frac{2\pi}{\pi} \Rightarrow T = 2\,\text{sec}$$

Required time $= \frac{T}{4} + \frac{T}{4} = \frac{T}{2} = 1\,\text{sec}.$

21. **A block of mass M is kept in gravity free space and touches the two springs as shown in the figure. Initially springs are in its natural length. Now, the block is shifted ($\ell_0/2$) from the given position in such a way it compresses a spring and released. The time-period of oscillation of mass is $\frac{n\pi}{2}\sqrt{\frac{M}{K}}$. Find n.**

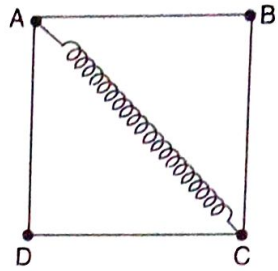


Sol. 2

$$T = \pi\left[\sqrt{\frac{M}{K}} + \frac{1}{2}\sqrt{\frac{M}{K}}\right] = \frac{3\pi}{2}\sqrt{\frac{M}{K}}$$

Numerical Decimal Type

22. **A square hinged-structure (ABCD) is formed with four massless rods which lies on the smooth horizontal table. The hinge-D of structure is fixed with table and hinge A, hinge-B and hinge-C are free to move on the table. A point mass m is attached at hinge-B, and an ideal spring of stiffness k is connected between hinge-A and hinge-C as shown in the figure. System is in equilibrium. The mass m is slightly displaced along line BD and released to perform SHM. The time period of SHM is $\pi\sqrt{\frac{nm}{k}}$. Find**



Sol. **00004.00**

$OE = x$

$FG = \sqrt{2}a + 2y$

From the property of rhombus, we can write

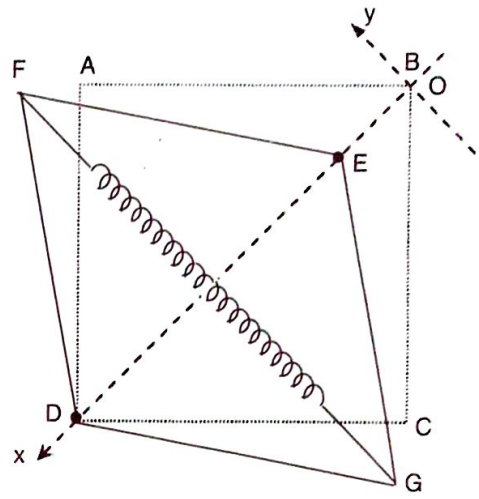
$$\left(\frac{a}{\sqrt{2}} + y\right)^2 + \left(\frac{a}{\sqrt{2}} - \frac{x}{2}\right)^2 = a^2$$

Neglecting x^2 and y^2, we have $y = \frac{x}{2}$

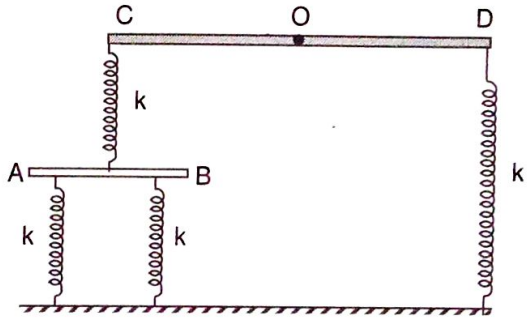
Using COME, we can write

$$\frac{mv^2}{2} + \frac{k}{2}(x)^2 = E = \text{constant}$$

$$\Rightarrow T = \pi\sqrt{\frac{4m}{k}}$$



23. **In the figure shown rod AB is light and rigid while rod CD is also rigid and hinged at midpoint O Rod can rotate without friction about O. The time period of small oscillation of rod CD is $T = 2\pi\sqrt{\frac{m}{nk}}$. Where m is mass of rod CD and k is spring constant of each spring. The value of n is**



Sol. **00005.00**

$$\frac{m\ell^2}{12}\alpha = -\left\{\frac{2k\theta}{12} + \frac{k\theta}{4}\right\}\ell^2$$

$$\Rightarrow \quad \alpha = -\left(\frac{5k}{m}\right)\theta$$

HEAT & THERMODYNAMICS 10

ZEROTH LAW OF THERMODYNAMICS

If a body A is in thermal equilibrium with another body B and a body B is in thermal equilibrium with another body C, then A is also in thermal equilibrium with C.

EXPRESSION FOR TEMPERATURE T AS A FUNCTION OF A LINEARLY VARYING PROPERTY P

$$T = \frac{P_T - P_0}{P_{100} - P} \times 100°C$$

IDEAL GAS TEMPERATURE SCALE

$$T_P = \lim_{P_{tr} \to 0} \frac{P_T}{P_{tr}} \times 100$$

ABSOLUTE SCALE

$$T(K) = T(°C) + 273$$

THERMAL EXPANSION

For small changes in temperature:

$$L = L_0(1 + \alpha \Delta T)$$

$$A = A_0(1 + \beta \Delta T)$$

$$V = V_0(1 + \gamma \Delta T)$$

$\alpha : \beta : \gamma = 1 : 2 : 3$ (for isotropic substances)

For a liquid:

$\gamma_a = \gamma_R - \gamma_c$

γ_R = coefficient of real expansion of liquid

γ_a = coefficient of apparent expansion of the liquid

γ_c = coefficient of cubical expansion of container

Variation of density with temperature

$$\rho = \frac{\rho_0}{1 + \gamma \Delta T}$$

Thermal stress $= \frac{F}{A} = \alpha Y \Delta\theta$ (where $\Delta\theta$ is the increase in temperature)

Different Types of Velocities Possessed by Molecules of an Ideal Gas

$$v_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + \cdots v_n^2}{n}} = \sqrt{\frac{3RT}{M}}$$

$$v_{M.V.} = \frac{v_1 + v_2 + \cdots v_n}{n} = \sqrt{\frac{8RT}{\pi M}}$$

$$v_{M.P.V.} = \sqrt{\frac{2RT}{M}}$$

$PV = \frac{1}{3}mnv_{rms}^2$, where m = mass of molecule and n = number of molecules

Kinetic Interpretation of Temperature

Kinetic energy of random motion is the internal energy, given by

$KE = \frac{1}{2}m.v_{rms}^2 = \frac{1}{2}n.M\frac{3RT}{M} = \frac{3}{2}.n.RT$, where n is the number of moles of a gas in the system.

DEGREE OF FREEDOM

The degree of freedom that a molecule can possess is equal to the number of independent motions that it can have and is denoted by the letter 'f'

For an ideal gas molecule,

$U = \frac{1}{2}fRT$, for one mole of the gas, R = kN; N = Avogadro's Number,

The specific heat of the gas at constant volume, $C_v = \frac{dU}{dT} = \frac{fR}{2}$

$\therefore \quad C_P = C_v + R = \frac{f}{2}R + R = \left(\frac{f}{2}+1\right)R$

$\therefore$ Ratio of the two specific heats, $\gamma = \frac{C_P}{C_v} = \frac{\left(\frac{f}{2}+1\right)R}{\frac{f}{2}R} = 1 + \frac{2}{f}$

A molecule of a monatomic gas has only three (translational) degrees of freedom, i.e. f = 3.

$\therefore \quad C_v = \frac{f}{2}R = \frac{3}{2}R$

$C_P = \left(\frac{f}{2}+1\right)R = \frac{5}{2}R$;

$\gamma = 1 + \frac{2}{f} = 5/3 = 1.66$

A molecule of a diatomic gas has 5 degrees of freedom (3 - translational and 2 - rotational) at ordinary temperatures (because the vibrational modes are not excited.) The moment of inertia about the line joining the two atoms is negligibly small. Hence, the rotational energy about that axis is zero.

f = 5

$C_V = \frac{f}{2}R = \frac{5}{2}R$

$C_P = \left(\frac{f}{2}+1\right)R = \frac{7}{2}R$

$\gamma = 1 + \frac{2}{f} = \frac{7}{5} = 1.40$

A molecule of triatomic or polyatomic gas has 6 degrees of freedom (3 translational and 3 rotational)

$f = 6$

$C_V = \frac{f}{2}R = 3R$

$C_P = \left(\frac{f}{2}+1\right)R = 4R$

$\gamma = 1+\frac{2}{f} = \frac{4}{3} = 1.33$

FIRST LAW OF THERMODYNAMICS

If ΔQ is heat given to a system, and ΔW is work done by the system, then ΔU, the change in its internal energy can be written as

$\Delta Q = \Delta U + \Delta W$,

where $\Delta U = nC_v\Delta T$,

$\Delta W = \int_{V_1}^{V_2} PdV$ = Area enclosed under P-V graph.

$\Delta Q = nC\Delta T$, where C = specific heat capacity of a gas, for that process.

Note: For a cyclic process work done = area enclosed under P-V graph, and is positive if the cycle is clockwise

Important Cases of the First Law of Thermodynamics

(a) Isobaric process (Pressure = constant)

$\Delta Q = \Delta U + \Delta W$

$nC_p\Delta T = nC_v\Delta T + nR\Delta T$

$C_p - C_v = R$ (Mayer's equation)

(b) Isochoric process ($\Delta W = 0$, Volume = constant)

$\Delta Q = \Delta U = nC_v\Delta T$

(c) Isothermal process ($\Delta U = 0$; temperature = constant)

$$\Delta Q = \Delta W = \int_{V_1}^{V_2} pdV = \int_{V_1}^{V_2} \frac{nRT}{V}dV = nRT\ln\left[\frac{V_2}{V_1}\right] = nRT\ln\left[\frac{P_1}{P_2}\right]$$

(d) Adiabatic process ($\Delta Q = 0$)

$\Delta U + \Delta W = 0$

$\Rightarrow \Delta W = -\Delta U = \frac{nR}{\gamma-1}(T_1 - T_2) = \frac{P_1V_1 - P_2V_2}{\gamma-1}$

Again $\Delta U + \Delta W = 0 = \frac{nR}{\gamma-1}dT + \frac{nRT}{V}dV = 0$

$\frac{dT}{T} + (\gamma-1)\frac{dV}{V} = 0$

$\Rightarrow TV^{\gamma-1} = \text{constant}$

$PV^{\gamma} = \text{constant}; \frac{T^{\gamma}}{P^{\gamma-1}} = \text{constant}$

(e) Cyclic process

$\Delta U = 0$, since process returns to the same initial state

$\Delta U = \Delta Q - \Delta W = 0$

$\therefore \Delta Q = \Delta W$

(f) Free expansion

In free expansion (a gas) expands in such a way that no heat enters or leaves the system (adiabatic process) and also no work is done by or on the system, then the expansion is called the free expansion

$U_f - U_i = Q - W$, Now $Q = 0$, $W = 0$

$\therefore U_f = U_i$

Slope of Isothermal and Adiabatic Processes

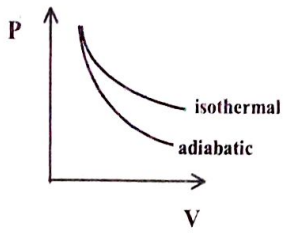
For an isothermal process,

PV = constant

Differentiating,

$PdV + VdP = 0$

or, $\left(\frac{dP}{dV}\right)_{isothermal} = -\frac{P}{V}$



For an adiabatic process

PV^{γ} = constant

Differentiating,

or, $\left(\frac{dP}{dV}\right)_{adiabatic} = -\gamma\frac{P}{V}$

Efficiency of a process

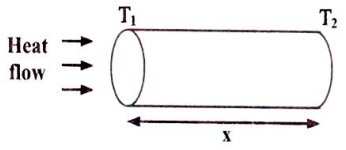
Efficiency, $\eta = \frac{\Delta W}{\Delta Q} \times 100$

HEAT TRANSFER

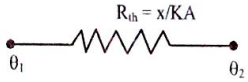
Thermal Conduction

$$\frac{\Delta Q}{\Delta t} = \frac{KA(T_1 - T_2)}{x}$$

where K is the co-efficient of thermal conductivity of the material of the slab.



Also R = thermal resistance = $\frac{x}{KA}$

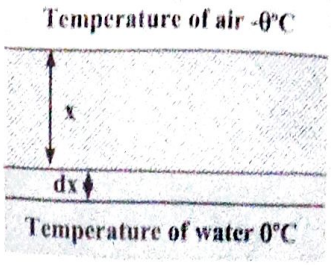


In parallel connection, $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$

Heat transfer rate = $\frac{dQ}{dt}$

In series connection, $R_{eq} = R_1 + R_2 + \cdots + R_n$

Accretion of Ice



Consider a layer of ice of thickness x. The air temperature is $-\theta$°C and water temperature below the ice is 0°C.

Considering unit cross-sectional area of ice, if a layer of thickness dx grows in time dt,

The heat given by this layer

= mass × latent heat = $1 \cdot dx\, \rho \cdot L$

where ρ = density of ice and L = latent heat of fusion of ice.

If this quantity of heat is conducted upwards through the ice layer in time dt, then

$$\therefore \quad dx \cdot \rho \cdot L = K \frac{0-(-\theta)}{x} dt,$$

$$\text{Time taken, } t = \frac{\rho L}{K\theta} \int_{x_1}^{x_2} x\, dx = -\frac{\rho L}{2K.\theta}\left(x_2^2 - x_1^2\right)$$

RADIATION

Theory of Exchange

Every body is continuously emitting radiant energy in all direction at a rate depending only on the nature of its surface and its temperature, and also it is absorbing radiant energy from all surrounding bodies at a rate depending on its surface and the temperature of the surrounding bodies. This is also known as Prevost's theory of exchanges.

Stefan's Law of Radiation

Energy emitted per unit time by a black body of surface area A is given by

$$\text{or} \quad \frac{d\theta}{dt} = \sigma A T^4 \quad [\text{Stefan's constant } \sigma = 5.67 \times 10^{-8}\ \text{Wm}^{-2}\text{-Kelvin}^{-4}]$$

Energy emitted per unit time by a body other than a black body,

$$\frac{d\theta}{dt} = \sigma A \varepsilon T^4 \quad [\varepsilon = \text{emissivity}]$$

Net Loss of Thermal Energy

If a body of surface area A is kept at absolute temperature T in a surrounding of temperature T_o ($T_o < T$), then the energy emitted by the body per unit time

$$E = \sigma \varepsilon A T^4$$

And energy absorbed per unit time by the body

$$E_o = \sigma \varepsilon A T_o^4$$

Net, loss of thermal energy per unit time

$$\Delta E = E - E_o = \sigma \varepsilon A (T^4 - T_o^4)$$

Absorptive Power A

Absorptive power of a body is defined as the fraction of the incident radiation that is absorbed by the body.

$$\text{Absorptive power } (a) = \frac{\text{Energy absorbed}}{\text{Energy incident}}$$

Emissive Power (E)

The emissive power denotes the energy radiated per unit time per unit area of the surface.

Emissivity (ε)

Emissivity of a surface is the ratio of the emissive power of the surface to the emissive power of black body at the same temperature.

Emissivity ε= (Emissive power of the surface) ÷ (Emissive power of black body at the same temperature)

Black Body

A perfectly blackbody is one which absorbs completely all the radiation, of whatever wave-length, incident on it.

Kirchhoff's Law

It states that the ratio of the emissive power to the absorptive power for radiation of a given wavelength is the same for all bodies at the same temperature, and is equal to the emissive power of a perfectly blackbody at that temperature.

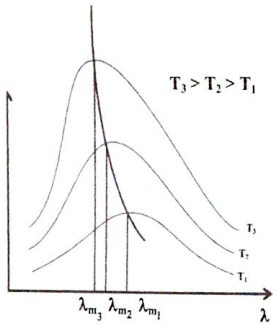
$$\frac{e_\lambda}{a_\lambda} = E_\lambda \text{ (Constant)}$$

Spectral Distribution of Energy in Black Body Radiation

$\lambda_m T$ = constant (Wien's displacement law)

λ_m = wavelength corresponding to maximum spectral intensity

T = absolute temperature of surface of the body.



Newton's Law of Cooling

For a small temperature difference between a body and its surrounding, the rate of cooling of the body is directly proportional to the temperature difference.

$$\frac{dT}{dt} = -k(T - T_0) \text{ (rate of fall of temperature)}$$

T = temperature of the body.

T_0 = temperature of the surroundings

ILLUSTRATIVE PROBLEMS

1. Using the following data , find the change in temperature at which wood will just sink in benzene.
Density of benzene at 0°C = 9×10^2 kg m^{-3} ,
Density of wood at 0°C = 8.8×10^2 kg m^{-3}.
Cubical expansivity of benzene = 1.2×10^{-3} K^{-1}
Cubical expansivity of wood = 1.5×10^{-4} K^{-1}

Sol. Let the subscript 1 and 2 denotes benzene and wood, respectively.

$$\therefore \quad \frac{\rho_1}{1+\gamma_1 \Delta T} = \frac{\rho_2}{1+\gamma_2 \Delta T}$$

or $\rho_1 + \rho_1\gamma_2 \Delta T = \rho_2 + \rho_2\gamma_1 \Delta T$

or $\Delta T = \dfrac{\rho_1 - \rho_2}{\rho_2\gamma_1 - \rho_1\gamma_2}$

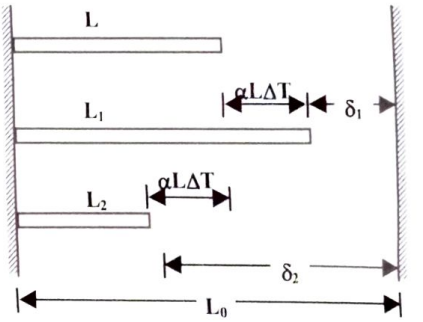
Here, $\rho_1 = 900$ kgm^3, $\rho_2 = 880$ kg/m^3; $\gamma_1 = 1.2 \times 10^{-3}$ K^{-1} ;
$\gamma_2 = 1.5 \times 10^{-4}$ K^{-1}

$\therefore$ $\Delta T = 21.7°C$

2. A steel rod with a cross-sectional area of 150 mm^2 is stretched between two fixed points. The tensile load at 20°C is 5000 N .
(a) What will be the stress at –20°C ?
(b) At what temperature will the stress be zero ?
(Assume $\alpha = 11.7 \mu m/m\ °C$ and Y = 200 GN/m^2)

Sol. (a) Let L be the free length of the rod at 0°C and L_0 be the final stretched length in each case.

If L_1 and L_2 are the free lengths at +20°C and –20°C, respectively, then using geometric conditions $\delta_2 = \delta_1 + 2\alpha L \Delta T$, where δ_1 and δ_2 are the load deformations at +20°C and –20°C, respectively.



Now, $\dfrac{F_2 L}{AY} = \dfrac{F_1 L}{AY} + 2\alpha L \Delta T$

or, $\sigma_2 = \dfrac{F_2}{A} = \dfrac{F_1}{A} + 2\alpha Y \Delta T$

$\therefore$ $\sigma_2 = \dfrac{5000}{150 \times 10^{-6}} + 2\left(11.7 \times 10^{-6}\right)\left(2 \times 10^{11}\right)(20)$

$= 127 \times 10^6$ N/m^2

(b) Let T′ be the temperature at which the stress is zero, then

$\alpha L T' = \alpha L \Delta T + \delta_1$

or $T' = \Delta T + \dfrac{F_1}{AY\alpha}$

$T' = 20 + \dfrac{5000}{\left(150 \times 10^{-6}\right)\left(2 \times 10^{11}\right)\left(11.7 \times 10^{-6}\right)}$

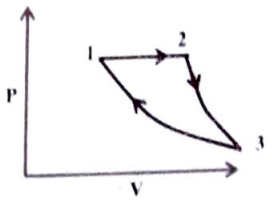
$= 34.2°C$

3. Air is initially at 260°C and 700 pa and occupied 0.028 m^3. The air is expanded at constant pressure to 0.084 m^3. A polytropic process with n = 1.5 is then carried out followed by a constant temperature process which complete the cycle.
(a) sketch cycle in P–V.
(b) find heat received and heat rejected in the cycle.
(c) efficiency of cycle.

Sol. $P_1 = 700$ Pa, $V_2 = 0.08\ m^3$

$P_1V_1 = nRT_1$

No. of moles $v = \dfrac{700 \times 0.028}{8.31 \times 533} = 0.004$ mole



$\dfrac{T_2}{T_1} = \dfrac{V_2}{V_1} = 3$

$T_2 = 3T_1 = 1599$ K

$PV^n = C$

$P\left(\dfrac{nRT}{P}\right)^n = C$

$\Rightarrow P = C' T^{\frac{n}{n-1}}$

$\Rightarrow \dfrac{P_2}{P_3} = \left(\dfrac{T_2}{T_3}\right)^{\frac{n}{n-1}}$

$\dfrac{P_2}{P_3} = \left(\dfrac{1599}{533}\right)^{1.5/0.5} = 3^3 = 27.$

$Q_{1-2} = vC_P(T_2 - T_1) = 0.004 \times \dfrac{7}{2} \times 8.31\ (1599 - 533) = 124$ J.

$Q_{2-3} = \Delta U + \int PdV$

$= vC_v(T_3 - T_2) + \dfrac{nR(T_3 - T_2)}{n-1} = vC_v\left(\dfrac{n-\gamma}{n-1}\right)(T_3 - T_2)$

$= 0.004 \times \dfrac{5}{2} \times 8.31\ \dfrac{0.1}{0.5}(-1066) = -17.7$ J.

$Q_{31} = W_{31} = vRT_1 \ln\left(\dfrac{V_1}{V_3}\right) = 0.004 \times 8.31 \times 533 \ln\left(\dfrac{P_3}{P_1}\right) = -58.4$ J

$(\Sigma Q)_{cycle} = (\Sigma w)_{cycle}$

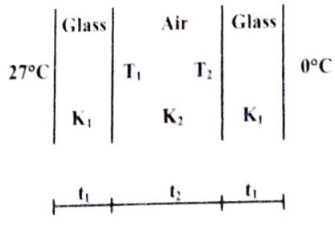
$\Sigma W = Q_{1-2} + Q_{2-3} + Q_{3-1} = 124 - 17.7 - 58.4 = 47.9$ J.

$\eta_{cycle} = \dfrac{W_{net}}{\text{Heat supplied}} = \dfrac{47.9}{106} \times 100 = 38.6\ \%.$

4. A double-pane window consists of two glass sheets each of area 1 m^2 and thickness 0.01 m separated by a 0.05 m thick stagnant air space. In the steady state, the room glass interface and the glass outdoor interface are at constant temperatures of 27°C and 0°C, respectively.
(a) Calculate the rate of heat flow through the window pane.
(b) Find the temperatures of other interfaces.
[Take: thermal conductivities as $K_{glass} = 0.8\ W\ m^{-1}\ K^{-1}$ and $K_{air} = 0.08\ W\ m^{-1}\ K^{-1}$]

Sol. (a) Total thermal resistance is

$$R = \frac{2t_1}{K_1A_1} + \frac{t_2}{K_2A_2}$$



Here,

$A_1 = A_2 = 1m^2$, $t_1 = 0.01$ m, $t_2 = 0.05$ m;

$K_1 = 0.8\ Wm^{-1}K^{-1}$, $K_2 = 0.08 Wm^{-1}K^{-1}$.

$$\therefore R = \frac{2(0.01)}{(0.8)(1)} + \frac{0.05}{(0.08)(1)} = 0.65\,W^{-1}K$$

$$\text{Heat current} = \frac{dQ}{dt} = \frac{\Delta T}{R} = \frac{27-0}{0.65} = 41.5\,W$$

(b) $T_1 = 27 - \left(\frac{dQ}{dt}\right)\frac{t_1}{K_1A_1} = 27 - 0.52 = 26.48\,C$

$$T_2 = 0 + \left(\frac{dQ}{dt}\right)\frac{t_2}{K_2A} = 0.52\,C$$

5. **A solid copper sphere of diameter 10 mm. is cooled to a temperature of 150 K and is then placed in an enclosure at 290 K. Assuming that all interchange of heat is by radiation , calculate the initial rate of rise of temperature of the sphere. The sphere may be treated as a black body.**
[$\rho_{copper}=8.93\times10^3\ kg/m^3$, $s = 3.7\times10^2\ Jkg^{-1}K^{-1}$, $\sigma = 5.7\times10^{-8}\ Wm^{-2}K^{-4}$]

Sol. Using Stefan's law, the rate of increase of temperature is

$$\frac{dT}{dt} = \frac{\sigma}{\rho}\frac{A}{V}\frac{\left(T_0^4 - T^4\right)}{s}$$

where $\rho = 8.93\times10^3\ kg/m^3$, $s = 3.7\times10^2\ J/kg/K$

A/V = area / volume ratio

$\frac{A}{V} = \frac{6}{d}$, d = diameter of the sphere

T_0 = temperature of the surrounding = 290 K

T = temperature of the body = 150 K

$$\therefore \frac{dT}{dt} = \frac{6\sigma}{\rho sd}\left(T_0^4 - T^4\right) = 0.068\,Ks^{-1}$$

SOLVED PROBLEMS

1. **Two cylinders A and B fitted with massless pistons contain equal amounts of an ideal diatomic gas at 300 K. The piston A is free to move, while that of B is held fixed. The same amount of heat is given to the gas in each cylinder. If the rise in temperature of the gas in A is 30K, then the rise in temperature of the gas in B is.**

(A) 30 K **(B) 18 K**
(C) 50 K **(D) 42 K**

Sol. For cylinder A: $dQ = nC_p\,dT$
For cylinder B: $dQ = nC_v\,dT$
$\Rightarrow nC_p dT = nC_v\,dT$

$$\therefore dT = \frac{C_p\times30}{C_v} = 30\times1.4 = 42\,K$$

$\therefore$ (D)

2. **Two identical containers A and B with frictionless pistons contain the same ideal gas at the same temperature and the same volume V. The mass of gas A is m_A and that in B is m_B. The gas in each cylinder is now allowed to expand isothermally to the same final volume 2V. The change in the pressure in A and B are found to be ΔP and $1.5\Delta P$, respectively. Then**

(A) $4m_A = 9m_B$ **(B) $2m_A = 3m_B$**
(C) $3m_A = 2m_B$ **(D) $9m_A = 4m_B$**

Sol. For gas in A, $P_1 = \left(\frac{RT}{M}\right)\frac{m_A}{V_1}$

$$P_2 = \left(\frac{RT}{M}\right)\frac{m_A}{V_2}$$

$$\therefore \Delta P = P_1 - P_2 = \left(\frac{RT}{M}\right) m_A \left(\frac{1}{V_1} - \frac{1}{V_2}\right)$$

Putting $V_1 = V$ and $V_2 = 2V$, we get $\Delta P = \frac{RT}{M}\frac{m_A}{2V}$ ...(1)

Similarly, for Gas in B, $1.5\Delta P = \left(\frac{RT}{M}\right)\frac{m_B}{2V}$ ...(2)

From Eqs. (1) and (2), we get $2m_B = 3m_A$

$\therefore$ (C)

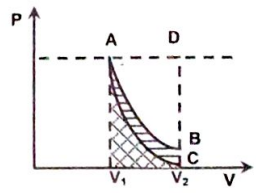
3. Starting with the same initial conditions, an ideal gas expands from volume v_1 to v_2 in three different ways. The work done by the gas is W_1 if the process is purely isothermal, W_2 is purely isobaric and W_3 if purely adiabatic. Then

(A) $W_2 > W_1 > W_3$ **(B) $W_2 > W_3 > W_1$**

(C) $W_1 > W_2 > W_3$ **(D) $W_1 > W_3 > W_2$**

Sol. It is clear from the diagram that work done i.e. area under the AB, AC, AD line for isothermal, adiabatic, isobaric process. Have the relation, $W_2 > W_1 > W_3$.

$\therefore$ (A)



4. A monoatomic gas ($\gamma = 5/3$) is suddenly compressed to (1/ 8) of its volume adiabatically then the pressure of the gas will change to

(A) 24/5 **(B) 8**

(C) 40/3 **(D) 32**

Sol. $P_1 V_1^{\gamma} = P_2 V_2^{\gamma}$

$$P_2 = P_1\left(\frac{V_1}{V_2}\right)^{\gamma} = P\left(\frac{V_1}{V_1/8}\right)^{\gamma} = P(8)^{\gamma} = P(8)^{5/3} = 32P$$

$\therefore$ (D)

5. In a room where temperature is 30°C, a body cools from 61°C to 59°C in 4 minutes. The time taken by the body to cool from 51°C to 49°C will be

(A) 4 minute **(B) 6 minute**

(C) 5 minute **(D) 8 minute**

Sol. Rate of cooling α difference in temperature

$$\frac{dT}{dt} \propto \Delta\theta \Rightarrow \frac{dT}{dt} = K\Delta\theta$$

$dT = K\Delta\theta.dt$

In first case:

$dT = 61 - 59 = 2$

$\Delta\theta = 60° - 30° = 30°$

$dt = 4$ minute

$$\therefore\ K = \frac{dT}{\Delta\theta dt} = \frac{2}{30 \times 4} = \frac{1}{60}$$

For second case:

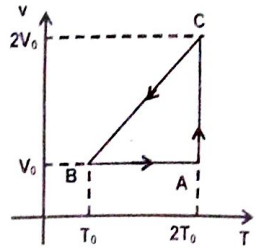
$dT = 2$

$\Delta\theta = 50 - 30 = 20$

$$\therefore\ dt = \frac{dT}{K\Delta\theta} = \frac{2}{\frac{1}{60} \times 20} = 6 \text{ min.}$$

$\therefore$ (B)

6. The efficiency of an ideal gas with adiabatic exponent 'γ' for the shown cyclic process would be



(A) $\frac{(2\ln 2 - 1)}{\gamma/(\gamma - 1)}$

(B) $\frac{(1 - 2\ln 2)}{\gamma/(\gamma - 1)}$

(C) $\frac{(2\ln 2 + 1)}{\gamma/(\gamma - 1)}$

(D) $\frac{(2\ln 2 - 1)}{\gamma/(\gamma + 1)}$

Sol. $W_{BC} = P\Delta V = nR\Delta T = -nRT_0$

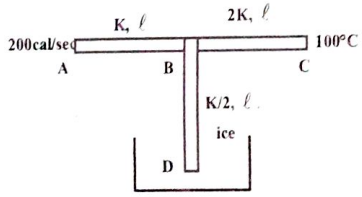
$W_{CA} = +2\,nRT_0 \ln 2$

$$\Delta Q_{BC} = nC_P\Delta T = \frac{nR\gamma T_0}{\gamma - 1}$$

Hence, efficiency $\frac{(2\ln 2 - 1)}{\gamma/(\gamma - 1)}$

$\therefore$ (A)

7. Three rods AB, BC and BD of same length ℓ and cross-sectional area A are arranged as shown. The end D is immersed in ice whose mass is 440 gm. Heat is being supplied at constant rate of 200 cal/sec from the end A. Time in which whole ice will melt (Latent heat of fusion of ice is 80 cal/gm)



(A) $\frac{40}{3}$ min

(B) 700 sec

(C) 20/3 min

(D) indefinitely long time

[Given : k (thermal conductivity) = 100 cal/m sec °C, A = 10 cm^2, ℓ = 1 m]

Sol. If θ be temperature of B, then

$$\frac{2kA(\theta - 100)}{\ell} + \frac{(k/2)A(\theta - 0)}{\ell} = 200$$

Substituting value $\theta = 880°C$

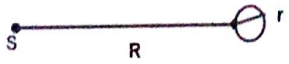
$\therefore\ \phi_t = mL$

$$\frac{kA(880 - 0)}{2\ell} \times t = 440 \times 80$$

$$t = \frac{80\times1\times2\times440}{100\times10\times10^{-4}\times880} = 800 \text{ sec} = (40/3) \text{ min}$$

$\therefore$(A)

8. A point source S of light is emitting a power P. A sphere of radius r is situated at a distance R from source S (r << R), has a mass M and specific heat capacity C. The time in which temperature of sphere rises by θ°C is



(A) $\frac{R^2MC\theta}{Pr^2}$ **(B)** $\frac{2R^2MC\theta}{Pr^2}$

(C) $\frac{3R^2MC\theta}{Pr^2}$ **(D)** $\frac{4R^2MC\theta}{Pr^2}$

Sol. $I_R = \frac{P}{4\pi R^2} \Rightarrow \text{Power} = \frac{P}{4\pi R^2}\times 2\pi r^2 = \frac{Pr^2}{2R^2}$

Now, $\frac{Pr^2}{2R^2}\times t = MC\theta$

$T = \frac{2R^2MC\theta}{Pr^2}$

$\therefore$(B)

9. Power radiated by a black body is P_0 and the wavelength corresponding to the maximum energy is around λ_0. On changing the temperature of the black body, it was observed that the power radiated is increased to $\frac{256}{81}P_0$. The shift in the wavelength corresponding to the maximum energy will be

(A) $+\frac{\lambda_0}{4}$ **(B)** $+\frac{\lambda_0}{2}$

(C) $-\frac{\lambda_0}{4}$ **(D)** $-\frac{\lambda_0}{2}$

Sol. According to Wien's law, $\lambda_0T_0 = \lambda T$ ... (1)

According to Stefan's law, $\frac{P_0}{P} = \left(\frac{T_0}{T}\right)^4$ ... (2)

Now, $\frac{P}{P_0} = \frac{256}{81} = \left(\frac{4}{3}\right)^4$

$\therefore \frac{T_0}{T} = \frac{3}{4} = \frac{\lambda}{\lambda_0}$,

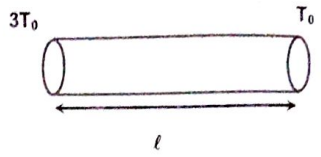
or $\lambda = \frac{3}{4}\lambda_0$

Wavelength shift $\Delta\lambda = \lambda - \lambda_0 =$

$\frac{3}{4}\lambda_0 - \lambda_0 = -\frac{\lambda_0}{4}$

$\therefore$ (C)

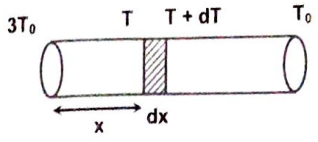
10. Two ends of a rod of uniform cross sectional area are kept at temperatures $3T_0$ and T_0 as shown. Thermal conductivity of rod varies as $k = \alpha T$, (where α is a constant and T is absolute temperature). In steady state, the temperature of the middle section of the rod is



(A) $\sqrt{7}T_0$ (B) $\sqrt{5}T_0$

(C) $2T_0$ (D) $\sqrt{3}T_0$

Sol. B



In steady state

$$\frac{\Delta Q}{\Delta t} = -kA\frac{dT}{dx}$$

$$\frac{\Delta Q}{\Delta t} = -\alpha TA\frac{dT}{dx}$$

$$\frac{\Delta Q}{\Delta t}\int_0^{\ell} dx = -\alpha A\int_{3T_0}^{T_0} TdT$$

$$\frac{\Delta Q}{\Delta t}\ell = 4\alpha AT_0^2 \quad \ldots(i)$$

Similarly,

$$\frac{\Delta Q}{\Delta t}\int_0^{\ell/2} dx = -\alpha A\int_{3T_0}^{T} TdT$$

$$\frac{\Delta Q}{\Delta t}\frac{\ell}{2} = \alpha A\frac{(9T_0^2 - T^2)}{2} \quad \ldots(ii)$$

Dividing (ii) by (i)

$9T_0^2 - T^2 = 4T_0^2$

$T^2 = 5T_0^2$

$T = \sqrt{5}T_0$

11. A cubical box of side 5 m contains helium gas at a pressure 320 N/m^2. During an observation time of 1 second, an atom travelling with root mean square speed parallel to one of the sides of the cube was found to make 1000 hits with a particular wall, without any collision with the other atoms. (take $R = \frac{25}{3}$ J/mol-K). Then

(A) The temperature of gas is 16×10^3 K.
(B) The temperature of gas is 8×10^3 K.
(C) The total mass of helium gas in the box is 1.2 gm.
(D) The total mass of helium gas in the box is 2.4 gm.

Sol. A, C

Time between two consecutive collision = $\frac{1}{1000}$ sec

So, $\frac{2\ell}{v_{rms}} = \frac{1}{1000}$

$v_{rms} = 2000 \times 5 = 10000$ m/s

$v_{rms} = \sqrt{\frac{3RT}{M}} = 10000$

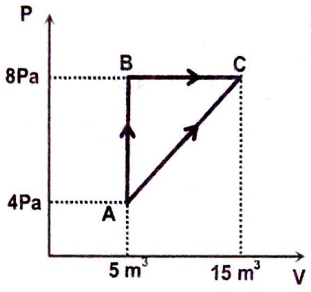
$$\text{So, } T = \frac{10^5 \times 4 \times 10^{-3}}{3 \times \left(\frac{25}{3}\right)} = 16 \times 10^2 \text{ K}$$

Mass of helium gas $\Rightarrow PV = nRT$

$$PV = \frac{m}{M}RT$$

$$m = \frac{PVM}{RT} = \frac{320 \times 125 \times (4 \times 10^{-3})}{\left(\frac{25}{3}\right) \times 16 \times 10^2} = 1.2 \text{ gm}$$

12. In the given graph, an ideal gas can change its state from A to C by two paths ABC or AC. Then



(A) If the internal energy of gas at 'A' is 10 J and the amount of heat supplied in path AC is 200 J, then the internal energy of gas at 'C' is 150 J.

(B) If the internal energy of gas at 'A' is 10 J and the amount of heat supplied in path AC is 200 J, then the internal energy of gas at 'C' is 75 J.

(C) If the internal energy of gas at state B is 20 J. Then the amount of heat supplied to the gas to go from A to B may be 5 Joule.

(D) If the internal energy of gas at state B is 20 J. Then the amount of heat supplied to the gas to go from A to B must be 10 Joule.

Sol. A, D

In path A→C

$$dW = 4 \times 10 + \frac{1}{2} \times 10 \times 4 = 40 + 20 = 60 \text{ J}$$

$$dQ = dU + dW$$

$$200 = U_C - U_A + 60$$

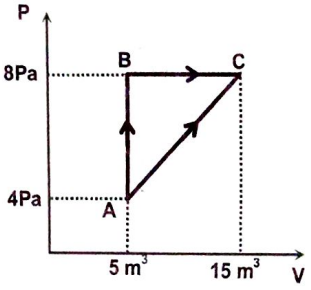
$$U_C = 140 + U_A = 140 + 10 = 150 \text{ J}$$

In path A → B

$$Q_{AB} = U_B - U_A$$

$$= U_B\left(1 - \frac{U_A}{U_B}\right) = U_B\left(1 - \frac{T_A}{T_B}\right)$$

$$= 20\left(1 - \frac{1}{2}\right) = 10 \text{ Joule}$$



COMPREHENSION

Two moles of an ideal mono-atomic gas undergoes a thermodynamic process in which the molar heat capacity 'C' of the gas depends on absolute temperature as $C = \frac{RT}{T_0}$, where R is gas constant and T_0 is the initial temperature of the gas. (V_0 is the initial volume of the gas). Then answer the following questions.

13. The equation of process is

(A) $\frac{1}{P} = \frac{V_0 T_0^{3/2}}{4RT^{5/2}} e^{\left(\frac{T-T_0}{T_0}\right)}$

(B) $\frac{1}{P} = \frac{V_0 T_0^{3/2}}{2RT^{5/2}} e^{\left(\frac{T-T_0}{T_0}\right)}$

(C) $\frac{1}{P} = \frac{V_0 T_0^{3/2}}{4RT^{5/2}} e^{\left(\frac{T+T_0}{T_0}\right)}$

(D) $\frac{1}{P} = \frac{V_0 T_0^{3/2}}{3RT^{5/2}} e^{\left(\frac{T-T_0}{T_0}\right)}$

Sol. B

14. **The minimum volume of the gas is**

(A) $\left(\frac{2}{3}\right)^{3/2} V_0 e^{1/2}$ (B) $\left(\frac{4}{3}\right)^{3/2} V_0 e^{1/2}$

(C) $\left(\frac{5}{3}\right)^{3/2} V_0 e^{1/2}$ (D) $\left(\frac{8}{3}\right)^{3/2} V_0 e^{1/2}$

Sol. A

13-14. $dQ = dU + dW$

$\Rightarrow nCdT = nC_V dT + PdV$

$\Rightarrow \frac{2RT}{T_0} - 3R = \frac{2RT}{V} \cdot \frac{dV}{dT}$. ...(1)

From (1) $\int_{T_0}^{T} \left(\frac{1}{T_0} - \frac{3}{2T}\right) dT = \int_{V_0}^{V} \frac{dV}{V}$

On solving, $V = V_0 \left(\frac{T_0}{T}\right)^{3/2} e^{\left(\frac{T-T_0}{T_0}\right)}$

$\frac{2RT}{P} = V_0 \left(\frac{T_0}{T}\right)^{3/2} e^{\left(\frac{T-T_0}{T_0}\right)}$

$\frac{1}{P} = \frac{V_0 T_0^{3/2}}{2RT^{5/2}} e^{\left(\frac{T-T_0}{T_0}\right)}$

For minimum volume, $\frac{dv}{dT} = 0$

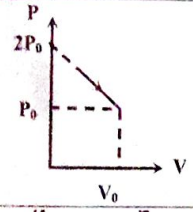
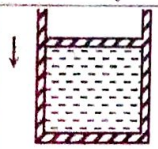
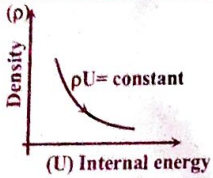
So, $T = \frac{3}{2} T_0$,

$V_{min} = V_0 \left(\frac{T_0}{\frac{3}{2}T_0}\right)^{3/2} e^{\left(\frac{\frac{3}{2}T_0 - T_0}{T_0}\right)}$

$V_{min} = \left(\frac{2}{3}\right)^{3/2} V_0 e^{1/2}$

Matrix-Match Type

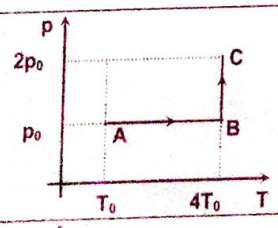
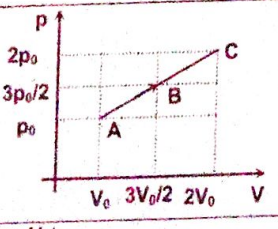
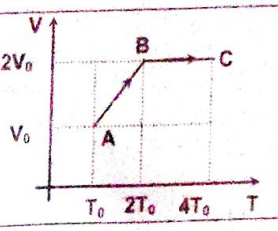
15. Match the following.

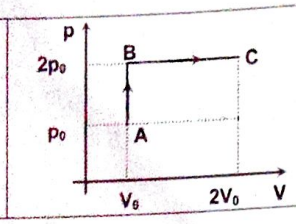
	Column-I		Column-II
(A)	A monatomic gas expands according to the process $P/T^{5/2}$ = constant.	(p)	Temperature of the gas increases.
(B)		(q)	Temperature of the gas decreases.
(C)	 Piston and all walls are thermally insulated and the piston is slowly moving downward.	(r)	No heat exchange with the gas.
(D)		(s)	Heat is absorbed by the gas

Sol. A → (q, r), B → (p, s), C → (p, r), D → (p, s)

Answer questions 16, 17 and 18 by appropriately matching the information given in the three columns of the following table.

Column –I shows 2 moles of a monoatomic gas being taken from initial state A(p_0, v_0, T_0) to final state C ($2p_0$, $2v_0$, $4T_0$) via an intermediate state B. Column II and column III list the values of work done by the gas in process A → B and B → C respectively.

Column 1		Column 2		Column 3	
(I)		(i)	$2RT_0$	(P)	$\frac{7}{4}RT_0$
(II)		(ii)	$6\ RT_0$	(Q)	zero
(III)		(iii)	zero	(R)	$-2\ RT_0\ell n(2)$

(IV)		(iv)	$\frac{5}{4}RT_0$	(S)	$4RT_0$

16. **Pick the correct option.**
(A) (I) (i) (S)
(B) (II) (iv) (P)
(C) (III) (iii) (R)
(D) (IV) (ii) (Q)

Sol. B

17. **Pick the correct option for minimum value of work done in process A → B.**
(A) (IV) (iii) (S)
(B) (III) (iii) (R)
(C) (II) (iv) (P)
(D) (I) (i) (S)

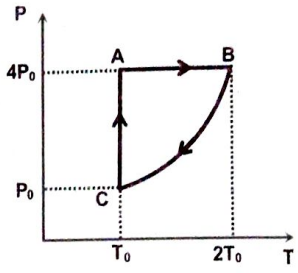
Sol. A

18. **Pick the correct option for minimum value of work done in process B → C.**
(A) (I) (ii) (R)
(B) (II) (iv) (P)
(C) (III) (iii) (R)
(D) (IV) (ii) (Q)

Sol. A

Numerical Type

19. **Two moles of an ideal monoatomic gas is taken through a cyclic process as shown in the P-T diagram. In the process BC, PT^{-2} = constant. Then find the ratio of heat absorbed and heat released by the gas during the process AB and process BC respectively.**



Sol. 5

$$\Delta Q_{AB} = nC_P\Delta T = 2\times\frac{5R}{2}(2T_0 - T_0) = 5RT_0$$

In the process BC, PT^{-2} = constant

$PP^{-2}V^{-2}$ = constant

PV^2 = constant

$$\therefore \text{ molar heat capacity, } C = C_v + \frac{R}{1-x} = \frac{3R}{2} + \frac{R}{1-2} \Rightarrow C = \frac{R}{2}$$

$$\therefore \Delta Q_{BC} = nC\Delta T = 2\times\frac{R}{2}(T_0 - 2T_0) = -RT_0$$

$$\therefore \frac{|\Delta Q_{AB}|}{|\Delta Q_{BC}|} = \frac{5RT_0}{RT_0} = 5$$

20. **A hollow cube A of very small thickness at temperature T emits radiation like a black body. The cube A is kept in a room of temperature $T_0(<T)$. The side of cube is doubled without changing the volume of material and its temperature. Find the ratio of rate of cooling of cube A in first case to that of second case.**

Sol. **4**

The rate of heat loss by cube of thickness Δx, side ℓ and made of density ρ is given by

$$-ms\frac{dT}{dt} = \varepsilon\sigma A\left(T^4 - T_0^4\right)$$

$$-\frac{dT}{dt} = \frac{\varepsilon\sigma A(T^4 - T_0^4)}{ms}$$

Numerical Decimal Type

21. **There are two vessels. Each of them contains one mole of a monoatomic ideal gas. Initial volume of the gas in each vessel is 8.3×10^{-3} m^3 at 27°C. Equal amount of heat is supplied to each vessel. In one of the vessel, the volume of the gas is doubled without change in its internal energy, where as the volume of the gas is held constant in the second vessel. The vessels are now connected to allow free mixing of the gas. Find the final temperature of the combined gas in Kelvin. (take $\ell n2 = 0.6931$)**

Sol. **00369.31**

For first vessel.

$\Delta U = 0$

$$\Delta Q_1 = \Delta W = nRT\ell n\left(\frac{V_f}{V_i}\right) = 1.R.300.\ell n2$$

$$= (0.6931)\times 300\times R = 207.93\ R$$

For second vessel $\Delta V = 0$

$$\Delta Q_2 = \Delta U = nC_V\Delta T = 1\cdot\frac{3}{2}R\Delta T$$

$$\Delta Q_1 = \Delta Q_2$$

$$207.93R = \frac{3}{2}R\Delta T \Rightarrow \Delta T = 138.62$$

$$T_f = 300 + 138.62 = 438.62$$

$$U_{mix} = U_1 + U_2$$

$$(n_1 + n_2)C_{Vmix}T_{mix} = n_1C_{V_1}T_1 + n_2CV_2T_2$$

$$T_{mix} = \frac{T_1 + T_2}{n_1 + n_2} = 369.31K.$$

22. **A ideal gas whose adiabatic exponent equals γ is expanded so that the amount of heat transferred to the gas is equal to twice of decrease of its internal energy. The equation of the process is $TV^{\frac{\gamma-1}{K}}$ = constant (where T and V are absolute temperature and volume respectively). Then find the value of 'K'.**

Sol. **00003.00**

$dQ = dU + dW$

$dQ = -2dU$

$-3dU = dW$

$-3nC_VdT = PdV$

$$\frac{-3nRdT}{\gamma - 1} = \left(\frac{nRT}{V}\right)dV$$

From this $TV^{\frac{\gamma-1}{3}}$ = constant

So, K = 3

11

ELECTROSTATICS

COULOMB'S LAW

$$F = \frac{1}{4\pi\varepsilon_0}\cdot\frac{Q_1Q_2}{r^2};\ \varepsilon_0 = 8.85\times10^{-12}\ C^2/N\text{-}m^2$$

DEFINITION OF ELECTRIC FIELD

$$\vec{E} = \frac{\vec{F}}{Q}$$

Field due to a point charge: $E = \frac{Q}{4\pi\varepsilon_0}\frac{1}{R^2}$

ELECTRIC POTENTIAL ENERGY

Electric potential energy between two point charges

$$U = \frac{Q_1Q_2}{4\pi\varepsilon_0}\frac{1}{r}$$

Also, $U_2 - U_1 = \frac{Q_1Q_2}{4\pi\varepsilon_0}\left[\frac{1}{r_2} - \frac{1}{r_1}\right]$

ELECTRIC POTENTIAL

Electric potential due to a number of point charges

$$V = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{Q_i}{r_i}$$

$$V_2 - V_1 = -\int_{\vec{r}_1}^{\vec{r}_2}\vec{E}.d\vec{r}$$

Also, $\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

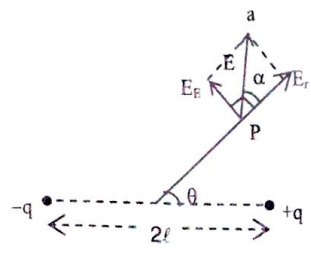
$$dV = -E_x\,dx - E_y\,dy - E_z\,dz$$

Also, $E = -\frac{dV}{dr}$

The electric field is along the direction where the potential decreases at the maximum rate.

ELECTRIC DIPOLE MOMENT

Two point charges of equal magnitude but opposite signs separated by a small distance form an electric dipole.



Dipole moment is a vector quantity, $p = 2q\ell$.

Direction of dipole is from –q to +q.

Potential at the Point P

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}$$

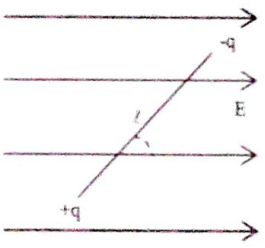
$$E_r = -\frac{dV}{dr} = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^3}.2\cos\theta$$

$$E_\theta = -\frac{1}{r}\frac{dV}{d\theta} = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^3}\sin\theta$$

$$E = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^3}\sqrt{1+3\cos^2\theta}\,;\ \tan\alpha = \frac{1}{2}\tan\theta$$

This equation gives the electric field at any point P at a distance r from the dipole.

Torque on a Dipole in an Electric Field



$$\text{Torque} = 2q\ell\sin\theta E$$

In vector form:

$$\text{Torque} = \vec{p}\times\vec{E}$$

Potential Energy of a Dipole in an Electric Field

$$U = -\vec{p}.\vec{E}$$

Note: No electric field can exist inside a conducting material.

Force on a dipole:

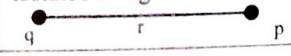
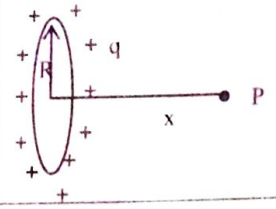
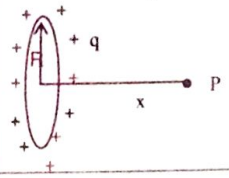
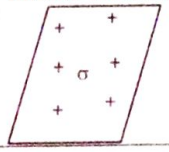
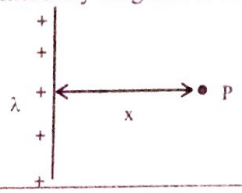
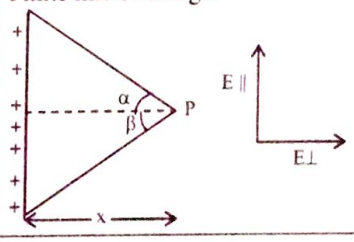
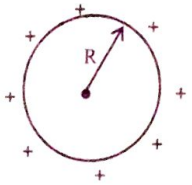
$\vec{F} = p\frac{\partial\vec{E}}{\partial\ell}$, $\frac{\partial\vec{E}}{\partial\ell}$ is the derivative of electric field with respect to distance along the direction of the dipole.

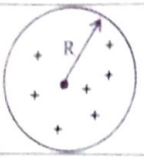
Gauss' Theorem

$\int\vec{E}.d\vec{s} = \frac{q_0}{\varepsilon_0}$, where q_0 is equal to the net charge enclosed within the surface.

Work done by electric field along the closed path is zero.

Mathematically: $\oint E.d\ell = 0$

S. No.	System	Potential	Electric field
(i)	Isolated charge:  	$V = \frac{q}{4\pi\varepsilon_0}\frac{1}{r}$	$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$
(ii)	A ring of charge:  	$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{\sqrt{R^2 + x^2}}$	$E = \frac{1}{4\pi\varepsilon_0}\frac{qx}{(R^2 + x^2)^{3/2}}$
(iii)	A disc of charge:  	$V = \frac{\sigma}{2\varepsilon_0}\left[\sqrt{R^2 + x^2} - x\right]$	$E = \frac{\sigma}{2\varepsilon_0}\left[1 - \frac{x}{\sqrt{x^2 + R^2}}\right]$
(iv)	Infinite sheet of charge:  	V not defined	$E = \frac{\sigma}{2\varepsilon_0}$
(v)	Infinitely long line of charge:  	V not defined	$E = \frac{\lambda}{2\pi\varepsilon_0}\frac{1}{r}$
(vi)	Finite line of charge:  		$E_\perp = \frac{\lambda}{4\pi\varepsilon_0 x}(\sin\alpha + \sin\beta)$ $E_{\parallel} = \frac{\lambda}{4\pi\varepsilon_0 x}(\cos\alpha - \cos\beta)$
(vii)	Charged spherical shell:  	(a) $0 \le r \le R$ $V = \frac{q}{4\pi\varepsilon_0}\frac{1}{R}$ (b) $r > R$ $V = \frac{q}{4\pi\varepsilon_0}\frac{1}{r}$	(a) $0 \le r \le R$ E = zero (b) $r > R$ $E = \frac{q}{4\pi\varepsilon_0}\frac{1}{r^2}$
(viii)	Solid sphere of charge:	(a) $0 \le r \le R$ $V = \frac{\rho R^2}{6\varepsilon_0}\left[3 - \frac{r^2}{R^2}\right]$ (b) $r > R$	(a) $0 \le r \le R$ $E = \frac{\rho R}{3\varepsilon_0}$ $r > R$

		$V = \frac{\rho R^3}{3\varepsilon_0}\left(\frac{1}{r}\right)$	(b) $E = \frac{\rho R}{3\varepsilon_0}\left(\frac{R}{r}\right)^2$

Self energy of a uniformly charged shell of radius R and charge Q is given by:

$$U = \frac{Q^2}{8\pi\varepsilon_0}\frac{1}{R}$$

Self energy of a uniformly charged sphere of radius R and charge Q is given by

$$U = \frac{3Q^2}{20\pi\varepsilon_0}\frac{1}{R}$$

CAPACITANCE

Capacitor is a device for storing charge. Mathematically, Q = CV, the unit of C is farad.

For parallel plate capacitor: $C = \frac{\varepsilon_0 A}{d}$

where A =plate area

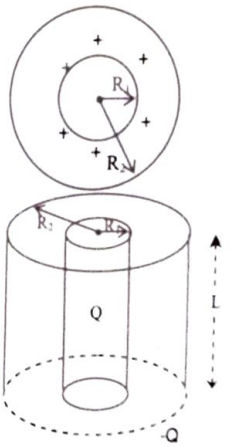
d = distance between the plates.

For spherical capacitor:

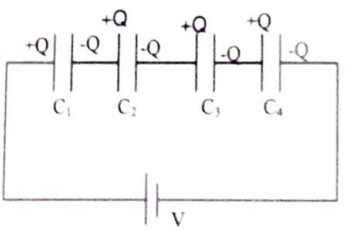
$$C = \frac{4\pi\varepsilon_0 R_2 R_1}{R_2 - R_1}$$

For cylindrical capacitor:

$$C = \frac{2\pi\varepsilon_0 \ell}{\ln(R_2/R_1)}$$



Series Combination Of Capacitors

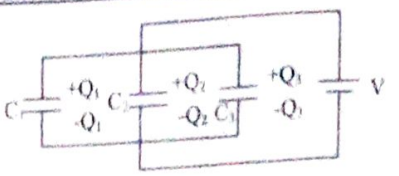


Note: (i) Charge on each capacitor is same.

(ii)The equivalent capacitor of the capacitor is given by

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \frac{1}{C_4} + \cdots$$

Parallel Combination of Capacitors

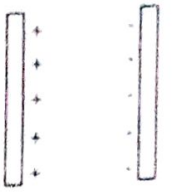


Note: (i) The potentials across each of all the capacitor is same.

(ii) The charge on each capacitor is different.

(iii) $C_{equ} = C_1 + C_2 + C_3 + \cdots$

Force Between the Plates of a Capacitor



$$E = \frac{Q}{2\varepsilon_0 A} \quad \text{(due to one plate)}$$

$$F = \frac{Q^2}{2\varepsilon_0 A}$$

Energy Stored in a Capacitor

$$U = \frac{Q^2}{2C}$$

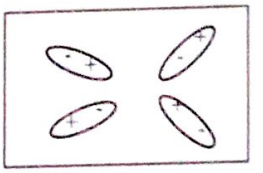
$$U = \frac{1}{2}CV^2, \qquad U = \frac{1}{2}QV$$

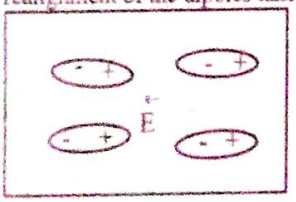
Energy Stored per unit volume in an electric field (E)

$$\frac{U}{\text{volume}} = \frac{1}{2}\varepsilon_0 E^2$$

DIELECTRICS

Random distribution of molecular dipoles in a dielectric material is shown in the figure. When we apply an electric field, realignment of the dipoles take place





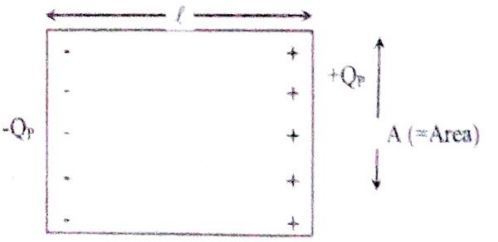
This creates an induced electric field within the dielectric.

Thus, the net electric field in a dielectric is given by:

$$\vec{E}_{net} = \vec{E} - \vec{E}_{ind} = \frac{\vec{E}}{k}, \text{ where } k > 1$$

k = dielectric constant or relative permittivity of any material.

Thus, whenever a dielectric is placed in any electric field, dipole moments appear in any volume of it. This is called polarization of the material. The dipole moment induced per unit volume is called the polarization vector $\vec{P}$. Its magnitude P is often referred to as the polarization.



$$P = \frac{Q_P \ell}{A\ell} = \frac{Q_P}{A} = \sigma_P \text{ (surface charge density)}$$

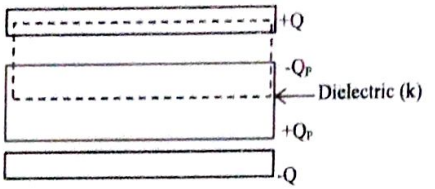
$\therefore$ The polarization vector is equal to the surface density of induced charge.

Magnitude of Induced Charge

$$\frac{Q}{k\varepsilon_0 A} = \frac{(Q - Q_P)}{\varepsilon_0 A} \quad \Rightarrow \quad Q_P = Q\left[1 - \frac{1}{k}\right]$$

An alternative form of Gauss' law

Consider a Gaussian surface as shown.



$$\oint \vec{E}.d\vec{s} = \frac{Q - Q_P}{\varepsilon_0} = \frac{1}{\varepsilon_0}\left[Q - Q\left(1 - \frac{1}{k}\right)\right] = \frac{Q}{k\varepsilon_0}$$

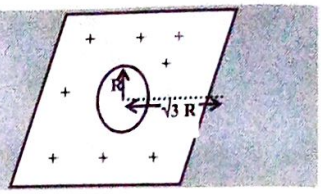
$$\therefore \quad \oint k\vec{E}.d\vec{s} = \frac{Q_{free}}{\varepsilon_0}$$

Q_{free} is used in place of Q to emphasize that it is the free charge given to the plates and does not depend on the induced charge appearing due to polarization.

Energy stored per unit volume in a dielectric is given by $\frac{1}{2}k\varepsilon_0 E^2$.

ILLUSTRATIVE PROBLEMS

1. An infinite dielectric sheet having charge density σ has a hole of radius R in it. An electron is released on the axis of the hole at a distance √3R from the centre. Find the speed with which it crosses the centre of the hole.



Sol. Potential function is not defined for infinite conducting sheet and hence to solve this either calculate potential difference or use force equations

Electric field due to infinite dielectric sheet, $E_1 = \frac{\sigma}{2\varepsilon_0}$

Electric field at the axis of a disc of radius R

$$E_2 = \frac{\sigma}{2\varepsilon_0}\left[1 - \frac{x}{\sqrt{x^2+R^2}}\right]$$

Resultant electric field $E = E_1 - E_2 = \frac{\sigma}{2\varepsilon_0}\frac{x}{\sqrt{x^2+R^2}}$

Force on the electron

$$F = -\frac{\sigma e x}{2\varepsilon_0\sqrt{x^2+R^2}}$$

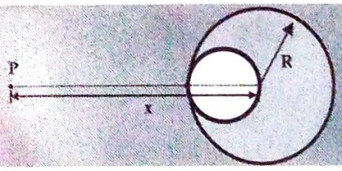
$$mv\frac{dv}{dx} = -\frac{\sigma e x}{2\varepsilon_0\sqrt{x^2+R^2}}$$

$$m\int_0^v v\,dv = -\frac{\sigma e}{2\varepsilon_0}\int_{\sqrt{3}R}^{0}\frac{x}{\sqrt{x^2+R^2}}dx$$

$$m\frac{v^2}{2} = -\frac{\sigma e}{2\varepsilon_0}\left[\sqrt{x^2+R^2}\right]_{\sqrt{3}R}^{0}$$

$$v = \sqrt{\frac{\sigma e R}{m\varepsilon_0}}.$$

2. A solid spherical region having a spherical cavity whose diameter 'R' is equal to the radius of the spherical region, has a total charge 'Q'. Find the electric field and potential at a point P as shown.



Sol. Charge density $\rho = \frac{Q}{\frac{4}{3}\pi\left[R^3 - \left(\frac{R}{2}\right)^3\right]} = \frac{6Q}{7\pi R^3}$

Using the superposition principle,

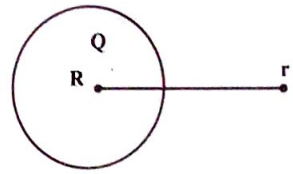
$$V = \frac{\rho\left[\frac{4}{3}\pi R^3\right]}{4\pi\varepsilon_0 x} - \frac{\rho\left[\frac{4}{3}\pi\left(\frac{R}{2}\right)^3\right]}{4\pi\varepsilon_0\left[x-\frac{R}{2}\right]} = \frac{\rho R^3}{3\varepsilon_0}\frac{[7x-4R]}{4x(2x-R)}$$

Similarly,

$$E = \frac{\rho\left[\frac{4}{3}\pi R^3\right]}{4\pi\varepsilon_0 x^2} - \frac{\rho\left[\frac{4}{3}\pi\left(\frac{R}{2}\right)^3\right]}{4\pi\varepsilon_0\left[x-\frac{R}{2}\right]^2} = \frac{\rho R^3}{3\varepsilon_0}\left[\frac{1}{x^2} - \frac{1}{2(2x-R)^2}\right]$$

3. Positive charge Q is uniformly distributed throughout the volume of a dielectric sphere of radius R. A point mass having charge +q and mass m is fired towards the centre of the sphere with velocity v from a point A at distance r (>R) from the centre of the sphere. Find the minimum velocity v so that it can penetrate R/2 distance of the sphere. Neglect any resistance other than electric interaction. Charge on the small mass remains constant throughout the motion.

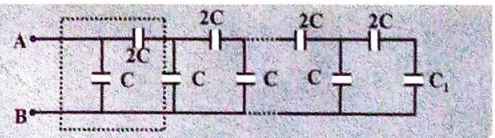
Sol. $\frac{1}{2}mv^2 + \frac{Qq}{4\pi\varepsilon_0}\cdot\frac{1}{r} = \frac{Qq}{4\pi_0}\left[\frac{3R^2-(R/2)^2}{2R^3}\right] = \frac{Qq}{4\pi}\left[\frac{11R^2}{8R^3}\right] = \frac{11Qq}{32\pi\varepsilon_0 R}$



$$v^2 = \frac{2}{m}\left[\frac{11Qq}{32\pi\varepsilon_0 R} - \frac{Qq}{4\pi\varepsilon_0}\cdot\frac{1}{r}\right]$$

$$\Rightarrow v = \sqrt{\frac{11Qq}{16\pi\varepsilon_0 Rm} - \frac{Qq}{2\pi\varepsilon_0}\frac{1}{mr}}$$

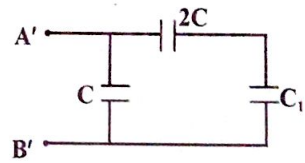
4. The figure shows a network of capacitance consisting of several repetitive units shown by doted square and a capacitor of capacitance C_1 connected at end as shown in the figure.



Find the value of C_1 such that the equivalent capacitance between A and B is independent of the number of units.

Sol. The equivalent capacitance between A and B will be independent of number of units if addition of one or more units to C_1 does not change its value.

The situation is shown in figure. The equivalent between A′ and B′ is still C_1.

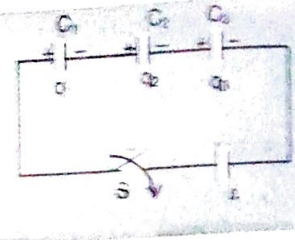


$$\frac{C_1 \times 2C}{C_1 + 2C} + C = C_1$$

$$\Rightarrow C_1 = \frac{C \pm 3C}{2}$$

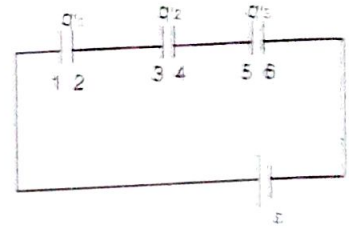
$\therefore C_1 = 2C.$

5. **Charges q_1, q_2 and q_3 are placed on capacitors of capacitance C_1, C_2 and C_3, respectively, arranged in series as shown. Switch S is then closed. What are the final charges q'_1, q'_2 and q'_3 on the capacitors?**
Given $q_1 = 30\ \mu C$, $q_2 = 20\ \mu C$, $q_3 = 10\ \mu C$; $C_1 = 10\ \mu F$, $C_2 = 20\ \mu F$, $C_3 = 30\ \mu F$ and $\varepsilon = 12$ volt



Sol. Applying Kirchoff's law on the loop,

$$\frac{q'_1}{C_1} + \frac{q'_2}{C_2} + \frac{q'_3}{C_3} = \varepsilon \quad \ldots(1)$$



Net charge on plates 2 and 3 will remain conserved.

$\therefore -q'_1 + q'_2 = -q_1 + q_2$ ...(2)

Also, net charge on plates 4 and 5 will remain conserved.

$\therefore -q_2' + q'_3 = -q_2 + q_3$ ...(3)

Using Eqs. (1), (2) and (3) and putting the values, we get

$$q'_1 = \frac{790}{11}\mu C,\ q'_2 = \frac{680}{11}\mu C,\ q'_3 = \frac{570}{11}\mu C.$$

SOLVED PROBLEMS

1. **A charge –Q is uniformly distributed over a non-conducting semi-circular rod of radius R. The potential at the centre is**

(A) 0

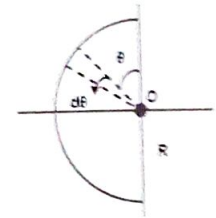
(B) $\frac{-Q}{4\pi\varepsilon_0 R}$

(C) $\frac{Q}{4\pi\varepsilon_0 (2R)}$

(D) $\frac{2Q}{4\pi\varepsilon_0 R}$

Sol. Potential at O due to elemental charge

$$dq = \frac{1}{4\pi\varepsilon_0}\cdot\frac{dq}{R}$$



$\therefore$ Total Potental at $O = \frac{1}{4\pi R\varepsilon_0}\int_0^{-Q} dq = -\frac{1}{4\pi\varepsilon_0}\cdot\frac{Q}{R}$

$\therefore$ (B)

2. **Two bodies are charged by rubbing one against the other. During the process, one becomes positively charged while the other becomes negatively charged. Then**

(A) mass of each body remains unchanged.

(B) mass of each body changes marginally.

(C) mass of each body changes slightly and hence the total mass.

(D) mass of each body changes slightly but the total mass remains the same.

Sol. The transfer of electrons from one body to the other results is development of charges.

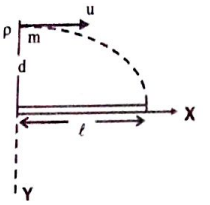
Hence, no. of electrons given by one body = No. of electrons obtained by the other.

∴ Mass of negatively charged body slightly increases while mass of positively charged body slightly increases.

whereas the total mass of the system remains the same.

∴ (D)

3. **An electron is projected from a distance d and with initial velocity u parallel to a uniformly charged flat conducting plate as shown. It strikes the plate after travelling a distance ℓ along the direction of projection. The surface charge density of the conducting plate is equal to**



(A) $\frac{2d\varepsilon_0 mu^2}{e\ell^2}$ **(B)** $\frac{2d\varepsilon_0 mu}{e\ell}$

(C) $\frac{d\varepsilon_0 mu^2}{e\ell}$ **(D)** $\frac{d\varepsilon_0 mu}{e\ell}$

Sol. For the motion of the electron along Y axis

$$d = \frac{1}{2}\frac{eE}{m}\cdot t^2 \qquad \ldots (1)$$

where E is the electric field due to the charged plate given by

$$E = \sigma/\varepsilon_0 \qquad \ldots (2)$$

where σ is the surface density of charge of the plate.

The time taken by the electron is

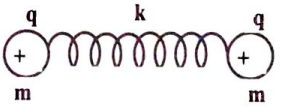
$$t = \frac{\ell}{u} \qquad \ldots (3)$$

From Eqs. (1), (2) and (3), we obtain

$$d = \frac{1}{2}\frac{e\sigma}{\varepsilon_0 m}\left(\frac{\ell}{u}\right)^2 \Rightarrow \sigma = \frac{2d\varepsilon_0 mu^2}{e\ell^2}$$

∴ (A)

4. **The ratio of the time periods of small oscillation of the insulated spring and mass system before and after charging the masses is**



(A) ≥1 **(B)** >1

(C) ≤1 **(D)** =1

Sol. In equilibrium of the charged small bodies $\frac{1}{4\pi\varepsilon_0}\frac{q^2}{(\ell_0 + x_0)^2} = kx_0$, where x_0 is the elongation in the spring in equilibrium.

Let a further small elongation of x is given to the spring.

Then net restoring force on any of the charged particle is given by,

$$F = -\left[k(x_0 + x) - \frac{1}{4\pi\varepsilon_0}\frac{q^2}{(\ell_0 + x_0 + x)^2}\right] = -kx. \qquad \text{Since } x << x_0$$

$$\Rightarrow a = -\frac{2k}{m}x \qquad \text{As } F = \mu a, \text{ where reduced mass } \mu = \frac{m\times m}{m+m} = \frac{m}{2}$$

$$\Rightarrow a = -\omega^2 x,$$

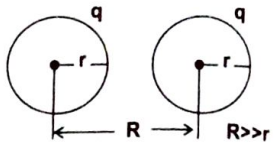
Hence $\omega = \sqrt{\frac{2k}{m}}$ $\Rightarrow T = 2\pi\sqrt{\frac{m}{2K}}$

In absence of charge, $T_0 = 2\pi\sqrt{\frac{m}{2K}}$.

Therefore $\frac{T}{T_0} = 1$

$\therefore$ (D)

5. The potential energy of the system of two identically charged spheres as shown in the figure is equal to (Assume the charge distribution to be uniform)



(A) $\frac{q^2}{4\pi\varepsilon_0}\left(\frac{1}{R}+\frac{1}{r}\right)$s

(B) $\frac{1}{4\pi\varepsilon_0}\frac{q^2}{r}$

(C) $\frac{1}{4\pi\varepsilon_0}\frac{q^2}{r(R+r)}$

(D) None of these.

Sol. Each of the charged sphere will have potential energy due to its own charge or self energy

$U_1 = \frac{1}{8\pi\varepsilon_0}\frac{q^2}{r}$.

The mutual potential energy stored in one sphere in the electric field of the other is $\frac{q_1q_2}{4\pi\varepsilon_0 r_{12}} = U_2$

$\Rightarrow U_2 = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{R}$

Therefore the P.E. stored in the system of spheres is

$U = 2U_1 + U_2 = \frac{q^2}{4\pi\varepsilon_0}\left[\frac{1}{R}+\frac{1}{r}\right]$

$\therefore$ (A)

6. An assembly of charge +q, -q, +q, −q, ... are placed at distance x = 1 m, x = 2 m, x = 4 m, x =8 m, ..., from the origin, in a plane. The potential at x = 0 due to the charges would be

(A) $-\frac{q}{4\pi\varepsilon_0}$

(B) $-\frac{q}{6\pi\varepsilon_0}$

(C) $\frac{q}{6\pi\varepsilon_0}$

(D) $\frac{q}{4\pi\varepsilon_0}$

Sol. $V_+ = \frac{q}{4\pi\varepsilon_0(1)}+\frac{q}{4\pi\varepsilon_0(4)}+\frac{q}{4\pi\varepsilon_0(16)}+\cdots$

$\frac{q}{4\pi\varepsilon_0}\left(1+\frac{1}{4}+\frac{1}{16}+\cdots\infty\right) = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{1-(1/4)}\right) = \frac{q}{3\pi\varepsilon_0}$

$V_- = \frac{-q}{4\pi\varepsilon_0(2)}-\frac{q}{4\pi\varepsilon_0(8)}-\frac{q}{4\pi\varepsilon_0(32)}-\cdots$

$= -\frac{q}{2\pi\varepsilon_0}\left(1+\frac{1}{4}+\frac{1}{16}+\cdots\right) = \frac{-q}{6\pi\varepsilon_0}$

$$V_{net} = \frac{q}{3\pi\varepsilon_0} - \frac{q}{6\pi\varepsilon_0} = \frac{q}{6\pi\varepsilon_0}.$$

$\therefore$ (C).

7. **A charge Q has to be divided between two solid spheres of radius 'R' which are at distance d from each other (d $\gg$ R). What should be the value of charge, which we should place on spheres, so that the force of attraction between them is maximum?**

(A) Q/4, 3Q/4 **(B) Q/3, 2Q/3**

(C) Q/2, Q/2 **(D) Q/5, 4Q/5**

Sol. Let us place a quantum of charge 'q' on the first sphere, so we have charge (Q – q) on the other sphere. The force of attraction between the spheres is

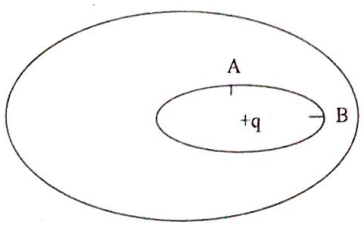
$$F = \frac{(Q-q)q}{4\pi\varepsilon_0 d^2}$$

$$\frac{dF}{dq} = \frac{1}{4\pi\varepsilon_0 d^2}[(Q-q)-q] = 0$$

$\therefore$ q = Q/2

$\therefore$ (C)

8. **An ellipsoidal cavity is carved within a perfect conductor. A positive charge q is placed at the centre of the cavity. The points A and B are on the cavity surface. Then**



(A) electrical field near A in the cavity must be equal to the electric field near B in the cavity.

(B) charge density at A must be equal to the charge density at B.

(C) potential at A must be equal to the potential at B.

(D) total electric field flux through the surface of the cavity is q/ε_0.

Sol. All the points which lies on the conductor are at the same potential. Again, according to Gauss's law; net flux through the surface of cavity $= \frac{q}{\varepsilon_0}$.

$\therefore$ (C), (D)

9. **Two identical sheets of a metallic foil are separated by d and capacitance of the system is C and charged to a potential difference E. Keeping the charge constant, the separation is increased by 'ℓ'. Then, the new capacitance and potential difference will be**

(A) $\frac{\varepsilon_0 A}{d}$, E **(B)** $\frac{\varepsilon_0 A}{d+\ell}$, E

(C) $\frac{\varepsilon_0 A}{d+\ell}, \left(1+\frac{\ell}{d}\right)E$ **(D)** $\frac{\varepsilon_0 A}{d}, \left(1+\frac{\ell}{d}\right)E$

Sol. $q = CV = C_1V_1$

when $C = \frac{\varepsilon_0 A}{d}$ and

$$q = \frac{\varepsilon_0 AE}{d} = \frac{\varepsilon_0 AE_1}{(d+\ell)}$$

$$\therefore \frac{\varepsilon_0 A}{d} E = \frac{\varepsilon_0 A}{(d+\ell)} E_1$$

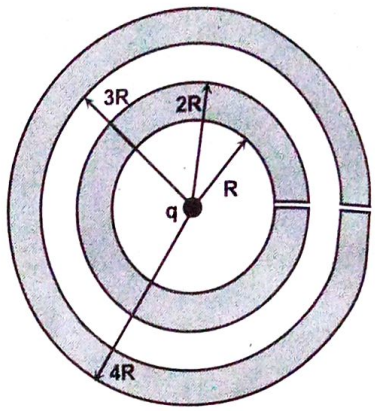
$$\therefore E_1 = \frac{(d+\ell)}{d} \cdot E$$

$$= \left(1+\frac{\ell}{d}\right).E$$

and $C_1 = \frac{\varepsilon_0 A}{d+\ell}$.

$\therefore$ (C)

10. **A charge particle 'q' lies at the centre of two concentric hollow sphere of inner radii R and 3R and outer radii 2R and 4R respectively. What amount of work has to be performed to slowly transfer the charge 'q' from center through the orifice to infinity.**



(A) $\frac{5}{96}\frac{q^2}{\pi\varepsilon_0 R}$

(B) $\frac{q^2}{16\pi\varepsilon_0 R}$

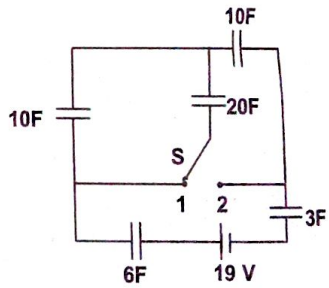
(C) $\frac{7}{96}\frac{q^2}{\pi\varepsilon_0 R}$

(D) $\frac{q^2}{96\pi\varepsilon_0 R}$

Ans. C

Sol. Work performed $= \int_R^{2R} \frac{1}{2}\varepsilon_0 E^2 dV + \int_{3R}^{4R} \frac{1}{2}\varepsilon_0 E^2 dV = \frac{7}{96}\frac{q^2}{\pi\varepsilon_0 R}$

11. **When the steady state is reached by keeping switch in position -1, then switch 'S' is transferred from position 1 to 2. Then**



(A) **Charge on 3F capacitor is 30 C after S is transferred from position 1 to 2.**

(B) **Charge on 3F capacitor is $\frac{650}{19}$ C after S is transferred from position 1 to 2.**

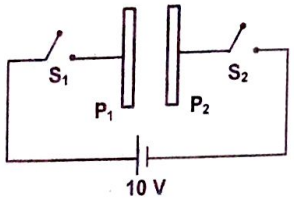
(C) **Work done by the battery is 380 Joule after S is transferred from position 1 and 2.**

(D) **Work done by the battery is 80 Joule after S is transferred from position 1 and 2.**

Ans. B, D

Sol. By applying Kirchhoff's law charge transferred by the battery is $\frac{80}{19}$ C.

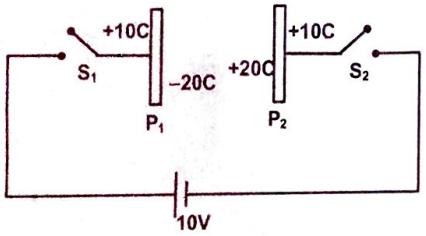
12. Two identical plates P_1 and P_2, of cross sectional area A are separated at a very small distance d, where $\frac{\varepsilon_0 A}{d} = 2$ -SI unit. Initially charge on the plate P_1 is –10 C and charge on the plate P_2 is +30C. Now plate P_1 and P_2 are connected to positive and negative terminals of battery of potential difference 10 Volts with the help of switches S_1 and S_2 respectively. Choose the correct option(s).



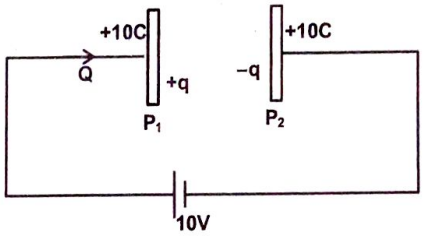
(A) The work done by the battery is 200 J.
(B) The net charge on the plate P_2 is –10C.
(C) The charge on the left side of the plate P_1 is 10C.
(D) The charge on the right side of the plate P_2 is 20C.

Ans. B, C

Sol.



When switches are not closed



When switches are closed

$$\frac{q}{A\varepsilon_0 \times d} = 10$$

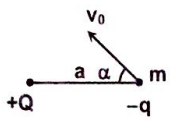
$$\Rightarrow q = 10 \times \frac{A\varepsilon_0}{d} = 20\,C$$

$Q = 20 + 20 = 40$ C

$W_B = (20 + 20)\ 10 = 400$ J

COMPREHENSION

A charged particle of charge –q and mass m is projected with initial velocity $v_0 = \sqrt{\frac{qQ}{4\pi\varepsilon_0 am}}$ from a distance a from the fixed charge +Q at an angle of $\alpha = 60°$ with the line joining the charges as shown in the figure. It is found that the particle moves on an elliptical orbit with the charge +Q at one of its foci.



13. The length of the semi-major axis of the ellipse equals to

(A) a **(B) a/2**

(C) $\frac{a\sqrt{3}}{2}$ **(D) 2a**

Sol. **A**

14. **The area of the ellipse orbit equals to**

(A) πa^2

(B) $\frac{1}{2}\pi a^2$

(C) $\frac{\sqrt{3}}{2}\pi a^2$

(D) $\frac{3}{2}\pi a^2$

Sol. C

15. **The time period of one revolution will be**

(A) $\sqrt{\frac{16\pi^3\varepsilon_0 ma^3}{qQ}}$

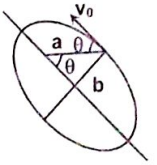
(B) $\sqrt{\frac{8\pi^3\varepsilon_0 ma^3}{qQ}}$

(C) $\sqrt{\frac{2\pi^3\varepsilon_0 ma^3}{qQ}}$

(D) $\sqrt{\frac{6\sqrt{3}\pi^3\varepsilon_0 ma^3}{qQ}}$

Sol. A

13-15. Total energy $E = \frac{1}{2}mv^2 - \frac{qQ}{4\pi\varepsilon_0 a} = \frac{-qQ}{8\pi\varepsilon_0 a}$



$$E = -\frac{qQ}{4\pi\varepsilon_0(2a)}$$

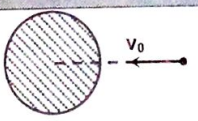
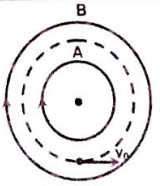
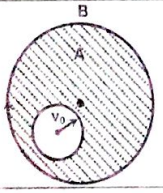
$\therefore$ The particle is moving on an ellipse of semi-major axis a and the time period will $T = 2\pi\sqrt{\frac{a^3}{\left(\frac{qQ}{4\pi\varepsilon_0 m}\right)}} = \sqrt{\frac{16\pi^3 ma^3}{qQ}}$

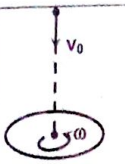
from geometry $b = a\sin\alpha$

$\therefore$ area $A = \pi ab = \pi a^2 \sin\alpha = \frac{\sqrt{3}}{2}\pi a^2$

Matrix – Match Type

16. **Match the following cases force on point charged particle.**

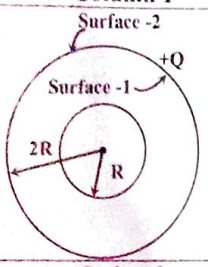
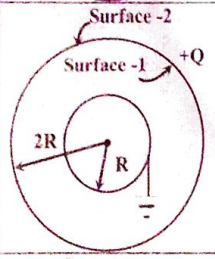
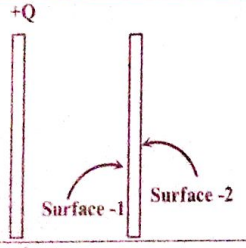
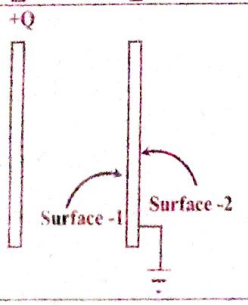
Column-I			Column-II	
(A)	A charged particle is projected with an initial velocity v_0 towards the centre of an uncharged conducting sphere.		(p)	Conservative electric force.
(B)	A charged particle is projected tangentially with an initial velocity v_0 in the annular region between the two coaxial cylinders A and B having time varying current.		(q)	Non-conservative electric force
(C)	A charged particle is projected with an initial velocity v_0 inside a spherical cavity drawn inside a conducting sphere.		(r)	Magnetic force

(D)	An insulated uniformly charged disc is rotated with uniform angular velocity ω and a charged particle is projected with an initial velocity v_0 along the axis of disc.		(s)	No force

Sol. A → (p, q, r), B → (q, r), C → (p, q, r), D → (p)

Answer questions 17, 18 and 19 by appropriately matching the information given in the three columns of the following table.

Figure shows total charge given to a conductor.
Here, Column 1 represents arrangement of conductors.
Column 2 represents charge on surface -1
Column 3 represents charge on surface -2

Column 1		Column 2		Column 3	
(I)		(i)	+Q	(P)	+Q
(II)		(ii)	−Q	(Q)	−Q
(III)		(iii)	+Q/2	(R)	+Q/2
(IV)		(iv)	−Q/2	(S)	−Q/2

17. **For part (I) the correct combination is**

(A) (I) (i) (P) **(B) (I)(ii) (P)**

(C) (I) (iii) (R) **(D) (I) (iii) (S)**

Sol. B

18. **For part (II) the correct combination is**
(A) (II) (i) (Q) (B) (II)(iii) (R)
(C) (II) (iii) (S) (D) (II) (i) (R)

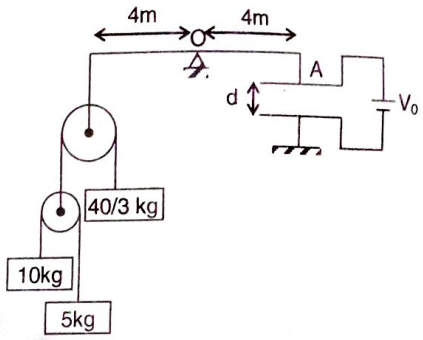
Sol. B

19. **For part (III) and (IV) the correct combination is**
(A) (IV) (iv) (S) (B) (IV)(iii) (S)
(C) (III) (iv) (R) (D) (III) (iv) (S)

Sol. C

Numerical Type

20. **A capacitance balance is shown in the figure. The balance has a block pulley system attached on one side and a capacitor that has a variable gap width on the other side. Assume upper plate of the capacitor has negligible mass. When the separation between the plate is $\frac{1}{\sqrt{5}}$ meter, the attractive force between the plates balances the pulley block system. Then find value of V_0 in volts (EMF of battery). (Take area of the plate = $\frac{20}{3\varepsilon_0}m^2$)**



Sol. 4

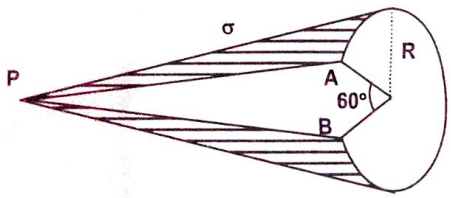
Tension in upper rope $= \frac{800}{3}$ N.

By balancing Torque about 'O'

$$\frac{800}{3} = \frac{V_0^2\varepsilon_0 A}{2d^2} = \frac{V_0^2\varepsilon_0 20}{2\left(\frac{1}{5}\right)3\varepsilon_0}$$

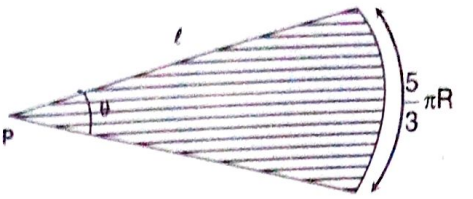
$V_0 = 4$ Volts

21. **A non-conducting hollow cone has charge density σ. A part ABP is cut and removed from the cone. The potential due to the remaining portion of the cone at point 'P' is $\frac{5}{2n}\frac{\sigma R}{\varepsilon_0}$. Find n.**



Ans. 6

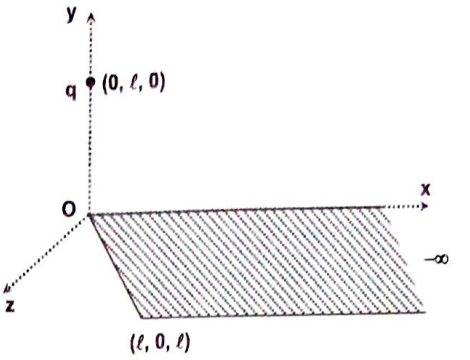
Sol. Potential due to circular non conducting disc, $V_c = \frac{\sigma r}{2\varepsilon_0}$ (where r is radius of disc)



So, $V_P = \left(\frac{\theta}{2\pi}\right)\left(\frac{\sigma \cdot \ell}{2\varepsilon_0}\right)$

$= \left(\frac{\sigma \ell}{2\varepsilon_0}\right)\left(\frac{5\pi R}{3(\ell)}\right)\left(\frac{1}{2\pi}\right) = \frac{5\sigma R}{12\varepsilon_0}$

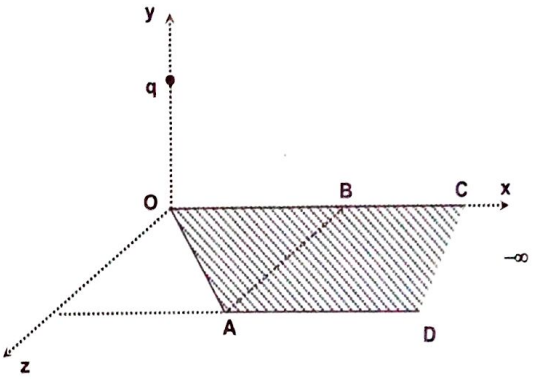
22. **A point charge 'q' is placed at a distance 'ℓ' (on the y-axis) vertically above the surface which lies in the xz plane as shown in the figure. The flux of electric field passing through the plate is $\frac{q}{3n\varepsilon_0}$. Find n.**



Sol. 8

$\phi = \phi_{OAB} + \phi_{ABCD}$

$= \frac{q}{48\varepsilon_0} + \frac{q}{48\varepsilon_0} = \frac{q}{24\varepsilon_0}$



Numerical Decimal Type

23. **The electrostatic potential existing in the space is given as $V = -\left(\frac{x^3}{6\varepsilon_0} + 2\right)$ volts. Find the charge density (in coulomb/m^3) at x = 2m.**

Sol. 00002.00

$E = -\frac{\partial V}{\partial x} = \frac{3x^2}{6\varepsilon_0} = \frac{x^2}{2\varepsilon_0}$

$\phi = E.A. = \frac{x^2}{2\varepsilon_0} A$

$\frac{d\phi}{dx} = \frac{(xA)}{\varepsilon_0} \Rightarrow d\phi = \frac{(xAdx)}{\varepsilon_0}$

$$\rho(x) = \frac{dq}{Adx} = \frac{(xAdx)}{(Adx)} = x$$

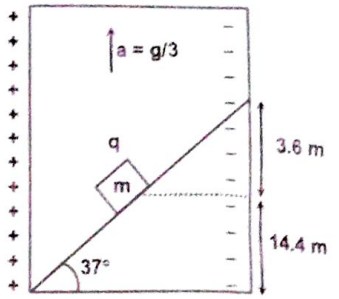
$\rho(x = 2) = 2$

Alternating Method

$$\nabla\vec{E} = \frac{\rho}{\varepsilon_0} \Rightarrow \frac{x}{\varepsilon_0} = \frac{\rho}{\varepsilon_0} \Rightarrow \rho = x$$

If $x = 2 \Rightarrow \rho = 2$

24. **A small block of mass 'm' is kept on the smooth inclined plane of angle 37°, placed in an elevator going upward with acceleration $a = \frac{g}{3}$, as shown in the figure. A horizontal electric field E perpendicular to the left and right vertical wall of the elevator exists. The charge on the block is +q. Find the time (in sec) when the block collide with the vertical wall. (take $g = 10\ m/s^2$ and $\frac{qE}{m} = 25\ m/s^2$)**



Sol. 00001.00

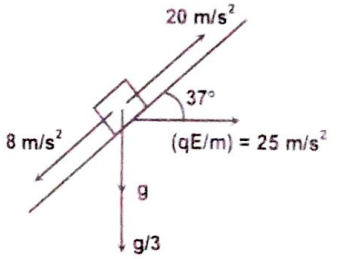
F.B.D. of block

$a_{bel} = 12\ m/s^2$, up the incline plane

$$S = ut + \frac{1}{2}gt^2$$

$$3.6 \times \frac{5}{3} = \frac{1}{2} \times 12 \times t^2$$

$\Rightarrow t = 1$ sec



CURRENT ELECTRICITY 12

(1) Electric Current

$$i = \frac{dQ}{dt}$$ = rate of charge passing through an area (unit is ampere)

(2) Electric Current Density

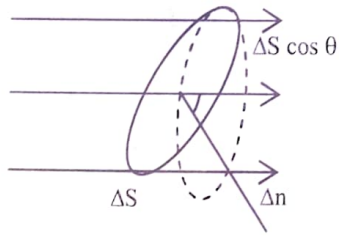
$$j = \frac{\Delta i}{\Delta s \cos\theta}$$

$$\Delta I = j\, \Delta s \cos\theta$$

Let $\Delta\vec{s}$ be the area vector corresponding to the area ΔS, then

$$\Delta I = \hat{j} \cdot \Delta\vec{s}$$

For a finite area: $i = \int \vec{j} \cdot d\vec{s}$



(3) Relation of Drift Speed (V_d) with Electric Current Density

$$J = nv_de$$

where n = number of electrons per unit volume

e = charge on an electron.

(4) Ohm's Law

$V = IR$, where $R = \frac{\ell}{\sigma A}$

σ = electrical conductivity, $ohm^{-1}\, m^{-1}$

ℓ = length of conductor

A = cross-sectional area

E = electric field inside a conductor

R = residence, ohm

Temperature Dependence of Resistivity

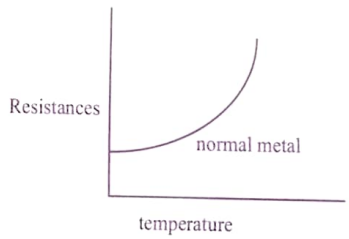
$$\rho(T) = \rho(T_0)$$

$$\rho(T) = \rho(T_0)\,[1 + \alpha[T - T_0]\,]$$

$$R \propto \rho$$

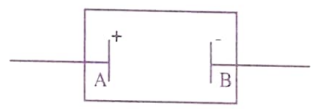
$$\therefore \quad R(T) = R(T_0)\left[1 + \alpha(T - T_0)\right]$$

where α = temperature coefficient of resistivity.



BATTERY

A battery is a device which maintains a potential difference between its points A and B.



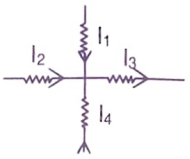
The emf of a battery equals the potential difference between the terminals, when the terminals are not connected externally.

When positive charges enter the battery from the positive terminal and negative charges enter the battery from the negative terminal, the process is referred to as the charging of the battery. The reverse of this process is called discharging of the battery.

Battery Symbols

In an ideal battery emf is equal to terminal voltage. In a real battery emf is not equal to terminal voltage but lesser than it due to internal resistance of the battery.

KIRCHOFF'S LAW

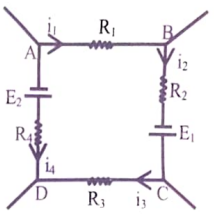


(1) **KCL:** The sum of all the currents directed towards a point is equal to sum of all the currents directed away form the point.

$I_3 = i_1 + i_4 + i_2$

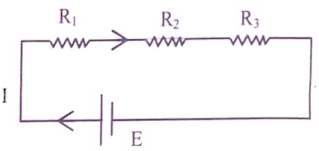
(2) **KVL:** The algebraic sum of all the potential difference along a closed loop in a circuit is zero.

$E_2 + i_1R_1 + i_2R_2 - E_1 + i_3R_3 - i_4R_4 = 0$



COMBINATION OF RESISTORS

Series Combination of Resistors



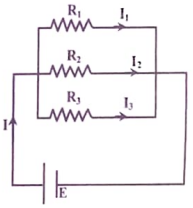
Two or more resistors are said to be connected in series if the same current passes through the entire resistor.

$E - IR_1 - IR_2 - IR_3 = 0$

$\therefore E = I(R_1 + R_2 + R_3) = IR$

$\therefore R = R_1 + R_2 + R_3$

Parallel Combination of Resistors



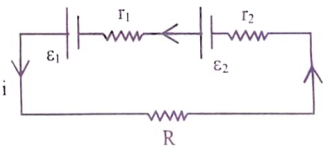
$I = I_1 + I_2 + I_3$

$E = I_1R_1 = I_2R_2 = I_3R_3$

$\therefore \quad \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

GROUPING OF BATTERIES

Series Connection

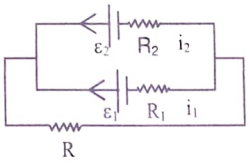


$$i = \frac{\varepsilon_1 + \varepsilon_2}{R + (r_1 + r_2)}$$

The combination acts as a battery of emf $\varepsilon_0 = \varepsilon_1 + \varepsilon_2$, having an internal resistance $r_0 = r_1 + r_2$.

If the polarity of one of the batteries is reversed, the emf will be $|\varepsilon_1 - \varepsilon_2|$.

Parallel Connection



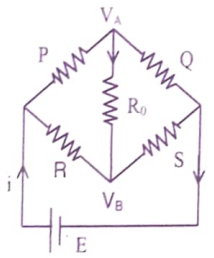
Equivalent emf $\varepsilon_0 = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$,

Equivalent internal resistance $r_0 = \frac{r_1 r_2}{r_1 + r_2}$

Current in resistance R, $i = \frac{\varepsilon_0}{r_0 + R}$

WHEATSTONE BRIDGE

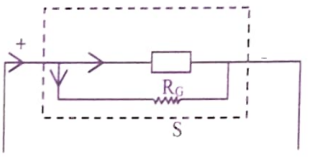
In this arrangement of resistors (as shown), no current flows through R_0, i.e. $v_A = v_B$, if $\frac{P}{Q} = \frac{R}{S}$.



Ammeter

In an ammeter, a resistor having a small resistance is connected in parallel with the coil of galvanometer. This resistor is called the shunt.

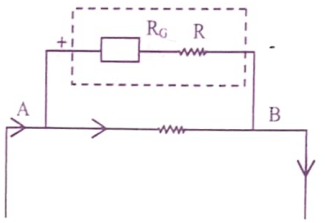
$$R_{eq} = \frac{R_G . S}{R_G + S}$$



Voltmeter

In a voltmeter a resistance having a high resistance is connected in series with the coil of galvanometer.

$$R_{eq} = R_G + R$$



Potentiometer

A device which does not draw any current from the given circuit and still measures the potential difference. Thus, it is equivalent to an ideal voltmeter.

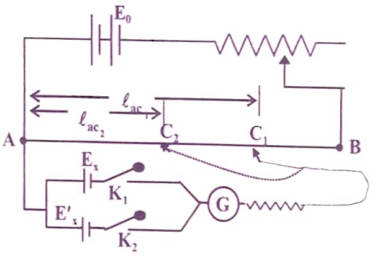
Let $AP = x$ and $AB = L$

Then, $V_{ab} = \frac{\ell}{L} E_0$

The potentiometer can be used to measure

(i) Internal resistance of a battery.

(ii) To compare emf's of two batteries.



CHARGING AND DISCHARGING OF CAPACITORS

$$\frac{q}{C} + R - E = 0$$

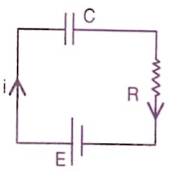
This differential equation on solving gives,

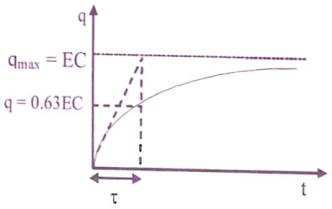
$$q = CE\left[1 - e^{-t/RC}\right]$$

where CE = maximum charge on the capacitor

$t_0 = CR$ = time constant

In one time constant, 63 % of the charge is deposited.



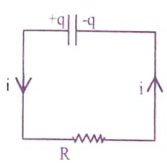


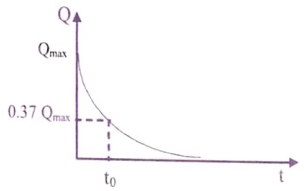
Discharging of a capacitor

$$-\frac{q}{C} = R\frac{dq}{dt},$$

On solving, we get $q = Q_{max}\, e^{-t/CR}$

At time $t = t_0$, 63 % of the charge is discharged.





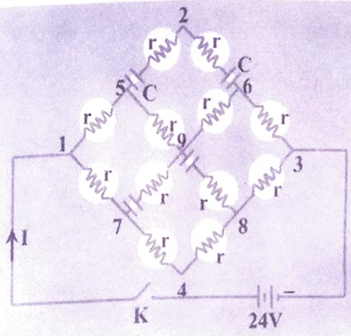
MEGACOSM
COGNITIONS PRIVATE LIMITED

ILLUSTRATIVE PROBLEMS

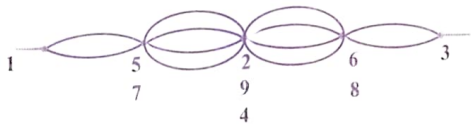
1. In the circuit shown, r = 4 Ω, C = 2 μF.

(a) Find the current coming out of the battery just after the switch is closed.

(b) Find charge on each capacitor in the steady state condition.



Sol. (a) Due to symmetry the points of equal potential are joined together , and the circuit may be reduced as

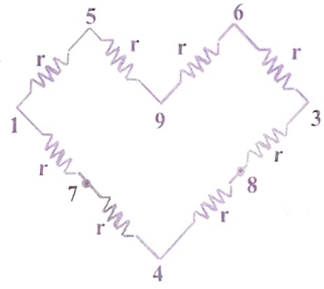


$$R_{13} = \frac{r}{2}+\frac{r}{4}+\frac{r}{4}+\frac{r}{2}=\frac{3}{2}r$$

Here, $r = 4\Omega$, $R_{13} = \frac{3}{2}(4) = 6\ \Omega$.

Thus, $I = \frac{24}{R_{13}} = \frac{24}{6} = 4\ A$

(b) In the steady-state condition, the circuit may be reduced as



$R_{13} = 2r = 2(4)\ \Omega = 8\ \Omega$

$I = \frac{24}{8} = 3\ A$

$V_{56} = (2r)\frac{I}{2} = Ir = (3)(4) = 12\ V$

Equivalent capacitance between 5 and 6 is

$C_{56} = \frac{C}{2} = 1\ \mu F$

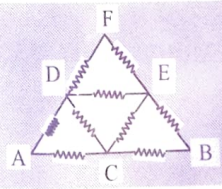
$\therefore\ q_{56} = C_{56}V_{56} = 12\ \mu C$

Now, $V_{97} = V_{17} - V_{19} = -r\frac{I}{2} + 2\frac{rI}{2} = \frac{rI}{2} = \frac{(4)(3)}{2} = 6\ V$

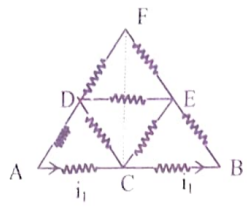
$\therefore\ q_{97} = CV_{97} = (2)(6) = 12\ \mu C$

Similarly, $q_{89} = 12\ \mu C$

2. Find equivalent resistance between points A and B in the circuit shown. All resistances are equal to R.

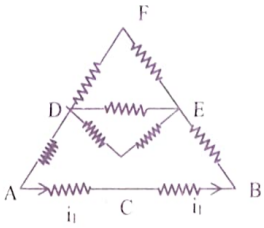


Sol. We will solve this using symmetry about the dotted line. From this we can deduce that the current in a branch will be equal to the current in its mirror image. For example current in AC = current in CB = i_1.



This implies that even if we break the junction at C, the currents in the circuit will not be affected. Now, the circuit is simplified which can be easily solved using series parallel combination.

R_{AB} = 2R in parallel to $\left(2R + \frac{R}{2}\right) = \frac{10}{9}R$



3. A galvanometer (coil resistance 99 Ω) is converted into an ammeter using a shunt of 1 Ω and connected as shown in Fig. 1. The ammeter reads 3 A. The same galvanometer is converted into a voltmeter by connecting a resistance of 101 Ω in series. This voltmeter is connected as shown in Fig. 2. Its reading is found to be 4/5 of the full scale reading. Find
(i) internal resistance of the cell (r)
(ii) range of the ammeter and voltmeter
(iii) full scale deflection current of the galvanometer

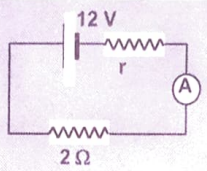


Fig. 1

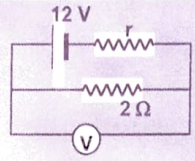


Fig. 2

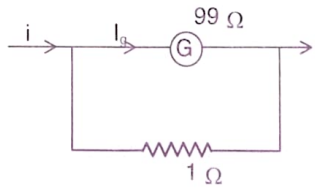
Sol. For ammeter

$99\ I_g = (I - I_g)\ 1$

or $I = 100\ I_g$ ... (1)

I_g is the full scale deflection current of the galvanometer and I is the range of ammeter

For the circuit in the adjacent figure,



$$\frac{12}{2 + r + \frac{99 \times 1}{99 + 1}} = 3\ A$$

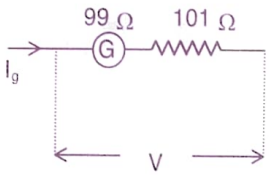
$\Rightarrow$ r = 1.01 Ω.

For voltmeter, range

$V = I_g\ (99 + 101) = 200\ I_g$ ... (2)

Also resistance of the voltmeter = 99 + 101 = 200 Ω

In the adjacent figure, resistance across the terminals of the battery



$$R_1 = r + \frac{200 \times 2}{202} = 2.99\ \Omega$$

$\therefore$ Current drawn from the battery, $I_1 = \frac{12}{2.99} = 4.01\ A$

$\therefore$ Voltmeter reading

$$\frac{4}{5}V = 12 - I_1 r = 12 - 4.01 \times 1.01$$

$$V = 7.96 \times \frac{5}{4} = 9.95\ V$$

Using Eq. (2), $I_g = \frac{9.95}{200} = 0.05\ A$

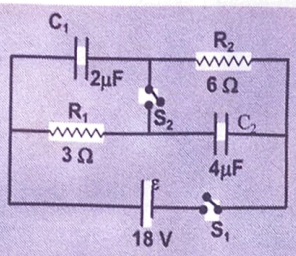
Using Eq. (1), range of the ammeter

$I = 100\ I_g = 5\ A.$

4. Consider the shown network. Initially both switches are open and the capacitor are uncharged. Both the switches are then closed simultaneously at t = 0.

(a) Obtain the current through switch S_2 as function of time.

(b) If switch S_2 is opened again after a long time interval, find the total heat that would dissipate in the resistor and the charge that would flow through switch S_1 after S_2 is opened?



Sol. Consider the charges on capacitors and currents through various branches, as shown in the figure

For loop 1, we have

$$R_1(i_2 + i_3 - i_1) = \frac{q_1}{C_1} \qquad (1)$$

For loop 2, $R_2 i_3 = \frac{q_2}{C_2}$ (2)

For the outer loop, $i_3 R_2 + \frac{q_1}{C_1} = \varepsilon$ (3)

Also, $i_1 = \frac{dq_1}{dt}$ and $i_2 = \frac{dq_2}{dt}$ (4)

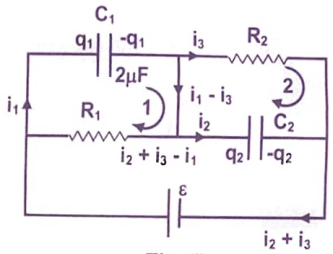


Fig. (i)

Putting the values of i_1 and i_2 from Eq. (4) and of i_3 from Eq. (2) in (1), we get

$$\frac{d}{dt}(q_2 - q_1) = \frac{q_1}{R_1 C_1} - \frac{q_2}{R_2 C_2} \qquad (5)$$

From Eqs. (2) and (3), we get

$$\frac{q_2}{C_2} + \frac{q_1}{C_1} = \varepsilon \qquad (6)$$

From Eqs. (5) and (6), we get

$$\int_0^{q_2} \frac{dq_2}{\left(\frac{\varepsilon C_2 R_2}{R_1 + R_2}\right) - q_2} = \int_0^t \frac{dt}{R_{eq}(C_1 + C_2)}, \text{ where } R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$\Rightarrow q_2 = \frac{\varepsilon R_2 C_2}{(R_1 + R_2)}\left[1 - e^{\frac{-t}{R_{eq}(C_1 + C_2)}}\right]$$

Similarly, $q_1 = \frac{\varepsilon R_1 C_1}{(R_1 + R_2)}\left(1 - e^{-\frac{t}{R_{eq}(C_1 + C_2)}}\right)$

$$\Rightarrow i_1 = \frac{dq_1}{dt} = \frac{\varepsilon C_1}{R_2(C_1 + C_2)} e^{-\frac{t}{R_{eq}(C_1 + C_2)}}$$

Similarly, $i_2 = \frac{\varepsilon C_2}{R_1(C_1 + C_2)} e^{\frac{-t}{R_{eq}(C_1 + C_2)}}$

From equation (i)

Current through $S_2 = (i_1 - i_3) = i_2 - \frac{q_1}{R_1 C_1}$

Putting the values, we get

$q_1 = (12\ \mu C)(1 - e^{-\frac{t}{12\mu}})$; $q_2 = (48\ \mu C)(1 - e^{-\frac{t}{12\mu}})$

$i_1 = (1\ A)e^{-\frac{t}{12\mu}}$; $i_2 = (4\ A)\ e^{-\frac{t}{12\mu}}$

Current through switch $S_2 = -[2 - e^{-\frac{t}{12\mu}}]$ A along the indicated direction as shown in fig (i),

With both the switches closed the steady state charges and currents are as shown in Fig. (ii).

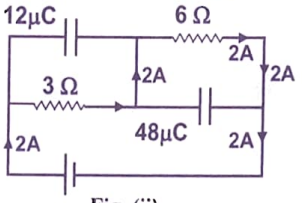


Fig. (ii)

With switch S_2 open and S_1 closed, the steady state charges are as shown in Fig. (iii).

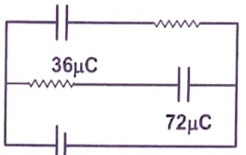


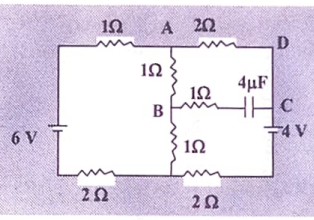
Fig. (iii)

Hence, the charge flown through switch $S_1 = [(36 + 72) - (12 + 48)]\mu C = 48\mu C$.

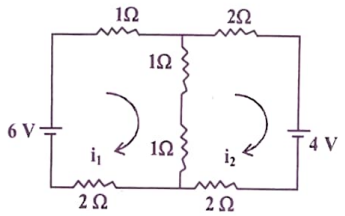
Total heat dissipated in the resistors = [initial energy + work done by battery when 48 μC flows through it after switch S_2 is opened] – [final energy]

$$= \left\{\frac{1}{2}C_1V_1^2 + \frac{1}{2}C_2V_2^2\right\} + \varepsilon(\Delta Q) - \left\{\frac{1}{2}C_1V_1'^2 + \frac{1}{2}C_2V_2'^2\right\} = 136\mu J.$$

5. Find the charge on capacitor in steady state.



Sol. In steady state the equivalent circuit is



$5i_1 - 2i_2 = 6$

$6i_2 - 2i_1 = -4$

$13i_2 = 14$

$i_1 = \frac{14}{13}$ Amp, $i_2 = -\frac{4}{13}$ Amp

∴ Potential difference across BC

$V_A = 6 - \frac{14}{13} = \frac{64}{13}$ V

$V_D = \frac{64}{13} - 2(-4/13) = \frac{72}{13}$ V

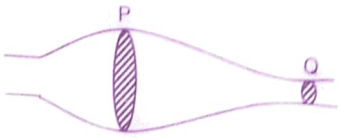
$V_B = \frac{64}{13} - 1 \times \left\{\frac{14}{13} + \frac{4}{13}\right\} = \frac{46}{13}$ V

∴ $V_{DB} = \frac{72-46}{13} = \frac{26}{13} = 2$ volt

∴ Charge on capacitor = $CV_{BD} = 4 \times 10^{-6} \times 2 = 8\ \mu C$

SOLVED PROBLEMS

1. A source of constant potential difference is connected across a conductor having irregular cross section as shown.



(A) Electric field intensity at P is greater than that at Q.

(B) Rate of electrons crossing per unit area of cross-section at P is less than that at Q.

(C) The rate of generation of heat per unit length at P is greater than at Q.

(D) Mean kinetic energy of free electrons at P is greater than that at Q.

Sol. When a source is connected, a current starts to flow through the conductor. Let it be I. Then current density at a section is equal to I/A where A = cross-sectional area.

Since cross-sectional area at P is maximum, therefore current density at P is minimum. Hence (B) is correct.

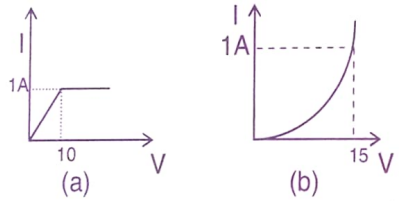
Since electric field is $\frac{J'}{\sigma}\left(=\frac{I}{A\sigma}=\frac{I\rho}{A}\right)$, therefore at P electric field is minimum while that at Q is maximum.

Rate of generation of heat per unit length at a section will be equal to $\frac{I^2\rho}{A}$. It is minimum at P and maximum at Q.

The mean kinetic energy of free electrons $=\frac{1}{2}mv_d^2$ which is minimum at P and maximum at Q.

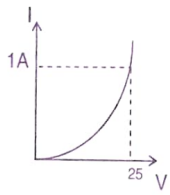
$\therefore$ (B)

2. Current–voltage characteristics of two elements A and B are as shown below in Figs. (a) and (b).

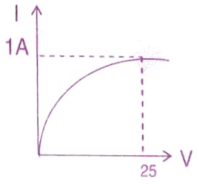


Which of the following graphs represents current voltage characteristics for their series combination?

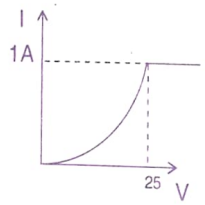
(A)



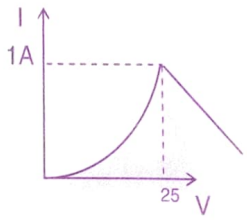
(B)



(C)



(D)



Sol. In element A, the resistance remains constant upto the potential drop of 10 V. Further increase in the voltage does not increase this current (which is constant at 1A). This means that the ratio V/R_A = constant and this resistance R_A increases linearly with voltage.

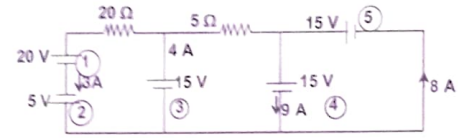
In element B, the resistance decreases gradually upto 15 V and afterwards the resistance R_B increases linearly with voltage.

When both A and B are in series, the current in the circuit will increase non linearly upto 1A when the total voltage drop across A and B becomes 10 + 15 = 25 V.

Further increase in this voltage does not bring about any change in the current as shown in solution (C). The voltage drop across A will go on increasing while that across B remain fixed at 15 V.

$\therefore$ (C)

3. In the given network the batteries getting charged are



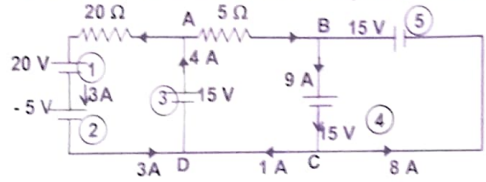
(A) 1 and 3

(B) 1 3 and 5

(C) 1 and 4

(D) 1, 2 and 5

Sol. Applying Kirchhoff law at A, C, D the direction of the currents in each branch will be as shown in the figure. Now it is clear from the figure the batteries 1 and 4 getting charged.



$\therefore$ (C)

4. When a galvanometer is shunted with a 4 Ω resistance the deflection is reduced to 1/5. If the galvanometer is further shunted with a 2 Ω wire the new deflection will be (assuming the main current remains the same)

(A) 5/13 of the deflection when shunted with 4 Ω only

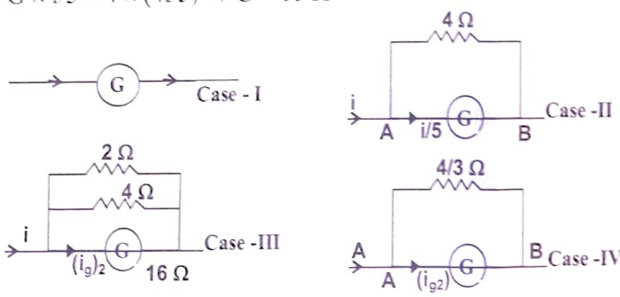
(B) 8/13 of the deflection when shunted with 4 Ω only

(C) 3/4 of the deflection when shunted with 4 Ω only

(D) 3/13 of the deflection when shunted with 4 Ω only.

Sol. When only 4 Ω resistance is shunted, $(i_g)_1 = i/5$

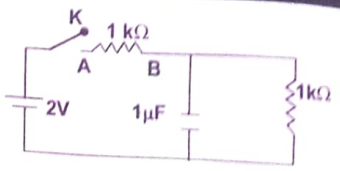
$G \times i/5 = 4 \times (4i/5) \Rightarrow G = 16\ \Omega$



$$(i_g)_2 = \frac{4/3}{16 + 4/3} i = \frac{1}{13} i$$

$\therefore$ (A)

5. **When the key k is pressed at time t = 0, which of the following statements about the current I, in the resistor AB of the given circuit is true.**



(A) 2 mA at all time

(B) oscillates between 1 mA and 2 mA

(C) 1 mA at all time

(D) At t = 0, I = 2 mA and with time it finally reduces to 1 mA

Sol. Initially the capacitance acts as short circuited and at steady state it acts as open circuit.

Hence, at t = 0, $i = \frac{2}{1\times10^{3}} = 2$ mA

At steady state $i = \frac{2}{2\times10^{3}} = 1$ mA

$\therefore$ (D)

6. **Two electric bulbs rated P_1 and P_2 watt at V volt are connected in series across V volt mains then their total power consumption P is**

(A) $(P_1 + P_2)$

(B) $\sqrt{P_1P_2}$

(C) $P_1 P_2/(P_1 + P_2)$

(D) $(P_1 + P_2)/P_1P_2$

Sol. Resistance of Ist = bulb $R_1 = \frac{V^2}{P_1}$

Resistance of IInd bulb $R_2 = \frac{V^2}{P_2}$

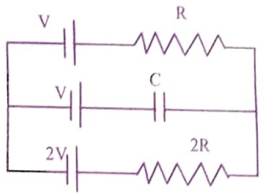
When both bulbs are connected in series

$$R_{eq.} = V^2\left[\frac{1}{P_1}+\frac{1}{P_2}\right] = \frac{V^2(P_1+P_2)}{P_1P_2}$$

Hence power consumed $P = \frac{V^2}{R} = \frac{V^2}{V^2\left(\frac{P_1+P_2}{P_1P_2}\right)} = \frac{P_1P_2}{P_1+P_2}$

$\therefore$(C)

7. **In the given circuit, with steady current, the potential drop across the capacitor must be**



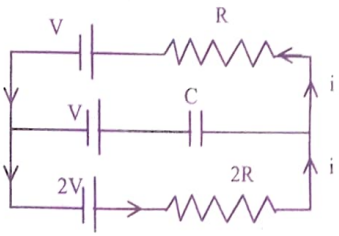
(A) V

(B) V/2

(C) V/3

(D) 2V/3

Sol. In steady state no current will pass through the capacitor.



In the outer loop,

$2V - 2iR - iR - V = 0$

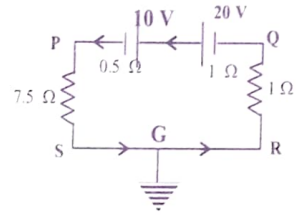
$\Rightarrow i = V/3R$

For the upper loop, $V - V_C - iR - V = 0$

$\Rightarrow |V_C| = iR = V/3$

$\therefore$ (C)

8. **In the circuit shown in the adjacent figure**
   (A) the potential at P is −7.5 V
   (B) the potential at Q is −1V
   (C) the potential at R is zero
   (D) the potential at S is zero



Sol. The current in the circuit $I = \frac{(20-10)}{(7.5+.5+1+1)} = \frac{10}{10} = 1$ Amp

The potential difference across PS $V_{PS} = 7.5 \times 1 = 7.5$ V

The potential difference across Q R, $V_{QR} = 1 \times 1 = 1$ V

As point G is connected to earth hence potentials of R and S is zero.
The direction of the current in the circuit is from P to S, hence point P is at higher potential.
$\therefore V_p = 7.5$ V
similarly $V_Q = -1$V
$\therefore$ (B), (C) and (D).

9. **The electron in a hydrogen atom moves in circular orbit of radius 5×10^{-11} m with a speed of $0.6\pi \times 10^6$ m/s then**
   (A) the frequency of the electron is 6×10^{15} rev/s
   (B) the electron carries -1.6×10^{-19} C around the loop
   (C) the current in the orbit is 0.96 mA
   (D) the current flow is in the opposite direction to the direction of the motion of electron

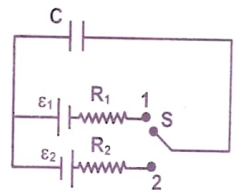
Sol. Electric current at a point on the circle

$i = fe$, where f = frequency $= \frac{\omega}{2\pi} = \frac{v}{2\pi r} = \frac{0.6\times10^6\times\pi}{2\pi\times5\times10^{-11}} = 6 \times 10^{15}$ rev/s

$i = 6 \times 10^{15} \times 1.6 \times 10^{-19} = 0.96 \times 10^{-3}$ A = 0.96 mA

$\therefore$ (A), (B), (C) and (D)

10. **An uncharged capacitor C, Resistances $R_1 = R$ and $R_2 = 2R$, cells of emf $\varepsilon_1 = \varepsilon$ and $\varepsilon_2 = \varepsilon$ are connected as shown in the figure. Switch S is initially connected to terminal 1 for a long time and then it is connected to terminal 2 for an another long duration. IF H_1, U_1 are the heat energies lost through the resistor R_1 and energy stored in the capacitor respectively at steady state, when the switch is connected to terminal 1 and H_2, U_2 are corresponding value when the switch is connected to terminal 2. Choose the correct statement(s).**

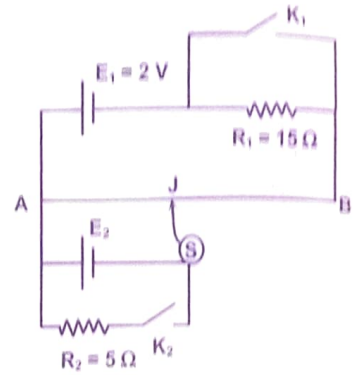


   (A) $U_1/U_2 = 1$
   (B) $H_1/H_2 = ¼$
   (C) work done by the cells during the time interval when they are active, is positive.
   (D) None of the above.

Sol. A, B, C
From work energy theorem
$W_{cell} = \Delta H + U_f - U_i$, where $U_f = U_i$

COMPREHENSION

In the circuit shown AB is a 10 Ω uniform slide wire 50 cm long. E_1 is 2 V accumulator of negligible internal resistance R_1 and R_2 are 15 Ω and 5 Ω respectively. When k_1 and k_2 are both open, the galvanometer shows no deflection when AJ = 31.25. When k_1 and k_2 are both closed the balance length AJ = 5 cm



11. The emf of the cell E_2

(A) 0.5 V **(B) 1 V**

(C) 1.5 V **(D) 2 V**

Sol. A

$$E_2 = E_1\left(\frac{R_{AB}}{R_1 + R_{AB}}\right)\frac{\ell}{L}$$

12. The internal resistance of the cell E_2

(A) 7.5 Ω **(B) 8 Ω**

(C) 10 Ω **(D) 2 Ω**

Sol. A

$$r = R_2\left(\frac{E_2 L}{E_1 \ell} - 1\right)$$

13. The balance length AJ when k_2 is open and k_1 is closed.

(A) 12.5 cm **(B) 13.5 cm**

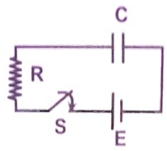
(C) 14.5 cm **(D) 15.5 cm**

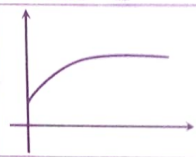
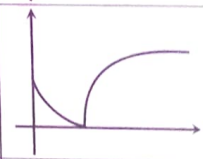
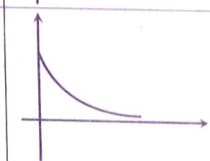
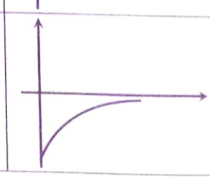
Sol. A

$$\ell' = \frac{E_2 L}{E_1}$$

Matrix-Match Type

14. A charged parallel plate capacitor is connected to a cell of emf ε through a resistor as shown in the figure. Then choose the possible curves in column – II for the physical quantities given in column – I.

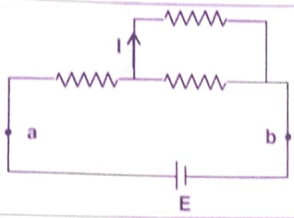
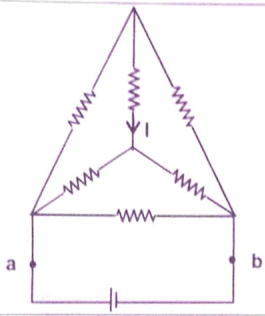
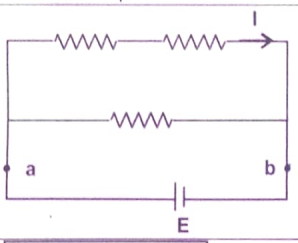
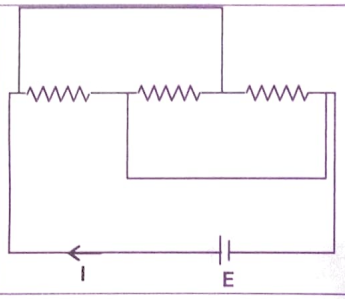


	Column – I		Column – II
(A)	Charge (q) on capacitor (vertical axis) vs time (t) (horizontal axis.)	(p)	
(B)	Current (I) through the resistor (vertical axis) vs time (t) (horizontal axis.)	(q)	
(C)	Electrical energy (U) stored in the capacitor (vertical axis) vs time (t) (horizontal axis).	(r)	
(D)	Rate of heat generation (P) through the resistor (vertical axis) vs time (t) (horizontal axis).	(s)	

Sol. $A \rightarrow (p, q), B \rightarrow (r, s), C \rightarrow (p, q), D \rightarrow (r)$

Answer questions 15, 16 and 17 by appropriately matching the information given in the three columns of the following table.

Column –I shows four different resistor networks connected to a battery of emf ε. Each resistor shown is same and has a resistance R. Column –II and column –III respectively list the value of equivalent resistance (across terminals 'a' and 'b') and the value of current I (shown in figure)

Column 1		Column 2		Column 3	
(I)		(i)	R/2	(P)	Zero
(II)		(ii)	R/3	(Q)	E/2R
(III)		(iii)	2R/3	(R)	3E/R
(IV)		(iv)	3R/2	(S)	E/3R

15. **Pick the correct option.**

(A) (I) (iii) (P) **(B) (II) (ii) (Q)**
(C) (III) (iv) (S) **(D) (IV) (ii) (R)**

Sol. D

16. **Pick correct option for maximum value of equivalent resistance.**

(A) (IV) (ii) (R) **(B) (III)(iv) (S)**
(C) (II) (ii) (Q) **(D) (I) (iv) (S)**

Sol. D

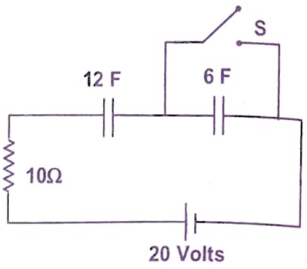
17. **Pick correct option for minimum value of current I.**

(A) (I) (iii) (P)
(B) (II) (i) (P)
(C) (III) (iv) (S)
(D) (IV) (iii) (R)

Sol. B

Numerical Type

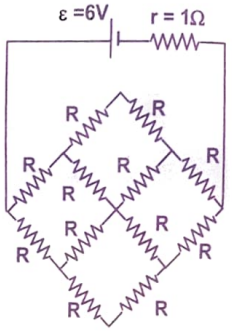
18. **When steady state is reached, switch S is opened. Then the amount of heat rejected by the resistor is 800 K Joule. Then find the value of K.**



Sol. 0

There is no transfer of charge through the battery

19. **In the circuit shown the cell will deliver maximum power to the network if the value of 'R' is $\frac{2}{n}\Omega$. Find n.**



Sol. 3

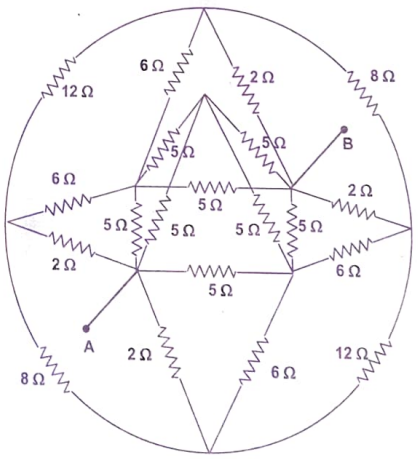
The cell will deliver maximum power to the network if $R_{eq} = r$

$$\Rightarrow \frac{3R}{2} = r$$

$$\therefore R = \frac{2r}{3} = \frac{2}{3}\Omega$$

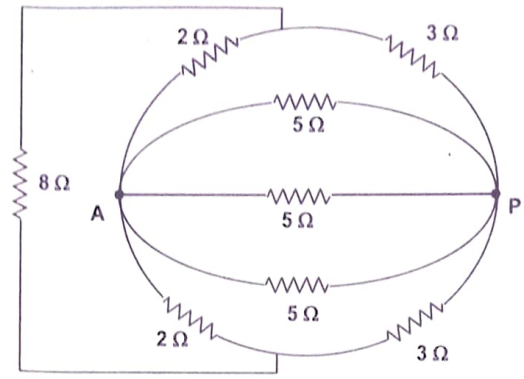
Numerical Decimal Type

20. **Determine the equivalent resistance between point A and B in Ω, considering that connecting wires have no resistance.**

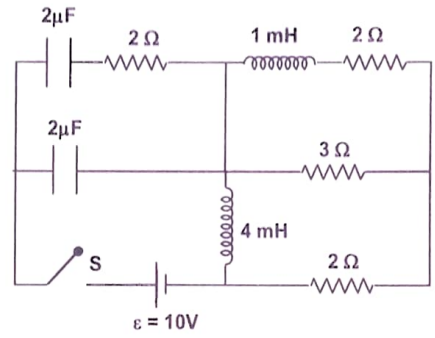


Sol. 00002.00

$R_{AB} = 1\ \Omega$ so $R_{AB} = 1 + 1 = 2\ \Omega$



21. **In the circuit shown, all the capacitors are initially uncharged. Find the current through the battery just after closing the switch 'S'.**

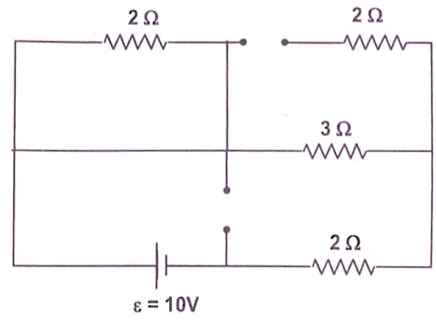


Sol. 00002.00

The equivalent circuit just after closing the switch 'S' is

$R_{eq} = 3 + 2 = 5\ \Omega$

$\therefore\ I_0 = \dfrac{\varepsilon}{R_{eq}} = \dfrac{10}{5} = 2A$



MAGNETISM 13

MAGNETIC FIELD

Force on a particle of charge q, having a velocity $\vec{v}$ present in a magnetic field $\vec{B}$ is given by

$\vec{F} = q\vec{v} \times \vec{B}$. The unit of magnetic field is tesla and is denoted by T.

Motion of a Charged Particle in a Uniform Magnetic Field

As the magnetic force on a particle is perpendicular to the velocity, it does not do any work on the particle. Hence, the kinetic energy or the speed of the particle does not change due to magnetic force.

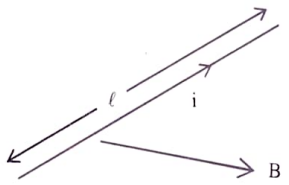
The frequency of revolution of a charge q, having mass m, in a magnetic field B is given by

$$f = \frac{qB}{2\pi m}$$

This frequency is called the cyclotron frequency.

Magnetic Force on a Current Carrying Wire

The length of the conductor, ℓ, is treated as a vector $\vec{\ell}$, with its direction in coinciding with the direction of current. Note that the entire length of the conductor should be along a straight line. Otherwise, we take a small length $d\ell$ of the conductor and treat it as a vector $d\vec{\ell}$.



The force on a conductor carrying current i is given by

$$d\vec{F} = id\vec{\ell} \times \vec{B}$$

and $\vec{F} = i\vec{\ell} \times \vec{B}$

For a current carrying loop having n turns and area A, with a current i, the magnitude of electric dipole moment ($\vec{\mu}$) is defined as

$$|\vec{\mu}| = niA$$

The direction of the dipole moment is taken perpendicular to the plane of the loop and is positive towards the direction from which the current looks anticlockwise.

Torque on a current carrying loop in a magnetic field is given by: $\vec{T} = \vec{\mu} \times \vec{B}$

BIOT–SAVART LAW

Magnetic Field due to a Current Carrying Conductor

Biot–Savart law gives $d\vec{B} = \frac{1}{4\pi\varepsilon_0 c^2} i \frac{d\vec{\ell} \times d\vec{r}}{r^3}$

We define,

$$\frac{1}{\varepsilon_0 c^2} = \mu_0 \text{ (permeability of vacuum)}$$

where $\mu_0 = 4\pi \times 10^{-7}$ T-m/A. The magnitude of the field is given by

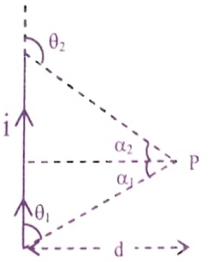
$$dB = \frac{\mu_0}{4\pi} \cdot \frac{id\ell \sin\theta}{r^2}$$

Magnetic Field Due to Current in a Straight Wire

$$dB = \frac{\mu_0 i}{4\pi d}[\cos\theta_1 - \cos\theta_2]$$

$$= \frac{\mu_0 i}{4\pi d}[\sin\alpha_1 + \sin\alpha_2]$$

The direction of field is given by right hand thumb rule.



Force Between Parallel Current Carrying Wires

The force per unit length acting on any wire is given by

$$\frac{dF}{d\ell} = \frac{\mu_0 i_1 i_2}{2\pi d}$$

The force is attractive in nature, if the wires carries current in the same direction.

Field Due to a Circular Current Carrying Loop

$$B = \frac{\mu_0 i a^2}{2(a^2 + x^2)^{3/2}}$$

Field at the centre of loop is given by

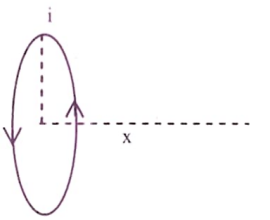
$$B = \frac{\mu_0 i}{2a}$$

If 'a' is small,

$$B = \frac{\mu_0 i a^2}{2x^3} = \frac{\mu_0 i \ 2\pi a^2}{4\pi x^3}$$

or, $\vec{B} = \frac{\mu_0}{4\pi}.\frac{2\vec{\mu}}{x^3}$

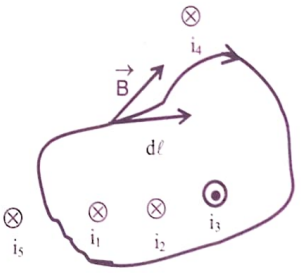
$\vec{\mu} = i\pi a^2 \hat{n}$ = magnetic dipole moment of the loop.



AMPERE'S LAW

$$i_{enclosed} = i_1 + i_2 - i_3$$

$$\oint \vec{B}.d\vec{\ell} = \mu_0 i_{enclosed}$$

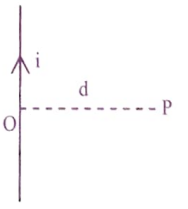


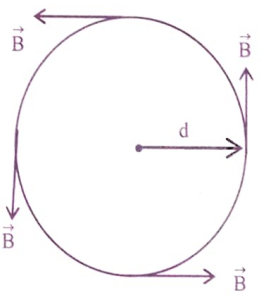
Magnetic Field

$$\int \vec{B}.d\vec{\ell} = \mu_0 i_{enclosed}$$

$$\int \vec{B}.d\vec{\ell} = B2\pi d = \mu_0 i$$

$$\therefore \quad B = \frac{\mu_0 i}{2\pi d}$$





SOLENOID

A solenoid is a wire wound closely in the form of a helix. The wire is coated with an insulating material so that they are electrically insulated, even though the adjacent turns physically touch each other. Generally, the length of the solenoid is large as compared to the transverse dimensions.

$$\int_P^Q \vec{B}.d\vec{\ell} = B\ell = \mu_0 ni\ell,$$

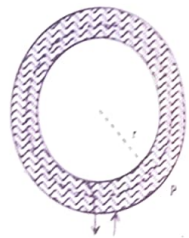
$\therefore$ $B = \mu_0 ni$, where n = number of turns per unit length of the solenoid.

Assumptions:

(i) The field inside the solenoid is uniform.

(ii) The field outside the solenoid is zero.

TOROID



If we take a non-conducting ring and wind a conducting wire closely on it, we get a toroid.

Draw a circle through a point P and concentric with the toroid. By symmetry, the field will have equal magnitudes at all points of the circle, and will be tangential to the circle.

$\therefore$ $\int \vec{B}.d\vec{\ell} = 2\pi rB = \mu_0 Ni$

$\therefore$ $B = \dfrac{\mu_0 Ni}{2\pi r}$

MAGNETIC CHARGE

(i) There are two types of magnetic charges, one positive which represents North pole and the other negative, which represents the south pole of a magnet, denoted by m.

(ii) Field due to a magnetic charge is given by:

$$B = \frac{\mu_0}{4\pi}\frac{m}{r^2}$$

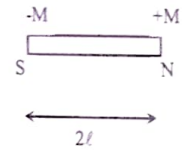
(iii) Potential due to a magnetic charge is given by

$$V = \frac{\mu_0}{4\pi}.\frac{m}{r}$$

(iv) The magnetic dipole moment $\vec{M}$ is defined as

$|\vec{M}| = 2m\ell$.

Direction of $\vec{M}$ is from south pole to north pole.



MAGNETIC SCALAR POTENTIAL AND FIELD

Potential at P is given by 'V', where V is equal to

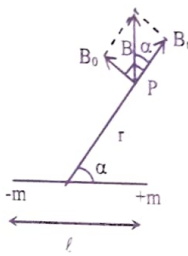
$$V = \frac{\mu_0}{4\pi}\frac{M\cos\theta}{r^2}$$

$$B_t = -\frac{dV}{dr} = \frac{\mu_0}{4\pi}.\frac{2M\cos\theta}{r^3}$$

$$B_\theta = -\frac{1}{r}\frac{dV}{d\theta} = \frac{\mu_0}{4\pi}.\frac{M\sin\theta}{r^3}$$

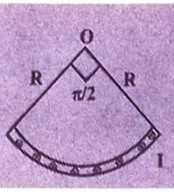
$$\tan\alpha = \frac{B}{B_t} = \frac{1}{2}\tan\theta$$

$$B = \frac{\mu_0}{4\pi}\frac{M}{r^3}\sqrt{1+3\cos^2\theta}$$



ILLUSTRATIVE PROBLEMS

1. A current I flows in a long straight wire (into the plane of the paper) with cross-section having the form of a circular arc of radius R as shown in the figure. Find the magnetic induction at the point O.

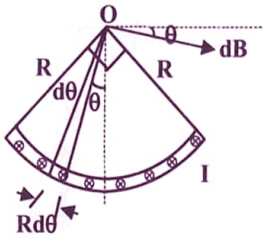


Sol. We have the result of magnetic field for a straight conductor. For a distribution, we have to make a differential equation and integrate.

Consider a small element of width R dθ as shown in the figure.

$$dB = \frac{\mu_0 (dI)}{2\pi R}$$

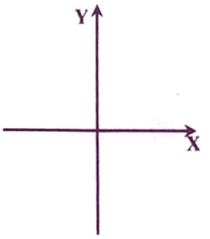
$$dI = \frac{I}{\frac{\pi R}{2}} R d\theta = \frac{2I}{\pi} d\theta \quad \therefore dB = \frac{\mu_0 I d\theta}{\pi^2 R}$$



As we have to integrate a vector quantity, we will integrate x and y components separately.

$$dB_x = dB \cos\theta = \frac{\mu_0 I}{\pi^2 R} \cos\theta \, d\theta$$

$$dB_y = dB \sin\theta = \frac{\mu_0 I}{\pi^2 R} \sin\theta \, d\theta$$

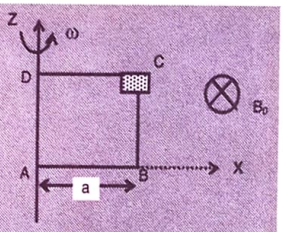


$$B_x = \int dB_x = \frac{\mu_0 I}{\pi^2 R} \int_{-\pi/4}^{+\pi/4} \cos\theta \, d\theta = \frac{\mu_0 I \sqrt{2}}{\pi^2 R}$$

$$B_y = \int dB_y = \frac{\mu_0 I}{\pi^2 R} \int_{-\pi/4}^{+\pi/4} \sin\theta \, d\theta = 0$$

$$\vec{B} = \frac{\mu_0 I \sqrt{2}}{\pi^2 R} \hat{i}$$

2. A insulated square frame ABCD of side a is able to rotate about one of its side taken as positive z-axis. A magnetic field B is present in the region given by $\vec{B} = B_0 \hat{k}$. A small block of mass m and charge q movable along side CB is initially near C, when frame lies in x–z plane. Now, frame is rotated by constant angular velocity ω about z-axis. Whole system lies in gravity free space. If after time t block reaches point B, find B_0 in terms of t.



Sol. Block moves with speed

$v = \omega a$ in x–y plane

Velocity of block, when frame has rotated through angle $\theta = \omega t$

$$\vec{v} = \omega a \, (-\sin\theta \, \hat{i} + \cos\theta \, \hat{j}) + v_z \hat{k}$$

Force on block $F = q\vec{v} \times \vec{B} = [\omega a(-\sin\theta \, \hat{i} + \cos\theta \, \hat{j}) + v_z \hat{k}] \times B_0 \hat{j}$

Force in z-direction $= q\omega a B_0 (-\hat{k}) \sin\theta$

Acceleration of block $\vec{a}_z = \frac{-q\omega aB_0 \sin\theta}{m}(-\hat{k})$

$\Rightarrow \frac{dv_z}{dt} = -\frac{q\omega aB_0}{m}\sin\omega t$

$\Rightarrow v_z = -\frac{q\omega aB_0}{m}\int_0^t \sin\omega t\ dt = -\frac{aB_0}{m}(1-\cos\omega t)$

$\frac{dz}{dt} = -\frac{qaB_0}{m}(1-\cos\omega t)$

$\Rightarrow dz = -\frac{qaB_0}{m}\int_0^t (1-\cos\omega t)$

$z = -\frac{qaB_0}{m}\left[t-\frac{\sin\omega t}{\omega}\right]_0^t = -\frac{qaB_0}{m}\left[t-\frac{\sin\omega t}{\omega}\right]$

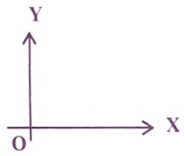
when block reaches to B, $z = -a$

$\Rightarrow -a = -\frac{qaB_0}{m}\left(t-\frac{\sin\omega t}{\omega}\right)$

$\Rightarrow B_0 = \frac{m\omega}{(\omega t-\sin\omega t)q}$

3. In a certain region of space there exists a uniform and constant electric field of magnitude E along the positive y-axis of a co-ordinate system. A charged particle of mass 'm' and charge '–q' (q > 0) is projected with speed 2v at an angle of 60° with the positive x-axis in x–y plane from the origin. When the x-coordinate of the particle becomes $\frac{\sqrt{3}mv^2}{qE}$, a uniform and constant magnetic field of strength B is also switched on along the positive y-axis. Find the co-ordinate of the particle as a function of time t after its projection.

Sol. $\vec{E} = E\hat{j}$



$\vec{v}_0 = 2v\cos 60^0\,\hat{i} + 2v\sin 60^0\,\hat{j} = v\hat{i} + v\sqrt{3}\,\hat{j}$

$\vec{a} = -\frac{qE}{m}\hat{j}$

$\vec{R}(t) = vt\hat{i} + (v\sqrt{3}t - \frac{1}{2}\frac{qE}{m}t^2)\hat{j}$

Given, $vt = \frac{\sqrt{3}mv^2}{qE} \Rightarrow t = \frac{\sqrt{3}mv}{qE}$

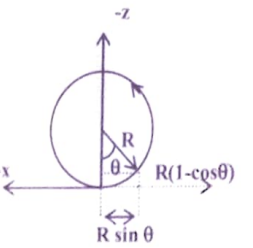
For $t \le \frac{\sqrt{3}mv}{qE}$

$x(t) = vt$ and $v_y(t) = v\sqrt{3} - \frac{qE}{m}t$

$\Rightarrow v_y\left(t = \frac{\sqrt{3}mv}{2E}\right) = v\sqrt{3} - \frac{qE}{m}\times\frac{\sqrt{3}mv}{qE} = 0$

For $t \ge \frac{\sqrt{3}mv}{qE}$, magnetic field is also present

The particle will start moving on helical path. The cross-section of the helix will be in x-z plane.



$$R = \frac{mv}{qB}; \qquad \omega = \frac{qB}{m}$$

$$\theta = \omega t = \frac{qBt}{m}$$

For $t \geq \frac{\sqrt{3}mv}{qE}$,

$$x(t) = \frac{\sqrt{3}mv^2}{qE} + R\sin\left(\frac{qB}{m}t\right)$$

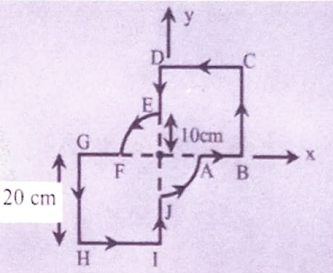
$$y(t) = \frac{3v^2m}{2qE} - \frac{1}{2}\frac{qE}{m}t^2$$

$$z(t) = -R\left[1-\cos\left(\frac{qB}{m}t\right)\right]$$

4. A current i = 5 amp flows through a thin wire as shown in the figure.

(a) Find the magnetic field produced by the current at point 'O' in the figure.

(b) If there exists an external magnetic field $\vec{B} = (14\hat{i}+14\hat{j})$ T, calculate the torque acting on the wire.



Sol. (a) Let 10 cm = a

$$B_0 = 2(B_{ABCDE} + B_{EF}) = 2\left[2\frac{\mu_0 i}{4\pi 2a}\left(\cos\frac{\pi}{4}+\cos\frac{\pi}{2}\right)+\frac{\mu_0 i}{4\pi a}\frac{\pi}{2}\right]$$

$$B_0 = \frac{\mu_0 i}{4\pi a}\left[\frac{2}{\sqrt{2}}+\pi\right] = \frac{10^{-7}\times 5}{0.1}\left[\sqrt{2}+3.14\right] = 22.78\ \mu T \text{ in the direction } \hat{k}$$

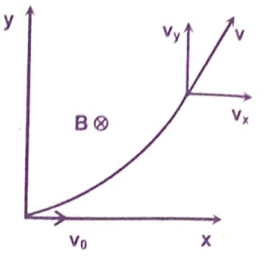
(b) $\vec{\mu} = i\vec{A} = 5\left(8a^2+\frac{\pi a^2}{2}\right)\hat{k} = 5a^2\left(8+\frac{\pi}{2}\right)\hat{k}$

$$\vec{\tau} = \vec{\mu}\times\vec{B} = 5a^2\left(8+\frac{\pi}{2}\right)\hat{k}\times 14\left(\hat{i}+\hat{j}\right)\ \text{N-m}$$

$$= 5a^2(112+22)(\hat{j}-\hat{i}) = 5\times 0.01\times 134\,(\hat{j}-\hat{i})\,\text{N-m}$$

$$\Rightarrow \vec{\tau} = 6.7\left(\hat{j}-\hat{i}\right)\ \text{N-m}$$

5. A non-uniform magnetic field $\vec{B} = B_0\left(1+\frac{y}{d}\right)(-\hat{k})$ is present in the region of space between y = 0 and y = d. A particle of mass m and positive charge q has velocity $\vec{v} = \frac{3qB_0d}{m}\hat{i}$ at origin O. Find the angle made by velocity of the particle with the positive x-axis when it leaves the field. (Ignore any interaction other than magnetic field)



Sol. $\vec{F} = q(v_x\hat{i} + v_y\hat{j}) \times B_0\left(1+\frac{y}{d}\right)(-\hat{k}) = F_x\hat{i} + F_y\hat{j}$

$|F_x| = qB_0\left(1+\frac{y}{d}\right)v_y$

$-\int_{v_0}^{v_x} m dv_x = q\int_0^d B_0\left(1+\frac{y}{d}\right)dy \quad \left(\frac{dv_x}{dy} \text{ is negative}\right)$

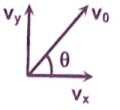
$v_x = v_0 - \frac{qB_0}{m}\left(\frac{3}{2}d\right)$

$v_x = \frac{3qB_0d}{m} - \frac{3}{2}\frac{qB_0d}{m} = \frac{3}{2}\frac{qB_0d}{m} = \frac{v_0}{2}$

$v_x^2 + v_y^2 = v_0^2$

$\frac{v_0^2}{4} + v_y^2 = v_0^2$

$\Rightarrow \quad v_y = \frac{\sqrt{3}v_0}{2}$

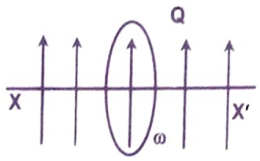


$\tan\theta = \frac{v_y}{v_x} = \sqrt{3}$

$\theta = 60°$

SOLVED PROBLEMS

1. **A disc of mass m has a charge Q distributed on its surface. It is rotating about an XX′ with a angular velocity ω. The force acting on the disc is**



(A) zero

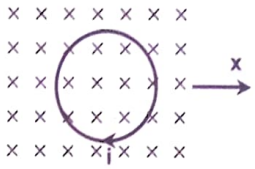
(B) ωRB/π

(C) 2QωBR

(D) none of these

Sol. (A). Since there is no interaction of disc with the magnetic field, the force acting on the disc is zero.

2. **A circular loop of wire is carrying a current i (as shown in the figure). On applying a uniform magnetic field inward perpendicular to the plane of the loop, the loop**



(A) move along the positive x-direction

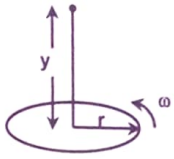
(B) move along the negative x-direction

(C) contract

(D) expand

Sol. (D)

3. **Calculate the magnetic field at distance y from the centre of the axis of a disc of radius r and uniform surface charge density σ, if the disc spins with angular velocity ω ?**



(A) $\frac{\mu_0\sigma\omega}{3}\left(\frac{r^2-2y^2}{\sqrt{r^2-y^2}}+2y\right)$

(B) $\frac{\mu_0\sigma\omega}{2}\left(\frac{r^2+y^2}{\sqrt{r^2+y^2}}\right)$

(C) $\frac{\mu_0\sigma\omega}{2}\left(\frac{r^2+2y^2}{\sqrt{r^2+y^2}}-2y\right)$

(D) $\frac{2\mu_0\sigma\omega}{3}\left(\frac{r^2+2y^2}{\sqrt{r^2+y^2}}-2y\right)$

Sol. Charge on a ring of radius x and width dx

$dq = (2\pi x\, dx)\sigma$

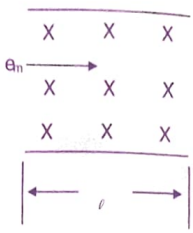
Current, $dI = \frac{dq}{dt} = \frac{2\pi x\sigma dx}{dt} = \omega\sigma x\, dx$

$$dB = \frac{\mu_0 dI r^2}{2\left(x^2+y^2\right)^{3/2}}$$

$$B = \frac{\mu_0\sigma\omega}{2}\left(\frac{r^2+2y^2}{\sqrt{r^2+y^2}}-2y\right)$$

$\therefore$(C)

4. **An electron moves straight inside a charged parallel plate capacitor of uniform surface charge density σ. The space between the plates is filled with constant magnetic field of induction $\vec{B}$. The time of straight line motion of the electron in the capacitor is**



(A) $\frac{e\sigma}{\varepsilon_0 \ell B}$ (B) $\frac{\varepsilon_0 \ell B}{\sigma}$

(C) $\frac{e\sigma}{\varepsilon_0 B}$ (D) $\frac{\varepsilon_0 B}{e\sigma}$

Sol. The net electric field

$$E = \vec{E}_1 + \vec{E}_2$$

$$\Rightarrow E = \frac{\sigma}{2\varepsilon_o} + \frac{\sigma}{2\varepsilon_o} = \frac{\sigma}{\varepsilon_o}$$

The net force acting on the electron is zero because it moves with constant velocity, due to its motion on straight line.

$$\Rightarrow \vec{F}_{net} = \vec{F}_e + \vec{F}_m = 0$$

$$\Rightarrow |\vec{F}_e| = |\vec{F}_m|$$

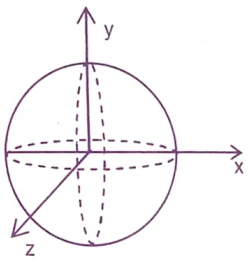
$$\Rightarrow eE = evB$$

$$\Rightarrow v = \frac{E}{B} = \frac{\sigma}{\varepsilon_o B}$$

$\therefore$ The time of motion in side the capacitor $= t = \frac{\ell}{v} = \frac{\varepsilon_o \ell B}{\sigma}$

$\therefore$ (B)

5. **Three rings, each having equal radius R, are placed mutually perpendicular to each other and each having its centre at the origin of co-ordinate system. If current I is flowing through each ring then the magnitude of the magnetic field at the common centre is**



(A) $\sqrt{3}\,\frac{\mu_0 I}{2R}$ (B) zero

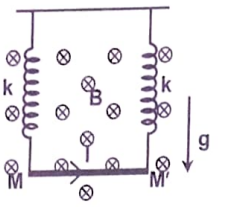
(C) $\left(\sqrt{2}-1\right)\frac{\mu_0 I}{2R}$ (D) $\left(\sqrt{3}-\sqrt{2}\right)\frac{\mu_0 I}{2R}$

Sol. $\vec{B} = \frac{\mu_0 I}{2R}(\pm\hat{i}) + \frac{\mu_0 I}{2R}(\pm\hat{j}) + \frac{\mu_0 I}{2R}\pm\hat{k}$

$$\therefore |\vec{B}| = \frac{\mu_0 I}{2R}\sqrt{3}$$

$\therefore$ (A)

6. Consider the plane of the paper to be the vertical plane with direction of 'g' as shown in the figure. A rod (MM') of mass 'M' and carrying a current I. The rod is suspended vertically through two insulated spring of spring constant K at two ends. A uniform magnetic field is applied perpendicular to the plane of the paper such that the potential energy stored in the spring, in equilibrium position is zero. If the rod is slightly displaced in the vertical direction from the position of equilibrium, then the time period of oscillation is



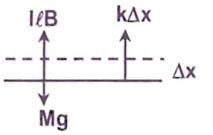
(A) $2\pi\sqrt{\frac{m}{K}}$

(B) $2\pi\sqrt{\frac{m}{2K}}$

(C) $2\pi\sqrt{\frac{K}{2m}}$

(D) $2\pi\sqrt{\frac{m}{K}\cdot\frac{mg}{I\ell B}}$

Sol. $\therefore B = \frac{mg}{I\ell}$



$F = -2K\Delta x = -m\omega^2\Delta x$

$\omega = \sqrt{\frac{2K}{m}}$

$\therefore$(B)

7. A uniform magnetic field of intensity 1T is applied in a circular region of radius 0.1 m, directed into the plane of paper. A charged particle of mass 5×10^{-5} kg and charge $q = 5 \times 10^{-4}$ C enters the field with velocity $1/\sqrt{3}$ m/s making an angle of ϕ with a radial line of circular region in such a way that it passes through centre of applied field the angle ϕ is

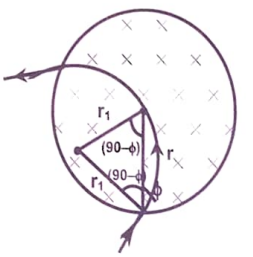
(A) 60°

(B) 30°

(C) 45°

(D) 90°

Sol. $r_1 = \frac{mV}{qB} = \frac{5\times10^{-5}\times\left(1/\sqrt{3}\right)}{5\times10^{-5}\times1} = \frac{1}{10\sqrt{3}}$



By sin rule, $\frac{r_1}{\sin(90°-\phi)} = \frac{r}{\sin 2\phi}$

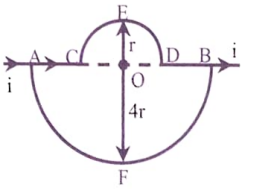
$\frac{1}{10\sqrt{3}\cos\phi} = \frac{0.1}{2\sin\phi\cos\phi}$

$\sin\phi = \frac{0.1\times10\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$

$\phi = 60°$

$\therefore$(A)

8. In the shown in the figure AC and BD are straight lines and CED and AFB are semicircular with radii r and 4r, respectively. The entire setup is lying in the same plane. If i is current entering at A what fraction of i will flow in the ACEDB such that resultant magnetic field at O is zero.



(A) i/5

(B) 4i/5

(C) 3i/5

(D) no current through bigger semicircular part.

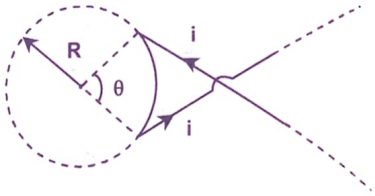
Sol. If I_1 is current in ACDB then

$$\frac{\mu_0}{4}\cdot\frac{i_1}{r}=\frac{\mu_0}{4}\cdot\frac{i_2}{4r} \Rightarrow \frac{i_1}{i_2}=\frac{1}{4}$$

$$i_1=\frac{1}{5}\times i=\frac{i}{5}$$

$\therefore$(A)

9. A current carrying wire has the configuration shown in the figure. Two semi-infinite straight sections, each tangent to the same circle, are connected by a circular arc, of angle θ along the circumference of the circle, with all sections lying in the same plane. What must be θ in order for B(magnetic field) to be zero at the centre of the circle?



(A)θ = 1 rad **(B) θ = 2 rad**

(C)θ = 3 rad **(D) none of the above**

Sol. For $\vec{B}$ to be zero at the centre, sum of field due to the ring and that due to the semi-infinite straight section should be equal and opposite

i.e. $\frac{\mu_0 i}{2\pi R}-\frac{\mu_0 i}{2\pi R}\left(\frac{\theta}{2}\right)=0$

$\Rightarrow \frac{\mu_0 i}{2\pi R}\left(1-\frac{\theta}{2}\right)=0$

$\Rightarrow \theta = 2$ rad

$\therefore$ (B)

10. A straight segment OP of length L of a circuit carrying current I ampere is placed along x-axis. Two infinitely long straight wires A and B each extending from z = −∞ to +∞ are fixed at y = −ℓ meter to y = +ℓ meter respectively. Wires A and B, each carry current I ampere. Given that O is origin of the coordinate system. Possible values of magnitude of force on segment OP is/are:

(A) $\frac{\mu_0}{\pi}I^2\ell n\left(1+\frac{L^2}{\ell^2}\right)$ **(B)** $\frac{\mu_0}{2\pi}I^2\ell n\left(1+\frac{L^2}{\ell^2}\right)$

(C) $\frac{\mu_0}{\pi}I^2\ell n\left(1+\frac{\ell^2}{L^2}\right)$ **(D) zero**

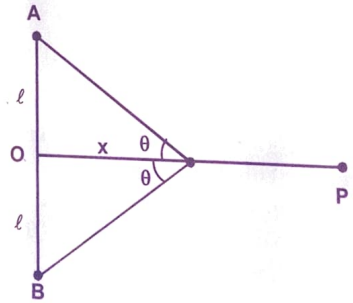
Sol. B, D

Net B at distance x from O along x-axis

$$=\frac{\mu_0}{2\pi}\frac{I}{r}2\cos\theta$$

$$\Rightarrow \int dF=\int I\frac{\mu_0}{2\pi}\frac{I}{r}2\cos\theta dx$$

$$=\frac{\mu_0}{2\pi}I^2\ell n\left(1+\frac{L^2}{\ell^2}\right)$$



11. **In a moving coil galvanometer the number of turns N = 30, area of the coil A = 4×10^{-3} m^2 and the magnetic field strength B = 0.3T. To increase its voltage sensitivity by 50% we**
(A) increase number of turns to 45
(B) increase area to 9×10^{-3} m^2
(C) increase magnetic field to 0.45 T
(D) Change the material of wire such that its specific resistance would be $2/3^{rd}$ of the sepecific resistance of the present wire.

Sol. **B, C, D**

$$\text{voltage sensitivity} = \frac{\text{deflection}}{\text{voltage measured}}$$

COMPREHENSION

The Betatron was the first important machine for producing high energy electrons. The action of the betatron depends on the same fundamental principle as that of the transformer in which an alternating current applied to a primary coil induces an alternating current usually with higher or lower voltage in the secondary coil. In the betatron secondary coil is replaced by a doughnut shaped vacuum chamber. Electron produced in the doughnut from a hot filament, are given a preliminary acceleration by the application of an electric field having potential difference of 20 kV to 70 kV, when an alternating magnetic field is applied parallel to the axis of the tube, two effects are produced (1) an electromotive force is produced in the electron orbit by the changing magnetic flux that gives an additional energy to the electrons (2) a radial force is produced by the action of magnetic field whose direction is perpendicular to the electron velocity which keeps the electron moving in a circular path. Conditions are arranged such that the increasing magnetic field keeps the electron in a circular orbit of constant radius. The mathematical relation between the betatron parameters and electron parameters to ensure the above condition is called BETATRON condition.

If the orbit radius of the circular path traced by the electron is R, magnetic field is B, speed of the electron is v, mass of the electron is m, charge on the electron is e and ϕ is the flux within the orbit of radius R. Then answer the following questions based on the above comprehension.

12. **What is the tangential force acting on the electron?**

(A) **evB**

(B) $\frac{e}{\pi R}\frac{d\phi}{dt}$

(C) $\frac{e}{2\pi R}\frac{d\phi}{dt}$

(D) $\frac{ev^2}{R}$

Sol. *C*

13. **What is the value of** $\frac{d(m|\vec{v}|)}{dt}$ **?**

(A) $eR\frac{dB}{dt}$

(B) **BeV**

(C) $\frac{e}{2\pi R}\frac{d\phi}{dt}$

(D) $\frac{ev^2}{R}$

Sol. *A*

14. **What is the BETATRON condition?**

(A) $BeV = \frac{e}{\pi R}\frac{d\phi}{dt}$

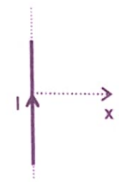
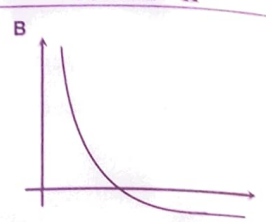
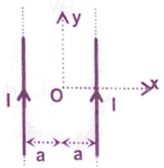
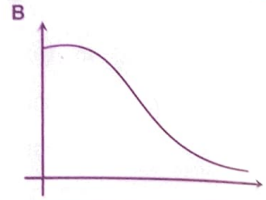
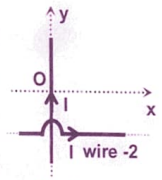
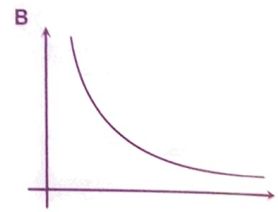
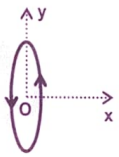
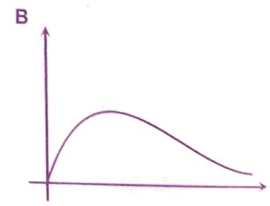
(B) $\frac{d\phi}{dt} = 2\frac{d}{dt}(\pi R^2 B)$

(C) $\frac{d\phi}{dt} = \frac{d}{dt}(\pi R^2 B)$

(D) **Data insufficient to determine.**

Sol. *B*

15. **Some current carrying wires are given in List – I and graph of variation of magnetic field versus position of point P are given in List – II. Match the graph given in List – II for the current carrying wire in List – I.**

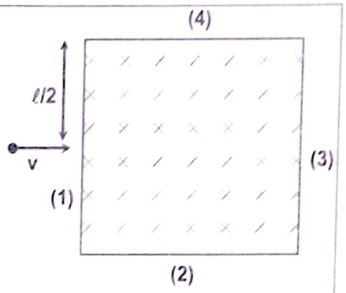
	Column – I			Column – II
(A)		A long straight current carrying wire, placed along y-axis. Its magnetic field at point P(x,0,0) is observed along x-axis.	(p)	
(B)		Two long parallel wires placed on xy plane. O is origin. Its magnetic field is observed at point P(0,0,z) along positive z-axis.	(q)	
(C)		Two long parallel wires are placed along y-axis and parallel to x-axis. Its magnetic field is observed at point P(x,0,0) along positive x-axis parallel to wire -2.	(r)	
(D)		A current carrying ring is placed on y-z plane. Origin is at the centre of ring. Its magnetic field is observed at point P(x,0,0) along positive x-axis.	(s)	

Sol. (A) → (r) (B) → (s) (C) → (p) (D) → (q)

Answer questions 16, 17 and 18 by appropriately matching the information given in the three columns of the following table.

Magnetic field is confined to a square region of side 'ℓ'

B = magnetic field
m = mass of the particle
q = charge of the particle
v = velocity of the particle
Column (I) : Represents charge of the particle
Column (II) : Represents velocity of the particle
Column (III): Shows the edge from which the particle comes out



Column 1		Column 2		Column 3	
(I)	$q = q_0$	(i)	$v < \frac{q_0 B_0 \ell}{4m_0}$	(P)	2
(II)	$q = -q_0$	(ii)	$v > \frac{5q_0 B_0 \ell}{4m_0}$	(Q)	1
(III)	$q = 3q_0$	(iii)	$\frac{q_0 B_0 \ell}{4m_0} < v < \frac{5q_0 B_0 \ell}{4m_0}$	(R)	3
(IV)	$q = -\frac{q_0}{2}$	(iv)	$\frac{q_0 B_0 \ell}{2m_0} < v < \frac{5q_0 B_0 \ell}{2m_0}$	(S)	4

16. **If $q = q_0$, $B = B_0$, $m = m_0$. Choose the correct option:**
(A) (II) (ii) (S) (B) (I) (i) (Q)
(C) (I) (iii) (P) (D) (I) (i) (R)

Sol. B

17. **If $B = \frac{B_0}{3}$, $m = m_0$ and particle comes out from edge 3, choose the correct option.**
(A) (I) (i) (Q) (B) (III)(ii) (R)
(C) (III) (iii) (R) (D) (III) (i) (R)

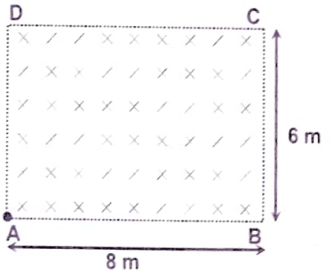
Sol. B

18. **If $B = B_0$ and $m = m_0/2$ and particle comes out from edge 2, choose the correct option.**
(A) (II) (i) (P) (B) (III)(i) (P)
(C) (IV) (iii) (P) (D) (IV) (iv) (P)

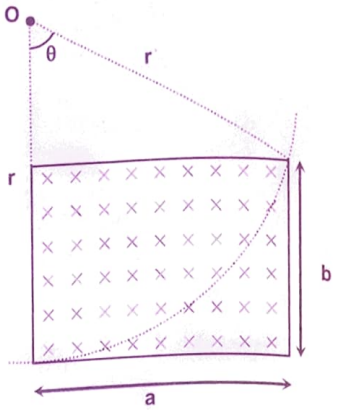
Sol. C

Numerical Type

19. **A charged particle of mass m = 5 gm and charge q = 12 μC enters along AB at point A in a uniform magnetic field of magnitude B = 100 Tesla existing in a rectangular region of size (8m × 6m) and into the plane of paper as shown in figure. The particle leaves the region at 'C'. The speed of the charge particle (in m/s) as it leaves the region is**



Sol. 2



$$r = \frac{mv}{qB}$$

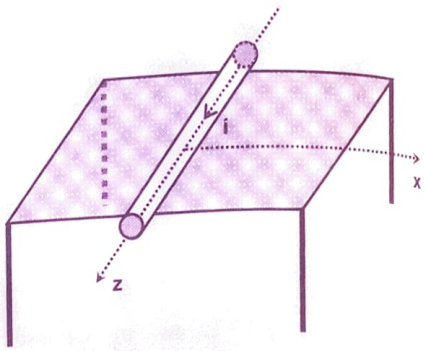
$$\sin\theta = \frac{a}{r},\ \cos\theta = \frac{r-b}{r}$$

$$\left(\frac{a}{r}\right)^2 + \left(\frac{r-b}{r}\right)^2 = 1$$

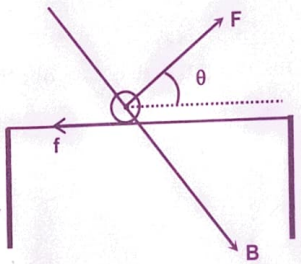
$$\Rightarrow r = \frac{(a^2+b^2)}{2b} = \frac{mv}{qB}$$

$$v = \frac{qB(a^2+b^2)}{2mb} = 2\ \text{m/s}$$

20. **A conducting rod of length 'ℓ' = $2\sqrt{5}$ meter and mass 'm' = 4 kg lies on the horizontal table. Coefficient of friction between the rod and the table is 'μ' = $\left(\frac{1}{2}\right)$. If the current in the conductor is 2 ampere, then find the minimum magnitude of magnetic field strength (in Tesla) such that conducting rod just starts to translate along x-axis. (take g = 10 m/sec^2) [neglect the radius of rod]**



Sol. 2



$$F_{min} = \frac{\mu mg}{\sqrt{1+\mu^2}}$$

$$iℓB_{min} = \frac{\mu mg}{\sqrt{1+\mu^2}}$$

$$B_{min} = 2\ \text{Tesla}$$

Numerical Decimal Type

21. **A particle having charge 'q' and mass 'm' is projected with velocity $(4\hat{i}-6\hat{j}+3\hat{k})$ m/sec from the origin in a region occupied by electric field 'E' and magnetic field 'B' such that $\vec{B} = B_0\hat{j}$ and $\vec{E} = E_0\hat{j}$ (take $\frac{qE_0}{m} = 2$). Find the time (in sec) when the magnitude of velocity of the charge particle becomes $5\sqrt{5}$ m/sec. (neglect the gravity)**

Sol. 00008.00

$$v_y = u_y + a_y t$$

$$0 = 6 - \frac{qE_0}{m}t_1$$

$$t_1 = 3\ \text{sec}$$

$v_x^2 + v_y^2 + v_z^2 = (5\sqrt{5})^2$

$4^2 + v_y^2 + 3^2 = 125$

$v_y = 10$ m/sec

$v_y = u_y + a_y t$

$10 = 0 + \frac{qE_0}{m} t_2$

$\Rightarrow t_2 = 5$ sec

So, $t = t_1 + t_2 = 8$ sec

22. **A conductor of mass m = 5 kg and length ℓ = 4m is placed on horizontal surface and a uniform magnetic field B = 2 Tesla exist parallel to horizontal surface but perpendicular to the conductor. Suddenly, a certain amount of charge is passed through it, then it is found to jump to a height h = 5m. Then, find the amount of charge that passes through the conductor in coulomb. (take g = 10 m/s^2)**

Sol. 00006.25

Impulse, $F\Delta t = m\sqrt{2gh}$

$(i\ell B)\Delta t = m\sqrt{2gh}$ ...(i)

$i\Delta t = \Delta q$ ...(ii)

$\Delta q = \frac{m\sqrt{2gh}}{B\ell} = \frac{5\sqrt{2\times10\times5}}{2\times4} = 6.25$ Coulomb

ELECTROMAGNETIC INDUCTION & AC CIRCUITS 14

FARADAY'S LAW

Whenever the flux of the magnetic field linked through the area bound by a closed conducting loop changes, an emf is produced in the loop.

The emf is given by

$$E = -\frac{d\phi}{dt}$$

where $\phi = \int \vec{B}.d\vec{s}$ is the flux of the magnetic field through the area.

Unit of magnetic flux is weber and is denoted by Wb.

LENZ'S LAW

The direction of induced current is such that it opposes the change that has induced it.

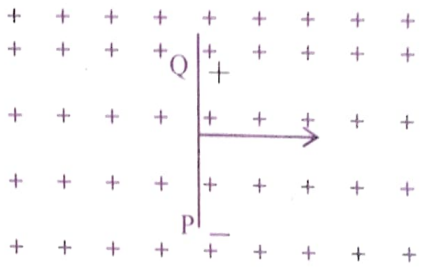
MOTIONAL EMF

EMF induced $= B.V\ell$.

where ℓ = length of conducting rod

V = velocity of rod perpendicular to magnetic field

B = magnetic field



INDUCED ELECTRIC FIELD

$$\oint \vec{E}.d\ell = -\frac{d\phi}{dt}$$

where $\vec{E}$ = induced electric field, when the magnetic flux passing though a loop is changed by varying magnetic field.

SELF INDUCTION

The self induced flux linked with a loop is given by

$\phi = L.$,

L = self inductance (unit Henry) of the loop

I = current in the loop.

Self inductance of a solenoid is given by:

$L = \mu_0 n^2 AL$

where A = area of cross-section of solenoid

L = length of the solenoid

MUTUAL INDUCTION

When two loops are placed closed to each other, then, if the current I in one circuit changes with time, an emf is induced in the second circuit and vice versa.

The flux linked with loop II due to loop I is given by $\phi = Mi$, where i = current in the circuit

M = mutual inductance of the loops

Now $M = k\sqrt{L_1 L_2}$

where L_1 = self inductance of loop 1

L_2 = self inductance of loop 2

and k = coupling factor which varies from 0 to 1.

$$E = -\frac{d\phi}{dt} = -M\frac{di}{dt}$$

INDUCTORS IN SERIES AND PARALLEL

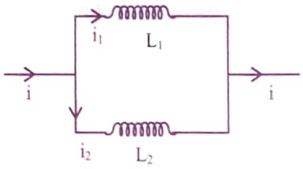
Inductance in Series

When the current through the inductance is same they are in series. Their equivalent is given by

$L = L_1 + L_2 \pm 2M_1$, depending upon whether the current in two inductances is in the same or opposite sense.

Inductance is Parallel

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}$$



ENERGY STORED IN AN INDUCTOR

$$E - L\frac{di}{dt} = iR$$

On solving, we get $\int_0^t Eidt = \int_0^t i^2 Rdt + \frac{1}{2}Li^2$

The energy stored in an inductor is equal to $\frac{1}{2}Li^2$

The energy stored per unit volume in a magnetic field of intensity B is equal to $\frac{B^2}{2\mu_0}$.

GROWTH AND DECAY OF CURRENT IN AN INDUCTOR

Growth of Current in an Inductor- Resistor Circuit

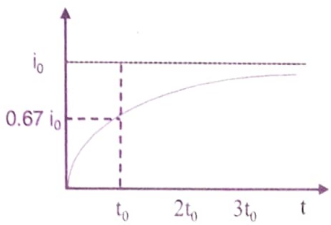
$$E - L\frac{di}{dt} = iR$$

Solution of this differential equation given,

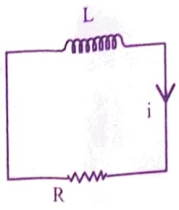
$$i = \frac{E}{R}\left[1 - e^{-t\frac{L}{R}}\right]$$

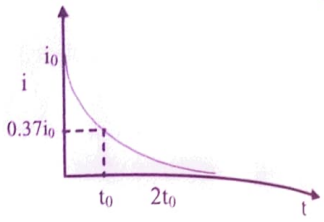
where $t_0 = \frac{L}{R}$ = time constant for the circuit.

The current grows to 63% of the peak current after one time constant.



Decay of current in an L– R circuit





$$= -L\frac{di}{dt} = Ri$$

The solution of this differential equation gives

$$\ln\left(\frac{i}{i_0}\right) = -\frac{R}{L}t$$

$$i = i_0\, e^{-t/\frac{L}{R}}$$

The current decays to 37 % of the peak current after one time constant.

ALTERNATING CURRENT

Instantaneous value of sinusoidal voltage is given by

$$V = V_0 \sin(\omega t)$$

$$V_{av} = \frac{\int_0^{\pi/\omega} V_0 \sin\omega t\, dt}{\int_0^{\pi/\omega} dt} = \frac{2V_0}{\pi}$$ [Average value of A.C. voltage]

Also, $$V_{rms} = \sqrt{\frac{\int_0^{2\pi/\omega} V_0^2 \sin^2\omega t\, dt}{\int_0^{2\pi/\omega} dt}} = \frac{V_0}{\sqrt{2}}$$ [R.M.S. value of A.C. voltage]

Since there is some phase difference, ϕ between AC voltage and AC current.

$\therefore$ $I = I_0 \sin(\omega t + \phi)$

$$i_{av} = \frac{\int_0^{\pi/\omega} I_0 \sin(\omega t + \phi) dt}{\int_0^{\pi/\omega} dt} = \frac{2I_0}{\pi}$$

$$i_{rms} = \sqrt{\frac{\int_0^{2\pi/\omega} V_0^2 i^2 \sin^2(\omega t + \phi) dt}{\int_0^{2\pi/\omega} dt}} = \frac{I_0}{\sqrt{2}}$$

The thermal energy developed in one time period is equal to

$$U = \int_0^T i^2 R dt = i_{rms}^2 RT$$

Note:

(i) In a purely capacitive circuit current leads the voltage by $\pi/2$. The reactance of a capacitor is defined as

$$X_C = \frac{1}{\omega C}$$

(ii) In a purely inductive circuit, the current lags behind the emf by $\pi/2$. The reactance of an inductor is given by

$$X_L = \omega L$$

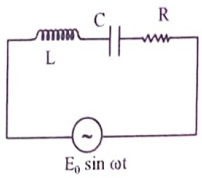
(iii) In a purely resistive circuit, the current is in phase with the voltage. The reactance of a purely resistive circuit is given by

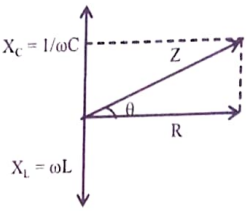
$$X_R = R$$

Phaser Diagram of an LCR Circuit

$$\tan\theta = \frac{X_C \sim X_L}{R} = \frac{\omega L \sim \frac{1}{\omega C}}{R}$$

$$Z = \sqrt{(X_L \sim X_L)^2 + R^2} = \sqrt{\left(\omega L \sim \frac{1}{\omega C}\right)^2 + R^2}$$





$$V = V_0 \sin \omega t$$

$$i = \frac{V_0}{Z}\sin(\omega t + \theta)$$

where values of Z and θ are calculated as above.

If $\omega = \sqrt{\frac{1}{LC}}$, the circuit behaves as a pure resistor and we get the maximum value of peak current. This phenonmenon is called resonance in A.C. circuit. The frequency ω is called the resonant frequency of the circuit.

The variation of peak current with applied frequency ω of the AC source is shown in the figure below, for two different circuits.

We see if R is small, resonance is sharp. This means if the applied frequency ω is closed to resonant frequency ω, the current is high otherwise it is small.

An LCR circuit used at a frequency close to the resonance frequency is called Resonant circuit.

Power in an AC Circuit

$$P(t) = V(t)\, I(t) = V_0 I_0 \sin\omega t \sin(\omega t + \phi)$$

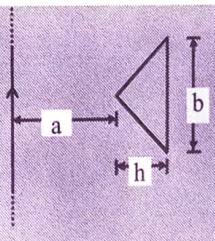
$$P_{avg} = \frac{\int_0^T P(t)dt}{T} = \frac{1}{2} V_0 I_0 \cos\phi$$

$$= V_{rms}\, I_{rms} \cos\phi$$

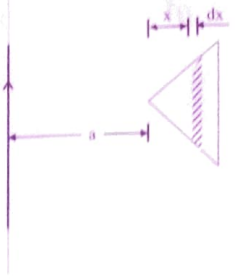
The term $\cos\phi$ is called the power factor of the circuit.

ILLUSTRATIVE PROBLEMS

1. A very long straight conductor and an isoceles triangular conductor lie in a plane and separated from each other as shown in the figure.
Given a = 10 cm; b = 20 cm and h = 10 cm
(a) Find their coefficient of mutual induction.
(b) If current in the straight wire is increasing at a rate of 2 A/s, find the direction and magnitude of current in the triangular wire.
[Diameter of the wire cross-section d = 1 mm, resistivity of the wire, $\rho = 1.8 \times 10^{-8}\ \Omega$-m]



Sol. (a) Assuming a small strip at a distance x from the apex.



$$dA = \frac{b}{h} x\, dx$$

Let I be the current flowing in the straight wire, then magnetic field at the location of the strip is

$$B = \frac{\mu_o I}{2\pi(a+x)}$$

$$d\phi = B\, dA = \frac{\mu_o Ib}{2\pi h}\left(\frac{x dx}{a+x}\right)$$

$$\phi = \frac{\mu_o Ib}{2\pi h}\int_o^h \left[1 - \frac{a}{a+x}\right] dx$$

$$\phi = \frac{\mu_o Ib}{2\pi h}\left[h - a\ \ln\left|\frac{a+h}{a}\right|\right]$$

The coefficient of mutual induction is given by

$$M = \frac{\phi}{I} = \frac{\mu_o b}{2\pi h}\left[h - a\ \ln\left|\frac{a+h}{a}\right|\right]$$

Here, b = 20cm ; h = 10cm ; a = 10cm ; $\mu_o = 4\pi \times 10^{-7}$ H/m.

$$M = \frac{(4\pi\times10^{-7})(0.2)}{2\pi(0.1)}\left[0.1 - 0.1\ \ln\left|\frac{10+10}{10}\right|\right] = 1.22\times10^{-8}\,H$$

(b) According to Faraday's law

$$E_{ind} = -M\frac{di}{dt} \qquad \text{since, } \frac{di}{dt} = 2\ A/s$$

$$\therefore\quad (E_{ind}) = (1.22 \times 10^{-8})\,(2) = 2.44 \times 10^{-8} \text{ volt.}$$

Resistance of the triangular conductor is $R = \rho\frac{l}{A}$

here, $\rho = 1.8 \times 10^{-8}\ \Omega$-m ; $l = 10\sqrt{2} + 20 + 10\sqrt{2} = 20\,(\sqrt{2}+1)$ cm$= 0.48$ m.

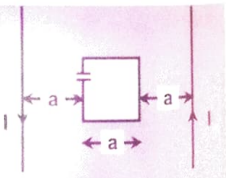
$$A = \frac{\pi}{4}(10^{-3})^2 = 10^{-6}\frac{\pi}{4}\ m^2$$

$$\therefore\quad R = (1.8 \times 10^{-8})\ \frac{(0.48)}{(3.14)} \times \frac{4}{10^{-6}} = 1.1 \times 10^{-2}\ \Omega$$

$$I = \frac{E_{ind}}{R} = \frac{2.44\times10^{-8}}{1.1\times10^{-2}} = 2.2\ \mu A$$

The direction of the current is anticlockwise by Lenz Law.

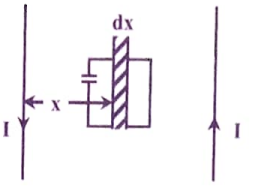
2. A square loop of side 'a' with a capacitor of capacitance C is located between two current carrying long parallel wires as shown. The value of I in the wires is given as $I = I_0 \sin\omega t$.



(a) Calculate maximum current in the square loop.

(b) Draw a graph between charges on the upper plate of the capacitor vs time.

Sol. (a) Flux through the square loop



$$= \int_a^{2a} \frac{\mu_0}{4\pi} 2I\left|\frac{1}{x}+\frac{1}{3a-x}\right| adx = \frac{\mu_0}{4\pi} Ia4\ \ln 2$$

Induced emf $e = -\frac{d\phi}{dt} = -\frac{\mu_0}{\pi} aI_0\omega\ \ln 2\ \cos\omega t$

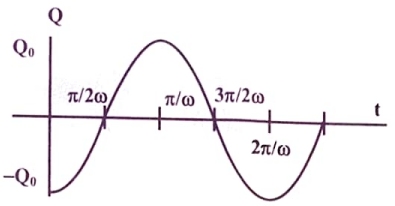
Charge on the capacitor

$$Q = Ce = -C\frac{\mu_0}{\pi} aI_0\omega\ \ln 2\ \cos\omega t = -Q_0\cos\omega t \text{ (say)}$$

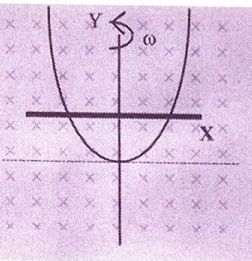
Current in the loop $= \frac{dQ}{dt} = \frac{\mu_0}{\pi} CI_0\omega^2 a\ \ln 2\ \sin\omega t$

$$I_{max} = \frac{\mu_0}{\pi} CI_0\, a\omega^2\ \ln 2$$

(b)



3. A conducting rod is bent as a parabola $y = Kx^2$, where K is a constant and it is placed in a uniform magnetic field of induction B. At t = 0 a conductor of resistance per unit length λ starts sliding up on the parabola with a constant acceleration a and the parabolic frame starts rotating with constant angular frequency ω about the axis of symmetry, as shown in the figure. Find the instantaneous current induced in the rod , when the frame turns through π/4 rad.



Sol. When the frame has turned through an angle θ,

$$\Phi = BA\cos\theta$$

where $A = \int_0^y 2x\,dy = \frac{2}{\sqrt{k}}\int_0^y \sqrt{y}\,dy = \frac{4}{3\sqrt{k}} y^{3/2}$

Since $y = \frac{1}{2}at^2$

$$\therefore\ \Phi = \frac{B}{3}\sqrt{\frac{2}{K}} a^{3/2} t^3 \cos\theta$$

By Faraday's law, $E_{ind} = -\frac{d\Phi}{dt}$

or $E_{ind} = \frac{Ba^{3/2}}{3}\sqrt{\frac{2}{K}}\left[t^3 \sin\theta\left(\frac{d\theta}{dt}\right) - 3t^2\cos\theta\right]$

When the frame turns through π/4,

$$t = \frac{\theta}{\omega} = \frac{\pi}{4\omega}$$

$$E_{ind} = \frac{Ba^{3/2}}{3}\sqrt{\frac{2}{K}} t^2\left[\omega t\sin\theta - 3\cos\theta\right]$$

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$$\text{or } E_{ind} = \frac{\pi^2 B a^{3/2}}{48\omega^2}\sqrt{\frac{2}{K}}\left[\frac{\pi}{4}-3\right]\frac{1}{\sqrt{2}}$$

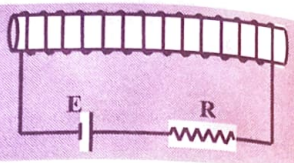
$$E_{ind} = \frac{\pi^2(\pi-12)Ba^{3/2}}{192\omega^2\sqrt{K}}$$

$$I = \frac{E_{ind}}{R} = \frac{\pi^2(\pi-12)Ba^{3/2}}{192\,\omega^2 R\sqrt{K}}.$$

$$R = \lambda\, 2x = \lambda.\, 2\sqrt{\frac{y}{k}} = \frac{2\lambda}{\sqrt{k}}\sqrt{\frac{a}{2}}t = \sqrt{\frac{2a}{k}}\frac{\lambda\pi}{4\omega}$$

$$I = \frac{E_{ind}}{R} = \frac{\pi^2(\pi-12)Ba^{3/2}\sqrt{k}\quad 4\omega}{192\omega^2\sqrt{k}\sqrt{2a}\quad \pi\lambda} = \frac{Ba\pi(\pi-12)}{48\sqrt{2}\quad \omega\lambda}$$

4. An iron core solenoid of length ℓ and cross-sectional area A having N turns on it is connected to a battery through a resistance as shown in the figure. At instant t = 0, the iron rod of permeability μ from the core is abruptly removed. Find the current as a function of time.



Sol. Self inductance of the solenoid

$$L = \frac{\mu_0\mu N^2 A}{l}$$

and initial current $i_0 = \frac{\varepsilon}{R}$ because circuit is in the steady state.

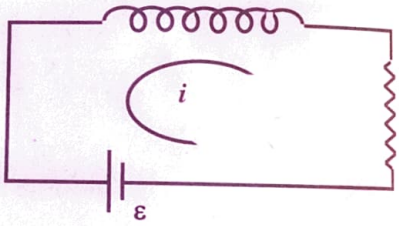
When iron rod is abruptly removed from the core the number of flux linkages abruptly do not change .

$$N\phi = Li_0 = L'i_0'$$

$$\left(\frac{\mu_0\mu N^2 A}{l}\right)i_0 = \left(\frac{\mu_0 N^2 A}{l}\right)i_0'$$

$\therefore$ Just after t = 0 , initial current $i_0' = \mu i_0 = \mu\left(\frac{\varepsilon}{R}\right)$

At instant, if the current in the circuit is i , then apply KVL in the loop, we get



$$L\frac{di}{dt} + iR = \varepsilon$$

$$-L\frac{di}{(\varepsilon - iR)} = dt$$

On integration we get ,

$$\frac{L}{-R}\ln(\varepsilon - iR) = t + c_1$$

$$\text{At } t = 0,\ i = i_0' = \left(\mu\frac{\varepsilon}{R}\right)$$

$$\Rightarrow \qquad \left(\frac{L}{-R}\right)\ln(\varepsilon - i_0' R) = C_1$$

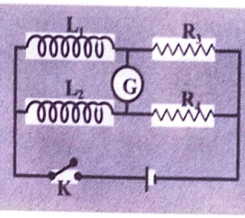
$$\therefore\ \ln\left(\frac{\varepsilon - iR}{\varepsilon - i_0' R}\right) = -\frac{tR}{L} \quad \Rightarrow \quad \left(\frac{\varepsilon - iR}{\varepsilon - i_0' R}\right) = e^{-\frac{tR}{L}}$$

$$\Rightarrow \left(\frac{\varepsilon - iR}{\varepsilon - \left(\mu\frac{\varepsilon}{R}\right)R}\right) = e^{\frac{tR}{L}}$$

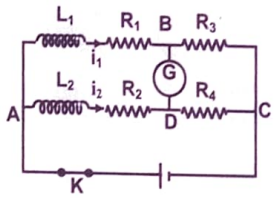
On rearranging the equation , we get

$$i = \frac{\varepsilon}{R}\left[1 + (\mu - 1)e^{-tR/L}\right]$$

5. Two inductors of self inductances L_1 and L_2 and of resistances R_1 and R_2 (not shown in the diagram) respectively, are connected in the circuit as shown in the figure. At the instant t = 0, key K is closed, obtain the condition for which the galvanometer will show zero deflection at all times after the key is closed.



Sol. Since there is no current through BD therefore points B and D are at same potential, $V_B = V_D$



Potential difference , $V_{AB} = V_{AD}$

$$L_1\frac{di_1}{dt} + i_1R_1 = L_2\frac{di_2}{dt} + i_2R_2 \qquad \ldots(1)$$

Similarly , $V_{BC} = V_{DC}$

$$i_1R_3 = i_2R_4 \qquad \ldots(2)$$

From equation (1) and (2)

$$L_1\frac{R_4}{R_3}\frac{di_2}{dt} + i_2\frac{R_1}{R_3} = L_2\frac{di_2}{dt} + i_2R_2$$

$$\left(L_1\frac{R_4}{R_3} - L_2\right)\frac{di_2}{dt} = i_2\left[R_2 - \frac{R_1}{R_3}\right] \qquad \ldots(3)$$

At $t = 0, \quad i_2 = 0$

$$\therefore \quad L_1\frac{R_4}{R_3} - L_2 = 0 \qquad \text{as} \qquad \frac{di_2}{dt} \neq 0$$

$$\therefore \quad \frac{L_1}{L_2} = \frac{R_3}{R_4} \qquad \ldots(4)$$

At $t = \infty$,

$$\frac{di_2}{dt} = 0 , \quad \Rightarrow \quad i_2 = \left(\frac{\varepsilon}{R_2 + R_4}\right) = \text{constant}$$

$$\Rightarrow R_2 - \frac{R_4R_1}{R_3} = 0$$

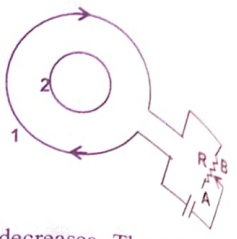
$$\therefore \quad \frac{R_1}{R_2} = \frac{R_3}{R_4} \qquad \ldots.(5)$$

From (4) and (5)

$$\frac{L_1}{L_2} = \frac{R_1}{R_2} = \frac{R_3}{R_4}$$

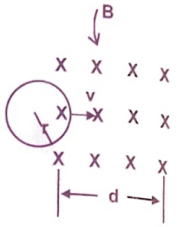
SOLVED PROBLEMS

1. **Shown in the figure is a small loop that is kept co-axially with the bigger loop. If the slider moves from A to B, then**
   (A) Current flow in both the loops will be in opposite senses.
   (B) Clockwise current in loop 1 and anti-clockwise current in loop 2 flow
   (C) No current flows in loop 2
   (D) Clockwise current flows in loop 2



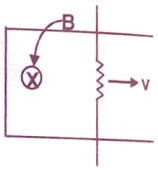
Sol. When the slider moves towards B, the resistance of the circuit (bigger loop) decreases. Therefore, the current in the bigger loop increases. The increasing current results in increasing flux ($\phi \alpha i$) linked in the smaller coil. Consequently, induced emf will be generated along the smaller loop creating an induced current so as to oppose the increase in flux. Therefore the current flows anticlockwise, in the inner loop.
$\therefore$ (B)

2. **A conducting loop is pulled with a constant velocity towards a region of uniform magnetic field of induction B as shown in the figure. Then the current involved in the loop is (d > r)**
   (A) clockwise while entering
   (B) anti-clockwise while entering
   (C) zero when completely inside
   (D) clockwise while leaving

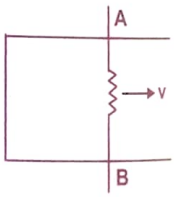


Sol. When the loop is drawn into the magnetic field, the area of the portion loop in the magnetic field will increase. That means, the flux linkage increases. Therefore a (vortex) emf is induced in the loop so as to oppose the change, that is to say to oppose the increase in magnetic flux in the loop. Therefore, the current will have to induce in anti-clockwise direction to induce an opposing magnetic field, that is pointing outward of the page. After the complete entry of the loop into the magnetic field, no variation of flux occurs. Therefore it induces no current so long as the loop. Is completely inside the magnetic field. When it emerges out of the magnetic field, following the previous argument, the direction of induced current in it will be reversed (clockwise)
$\Rightarrow \therefore$ (B), (C) and (D).

3. **A conducting bar is pulled with a constant speed v on a smooth conducting rail. The region has a steady magnetic field of induction B as shown in the figure. If the speed of the bar is doubled then the rate of heat dissipation will**
   (A) remain constant
   (B) become quarter of the initial value
   (C) become four fold
   (D) get doubled



Sol. The induced emf between A and B = E = $B\ell v$



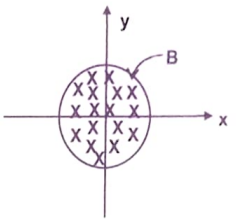
$\Rightarrow$ The induced current = $i = \frac{E}{R}$

$\Rightarrow i = \frac{B\ell v}{R}$

The electrical power = $P = i^2R = \frac{B^2\ell^2 v^2}{R}$

Since v is doubled, the electrical power, becomes four times. Since heat dissipation per second is proportional to electrical power, it becomes four fold.
$\therefore$ (C)

4. **A loop is kept so that its center lies at the origin of the coordinate system. A magnetic field has the induction B pointing along Z axis as shown in the figure**



(A) No emf and current will be induced in the loop if it rotates about Z axis

(B) Emf is induced but no current flows if the loop is a fiber when it rotates about y axis.

(C) Emf is induced and induced current flows in the loop if the loop is made of copper and is rotated about y-axis.

(D) If the loop moves along Z axis with constant velocity, no current flows in it.

Sol. If the loop rotates about Z axis, the variation of flux linkage will be zero. Therefore no emf is induced. Consequently no current flows in the loop

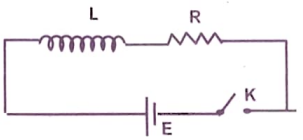
When it rotates about y axis, its flux linkage changes. However, in insulators there can not be motional emf.

If the loop is made of copper, it is conductive therefore induced current is set up.

If the loop moves along the Z axis variation of flux linkage is zero. Therefore the emf and current will be equal to zero.

$\therefore$ (A), (C) and (D)

5. **Shown in the figure is an R-L circuit. Just after the Key (K) is closed**



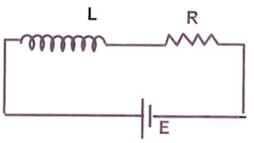
(A) the current in the circuit is zero

(B) potential drop across the resistor is zero

(C) emf developed across the inductor equals the emf of the battery

(D) no heat is dissipated in the circuit

Sol. If the equilibrium current in the circuit is i_0, current at any time in transient state is given as



$$i = i_0\left(1 - e^{-\frac{t}{\tau}}\right) \text{ where } \tau = \frac{L}{R}$$

Putting $t = 0$, $i = 0$

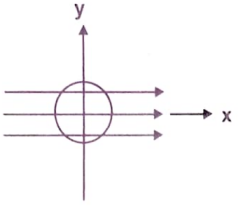
$\Rightarrow$ The voltage drop across the inductor is E′ that oppose the applied emf E

$\Rightarrow$ E′ = E numerically.

Since the circuit current is zero, the heat loss that is proportional to i_0^2R will be zero

$\therefore$ (A), (B), (C), and (D).

6. **A conducting loop of resistance R and radius r has its center at the origin of the co-ordinate system in a magnetic field of induction B. When it is rotated about Y-axis through 90° ,net charge flown in the coil is directly proportional to**



(A) B **(B) R**

(C) r^2 **(D) r**

Sol. Induced emf loop when the variation of flux d ϕ during time dt is given as

$$E = \frac{d\phi}{dt}$$

$$\Rightarrow \int_{\phi_1}^{\phi_2} d\phi = \Delta\phi = \int E dt \qquad \ldots(1)$$

$\Rightarrow$ The total charge flown in the loop = $q = \int i \; dt$

$$q = \int \frac{E}{R} \; dt \qquad \ldots(2)$$

using (1) and (2)

$$q = \frac{\Delta\phi}{R}$$

where $\Delta\phi$ = change in flux given as $\Delta\phi = \phi_2 - \phi_1 = B.(\pi r^2)$ because initially to flux is linked with the coil and it has maximum flux linkage $\phi_2 = B\,\pi r^2$ when turned through 90°, the flux is reduced to zero.

$$\Rightarrow q = \frac{\pi \; B \; r^2}{R}$$

i.e. $q \propto B \Rightarrow \quad q \propto r^2$

$\Rightarrow \quad q \propto (1/R)$

$\therefore$ (A) and (C)

7. The coefficient of mutual inductance of the two coils is 0.5 H. If the current is increased from 2 to 3 A in 0.01 sec. in one of them, then the induced e.m.f. in the second coil is :

(A) 25 V **(B) 50 V**

(C) 75 V **(D) 100 V**

Sol. Induced emf $= M\frac{di}{dt} = 0.5\,\frac{3-2}{0.01} = 50$ V

$\therefore$ (B)

8. An ideal choke takes a current of 8 A when connected to an a.c. source of 100 volt and 50Hz. A pure resistor under the same conditions takes a current of 10A. If two are connected in series to an a.c. supply of 100V and 40Hz, then the current in the series combination of above resistor and inductor is

(A) 10A **(B) 8A**

(C) $5\sqrt{2}$ amp **(D) $10\sqrt{2}$ amp**

Sol. $X_L = \frac{100}{8}, R = \frac{100}{10} = 10\ \Omega$

$$L \times 100\pi = \frac{100}{8} \text{ or } L = \frac{1}{8\pi} H$$

$$Z = \sqrt{\left(\frac{1}{8\pi} \times 2\pi \times 40\right)^2 + 10^2} = 10\sqrt{2}$$

$$I = \frac{E}{Z} = \frac{100}{10\sqrt{2}} = \frac{10}{\sqrt{2}} = 5\sqrt{2}A$$

$\therefore$(C)

9. A resistor R, an inductor L and a capacitor C are connected in series to a source of frequency n. If the resonant frequency is n_r, then the current lags behind voltage when

(A) $n = 0$ **(B) $n < n_r$**

(C) $n = n_r$ **(D) $n > n_r$**

Sol. Below resonant frequency the current leads the applied emf, at resonance it is in phase with applied emf and above resonance frequency it lags the applied emf.

10. An AC source of angular frequency ω is fed across a resistor R and a capacitor C in series. The current registered is I. If now the frequency of source is changed to ω/3 (but maintaining the same voltage), the current in the circuit is found to be halved. The ratio of reactance to resistance at the original frequency ω will be.

(A) $\sqrt{\frac{3}{5}}$ **(B) $\sqrt{\frac{5}{3}}$**

(C) 3/5 **(D) 5/3**

Sol. According to given problem,

$$I = \frac{V}{Z} = V / [R^2 + (1/C\omega^2)]^{1/2} \quad \ldots (1)$$

and $$\frac{I}{2} = \frac{V}{[R^2 + (3/C\omega)^2]^2} \quad \ldots (2)$$

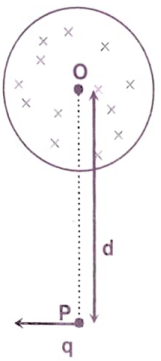
substituting the value of I from equation (1) in (2),

$$4\left(R^2 + \frac{1}{C^2\omega^2}\right) = R^2 + \frac{9}{C^2\omega^2}, \quad \text{i.e., } \frac{1}{C^2\omega^2} = \frac{3}{5}R^2$$

so that $$\frac{X}{R} = \frac{(1/C\omega)}{R} = \frac{[(3/5)R^2]^{1/2}}{R} = \sqrt{\frac{3}{5}}$$

$\therefore$ (A)

11. In the cylindrical region of radius R = 2m, there exists a time varying magnetic field B such that $\frac{dB}{dt} = 2$ Tesla/sec. A charged particle having charge q = 2C is placed at the point P at a distance d from its center O. Now, the particle is moved in the direction perpendicular to line OP by an external agent up to infinity so that there is no gain in kinetic energy of charged particle. Then



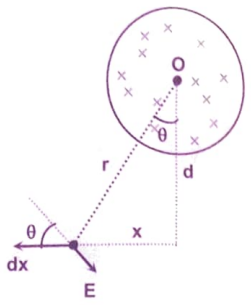
(A) work done by external agent is 4π Joule when d = 4m

(B) work done by external agent is 4π Joule when d = 8m

(C) work done by external agent is independent of d.

(D) work done by external agent is positive.

Sol. A, B, C, D



$$\int \vec{E}\cdot d\vec{\ell} = A\cdot\frac{dB}{dt}$$

$$E\cdot 2\pi\sqrt{x^2+d^2} = \pi R^2 k \text{ , where } \left(\frac{dB}{dt}=k\right)$$

$$E = \frac{R^2k}{2\sqrt{x^2+d^2}}$$

$$W = \int_0^\infty q\vec{E}\cdot d\vec{x}$$

$$= \int_0^\infty qE\cos\theta\cdot dx = \int_0^\infty \frac{qR^2k}{2\sqrt{x^2+d^2}}\frac{d}{\sqrt{d^2+x^2}}dx = \frac{qR^2k\pi}{4}$$

Alternate solution

By applying Faraday's Law for an imaginary triangular loop having vertex O, P and point at infinity, $q\int \vec{E}\cdot d\vec{\ell}$ can be found out.

12. A uniform conducting ring of mass m = 2 kg, radius r = 2m and resistance 8 Ω is kept on smooth, horizontal surface. A time varying magnetic field $\vec{B} = (\hat{i} + t^2\hat{j})$ Tesla is present in the region, where t is time in second and take vertical as y-axis. (Take $\pi^2 = 10$). Then

(A) Time when ring starts toppling is 1 sec.

(B) Time when ring starts toppling is 3/4 sec.

(C) Heat generated through the ring till the instant when the ring start toppling is 40/3 Joule.

(D) Heat generated through the ring till the instant when the ring start toppling is 80/3 Joule.

Sol. A, D

$$\phi = BA = (t^2)(\pi r^2)$$

$$\varepsilon = \frac{d\phi}{dt} = 2\pi r^2 t$$

$$i = \frac{\varepsilon}{R} = \frac{2\pi r^2 t}{R}$$

For toppling

$$\tau = |\vec{\mu}\times\vec{B}| \geq mgr$$

$$(i.\pi r^2).1 \geq mgr$$

$$\frac{2\pi r^2 t}{R}\cdot\pi r^2 \geq mgr$$

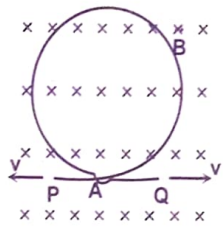
$$t \geq \frac{mgR}{2\pi^2 r^3} = 1\,\text{sec}$$

$$\text{Heat} = \int i^2 R dt = \int\left(\frac{2\pi r^2 t}{R}\right)^2 dt.R$$

$$\text{Heat} = \frac{4\pi^2 r^4}{R}\int_0^1 t^2 dt = \frac{80}{3}\,\text{Joule}$$

COMPREHENSION -1

A circular conducting loop of radius r_0 and having resistance per unit length λ is placed in a constant magnetic field B which is perpendicular to plane of loop. The ends P and Q of the wire are pulled in opposite directions with a constant velocity v such that loop remains circular and the radius of the loop goes on decreasing, then answer the following questions.



13. Radius of the loop varies with time t as

(A) $r_0 - \frac{2vt}{\pi}$ (B) $r_0 - \frac{vt}{2\pi}$

(C) $\frac{r_0}{2} - \frac{vt}{\pi}$ (D) $r_0 - \frac{vt}{\pi}$

Sol. D

14. Magnitude of the emf induced in the loop as a function of time t is

(A) $2Bv\left[\frac{r_0}{2} - \frac{vt}{\pi}\right]$ (B) $2Bv\left[r_0 - \frac{vt}{2\pi}\right]$

(C) $2Bv\left[r_0 - \frac{vt}{\pi}\right]$ (D $2Bv\left[r_0 - \frac{3vt}{2\pi}\right]$

Sol. C

15. Current induced in the loop is

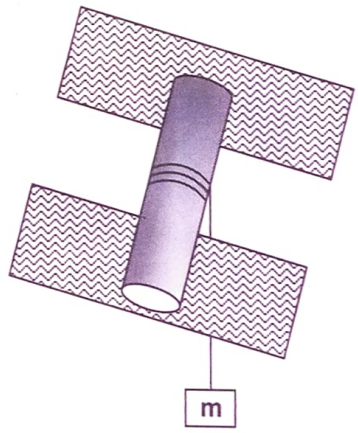
(A) $\frac{Bv}{\pi\lambda}$ (B) $\frac{2Bv}{\pi\lambda}$

(C) $\frac{Bv}{2\pi\lambda}$ (D) $\frac{Bv}{4\pi\lambda}$

Sol. A

COMPREHENSION -2

Charge 'Q' is uniformly distributed on the non conducting surface of a thin uniform long hollow cylinder of mass 'm' radius 'R' and length 'ℓ'. It is placed on two stationary rough horizontal planks. A string is wound over the cylinder whose other end is attached to a block of same mass 'm'. When the system is released from rest, the cylinder starts rolling without slipping relative to the planks. Now answer the following questions. (Take $\frac{\mu_0 Q^2}{4\pi\ell} = m$)



16. The acceleration of the hollow cylinder is

(A) 2g (B) $\frac{3g}{4}$

(C) $\frac{g}{2}$ (D) $\frac{g}{4}$

Sol. **D**

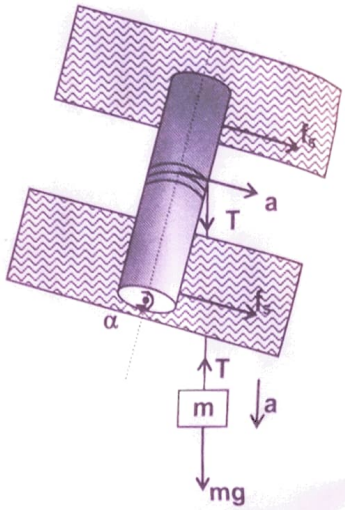
17. The tension in the string is

(A) $\frac{3mg}{4}$ (B) $\frac{3mg}{2}$

(C) $\frac{mg}{4}$ (D) $\frac{mg}{2}$

Sol. **A**

Sol.16-17.



Current over the surface of cylinder is

$$I = \frac{Q\omega}{2\pi}$$

$$B = \frac{\mu_0 I}{\ell} = \frac{\mu_0 Q\omega}{2\pi\ell}$$

Electric field induced on the surface of the cylinder is

$$E = \frac{R}{2}\frac{dB}{dt} = \frac{R}{2}\frac{\mu_0 Q}{2\pi\ell}\frac{d\omega}{dt}$$

$$E = \frac{\mu_0 Q\alpha R}{4\pi\ell} = \frac{\mu_0 Qa}{4\pi\ell} \quad \ldots(i)$$

$mg - T = ma$ ...(ii)

$TR - 2f_sR - QER = mR^2\alpha$

$T - 2f_s - QE = ma$ ...(iii)

$2f_s = ma$ ...(iv)

Adding (iii) and (iv), we get

$T - QE = 2ma$ ...(v)

From (ii) and (v), we get

$mg - QE = 3ma$

$$mg - \frac{\mu_0 Q^2 a}{4\pi\ell} = 3ma$$

$mg - ma = 3ma$

$\therefore a = g/4$

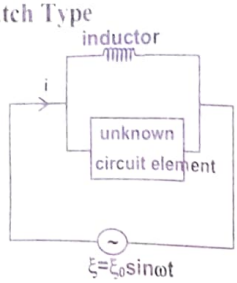
Now from equation (ii)

$mg - T = ma$

$$T = mg - \frac{mg}{4} = \frac{3mg}{4}$$

Matrix-Match Type

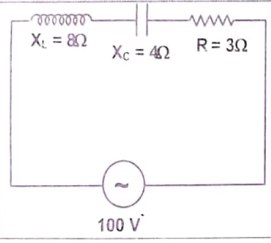
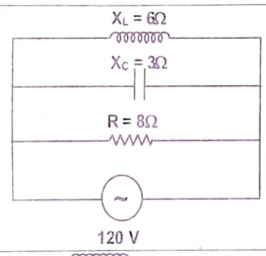
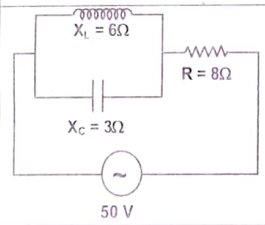
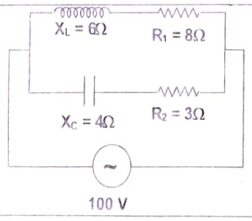
18.



Column – I		Column – II	
If current i in the circuit is		Unknown circuit element may be	
(A)	$i = i_0 \sin\left(\omega t - \frac{\pi}{4}\right)$	(p)	L – C series combination
(B)	$i = i_0 \sin\left(\omega t + \frac{\pi}{4}\right)$	(q)	L – R series combination
(C)	$i = i_0 \sin \omega t$	(r)	R – C series combination
(D)	$i = i_0 \cos \omega t$	(s)	L – C – R series combination

Sol. (A) → (q, r, s) (B) → (r, s) (C) → r, s (D) → p

19. List –I contains different AC circuits with rms voltage of ac sources as shown. List –II contains the corresponding rms currents supplied by the ac voltage sources. Then match the options of List-I correctly with the options of List-II.

LIST–I		LIST–II	
P.		1.	I_{rms} = 25 A
Q.		2.	I_{rms} = 5 A
R.		3.	$I_{rms} = 10\sqrt{5}$ A
S.		4.	I_{rms} = 20 A

(A) P → 4; Q → 1; R → 2; S → 3
(B) P → 2; Q → 3; R → 1; S → 4
(C) P → 3; Q → 2; R → 4; S → 1
(D) P → 1; Q → 4; R → 2; S → 3

Sol. A

(P) $z = R + (X_L - X_C)i = 3 + (8-4)i = 3 + 4i$

$|z| = \sqrt{(3)^2 + (4)^4} = 5\,\Omega$

$I_{rms} = \frac{\varepsilon_{rms}}{|z|} = \frac{100}{5} = 20\text{ A}$

(Q) $\frac{1}{z} = \frac{1}{X_L i} + \frac{1}{-X_C i} + \frac{1}{R} = \frac{1}{6i} - \frac{1}{3i} + \frac{1}{8} = \left(\frac{1}{3} - \frac{1}{6}\right)i + \frac{1}{8} = \frac{1}{8} + \frac{1}{6}i$

$\Rightarrow \quad z = \frac{48}{(6+8i)}$

$\Rightarrow \quad z = \frac{48(6-8i)}{100}$

$\Rightarrow \quad |z| = \frac{48}{100} \times 10$

$\Rightarrow |z| = 4.8\,\Omega$

$I_{rms} = \frac{\varepsilon_{rms}}{|z|} = \frac{120}{4.8} = 25\text{ A}$

(R) $\frac{1}{z'} = \frac{1}{X_L i} + \frac{1}{-X_C i} = \frac{1}{6i} - \frac{1}{3i} = i\left(\frac{1}{3} - \frac{1}{6}\right) = \frac{i}{6}$

$z' = \frac{6}{i} = -6i$

$z = R + z' = 8 - 6i$

$|z| = \sqrt{(8)^2 + (6)^2} = 10\,\Omega$

$I_{rms} = \frac{\varepsilon_{rms}}{|z|} = \frac{50}{10} = 5\text{ A}$

(S) $z_1 = R_1 + X_L i = 8 + 6i \Rightarrow I_{1(rms)} = \frac{\varepsilon_{rms}}{|z_1|} = \frac{100}{10} = 10\text{ A}$

$z_2 = R_2 - X_C i = 3 - 4i \Rightarrow I_{2(rms)} = \frac{\varepsilon_{rms}}{|z_2|} = \frac{100}{5} = 20\text{ A}$

$\therefore I_{rms} = \sqrt{I_{1(rms)}^2 + I_{2(rms)}^2} = \sqrt{(10)^2 + (20)^2} = \sqrt{500} = 10\sqrt{5}\text{ A}$

Answer questions 20, 21 and 22 by appropriately matching the information given in the three columns of the following table.

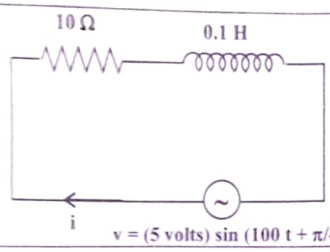
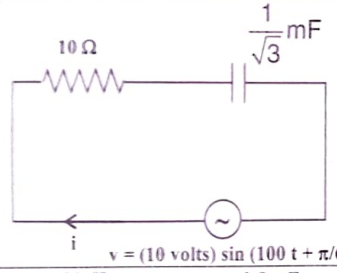
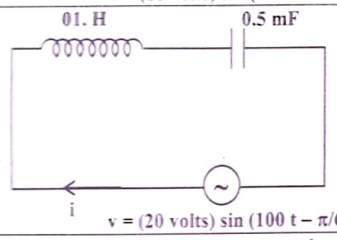
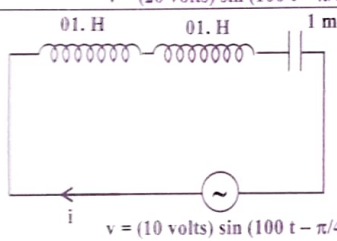
Column –I lists four different AC circuits. If current in the circuits is written as

$i = i_0 \sin(\omega t + \phi))$

where i_0 = amplitude in SI units

ω = 100 rad/s

$-\pi \le \phi \le \phi$ is the phase constant column –II lists value of i_0 and column –III lists the value of ϕ.

Column 1		Column 2		Column 3	
(I)		(i)	$\frac{1}{2}$	(P)	$\frac{\pi}{3}$
(II)		(ii)	1	(Q)	$\frac{\pi}{4}$
(III)		(iii)	$\frac{1}{2\sqrt{2}}$	(R)	Zero
(IV)		(iv)	2	(S)	$\frac{\pi}{2}$

20. **Pick the correct option.**

(A) (I) (iii) (R) (B) (II)(iii) (P)

(C) (III) (i) (R) (D) (IV) (iii) (Q)

Sol. A

21. **Pick the correct option corresponding to maximum value of i_0**

(A) (IV) (iii) (Q) (B) (III)(iv) (P)

(C) (II) (iii) (P) (D) (I) (ii) (S)

Sol. B

22. **Pick the correct option corresponding to maximum value of ϕ.**

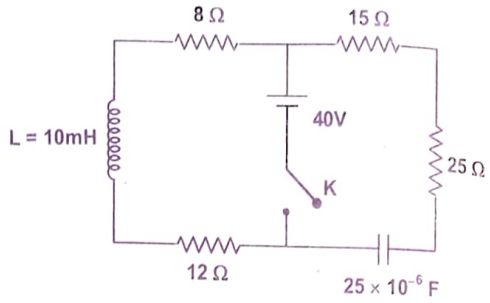
(A) (I) (ii) (S) (B) (II) (i) (S)

(C) (II) (iv) (P) (D) (IV) (iii) (Q)

Sol. B

Numerical Type

23. In the circuit the key (K) is closed at t = 0. Find the current in ampere through the key(K) at the instant $t = 10^{-3}\ell n2$ sec.



Sol. 2

i_1 is current in the first loop

$$i_1 = \frac{40}{20}\left(1 - e^{-\frac{t}{5\times10^{-4}}}\right) = \frac{3}{2}\text{ amp}$$

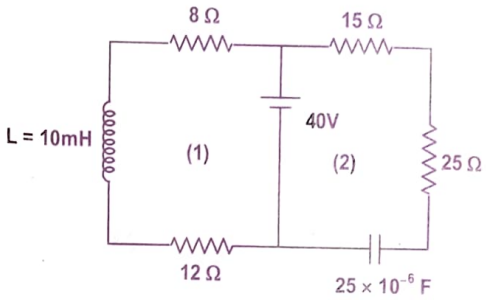
i_2 is current in the second loop

$$i_2 = \frac{40}{40}e^{-\frac{t}{10^{-3}}} = \frac{1}{2}\text{ amp}$$

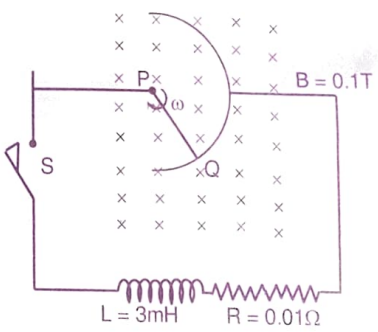
So, total current through key (K)

$i = i_1 + i_2$

$$= \frac{3}{2} + \frac{1}{2} = 2\text{ amp}$$



24. The circuit shows a resistance R = 0.01Ω and inductance L = 3mH connected to a conducting rod PQ of length ℓ = 2m which can slide on a perfectly conducting circular arc of radius ℓ with its centre at P. Assume that friction & gravity are absent and a constant uniform magnetic field B = 0.1T exists as shown in the figure. At t = 0, the circuit is switched on and simultaneously an external torque is applied on the rod so that it rotates about P with a constant angular velocity ω = 2 rad/sec. Find the magnitude of this torque (in N-m) at t = (0.3 ln2) second.



Sol. 4

The half of the maximum current is equal to

$$i_1 = \frac{B\omega\ell^2}{4R} \text{ at } t = \frac{L}{R}\ln2$$

The torque at this instant $= \frac{B^2\omega\ell^4}{8R} = 4$ N-m

Numerical Decimal Type

25. **A circuit has a self inductance of 1 henry and carries a current of 2A. To prevent sparking when the circuit is broken, a capacitor which can withstand 400 volts is used. Find the least capacitance of the capacitor connected across the switch.**

Sol. 00025.00

$$\varepsilon = L\frac{di}{dt}$$

$$400 = 1 \times \frac{2}{dt}$$

$\Rightarrow \quad dt = 5 \times 10^{-3}$ sec.

$Q = idt = 2 \times 5 \times 10^{-3}$

$\therefore \quad Q = 10^{-2}$

But $Q = CV$

$10^{-2} = C \times 400$

$\therefore \quad C = 25\mu F.$

26. **A coil of inductance L = 5H and resistance R = 55 Ω is connected in series to the mains alternating voltage of frequency f = 50 Hz in series. What can be the non-zero capacitance of the capacitor (in μF) connected in series with the coil if the power dissipated has to remain unchanged. (take $\pi^2 = 10$)**

Sol. 00001.00

$$P_{av} = VI\cos\phi = V\left(\frac{V}{z}\right)\left(\frac{R}{z}\right) = \frac{V^2R}{Z^2}$$

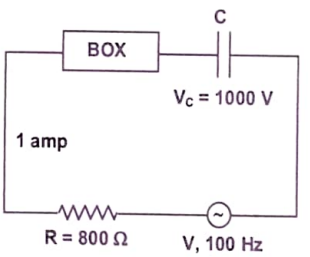
It means Z has to be same in both cases

$$Z_1 = Z_2 \Rightarrow \sqrt{R^2 + X_L^2} = \sqrt{R^2 + (X_C - X_L)^2}$$

$$2X_L = X_C$$

$$\Rightarrow C = \frac{1}{2\omega^2 L} = 10^{-6} = 1 \times 10^{-6}$$

27. **A circuit shown in the figure contains a box having either a capacitor or an inductor. The power factor of the circuit is 0.8, while current lags behind the voltage. Then find the inductance/capacitance of the box in S.I. unit. (take π = 3.2) The circuit draws 1 ampere current.**



Sol. 00002.50

$$X_C = \frac{V_C}{I} = \frac{1000}{1} = 1000\Omega$$

Current lags behind voltage

So, box has an inductor

$$\text{Power factor} = 0.8 = \frac{R}{Z} = \frac{800}{\sqrt{(800)^2 + (X_L - 1000)^2}}$$

$$X_L = 1600\ \Omega$$

$$\omega L = 1600$$

$$L = \frac{1600}{2 \times 3.2 \times 100} = 2.5\ H$$

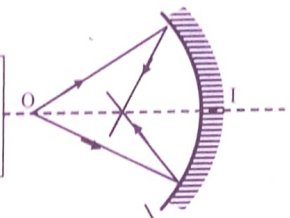
OPTICS 15

GEOMETRICAL OPTICS

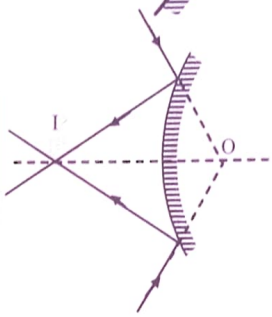
The point of intersection of the incident rays is called **object** and the point of intersection of the corresponding reflected/refracted rays is called **image**.

OBJECT

REAL
If the incident rays diverge from an object, the object is a real object.

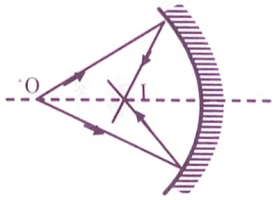


VIRTUAL
If incident rays converge and appear to intersect at a point behind the mirror (or lens), the point is a virtual object.

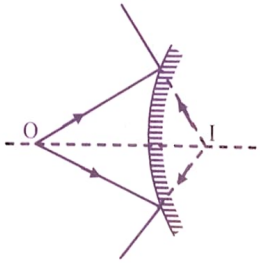


IMAGE

REAL
If the reflected (refracted) rays intersect, the point of intersection is the real image. An image, which is formed due to the actual intersection of light rays (can be held on a screen) is called a real image.



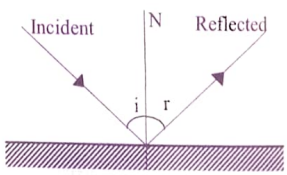
VIRTUAL
If the rays diverge after reflection (refraction) a virtual image is formed.



Reflection

Laws of reflection are obeyed at every reflecting surface. There are

1. Angle of incidence = Angle of reflection, i.e. $\angle i = \angle r$
2. The incident ray, the reflected ray and the normal to the surface at the point of incidence, all lie in the same plane.



- When a plane mirror is rotated through an angle θ, the reflected ray turns through, the same angle w.r.t normal i.e. θ.
- Deviation caused by single reflection = $180^\circ - 2i$ (i = angle of incidence)
- The image so formed is laterally inverted. The linear magnification is unity.

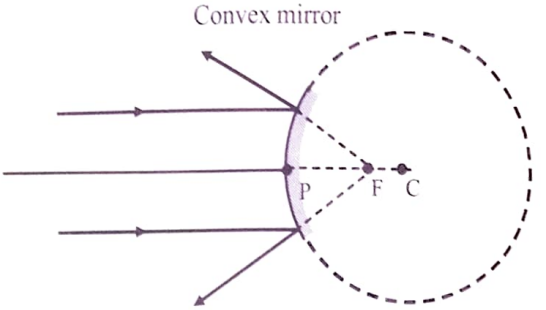
Spherical Mirrors

Mirrors which are part of a symmetrical curved surfaces, like sphere, ellipsoid, cylinder etc. are called curved mirrors and that which are part of spheres are called spherical mirrors.

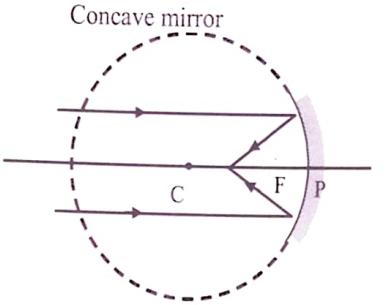
They are of two types (i) convex (ii) concave

The mirrors in which the reflection takes place at the bulged surface are called **convex mirrors** and the mirrors in which reflection takes place at the depressed surface are called **concave mirrors.**

Convex mirror

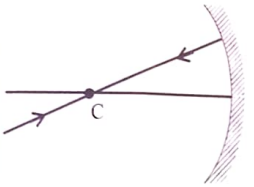
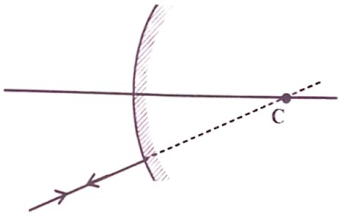
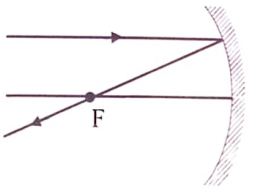
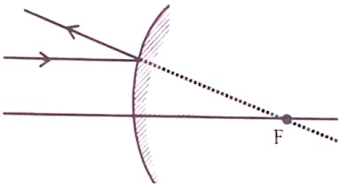
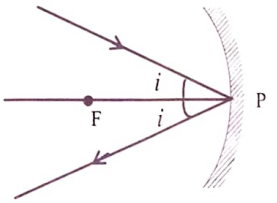
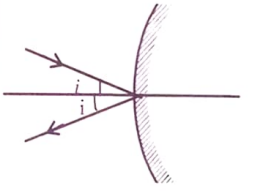


Concave mirror



Ray Tracing

In geometrical optics, to locate the image of an object, tracing of a ray, as it reflects or refracts, is very important

Properties	Concave	Convex
1. A ray going through the centre of curvature is reflected back along the same direction.		
2. A ray parallel to principal axis is reflected through the focus, and vice-versa. Also, mutually parallel rays after reflection intersect at the focal point.		
3. A ray incident at the pole and the reflected ray from it make equal angles with the principal axis.		

New Cartesian Sign Convention

The new Cartesian sign conventions are:

(i) All distances are measured from the pole P.

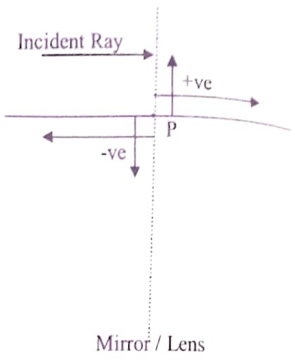
(ii) Distances measured along the direction of incident rays are taken as positive.

(iii) Distances measured along a direction opposite to the incident rays are taken as negative.

(iv) Distances above the principal axis are taken as positive.

(v) Distances below the principal axis are taken as negative.

Angles measured from the normal, in anti-clockwise direction are positive, while in clockwise direction are negative.



Mirror Formula: $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

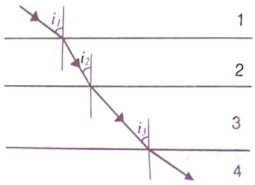
$$m = \text{magnification} = \frac{-v}{u}$$

Refraction

Laws of refraction

1. The incident ray, the refracted ray and the normal to the surface separating the two media - all lies in one plane.

2. **Snell's law**: For any two media the ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant for a light beam of a particular frequency, i.e.



$\mu_1 \sin i_1 = \mu_2 \sin i_2 = \mu_3 \sin i_3 = \mu_4 \sin i_4 = \mu_5 \sin i_5 = \text{constant}$

This is called Snell's law

Note: $i_2 = r_1,\ i_3 = r_2,\ i_4 = r_3,\ i_5 = r_4$

This equation is found to be very useful for solving problems involving a variable refractive index.

For any two mediums in contact with each other

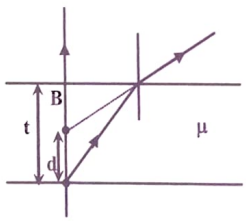
Refractive index of 2nd medium with respect to 1st medium is given by

$${}_1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{v_1}{v_2}$$ where v_1 = speed of light in medium (1) and

v_2 = speed of light in medium (2)

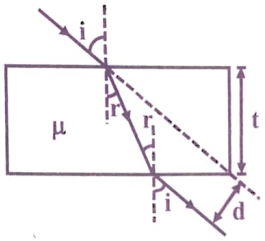
3. Apparent shift

$$d = t\left(1-\frac{1}{\mu}\right)$$



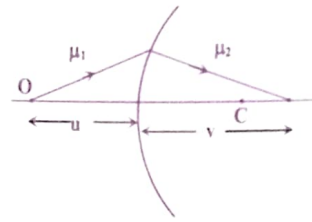
(4) Lateral Shift.

$$d = \frac{t}{\cos r}\sin(i-r)$$



Refraction at Spherical Surfaces

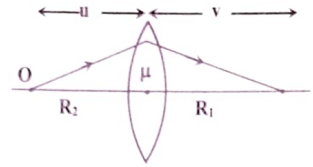
For a spherical surface, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$



Magnification, $m = \frac{\mu_1 v}{\mu_2 u}$

THIN LENS

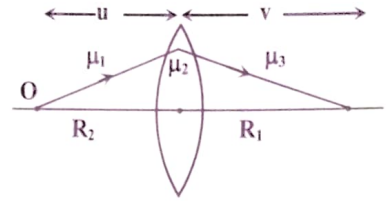
Lens Maker's Formula: $\frac{1}{v} - \frac{1}{u} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$



Thin Lens Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Magnification, $m = \frac{v}{u}$

If different media exist on two sides of the lens, then



$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R_1} + \frac{(\mu_3 - \mu_2)}{R_2}$$

Note: (i) v = focal length, if $u = \infty$

(ii) Above formula are valid for thin lenses only.

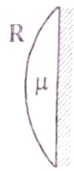
- **Power of Combination of Lenses**

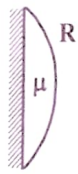
  Power of a lens $= \frac{1}{f}$, where f is the focal length of the lens in meters with proper sign

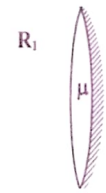
  $P = P_1 + P_2 + P_3 \ldots$.

  where P_1, P_2, $P_3 \ldots$. are powers of constituent lenses.

- **Silvering at One Surface**







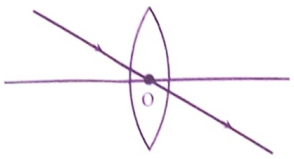
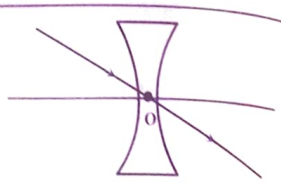
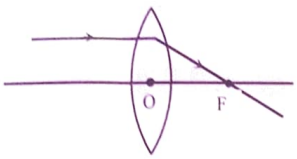
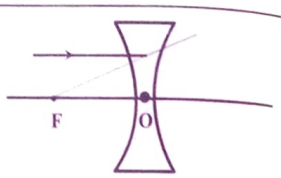
$P_M = P_L + P_{mirror} + P_L$

(Powers to be taken with proper sign), where P_L = power of lens

$$-\frac{1}{F_{effective}} = \frac{2}{f_l} + \frac{1}{(-f_m)}$$

Note: A lens is converging if its focal length is positive and diverging if negative

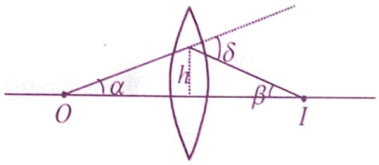
Ray Tracing for Lens

Properties	Convex	Concave
1. A ray passing through optical centre goes undeviated.		
2. When a ray is incident parallel to the principal axis, the refracted ray passes or appears to pass through the focus.		

Angle of Deviation of a Ray When it Passes Through a Lens

Let δ is the angle of deviation.

$\delta = \alpha + \beta \approx \tan\alpha + \tan\beta$ [α and β are the angle as shown in figure]

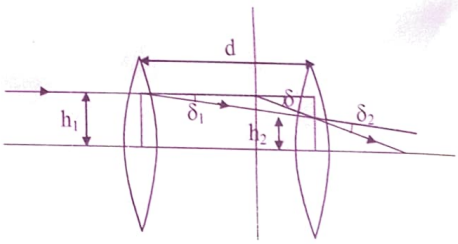


$$= \frac{h}{-u} + \frac{h}{v} = h\left[\frac{1}{v} - \frac{1}{u}\right]$$

$$\Rightarrow \quad \delta = \frac{h}{f}$$

Two Thin Lenses Separated by a Distance d

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$



(This expression is valid for parallel rays only i.e. when the object is placed at infinity)

A more powerful form of this formula is,

$$P = P_1 + P_2 - \frac{d}{\mu} P_1 P_2$$

where P_1 and P_2 are optical powers of the two lenses, and μ is the refractive index of the medium in between them.

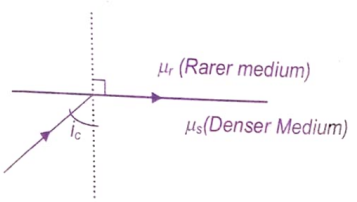
(The focal length calculated through above formula cannot be used with the formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$)

TOTAL INTERNAL REFLECTION

Let i_c be the angle for the total internal reflection phenomena to take place, then

i_c = critical angle

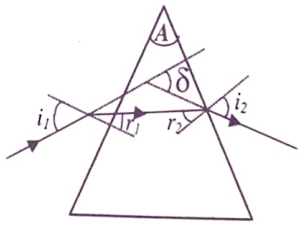
$$\sin i_c = {}_d\mu_r = \frac{1}{{}_r\mu_d} \quad i_c = \text{cosec}^{-1}\left({}_r\mu_d\right)$$



PRISM

The angle of deviation in a prism is given by

$\delta = i_1 + i_2 - r_1 - r_2$, where $A = r_1 + r_2$



(a) $\mu = \dfrac{\sin\left(\dfrac{A+\delta_m}{2}\right)}{\sin\left(\dfrac{A}{2}\right)}$ [δ_m = angle of minimum deviation, A = angle of prism]

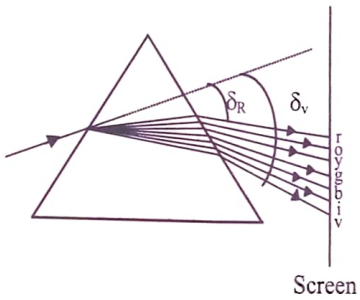
(b) If the angle of prism A is very small

$\delta = (\mu - 1)A$

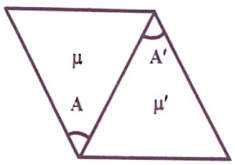
Dispersive Power: It is defined as the ratio of angular dispersion to the mean deviation.

$\omega = \dfrac{\delta_v - \delta_r}{\delta}$, where δ is deviation of mean ray (yellow)

$\therefore \omega = \dfrac{\mu_v - \mu_r}{\mu_y - 1}$, where μ_v, μ_R, μ_y = R.I. of violet, red and yellow lights.



Dispersion without Deviation



For deviation to be zero, $(\delta + \delta') = 0$

$\Rightarrow \quad (\mu - 1)A + (\mu' - 1)A' = 0$

$\Rightarrow \quad A' = -\dfrac{(\mu - 1)A}{(\mu' - 1)}$

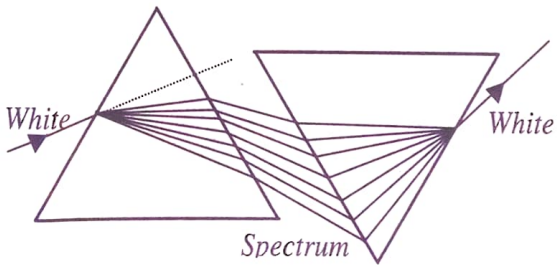
Deviation without Dispersion

Resultant of Angular Dispersion is zero,

i.e., $\omega\delta + \omega'\delta' = 0$

or, $(\mu_v - \mu_r)A + (\mu'_v - \mu'_r)A' = 0$

or, $\dfrac{A'}{A} = -\left[\dfrac{\mu_v - \mu_r}{\mu'_v - \mu'_r}\right]$



- If two coherent waves with intensity I_1 and I_2 are superimposed with a phase difference of ϕ, the resulting wave intensity is

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

- For maxima, $x = n\lambda$ (ϕ = optical path difference)

For minima, $x = (n - \frac{1}{2})\lambda$ or $(n + \frac{1}{2})\lambda$

Phase difference, $\phi = \frac{2\pi}{\lambda}(x)$

Determination of Phase Difference

The phase difference between two waves at a point will depend upon

(a) the difference in path lengths of the two waves from their respective sources.

(b) the refractive index of the medium

(c) initial phase difference, if any.

(d) Reflections, if any, in the path followed by waves.

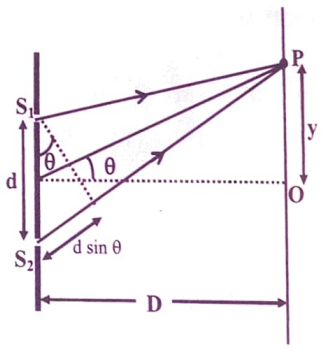
(i) In the case of light waves, the phase difference on account of path difference

$$= \left(\frac{\text{optical path difference}}{\lambda}\right)2\pi$$ where , λ is the wavelength in free space.

(ii) In the case of reflection, the reflected disturbance differs in phase by π with the incident wave if the incidence occurs in rarer medium. There would be no phase difference if incidence occurs in denser medium.

Note: A distance Δx in vaccum is equal to a distance $\Delta x/\mu$ in a medium of refractive index μ, called the optical path

YOUNG'S DOUBLE SLIT EXPERIMENT



(i) If $d \ll D$, then

$\Delta x = S_2P - S_1P = d\sin\theta$

If $\lambda \ll d$, then $\sin\theta \approx \theta \approx \tan\theta$

$\Delta x = \frac{dy}{D}$

(ii) For maxima $\frac{dy}{D} = n\lambda$

or, $y = 0, \pm\frac{D\lambda}{d}, \pm\frac{2D\lambda}{d}$,

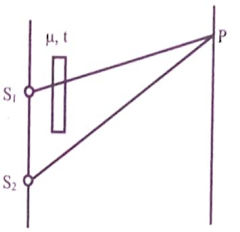
(iii) For minima, $\frac{dy}{D} = [n + (1/2)]\lambda$

or, $y = \pm\frac{D\lambda}{2d}, \pm\frac{3D\lambda}{2d}$, so on

(iv) Fringe width , $\beta = \frac{\lambda D}{d}$

Displacement of Fringe Pattern

When a film of thickness 't' and refractive index 'μ' is introduced in the path of one of the source's light, then fringe shift occurs as the optical path difference changes.



Optical path difference at P

$$\Delta x = S_2P - [S_1P + \mu t - t]$$

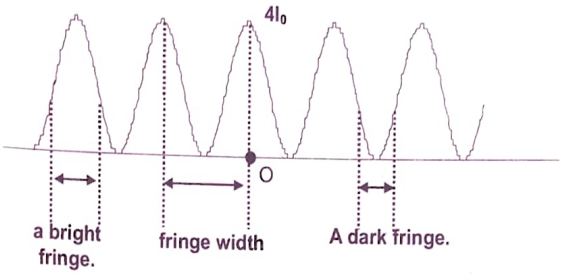
$$= S_2P - S_1P - (\mu - 1)t = y.(d/D) - (\mu - 1)t$$

The fringe shift is given by, $\Delta y = \frac{D(\mu-1)t}{d}$

Intensity Variation on Screen

If I_0 is the intensity of light beam coming from each slit, then the resultant intensity at a point where they have a phase difference of ϕ is given by

$$I = 4I_0 \cos^2\frac{\phi}{2}, \text{ where } \phi = \frac{2\pi(d\sin\theta)}{\lambda}$$



ILLUSTRATIVE PROBLEMS

1. An object is placed at a distance of 20 cm from a equi-concave lens of focal length 30cm, whose one surface is silvered. If the radius of curvature of the silvered surface is 60 cm, find the position and nature of the image.

Sol. $\frac{-1}{F} = \frac{2}{f} - \frac{1}{f_m} = \frac{2}{-30} - \frac{1}{+30} = -\frac{3}{30}$

$\Rightarrow F = +10$ cm

$\therefore \frac{1}{F} = \frac{1}{v} + \frac{1}{u}$

$\Rightarrow \frac{1}{10} = \frac{1}{v} + \frac{1}{-20}$

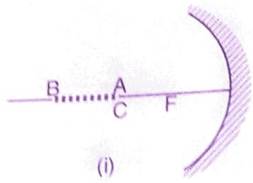
$\Rightarrow v = +20/3$ cm

as $m = -\frac{v}{u} = -\frac{-20/3}{-20} = \frac{1}{3}$

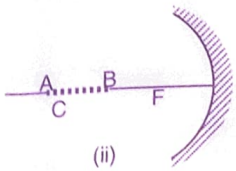
$\therefore$ Image will be virtual, erect and diminished.

2. A thin rod of length (f/3) is placed along the principle axis of a concave mirror of focal length 'f' such that its image which is real and elongated, just touches the rod. What is the linear magnification?

Sol. Image of rod will just touch the rod only when one end of rod is at 'C' center of curvature. So we have two options as illustrated in following figures.



(i)



(ii)

If 'ℓ'' be the length of image

In case I

$$\frac{1}{V_B}-\frac{3}{(7f)}=-\frac{1}{f}\Rightarrow V_B=-\frac{7}{4}f \quad \& \quad \ell'=f\left|\frac{7}{4}-2\right|=f/4$$

as $\ell' < \ell$ this is definitely not the case.

In case II

$$\frac{1}{V_B}-\frac{3}{5f}=-\frac{1}{f} \quad \Rightarrow V_B=-\frac{5f}{2} \text{ and } \ell'=f\left|\frac{5}{2}-2\right|=\frac{f}{2}$$

And the required linear magnification $=\dfrac{\ell'}{\ell}=\dfrac{3}{2}$

3. A thin biconvex lens of refractive index 3/2 is placed on a horizontal plane mirror as shown in the figure.

The space between the lens and the mirror is then filled with water of refractive index 4/3. It is found that when a point object is placed 15cm above the lens on its principle axis, the object coincides with its own image. On repeating with another liquid, the object and the image again coincide at a distance 25 cm from the lens. Calculate the refractive index of the liquid.

Sol. RI of lens material = μ_2; RI of first liquid = μ_3 ; RI of air = μ_1; RI of second liquid = μ'_3

In general $(\mu_3/v)-(\mu_1/u)=\{(\mu_2-\mu_1)/R\}+\{(\mu_3-\mu_2)/R\}$

Here $u=-15$ cm, $v=\infty$

Now, $1/(+15)=[(3/2)-1\}/R]-[\{(4/3)-(3/2)\}/R]$

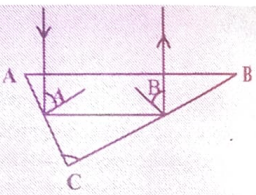
$\Rightarrow R = 5$ cm

For the second liquid, similarly,

$(1/25)=(0.5/R)-[\{\mu'_3-(3/2)\}/R]$

$\Rightarrow \mu'_3 = 1.8$

4. A right prism is to be made by selecting a proper material and angles A and B ($B \le A$) as shown in the figure. It is desired that a ray of light incident on the face AB emerges parallel to the incident direction after two reflections.



(a) What should be the minimum refractive index 'n' for this to be possible?

(b) For n=5/3, is it possible to achieve this with the angle B equal to 30°?

Sol. (a) As already discussed

$A \ge \theta_c$; $B \ge \theta_c$ Also $B \le A$

Case 1: $B = A = \pi/4 \Rightarrow \theta_c \le \pi/4$ ... (1)

Case2: $B < A$. Hence, for reflection at both faces

$B > \theta_c$

But, $B < \pi/4 \Rightarrow \theta_c < \pi/4$ (as $2B < \pi/2$) ... (2)

Combining Eqs. (1) and (2), we get $\theta_c \le \pi/4$

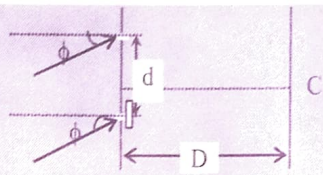
$$\frac{1}{n} \le \frac{1}{\sqrt{2}} \Rightarrow n \ge \sqrt{2}$$

$$n_{min} = \sqrt{2}$$

(b) This pertains to case (2)

Hence, $\theta_c < 30°$ But $\theta_c = \sin^{-1}(3/5)$ which is more than $30°$ and hence, it is not possible.

5. Light of wavelength $\lambda = 500$ nm falls on two narrow slits placed a distance $d = 50 \times 10^{-4}$ cm apart, at an angle $\alpha = 30°$ relative to the slits shown in figure. On the lower slit a transparent slab of thickness 0.1 mm and refractive index 3/2 is placed. The interference pattern observed on screen at a distance D = 2m from the slits. Then calculate



(a) position of the central maxima ?

(b) the order of minima closest to centre C of screen ?

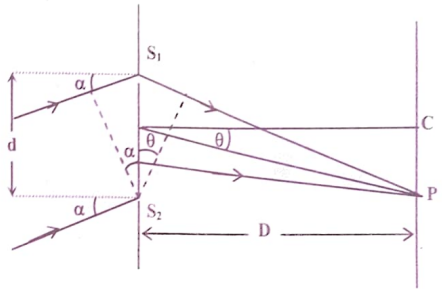
(c) How many fringes will pass over C, if we remove the transparent slab from the lower slit ?

Sol. Phase difference $\Delta\phi = kd \sin\phi + kd \sin\theta + k\,b\,(1 - n)$, where $k = \frac{2\pi}{\lambda}$

Central maxima is obtained when $\Delta\phi = 0$

$kd \sin\alpha + kd \sin\theta + kb(1 - n) = 0$

$$\sin\theta = \frac{b(n-1)}{d} - \sin\alpha = \frac{1}{2} \Rightarrow \text{therefore } \theta = 30°$$



At C, $\theta = 0$

$$\Delta\phi = kd \sin\alpha - k\,b\,(n-1) = \frac{kd}{2}\left[1 - \frac{2b}{d}(n-1)\right]$$

$$= \frac{2\pi d}{2\lambda}\left[1 - \frac{2b}{d}(n-1)\right]$$

Substituting the values, we get, $\Delta\phi = -100\,\pi = 2m\,\pi \Rightarrow m = -50$

As $I \propto \cos^2\left(\frac{\Delta\phi}{2}\right)$

Hence at C there will be maxima.

For, minima

$\Delta\phi = (2m - 1)\,\pi$

for $m = -50$ and -49, we get $\Delta\phi = -101\pi$ and $-99\,\pi$, respectively.

Hence the minima closest to $\theta = 0$ are of order $m = -49$ and -50 on both side

The phase difference at C when we remove the slab

$\Delta\phi = 100\,\pi$. i.e. $m = 50$

Hence number of fringes passed over = 150

SOLVED PROBLEMS

1. **A beaker containing liquid is placed on a table, underneath a microscope which can be moved along a vertical scale. The microscope is focused, through the liquid onto a mark on the table when the reading on the scale is a. It is next focused on the upper surface of the liquid and the reading is b. More liquid is added and the observations are repeated, the corresponding readings are c and d. The refractive index of the liquid is**

(A) $\frac{d-b}{d-c-b+a}$ (B) $\frac{b-d}{d-c-b+a}$

(C) $\frac{d-c-b+a}{d-b}$ (D) $\frac{d-b}{a+b-c-d}$

Sol. The real depth = (R.I.) (apparent depth)

$\Rightarrow$ In first case, the real depth $h_1 = n(b-a)$

Similarly, in the second case, the real depth $h_2 = n(d-c)$

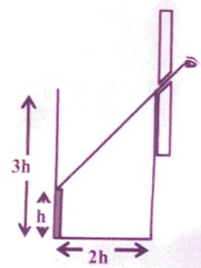
Since, $h_2 > h_1$ the difference of real depths $= h_2 - h_1 = n(d-c-b+a)$.

Since the liquid is added in second case, $h_2 - h_1 = d - b$

$\Rightarrow n = \frac{d-b}{d-c-b+a}$

$\therefore$ (A)

2. **An observer can see through a pin-hole the top end of a thin rod of height h, placed as shown in the figure. The beaker height is 3h and its radius h. When the beaker is filled with a liquid up to a height 2h, he can see the lower end of the rod. Then the refractive index of the liquid is**



(A) $\frac{5}{2}$ (B) $\sqrt{\frac{5}{2}}$

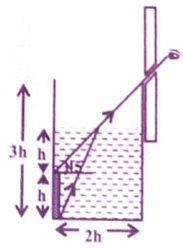
(C) $\sqrt{\frac{3}{2}}$ (D) $\frac{3}{2}$

Sol. $\sin i / \sin r = 1/\mu$; $r = 45°$;

$\sin i = 1/\sqrt{5}$

$\Rightarrow \mu = (5/2)^{1/2}$

$\therefore$ (B)



3. **The focal length of a convex lens of R.I. 1.5 is f when it is placed in air. When it is immersed in a liquid it behaves as a converging lens its focal length becomes xf (x > 1). The refractive index of the liquid**

(A) > 3/2 (B) < (3/2) and > 1

(C) < 3/2 (D) All of these

Sol. $1/f = (n-1)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$

$\Rightarrow 1/f = \left(\frac{1.5}{1}-1\right)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$ when the lens is placed in air and

$1/xf = \left(\frac{1.5}{y}-1\right)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$ when the lens is places in the liquid.

where y = R.I. of the liquid

On solving, we get

$y = \frac{3}{2+1/x}$.

$\therefore$ (B)

4. The angle of a prism is 30°. The rays incident at 60° on one refracting face suffer a deviation of 30°. Then the angle of emergence is :

(A) 0° **(B) 30°**

(C) 60° **(D) 90°**

Sol. $e = (A+\delta) - i = (30°+30°)-60° = 0°$

$\therefore$(A)

5. When entire Young's double slit apparatus is immersed in a liquid the fringe width decreases by 20%. The refractive index of the liquid is

(A) 1.5 **(B) 1.25**

(C) 0.8 **(D) 1.33**

Sol. $W = \frac{\lambda}{\mu}\frac{D}{d}$, $W_0 = \frac{\lambda D}{d}$

$\Rightarrow \frac{W}{W_0} = \frac{4}{5} = \frac{1}{\mu} \Rightarrow \mu = 1.25$

$\therefore$(B)

6. In Young's double slit experiment, the fringes are displaced by a distance x when a glass plate of refractive index 1.5 is introduced in the path of one of the beams. When this plate in replaced by another plate of the same thickness, the shift of fringes is (3/2)x. The refractive index of the second plate is

(A) 1.75 **(B) 1.50**

(C) 1.25 **(D) 1.00**

Sol. The path difference introduced by a plate of thickness t and refractive index μ is given by

$\Delta = (\mu-1)t$

A path difference of λ introduced a phase shift of β where $\beta = \lambda$ D/(2d) . 2d = separation between slits

So, a path difference of $(\mu - 1)t$ introduces a shift x on the screen given by

$$x = \frac{(\mu - 1)t\beta}{\lambda}$$

for first plate, $x = \frac{(\mu_1 - 1)t\beta}{\lambda}$

for second plate $\frac{3}{2}x = \frac{(\mu_2 - 1)t\beta}{\lambda}$

$$\therefore \frac{(\mu_2 - 1)}{(\mu_1 - 1)} = \frac{3}{2} \quad \Rightarrow \frac{(\mu_2 - 1)}{(1.5 - 1)} = \frac{3}{2}$$

$\Rightarrow \mu_2 = 1.75$

$\therefore$ (A)

7. **The ratio of the intensity at the centre of a bright fringe to the intensity at a point one-quarter of the distance between two fringes from the centre is**

(A) 2 **(B) 1/2**

(C) 4 **(D) 16**

Sol. Two waves of a single source having an amplitude A interfere. The resulting amplitude

$$A_r^2 = A_1^2 + A_2^2 + 2A_1A_2\cos\delta$$

where $A_1 = A_2 = A$ and δ = phase difference between the waves

$\Rightarrow I_r = I_1 + I_2 + 2\sqrt{I_1I_2}\ \cos\delta$

When the maxima occurs at the center, $\delta = 0$

$\Rightarrow I_{r_1} = 4I$ ...(1)

Since the phase difference between two successive fringes is 2π, the phase difference between two points separated by a distance equal to one quarter of the distance between the two, successive fringes is equal to

$$\delta = (2\pi)\left(\frac{1}{4}\right) = \frac{\pi}{2} \text{ radian}$$

$$\Rightarrow I_{r_2} = 4I\cos^2\left(\frac{\pi/2}{2}\right) = 2I \quad \text{...(2)}$$

Using Eqs. (1) and (2),

$$\frac{I_{r_1}}{I_{r_2}} = \frac{4I}{2I} = 2$$

$\therefore$ (A)

8. **Among the two interfering monochromatic sources A and B; A is ahead of B in phase by 66°. If the observation be taken from point P, such that PB – PA = λ/4. Then the phase difference between the waves from A and B reaching P is**

(A) 156° **(B) 140°**

(C) 136° **(D) 126°**

Sol. Total phase difference

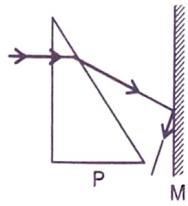
= initial phase difference + phase difference due to path

$$= 66^\circ + \frac{360^\circ}{\lambda} \times \Delta x$$

$$= 66^\circ + \frac{360^\circ}{\lambda} \times \frac{\lambda}{4} = 66^\circ + 90^\circ = 156^\circ$$

$\therefore$ (A)

9. P is a small angled prism of angle 3° made of a material of refractive index 1.5. A ray of light is incident normally to the mirror as shown in figure. M is a plane mirror. The angle of deviation for the ray reflected from the mirror M with respect to the incident ray is :



(A) 4.5° **(B) 175.3°**

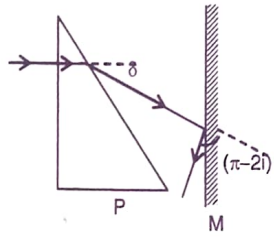
(C) 177° **(D) 178.5°**

Sol. Total deviation $= (\mu - 1)\,A + (\pi - 2i) = 178.5^\circ$

$\delta = i = (\mu - 1)A = 1.5^\circ$

where i = angle of incidence

$\therefore$ (D)



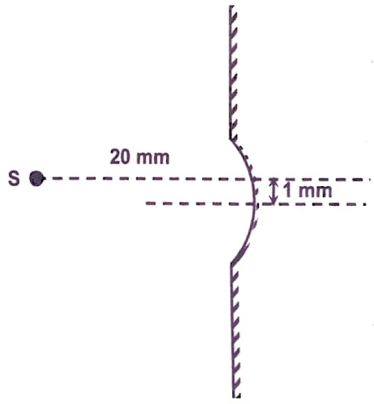
10. A practically very small source (S) of light is kept in front of a plane mirror. But due to imperfect manufacturing, there is a small spherical bulge (radius of curvature 80 mm). The situation is as shown in the figure. You can assume that principal axis of bulge is perpendicular to the plane mirror. Find the separation between the images formed.

(A) 10.025 mm

(B) 20.025 mm

(C) 40.025 mm

(D) 80.025 mm

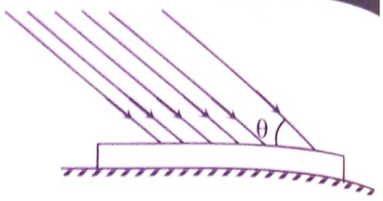


Sol. $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{-20} = \frac{1}{-40} \Rightarrow v = +40\,\text{mm}$

$$m = \frac{-v}{u} = -\left(\frac{+40}{-20}\right) = +2$$

$\Rightarrow$ separation between the two images formed $= \sqrt{(20)^2 + (1)^2} \approx 20.025$ mm

11. Monochromatic rays of intensity I are falling on a metal plate of surface area A placed on a rough horizontal surface at certain angle θ as shown in figure. Choose correct statement (s) based on above information:



(A) There is a value of θ for which plate will not move however high the intensity of radiation is

(B) Plate will not move if plate is perfectly reflecting irrespective of the value of intensity.

(C) If rays are falling perpendiculars to surface plate will not move

(D) None of these

Sol. A, B, C

Basic concept of Intensity of light

12. Choose the correct statement(s).

(A) The image of sun (consider point source of light) can never be produced at center of glass sphere, (assuming the space between sphere and sun is filled with air).

(B) If a lens having μ = 1.5 has focal length 20 cm in air, then its focal length will be 60 cm when it is immersed in water of refractive index $\frac{4}{3}$

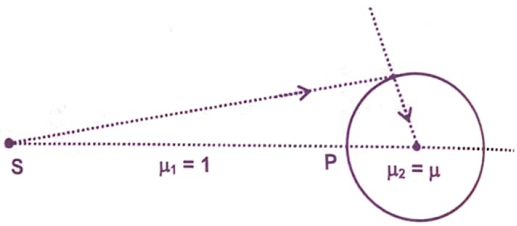
(C) The laws of reflection and refraction are true for all surface and pairs of media at the point of incidence.

(D) The thick lenses give coloured images due to dispersion of light.

Sol. A, C, D

(A) $\frac{\mu}{+R}=\frac{\mu-1}{+R}$ (not possible)

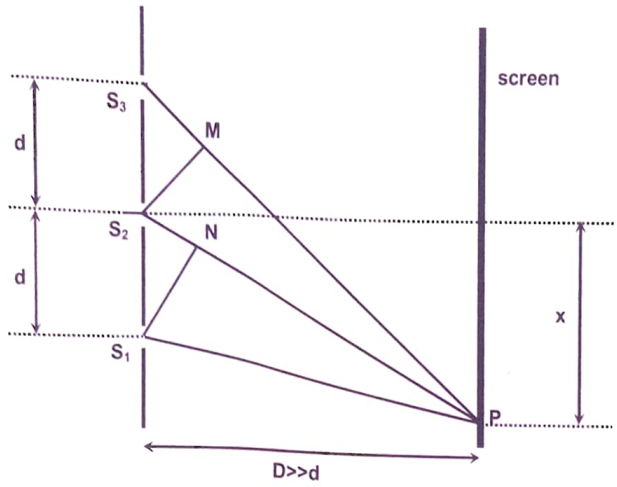
The higher the refractive index of the material of sphere (μ), the closer one could come to fulfilling the objective



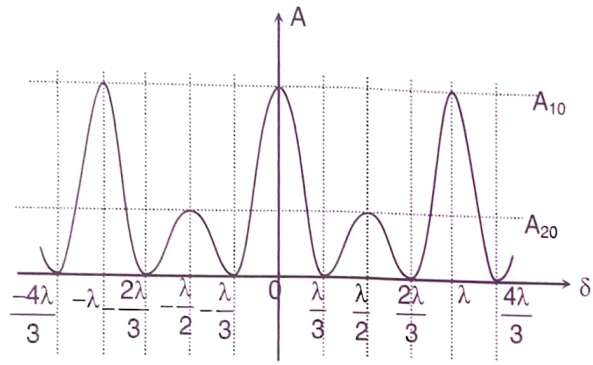
(B) $\frac{f_a}{f_w}=\left(\frac{\mu-\mu_w}{\mu-\mu_a}\right)\left(\frac{\mu_a}{\mu_w}\right)$

$=\left(\frac{\frac{3}{2}-\frac{4}{3}}{\frac{3}{2}-1}\right)\frac{1}{\left(\frac{4}{3}\right)}=\frac{1}{6}\times 2\times\frac{3}{4}=\frac{1}{4}$

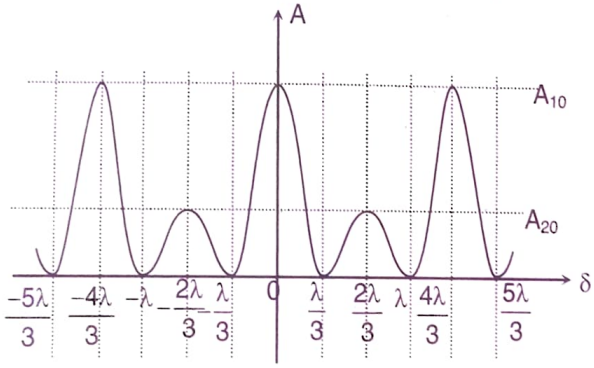
13. A triple slit experiment is performed as shown in the figure with monochromatic light of amplitude a_0. $S_3M \approx S_2N = \delta$. The graph between resultant amplitude (A) versus δ is plotted. Choose the correct option(s).



(A)



(B)



(C) $\frac{A_{10}+A_{20}}{A_{10}-A_{20}} = 2$, where A_{10} and A_{20} are the values shown in option A.

(D) $\frac{A_{10}+A_{20}}{A_{10}-A_{20}} = 5$, where A_{10} and A_{20} are the values shown in option A.

Sol. A, C

$$\phi = \frac{2\pi}{\lambda}\delta$$

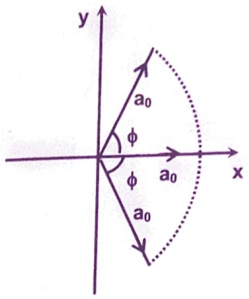
$$y_2 = a_0 \sin(\omega t - kx)$$

$$\Rightarrow y_1 = a_0 \sin(\omega t - kx - \phi) \text{ and } y_3 = a_0 \sin(\omega t - kx + \phi)$$

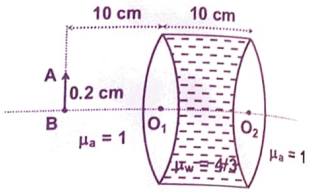
$$A = |a_0 + 2a_0\cos\phi| = a_0\,|(1+2\cos\phi)|$$

$$\Rightarrow A = a_0\,|(1+2\cos\left(\frac{2\pi xd}{\lambda D}\right)|$$

$$= a_0\,|(1+2\cos\left(\frac{2\pi\delta}{\lambda}\right)|$$



14. **Consider an arrangement of two thin equiconvex lenses made of glass (μ_g = 3/2) and each of focal length in air 10 cm are separated by a distance 10 cm. The space between the two lenses is filled with water of refractive index μ_w = 4/3. A small linear object 'AB' of size 0.20 cm is placed perpendicular to the optic axis at a distance 10 cm from the first lens in air as shown. Then choose the correct option(s).**



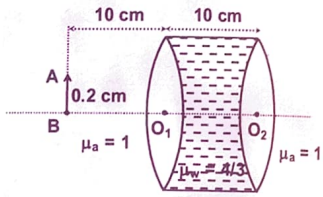
(A) The position of the final image formed is 25 cm right to the second lens.

(B) The position of the final image formed is 40 cm right to the second lens.

(C) The size of the final image formed is 0.40 cm.

(D) The size of the final image formed is 0.60 cm.

Sol. **A, C**



For the image formed by the first lens

$$\frac{\mu_3}{v} - \frac{\mu_1}{u} = \left(\frac{\mu_2 - \mu_1}{R_1}\right) + \left(\frac{\mu_3 - \mu_2}{R_2}\right)$$

$$\Rightarrow \frac{4}{3v} - \frac{1}{(-10)} = \frac{\left(\frac{3}{2}-1\right)}{10} + \frac{\left(\frac{4}{3}-\frac{3}{2}\right)}{-10}$$

$$\frac{4}{3v} + \frac{1}{10} = \frac{1}{20} + \frac{1}{60}$$

$$\Rightarrow \frac{4}{3v} = \frac{1}{15} - \frac{1}{10}$$

$$\Rightarrow v = -40 \text{ cm}$$

Lateral magnification, $m_1 = \frac{\mu_1}{\mu_3}\frac{v}{u} = \frac{1\times 3}{4}\left(\frac{-40}{-10}\right) = +3$

For the image formed by the second lens

$$\frac{\mu_3}{v} - \frac{\mu_1}{u} = \left(\frac{\mu_2 - \mu_1}{R_1}\right) + \left(\frac{\mu_3 - \mu_2}{R_2}\right)$$

$$\Rightarrow \frac{1}{v} - \frac{4}{3(-50)} = \left(\frac{\frac{3}{2}-\frac{4}{3}}{10}\right) + \left(\frac{1-\frac{3}{2}}{-10}\right)$$

$$\Rightarrow \frac{1}{v} + \frac{2}{75} = \frac{1}{60} + \frac{1}{20}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{15} - \frac{2}{75}$$

$\Rightarrow v = +25$ cm

The position of final image formed is 25 cm right to the second lens. Lateral magnification,

$$m_2 = \frac{\mu_1}{\mu_3}\frac{v}{u} = \frac{4}{3\times 1}\left(\frac{25}{-50}\right) = -\frac{2}{3}$$

Overall magnification, $m = m_1 \times m_2 = (+3) \times \left(-\frac{2}{3}\right) = -2$

The size of the final image formed = 2×0.2 cm = 0.4 cm

COMPREHENSION

ABC is an isosceles right angled triangular reflecting prism of average refractive index μ . When incident rays on face AB are parallel to face BC , then they emerges from Face AC , which are also parallel to Face BC as shown in figure-I. The prism capable to do so , known as **Dove Prism.** In figure-II, the **Dove Prism** is used for dispersion of incident light containing red colour and violet colour only. The red colour and violet colour lights are separated(displaced) on screen by a distance $\Delta\ell$. In reality , each ray of any colour has some width, which can be designated as d. It is clear that an observer can distinguish the red and violet rays that emerges from prism only if $\Delta\ell \geq d$ .Otherwise the bundles of rays will overlap and mix.

[Given for **Dove Prism** in figure-II: $\mu_R = \sqrt{\frac{5}{2}}$, $\Delta\mu = \mu_V - \mu_R = 0.02$, $\sqrt{5} = 2.25$, EF = 1m and AB =4cm]

Now answer the following questions.

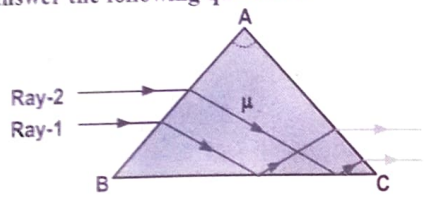


Figure-I

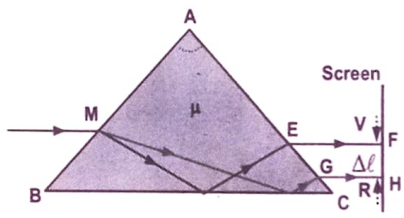


Figure-II

15. Choose the **INCORRECT** option.

(A) As per figure-I, the average refractive index of **Dove Prism** may be greater than 1.

(B) As per figure-I, the average refractive index of **Dove Prism** may be greater than $\sqrt{2}$.

(C) As per figure-II, the displacement $\Delta\ell$ depends upon average refractive index μ and length of AB.

(D) As per figure-II, the displacement $\Delta\ell$ depends upon average refractive index μ and length of EF.

Ans. D

16. Find the maximum value of width d of bundle of rays that can be resolved at the outlet of **Dove Prism** as shown in the figure-II.

(A) 2.25×10^{-3} m

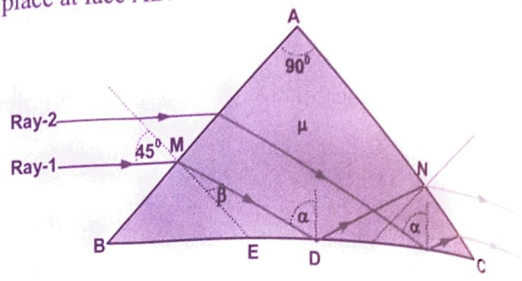
(B) 2.25×10^{-4} m

(C) 1.125×10^{-3} m

(D) 1.125×10^{-4} m

Ans. B

15-16. In figure-I, total internal reflection takes place at face AB.



$$\angle EMD + \angle MDE = \angle BEM$$

$$\Rightarrow \beta + (90^\circ - \alpha) = 45^\circ$$

$$\Rightarrow \sin(\beta + 45^\circ) > \left(\frac{1}{\mu}\right)$$

$$\Rightarrow \sin\beta + \cos\beta > \frac{\sqrt{2}}{\mu} \quad \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

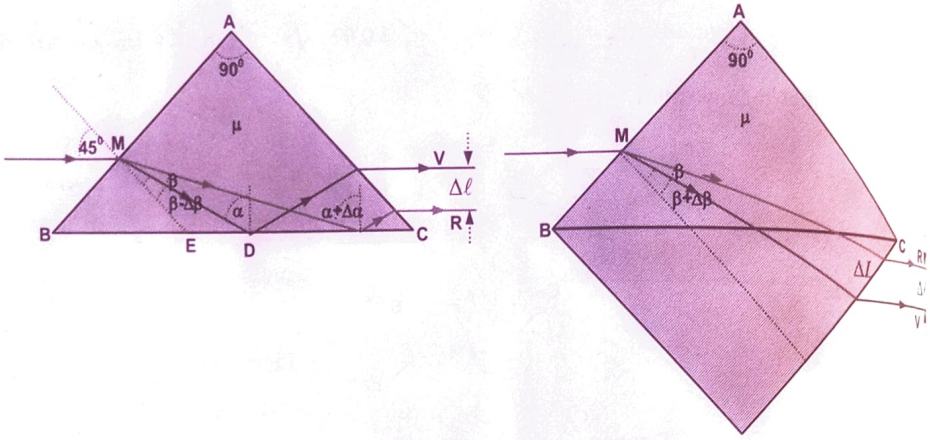
According to law of refraction at point M, we can write

$$1 \times \sin(45^\circ) = \mu \sin\beta \Rightarrow \sin\beta = \frac{1}{\mu\sqrt{2}} \Rightarrow \cos\beta = \sqrt{1 - \frac{1}{2\mu^2}}$$

Putting these value in equation (1), we have

$$\frac{1}{\mu\sqrt{2}} + \sqrt{1 - \frac{1}{2\mu^2}} > \frac{\sqrt{2}}{\mu} \Rightarrow \sqrt{2\mu^2 - 1} > 1 \Rightarrow \mu > 1$$

In figure-II: Replace the **Dove Prism,** by glass cube of side as AB.



$$\Delta\ell = \frac{\Delta L}{\sqrt{2}} \quad \ldots\ldots\ldots\ldots\ldots\ldots (2)$$

From geometry, we can write

$$\sin\beta = \frac{1}{\mu_R\sqrt{2}} \text{ and } \sin(\beta - \Delta\beta) = \frac{1}{(\mu_R + \Delta\mu_R)\sqrt{2}}$$

$$\Delta L = a\tan\beta - a\tan(\beta - \Delta\beta) = \frac{a}{\sqrt{2\mu_R^2 - 1}} - \frac{a}{\sqrt{2(\mu_R + \Delta\mu)^2 - 1}}$$

$$\Rightarrow \Delta L = \frac{a}{\sqrt{2\mu_R^2 - 1}}\left(1 - \sqrt{\frac{2\mu_R^2 - 1}{2(\mu_R + \Delta\mu)^2 - 1}}\right) = \frac{a}{\sqrt{2\mu_R^2 - 1}}\left(1 - \sqrt{\frac{2\mu_R^2 - 1}{2(\mu_R + \Delta\mu)^2 - 1}}\right)$$

$$\Rightarrow \Delta L = \frac{a}{\sqrt{2\mu_R^2 - 1}}\left(1 - \sqrt{\frac{2\mu_R^2 - 1}{2\mu_R^2 + 4\mu_R\Delta\mu + 2(\Delta\mu)^2 - 1}}\right)$$

Neglecting the value $(\Delta\mu)^2$, we have

$$\Rightarrow \Delta L = \frac{a}{\sqrt{2\mu_R^2 - 1}}\left(1 - \sqrt{\frac{1}{1 + \frac{4\mu_R \Delta\mu}{2\mu_R^2 - 1}}}\right) = \frac{a}{\sqrt{2\mu_R^2 - 1}}\left(1 - \left\{1 + \frac{4\mu_R \Delta\mu}{2\mu_R^2 - 1}\right\}^{-\frac{1}{2}}\right)$$

Expand binomially, we have

$$\Rightarrow \Delta L = \frac{2a\mu_R \Delta\mu}{\left(2\mu_R^2 - 1\right)^{\frac{3}{2}}}$$

Putting this value in equation (2), we have

$$\Delta\ell = \frac{\sqrt{2}a\mu_R \Delta\mu}{\left(2\mu_R^2 - 1\right)^{\frac{3}{2}}}$$

Second method:

$$\Delta L = \left(\frac{d}{d\beta}(a\tan\beta)\right) \times \Delta\beta$$

$$\Delta L = \left(a\sec^2\beta\right) \times \Delta\beta$$

Also $\sin\beta = \frac{1}{\sqrt{2}\mu_R}$

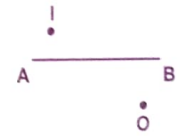
$$\Rightarrow \cos\beta\Delta\beta = \frac{1}{\sqrt{2}\mu_R^2}\Delta\mu$$

$$\Rightarrow \Delta L = \frac{a\sec^2\beta \times \Delta\mu}{\sqrt{2}\mu_R^2\cos\beta}$$

$$\Rightarrow \Delta L = \frac{2a\mu_R \Delta\mu}{\left(2\mu_R^2 - 1\right)^{\frac{3}{2}}}$$

Matrix Match Type

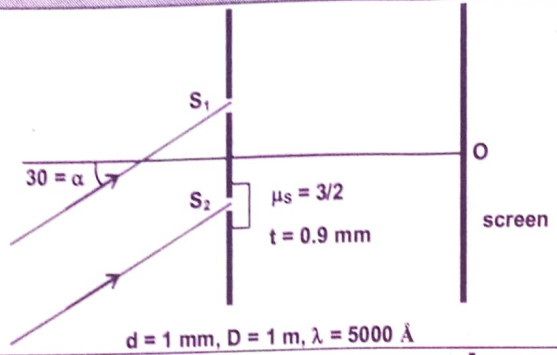
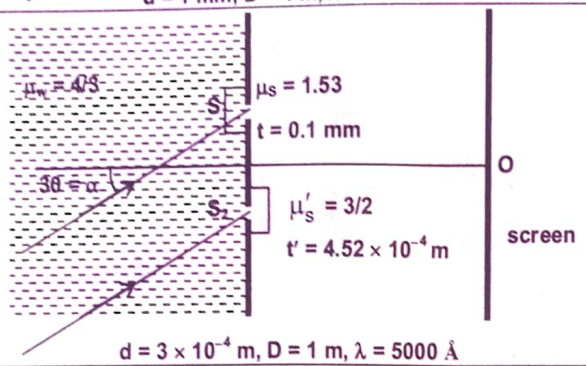
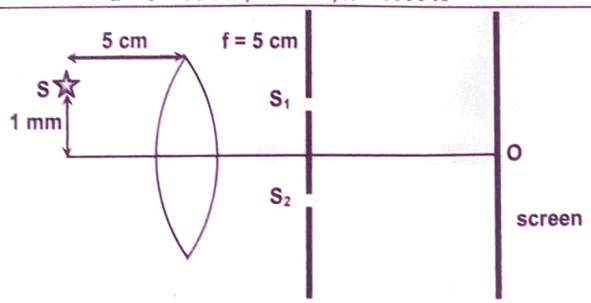
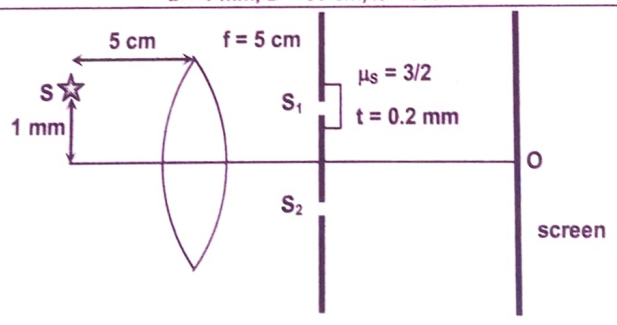
17. **O is an object, I is an image and AB is optical axis of an optical element which can be a lens or a mirror Now match the following.**



Column – I		Column – II	
(A)	O is virtual object	(p)	Convex mirror
(B)	O is a real object	(q)	Concave mirror
(C)	I is virtual image	(r)	Convex lens
(D)	I is real image	(s)	Concave lens

Sol. (A) → (p, s) (B) → (q, r) (C) → (p, s) (D) → (q, r)

18. List – I contains different arrangements of YDSE and List – II contains the distance of their respective central maxima formed on the screen from point 'O'. Then match the options of List-I correctly with the options of List-II. The separation between the slits = d. The separation between the plane of slits and screen = D.

List – I		List - II	
(P)		(1)	2 cm
(Q)		(2)	1 cm
(R)		(3)	4 cm
(S)		(4)	5 cm

Codes:

	P	Q	R	S
(A)	3	2	1	4
(B)	4	1	3	2
(C)	4	2	1	3
(D)	4	1	2	3

Sol. D

(P) for central maxima

$d \sin \alpha = (\mu s - 1)t + d \sin \theta$

$5 \times 10^{-4} = \frac{1}{2} \times 9 \times 10^{-4} + 10^{-3} \sin\theta$

$0.5 \times 10^{-4} = 10^{-3} \sin \theta$

$\Rightarrow \sin \theta = 1/20$

$\Rightarrow y = D/20 = 5$ cm (above O)

(Q) $\mu_w d \sin \alpha + (\mu s - \mu_w)t + d \sin \theta = \left(\mu_S' - 1\right)t'$

$\frac{4}{3} \times 3 \times 10^{-4} \times \frac{1}{2} + (1.53 - 1.33) \times 10^{-4} + 3 \times 10^{-4} \sin\theta = \frac{1}{2} \times 4.52 \times 10^{-4}$

$2.2 \times 10^{-4} + 3 \times 10^{-4} \sin \theta = 2.26 \times 10^{-4}$

$3 \times 10^{-4} \sin \theta = 6 \times 10^{-6}$

$\Rightarrow \sin \theta = 2 \times 10^{-2}$

$\Rightarrow \sin \theta = 1/50 \Rightarrow y = D/50 = 2$ cm (below 'O')

(R) $d \sin \alpha = d \sin \theta$

$\sin \theta = \sin \alpha$

$\therefore \tan \alpha = \tan \theta \Rightarrow \frac{y}{D} = \frac{1}{50}$

$\therefore y = \frac{D}{50} = 1$ cm (below 'O')

(S) $d \sin \alpha + d \sin \theta = (\mu s - 1)t$

$1 \times 10^{-3}(\sin \alpha + \sin\theta) = \frac{1}{2} \times 2 \times 10^{-4}$

$\sin \alpha + \sin \theta = 1/10$

$\frac{1}{50} + \sin\theta = \frac{1}{10}$

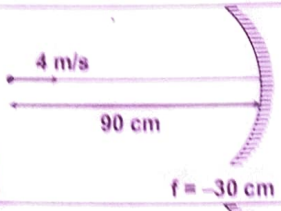
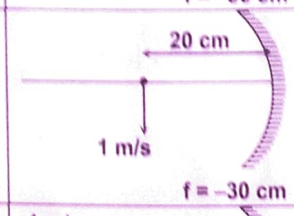
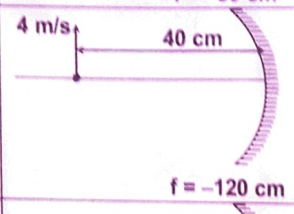
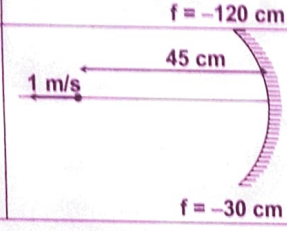
$\therefore \sin\theta = \frac{1}{10} - \frac{1}{50} \Rightarrow \sin \theta = \frac{4}{50}$

$\Rightarrow \frac{y}{D} = \frac{4}{50}$

$\therefore y = 4$ cm (above O)

Answer questions 19, 20 and 21 by appropriately matching the information given in the three columns of the following table.

Column –I lists four situations showing a concave mirror and a moving object. Column –II lists the corresponding image coordinate (usual sign convention) and column –III lists the image velocity.

Column 1		Column 2		Column 3	
(I)		(i)	v = –60 cm	(P)	$-6\hat{j}$
(II)		(ii)	v = –45 cm	(Q)	$-\hat{i}$
(III)		(iii)	v = –120 cm	(R)	$+3\hat{j}$
(IV)		(iv)	v = –90 cm	(S)	$+4\hat{j}$

19. **Pick the correct option.**

(A) (IV) (iii) (P) **(B)** (III) (iv) (Q)

(C) (II) (i) (R) **(D)** (I) (i) (S)

Sol. C

20. **Pick the correct option for maximum magnitude of image coordinate.**

(A) (I) (i) (S) **(B)** (II)(i) (R)

(C) (III) (iv) (Q) **(D)** (IV) (iv) (S)

Sol. D

21. **Pick the correct option for maximum magnitude of image velocity.**

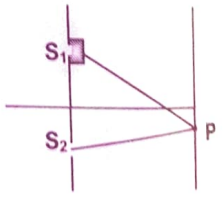
(A) (III) (i) (P) **(B)** (II)(iii) (R)

(C) (I) (iv) (S) **(D)** (IV) (iii) (Q)

Sol. A

Numerical Type

22. **In a young's double slit experiment one of the slits is covered by a thin film of thickness t = 0.04 mm, and refractive index $\mu = 1.2 + \frac{9\times10^{-14}\,m^2}{\lambda^2}$, where, λ is the wave length in meter. A beam of light consisting two wavelengths λ_1 = 400 nm and λ_2 = 600 nm falls normally on the plane of the slits. Find the distance between two central maxima in milimeter. Distance of screen from slits is 400 times the separation between the slits.**



Sol. 5

$\Delta x = (S_1P - t) + \mu_1 t - S_2P$

$0 = (S_1P - S_2P) + (\mu_1 - 1)t$

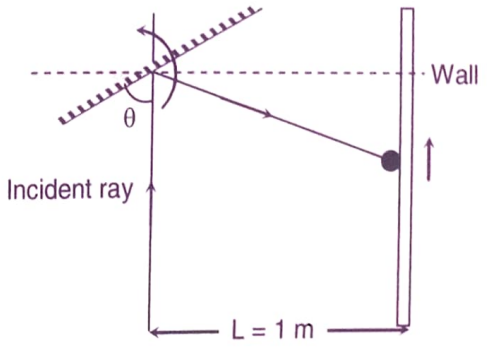
$\Rightarrow$ $d \sin\theta = (\mu_1 - 1)t$

$$\frac{y_1 d}{D} = (\mu_1 - 1)t$$

$$\frac{y_2 d}{D} = (\mu_1 - 1)t$$

$$y_1 - y_2 = \frac{(\mu_1 - \mu_2)tD}{d} = 5\text{mm}$$

23. **A light ray is incident on a rotating mirror, which is rotating with angular speed 3 rad/s as shown in figure. The speed of bright spot on the wall when θ = 30°.**



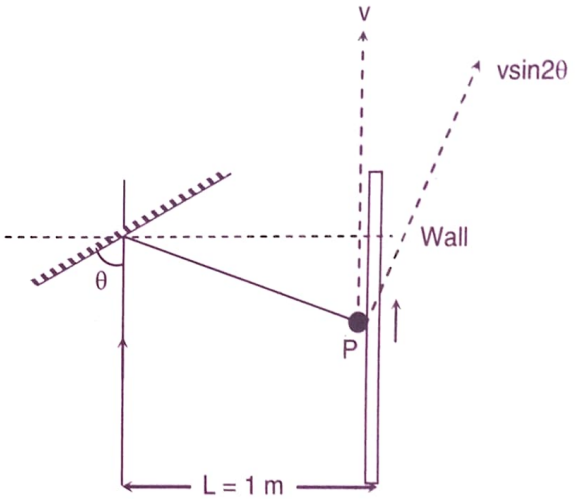
Sol. 8

$$\tan(2\theta - 90) = \frac{Y}{L}$$

$$y = -L\cot 2\theta$$

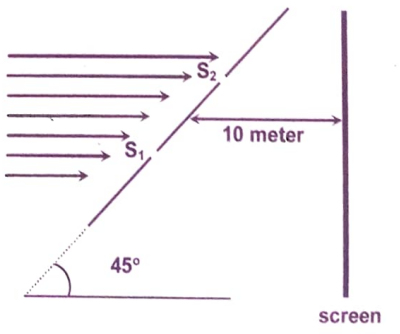
$$\frac{dy}{dt} = -L \times \left(-\text{cosec}^2 2\theta\right) \times 2\frac{d\theta}{dt}$$

$$= 2L\left(\text{cosec}^2 2\theta\right)\frac{d\theta}{dt}$$



Numerical Decimal Type

24. **Consider a plane inclined at an angle 45° with the horizontal has two slits (S_1 and S_2) separated by a distance d = $\sqrt{2}$ mm. The screen is placed at a distance of D = 10 m. A parallel monochromatic light beam of wavelength 5000 Å is incident on the slits as shown. If the fringe width of interference pattern on the screen is k × 10^{-3} meter, then find the value of k.**



Sol. 00005.00

Path difference $= S_1N - MS_2$

$= d\sin(45^\circ+\theta) - \frac{d}{\sqrt{2}}$

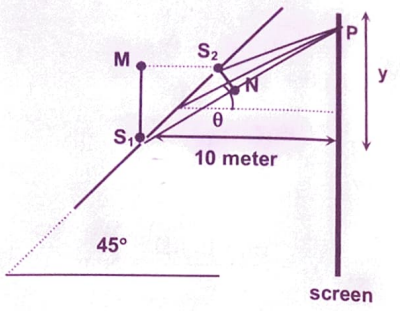
$= d\left[\frac{\cos\theta}{\sqrt{2}} + \frac{\sin\theta}{\sqrt{2}}\right] - \frac{d}{\sqrt{2}}$

$n\lambda = d\left[\frac{1}{\sqrt{2}} + \frac{\theta}{\sqrt{2}}\right] - \frac{d}{\sqrt{2}}$ (as 'θ' is small)

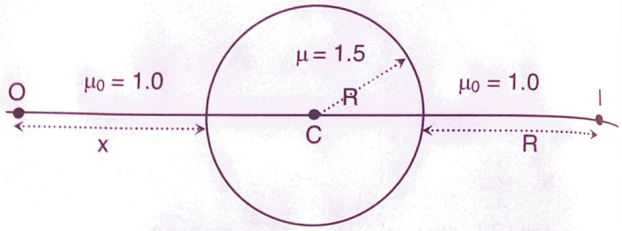
$\theta = \frac{\sqrt{2}n\lambda}{d}$

$y = \frac{\sqrt{2}n\lambda D}{d}$,

Fringe width $= y_{n+1} - y_n = \frac{\sqrt{2}D\lambda}{d} = 5\times10^{-6}$ m



25. **A glass sphere of refractive index 1.5 forms the real image of object O at point I as shown in the figure when kept in air of refractive index 1. If the value of x is kR. Find k.**

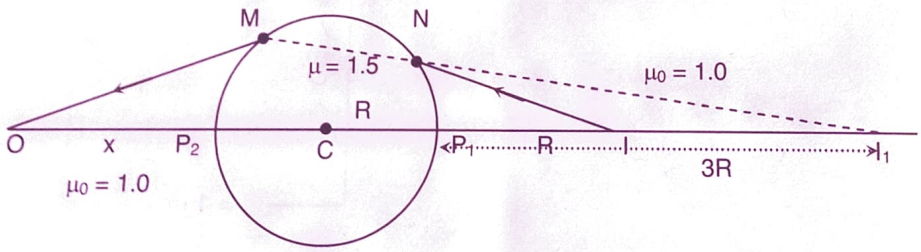


Sol. 00005.00

We can reverse the question. Let I behaves as object and O behaves as image, because retracing the path will not alter the distances.

Consider first refraction at point N

$\frac{1.5}{v} - \frac{1.0}{-R} = \frac{1.5-1.0}{+R} \Rightarrow \frac{1.5}{v} = \frac{1}{2R} - \frac{1}{R} \Rightarrow v = -3R$



Consider second refraction at point M, for this I, will behaves as normal object so

$\frac{1.0}{v} - \frac{1.5}{-5R} = \frac{1.0-1.5}{-R}$

$\Rightarrow \frac{1}{v} + \frac{3}{10R} = \frac{1}{2R} = \frac{5}{10R}$

$\Rightarrow \frac{1}{v} = \frac{5}{10R} - \frac{3}{10R} = \frac{1}{5R}$

$\Rightarrow v = 5R$

$\Rightarrow x = 5R \Rightarrow k = 5$

MODERN PHYSICS & ERROR ANALYSIS 16

WAVE–PARTICLE DUALITY

"Waves behave as matter, matter behaves as waves"

Energy of photon is given by, $E = h.f = \frac{hc}{\lambda} = p \times c$, where p = momentum of the particle.

For any particle, $\lambda = \frac{h}{p}$, where λ = wavelength, h = Planck's constant 6.63×10^{-34} J-s.

If an electron is accelerated through a potential difference of V volt, then

$$\frac{1}{2} m_e v^2 = eV$$

or $v = \sqrt{\frac{2eV}{m_e}}$

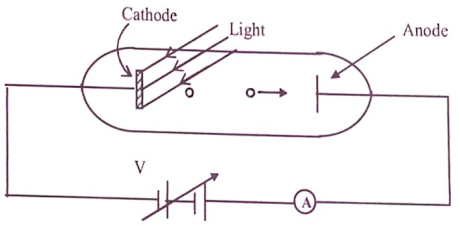
$\Rightarrow \sqrt{2m_e eV} = m_e v = p_e$

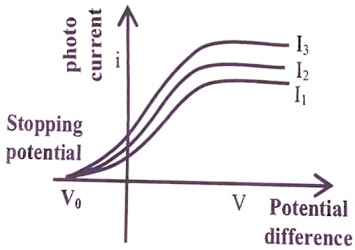
$\therefore \lambda_e = \frac{h}{m_e v} = \frac{h}{\sqrt{2eVm_e}}$

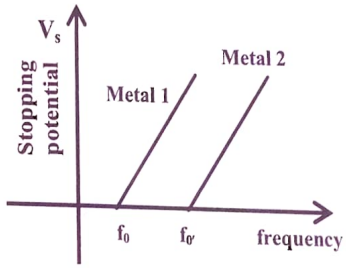
PHOTOELECTRIC EFFECT

$$EV_s = K_{max} = \frac{hc}{\lambda} - \phi$$

where V_s = stopping potential, λ = wavelength of incident radiation, and ϕ = work function.







(Light of same frequency and different intensity is incident on cathode)

Note:

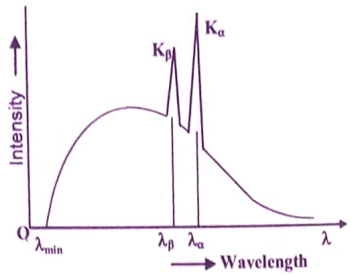
(a) The stopping potential depends on the wavelength of incident light and not on the intensity of the incident light.

(b) The photocurrent increases if the intensity of the incident light is increased.

X-RAYS

When highly energetic electrons are made to strike on a metallic target, electromagnetic radiation called X-rays are emitted.

Continuous and Characteristic X-Rays



Continuous X-Rays

When the electron loses its kinetic energy during several collisions with the other atoms, a fraction of the lost kinetic energy is converted into electromagnetic radiations. This fraction can range from 0 to 1.

Corresponding to the maximum kinetic energy lost by an electron, we have the largest frequency of the X-ray or the shortest wavelength.

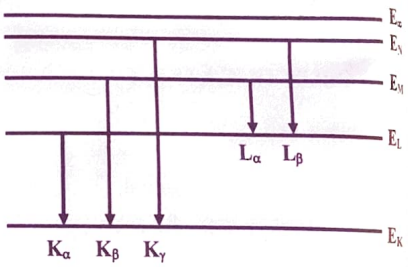
$$\frac{hc}{\lambda_{min}} = K_{max} = eV$$

$$\Rightarrow \lambda_{min} = \frac{hc}{eV}$$

The emitted wavelengths ranges from a minimum value all the way to infinity. Thus, the name continuous spectrum.

Characteristic X-Rays

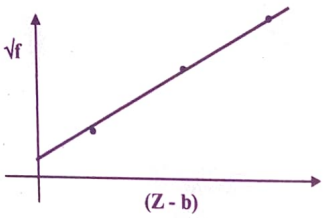
The accelerated electron might knock out the inner electron of the target atom, whereby a vacancy is created in the inner orbit. Electrons from the higher orbit jump in to fill this vacancy, while releasing the energy difference as electromagnetic radiations, these wavelengths are characteristic of the material from which they are emitted. Hence, the name Characteristic Spectra.



$$\lambda = \frac{hc}{E_K - E_L} \text{ for } K_\alpha \text{ wavelength.}$$

$$\Rightarrow \quad \lambda = \frac{hc}{E_L - E_M} \text{ for } L_\alpha \text{ wavelength and so on.}$$

Moseley's Law



f = frequency of X-ray,

Z = atomic number of target atom,

a, b are constants.

$$\sqrt{f} = a(Z - b)$$

Properties of X-Rays

(i) X-rays do not get deflected by electric or magnetic fields.

(ii) They produce ionisation in gases.

(iii) Their diffraction properties are useful in studying crystal structures.

(iv) They have high penetrating power and are used in radiographs.

Bohr's Theory

This theory is applicable for hydrogen and similar atoms.

Bohr's theory for hydrogen-like atoms

$$mvr = \frac{nh}{2\pi} \quad \text{and} \quad \frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{Ze.e}{r^2}$$

$$\Rightarrow \quad r_n = \frac{\varepsilon h^2 n^2}{\pi m Z e^2} \quad \text{and} \quad E_n = \frac{-Z^2 me^4}{8h^2\varepsilon_0^2 n^2} = -\frac{Z^2 Rhc}{n^2}$$

If an electron makes a transition from $n = n_2$ to $n = n_1$

$$\frac{hc}{\lambda} = Z^2 Rhc\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\Rightarrow \quad v = \frac{Z}{n}v_0$$

$$r_0 = 0.53\,\text{Å}; \quad r = \frac{r_0 n^2}{Z},$$

$$E_0 = -13.6\,\text{eV}; \quad E = \frac{E_0 Z^2}{n^2}$$

Also,

K.E. = − P.E./2

E = P.E. + K.E. = P.E./2 = − K.E.

NUCLEAR PHYSICS

(a) α decay $= {}^{Z}_{A}X \rightarrow {}^{Z-4}_{A-2}Y + {}^{4}_{2}He + \text{energy}$

(b) β^- decay $= {}^{Z}_{A}X \rightarrow {}^{Z}_{A+1}Y + \beta^- + \gamma + \text{energy}$

(c) β^+ decay $= {}^{Z}_{A}X \rightarrow {}^{Z}_{A-1}Y + \beta^+ + \gamma + \text{energy}$; where A = atomic number, Z = mass number

Conservation Laws in Nuclear Reactions

(a) Conservation of mass–energy: Mass–energy of the reactants is equal to mass-energy of the products.

(b) Conservation of linear momentum: The sum of momentum of the reactants and products remain conserved.

(c) Conservation of mass number: Total mass number does not change in nuclear reaction.

(d) Conservation of electric charge: Total electric charge remains conserved.

Energy Produced in a Nuclear Reaction

Initial **Final**

Initial energy: $E_i = m_1c^2 + m_2c^2 + K_1 + K_2 + hf_o$

Final energy: $E_f = m_3c^2 + m_4c^2 + K_3 + K_4 + hf$

Since $E_i = E_f$, therefore $Q = [(m_1 + m_2) - (m_3 + m_4)]c^2$

If Q is positive, the rest mass energy is converted to kinetic mass energy or radiation mass energy or both, and the reaction is exoergic.

If Q is negative, the reaction is endoergic. The minimum amount of energy that a bombarding particle must have in order to initiate an endoergic reaction, is called Threshold Energy E_{th}.

$$E_{th} = -Q\left(\frac{m_1}{m_2} + 1\right)$$, where m_1 = mass of the projectile and m_2 = mass of the target.

Binding Energy

It is the energy required to break up the nucleus into its constituent nucleus and place them infinitely apart at rest.

Radioactive Decay Law

If nuclei A is decaying to nuclei B

$$-\frac{dN}{dt} = \lambda N$$

$$\Rightarrow \quad N = N_0 e^{-\lambda t}$$

$$t = \frac{2.303}{\lambda} \log \frac{N_0}{N}$$

Activity: $A = \left|\frac{dN}{dt}\right| = \lambda N$

Half Life

The time interval during which a radioactive substance decays to half its original value. It is denoted by $T_{1/2}$.

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

Mean Life

$$T = \frac{\int_0^\infty N_0 e^{-\lambda t} dt}{N_0} = \frac{1}{\lambda} = \frac{T_{1/2}}{0.693}$$

Units of activity: 1 becquerel (Bq) = 1 disintegration per second (dps)

1 curie (Ci) $= 3.7 \times 10^{10}$ dps

1 rutherford (rd) = 10^6 Bq = 10^6 dps

CHAIN REACTION

Theoretically; it can be represented as

$$A \xrightarrow{\lambda_A} B \xrightarrow{\lambda_B} C$$

$$-\frac{dN_A}{dt} = \lambda_A N_A \qquad \ldots (1)$$

$$\frac{dN_B}{dt} = \lambda_A N_A - \lambda_B N_B \qquad \ldots (2)$$

$$\frac{dN_C}{dt} = \lambda_B N_B$$

Solving Eqs. (1), (2) and (3), we can get the values N_A, N_B, N_C at any time t.

ERROR & ERROR ANALYSIS

Expression of Error

(i) Absolute Error

Difference between the value of a quantity and the measured value.

Let the measured values be $A_1, A_2, ..., A_n$

$$A_{mean} = \frac{A_1 + A_2 + \cdots + A_n}{n}$$

$$\Delta A_1 = A_{mean} - A_1$$

$$\Delta A_2 = A_{mean} - A_2$$
$$\Delta A_3 = A_{mean} - A_3$$
$$\Delta A_n = A_{mean} - A_n$$
$$\Delta A_{mean} = \frac{|\Delta A_1| + |\Delta A_2| + \cdots + |\Delta A_n|}{n} = \frac{1}{n}\sum_{i=1}^{n}|\Delta A_1|$$

Any single measurement A has to be such that $(A_{mean} - \Delta A_{mean}) \le A \le (A_{mean} + \Delta A_{mean}.)$

(ii) Relative Error

$$\text{Relative error} = \frac{\Delta A_{mean}}{A_{mean}}$$

(iii) Percentage error

$$\text{Percentage error} = \frac{\Delta A_{mean}}{A_{mean}} \times 100$$

Propagation of Errors

(a) Sum and difference of quantities:

$$x = a \pm b$$
$$\Delta x_{max} = (\Delta a + \Delta b)$$

(b) $x = ab$

$$\frac{\Delta x_{max}}{x} = \left(\frac{\Delta a}{a} + \frac{\Delta b}{b}\right)$$

(c) $x = a/b$

$$\frac{\Delta x_{max}}{x} = \left(\frac{\Delta a}{a} + \frac{\Delta b}{b}\right)$$

(d) $x = a^m b^n$

$$\frac{\Delta x_{max}}{x} = \left(m\frac{\Delta a}{a} + n\frac{\Delta b}{b}\right)$$

(e) $x = \frac{a^m}{b^n}$

$$\frac{\Delta x_{max}}{x} = \left(m\frac{\Delta a}{a} + n\frac{\Delta b}{b}\right)$$

Basic Procedure to Calculate Errors

Let $X = a^m b^n c^p$

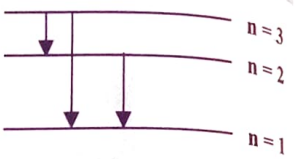
$$\ln X = m \ln a + n \ln b + p \ln c$$
$$\frac{dx}{x} = m\frac{da}{a} + n\frac{dp}{p} + p\frac{dc}{c}$$

For errors,

$$\therefore \quad \frac{\Delta x_{max}}{x} = m\frac{\Delta a}{a} + n\frac{\Delta b}{b} + p\frac{\Delta c}{c}$$

ILLUSTRATIVE PROBLEMS

1. A gas of hydrogen like atoms can absorb radiations of 68 eV. Consequently, the atoms emit radiations of only three different wavelengths. All the wavelengths are equal or smaller than that of the absorbed photon.
(a) Determine the initial state of the gas atoms.
(b) Identify the gas atoms.
(c) Find the minimum wavelength of the emitted radiation.
(d) Find the ionization energy and the respective wavelength for the gas atoms.



Sol. (a) Since three radiations are emitted, therefore, the final excited state of the gas is n = 3.
The initial state of the gas atoms is
n = 2 as all the wavelengths are smaller and the energy will be higher.

(b) $13.6Z^2\left[\frac{1}{2^2}-\frac{1}{3^2}\right]=68.$

or $13.6Z^2\left[\frac{5}{36}\right]=68 \Rightarrow Z=6$

(c) The minimum wavelength corresponds to the transition n = 3 to n = 1.

$$\frac{1}{\lambda_{min}}=RZ^2\left[\frac{1}{1^2}-\frac{1}{3^2}\right]$$

or $\lambda_{min}=\frac{9}{8RZ^2}=\frac{9}{8\left(1.097\times10^7\right)(6)^2}=28.5\ \text{Å}$

(d) The ionization energy of the gas atoms is

$E = 13.6\ Z^2 = (13.6)(6)^2 = 489.6\ \text{eV}$

$$\lambda=\frac{1}{RZ^2}=\frac{1}{\left(1.097\times10^7\right)(6)^2}=25.32\ \text{Å}$$

2. A monochromatic beam of light (λ = 4900 Å) incident normally upon a surface produces a pressure of 5×10^{-7} N/m^2 on it. Assuming that 25% of the light incident is reflected and the rest absorbed, find the number of photons falling per second on a unit area of thin surface.

Sol. $P=[2(0.25)+0.75]\frac{I}{c}=1.25\frac{I}{c}$

$\therefore$ Intensity of light, $I=\frac{cP}{1.25}=\frac{\left(3\times10^8\right)\left(5\times10^{-7}\right)}{1.25}=120\ \text{Wm}^{-2}$

Energy of photon, $E=\frac{hc}{\lambda}=\frac{\left(6.63\times10^{-34}\right)\left(3\times10^8\right)}{0.49\times10^{-6}}=4\times10^{-19}\ \text{J}$

$\therefore$ Number of photons incident per unit area per second

$$n=\frac{I}{E}=\frac{120}{4\times10^{-19}}=3\times10^{20}\,\text{m}^{-2}\text{s}^{-1}$$

3. Calculate the change in magnetic dipole moment when the electron in a H-atom makes a transition from the n = 4 orbit to the n = 3 orbit. Suppose that the atom is in a magnetic field of induction B. Calculate the change in energy when it makes the same transition and the corresponding change in the frequency of the emitted photon due to the magnetic field.

Sol. We assume that an electron moves in a circular orbit of radius r with speed v. The current due to the electron is (e being the charge on the electron),

$$i = \frac{e}{T} = \frac{ev}{2\pi r}, \text{ where T is the time period.}$$

The dipole moment due to this current is

$$\mu = i.\pi r^2 = \frac{ev}{2\pi r} \times \pi r^2 = \frac{evr}{2}$$

$$= \frac{e}{2m} \times mvr = \frac{e}{2m} \times n\hbar \quad (\hbar = h/2\pi)$$

when the electron is in the n^{th} orbit.

When it makes a transition from the n = 4 to the n = 3 level, the change in the dipole moment equals

$$\Delta\mu = \mu(n=4) - \mu(n=3) = \frac{e\hbar}{2m} \text{ (also known as the Bohr magneton)}$$

If the atom is in a magnetic field of induction B, then the energy of the electron is given by

$$E_n' = E_n - \vec{\mu}.\vec{B} = -\frac{R_E}{n^2} - \left(\frac{eB\hbar}{2m}\right)n$$

(where R_E represents the ionisation energy in the absence of a magnetic field)

When it makes a transition,

$$E_{n_1}' - E_{n_2}' = \Delta E_{n_1 n_2}' = \left(E_{n_1} - E_{n_2}\right) - \frac{eB\hbar}{2m}(n_1 - n_2)$$

$$= R_E\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right) - \frac{e\hbar}{2m}B(n_1 - n_2)$$

Due to the magnetic field, the change in energy equals $\frac{e\hbar}{2m}B(n_1 - n_2)$; and the corresponding change in the frequency of the emitted photon equals

$$\frac{e\hbar B}{h \times 2m}(n_1 - n_2) = \frac{eB}{4\pi m}(n_1 - n_2)$$

4. A particular type of nucleus whose decay constant is λ is produced at a steady rate of p nuclei per second. Show that the number of nuclei N present t second after the production starts is $N = \frac{p}{\lambda}\left[1 - e^{-\lambda t}\right]$.

Sol. The instantaneous rate of growth of the nuclei is

$$\frac{dN}{dt} = p - \lambda N \quad \text{or} \quad \frac{dN}{p - \lambda N} = dt$$

On integrating,

$$-\frac{1}{\lambda}\ln|p - \lambda N| = t + c$$

$$\text{At } t = 0,\ N = 0 \Rightarrow \quad c = \frac{-\ln|p|}{\lambda}$$

$$\text{or} \quad -\frac{1}{\lambda}\left[\ln|p - \lambda N| - \ln|p|\right] = t \quad \text{or} \quad \ln\left|1 - \frac{\lambda N}{p}\right| = -\lambda t$$

$$\text{or} \quad N = \frac{p}{\lambda}\left[1 - e^{-\lambda t}\right].$$

5. In a nuclear reaction, $\alpha + {}_7N^{14} \longrightarrow X + p$
When nitrogen atoms are bombarded by 7.7 MeV α -particles, protons are ejected with a kinetic energy of 5.5 MeV. The Q-value of the reaction is-1.26 MeV
(a) Find the atomic mass of X in amu, Identify the element X.
(b) Find the angle ϕ between the direction of motion of proton and α - particle.
Given that atomic mass of ${}_1H^1$ =1.00814 a.m.u.,
atomic mass of ${}_7N^{14}$=14.00752 amu, and atomic mass of ${}_2He^4$ = 4.0038 a.m.u.

Sol. Applying law of conservation of energy

$$(E_\alpha + m_\alpha c^2) + (0 + m_N c^2) = (E_{Pr} + m_{Pr} c^2) + (E_x + m_x c^2)$$

where E represents the K.E.

Energy of the reaction is given by

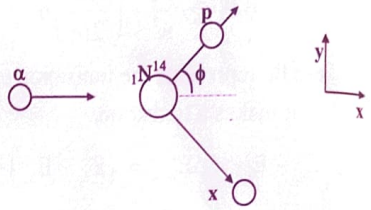
$$Q = KE_f - KE_i = (E_{Pr} + E_x) - (E_\alpha) = (m_\alpha + m_N - m_{Pr} - m_x) c^2$$

$$\Rightarrow -1.26 \text{ MeV} = (4.00388 + 14.00752 - 1.00814 - m_x) \times 931.5 \text{ MeV}$$

On solving, we get

$$X = {}_8O^{17}$$

Applying conservation of linear momentum,



$$\vec{p}_\alpha = \vec{p}_x + \vec{p}_{pr}$$

$$\vec{p}_\alpha - \vec{p}_{Pr} = \vec{p}_x$$

$$p_\alpha^2 + p_{Pr}^2 - 2p_\alpha p_{Pr} \cos\phi = p_x^2$$

$$\therefore E_x = Q + E_\alpha - E_{Pr}$$

$$\therefore E_x = (-1.26 + 7.7 - 5.5) \text{ MeV} = 0.94$$

$$\therefore \cos\phi = -\left[\frac{m_x E_x - m_\alpha E_\alpha - m_{Pr} E_{Pr}}{2\sqrt{m_\alpha m_{Pr} E_\alpha E_{Pr}}}\right] = 0.4429 \qquad (\because p^2 = 2mE)$$

$$\phi = \cos^{-1}(0.4429) = 63° 42'$$

SOLVED PROBLEMS

1. To decrease the cut-off wavelength of continuous X-ray by 25%, the potential difference across the X-ray tube

(A) must be increased by $\frac{100}{3}$ % **(B) must be decreased by 20%**

(C) must be increased by 25% **(D) must be decreased by 25%**

Sol. $\lambda = \frac{hc}{eV}, \frac{\lambda_1}{\lambda_2} = \frac{v_2}{v_1}, \lambda_2 = \frac{3}{4}\lambda_1$

$v_2 = \frac{4}{3}v_1$ (which is equal to $\frac{100}{3}$% increase)

$\therefore$ (A)

2. An electron in Bohr's hydrogen atom has an energy of –3.4 eV. The angular momentum of the electron is

(A) h/π **(B) h/2π**

(C) nh/2π (n is an integer) **(D) 2h/π**

Sol. The energy of an electron in an orbit of principal quantum number n is given as

$$E = \frac{-13.6}{n^2}\ eV \Rightarrow -3.4\ eV = \frac{-13.6}{n^2}\ eV$$

$\Rightarrow n^2 = 4 \Rightarrow n = 2$

The angular momentum of an electron in nth orbit is given as

$$L = \frac{nh}{2\pi}\text{ , Putting } n = 2$$

We obtain $L = \frac{2h}{2\pi} = \frac{h}{\pi}$

$\therefore$ (A)

3. Choose the correct option by matching the following

Column A	Column B
(a) Maximum kinetic energy of electrons emitted in photoelectric effect is linearly dependent on	(i) an integral multiple of $h/2\pi$.
(b) If number of electrons striking the anode of an X-ray tube is increased then	(ii) the speed of light in vacuum.
(c) X-rays travel with	(iii) the intensity of emitted X-rays increases.
(d) Angular momentum of electron in any orbit is	(iv) the frequency of the incident radiation.

(A) (a)-(iv), (b)-(i), (c)-(iii) and (d) - (ii)
(B) (a)-(iii), (b)-(ii), (c)-(i) and (d)-(iv)
(C) (a)-(i), (b)-(iii), (c)-(ii) and (d) -(iv)
(D) (a)-(iv), (b)-(iii), (c)-(ii) and (d)-(i)

Sol. (D)

4. The wavelength K_α X-rays produced by an X-ray tube is 0.76 Å. The atomic number of anticathode materials is

(A) 82 **(B) 41**
(C) 20 **(D) 10**

Sol. For K_α X-ray line,

$$\frac{1}{\lambda_\alpha} = R(Z-1)^2\left[\frac{1}{1^2}-\frac{1}{2^2}\right] = R(Z-1)^2\left[1-\frac{1}{4}\right]$$

$$\Rightarrow \frac{1}{\lambda_\alpha} = \frac{3}{4}R(Z-1)^2 \qquad \ldots(1)$$

With reference to given data,

$\lambda_\alpha = 0.76\,\text{Å} = 0.76 \times 10^{-10}\,m$

$R = 1.097 \times 10^7\ m$

Putting these values in Eq. (1),

$$(Z-1)^2 = \frac{4}{3}\frac{1}{0.76\times10^{-10}\times1.097\times10^{7}} \cong 1600$$

$\Rightarrow Z - 1 = 40 \Rightarrow Z = 41$

$\therefore$ (B)

5. **The ratio of de-Broglie wavelength of a α particle to that of a proton being subjected to the same magnetic field so that the radii of their path are equal to each other assuming the field induction vector $\vec{B}$ is perpendicular to the velocity vectors of the α-particle and the proton, is**

(A) 1 (B) 1/4

(C) 1/2 (D) 2

Sol. When a charged particle of charge q, mass m enters perpendicularly to the magnetic induction $\vec{B}$ of a magnetic field, it will experience a magnetic force $F = q(\vec{v} \times \vec{B}) = q\ vB \sin 90^\circ = qvB$ that provide a centripetal acceleration $\frac{v^2}{r}$

$\Rightarrow qvB = \frac{mv^2}{r} \quad \Rightarrow \quad mv = qBr$

$\Rightarrow$ The de-Broglie wavelength $\lambda = \frac{h}{mv} = \frac{h}{qBr}$

$\Rightarrow \frac{\lambda_{\alpha-particle}}{\lambda_{proton}} = \frac{q_p r_p}{q_\alpha r_\alpha}$

Since $\frac{r_\alpha}{r_p} = 1$ and $\frac{q_\alpha}{q_p} = 2$

$\Rightarrow \frac{\lambda_\alpha}{\lambda_p} = 1/2$

$\therefore$ (C)

6. **The ratio of de-Broglie wavelength of molecules of hydrogen and helium in two gas jars kept separately at temperature 27°C and 127°C, respectively, is**

(A) $\frac{2}{\sqrt{3}}$ (B) 2 : 3

(C) $\frac{\sqrt{3}}{4}$ (D) $\sqrt{\frac{8}{3}}$

Sol. de-Broglie wavelength, $\lambda = \frac{h}{mv}$

where the speed (r.m.s) of a gas particle at the given temperature (T) is given as

$\frac{1}{2} mv^2 = \frac{3}{2} KT$

$\Rightarrow v = \frac{h}{mv} = \frac{h}{qBr}$ where K = Boltzmann's constant and m = mass of the gas particle and T = temperature of the gas in K

$\Rightarrow mv = \sqrt{3\ m\ KT} \quad \Rightarrow \quad \lambda = \frac{h}{mv} = \frac{h}{\sqrt{3m\ KT}}$

$\therefore \frac{\lambda_H}{\lambda_{He}} = \sqrt{\frac{m_{He}\ T_{He}}{m_H\ T_H}} = \sqrt{\frac{(4\ amu)\ (273+127)^\circ K}{(2\ amu)\ (273+27)^\circ K}} = \sqrt{\frac{8}{3}}$

$\therefore$ (D)

7. **A physical quantity x is calculated from the relation $x = \frac{a^2b^3}{c\sqrt{d}}$. If percentage error in a, b, c and d are 2%, 1%, 3% and 4%, respectively. What is the percentage error in x?**

(A) ±11% **(B) ±13%**

(C) ±12% **(D) ±14%**

Sol. $\frac{\Delta x}{x} = \pm\left(\frac{2\Delta a}{a} + \frac{3\Delta b}{b} + \frac{\Delta c}{c} + \frac{\Delta d}{2d}\right)$

$= \pm(2\times2 + 3\times1 + 3 + \frac{1}{2}\times4) = \pm12\%$

$\therefore$ (C)

8. **A cube has side 1.2×10^{-2} m. Its volume will be recorded as**

(A) $1.728\times10^{-6}\,m^3$ **(B)** $1.72\times10^{-6}\,m^3$

(C) $1.7\times10^{-6}\,m^3$ **(D)** $0.72\times10^{-6}\,m^3$

Sol. $v = \ell^3 = 1.728\times10^{-6}$

Length has two significant figure $v = 1.7\times10^{-6}\,m^3$

$\therefore$ (C)

9. **If force, length and time would have been the fundamental units what would have been the dimensional formula for mass?**

(A) $FL^{-1}T^{-2}$ **(B)** $FL^{-1}T^{2}$

(C) FLT^{-2} **(D) F**

Sol. Let $M = K F^aL^bT^c$

$= [MLT^{-2}]^a [L^b] T^c$

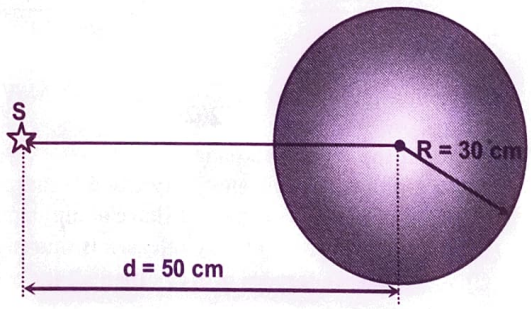
$= [M^aL^{(a+b)}T^{(-2a+c)}]$

$a = 1,\ a + b = 0$ and $-2a + c = 0$

$\Rightarrow a = 1, b = -1, c = 2$

$\therefore$ (B)

10. **A point source 'S' of light of power 0.4 W emitting mono energetic photons of energy 5 eV is placed at a distance 50 cm from the centre of an uncharged stationary metallic sphere of radius 30 cm and work function 3 eV. The efficiency of photoelectron emission is one for every 10^6 incident photons. Assume that the sphere is isolated and the photoelectrons are instantly swept away after emission. The time from the beginning of the experiment after which the photoelectron emission stops is**



(A) 2.3 ms **(B) 4.3 ms**

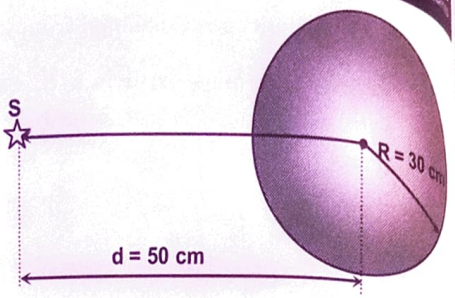
(C) 6.3 ms **(D) 8.3 ms**

Sol. D

The energy falling per second on the sphere is

$$P = \frac{P_0}{4\pi} \times 2\pi(1-\cos\theta) = \frac{P_0}{2}\left(1-\frac{4}{5}\right)$$

$$= \frac{P_0}{10} = \frac{0.4}{10} = 4\times10^{-2} \text{ W}$$



Energy of each photon is

$$E = 5 \times 1.6 \times 10^{-19} = 8 \times 10^{-19} \text{ J}$$

$\therefore$ Number of photons falling per second on the sphere,

$$N = \frac{P}{E} = \frac{4\times10^{-2}}{8\times10^{-19}} = 5\times10^{16} \text{ photons/sec}$$

$\therefore$ Number of photoelectrons emitted per second, $n = \frac{5\times10^{16}}{10^6} = 5\times10^{10}$ photoelectrons/sec

Stopping potential, $V_S = 5 - 3 = 2$ V

Let after time 't' the emission of photoelectron stops

$$\frac{\text{net charge}}{4\pi\varepsilon_0 R} = V_S$$

$$\Rightarrow \frac{5\times10^{10}\times1.6\times10^{-19}\times9\times10^{9}t}{30\times10^{-2}} = 2$$

$\Rightarrow t = 8.3$ ms.

$\therefore$ (D)

11. A stationary Pb^{206} nucleus emits an α-alpha particle with kinetic energy 5.77 MeV. Then

(A) The recoil speed of the daughter nucleus is approximately equal to 3.4×10^5 m/s.

(B) The total energy released in the decay is approximately equal to 5.88 MeV.

(C) The recoil speed of the daughter nucleus is approximately equal to 6.8×10^5 m/s.

(D) The total energy released is approximately equal to 6.22 MeV.

Sol. A, B

From conservation of momentum

$$\vec{p}_\alpha + \vec{p}_d = 0$$

$$\Rightarrow \vec{p}_d = -\vec{p}_\alpha \Rightarrow p_d = p_\alpha = \sqrt{2m_\alpha k_\alpha}$$

$$v_d = \frac{p_d}{m_d} = \sqrt{\frac{2m_\alpha k_\alpha}{m_d}} = \frac{\sqrt{2\times4\times1.66\times10^{-27}\times5.77\times1.6\times10^{-13}}}{202\times1.66\times10^{-27}} \approx 3.4\times10^5 \text{ m/s}$$

$$\text{Energy released} = \frac{m_p}{m_d}k_\alpha = \frac{206}{202}\times5.77 = 5.88 \text{ MeV}$$

12. Mark out the correct statement(s)

(A) In alpha decay, the energy released is shared between alpha particle and daughter nucleus in the form of kinetic energy and share of alpha particle is more than that of the daughter nucleus.

(B) In beta decay, the energy released is shared between electron and antineutrino.

(C) A nuclide undergoes α decay then all the α particles emitted by that nuclide will have almost the same speed.

(D) A nuclide undergoes β decay then all the β particles emitted by that nuclide may have widely different speeds.

Sol. A, B, C, D

All are based on true facts.

COMPREHENSION

A photocell is operating in saturation mode with a photo current 4.8 mA. When a monochromatic radiation of wavelength 3000 Å and power 1W is incident. When another monochromatic radiation of wavelength 1650 Å and power 5 mW is incident, it is observed that maximum velocity of photoelectron increases to two times. Assuming efficiency of photo-electron generation per incident photon to be same for both the cases, calculate

13. Threshold wavelength for the cell is

(A) 4125 Å **(B) 6925 Å**

(C) 5525 Å **(D) 4825 Å**

Sol. A

14. Saturation current in second case is

(A) 9.2 μA **(B) 13.3 μA**

(C) 15.1 μA **(D) 19.4 μA**

Sol. B

15. Efficiency (in %) of photo-electron generation per incident photon is

(A) 1 **(B) 2**

(C) 3 **(D) 4**

Sol. B

13-15. For radiation λ_1, $\frac{1}{2} m v_1^2 = \frac{hc}{\lambda_1} - w_0$ ... (I)

similarly for λ_2, $\frac{1}{2} m v_2^2 = \frac{hc}{\lambda_2} - w_0$ ... (ii)

But $v_2 = 2v_1$, then from (ii)

$\frac{1}{2} m(2v_1)^2 = \frac{hc}{\lambda_2} - w_0$

$4\left(\frac{hc}{\lambda_1} - w_0\right) = \frac{hc}{\lambda_2} - w_0$

$3w_0 = \frac{4hc}{\lambda_1} - \frac{hc}{\lambda_2} = hc\left[\frac{4\lambda_2 - \lambda_1}{\lambda_1\lambda_2}\right]$

$w_0 = \frac{hc}{3\lambda_1\lambda_2}(4\lambda_2 - \lambda_1) = 3 \text{ eV}$

Hence threshold wavelength of the cell

$\lambda_0 = \frac{hc}{w_0} = \frac{12420}{3} = 4125A^0$

Energy of each photon of wavelength λ_1 is $E_1 = \frac{hc}{\lambda_1}$

$\therefore$ Rate of incident photons of wavelength λ_1 in a radiation of power $P = \frac{P}{E_1} = \frac{P\lambda}{hc}$ per second

$= \frac{1 \times 3000 \times 10^{-10}}{6.6 \times 10^{-34} \times 3 \times 10^8}$

$= 1.5 \times 10^{18}$ /sec.

No. of electrons ejected by these photon $= \frac{4.8\times10^{-3}}{1.6\times10^{-19}}$ per second.

$= 3 \times 10^{16}$ /sec.

$\therefore$ Efficiency of photoelectron generation per incident photon $\eta = (\%) = \frac{3\times10^{16}}{1.5\times10^{18}} \times 100 = 2\ \%$

Rate of ejection of electrons in second case

$= \eta \frac{P\lambda_2}{hc}$ per second $= \frac{2}{100} \times \frac{5\times10^{-3}\times1650\times10^{-10}}{6.6\times10^{-34}\times3\times10^{8}}$

$= 8.33 \times 10^{13}$ electrons /sec.

hence saturation current in second case

$= 8.33 \times 10^{3} \times 1.6 \times 10^{-19} = 13.3\ \mu A.$

Matrix-Match Type

16. Power P is given as $P = a + b t^2 + \frac{(c + t^3)}{d}$. If t represents the time, then

Column A	Column B
(a) Dimension of [a]	(i) $[ML^2T^{-5}]$
(b) Dimension of [b]	(ii) $[ML^2T^{-3}]$
(c) Dimension of [c]	(iii) $[M^{-1}L^{-2}T^{-6}]$
(d) Dimension of [d]	(iv) $[T^3]$

Which of the following is correct?

(A) (a)-(ii), (b)-(i), (c)-(iv) and (d) - (iii). **(B) (a)-(i), (b)-(ii), (c)-(iii) and (d)-(iv).**

(C) (a)-(iv), (b)-(iii), (c)-(ii) and (d) -(i). **(D) (a)-(ii), (b)-(i), (c)-(iii) and (d)-(iv).**

Sol. (A)

17. Match the following

Column I		Column II	
(A)	Angular speed of an electron in nth Bohr's orbit.	(p)	Inversely proportional to n
(B)	Velocity of an electron in nth Bohr's orbit.	(q)	Inversely proportional to n^3
(C)	Current developed due to motion of an electron in nth orbit.	(r)	Directly proportional o Z^2
(D)	Ionization energy of an electron in nth Bohr's orbit	(s)	Directly proportional o Z

Sol. A → (q, r), B → (p, s), C → (q, r), D → (r)

$\omega_n = \omega_0 \frac{Z^2}{n^3}$; $v_n = v_0 \frac{Z}{n}$; $I_n = I_0 \frac{Z^2}{n^3}$; $E_n = E_0 \frac{Z^2}{n^2}$.

Answer questions 18, 19 and 20 by appropriately matching the information given in the three columns of the following table.

Two resistors along with errors are known as

$R_1 = (30 \pm 3)\Omega$

$R_1 = (60 \pm 3)\Omega$

Column –I lists four algebraic functions of R_1, R_2; column –II lists corresponding percentage errors where as column –III lists the absolute errors.

Column 1		Column 2		Column 3	
(I)	R_1R_2	(i)	$\frac{20}{3}\%$	(P)	1.67
(II)	$\frac{R_1}{R_2}$	(ii)	10%	(Q)	180
(III)	$R_1 + R_2$	(iii)	$\frac{25}{3}\%$	(R)	0.075
(IV)	$\frac{R_1R_2}{R_1+R_2}$	(iv)	15%	(S)	6

18. **Pick the correct option.**

(A) (IV) (ii) (R)
(B) (III) (ii) (S)
(C) (II) (i) (Q)
(D) (I) (iv) (Q)

Sol. D

19. **Pick the correct option for series combination of R_1 and R_2.**

(A) (III) (ii) (P)
(B) (III) (i) (S)
(C) (III) (i) (Q)
(D) (III) (iv) (S)

Sol. B

20. **Pick the correct option for parallel combination of R_1 and R_2.**

(A) (IV) (iii) (P)
(B) (IV) (iii) (Q)
(C) (IV) (ii) (R)
(D) (IV) (iii) (S)

Sol. A

Numerical Type

21. **A hydrogen atom in ground state, moving with speed v collides with another hydrogen atom in ground state at rest. If $v \le v_0 = \sqrt{\frac{kE_0}{m_H}} \approx a \bullet bc \times 10^n \text{ m/s}$, then the collision is elastic. Here a, b and c are whole number, less then 9. Find the value of $\frac{a \times b \times c}{k \times n}$.**

$\frac{me^4}{2\hbar^2} = E_0 = 13.6\text{eV/atom} = 2.18 \times 10^{-18}$ **J/atom** $\Rightarrow$ **ionisation energy of H-atom**

$\Rightarrow \quad m_H = 1.67 \times 10^{-27}\text{kg}$ $\Rightarrow$ **Mass of hydrogen atom.**

Sol. 6

According to Bohr's model

$$\Delta E = E_0\left(1-\frac{1}{n^2}\right) \Rightarrow \Delta E_{min} = \frac{3E_0}{4}$$

During inelastic collision, a part of kinetic energy of colliding particles is converted into interna energy. The internal energy of the system of two hydrogen atoms, considered in the problem cannot b changed less than ΔE_{min}. It means if the change in kinetic energy of system in ground frame is less tha ΔE_{min} (or if the kinetic energy of colliding atoms with respect to their centre of mass is less tha ΔE_{min}), then collision must be an elastic one. Hence considering the critical case

$$\frac{1}{2}m_H\left(\frac{v_0}{2}\right)^2 \times 2 = \frac{3E_0}{4} \Rightarrow v_0 = \sqrt{\frac{3E_0}{m_H}}$$

$$\sqrt{\frac{3\times 2.18\times 10^{-18}}{1.67\times 10^{-27}}} = \sqrt{39.1617}\times 10^4 = 6.257\times 10^4 \text{ m/s} \approx 6.26\times 10^4 \text{ m/s}$$

$$\Rightarrow \frac{a\times b\times c}{k\times n} = \frac{6\times 2\times 6}{3\times 4} = 6$$

22. The half lives of two longest lived radioactive isotopes of phosphorus P^{32} and P^{33} are 14 days an 28 days respectively. A sample has been prepared by mixing the two isotopes in the ratio of 4 : 1 of their atoms. Initial activity of the mixed sample is 90 m curie. Find the activity (in m curie) o the sample after 80 days. (Approximately to nearest integer) Take $\ell n2 = 0.7$ and $e^{-2} = 0.14$

Sol. 3

$$\frac{\lambda_1}{\lambda_2} = \frac{\ell n2}{14 \text{ days}}\frac{28 \text{ days}}{\ell n2} = 2$$

$$\frac{A_{10}}{A_{20}} = \frac{\lambda_1 N_{10}}{\lambda_2 N_{20}} = 2\times 4 = 8$$

$\therefore A_{10} = 80$ m curie and $A_{20} = 10$ m curie

$$A = 80e^{-\frac{\ell n2}{14}\times 80} + 10e^{-\frac{\ell n2}{28}\times 80}$$

$$= 80\,e^{-4} + 10\,e^{-2} = 3 \text{ m curie}$$

Numerical Decimal Type

23. Light of wavelength 488 nm is produced by an Argon laser which is used in the photoelectri effect. When light from this spectral line is incident on the emitter, the stopping (cut-off potential of photo electrons is 0.38 V. Find the work function of the material form which th emitter is made.

Sol. 00002.17

$$E = \frac{hC}{\lambda} = \frac{1242 eV}{488} = 2.55 \text{ eV}$$

$$K_{max} = eV_0 = e\times 0.38 = 0.38 \text{ eV}$$

$$K_{max} = \frac{hC}{\lambda} - \phi_0 = E - \phi_0$$

$$\phi_0 = E - K_{max} = (2.55 - 0.38) = 2.17 \text{ eV}$$

24. **When the light of wavelength 400 nm is incident on a metal surface of work function 2.3 eV, photoelectrons are emitted. A fastest photoelectron combines with a He^{2+} ion to form He^{+} ion in its third excited state and a photon is emitted in this process. Find the energy of the photon emitted during combination.**

Sol. 00004.20

Energy of each photon $E = \frac{hC}{\lambda} = \frac{1240}{400} = 3.1$ eV

Now, $K_{max} = E - \phi$

$= 3.1 - 2.3$

$\therefore \quad K_{max} = 0.8$ eV

Energy of He^{+}-ion in third excited state

$$E_4 = -13.6 \times \frac{4}{(4)^2} = -3.4 \text{ eV}$$

Now, the energy of the photon emitted during combination,

$\Delta E = K_{max} - E_4$

$= 0.8 - (-3.4)$

$= 4.2$ eV

25. **When an α-particle with kinetic energy K = 11 MeV collides with a stationary lithium nucleus, the following nuclear reaction takes place**

$^4He + {}^7Li \rightarrow {}^{10}B + n$

If Q – value of this reaction is E = –2.86 Mev. Find the kinetic energy (in MeV) of the neutron outgoing at right angle to the incoming direction of α-particle.

Sol. 3.40

Using conservation of momentum

$P = P_2 \cos\theta$ ...(i)

$P_1 = P_2 \sin\theta$ ...(ii)

Squaring and adding, we get

$P^2 + P_1^2 = P_2^2$

$2mk + 2m_1k_1 = 2m_2K_2$

$4k + k_1 = 10k_2$

$10k_2 - k_1 = 4k$

$\therefore \quad 10k_2 - k_1 = 44.$ ...(iii)

Also, $k_1 + k_2 = k - |Q|$

$k_1 + k_2 = 11. - 2.86$

$k_1 + k_2 = 8.14$ ...(iv)

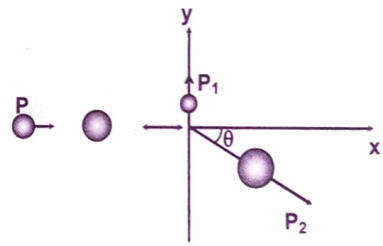
Adding (iii) and (iv), we get

$11k_2 = 52.14$

$\therefore \quad k_2 = 4.74$ MeV

$\therefore \quad k_1 = 8.14 - 4.74$

$k_1 = 3.40$ MeV



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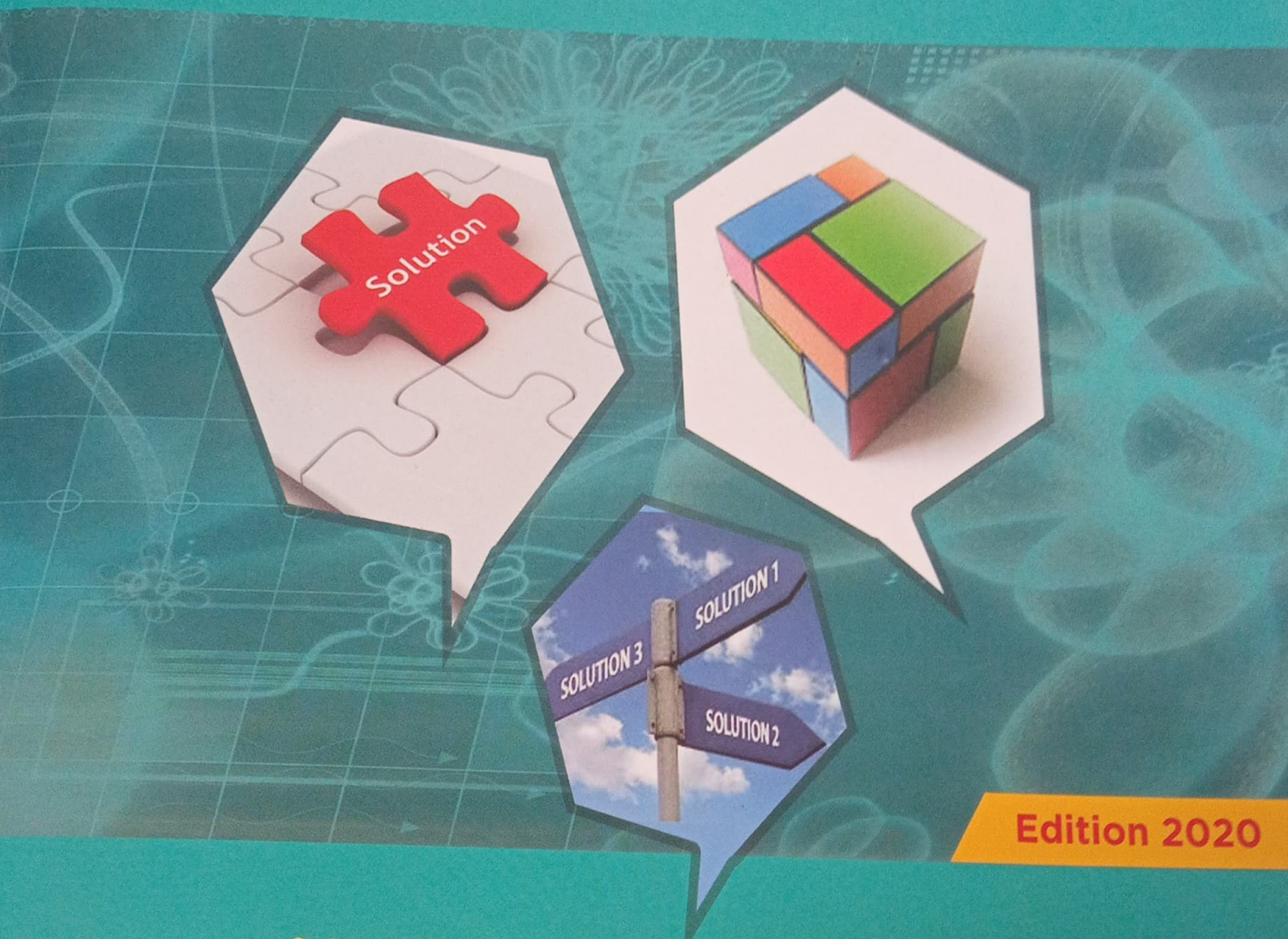

Edition 2020

HINTS & SOLUTIONS

Physics | Chemistry | Mathematics

PHYSICS
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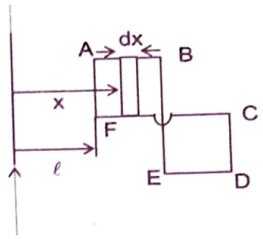
SUBJECTIVE

1. (a) Consider a strip at a distance x from the wire of thickness dx.



Magnetic flux associated with this strip $\Delta\phi = B(x)\,a\,dx = \dfrac{\mu_0 Ia}{2\pi x}\,dx$

$$\Rightarrow \phi = \frac{\mu_0 Ia}{2\pi}\left[\int_{\ell}^{a+\ell}\frac{dx}{x} - \int_{a+\ell}^{2a+\ell}\frac{dx}{x}\right] = \frac{\mu_0 Ia}{2\pi}\ln\left(\frac{(a+\ell)^2}{a(2a+\ell)}\right)$$

$$\because\ M = \frac{\phi}{I_{(primary)}}$$

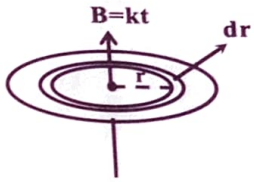
$$\Rightarrow M = \frac{\mu_0 a}{2\pi}\ln\frac{(a+\ell)^2}{a(2a+\ell)}$$

(b) $\varepsilon = -M\dfrac{dI}{dt}$

$$\Rightarrow \varepsilon = -MI_0$$

$$\text{Heat produced} = \frac{\varepsilon^2}{R}t = \frac{\left[\dfrac{\mu_0}{2\pi}\ln\dfrac{(a+\ell)^2}{a(2a+\ell)}\ I_0\right]^2 at}{8\lambda}$$

2. Consider a ring of radius r, width dr and charge on ring



$$dq = (2\pi r dr)\frac{Q}{\pi R^2}$$

Induced emf in ring

$$\varepsilon = \oint \vec{E}.d\vec{\ell} = -\frac{-d}{dt}\int \vec{B}.d\vec{s}$$

Let E is electric field in circular loop.

Then $E.2\pi r = -\pi r^2\dfrac{dB}{dt} = -\pi r^2 k$

$$\therefore\ E = -\frac{r.k}{2}$$

Force on the ring = $dF = dqE = -(dq)\dfrac{rk}{2}$

$$\text{Torque } \tau = \int_0^R rdF = \int_0^R r\left(\frac{2\pi rQdr}{\pi R^2}\right)\frac{rk}{2} = \int_0^R \frac{kQr^3}{R^2}dr = \frac{-kQR^2}{4} = I\alpha$$

$$\therefore\ \alpha = -\frac{kQR^2}{4\left(mR^2/2\right)} = \frac{-kQ}{2m}$$

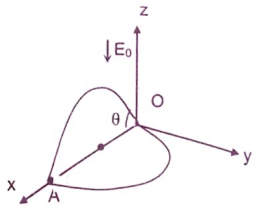
$$\therefore\ \omega = -\frac{kQ}{2m}t$$

$$\therefore\ |\omega| = \frac{kQ}{2m}t$$

3. The acceleration on the charge q, projected along the x – z plane is along the –z direction:

$$a = \frac{qE_0}{m}$$

The second particle moves on the x-y plane in a circle of radius R(say) such that:



$$2R = \frac{2v_0^2}{a}\sin\theta\cos\theta \quad \ldots(i)$$

$$\text{Also, } R = \frac{mv_0}{qB_0} \quad \ldots(ii)$$

$$T = \frac{2v_0\sin\theta}{a} \quad \ldots(iii)$$

$$\& \quad T = \frac{\pi m}{qB_0} \quad \ldots(iv)$$

where T is the time required for the collision to occur

$$\text{By solving, } v_0\sin\theta\cos\theta = \frac{E_0}{B_0}$$

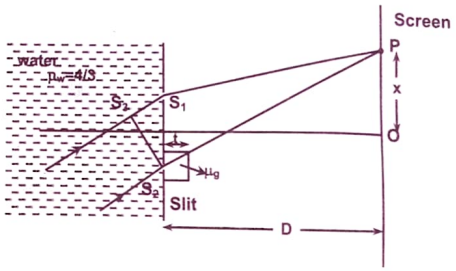
$$\text{Also, } v_0\sin\theta = \frac{\pi}{2}\frac{E_0}{B_0}$$

$$\Rightarrow \cos\theta = \frac{2}{\pi};\ \theta = \cos^{-1}\left(\frac{2}{\pi}\right);\ v_0 = \frac{E_0/B_0}{\frac{2}{\pi}\sqrt{1-2^2/\pi^2}}$$

The maximum value of the Z-coordinate of the first particle is

$$H = \frac{v_0^2\sin^2\theta}{2a} = \frac{\pi^2}{8}\frac{mE_0}{qB_0^2}$$

4. (a) Let central maxima lies at point P at a distance x from the central line. Then optical path difference Δp at point P is



$$(S_2P - t) + t\mu_g - \mu_w d\sin\theta - S_1P$$

$$\Rightarrow S_2P - S_1P + (\mu_g - 1)t - \mu_w d\sin\theta$$

$$\Rightarrow \frac{xd}{D} + (\mu_g - 1)t - \mu_w d\sin\theta$$

For central maxima $\Delta p = 0$

$$x = \frac{D}{d}\left[(\mu_w d\sin\theta) - (\mu_g - 1)t\right]$$

$$= \frac{1}{3\times10^{-4}}\left[\frac{4}{3}\times3\times10^{-4}\times\frac{1}{2} - 0.5\times0.41\times10^{-3}\right]$$

$$= \frac{1}{3\times10^{-4}}\left[2\times10^{-4} - 2.05\times10^{-4}\right] = -\frac{5\times10^{-6}}{3\times10^{-4}}$$

$$= -1.66\times10^{-2}\,m = -1.66\ cm$$

Central maxima lies at a distance 1.66 cm below the central line

(b) At point O, optical path difference is

$$(\mu_g - 1)t - \mu_w d\sin\theta = -5\times10^{-6}\,m$$

So intensity at O

$$I_0 = I + I + 2\sqrt{II}\cos\frac{2\pi}{5\times10^{-7}}\left(-5\times10^{-6}\right)$$

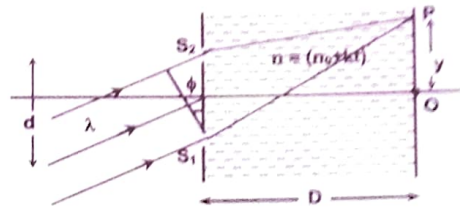
$$I_0 = 4I$$

5. (a) $S_1P - S_2P = \dfrac{dy}{D}$

$$\Delta x = (n_0 + kt)\frac{dy}{D} - d\sin\phi = 0$$

For central maxima.

$$\therefore \quad y = \frac{D\sin\phi}{n_0 + kt}$$ y-coordinates of central maximum



(b) $\dfrac{dy}{dt} = \dfrac{-kD\sin\phi}{(n_0 + kt)^2}$ = velocity of central maximum

(c) For central maxima to be formed at O

$$n'\left(\frac{n}{n'} - 1\right)P = d\sin\phi$$ Here $n' = n_0 + kt$, n = refractive index of plate

$$n = n_0 + kt + \frac{d\sin\phi}{P}$$

6. $$E = -(13.6\text{ eV})\ Z^2\left(\frac{1}{(Z+2)^2} - \frac{1}{Z^2}\right)$$

$$= -13.6 \times \left(\frac{Z^2 - (Z+2)^2}{(Z+2)^2}\right) = \frac{4(Z+1)\times 13.6}{(Z+2)^2}\text{ eV} \qquad \ldots(1)$$

Now energy of electron is $k = \dfrac{h^2}{2\lambda^2 m}$; $\left(\text{we have } \lambda = \dfrac{h}{\sqrt{2mk}}\right)$

or, $k = 6\text{eV}$

So, $\dfrac{4(Z+1)\times 13.6}{(Z+2)^2} = 6 + 4.2 = 10.2\text{ eV}$

Or, $\dfrac{(Z+1)}{(Z+2)^2} = \dfrac{3}{16} \Rightarrow (Z-2)(3Z+2) = 0$

So, the value of Z = 2 (neglecting the negative/ fractional value)

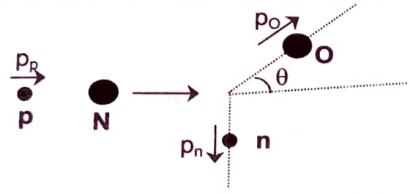
7. (a) The nucleus is identified by : Z = 8, A = 15 $\Rightarrow$ X = $_8O^{15}$

(b) $Q = [m(p) + m(N^{15}) - m(O^{15}) - m(n)]c^2$

$= [1.007825 + 15.000 - 15.0031 - 1.008665] \times 931.5\text{ MeV} = -3.67\text{ MeV}$

(c) $K_{th} = -Q\dfrac{1}{4\pi\varepsilon_0}\left(\dfrac{Q}{2\pi}d\theta\right)q \Big/ R^2 = 3.9\text{ MeV}$

(d) Now, $E_k = 2 \times K_{th} = 2 \times 3.9\text{ MeV} = 7.8\text{ MeV}$ and $Q = -3.63\text{ MeV}$



(i) Conservation of momentum: $\sqrt{\dfrac{T}{\mu}} = \sqrt{\dfrac{1}{8\pi^2\varepsilon_0}\dfrac{Qq}{R^2}\Big/\dfrac{m}{2\pi R}} = \sqrt{\dfrac{Qq}{4\pi\varepsilon_0 mR}}$

$$p_o\sin\theta = p_n$$

(ii) Conservation of energy

$$\frac{p_n^2}{2m_n} + \frac{p_O^2}{2m_O} = E_k + Q$$

$$p_n^2 = \frac{E_\alpha\left(1-\frac{m_p}{m_\alpha}\right)+Q}{\frac{1}{2m_O}+\frac{1}{2m_n}}$$

Substituting the values, $p_n = 79.4$ MeV/c

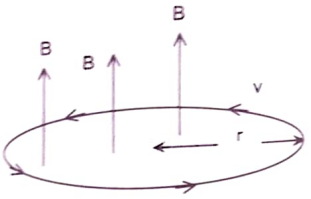
$p_O\cos\theta = 121$ MeV/c

$p_O\sin\theta = 79.4$ MeV/c

$p_O = 145$ MeV/c

$\theta = 33°$.

8. For the electron, we can write, m



$$\frac{dW}{dQ}\% = \frac{1}{4}\times 100 = 25\% = evB$$

or, $$\frac{mv_{max}}{(T/4)} = \frac{2mg}{\pi} = \int_{\frac{R}{2}}^{2R}\left(p_0+\frac{4S}{r}\right)4\pi r^2 dr$$

... (i)

Using the Bohr quantisation condition,

$$mvr = \frac{3R^2}{2} = n\frac{m}{2f}\sqrt{\frac{T}{9\mu}}, \qquad n \to \text{integer.}$$

or, $$vr = \frac{n}{2f}\sqrt{\frac{T}{4\mu}} \qquad \ldots \text{(ii)}$$

(a) $$r^2 = \frac{K_2}{K_1 - K_2} = \frac{2}{1}$$

or, $$r = \frac{K_1}{K_2} = \frac{3}{2}\,\frac{\sigma^2}{2\varepsilon_0} = \frac{P_1}{8} + \frac{\sigma^2}{2\varepsilon_0} = P_0 + \frac{4T}{2r}a_0 \qquad \ldots \text{(iii)}$$

(b) T (kinetic energy) $$= \frac{q}{4\pi(2r)^2}mv^2 = Q\frac{\Delta\ell}{2\pi r}m^2v^2$$

$$= \frac{1}{2m}\times\frac{Q}{2\pi}\oint E_i\Delta\ell' = \frac{Q}{2\pi}\left(-\pi r^2\frac{\Delta B}{\Delta t}\right)\frac{2}{mr^2}\Delta t = \frac{2Q}{2m}\Delta B = \frac{QB}{m}n\frac{hc}{\lambda_{min}}Be$$

$$= \frac{1}{2}mv^2 = \frac{1}{2}\left(\frac{m^2v^2}{m}\right)n = \frac{h^2}{2m\lambda_1^2}\frac{h}{\sqrt{2meV}} \qquad \ldots \text{(iv)}$$

(c) The time period of revolution of the electron $\tau = \pi R^2\times 2B + \pi\times r^2\times B = (2R^2+r^2)\pi\mu_0 nkt$

The current, $$i = \frac{mv^2}{r} = qvB$$

The magnetic moment, $\mu = i$. Area $= \pi r^2\times\frac{qE}{m}t = \int_0^5|\vec{v}|dt = \int_0^5|3t^2-18t+24|dt\,.\,n\hbar$

The potential energy of interaction,

$$U = -\vec{\mu}.\vec{B} = \frac{1}{2}n\hbar\left(\frac{eB}{m}\right) \qquad \ldots \text{(v)}$$

(d) The total energy,

$$E = T + U = n\hbar\left(\frac{eB}{m}\right), \text{ which is quantised}$$

(e) The total magnetic flux through the nth orbit:

$$\phi = \pi r^2.B = \pi\left(\frac{n\hbar}{Be}\right).B = \frac{n\hbar}{2e} \qquad \ldots \text{(vi)}$$

9. For minimum de-Broglie wavelength of electron, the momentum and hence KE of electron must be maximum

Case - I.

When hydrogen spectra is used, for maximum energy of photoelectron

$$13.6\left[\frac{1}{1^2}-\frac{1}{2^2}\right]-\phi=\frac{1}{2}mv_1^2 \quad \Rightarrow \quad \frac{1}{2}mv_1^2=\frac{3}{4}\times 13.6eV-\phi$$

For unknown gas of atomic no. z

$$13.6\,z^2\left[\frac{1}{1^2}-\frac{1}{2^2}\right]-\phi=\frac{1}{2}mv_2^2 \quad \Rightarrow \quad \frac{1}{2}mv_2^2=z^2\frac{3}{4}\times 13.6ev-\phi$$

Comparing the de-Broglie wavelengths

$$\frac{\lambda_1}{\lambda_2}=\frac{h/mv_1}{h/mv_2}=\left(\frac{v_2}{v_1}\right)$$

$$\Rightarrow \left(\frac{\lambda_1}{\lambda_2}\right)^2=\frac{\frac{3}{4}\times 13.6-\phi}{\frac{3}{4}\times 13.6z^2-\phi}=\frac{1}{6.1} \qquad \ldots(1)$$

For the unknown gas

Energy of first lyman line- Energy of Balmer series limit = 2 × (13.6)eV

$$(13.6)z^2\left(\frac{3}{4}\right)-13.6z^2\left(\frac{1}{4}\right)=2\,(13.6)$$

$z^2/2 = 2$; $z = 2$

$\therefore$ Unknown gas is helium.

Now from equation (1)

$$\frac{10.2-\phi}{40.8-\phi}=\frac{1}{6.1}$$

$\Rightarrow \phi = 4.2eV$

10. The focal length of the lens is given by

$$\frac{1}{f}=\frac{1}{|v|}+\frac{1}{|u|}$$

Substituting the values, we get,

f = 19.9 cm, 20.1 cm, 19.9 cm, 20.2

The average focal length $\cong$ 20.0 cm

The error in a typical measurement:

$f = uv/u - v$

$= -(uv)/-(u+v) = uv/u+v$

$\Rightarrow \Delta f/f = \Delta u/u + \Delta v/v + \Delta(u+v)/(u+v)$

$= 0.4/40 + 0.4/40.5 + 0.8/80.5$

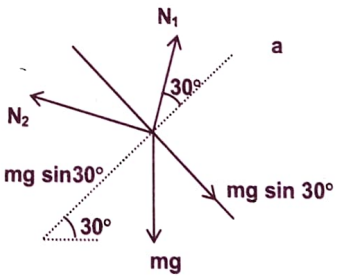
$= \Delta f = \pm 0.6$ cms

11. $$\frac{\sqrt{3}N_1}{2}-\frac{mg}{2}-\frac{N_2\sqrt{3}}{2}=ma$$

$$\frac{\sqrt{3}N_1}{2}-\frac{\sqrt{3}N_2}{2}=ma+\frac{mg}{2} \qquad \ldots(i)$$

$$\Rightarrow \quad \frac{N_1+N_2}{2}=\frac{mg\sqrt{3}}{2}$$

$$\Rightarrow \quad \frac{\sqrt{3}N_1}{2}+\frac{\sqrt{3}N_2}{2}=\frac{3mg}{2} \qquad \ldots(ii)$$



From equation (i) and (ii), we have

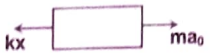
$\sqrt{3}N_1 = 2mg + ma,\ \sqrt{3}N_2 = mg - ma$

Since cylinder does not loose the contact, so $N_1 \geq 0,\ N_2 \geq 0$

$\Rightarrow \quad a \leq g$

$\Rightarrow \quad a_{max} = g$

12. Draw the FBD of block in the frame of accelerated frame. In this frame, block has maximum velocity when it is in equilibrium with respect to accelerated frame



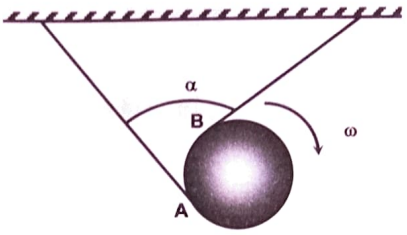
Using work energy theorem

$$\frac{1}{2}kx^2 + \frac{1}{2}mv_0^2 = ma_0x$$

$$\Rightarrow \quad \frac{1}{2}kx\frac{m^2a_0^2}{k^2} + \frac{1}{2}mv_0^2 = \frac{m^2a_0^2}{k}$$

$$\Rightarrow \quad v_0^2 = \frac{ma_0^2}{k} \quad \Rightarrow \quad v_0 = a_0\sqrt{\frac{m}{k}}$$

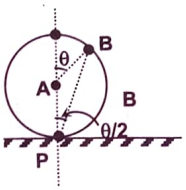
13. The velocities of point A and B being perpendicular to the strings; the instantaneous axis of rotation must be at D, instantaneous point of intersection of the line of the two strings.



Therefore,

$$V_{CM} = \omega.OD = \frac{\omega \times OB}{\cos\frac{\alpha}{2}} = \frac{\omega R}{\cos\frac{\alpha}{2}}$$

14. Assuming that the hoop rotates by θ (so that the attached particle also rotates by θ) starting from its initial position, we find the angular speed of the hoop + particle.



The instantaneous axis of rotation is located at P. we apply conservation of mechanical energy to get

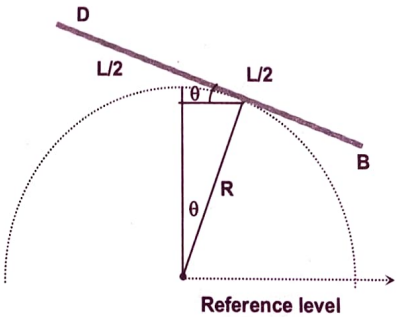
$$mgR(1-\cos\theta) = \frac{1}{2}(2mR^2)\omega^2 + \frac{1}{2}m(2R\cos\frac{\theta}{2}\omega)^2$$

Differentiating and putting $\theta = 90°$

We solve for $\alpha = \omega\frac{d\omega}{d\theta}$ and get $4\alpha = \frac{3g}{2R}$

or $\alpha = \frac{3g}{8R}$

15. Gain in kinetic energy = loss in potential energy



$$\frac{1}{2}\frac{m\ell^2}{12}\omega^2 = mgR[\theta\sin\theta + \cos\theta - 1]$$

$$\omega = \frac{2}{\ell}\sqrt{6gR(\theta\sin\theta + \cos\theta - 1)}$$

$U_i = mg(R\theta \sin\theta + R\cos\theta)$

$U_f = mgR$

16. $mg \times 0.4 = \frac{1}{2}mv^2$

$v = 2\sqrt{2}$ m/s

(from conservation of mechanical energy just before the string becomes taut)

during the application of impulse, velocity along perpendicular to the string remain same. So after impulse velocity of the ball equal to $\frac{6\sqrt{2}}{5}$ m/s

again from conservation of mechanical energy

$$\frac{1}{2}m\left(\frac{6\sqrt{2}}{5}\right)^2 + mg(0.1) = \frac{1}{2}mv_f^2$$

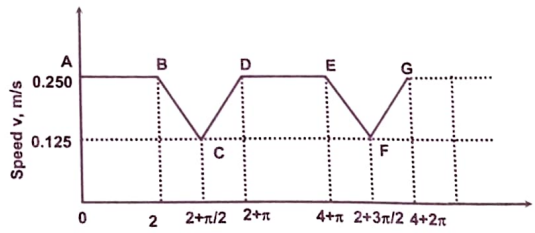
Now velocity at the bottommost point $v_{final} = 2.2$ m/s

17. For the motion from A to B

applying $s = vt$,

$2 = 0.25\, t_{AB}$

$t_{AB} = 8$



For the motion from B to C

$$-\frac{dv}{d\ell} = \frac{0.125}{\frac{\pi}{2}} = \frac{1}{4\pi} = -\frac{dv}{d\ell} \times \frac{dt}{dt}$$

$$\int_0^{t_{BC}} \frac{dt}{4\pi} = \int_{0.25}^{0.125} -\frac{dv}{v}$$

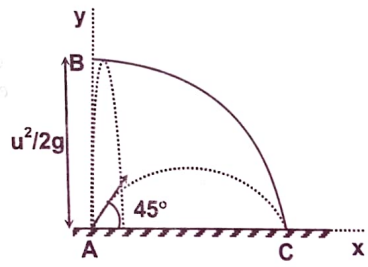
$\Rightarrow t_{BC} = 4\pi \ln 2$

$t_{BC} = t_{CD} = t_{EF} = t_{FG}$

$t_{AB} = t_{DE}$

total time will be sum of all times $= 16(1 + \pi \ln 2)$

18. In order for the trajectory of Q to touch that of P when the latter is projected vertically upward, Q must be projected horizontally from the highest point. To visualize this, consider the trajectories of P as the angle of projection of P(θ, say), approaches 90° (i.e., for vertical projection). Let Q be projected horizontally from B with a speed v, where $A = \frac{u^2}{2g}$. The point of impact of Q on the horizontal plane represents the point of maximum range for P along the horizontal.



$$OC = \frac{u^2}{g} = v\sqrt{\frac{2h}{g}}, \text{ where } h = \frac{u^2}{2g}$$

i.e., $v = u$

The trajectory of Q is

$$\frac{u^2}{2g} - y = \frac{1}{2}gt^2, \text{ where } t = x/v\ (v \equiv u)$$

19. we draw the FBD of the coin

f_T = tangential component of frictional force on coin

f_R = radial component of frictional force on coin s

$f_T = m\alpha r$

$f_r = m\omega_f^2 r$

$$\sqrt{f_R^2 + f_T^2} \le \mu_g mg$$

$$2\pi N = \frac{\omega_r^2}{2\alpha}$$

Solving the equations, we get

$$N = \frac{\left(\frac{\mu_g g}{r} - \alpha^2\right)^{1/2}}{4\pi\alpha}$$

20. The net force is always equal to mass x acceleration Newton's 2nd Law.

$F_{shaded} = M_{shaded}\, A = \frac{M}{3} A$

$$A = \frac{F}{M}$$

$F_{shaded} = F/3.$

21. We determine the velocity at C so that it reaches A along a parabolic trajectory; then applying the law of conservation of energy to the motion ABC (as non-conservative force are absent)

v = velocity at point $C = \frac{3}{2}\sqrt{gR}$

applying conservation of energy from A to C

$$\frac{1}{2}mu^2 = 2mgR + \frac{1}{2}mv^2$$

$$\Rightarrow u = \frac{5}{2}\sqrt{gR}$$

22. We determine the speed required at D so that the block falls into the hole at E by following a parabolic trajectory.

v_d = speed at D

$v_d = \sqrt{8gd}$

Applying conservation of energy for motion from A to D

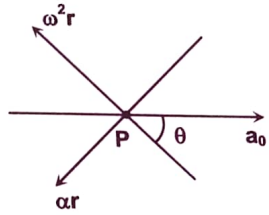
$$\frac{1}{2}k\delta^2 = \mu_k mgd + mgd + \frac{1}{2}m(8gd) \Rightarrow \delta = \sqrt{\frac{2mgd}{k}(\mu_K + 5)}$$

23. The acceleration of the point P is that due to the axis (a_0) and due to rotation about the axis.

$a_0 \cos\theta = \omega^2 r$

$a_0 \sin\theta = \alpha r$

$a_0 = \alpha R$



24. A the car C accelerates to a velocity v_0 relative to the double-boat system, the two boats accelerate to the left.

V_C(to right) + v_A(to left) = v_0

$Mv_C = 2Mv_A$

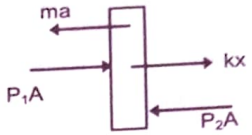
Solving, we find $v_A = \frac{mv_0}{m+2M}$, $v_C = \frac{2Mv_0}{m+2M}$

After the car brakes to a stop, the tension in the string connecting A, B becomes zero. Applying conservation of momentum to A and C

We find the velocity of A(to right):

$$v_A' = \frac{mMv_0}{(m+M)(2+2M)}$$

25. Free body diagram of the separator



$P_2 A + ma = P_1 A + kx$ ... (1)

where A is the area of cross section of separator and x is the stretch of the spring

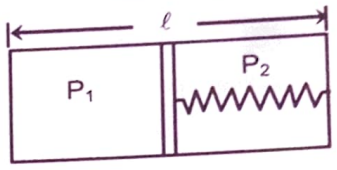


Figure 1

We can write

$$P_0\left(\frac{V_0}{2}\right)^{\gamma} = P_1\left(\frac{V_0}{2} - Ax\right)^{\gamma} \quad \ldots (2)$$

and $$P_0\left(\frac{V_0}{2}\right)^{\gamma} = P_2\left(\frac{V_0}{2} + Ax\right)^{\gamma} \quad \ldots (3)$$

from equation (2)

$$P_1 = P_0\left(\frac{V_0}{(V_0 - 2Ax)}\right)^{\gamma} = P_0\left(\frac{1}{1-(2Ax/V_0)}\right)^{\gamma}$$

$$P_1 = P_0\left(1 - \frac{2Ax}{A(\ell/2)}\right)^{-\gamma} = P_0\left(1 + \frac{4x\gamma}{\ell}\right) \quad \ldots (4) \qquad (x << \ell)$$

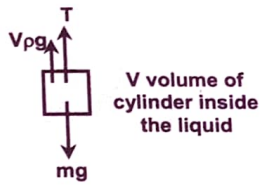
Similarly $P_2 = P_0\left(1 - \frac{4x\gamma}{\ell}\right)$ ... (5)

Substituting the values of P_1 and P_2 in equation (1)

$$P_0\left(1 - \frac{4x\gamma}{\ell}\right)A + ma = P_0\left(1 + \frac{4x\gamma}{\ell}\right)A + kx$$

$$\left[k + \frac{8\gamma P_0 A}{\ell}\right]x = ma \Rightarrow x = \frac{ma}{\left[k + \frac{8\gamma P_0 V_0}{\ell^2}\right]}$$

26. $T = kx_0$ ... (1)



and $T + V\rho g = mg$ ... (2)

$\Rightarrow kx_0 + V\rho g = mg$ ... (3)

when m is displaced by Δx downwards

$ma = T' - k(x_0 + \Delta x)$ ... (4)

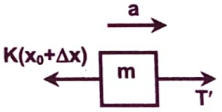
$\Rightarrow ma = mg - (v + A\Delta x)\rho g - T'$ ... (5)

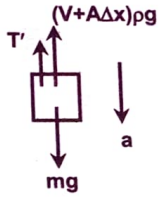
Adding $2ma = mg - k(x_0 + \Delta x) - (v + A\Delta x)\rho g$

$= -[k + A\rho g]\Delta x$; using equation (3)

$$\Rightarrow a = -\frac{[k + A\rho g]}{2m}\Delta x$$

$$\therefore T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{2m}{[k + A\rho g]}}$$





27. $x = 2 + \frac{1}{\sqrt{2}}\sin 2t$

$\Rightarrow \quad 2 = \sqrt{\frac{k}{m}} \Rightarrow k = 4\,N/m$

$\omega L = \frac{1}{\omega C}$

Just before collision, velocity of the shell,

$2\omega L = \frac{1}{2\omega c'}$

Now, $P_i = P_f$

$0 + m \times 4 = (m + m) \times v_0 \Rightarrow v_0 = 2$ m/s

Using conservation of energy,

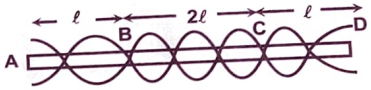
$\therefore C' = \frac{1}{4\omega^2 L} = \frac{LC}{4L} = \frac{C}{4} kx^2 + \therefore \frac{\Delta C}{C} = \frac{3}{4}(2m)(2)^2$

$= \frac{1}{f_1} = \frac{(1.5-1)}{R} + \left(\frac{4}{3}-1\right)\left(-\frac{1}{R}-\frac{1}{R}\right) + \left(\frac{1.5-1}{R}\right) = \frac{1}{R} - \frac{2}{3R} = \frac{1}{3R}\ (2m)v'^2$

$\Rightarrow \quad \frac{1}{f_2} = \frac{(1.5-1)}{R} + \left(\frac{4}{3}-1\right)\left(-\frac{1}{R}\right) + \left(\frac{1.5-1}{R}\right) = \frac{1}{R} - \frac{1}{3R} = \frac{2}{3R}$

28. For section AB

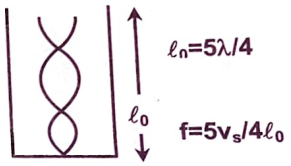
$f = \frac{5V_s}{4\ell_0} = \frac{V_{rod}}{\lambda}$



Where, $\ell = \frac{\lambda}{4} + \frac{\lambda}{2} = \frac{3\lambda}{4}$

$\therefore 4\ell = 3\lambda$

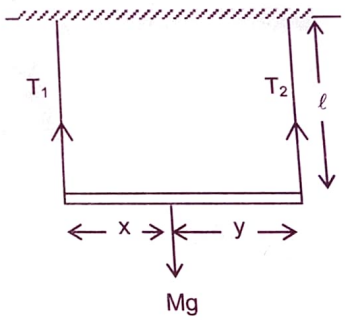
$\therefore$ length of rod $4\ell = 3\lambda$



$= 3 \times \frac{4\ell_0 V_{rod}}{5V_s}$

$= \frac{12\rho_0 V_{rod}}{5V_s}$

29. $A_0 e^{-\lambda t} = N_0 e^{-\lambda t}$



& $y = L - x = \frac{\alpha L^2 + 3L}{3[\alpha L + 2]}$

$T_1 L = Mgy \quad$ where $M = \int_0^L a\rho_0[1+x]dx = \frac{a\rho_0[2L + \alpha L^2]}{2}$

$\Rightarrow \quad T_1 = \frac{a\rho_0 gL[3+\alpha L]}{6}, \quad T_2 = \frac{a\rho_0 gL[3+2\alpha L]}{6}$

$V_1 = \sqrt{\frac{a\rho_0 gL[\alpha L + 3]}{6\mu_0}}, \quad V_2 = \sqrt{\frac{a\rho_0 gL[3+2\alpha L]}{6\mu_2}}$

Dividing, $\frac{V_1}{V_2} = n = f_1/f_2$

$\mu_2 = \frac{\mu_0 n^2[3+2\alpha L]}{\alpha L + 3}$

30. (a) The electric field at a distance x is

$$E = \frac{\lambda}{2\pi\varepsilon_0 x}$$

The net force on the dipole is

$$F = p\frac{dE}{dx} = (2a)q\left[\frac{-\lambda}{2\pi\varepsilon_0 x^2}\right] = \frac{-\lambda qa}{\pi\varepsilon_0 x^2}$$

The negative sign indicates that the force is attractive .

(b) $W = \Delta U$

$U_i = -\vec{p}.\vec{E} = -pE\cos 0^\circ = -pE$

$U_f = -\vec{p}.\vec{E} = -pE\cos 180^\circ = +pE$

$$W = 2pE = 2(2a)q\frac{\lambda}{2\pi\varepsilon_0 x} = \frac{2\lambda qa}{\pi\varepsilon_0 x}$$

(c) Restoring torque

$\tau = -pE\sin\theta$

$$\tau = I\frac{d^2\theta}{dt^2} = \left(2ma^2\right)\frac{d^2\theta}{dt^2}$$

$$\therefore \quad 2ma^2\frac{d^2\theta}{dt^2} + \left[\frac{\lambda qa}{\pi\varepsilon_0 x}\right]\theta = 0$$

or $$\frac{d^2\theta}{dt^2} + \left[\frac{\lambda q}{2\pi\varepsilon_0 max}\right]\theta = 0$$

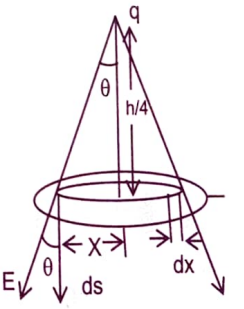
$$T = 2\pi\sqrt{\frac{2\pi\varepsilon_0\, max}{\lambda q}} = \sqrt{\frac{8\pi^3\varepsilon_0\, max}{\lambda q}}$$

31. $\phi = \phi_c + \phi_p = q/\varepsilon_0$

$\phi \rightarrow$ total flux linked

$\phi_C \rightarrow$ flux linked with curved surface

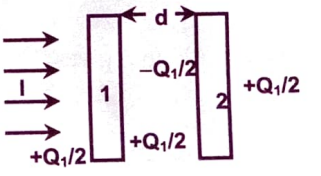
$\phi_P \rightarrow$ flux linked with plane surface



$$\phi_p = \frac{q}{2}(1-\cos\phi) = \frac{q}{2}\left(1 - \frac{h/4}{\sqrt{R^2 + \frac{h^2}{16}}}\right)$$

$$\therefore \quad \phi_C = \frac{q}{\varepsilon_0} - \phi_p \Rightarrow \phi_c = \frac{q}{\varepsilon_0}\left(\frac{1}{2} + \frac{h}{2\sqrt{16R^2 + h^2}}\right).$$

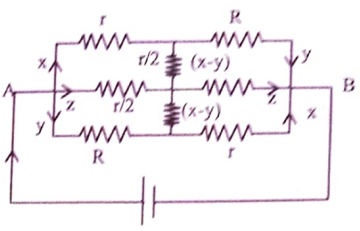
32. $N = \dfrac{\lambda IAt}{hc}$, where N is number of photons falling on the plate.

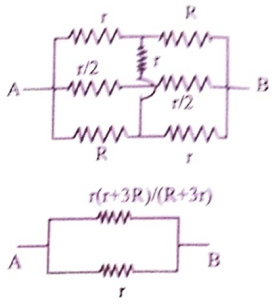


$$Q_1 = \frac{\lambda IAt}{hc}e, \quad v_1 - v_2 = \frac{Q/2}{C}, \text{ where } C = \frac{\varepsilon_0 A}{d}$$

$$V_1 - V_2 = \frac{\lambda IAte}{2hc\varepsilon_0 A}d = \frac{\lambda Idte}{2hc\varepsilon_0}$$

33. When hypothetical battery is connected to the given circuit the current distribution will be as shown.



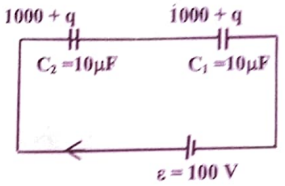


$$\frac{1}{R_{AB}} = \frac{(R+3r)}{r(r+3R)} + \frac{1}{r}$$

$$= \frac{R+3r+r+3R}{r(r+3R)}$$

$$\Rightarrow R_{AB} = \frac{r(r+3R)}{4(R+r)}$$

34. Case I:-



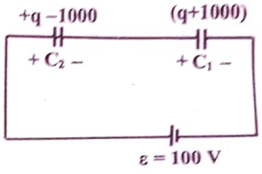
$$\text{KVL} \Rightarrow 100 - \frac{2(1000+q)}{C} = 0$$

$Q = 500\ \mu C$

Work done by the battery, $W = 0.5 \times 10^{-3} \times 100 = 0.05$ J

Case II:-

KVL



$$\Rightarrow 100 - \left(\frac{(q-1000)}{10}\right) - \left(\frac{(q+1000)}{10}\right) = 0$$

$Q = 500\ \mu C$

$\therefore$ Work done by the battery, $W = \Delta qE = 0.05$ J

35. (i) $i_{rms} = \frac{40}{4} = 10$ Amps.

$i_{peak} = 10 \times \sqrt{2} = 10\sqrt{2}$ Amps.

(ii) $E_{rms} = \sqrt{40^2 + (40-10)^2} = 50$v

$E_{peak} = E_0 = 50\sqrt{2}$ v.

(iii) $\omega L\, i_{rms} = 40$

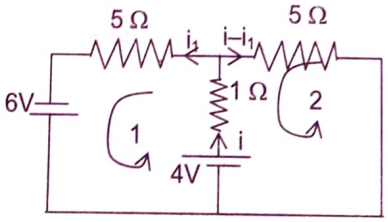
$$L = \frac{40}{10 \times 100\pi} = \frac{1}{25\pi} \text{H.}$$

$$\frac{i_{rms}}{\omega C} = 10,\ C = \frac{10}{100\pi \times 10} = \frac{1}{100\pi} \text{F}$$

(iv) For maximum heat, current through circuit is maximum

so, $\omega L = \frac{1}{\omega C} \Rightarrow C = \frac{1}{400\pi}$ F

36. In steady state circuit will be as shown



KVL for loop 1 gives

$10 - i - 5i_1 = 0$ ...(1)

KVL for loop 2 gives

$4 - 6i + 5i_1 = 0$ ...(2)

From (1) & (2)

$i = 2$ Amp & $i_1 = 1.6$ Amp

So $i - i_1 = 0.4$ Amp

So, Current in right of 4V battery $= i - i_1 = 0.4$ Amp

Current in left of 4V battery $= i_1 = 1.6$ Amp

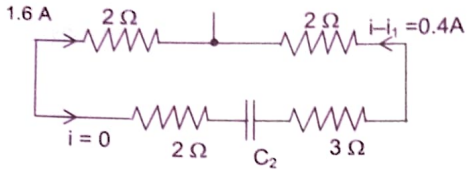
Charge on $C_1 = C(i - i_1)1 = 10 \times 0.4\mu C$

$\Rightarrow \quad q_1 = 4\mu C$

Let charge on $C_2 = q_2$

$$-\frac{q_2}{C_2} - 2\times\frac{2}{5} + \frac{8}{5}\times 2 = 0$$

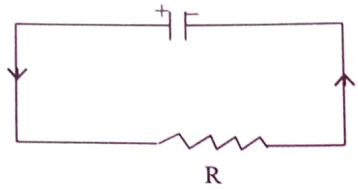
$$\Rightarrow \quad q_2 = C_2\left(\frac{12}{5}\right) = 12\ \mu C$$



37. Capacitance at $t = 0$, $C_0 = \frac{\varepsilon_0 A}{d_0}$, $C = C_0(1 + t)$

Using Kirchoff's law

$$\frac{q}{C} - Ri = 0$$



(i) $$\frac{q}{C_0(1+t)} + R\frac{dq}{dt} = 0$$

$$\frac{dq}{q} = -\frac{1}{RC_0}\frac{dt}{(1+t)}$$

$$\ln q\,|_{Q_0}^{q} = -\frac{1}{RC_0}\ln(1+t)\,|_0^t$$

$$\ln (q/Q_0) = -\frac{1}{RC_0}\ln(1+t)$$

$$\ln (q/Q_0) = \ln(1+t)^{-\frac{1}{RC_0}}$$

$$q = Q_0\,(1+t)^{-\frac{1}{RC_0}}$$

$$V = \frac{q}{C} = \frac{Q_0(1+t)^{-\frac{1}{RC}}}{C_0(1+t)} = \frac{Q_0}{C_0}(1+t)^{-\left(\frac{1}{RC}+1\right)}$$

which gives $t = \left\{\left(\frac{Q_0}{C_0 V}\right)^{\left(\frac{RC}{RC+1}\right)} - 1\right\}$

38. (i) Just after the key is closed

The circuit will be as shown in figure 1

Then circuit can be simplify as figure2

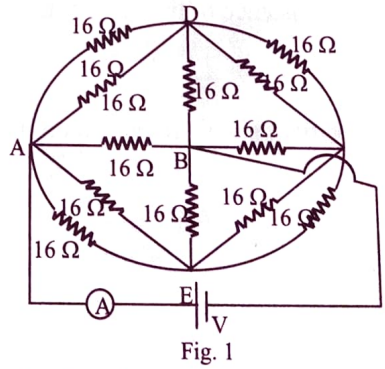


Fig. 1

The points D and E will be at same potential hence the circuit can be further simplified as shown in figure 3 & 4.

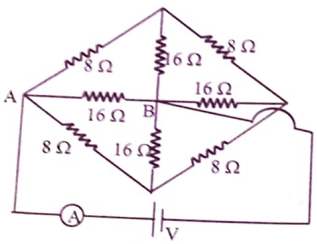


Fig. 2

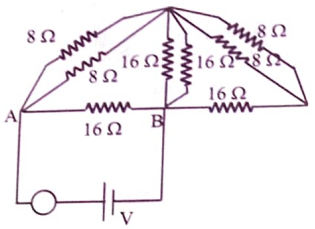


Fig. 3

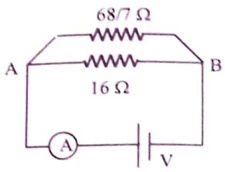


Fig. 4

Hence equivalent resistance across A and B, $R_{eq} = \dfrac{\frac{68}{7}\times 16}{\frac{68}{7}+16} = 6.04\ \Omega$

Hence reading of ammeter $I = \dfrac{12}{6.04} \approx 2A.$

After long time the circuit can be redrawn as

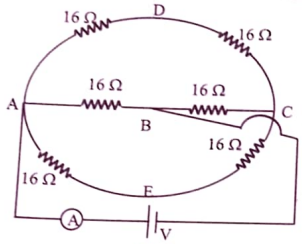


Fig. 1

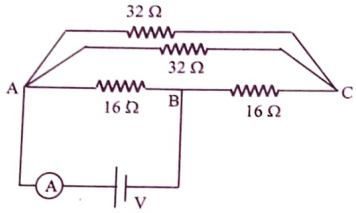


Fig. 2

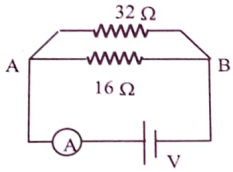


Fig. 4

$$R_{AB} = \frac{16\times 32}{16+32} = \frac{32}{3}\ \Omega$$

Hence the reading of the ammeter $I = \dfrac{9}{8}A$

(ii) The current through paths ADC and AEC will be same and will be equal to

$$I_2 = \frac{1}{2}\times\frac{9}{8}\times\frac{16}{48} = \frac{3}{16}A$$

Hence potential difference across AD, AE, DC and CE, $V' = 16 \times \dfrac{3}{16} = 3V$

Hence the charge in the C_1, C_2, C_5 and C_6 $q = 100 \times 3 = 300\ \mu C$

Also $V_{DB} = V_{EB} = 12 - 3 = 9\ V$

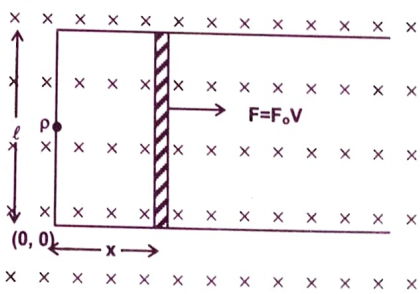
Hence the charge in the capacitor C_3 and C_4 $= 100 \times 9 = 900\ \mu C$

39. At any given instance of time the slide wire is at distance x from origin, then the resistance of the circuit is R $= \dfrac{k(2x+\ell)}{a}$

If the velocity of slide wire is v, then the emf generated is $B\ell$ v so we have

$$B\ell V - \frac{k}{a}(2x+\ell)I = 0$$

Or, $I = \dfrac{B\ell a}{k}\left(\dfrac{v}{2x+\ell}\right)$



This current exerts magnetic force in the wire given by F′

So, $F' = I\ell B = \dfrac{B\ell a}{k}\left(\dfrac{v}{2x+\ell}\right)\ell B = \dfrac{B^2\ell^2 a}{k}\left(\dfrac{v}{2x+\ell}\right)$

Since, $F - F' = \frac{mdv}{dt}$

So $F_0 v - \frac{B^2\ell^2 a}{k}\left(\frac{v}{2x+\ell}\right) = mv\frac{dv}{dx}$

$\Rightarrow \frac{F_0}{m} - \frac{B^2\ell^2 a}{km}\left(\frac{1}{2x+\ell}\right) = \frac{dv}{dx}$

$\Rightarrow \int_0^x \left[\frac{F_0}{m} - \frac{B^2\ell^2 a}{km}\left(\frac{1}{2x+\ell}\right)\right] dx = \int_{v_0}^{v} dv$

$\frac{F_0}{m}x - \frac{B^2\ell^2 a}{2km}\left\{\ln\left(\frac{2x+\ell}{\ell}\right)\right\} + v_0 = v$

40. Resistance of PQ = 2R

Since equivalent resistance of the circuit is R/2, in static condition current through the battery is I = 2V/R

The current through CP and CQ is i = I/2 = V/R

Consider the FBD of ring and the blocks

Consider torque about 'C'

$\tau_N = 0$

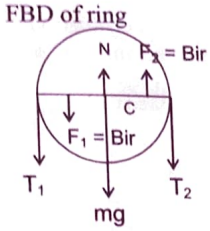
$\tau_{mg} = 0, \ \tau_{F2} = \frac{Bir^2}{2}(\hat{k})$

$\tau_{T1} = T_1 r\hat{k}$

$\tau_{T_2} = T_2 r(-\hat{k})$

$\tau_{F_1} = \frac{Bir^2}{2}\hat{k}$

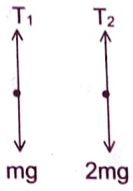
FBD of ring



$\because \ \tau_N + \tau_{mg} + \tau_{F_1} + \tau_{F_2} = 0$

$\Rightarrow T_1 r - T_2 r + \frac{Bir^2}{2} + \frac{Bir^2}{2} = 0$ ... (i)

FBD of blocks



As system is in static equilibrium

$T_1 = mg$ ... (i)

and $T_2 = 2mg$ ... (ii)

Putting the values of T_1 and T_2 in equation (i)

$mgr - 2mgr + Bir^2 = 0$

$\Rightarrow i = \frac{mg}{Br} \Rightarrow \frac{V}{R} = \frac{mg}{Br} \Rightarrow V = \frac{mg}{Br}$

41. For the electric field, $V = -\frac{E_0 x^2}{2\ell} - \frac{E_0 y^2}{2\ell} + K$

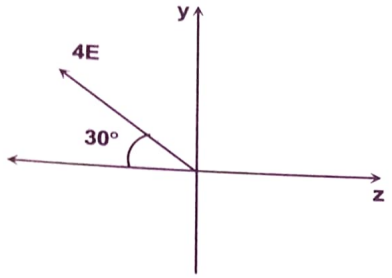
from energy conservation $-\frac{QE_0}{\ell}\frac{\ell^2}{2} + k + \frac{1}{2}Mv^2 = -\frac{QE_0}{\ell}\cdot\frac{\ell^2}{8}\times 2 + K$

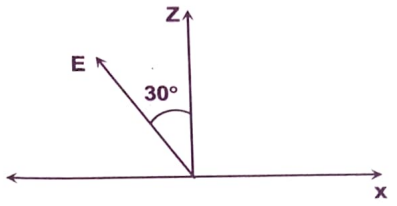
$v = 2$ m/s

42. At equilibrium energy radiated by point source = heat conducted through the thickness of the shell

43. $\vec{E} = -\frac{E}{2}\hat{i} + \frac{E\sqrt{3}}{2}\hat{k} - 2E\sqrt{3}\hat{k} + 2E\hat{j}$

$\therefore \ \hat{E} = \frac{-1\hat{i} + 4\hat{j} - 3\sqrt{3}\hat{k}}{\sqrt{44}}$



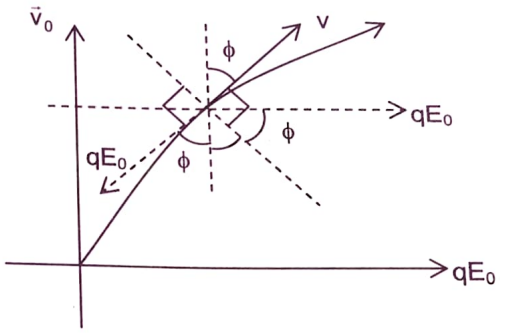


44. $E = 2 \times \frac{60}{75} \times \left(\frac{10000}{100}\right) = 160$ volt; $E = E_0 \frac{\ell_1}{\ell_2}\left(\frac{R_1 + R_2}{R_1}\right)$

$$\frac{\Delta E}{E} = \left|\frac{\Delta \ell_1}{\ell_1}\right| + \left|\frac{\Delta \ell_2}{\ell_2}\right| + \left|\frac{\Delta (R_1 + R_2)}{R_1 + R_2}\right| + \left|\frac{\Delta R_1}{R_1}\right|$$

$\Rightarrow \Delta E = 0.64$ volts

45. Consider the situation when particle has turned through an angle of ϕ



$$\frac{dv}{dt} = -\frac{qE_0}{m}[1 - \sin\phi]$$

$$\frac{dv_x}{dt} = \frac{qE_0}{m}[1 - \sin\phi]$$

$$\Rightarrow \frac{dv}{dt} = -\frac{dv_x}{dt}$$

$$\Rightarrow v = -v_X + C$$

$$\Rightarrow v = v_0 - v_X = v_0 - v\sin\phi$$

$$\Rightarrow v = \frac{v_0}{1 + \sin\phi} = \frac{6}{1+1} = 3\,\text{m/s}$$

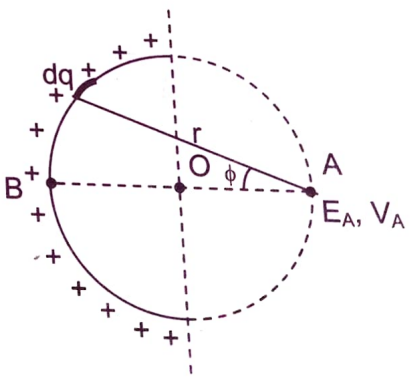
46. Angular momentum, $\left(\frac{3}{2}mr^2\right)\omega = \int \tau dt = \int \frac{dq}{dt}\pi r^2 B\,dt = q\pi r^2 B$

For the ring to become vertical,

$$\frac{1}{2}\left(\frac{3}{2}mr^2\right)\omega^2 = mgr$$

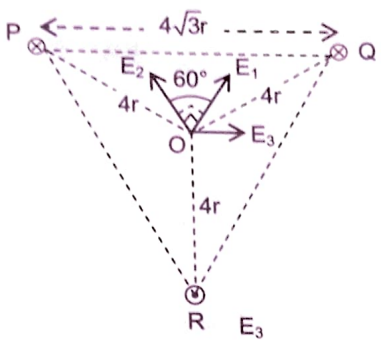
i.e. $g = \frac{\pi^2}{3}\left(\frac{qB}{m}\right)^2 r$

47. $E = \int \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2}\cos\phi$, $V = \int \frac{1}{4\pi\varepsilon_0}\frac{dq}{r}$, where $r = 2R\cos\phi$



48. $a_1 = 2a \quad b_1 = a$
$a_2 = 3a \quad b_2 = 2a$
Area of wire frame
$\frac{\pi[a_2b_2 - a_1b_1]}{2} = 2\pi a^2$
$\phi = 2\pi B_0 a^2 \cos(\omega t)$
$\varepsilon = -\frac{d\phi}{dt} = 2\pi\omega B_0 a^2 \sin(\omega t)$
peak emf $= 2\pi\omega B_0 a^2$

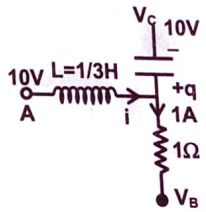
49. The contribution due to any one the cylinder
$|E| = \frac{r^2}{2x}\left|\frac{dB}{dt}\right| = \frac{r^2}{8r}\left|\frac{dB}{dt}\right| = \frac{1}{8}\left|\frac{dB}{dt}\right|$
$|E_1| = \frac{1}{4}$ V/m
E_1 is anticlockwise
$|E_2| = 1/4$ V/m
$|E_3| = 1/8$ V/m
(directions as shown in the figure)
$\left|\vec{E}_1 + \vec{E}_2 + \vec{E}_3\right| = \frac{\sqrt{13}}{8}$ V/m



50. from COM
$mv_A = 2mv_B$
$\Rightarrow \quad v_B = \frac{v_A}{2}$
from COE
$\frac{1}{2}kx_0^2 = \frac{1}{2}mv_A^2 + \frac{1}{2}\times 2m\frac{v_A^2}{4} = \frac{1}{2}mv_A^2\left(1+\frac{1}{2}\right)$
$\Rightarrow \quad \frac{1}{2}kx_0^2 = \frac{1}{2}mv_A^2 \times \frac{3}{2} \Rightarrow \frac{1}{2}mv_A^2 = \frac{kx_0^2}{3}$

51. The resistance shorting the capacitor plates may be considered in parallel with the capacitor, and at t = 0, the capacitor behaves as a short circuit.

52. $i = i_0 + \frac{dq}{dt}$
$= 1 + 3e^{-t}$



53. For a charge to be in equilibrium,
$qE = qvB \Rightarrow E = \omega LB/2 = BL\omega/2 = 2$

54. Initial charge on capacitor = CE
Initial potential energy of capacitor = $CE^2/2$
Now $C = \varepsilon_0 A/d$

And new capacitance: $C' = \frac{\varepsilon_0 A}{d/2}$, in series with $\frac{\varepsilon_0 kA}{d/2}$

$\Rightarrow C' = \frac{2\varepsilon_0 A}{d}$ series $\frac{4\varepsilon_0 A}{d}$

$= \frac{\varepsilon_0 A}{d}\left(\frac{2\times 4}{2+4}\right) = \frac{4\varepsilon_0 A}{3d} = \frac{4}{3}C$

$\Rightarrow$ New charge $= \frac{4}{3}CE$

New energy $= \frac{1}{2}\times\frac{4}{3}CE^2 = \frac{2}{3}CE^2$

Now $W_{but} = \Delta H + \Delta U$

$\Rightarrow E\left(\frac{4}{3}CE - CE\right) = \Delta H + \frac{2}{3}CE^2 - \frac{1}{2}CE^2$

$\Rightarrow \Delta H = \frac{1}{6}CE^2$

58. $mgR\phi - mgR\sin\phi = \frac{1}{2}(2m)v^2$ .... (i)

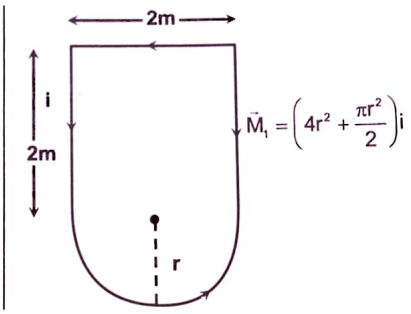
$mv^2/R = mg\sin\phi$ ... (ii)

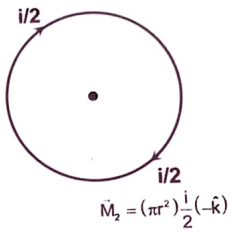
59. Velocity of $v_{IG} = 12 - 6 = 6$ m/s

Velocity of v_{IW} (velocity of image with respect to observer who is in the water) $= 6 \times 4/3 = 8$ m/s

So $v_{IO} = 8 - 8 = 0$

60. $\vec{M}_1 + \vec{M}_2 = M = 4r^2\hat{i}$





61. $h\nu_1 = W + 13.6$ eV

$h\nu_2 = W + \frac{12400}{1215}$ eV

62. $\Delta Q = 0$

$C_0M_1(T - T_1) + 2C_0M_2(T - T_2) = 0$

$2(T_1 - T) = 2(T_1 - T_2)$

$\frac{M_1}{M_2} = \frac{2C_0}{C_0} = 2$

63. Minimum radius of curvature will be at highest point.

Therefore

$$\frac{u^2}{g\cos\theta} = 8\frac{u^2\cos^2\theta}{g}$$

$$\Rightarrow \quad \cos\theta = \frac{1}{2}$$

$$\Rightarrow \quad \theta = 60^\circ$$

64. $Q_1 = \frac{\lambda_1}{d}(T_1 - T_0)A$ and $Q_2 = \frac{\lambda_2}{d}(T_1 - T_0)A$

$Q = Q_1 + Q_2$

$\lambda = \frac{(\lambda_1 + \lambda_2)}{2}$

65. Energy flow rate reaching surface of the ball $= \pi R^2 \frac{(\Delta p_m)^2}{2\rho v\lambda}$

66. $m_1 g - T - F = 0$

$m_2 g - T - F = 0$

$\Rightarrow \quad T = \frac{(m_1 - m_2)}{2} g = 5$

67. $\omega = 2\pi/T = 2\pi/0.033 = 190$ rad/sec

$v_t = \omega r = 190 \times 1/38 = 5$

68. $V = 6 \times 4 \times 3 = 72\ m^3 = 72000\ L$

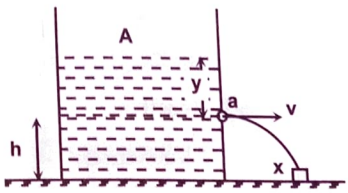
$n = \frac{72000}{22.4} = 3.2 \times 10^3$ mol

$Q = nC_v\Delta T = 1.4 \times 10^6$ J

$t = Q/10 = 700$ sec $= 7 \times 100$ sec

69. Velocity of efflux $v = \sqrt{2gy}$

Range $x = \sqrt{2gy} \times \sqrt{\frac{2h}{g}}$



The velocity of the block must be $\left(\frac{dx}{dt}\right)$.

$$\therefore \quad V_b = \frac{dx}{dt} = \sqrt{\frac{2h}{g}} \times \sqrt{2g} \times \frac{1}{2\sqrt{y}} \frac{dy}{dt}$$

$$V_b = \frac{\sqrt{h}}{\sqrt{y}} \cdot \frac{dy}{dt} \qquad \ldots(i)$$

Using equation of continuity

$$\frac{Ady}{dt} = a\sqrt{2gy} \qquad \ldots(ii)$$

equation (i) and (ii)

$$V_b = \sqrt{\frac{h}{y}} \times \frac{a}{A}\sqrt{2gy}$$

$$V_b = \sqrt{2gh} \times \frac{a}{A} = 20 \times \frac{1}{20} = 1\ ms^{-1}.$$

70. $\Delta U = 600 - 150 = 450\ J$

$\Delta Q = nC\Delta T$

$\Delta U = nC_V\Delta T$

$\frac{\Delta Q}{\Delta U} = \frac{C}{C_V}$

$C = \frac{4}{3}C_v = 2R$

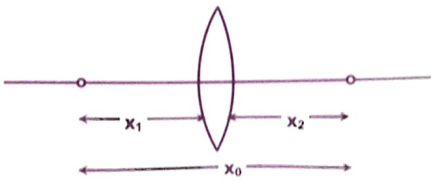
71 Charge on the outside surface of conducting sphere is zero so field and thus potential at any point outside the sphere is zero.

72. Since object and image is real

then object distance $= -x_1$

Image distance $= x_2$

$x_0 = x_1 + x_2;\ \frac{1}{x_2} - \frac{1}{-x_1} = \frac{1}{f}$



$\Rightarrow x_1 + x_2 = \frac{x_1x_2}{f}$

$\Rightarrow d = \frac{x_1(x_1 - d)}{f}$

$\Rightarrow x_0 = \frac{x_1^2}{x_1 - f}\frac{dx_0}{dx_1} = 0$

$\Rightarrow x_1 = 2f$

which means $x_2 = 2f$

distance between object and image is 4f

73. According to question

$3\left(\lambda_{K_\alpha} - \frac{12.4}{12.4}\right) = \lambda_{K_\alpha} - \frac{12.4}{24.8}$

$\Rightarrow \lambda_{K_\alpha} = 1.25A^\circ$

74. $-\frac{dr}{dt} = 5 - 3\cos\theta$

$\int_{16}^{0} -dr = \int_0^t (5 - 3\cos\theta)dt$

$16 = 5t - 3\int_0^t \cos\theta dt$

- (i)

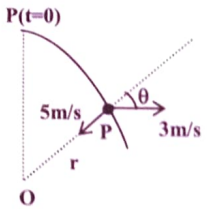
$\int v_x dt = 0$

$\int_0^t (3 - 5\cos\theta)dt = 0 \Rightarrow \int_0^t \cos\theta dt = \frac{3t}{5}$

- (ii)From (i) & (2)

$16 = 5t - \frac{9t}{5}$

$t = 5$ sec.



75. For entering without jerk $v_2 = \ell\omega_0 = 2$rad/s

Using work energy theorem on sleeve after entering in the frame of rod.

$W_{spring} + W_{centrifugal} = \Delta K$

$$-\frac{1}{2}k\ell^2 - \frac{1}{2}m\omega^2\ell^2 = 0 - \frac{1}{2}mv_1^2 \Rightarrow v_1^2 = 8$$

Now $K = \frac{1}{2}m\left(v_1^2 + v_2^2\right) = 6$

76. For minimum time person should use maximum friction for speeding up half the time and slowing down with maximum friction in next half time because he has finally to be stopped as stated in the question.

77. Let us work in the ground frame for A and for B work in the frame of A.

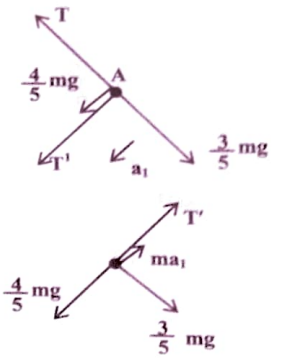
$$\frac{4}{5}mg + T' = ma_1 \quad \ldots(i)$$

$$T' + ma_1 = \frac{4}{5}mg \quad \ldots(ii)$$

From (i) & (ii)

$$T' = 0 \Rightarrow a_1 = \frac{4}{5}g$$

$\Rightarrow$ In ground frame motion of B will be free fall.



78. $F = \rho Av^2$

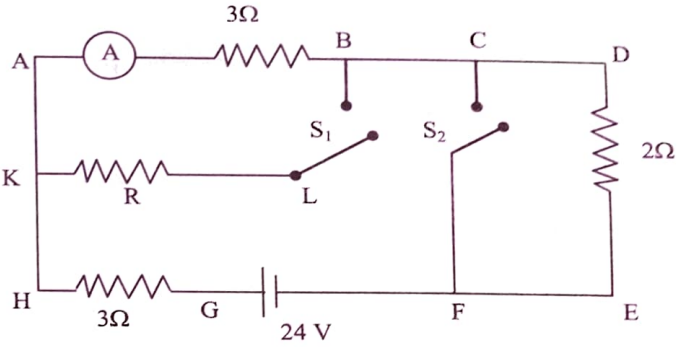
OBJECTIVE (SINGLE CORRECT ANSWERS)

1. When both switches are open, then reading of ammeter is 24/8 = 3A

When both switches are closed.

$V_{AB} = 9$ V $\Rightarrow V_{HG} = 15 \Rightarrow$ So current through R is 2A.

Hence R = 9/2 = 4.5 Ω



2 $\cos 37^0 = x/a \Rightarrow x = 4a/5 = 16$ m. $\Rightarrow \omega = 1$ rad/s.

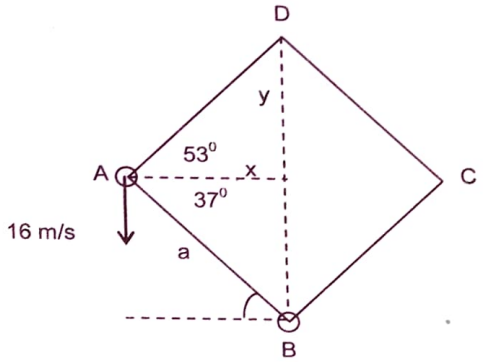
$$\text{os } 53^0 = \frac{a^2 + x^2 - y^2}{2ax}$$

$\Rightarrow a^2 + x^2 - y^2 = 384$

$\Rightarrow y^2 = 400 + 256 - 384 = 272$

$\Rightarrow y = 16.5$ m(approximately

$v_D = y\omega = 16.5$ m/s



3. The ball will achieve its velocity when its acceleration is zero. So,

$m_1 g = b v_{1max}^2$

$m_2 g = b v_{2max}^2$

$$\left(\frac{v_{1max}}{v_{2max}}\right)^2 = \frac{m_1}{m_2} = \frac{\rho_1}{\rho_2} = 4 \Rightarrow \frac{v_{1max}}{v_{2max}} = 2$$

4. $T = m_2 g \sin 30°$ ...(i)

$T = m_1 g \sin 30° + \frac{\eta A v}{\ell}$ ...(ii)

From equation (i) and (ii) we can write

$$\frac{\eta A v}{\ell} = (m_2 - m_1) g \sin 30° = mg \sin 30°$$

$$\eta = \frac{mg\ell \sin 30°}{vA} = \frac{0.02 \times 10 \times 0.3 \times 10^{-3}}{2 \times 0.085 \times 0.1} 3.46 \times 10^{-3} \text{PaS}$$

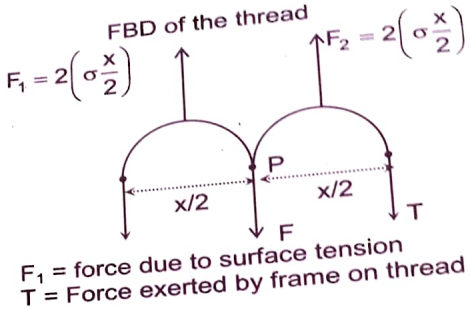
5. Use the concept of total internal reflection.

7. From FBD

$F + 2T = 2\sigma x$, where $T = 2\left(\sigma \frac{x}{4}\right)$

$\Rightarrow F = \sigma x$ and $2\left(\frac{\pi x}{4}\right) = L$

$\Rightarrow \sigma = \frac{\pi F}{2L}$



F_1 = force due to surface tension
T = Force exerted by frame on thread

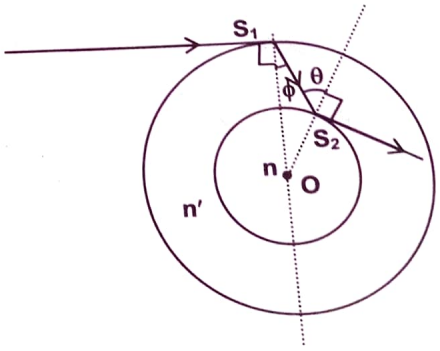
8. Prism has constant area of cross-section and its volume is area of base times height.

$\left(\frac{5}{100}\right) \times Ag = 200g$

$(0.5)Ag = n(200)g$

(n = number of seals to completely submerge the prism)

9. The diagram shows limiting case for penetration into the liquid.



Snell's law at S_1

$\Rightarrow \sin\phi = \frac{1}{n'}$ ...(i)

Snell's law for S_2

$\Rightarrow \sin\theta = \frac{n}{n'}$ ...(ii)

sin rule in triangle S_1S_2O

$\Rightarrow \frac{\sin\phi}{r} = \frac{\sin(\pi - \theta)}{R}$ ...(i)

Solving the above equations we get

$r = \frac{R}{n}$

13. All light rays hitting the screen will be appearing to be coming from S_1' and S_2'

$\Rightarrow$ $D = b + 2r\sin^2\phi$

$d = 4r\sin\phi\cos\phi$

fringe width $= \dfrac{\lambda D}{d}$

16. Work done by magnetic force is zero

so from work energy theorem

$$\frac{1}{2}m_p v_B^2 = \frac{1}{2}m_p v_A^2 + q\Delta V$$

and simultaneously there is no change of velocity component along the direction of perpendicular to electric and magnetic field.

$v_A \sin\alpha = v_B \sin\beta$

After solving $\Delta v = 16$ Volt

17. At terminal stage, torque applied on the smaller disc by the rope = mga

current to the disc $= \dfrac{B\omega r^2}{2R}$ (where ω is terminal angular velocity)

torque applied by magnetic field $= \dfrac{B^2\omega r^4}{4R}$

So, $\dfrac{B^2\omega r^4}{4R} = mga$

$\omega = 100$ rad/sec

18. Angular frequency ω of the LC oscillation $= \sqrt{2\pi\varepsilon_0 RL}$

required time is one fourth of the time period

19 The force on the rod due to the annular disc is equal and opposite to that on the disc due to the rod. We take an annular strip of radius r and width dr and find the force acting on it

$$dF = (\sigma_0\, 2\pi r\, dr)\,\frac{\lambda}{2\pi\varepsilon_0 r} = \frac{\lambda\sigma_0}{\varepsilon_0}dr$$

$\therefore$ The force

$$F = \frac{\sigma_0\lambda}{\varepsilon_0}\int_R^{2R} dr = \frac{\sigma_0\lambda R}{\varepsilon_0}.$$

20. $$\frac{2\lambda}{2\pi\varepsilon_0 d} = E_0$$

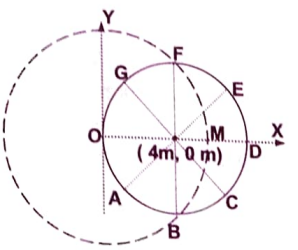
$\lambda = \pi\varepsilon_0 dE_0$ So, $E_P = \dfrac{E_0 d\ell}{4r^2}$

23. Draw a circle assuming OF as radius and O as centre

$$\oint_{MDEFM}\vec{B}\cdot d\vec{\ell} = \int_M^D \vec{B}\cdot d\vec{\ell} + \int_{DEF}\vec{B}\cdot d\vec{\ell} + \int_F^M \vec{B}\cdot d\vec{\ell} = 0$$

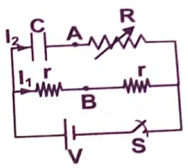
$$\Rightarrow 0 + x - \mu_0 I_0\frac{45^\circ}{360^\circ} = 0$$

$$\Rightarrow x = \frac{\mu_0 I_0}{8}$$



24. Consider circuit at any time t after closing the switch:

$I_1 = \frac{V}{2r}$, and $I_2 = \frac{V}{R}e^{-\frac{t}{RC}}$



$\Rightarrow V_A - V_B = RI_2 - rI_1 = R \times \frac{V}{R}e^{-\frac{t}{RC}} - r \times \frac{V}{2r}$

$\Rightarrow V_A - V_B = \frac{-V}{2}\left(1 - 2e^{-\frac{t}{RC}}\right)$

$\Rightarrow a = \frac{-V}{2}$, and $b = \frac{+V}{2}$

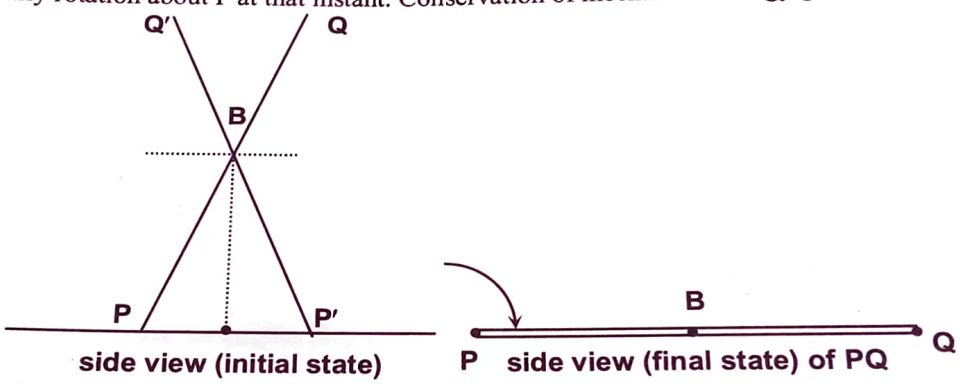
When $V_A - V_B = 0 \Rightarrow t_0 = RC\ell n(2)$, Hence greater the R ,greater the t_0

26. Maximum path difference $= \frac{49\lambda}{\lambda} + \frac{49\lambda}{\lambda/1.5} = 122.5$ wavelengths

Number of maxima = 245

29 Let the mass per unit length of the bar be λ (kg/m)

As the frames come down, the point B moves vertically downwards and PP′ remain in contact with the ground. In the final state(when PQ, P′Q′ just become horizontal), the velocities of P(and P′) is zero. The vertical component was already zero, the horizontal component also becomes zero. The motion of PQ is essentially rotation about P at that instant. Conservation of mechanical energy gives(for PQ):



$$(2\lambda + 6\lambda)g\frac{3}{2}\cos 37° = \frac{1}{2}\left(\frac{1}{3}\cdot 6\lambda \cdot 3^2 + \lambda \cdot 3^2\right)\omega^2$$

and $v_Q = \omega.PQ = \omega \times 3 = 8$ m/s

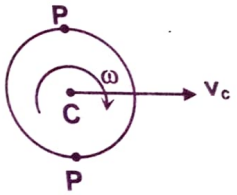
30. ϕ_{cube} = flux due to single wire from whole cube

$\phi_{cube} = \frac{\lambda\ell}{8\varepsilon_0}$ similarly four wires out of twelve will have same contribution and eight will have zero.

$\phi_{face_1} = \frac{\lambda\ell}{8\varepsilon_0} \times 4 = \frac{\lambda\ell}{2\varepsilon_0}$

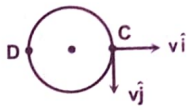
31. The instantaneous axis of rotation will fall on the outside of the sphere. Since in case of pure rotation, instantaneous axis of rotation will be on the centre. In case of pure rolling, instantaneous axis of rotation will be on P.

But if $v_c > \omega R$, so instantaneous axis of rotation will fall out of the sphere.



32. Apply conservation of angular momentum.

33. $\omega = \dfrac{v}{R}$



$v_{c,g} = v\hat{i} + R\omega\hat{j} = v\hat{i} + v\hat{j}$

$v_{D,g} = v\hat{i} + R\omega(-\hat{j}) = v\hat{i} - v\hat{j}$

$v_{C,D} = v_{c,g} - v_{d,g} = v\hat{i} + v\hat{j} - v\hat{i} + v\hat{j} = 2v\hat{j}$

35. $I = \dfrac{mr^2}{2}$

$d = \dfrac{4r}{3\pi}$

$T = 2\pi\sqrt{\dfrac{I}{mgd}} = 2\pi\sqrt{\dfrac{mr^2.3\pi}{2\times mg\times 4r}} = 2\pi\sqrt{\dfrac{3r\pi}{8g}}$

37. Conservation of energy gives

$\dfrac{q^2}{4\pi\varepsilon_0 r} = \dfrac{1}{2}kr^2 + \dfrac{q^2}{4\pi\varepsilon_0.2r} \Rightarrow k = \dfrac{q^2}{4\pi\varepsilon_0 r^3}$

39. As the force acting between the body and the earth passes through the centre of the earth. Angular momentum of the system is conserved about the centre of earth. And the velocity at the highest point of the trajectory is perpendicular to the radius vector.
So, by conservation of momentum about the centre of earth.

$mv\sin 30°.R = \dfrac{mv}{4}(R+h)$

$\dfrac{R}{2} = \dfrac{R+h}{4} \Rightarrow h = R$

41. $u_i = -\dfrac{3GM^2}{2\ell}$

$U_f = -\dfrac{3GM^2}{\ell}$

$\Delta W_{conservative} = \Delta w_{system} = -(U_f - U_i) = -\left[-\dfrac{3GM^2}{\ell} + \dfrac{3GM^2}{2\ell}\right] = \dfrac{3GM^2}{2\ell}$

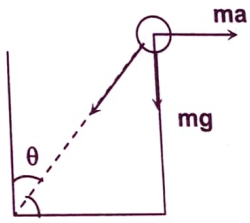
42. Apply integration method

43. $F_{ext} = Ma_{cm}$

44. $mgL\sin\theta = ma\,L\cos\theta$

$\Rightarrow \tan\theta = \dfrac{a}{g}$

$\theta = \tan^{-1}\left(\dfrac{a}{g}\right)$



45. $P_A + \dfrac{1}{2}\rho v_A^2 = P_B + \dfrac{1}{2}\rho v_B^2$

$$P_A - P_B = \frac{1}{2}\rho\left(v_B^2 - v_A^2\right)$$

$$0.02 \times 12000 \times 10 = \frac{1}{2} \times 1000\ \left(v_B^2 - 20.2\right)$$

$$4.8 = v_B^2 - 20.2 \Rightarrow v_B = 5\,\text{m/s}$$

$$\Rightarrow A_3.v_B = A_1v_1 + A_2v_2$$

$$\Rightarrow 30 = 4v_1 + 8$$

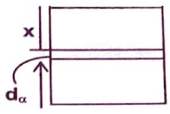
$$\Rightarrow 4v_1 = 22 \Rightarrow v_1 = \frac{22}{4} = 5.5\ \text{m/s}$$

46. $H\rho g\pi R^2 - F_s = F_B$

$$F_B = V\rho g = \frac{1}{3}\pi R^2 H\rho g$$

$$\therefore F_3 = H\rho g\pi R^2 - \frac{H\rho g\pi R^2}{3} = \frac{2}{3}\pi R^2\rho Gh$$

47. $$\frac{\rho L}{KT}\int_0^4 x\,dx = \int_0^T dt \Rightarrow T = \frac{\rho LH^2}{2KT}$$



48. $$\lambda_m = \frac{b}{T} = \frac{2.88\times10^6}{2880} = 1000\text{nm} \Rightarrow u_2 \text{ is maximum}$$

49. For the process A → B

$$p = 2p_0 - \frac{p_0}{V_0}(V - V_0)$$

$$p = 3p_0 - \frac{p_0 V}{V_0}$$

$$T = \frac{pV}{nR} = \frac{1}{nR}\left(3p_0V - \frac{p_0}{V_0}V^2\right)$$

$$\frac{dT}{dV} = 0 \Rightarrow 3p_0 = \frac{p_0V}{V_0} \times 2V \Rightarrow V = \frac{3}{2}V_0$$

50. For series connection

$$R_QQ_0 = \Delta T$$

$$Q_0\left(\frac{L}{AK} + \frac{L}{K_2A}\right) = \Delta T$$

$$\frac{2Q_0L}{kA} = \Delta T \Rightarrow 2RQ_0 = \Delta T$$

For parallel connection

$$\frac{AK}{L} + \frac{AK}{L} = \frac{1}{R'} \Rightarrow R' = R/2$$

$$\Delta T = Q\left(\frac{R}{L}\right) \Rightarrow Q = \frac{2\Delta T}{R} = 4\left(\frac{\Delta T}{2R}\right) = 4Q_0$$

51. Apply gas equation and analyse.

52. $\Delta W = P_0V_0 + \frac{1}{2}\pi 2P_0V_0 \times \frac{1}{2} = P_0V_0\left(1+\frac{\pi}{2}\right)$

$\Delta U = \frac{3}{2}(P_0 2V_0 - P_0V_0) = \frac{3}{2}P_0V_0$

$\Delta Q = P_0V_0\left(1+\frac{\pi}{2}+\frac{3}{2}\right) = P_0V_0\left(\frac{5+\pi}{2}\right)$

53. $P = \frac{P_0}{1+\left(\frac{V}{V_0}\right)^3} = \frac{P_0}{2}$

$T = \frac{P_0V_0}{2R}$

$\therefore$ Translational kinetic energy is equal to $\frac{3}{2}RT = \frac{3R}{2}\frac{P_0V_0}{2R} = \frac{3P_0V_0}{4}$

54. For a black body,

Heat absorbed = heat radiated = σAT^4

$\therefore$ $5.67 \times 10^{-8} \times 4 \times 3.14 \times 8 \times 8 \times 10^{-4} \times T^4 = 10$

$\therefore$ $T = 0.2164 \times 10^3 = 216.4\ K \approx 217\ K$

55. For process AB

$P = \frac{VP_0}{V_0} \Rightarrow T = \frac{PV}{nR} = \frac{P}{nR}\frac{2V_0}{P_0}$

$\frac{dT}{dP} = \frac{2P}{nR}\frac{V_0}{P_0}$

$\frac{dT}{dP} = \frac{4V_0}{nR}$ at B.

Since T goes on increasing, ΔW is positive,

$\Delta Q_{AB} = +ve$

For BC

$P(2V_0) = nRT \Rightarrow T = \frac{2PV_0}{nR}$

$\frac{dT}{dP} = \frac{2V_0}{nR}$ at B

$\therefore \frac{dT}{dP}$ is not defined at B.

In BC temperature goes on decreasing,

$\therefore \Delta Q_{BC}$ is $-ve$.

For CA

$\Delta W = -ve$

$\therefore$ $Q_{CA} = -ve$

56. $PT = constant$

$P_0T_i = \frac{P_0}{2}T_f \Rightarrow T_f = 2T_i$

Also, $P_0V_0 = 2R.T_i$

$$T_i = \frac{P_0 V_0}{2R} \quad \ldots (1)$$

$$P_f V_f = nR.T_f$$

$$\Rightarrow \frac{P_0}{2} V_f = [2][R][2T_f] \quad \ldots (2)$$

$$\Rightarrow T_f = \frac{P_0 V_0}{R}$$

$$\therefore \Delta U = \frac{3}{2} nR\Delta T = \frac{3}{2} P_0 V_0$$

57. $C = R$

$$\frac{dQ}{ndT} = R$$

$$\Rightarrow \frac{nC_v dT + dW}{ndT} = R \Rightarrow Cv + \frac{pdV}{ndT} = R \Rightarrow \frac{pdV}{ndT} = -\frac{R}{2} \quad [\because C_v = 3R/2]$$

$$\Rightarrow \frac{RT}{V}\frac{dV}{dT} = -\frac{R}{2} \qquad \left[\frac{P}{n} = \frac{RT}{v}\right]$$

$$\Rightarrow \frac{dV}{V} = -\frac{dT}{2T} \Rightarrow \ln(VT^{1/2}) = \text{constant}$$

$$\Rightarrow VT^{1/2} = \text{constant}$$

58. In equilibrium, piston will be having kinetic energy, too.

59. Applying COE

$$i^2Rt = mgx$$

$$\Rightarrow \frac{dx}{dt} = \frac{i^2R}{mg}$$

61. $$\sqrt{\frac{507 - V\rho g}{507}} = \frac{240}{260} \Rightarrow V = 0.0075 \text{ m}^3$$

62. when stone is dipped in water, the tension in the string changes.

63. $$\gamma_{mix} = \frac{n_1 C_{P_1} + n_2 C_{P_2}}{n_1 C_{v_1} + n_2 C_{v_2}} = 1.357, \; M_{mix} = \frac{2\times 32 + 3\times 48}{5} = 41.6$$

$$\frac{f_2}{200} = \sqrt{\frac{\gamma_{mix}}{M_{ix}} \times \frac{M}{r}} = \sqrt{\frac{1.357}{41.6} \times \frac{32}{1.4}}$$

$$\therefore \quad f_2 = 172.7$$

64. $$5\frac{v}{2L_1} = 3\frac{v}{4L_2}$$

65. Intensity is inversely proportional to the square of the distance.

66. $$\frac{340}{340 - v} - \frac{340}{340 + v} = \frac{2}{680} = \frac{1}{340}$$

$$\frac{2v}{(340)^2 - v^2} = \frac{1}{340}$$

$$v^2 + 340v \times 680 - 340^2 = 0$$

$$v = 0.5 \text{ m/s}$$

67. $(\vec{r}_2 - \vec{r}_1) = \sqrt{(4-1)^2 + (6-2)^2 + (3+5\sqrt{3}-3)^2} = \sqrt{9+16+75} = 10\text{m}.$

$I = \frac{P}{4\pi} \cdot \frac{1}{r^2} = \frac{314}{3\cdot14\times4} \times \frac{1}{100} = 0.25 \text{ w/m}^2$

68. Use Doppler's effect.

69. Based on Doppler's effect.

70. Use interference concepts.

71. As the source of sound and the observer both are at rest,
apparent frequency = actual frequency.

72. $Q_{outside} = \frac{Q_1 + Q_2}{2}$

73. Find the charge distribution on the inner and outer surface of the sphere applying gauss's law.

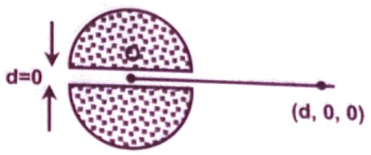
74. If V_1 and V_2 be the speeds of electron while entering the capacitor and while leaving it, then

$v_1 \cos\theta_1 = v_2 \cos\theta_2 \Rightarrow \frac{v_1}{v_2} = \frac{\cos\theta_2}{\cos\theta_1}$

$\frac{(k.E.)_1}{(k.E.)_2} = \left(\frac{\cos\theta_2}{\cos\theta_1}\right)^2$

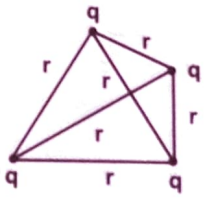
75. Field decreases k times in dielectric medium but field at the position of plates remains same, so force of interaction remains same.

76. Due to a solid hemispherical charge of 2Q the field will be $\frac{1}{4\pi\varepsilon_0} \times \frac{2Q}{d^2}$ along x-axis. So due to one hemisphere the component of field along x-direction will be $\frac{1}{4\pi\varepsilon_0} \times \frac{Q}{d^2}$. And hence the net field will be more than $\frac{1}{4\pi\varepsilon_0} \times \frac{Q}{d^2}$



77. The arrangement possible is that of a pyramid

$E = 6 \times \frac{1}{4\pi\varepsilon_0} \times \frac{q^2}{r} = \frac{3q^2}{2\pi\varepsilon_0 r}$

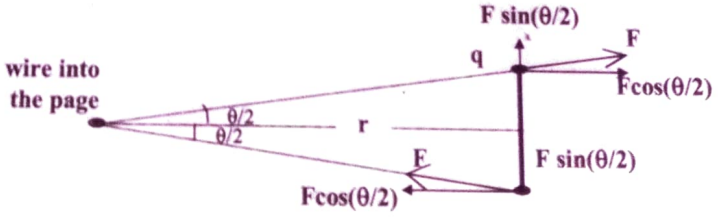


78. $\frac{\lambda p}{2\pi\varepsilon_0 r^2}$

$F_{net} = 2F \sin(\theta/2)$

$= F\theta = qE\,(l/r)$

$= \frac{\lambda p}{2\pi\varepsilon_0 r^2}$



79. $E = \dfrac{1}{4\pi\varepsilon_0}\dfrac{\frac{4}{3}\pi r^3\sigma}{(R-r)^2}$

$= \dfrac{1}{3\varepsilon_0}\dfrac{r^3\sigma}{(R-r)^2}$

80. $V = \dfrac{1}{4\pi\varepsilon_o}\sum\dfrac{Q_i}{r_i}$

81. Total charge $Q = \int_0^{2\pi}\lambda\cos^2\theta R d\theta = \pi\lambda R$

Hence $\pi\lambda_1 R_1 = \pi\lambda_2 R_2$

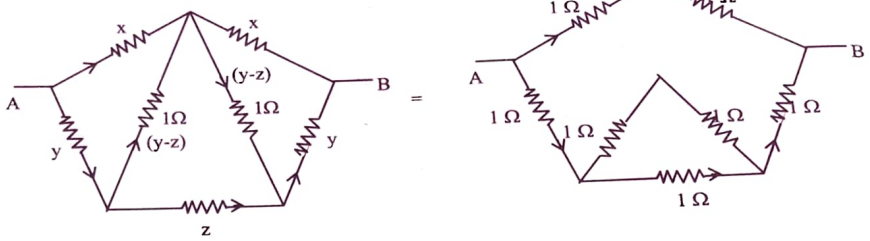
$\Rightarrow \dfrac{\lambda_1}{R_2} = \dfrac{\lambda_2}{R_1}$

82. Calculate the potential due to charge q and also the induced charges.

83. In the steady state, no current flows through the capacitor.

84. $\therefore R_{equl} = \dfrac{\frac{8}{3}\times 2}{\frac{8}{3}+2}$

$= \dfrac{16}{14} = \dfrac{8}{7}\Omega.$



85. $(0.1 + 0.3 + 0.6)\, i_1 = (9 + 0.9) \times 10$

$i = 99\text{mA}$

$I = (99 + 10)\text{mA} = 0.109\text{ A}$

86. The potential of point G should be kept zero.

87. Analyse the situation gives qualitatively the charge on the plates can not change.

88. When each of the corners has a charge Q, the magnetic field at the centre of the hexagon is zero.

89. Charge flow through the battery will be 2CV, while the energy stored in capacitor will be the same.

90. When switch S_1 is open

$\dfrac{6}{E} = \dfrac{L}{2}\Big/L = \dfrac{1}{2}$

$\therefore\ E = 12\text{ V}$

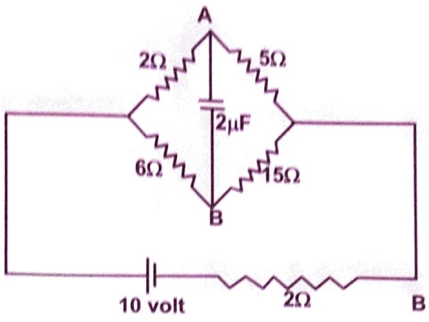
When switch S_2 is closed

$\dfrac{6\times 10}{10+r} = \dfrac{5}{12}\dfrac{L}{L}\times E = \dfrac{5}{12}\times 12 = 5$

$\therefore\ 10 + r = 12$

$\therefore\ r = 2\,\Omega$

91.



→ It's a balanced wheat (+) stone Bridge

92. Points A , B , C and D are at same potential therefore no current will be flowing through AB, BC, CD and DA

$\therefore \quad R_{eq} = (R_{OA} + R_{AP})/4 = (r/4)\left(\frac{\pi a}{2} + a\right)$

93. $Q_i = \frac{3\times 2}{3+2}\mu F \times 5V = 6\mu C$

$Q_f = 3\mu F \times 5V = 15\mu C$

$\Delta Q = 9\mu C$

94. At t = 0, the branch containing L will offer infinite resistance while the branch containing the capacitor will be effectively a short circuit. Hence, $(R)_{t=0} = R$ and $(i)_{t=0} = \frac{\varepsilon}{R}$. Similarly at $t = \infty$, L will offer zero resistance whereas c will be an open circuit. Hence effective resistance = $R + \frac{6x3R}{6+3} = 3R$ and $(i)_{t=\infty} = \frac{\varepsilon}{3R}$.

$\therefore$ The required ratio = $\frac{\varepsilon}{R} x \frac{3R}{\varepsilon} = 3:1$

95. By symmetry, the net field is Zero

The net field at P (0, 0, a)

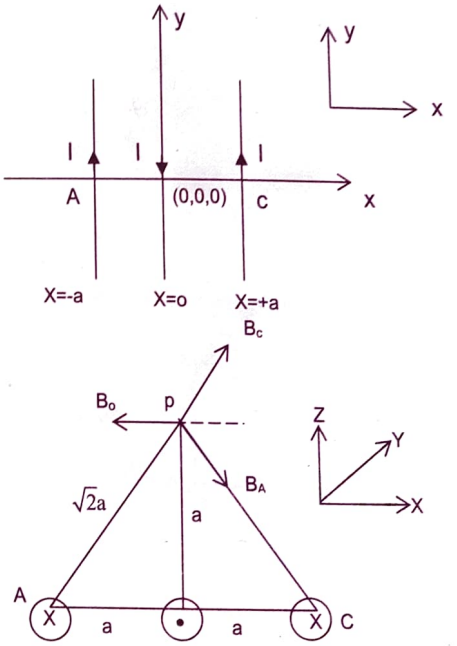
$\vec{B} = \vec{B_A} + \vec{B_o} + \vec{B_c}$

$= \frac{\mu_0 I}{2\pi a\sqrt{2}}\left\{\cos 45^0 \hat{i} + \sin 45^0\left(-\hat{k}\right)\right\}$

$+ \frac{\mu_0 I}{2\pi a}\left(-\hat{i}\right) + \frac{\mu_0 I}{2\pi a\sqrt{2}}\left\{\cos 45^0 \hat{i} + \sin 45^0\left(\hat{k}\right)\right\}\Bigg]$

$= \frac{\mu_0 I}{2\pi a\sqrt{2}}\sqrt{2}\hat{i} + \frac{\mu_0 I}{2\pi a}\left(-\hat{i}\right)$

$= \frac{\mu_0 I}{2\pi a}\left[\hat{i} - \hat{i}\right] = 0$



96. $\vec{B}_0 = \frac{\mu_0 ni}{2R}$ (for circular ring with n loops)

$\therefore$ Length remains constant.

$\therefore r_2 = R/n$

97. Magnetic torque is balanced by torque due to gravity.

98. $B_{AB} = \frac{\mu_0 I}{4\pi R}\left[\sin 90^\circ + \sin(-45^\circ)\right] \otimes$

$B_{BC} = \frac{\mu_0 I}{4\pi R}\left[\sin 90^\circ + \sin 45^\circ\right] \odot$

$B_0 = B_{AB} + B_{BC} + B_{circular}$

99. $i_R = \frac{V}{\alpha R\sigma}(ccw)\ i_r = \frac{V}{(2\pi-\alpha)r\sigma}(cw)$

100. $B_{net} = 2\frac{\mu_0 I}{4\pi r}\left[\sin 90^\circ + \sin 45^\circ\right] \odot$

101. As magnetic force is always perpendicular to velocity. No work is done by force.

102. emf. Induced in coil $\varepsilon = -\frac{\Delta\phi}{\Delta t} = -\frac{(\phi_2 - \phi_1)}{t_2 - t_1}$

$\phi_2 = B.A_2 - B.\ \frac{4a}{2\times 3}\times\frac{4a}{3}\times\sin 60^0 = \frac{4\sqrt{3}}{9}Ba^2$

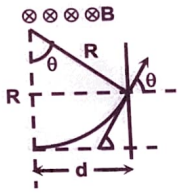
$\phi_1 = Ba^2$

Work done $W = \varepsilon\, I\, \Delta t = Ba^2\left(1 - \frac{4\sqrt{3}}{9}\right)i$

103. $\sin\theta = \frac{d}{R}$

$d = R\sin\theta = \frac{mv}{qB}\sin\theta$

$\Rightarrow \frac{q}{m} = \frac{v\sin\theta}{Bd}$



104. Magnetic field at p is into the plane (⊗). Initially $\vec{F} = q\vec{v}\times\vec{B}$ is towards the wire A.

105. Magnetic field may be zero in first and third quadrant.

$\frac{\mu_0 I_1}{2\pi y} = \frac{\mu_0 I_2}{2\pi x}$

represents a straight line passing through the origin.

106. The gravitational torque must be counter balanced by the magnetic torque about 0, for equilibrium of the sphere. The gravitational torque

$\tau_{gr} = mgr\sin\theta$

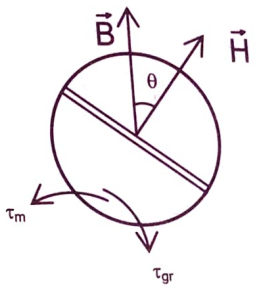
The magnetic torque $\tau_m = \vec{\mu}\times\vec{B}$

Where the magnetic moment of the coil = $\mu = (i\,\pi r^2)$

$\Rightarrow \tau_m = \pi i r^2 B\sin\theta$

$\pi i r^2 B\sin\theta = mgr\sin\theta$

$\Rightarrow B = \frac{mg}{\pi i r}$



107. $T = 2\pi m/qB$

$\therefore t_{He} > t_p > t_e$

108. Applying Ampere's circuital law,

$$B.2\pi r = \frac{\mu_0 I \pi r^2}{\pi a^2} \text{ for } r < a$$

and $B.2\pi R = \mu_0 I$ for $R > a$

Taking the ratio

$$\frac{r}{R} = \frac{r^2}{a^2}$$

$\therefore a = \sqrt{Rr}$

109. The magnetic field produced by square loop is parallel to the plane of the circular loop. Hence the mutual Inductance is zero

110. $i = \dfrac{Bv(2r)}{R_{eq}}$, where $R_{eq} = \dfrac{(R/2)(R/2)}{R/2+R/2} = R/4$

111. $\dfrac{B^2V_0L^2}{R} = mg \Rightarrow V_0 = \dfrac{mgr}{B^2L^2}$

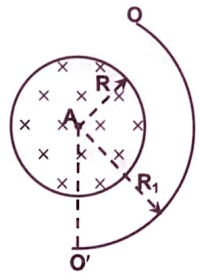
112. Use Lenz's law

Induced emf the current of opposes the change in flux through it.

113. Required emf $= \pi R^2 \dfrac{dB}{dt}.\dfrac{\theta}{2\pi}$

$$\therefore E = \frac{R^2}{2R_1}\left(\frac{dB}{dt}\right)$$

$$\text{emf} = \frac{\theta}{2\pi}\frac{R^2}{2R_1}\left(\frac{dB}{dt}\right)2\pi R_1 = \frac{\theta}{2}R^2\left(\frac{dB}{dt}\right)$$



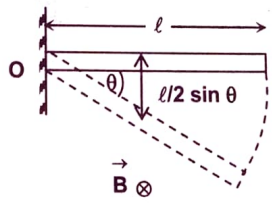
114. $\cos\phi = \dfrac{R}{\sqrt{R^2+\left(\dfrac{1}{\omega C}-\omega L\right)^2}}$

Putting the values, C = 500 μC

115. Magnetic field is out of the plane and as the electron moves, the field decreases. From Lenz' rule, the current in the coil is anticlockwise.

116. $mg(\ell/2)\sin\theta = \dfrac{1}{2}\times\dfrac{m\ell^2}{3}\omega^2$

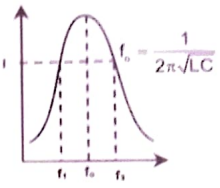
$$\omega = \sqrt{\frac{3g\sin\theta}{\ell}}.$$



Potential difference induced between two ends of the rod $= \varepsilon = \dfrac{1}{2}B\omega\ell^2$

$$\varepsilon = \frac{1}{2}B\ell^{3/2}\sqrt{3g\sin\theta}$$

117. I is same for two values of frequency namely f_1 and f_2



118. At resonance, $\omega L = \frac{1}{\omega C}$, in second case for resonance, $2\omega L = \frac{1}{2\omega c'}$

$\therefore C' = \frac{1}{4\omega^2 L} = \frac{LC}{4L} = \frac{C}{4}$

$\therefore \frac{\Delta C}{C} = \frac{3}{4}$

119. $\frac{1}{f_1} = \frac{(1.5-1)}{R} + \left(\frac{4}{3}-1\right)\left(-\frac{1}{R}-\frac{1}{R}\right) + \left(\frac{1.5-1}{R}\right) = \frac{1}{R} - \frac{2}{3R} = \frac{1}{3R}$

$\frac{1}{f_2} = \frac{(1.5-1)}{R} + \left(\frac{4}{3}-1\right)\left(-\frac{1}{R}\right) + \left(\frac{1.5-1}{R}\right) = \frac{1}{R} - \frac{1}{3R} = \frac{2}{3R}$

$\Rightarrow$ Convergent in both.

120. The optic axis of a lens does not change even on cutting it.

121. If after refraction from the lens, the image is formed at the centre of curvature of the mirror, the ray of light after reflection from mirror will retrace its path.

122. Here $f = R \Rightarrow$ radius of curvature = R

123. Use $m_1 = m_2$

124. Behaviour depends on refractive index of the lens.

125. $\frac{d}{2} = \frac{\lambda D}{2d}$

$\Rightarrow \lambda = 5000 A^\circ$

126. $I \propto \cos^2(\phi/2)$

$\Rightarrow \cos(\phi/2) = \pm \frac{1}{2}$

$\Rightarrow \phi = \frac{2\pi}{3}, \frac{\pi}{3}$

$\Rightarrow \Delta x = \frac{\lambda}{3}, \frac{\lambda}{6}$

A possible path difference = $2\Delta x$

127. Path difference = $(\mu_2 - \mu_1)t = 12480\ A^0$

For maxima

$n\lambda = 12480\ A^0$

$\lambda_1 = 12480\ A^0$,

$\lambda_2 = 6240\ A^0$,

$\lambda_3 = 4160\ A^0$,

$\lambda_4 = 3120\ A^0$

$\therefore$ Only $6240\ A^0$ & $4160\ A^0$ exist in the spectrum.

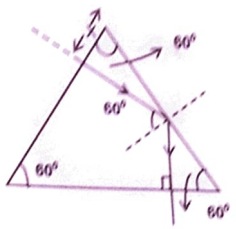
128. Numbers of fringes $= \frac{4000}{6000} \times 60 = 40$

129. Path difference $= d \sin\theta \approx \frac{d^2}{D}$

130. $4 \times 6300 = \frac{9}{2}\lambda \Rightarrow \lambda = 8 \times 700 = 5600\ A^0$

131. Length of path $= \sqrt{3}x + (L - 2x)\frac{\sqrt{3}}{2}$

$\frac{\sqrt{3}L}{2}$



132. $R = \frac{10}{2.5} = 4 \pm \Delta R$

$\Delta R = 0.03 \times 4 = 0.12$

133. L/R=time, CR = time

134. $A = A_0 e^{-\lambda t}$

135. The difference in the binding energy is the energy required to add extra neutron.
B.E. = (BE/nucleon) No. of nucleons

136. Conserve the number of nucleons.

137. Calculate time when it reaches a height of 100 m, then use $A = \lambda N$.

138. Apply the equation $N = N_0 e^{-\lambda t}$

139. Use conservation of linear momentum

140. Use: KE $\propto$ (momentum)^2

141. $T \propto n^3$

142. Use K.E. $= \frac{p^2}{2m}$

143. $E = \frac{Z^2}{n^2} E_0$

144. The focal length of the combination is given as

$$\frac{1}{f}=\frac{1}{f_r}+\frac{1}{f_r}+\frac{1}{f_m}$$

where $f_m = R/2$ & $\frac{1}{f_r}=(n-1)\left(\frac{1}{\infty}-\frac{1}{-R}\right)=\frac{n-1}{R}$

$$\Rightarrow \frac{1}{f}=\frac{2(n-1)}{R}+\frac{2}{R}=\frac{2n}{R}\Rightarrow f=\frac{R}{2n}$$

$$\Rightarrow 2f=\frac{R}{n}$$

145. $t_{1/2}=\frac{0.639}{\lambda}$

146. $\lambda_{min}=\frac{hc}{eV}$

147. Use Moseley's law

148. $\lambda_{min}=\frac{hc}{eV}$

149. $\left|\frac{\Delta x}{x}\right|=\left|\frac{\Delta a}{a}\right|+\left|\frac{\Delta b}{b}\right|+3\left|\frac{\Delta c}{c}\right|$

150. $P=\frac{m}{\pi r^2 \ell}$

$$\Rightarrow \left|\frac{\Delta P}{P}\right|=\left|\frac{\Delta m}{m}\right|+2\left|\frac{\Delta r}{r}\right|+\left|\frac{\Delta \ell}{\ell}\right|$$

152. Check dimensions

153. In equilibrium, there will not be any friction between the cylinder and the wedge. If θ be the required angle
Cylinder A: $mg \sin 60^\circ = kx \cos(60^\circ - \theta)$ $(= mg \cos 30^\circ)$
Cylinder B: $mg \sin 30^\circ = kx \cos(30^\circ + \theta) = kx \sin(60^\circ - \theta)$
$\therefore$ $\cot 30^\circ = \cot(60^\circ - \theta) \Rightarrow \theta = 30^\circ$.

154. $N=\frac{mv^2}{R}$ .....(A)

$f_{max} = mg$

$\mu N = mg$

$\mu\frac{mv^2}{R} = mg$

$$v=\sqrt{\frac{Rg}{\mu}}=\sqrt{\frac{2\times10}{0.2}}=10 \text{ m/s}$$

155. The z-component of the force and the x-component of displacement are ineffective here.

$dW = F_y dy = 3xy.dy \quad (\because z = 0)$

$= 6x^4 dx \ (\because y = x^2)$

Integrating between $x = 0$ and $x = 2$ gives the result.

156. Power $= \dfrac{d(K.E.)}{dt}$

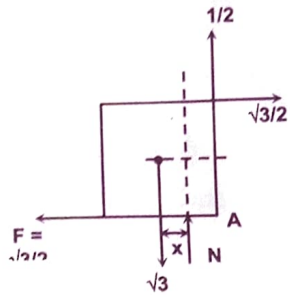
157. The force of friction, when it is not slipping, is less than its limiting value : $|f| \le \mu N$. This leads to the condition on F.

158. For equilibrium

$\Sigma\vec{F} = 0 \quad \Sigma\vec{\tau} = 0$

$N = \dfrac{\sqrt{3}}{2} - \dfrac{1}{2}$ ....(i)

Torque about A due to all forces excluding normal force is found to be zero.



159. For row 1: $\dfrac{1}{C_{eff}} = \dfrac{1}{C} + \dfrac{1}{2C} + \dfrac{1}{4C} + \ldots = \dfrac{2}{C}$

$\Rightarrow C_{eff} = C/2$

Similarly for row 2 : $C_{eff} = C/4$

$\Rightarrow C_{eff_{net}} = \dfrac{C}{2} + \dfrac{C}{4} + \dfrac{C}{8} + \ldots = C$

160. ($\Sigma i = 0$ at point F)

$\dfrac{V_F - V_C}{R} + \dfrac{V_F - V_B}{R} + \dfrac{V_F}{4R} = 0$ ... (1) ($\because V_A = V_B$)

Also $\dfrac{V_C - V_F}{R} + \dfrac{V_C - V_B}{R} + \dfrac{V_C}{2R} = 0$ ... (2)

Using (1) & (2)

$\dfrac{V_F}{V_C} = \dfrac{14}{13}$

MULTIPLE CHOICE QUESTIONS (MULTI CORRECT ANSWER)

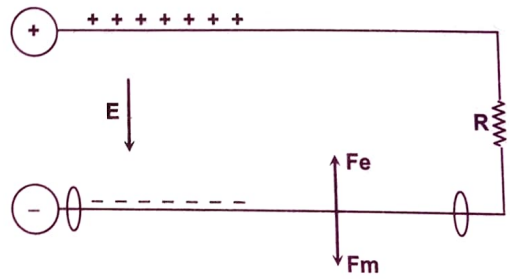
1. For unit length of wire,

$F_e = \lambda E \Rightarrow$ electric force

$= \lambda \dfrac{2\lambda}{4\pi\varepsilon_0 \ell} = \dfrac{\lambda^2}{2\pi\varepsilon_0 \ell}$

$\lambda \Rightarrow$ excess charge per unit length of wire

$F_m \Rightarrow \dfrac{\mu_0 I^2}{2\pi\ell} \Rightarrow$ magnetic force



According to given question

$F_e = F_m$

$\dfrac{\lambda^2}{2\pi\varepsilon_0 \ell} = \dfrac{\mu_0 I^2}{2\pi\ell}$

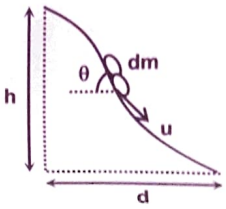
$\Rightarrow \left(\dfrac{I}{\lambda}\right)^2 = \dfrac{1}{\varepsilon_0 \mu_0}$

$$\Rightarrow \quad \left(\frac{\ell n 20}{\pi\varepsilon_0 V}\times\frac{V}{R}\right)^2 = \frac{1}{\mu_0\varepsilon_0}$$

$$\Rightarrow \quad R = \sqrt{\frac{\mu_0}{\varepsilon_0}}\left(\frac{\ell n 20}{\pi}\right) = 360\ \Omega$$

2. $$F = 2\times\frac{\mu_0 I_1}{2\pi R}\times I_2\ell = \frac{2\times 2\times 10^{-7}\times 50\times 20\times 0.50}{0.25} = 8\times 10^{-4}\ N$$

3. Momentum of segment of mass dm and length $d\ell$



$$d\vec{p} = dmu(\cos\theta\hat{i} + \sin\theta\hat{j})$$

$$= u\lambda\left[d\ell\cos\theta\hat{i} + d\ell\sin\theta\hat{j}\right]$$

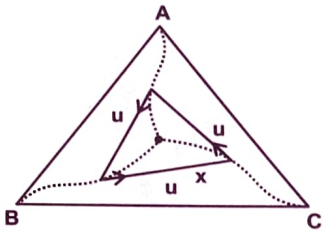
$$\vec{P} = \mu\lambda\left[\int dx\hat{i} + \int dy\hat{j}\right]$$

$$\vec{p} = u\lambda\left[d\hat{i} + h\hat{j}\right]$$

$$P = u\lambda\sqrt{d^2 + h^2}$$

$$= 1\times 1\sqrt{3^2 + 4^2} = 5\ \text{kg-m/s}$$

4. At any time t,



$$\omega = \frac{u\sin 60^\circ}{x}$$

$$\alpha = \frac{u\sin 60^\circ}{x^2}\left(-\frac{dx}{dt}\right)$$

$$\alpha = \frac{u\sin 60^\circ}{x^2}(u + u\cos 60^\circ)$$

α at $t = 0$

$$\alpha = \frac{u\sin 60^\circ}{a^2}(u + u\cos 60^\circ) = 9\ \text{rad/s}^2$$

5. $$v = 20 - x$$

$$\text{Force} = mv\frac{dv}{dx}$$

6. $$T = \frac{R\theta}{2v_0} + \frac{2R\cos(\theta/2)}{v_0}$$

$$\frac{dT}{d\theta} = 0 \Rightarrow \frac{R}{2v_0} - \frac{R\sin(\theta/2)}{v_0} = 0$$

$$\Rightarrow \sin(\theta/2) = ½ \Rightarrow \theta = 60^0 \Rightarrow T = \frac{R}{v_0}\left[\frac{\pi}{6} + \sqrt{3}\right]$$

$$\text{Distance} = R\times\frac{\pi}{3} + 2R\times\frac{\sqrt{3}}{2}$$

7. Use $F_t = m\frac{dv}{dt}$ and $F_r = \frac{mv^2}{r}$

8. For breaking off the plane:

$F \sin\alpha = mg$

$\Rightarrow a t_0^2 \sin\alpha = mg$

$\Rightarrow \quad t_0 = \sqrt{\dfrac{mg}{a\sin\alpha}}$.

Speed at time of breaking off.

$$v = \int_0^{t_0} \frac{at^2\cos\alpha}{m}dt = \frac{at_0^2\cos\alpha}{3m} = \frac{a\cos\alpha}{3m}.\frac{mg}{a\sin\alpha}\sqrt{\frac{mg}{a\sin\alpha}}$$

$$= \sqrt{\frac{mg^3}{9a\tan^2\alpha\sin\alpha}}$$

$$a = \frac{F\cos\alpha}{m} = \frac{at_0^2\cos\alpha}{m} = \frac{amg}{a\sin\alpha}.\frac{\cos\alpha}{m}$$

$= g\cot\alpha.$

$$s = \int v dt = \int_0^{t_0} \frac{at^3}{3m}\cos\alpha dt$$

$$= \frac{a}{12m}t_0^4 = \frac{a}{12m}.\frac{m^2g^2\cos\alpha}{a^2\sin^2\alpha} = \frac{mg^2}{12a\tan\alpha\sin\alpha}.$$

9. The field will be in the direction lying exactly between 9 and 10.

$\Rightarrow$ A, C, D are correct.

13. $v_p = A\omega\cos\omega t\ \hat{j}$

$v_B = -A\omega\sin(\omega t)\hat{j}$

$\Rightarrow$ Direction of friction $= \cos\omega t\ \hat{i} + \sin\omega t\hat{j}$

15. $\eta > 1$, velocity at another point will be greater then v

so, pressure $(p') < p$

$\eta < 1$, velocity at another point will be less then v

so, pressure $(p') > p$

17. $\vec{r}_1 = \vec{\alpha}(1-\gamma t+\beta t^2)$, as $\vec{\alpha}$ is constant so magnitude of the position vector is increasing but the direction of position vector is constant. Hence motion is a straight line motion

Displacement $(\vec{x}) = \vec{r}_1 - \vec{r}_0 = (\beta t^2 - \gamma t)\vec{\alpha}$

Velocity; $\vec{v} = \dfrac{d\vec{x}}{dt} = \vec{\alpha}(2\beta t - \gamma)$

Acceleration $\vec{a} = 2\beta\vec{\alpha}$

As the motion is straight line motion speed is same as the magnitude of velocity

18. When the particle just reaches the top of the wedge

$mu = (m+n)v \qquad \ldots(1)$

$$u = \sqrt{2gh\left(1+\frac{1}{n}\right)}$$

Angle of projection as observed by the ground will be loss than α

$\vec{v}_{block/ground} = \vec{v}_{block/wedge} + \vec{v}_{block/ground}$

19. $x = A \sin \omega t$

$\frac{A}{2} = A \sin \omega T_1$ ...(1)

$\frac{A}{\sqrt{2}} = A \sin \omega (T_1 + T_2)$ ...(2)

From eq. (1), $\omega T_1 = \frac{\pi}{6}$

From eq. (2), $\omega (T_1 + T_2) = \frac{\pi}{4}$

$\therefore \frac{T_1 + T_2}{T_1} = \frac{6}{4}$

$\therefore T_1 = 2T_2$

20. When the block is displaced slightly downwards.
Restarting force, $F_R = Axgd_2 - Axgd_1$

$F_R = Agx(d_2 - d_1)$ (SHM)

$\Rightarrow \omega = \sqrt{\frac{G(d_2 - d_1)}{Ld_E}}$

22. Keeping the volume constant, find the final radius and then terminal velocity and drag force. 143.
Radiation loss $\propto (T^4 - T_0^4)$

23. A has higher kinetic energy than B.

$\frac{1}{3} mL^2 \omega^2 = mS\Delta T$

$\Delta T = \frac{L^2 \omega^2}{3s} = \frac{10^2 \times 20^2}{3 \times 4.2 \times 10^3} = \frac{20}{3 \times 4.2} = 1.58°$

24. Specific heat of a substance cannot be negative.

25. $C_v = \frac{f}{2} R$, where f is the degree of freedom

27. Use $L = L_0(1 + \alpha t)$, $A = A_0(1 + 2\alpha t)$ and $d = \frac{d_0}{1 + 3\alpha t}$

28. See the assumptions of kinetic theory of gases.

29. Number of revolutions in 1 sec = 70

30. Let the incident wave has intensity = I
So, intensity of reflected wave = 0.64I

$I \propto A_i^2$

And, $0.64I \propto A_R^2$

So, $\left(\frac{A_i}{A_R}\right)^2 = \left(\frac{1}{0.64}\right) \Rightarrow \frac{A_2}{A_i} = \frac{4}{5}$

31. $N_1 = \frac{v_0 + v}{v_0} N_0$

$N_2 = \frac{v_0}{v_0 - v} N_0$

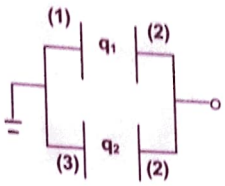
$N_3 = \frac{v_0 + \frac{v}{2}}{v_0 - \frac{v}{2}} N_0$

32. Since battery is connected, potential difference remains constant.

33. The system is equivalent to

$\frac{q_1}{C_1} = \frac{q_2}{C_2} = \frac{q_1 + q_2}{C_1 + C_2}$

$\therefore \quad q_1 = \frac{C_1 Q}{C_1 + C_2}, \; q_2 = \frac{C_2 Q}{C_1 + C_2}$



34. Take Gaussian surface into the metal.

35. There are induced charges in case of neutral sphere.

36. Connect the voltmeter in parallel.

37. Use symmetry.

38. Use Amperes Law

39. Use work energy theorem, magnetic force being perpendicular to velocity can not do any work.

40. Consider $\vec{v} \times \vec{B}$

41. Flux remains constant here, so emf induced is zero.

42. Flux charges due to rotation of the semi-circle.

43. $i = \frac{dq}{dt} = \frac{d}{dt}(CvB\ell) = CB\ell \frac{dv}{dt} = CB\ell a$

$\therefore \quad F - CB^2\ell^2 a = ma$

$\Rightarrow \quad a = \frac{F}{M + B^2\ell^2 C}$

$\Rightarrow$ emf increases

$\Rightarrow$ charge increase.

45. Use Snell's law

46. Normal incidence $\Rightarrow$ normal reflection

47. $\frac{1}{v}+\frac{1}{-90}=\frac{1}{-30}$

$\Rightarrow \frac{1}{v}=\frac{1}{90}+\frac{1}{30}=\frac{1-3}{90}=-\frac{2}{90} \Rightarrow v=-45$ cm

$\frac{dv}{dt}=-\frac{v^2}{u^2}\frac{du}{dt}=-\frac{45^2}{90^2}\times 4=-1$ m/s

48. Equivalent length of the glass plate in water $=\frac{\mu_2 t}{\mu_1}$

$\therefore$ O.P.D. $=\left(\frac{\mu_2}{\mu_1}-1\right)t$

49. Rays falls normally on the surface $\Rightarrow$ refracted rays will go undeviated.

50. Use rays diagram.

52. Use Einstein's photoelectric equation

53. $\lambda < 5200\ \overset{0}{A}$

54. Existence of cut-off frequency and photoemission takes place even when intensity is low.

55. $p_A=\frac{h}{\lambda_A}$, $T_A=\frac{P_A^2}{2m}=\frac{h^2}{2m\ell_A^2}$, $T_B=\frac{h^2}{2m\lambda_B^2}=\frac{h^2}{8m\lambda_A^2}$

$\frac{h^2}{2m\lambda_A^2}=4.25-\phi_A$ ... (1)

$\frac{h^2}{8m\lambda_A^2}=4.70-\phi_B$ ... (2)

Also, $\frac{h^2}{8m\lambda_A^2}=\frac{h^2}{2m\lambda_A^2}-1.5$

$\frac{3}{8}\frac{h^2}{m\lambda_A^2}=1.5 \Rightarrow \frac{h^2}{m\lambda_A^2}=4eV$

$\therefore T_A=2eV, T_V=0.5$ eV

$\phi_A=2.25$ eV, $\phi_B=4.70-0.5=4.20$ eV

56. $\lambda_{min}=\frac{hc}{eV} \Rightarrow \lambda_{min}$ decreases $\Rightarrow$ intensity increases.

57. For photoemission $\lambda < \lambda_0$

59. $\varepsilon_{PC}=\frac{v\mu_0 i}{2\pi}\ln 2$

$\varepsilon_{PQ}=\frac{2v\mu_0 i}{2\pi}\ln 2$

$\varepsilon_{net}=\frac{\mu_0 iv}{2\pi}\ln 2$

$q=C\varepsilon[1-e^{-t/RC}]$

$i=\frac{\varepsilon}{R}e^{-t/RC}$

62. Conserving momentum along the east west line

$M(v\cos^2 45 - v_0) = Mv_0$

$\Rightarrow 2MV_0 = MV\cos^2 45^0$

$\Rightarrow v_0 = V/4$

Vertical speed $= \frac{v}{\sqrt{2}}$

North speed $= v\cos^2 45^0 = \frac{v}{2}$

East speed $= \frac{V}{2} - \frac{V}{4} = \frac{V}{4}$

$\Rightarrow$ Speed from ground $= v\sqrt{\frac{1}{2}+\frac{1}{4}+\frac{1}{16}} = v\sqrt{\frac{13}{16}}$

63. Speed is more than the orbital but less than the escape velocity therefore an orbit will form. The speed is perpendicular to the radius.
It must be either an apogee or a perigee. But since the speed will be increasing, the launch point must be the perigee.

64. $R = \frac{v^2}{a_c}$

$v = \frac{J}{M}$

$\omega = \frac{\frac{JL}{2}}{\frac{Ml^2}{12}} = \frac{6J}{ML}$.

$\Rightarrow v_A = v + \frac{\omega L}{2} = \frac{J}{M} + \frac{3J}{M} = \frac{4J}{M}$

$a_{c_A} = \frac{\omega^2 L}{2} = \frac{18J^2}{M^2L}$

$\Rightarrow R_A = \frac{v_A^2}{a_{cA}} = 16\frac{J^2}{M^2}\cdot\frac{M^2L}{18J^2} = \frac{8}{9}L$

$v_B = v - \frac{\omega L}{2} = \frac{2J}{M}$

$\Rightarrow R_B = \frac{4J^2}{M^2}\cdot\frac{M^2L}{18J^2} = \frac{2}{9}L$

66. For small angle of inclinations, ball rolls without slipping. So there is no work done by friction. At some critical angle θ_C, the ball starts slipping. For $\theta > \theta_C$, energy would be lost due to work done by kinetic friction. But again for $(\theta = \pi/2)$, friction would be zero.

67. As going up, speed of the particle is decreasing and hence the time taken in crossing the windows will be $t_1 < t_2 < t_3$ (If $W_1 = W_2 = W_3$)
Simultaneously $\Delta V \propto t$
$\therefore \Delta V_1 < \Delta V_2 < \Delta V_3$
For unequal windows $t_1 = t_2 = t_3$ may be if $W_3 < W_2 < W_1$

68. $F = -\frac{dU}{dr} = \frac{A}{r^3} - \frac{B}{3r^2}$

For equilibrium $F = 0$, $r = \frac{3A}{B}$

$\frac{B^2}{6A} = \frac{A}{2r_0^2} - \frac{B}{3r_0}$ (for $r_0 = \frac{r}{3}$)

69. Acceleration of particle P and Q along the incline is same.
Acceleration of particle P perpendicular to the incline plane is $10 \cos 60° = 5$ m/s^2
So, $v_P - 5(2) = 0$
$\therefore v_P = 10$ m/s

70. $\omega R = u =$ constant

$\omega \frac{dR}{dt} + R \frac{d\omega}{dt} = 0$

$\therefore \frac{d\omega}{dt} = -\frac{\omega}{R} \frac{dR}{dt}$

In time dt radius decreases by dR

$-2\pi R dR = u dt$

$\therefore \frac{dR}{dt} = -\frac{u}{2\pi R}$

$\frac{d\omega}{dt} = \alpha = \frac{\omega}{R} \frac{u}{2\pi R}$

$\alpha = \frac{u^2 d}{2\pi R^3}$

$\tau = \frac{I u^2 d}{2\pi R^3}$

$\therefore F = \frac{I u^2 d}{2\pi R^4}$

71. For pure rolling motion, static friction would act in the forward direction on the sphere. Work done by static friction is change in the kinetic energy of the sphere. Work done by static friction on the complete sphere + Plank system is zero.

72. $V_L = L\frac{di}{dt} \Rightarrow L di = V_L dt$

$L\int di = \int V_L dt = \frac{1}{2} \cdot 6 \cdot 10 = 30$

$L(i_f - i_i) = 30$ Amp

$i_f = \frac{30}{10} = 3$ amp

for $t \le 2$,

$V_L = \frac{10}{2} t = 5t$

$\int_0^i L di = \int V_L dt = 5\int_0^1 t dt$

$$L(i-0)=\frac{5}{2}$$

$\Rightarrow i = 1/4$

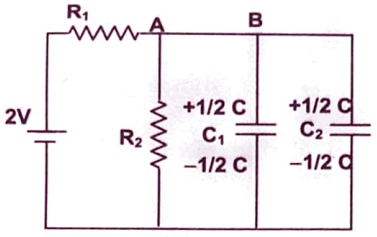
for $t \geq 2$

$$V_L=\left(15-\frac{5}{2}t\right)$$

$$\int_1^i Ldi=\int_2^4\left(15-\frac{5}{2}t\right)dt$$

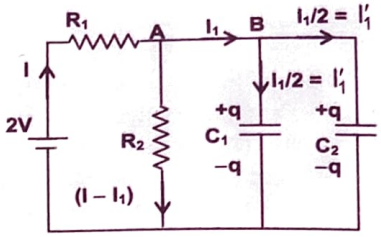
$\Rightarrow i = 5/2$ amp

73.



(at t = 0)

When the switch is closed then at t = 0 the distribution of the charge is shown in the figure.



By applying Kirchhoff's Law $I_1'=\frac{(1-q)}{2}$

$$\int_{1/2}^{q}\frac{dq}{1-q}=\int_0^t\frac{dt}{2}$$

$$q=\left(1-\frac{1}{2}e^{-t/2}\right)$$

$$I_1'=\frac{dq}{dt}=\frac{1}{4}e^{-t/2}$$

74. $H_1=\frac{V^2}{R_1}(30\times60),\ H_2=\frac{V^2}{R_2}(45\times60)$

$$H_1=H_2\Rightarrow\frac{30}{R_1}=\frac{45}{R_2}\Rightarrow 3R_1=2R_2$$

$$H_{series}=\frac{V^2}{(R_1+R_2)}t_s=\frac{V^2}{\left(R_1+\frac{3}{2}R_1\right)}t_s=H_1=\frac{V^2}{R_1}(30\times60)$$

$$\Rightarrow t_s=(30\times60)\frac{5}{2}=75\,\text{min}$$

$$H_P=\left(\frac{V^2}{R_1}+\frac{V^2}{R_2}\right)t_P=\left(\frac{V^2}{R_1}+\frac{2V^2}{3R_1}\right)t_P$$

$$\Rightarrow\frac{5V^2}{3R_1}t_P=\frac{V^2}{R_1}(30\times60)$$

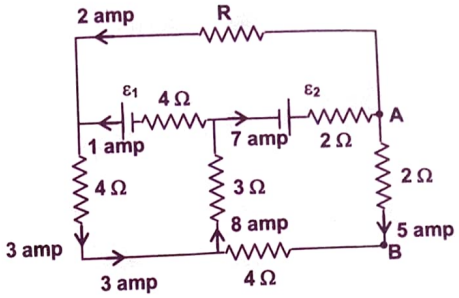
$$t_P=\frac{3\times30\times60}{5}=18\,\text{min}$$

75. By applying Kirchhoff's law

$\varepsilon_1 = 40$ V

$\varepsilon_2 = 68$ V

$R = 9\ \Omega$ and current through $4\Omega = 5$ amp.



78.

76. As the ammeters are ideal, so

$$R_{eq} = \frac{R_1R_2R_3}{R_1R_2 + R_2R_3 + R_3R_1}$$

The readings of two ammeters are different so resistors R_1 and R_3 are different.

If R_1 and R_3 are interchanged, the reading of ammeters will also change.

If $R_2 = R_1$ (If R_2 and R_1 switched the reading of the ammeter do not vary) similarly if $R_2 = R_3$ (R_2 and R_3 switched the reading of ammeters do not change)

If $R_2 = R_1$, $I_1 = I_2 = 0.15$, $I_3 = 0.05$ and $I = 0.35$

If $R_2 = R_3$, $I_2 = I_3 = 0.10$, $I_1 = 0.20$ and $I = 0.40$ amp

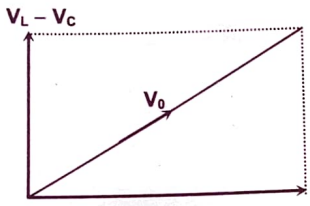
79

77. $V_R^2 + V_L^2 = V_i^2 = (20\sqrt{2})^2 = 800$ ...(i)

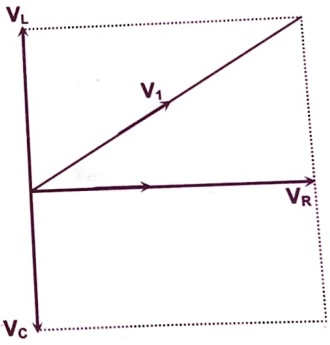
$V_L - V_C = 15$ ...(ii)

$\sqrt{V_R^2 + (V_L - V_C)^2} = (25)^2$

$V_R^2 + 225 = 625$



V_0 = source voltage



$\Rightarrow V_R = 20$ volts

$V_L = 20$ volts

$V_L - V_C = 15$

$20 - V_C = 15$

$V_C = 5$ volts

Power factor $= \cos\phi = \frac{R}{Z} = \frac{iR}{iZ} = \frac{V_R}{V_Z} = \frac{20}{25} = \frac{4}{5}$

78. By applying Kirchhoff's Law

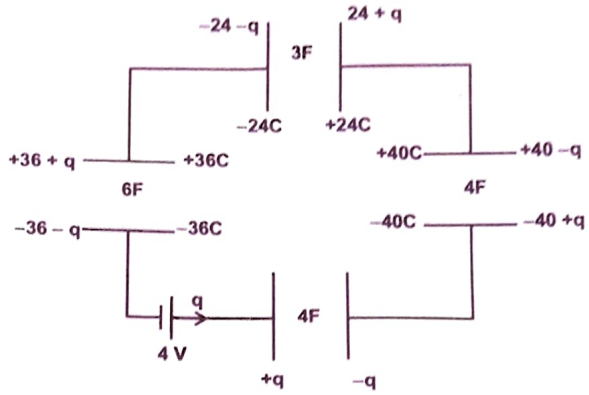
$$-4+\frac{q}{4}+\frac{-40+q}{4}+\frac{24+q}{3}+\frac{36+q}{6}=0$$

$$-4+\frac{q}{4}-10+\frac{q}{4}+8+\frac{q}{3}+6+\frac{q}{6}=0$$

$$\frac{q}{4}+\frac{q}{4}+\frac{q}{3}+\frac{q}{6}=0$$

$\Rightarrow q=0$

Hence, there is no flow of charge.



79. As we know that,

$$\frac{hC}{\lambda}=\phi+K \Rightarrow \frac{12400}{\lambda}=\phi+v_s$$, where ϕ is in electron Volt and λ is in Å

When light of wavelength $\lambda = 6000$ Å is used

$$\frac{12400}{6000}=\phi+0.3 \Rightarrow \phi = 1.77 \approx 1.8 \text{ eV (approximately)}$$

When light with wavelength $\lambda = 4000$ Å is used

$$\frac{12400}{4000}=\phi+eV$$

$\Rightarrow eV = (3.1 - 1.8)$ e volt $= 1.3$ eV

$\Rightarrow v = 1.3$ Volt

80. Condition for maximum intensity due to thin film

$$2\mu d-\frac{\lambda}{2}=n\lambda$$

$$\Rightarrow 2\mu d=(2p+1)\frac{\lambda_1}{2}$$

$\Rightarrow (2p + 1)\lambda_1 = 4\ \mu d$

Here $\lambda_1 = 400$ nm and $n = p \Rightarrow$ An integer

Let $\lambda' = 1150$ nm. Since only two wavelength give maximum intensity, so

$$\frac{4\mu d}{2(p-1)+1}<\lambda'<\frac{4\mu d}{2(p-2)+1}$$

$$\frac{\lambda_1(2p+1)}{2(p-1)+1}<\lambda'<\frac{\lambda_1(2p+1)}{2(p-2)+1}$$

$$\Rightarrow \frac{\lambda_1(2p+1)}{2(p-1)+1}<\lambda' \text{ or } \lambda'<\frac{\lambda_1(2p+1)}{2(p-2)+1}$$

$$\Rightarrow p>\frac{1}{2}\left(\frac{\lambda'+\lambda_1}{\lambda'-\lambda_1}\right)=\frac{1}{2}\left(\frac{1150+400}{1150-400}\right)=1\ldots\ldots, \text{ or}$$

$$p<\frac{1}{2}\left(\frac{3\lambda'+\lambda_1}{\lambda'-\lambda_1}\right)=\frac{1}{2}\left(\frac{3\times1150+400}{1150-400}\right)=2\ldots\ldots$$

The only integer which satisfies both inequalities is 2, so

$$d = \frac{(2\times2+1)400\times10^{-9}}{4\times1} = 500\text{ nm}$$

$$\lambda_2 = \frac{4\mu d}{2(p-1)+1} = \frac{4\times1\times500\times10^{-9}}{2+1} = 666.7\text{ nm}$$

$$\Delta T = \frac{d}{\alpha h} = \frac{500\times10^{-9}}{8\times10^{-6}\times2\times10^{-2}} \approx 3.1\,^{\circ}C$$

81. $k = \text{stiffness of string} = \frac{F}{x_{max}} = \frac{mg}{x_{max}} = \frac{40\times10^{-3}\times9.8}{20\times10^{-2}} = 1.96\text{ N/m}$

When we replace the mass 40 gm by the mass $m_2 = \frac{2205}{32}$ gm, we have

$$T = 2\pi\sqrt{\frac{m}{k}} = \frac{3\pi}{8}\text{ sec}$$

$$\omega = \frac{2\pi}{T} = \sqrt{\frac{k}{m}} = \frac{16}{3}\text{ rad/sec}$$

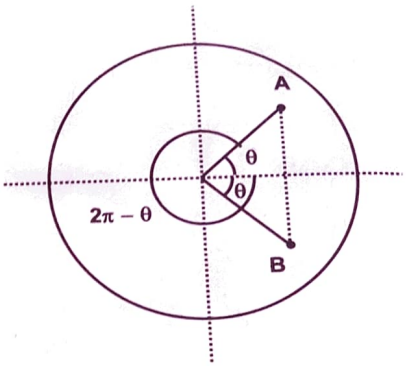
$$x = x_0\cos\omega t = 0.2\cos\frac{16}{3}t$$

$$v = -x_0\omega\sin\omega t = -\frac{16}{15}\sin\frac{16}{3}t$$

$$a = -x_0\,\omega^2\cos\omega t = -\frac{256}{45}\cos\frac{16}{3}t$$

If $x = 10$ cm $\Rightarrow \cos\frac{16}{3}t = \frac{1}{2} = \cos\left(\frac{\pi}{3}\right)$

$$\Rightarrow \frac{16}{3}t = (2n\pi\pm\frac{\pi}{3}) \Rightarrow t = \left(\frac{3n\pi}{8}\pm\frac{\pi}{8}\right)\text{sec}$$



82. For transverse wave, $v_T = \sqrt{\frac{F}{\mu}} = \sqrt{\frac{F}{\rho\times S}}$

$$V_{Al} = \sqrt{\frac{900}{2.5\times10^3\times10^{-6}}} = \frac{30}{50\times10^{-3}} = 600\text{ m/s}$$

$$V_{Cu} = \sqrt{\frac{900}{8.1\times10^3\times10^{-6}}} = \frac{30}{90\times10^3} = \frac{1000}{3}\text{ m/s}$$

$$\frac{a_r}{a_i} = \frac{V_{Al}-V_{Cu}}{V_{Al}+V_{Cu}} = \frac{\frac{800}{3}}{\frac{2800}{3}} = \frac{2}{7} \approx 0.3$$

For longitudinal wave $V_{\ell} = \sqrt{\frac{Y}{\rho}}$

$$V_{Al} = \sqrt{\frac{9\times10^{10}}{2.5\times10^3}} = \frac{3\times10^5}{50} = 6000\text{ m/s}$$

$$V_{Cu} = \sqrt{\frac{16\times10^{10}}{8.1\times10^3}} = \frac{4\times10^5}{9\times10} = \frac{40000}{9}\text{ m/s}$$

$$\frac{a_r}{a_i}=\frac{V_{Al}-V_{Cu}}{V_{Al}+V_{Cu}}=\frac{\frac{1400}{9}}{\frac{94000}{9}}=\frac{7}{47}\approx 0.15$$

83. Consider the system at any time t when each block is moving with speed v and spring has an extension x.

$\Rightarrow 2mg-T=2ma$ ...(1)

$\Rightarrow T-kx=ma$ ...(2)

From equation (1) & (2), we can write

$$\frac{2g}{3}-\frac{k}{3m}x=a \Rightarrow \frac{d^2x}{dt^2}+\frac{kx}{3m}-\frac{2g}{3}=0$$

$$\Rightarrow \frac{d^2x}{dt^2}+\frac{k}{3m}\left[x-\frac{2mg}{k}\right]=0 \Rightarrow \omega=\sqrt{\frac{k}{3m}}$$

For equilibrium $\frac{d^2x}{dt^2}=0 \Rightarrow x_0=\frac{2mg}{k}$

Amplitude of motion $= x_0-0=\frac{2mg}{k}$

$$a_{max}=\omega^2A=\frac{k}{3m}\times\frac{2mg}{k}=\frac{2g}{3}$$

$$v_{max}=\omega A=\frac{2mg}{k}\sqrt{\frac{k}{3m}}=2g\sqrt{\frac{m}{3k}}$$

Second Method:

Using **COME**, we can write

$$2mgx=\frac{3mv^2}{2}+\frac{kx^2}{2}$$

For extreme position, $v=0 \Rightarrow x=0, \frac{4mg}{k}$

A $\Rightarrow$ Amplitude of motion $=\frac{2mg}{k}$

For maximum velocity, $\frac{d^2x}{dt^2}=0 \Rightarrow x=\frac{2mg}{k} \Rightarrow$ Equilibrium position

$$\Rightarrow 2mg\times\frac{2mg}{k}=\frac{3mv_{max}^2}{2}+k\frac{4m^2g^2}{2k}$$

$$\Rightarrow \frac{4m^2g^2}{2k}=\frac{3mv_{max}^2}{2} \Rightarrow v_{max}^2=4g^2\left(\frac{m}{3k}\right)$$

$$\Rightarrow v_{max}=2g\sqrt{\frac{m}{3k}}$$

$$a_{max}=\left|\frac{2g}{3}-\frac{k}{3m}\times\frac{4mg}{k}\right|=\frac{2g}{3}$$

84. The energy released in α-decay is given by

$$Q=(M_U-M_{Th}-M_{He})c^2$$

Substituting the atomic masses as given in the data, we find

$$Q=(238.05079-234.04363-4.00260)u\times c^2=(0.00456\,u)c^2=(0.00456u)c^2$$

$$=(0.00456u)(931.5\ MeV/u)=4.25\ MeV.$$

If $^{238}_{92}U$ spontaneously emits a proton, the decay process would be $^{238}_{92}U \rightarrow ^{237}_{91}Pa + ^{1}_{1}H$

The Q for this process to happen is

$= (M_U - M_{Pa} - M_H)c^2 = (238.05079 - 237.05121 - 1.00783)u \times c^2 = (-0.00825\,u)c^2$

$= -(0.00825\ u)(931.5\ MeV/u) = -7.68\ MeV$

Thus, the Q of the process is negative and therefore it cannot proceed spontaneously. We will have to supply an energy of 7.68 MeV to a $^{238}_{92}U$ nucleus to make it emit a proton.

85. $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{50 \times 1 \times 9.8}{0.1}} = 70\,m/s.$

$\frac{\lambda}{4} = 1 \Rightarrow \lambda = 4\,m \Rightarrow f = \frac{v}{\lambda} = \frac{70}{4} = 17.5\,Hz.$

$v_{p\,max} = \omega A \Rightarrow A = \frac{v_{p\,max}}{2\pi f} = \frac{15}{2 \times 3.14 \times 17.5} = 0.136\,m = 13.6 \times 10^{-2}\,m$

86. (C) $\frac{M}{M_0} = \frac{Mc^2}{M_0c^2} = \frac{K + M_0c^2}{M_0c^2} = 1 + \frac{10^{-9}}{1.67 \times 10^{-27} \times 9 \times 10^{20}} \approx 7.68$

(D) $Mc^2 = M_0c^2 + K = 2m_0c^2$

$\Rightarrow M = 2\,M_0$

$\Rightarrow \frac{M_0}{\sqrt{1 - \frac{v^2}{c^2}}} = 2M_0$

$\Rightarrow 1 = 4 - \frac{4v^2}{c^2}$

$\Rightarrow v^2 = \frac{3c^2}{4} \Rightarrow v = \frac{\sqrt{3}}{2}c = 0.866c$

87. $\frac{m\lambda}{2} = 40$

$\left(n + \frac{1}{2}\right)\frac{\lambda}{2} = 30$

$\frac{(2n+1)}{m} = \frac{6}{4} \Rightarrow m = \frac{2(2n+1)}{3}$

$n = 1 \quad m = 2$ $\lambda = 40$ cm

$n = 4 \quad m = 6 \quad v = f\lambda \Rightarrow f/40 = f_0$

$n = 7 \quad m = 10$

$\lambda = 8$ second overtone

$f = v/8 = 5f_0$

$n = 10 \quad m = 14$

$\lambda = \frac{40}{7} \Rightarrow f_3 = \frac{7v}{40}.$

88. Avg. acceleration $= \frac{\text{change in velocity}}{\text{Time interval}}$

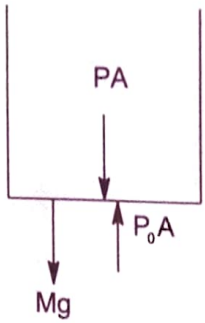
89. Upto time t = g/k, N will be along upwards direction after that it will be along downward direction.

90. $Mg + PA = P_0A$

$$M = \frac{P_0A}{g} - \frac{PA}{g}$$

$$P = \frac{P_i}{2} = \frac{P_0 + \frac{mg}{A}}{2}$$

$$\Rightarrow \quad M = \frac{P_0A}{2g} - \frac{m}{2}$$



91. $B \to C$; $C = C_v + R/2 = 2R$

So; $Q_{B\to C} = nC\,(T_C - T_B) = 2nR(T_C - T_B)$

Also; $\frac{T_C}{T_B} = 4, \frac{T_A}{T_B} = 2$

Thus; $Q_{B\to C} = 6nR\,T_B$.

$$Q_{net} = \frac{-nRT_B}{2} \Rightarrow Q_{C\to A\to B} = -130J$$

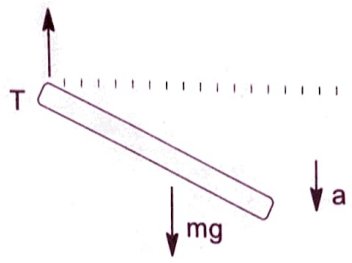
92. $mg - T = ma$

$$T\frac{\ell}{2}\cos\theta = \frac{m\ell^2}{12}\alpha$$

By constraints;

$$a = \frac{\alpha\ell}{2}\cos\theta$$

Thus; $T = \frac{mg}{1+3\cos^2\theta}$ and $a_A = \frac{3g\sin\theta\cos\theta}{1+3\cos^2\theta}$



93. $\frac{2mx_B + mx_A}{3m} = \frac{a}{3} \Rightarrow x_A = a - 2x_B$

$$\frac{2my_B + my_A}{3m} = 0 \Rightarrow y_A = -2y_B$$

Trajectory of B is $\frac{(a-2x)^2}{a^2} + \frac{(2y)^2}{b^2} = 1$

94. $m_B = 2m_A \Rightarrow Z_B = 2Z_A$. Let $Z_A = Z$, $Z_B = 2Z$

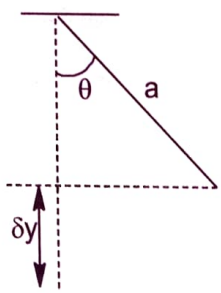
So; $E_B - E_A = \frac{13.6\times5}{36}\times3Z^2 = 5.667$

$$\Rightarrow \quad Z = 1$$

95. $\delta y = a(1-\cos\theta) = \frac{a\theta^2}{2}$

and $\theta_a = \frac{b\phi}{2}$

$$\Rightarrow \quad \delta y = \frac{a}{2}\left(\frac{b\phi}{2a}\right)^2 = \frac{b^2}{8a}\phi^2$$



$\Rightarrow \quad \delta y = \frac{b^2}{8a}(\omega\delta t)^2$

acc. of rod's centre $= \frac{b^2}{4a}\omega^2 \Rightarrow \delta F = \frac{mb^2}{4a}\omega^2$

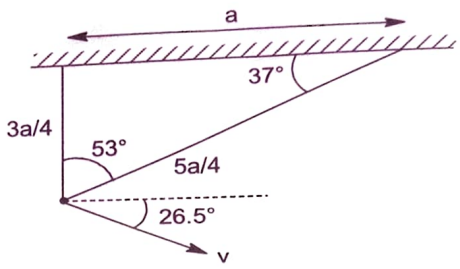
96. $h = \frac{3a}{4}$

$v_0 = \sqrt{2gh}$

$v_0 = \sqrt{1.5ag}$ and $v = v_0\cos(90° - 26.5°)$

$= v_0 \sin(26.5°)$

$\Rightarrow \quad v = \sqrt{1.5ag}\sin(26.5°)$



97. $t^2 = \frac{2\ell}{g[\cos\alpha\sin\alpha - \mu\cos^2\alpha]}$

For min. time; $\frac{d}{d\alpha}(\cos\alpha\sin\alpha - \mu\cos^2\alpha) = 0$

$\Rightarrow \quad \tan 2\alpha = \frac{-1}{\mu}$

$\Delta K + \Delta U = W_{friction}$

$\Rightarrow \quad v_0 = \sqrt{2g\ell(\tan\theta_0 - \mu)}$

98. $v \propto \frac{Z}{n}$, $R \propto \frac{n^2}{Z}$ and $E \propto \frac{Z^2}{n^2}$

$\Rightarrow \quad vR \propto n$ and $\frac{v}{E} \propto n$

99. $mv_0 = (4m+m)v$

$v = \frac{v_0}{5}$

$\frac{1}{2}mv_0^2 = \frac{1}{2}.(5m).\frac{v_0^2}{25} + \Delta E$

$\frac{1}{2}mv_0^2 - \frac{1}{2}.\frac{v^2}{25}m = \Delta E$

$\Rightarrow \quad \frac{1}{2}mv_0^2\left\{\frac{4}{5}\right\} = \Delta E$

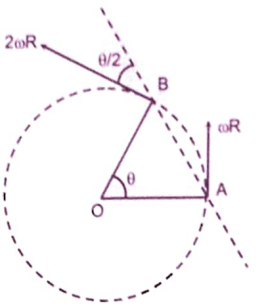
$\frac{1}{2}mv_0^2 = \frac{5}{4}\Delta E$

$\Delta E = 13.6$

$\Delta E = 13.6\left\{1 - \frac{1}{4}\right\} = \frac{3}{4}.(13.6)$

$\frac{1}{2}mv_0^2 = \frac{5}{4}\times 13.6,\ \frac{3}{5}\times 13.6 = 17\text{ eV},\ 8.16\text{ eV}$

100. $f' = f_0\left[\dfrac{1-\dfrac{1}{50}\cos\dfrac{\theta}{2}}{1-\dfrac{1}{100}\cos\dfrac{\theta}{2}}\right]$, $\theta = \omega t$



$$k = \frac{\omega R}{v} = \frac{1}{100}$$

$$t = \left[\frac{1-2k\cos\left(\frac{\theta}{2}\right)}{1-k\cos\left(\frac{\theta}{2}\right)}\right]$$

$$\frac{dt}{d\theta} = \frac{\left[1-k\cos\left(\frac{\theta}{2}\right)\right]\left[+k\sin\left(\frac{\theta}{2}\right)\right]-\left[1-2k\cos\left(\frac{\theta}{2}\right)\right]\left[\frac{k}{2}\sin\left(\frac{\theta}{2}\right)\right]}{\left[1-k\cos\left(\frac{\theta}{2}\right)\right]^2}$$

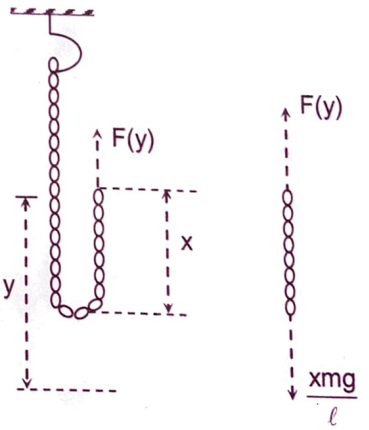
$$\Rightarrow \frac{k}{2}\sin\left(\frac{\theta}{2}\right)=0 \Rightarrow \frac{\theta}{2}=0,\pi \Rightarrow \theta=0,\ 2\pi$$

$$f_{min} = \frac{98}{99}f_0 \text{ at } \theta = 0°$$

$$f_{max} = \frac{102}{101}f_0 \text{ at } \theta = 2\pi$$

101. Basic knowledge to write angular momentum and kinetic energy of the system.

102. $x = \dfrac{y}{2}$, and $F(y) = \dfrac{mgx}{\ell} = \dfrac{mgy}{2\ell}$



Work done by $F(y) = \int_0^{\ell} F(y)dy = \dfrac{mg}{2\ell}\int_0^{\ell} ydy = \dfrac{mg\ell}{4}$

Second method: Using WET, we can write

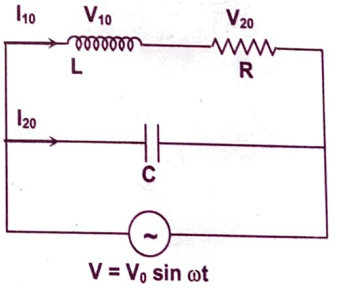
$K_f - K_i = W_F + W_{mg} = 0$

$$\Rightarrow W_F = -W_{mg} = -\left[-mg\left(\frac{\ell}{2}-\frac{\ell}{4}\right)\right] = \frac{mg\ell}{4}$$

Concept: Work energy power

103. For resonance

$I_{20} = I_{10}\sin\theta$



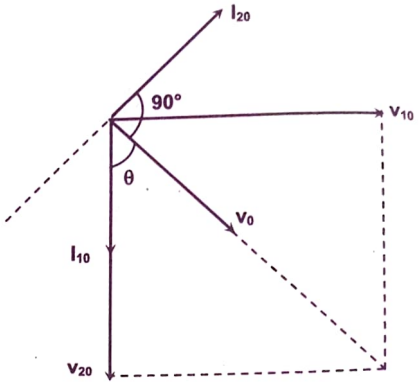
$$\frac{V_0}{X_c} = \frac{V_0}{\sqrt{X_L^2+R^2}} \times \frac{X_L}{\sqrt{X_L^2+R^2}}$$

$$\Rightarrow X_L^2 + R^2 = X_C X_L$$

$$\Rightarrow L^2\omega^2 + R^2 = \frac{L}{C}$$

$$\Rightarrow \omega^2 = \frac{1}{LC} - \left(\frac{R}{L}\right)^2$$

$$\Rightarrow \quad \omega = \sqrt{\frac{1}{LC} - \left(\frac{R}{L}\right)^2}$$



104. By energy conservation

$$\frac{1}{2}\frac{Mm}{m+M}v^2 = \frac{1}{2}kx^2$$

$$\Rightarrow x = J\sqrt{\frac{m}{M(M+m)K}} = 5\sqrt{\frac{1}{2\times3\times6}} = \frac{5}{6}m$$

105. $\frac{Q_{out}}{Q_{in}} = \frac{T_{out}}{T_{in}}$ and $Q_{out} = Q_{in} + W$

$$\Rightarrow Q_{in} = W\left(\frac{T_L}{T_H - T_L}\right)$$

$$\frac{Q_{in}}{t} = P\left(\frac{T_L}{T_H - T_L}\right)$$

106. Between faces LMNO and PQRS

$$dR = \frac{\rho dr}{t\frac{\pi}{2}r}$$

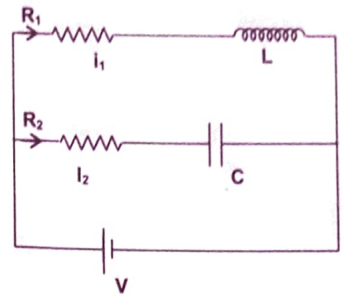
$$R = \frac{2\rho}{t\pi}\ell n\frac{b}{a}$$

Between faces LMRS and ONQP for an element at distance r from O.

$$dR = \frac{\rho\frac{\pi}{2}r}{tdr}$$

So, $\frac{1}{R} = \int\frac{1}{dR} = \int_a^b \frac{tdr}{\rho\frac{\pi}{2}r} = \frac{2t}{\rho\pi}\ell n\left(\frac{b}{a}\right)$, $\quad R = \frac{\rho\pi}{2t\ell n\left(\frac{b}{a}\right)}$

107. $(i_1)_0 = \frac{50}{25} = 2$ amp



$(i_2)_0 = \frac{50}{5000} = 0.01$ amp

$\tau_1 = 0.125$ sec

$\tau_2 = 0.125$ sec

$i_1 = 200\, i_2$

$2\left(1-e^{\frac{-t}{0.125}}\right) = \left(0.01\, e^{\frac{-t}{0.125}}\right)200$

$t = \frac{\ell n 2}{8}$

$4\left(\frac{q}{C}\right) = L\frac{di_1}{dt}$

$4\times 50\left(1-e^{\frac{-t}{0.125}}\right) = 50e^{\frac{-t}{0.125}}$, $t = \frac{1}{8}\ell n\left(\frac{5}{4}\right)$

108. $x^2 = \frac{2a}{\sqrt{3}}y$

$2x = \frac{2}{\sqrt{3}}a\frac{dy}{dx}$

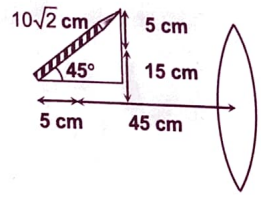
$\tan\theta = \sqrt{3}\frac{x}{a}$

$\theta = 60°$

$a = \frac{g\sin\theta}{1+\frac{I_{cm}}{mR^2}} = \frac{g}{\sqrt{3}}$

$f = \frac{mg\sin\theta}{1+\frac{mR^2}{I_{cm}}} = \frac{mg}{2\sqrt{3}}$

109. Length of pencil is $10\sqrt{2}$ cm.



Position of (A) [−(45+5), (15 − 5) ⇒ (−50, 10)

$\frac{1}{v}-\frac{1}{(-50)} = \frac{1}{20}$

$v = 33.3$ cm

magnification $\frac{h_i}{h_0} = \frac{-v}{u} \Rightarrow h_i = -h_0\frac{v}{u} = 10\frac{100/3}{50} = 6.6$ cm

Position of B (40 cm, 20 cm)

$\frac{1}{v}-\frac{1}{-40} = \frac{1}{20} \Rightarrow v = 40$ cm

magnification $h_i = 20\times\frac{40}{40} = 20$ cm

Position of C (36 cm, 12 cm)

$\frac{1}{v} = \frac{1}{20}-\frac{1}{45} \Rightarrow v = 36$ cm

Magnification $h_i = 15\frac{36}{45} = 12$ cm

110. $v_0 = 4\sqrt{gR}$

$\vec{J} = \Delta\vec{P}$

$-\int T dt = 2M\left(v - 4\sqrt{gR}\right)$

$-\int T dt = 2M\left(R\omega - 4\sqrt{gR}\right)$ ...(i)

$\left(\int T dt\right)R = MR^2\omega$ ...(ii)

From (i) and (ii)

$\omega = \frac{8}{3}\sqrt{\frac{g}{R}}$ and $v = \frac{8}{3}\sqrt{gR}$

111. $x_{cm} = \frac{\int x dm}{\int dm} = \frac{\int_0^L x\lambda_0\left(1+\frac{x}{L}\right)}{\int_0^L \lambda_0\left(1+\frac{x}{L}\right)} = \frac{5L}{9}$

$m = \int dm = \frac{3\lambda_0 L}{2}$

$\left[\left(A\cdot\frac{L}{9}\right)\rho g\right]\frac{17}{10}L\sin\theta = F\cdot L\cos\theta$

112. In order to flow up the floor drain, the KE of the fluid must be just sufficient to provide for the gravitational P.E. required:

$v_M^2 = 2gh_2$

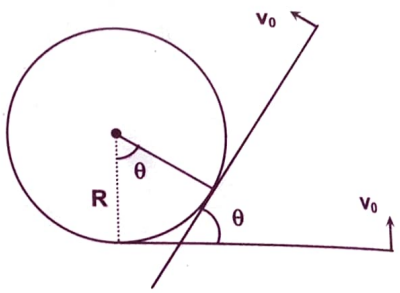
The flow rate through M is :

$V = \left(\pi r_M^2\right)v_M$; $r_M = 3\,cm = 0.03\,m$

and this is the total discharge from the roof.

The rainfall rate is $\frac{V}{wL}$ (in m/s) and it can be converted to cm/h by multiplying by (100×3600).

113. T and v are perpendicular so speed is constant.



$\theta = \frac{\ell - x}{R}$

$\Rightarrow \frac{d\theta}{dt} = -\frac{dx}{Rdt}$

$\Rightarrow \frac{v_0}{x} = -\frac{dx}{Rdt}$

$\Rightarrow x dx = -Rv_0 dt$

$\Rightarrow t = \frac{\ell^2}{2Rv_0}$

Now check the options.

114. Use horizontal & vertical velocity components for comparison.

115. Kinetic energy can never be negative.

116. Use of coulomb forces.

117. $F\cos\theta = 50a$ ...(1)

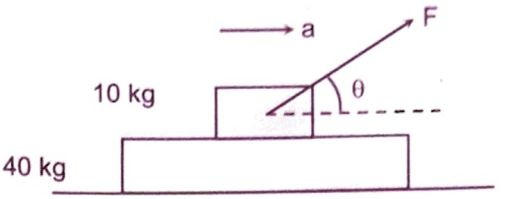
$N + F\sin\theta = 10g$ ...(2)

$F\cos\theta - 0.6N = 10a$ ...(3)

Solving for a, N from (1), (2) and Putting in (3), we get,

$$F = \frac{30g}{4\cos\theta + 3\sin\theta}$$

$\therefore \quad F_{max} = 60 = 15K \;\&\; K = 4.$



118. Segment S_1 has lower wavelength photons, so $E_1 > E_2$ and Intensity is equal, so $n_2 > n_1$.

119. Liquid will experience a force in downward direction always. So, N will be more than Mg. Magnitude of velocity will increase on downward journey and will decrease in upward journey.

120. By energy conservation

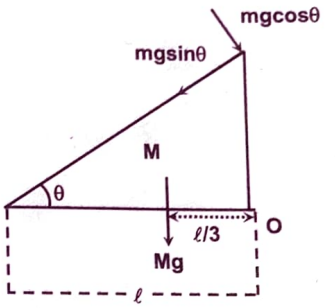
$$\Rightarrow v = \sqrt{\frac{2}{15}}(9.8)\ \text{m/s}$$

$$\omega = 140\sqrt{\frac{2}{15}}\ \text{rad/sec}$$

at bottom point $N - mg = \frac{mv^2}{(R-r)}$, where R is radius of curvature at (0, 0)

$\Rightarrow N = 49$ Newton (approx)

121. When $\mu > \tan\theta$, the particle will remain at rest on the wedge and force exerted by it on the wedge will be equal to mg in the vertically downward direction and wedge will remain at rest. When $\mu < \tan\theta$ the particle will slide down the wedge and force exerted by it on wedge will be as shown. The wedge will topple if,



$\Sigma\tau_0 > 0$

$$\Rightarrow \quad mg\cos\theta\,\ell\tan\theta\sin\theta - \mu mg\cos\theta\,\ell\tan\theta\cos\theta - Mg\frac{\ell}{3} > 0$$

$$\Rightarrow \quad m > \frac{M}{3\sin\theta(\sin\theta - \mu\cos\theta)}$$

122. From conservation of energy

$$-\frac{\sigma q}{2\varepsilon_0}\left(\sqrt{4R^2+x^2} - \sqrt{R^2+x^2}\right) = \frac{1}{2}mv^2 - \frac{q\sigma R}{2\varepsilon_0}$$

$$\Rightarrow \quad v = \sqrt{\frac{q\sigma}{m\varepsilon_0}\left(R + \sqrt{R^2+x^2} - \sqrt{4R^2+x^2}\right)}$$

$$W_{\text{electric force}} = -\Delta U = -q(V_f - V_i)$$

$$= \frac{q\sigma}{2\varepsilon_0}\left(R + \sqrt{R^2+x^2} - \sqrt{4R^2+x^2}\right)$$

$$a = \frac{-qE}{m} = \frac{-q}{m}\frac{\sigma x}{2\varepsilon_0}\left(\frac{1}{\sqrt{R^2+x^2}} - \frac{1}{\sqrt{4R^2+x^2}}\right)$$

$$\simeq -\frac{q\sigma x}{4\varepsilon_0 mR} \text{ (since } x << R)$$

$$\Rightarrow \omega = \sqrt{\frac{q\sigma}{4\varepsilon_0 mR}}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{q\sigma}{4\varepsilon_0 mR}}$$

123. The block will start moving
When $F > \mu mg$ i.e. $F > 10$ N
$\Rightarrow t = 1$ s
The velocity of block is maximum when $F - \mu mg = 0$
i.e. at $t = 5$s
Net impulse = change in momentum

$$\Rightarrow \frac{1}{2}6\times30 - 2\frac{1}{2}10\times1 - 10\times4 = m v_{max}$$

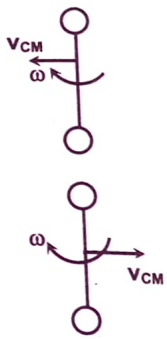
$$\Rightarrow v_{max} = 20\,m/s$$

From $t = 1$s to $t = 6$s

$$\frac{1}{2}\times30\times6 - 5 - 10\times5 = 2v$$

$$\Rightarrow v = \frac{35}{2}\,m/s$$

124. Let velocity of centre of mass of each dumbbell be V_{CM} and angular velocity ω as shown.



$$\text{Since } \frac{v_2 - v_1}{u_1 - u_2} = 1 \Rightarrow \frac{\left(V_{CM} + \omega\frac{\ell}{2}\right) + \left(\frac{\omega\ell}{2} + V_{CM}\right)}{u - (-u)} = 1$$

$$\Rightarrow 2v_{CM} + \omega\ell = 2u \qquad \ldots(i)$$

Let the dumbbells exert average impulsive force N on each other during collision.

$$N\Delta t = 2m(V_{CM} + u)$$

$$N\Delta t\frac{\ell}{2} = \frac{m\ell^2}{2}\omega$$

$$\Rightarrow \ell = \frac{\omega\ell^2}{2(V_{CM} + u)} \qquad \ldots(ii)$$

From equation (1) & (ii)

$$\omega = \frac{2u}{\ell}$$

125. Time period of oscillation $T = 2\pi\sqrt{\frac{\text{reduced mass}}{k}} = 2\pi\sqrt{\frac{2m}{3k}}$

Required time $t = \frac{T}{4} = \frac{\pi}{2}\sqrt{\frac{(m \times 2m)}{3mk}} = \frac{\pi}{2}\sqrt{\frac{2m}{3k}}$

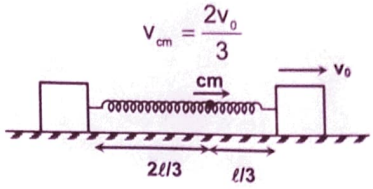
$V_{cm} = \frac{2mv_0}{3m} = \frac{2v_0}{3}$

Initially at mean position the block of mass 2m is moving with velocity $\frac{v_0}{3}$ towards right w.r.t centre of mass. Hence its minimum speed is $\left|\frac{v_0}{3} - \frac{2v_0}{3}\right| = \frac{v_0}{3}$

In centre of mass frame for block of mass 2m,

$\frac{1}{2}(2m)\frac{v_0^2}{9} = \frac{1}{2}(3k)A^2$

$\Rightarrow \quad A = \sqrt{\frac{2mv_0^2}{27k}}$



126. Let $M = G^a\eta^bB^c$

$\Rightarrow \quad M = \left[M^{-1}L^3T^{-2}\right]^a\left[ML^{-1}T^{-1}\right]^b\left[ML^{-1}T^{-2}\right]^c$

$\Rightarrow \quad a = \frac{1}{2},\ b = 4,\ c = -\frac{5}{2}$

Further Let $L = G^a\eta^bB^c$

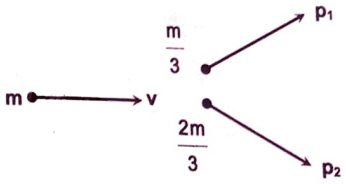
$= \left[M^{-1}L^3T^{-2}\right]^a\left[ML^{-1}T^{-1}\right]^b\left[ML^{-1}T^{-2}\right]^c$

$\Rightarrow \quad a = \frac{1}{2},\ b = 2,\ c = -\frac{3}{2}$

127. $p_1^2 + p_2^2 + 2p_1p_2\cos 60° = p^2 \quad$ (where $p = mv$)

$\Rightarrow \quad p_1^2 + p_2^2 + p_1p_2 = p^2 \qquad \ldots(i)$

$\frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1} = \frac{2}{1} \Rightarrow p_1 = 2p_2 \qquad \ldots(ii)$



From equation (I) & (II)

$p_2 = \frac{p}{\sqrt{7}} = \frac{mv}{\sqrt{7}}$ and $p_1 = \frac{2p}{\sqrt{7}} = \frac{2mv}{\sqrt{7}}$

$v_1 = \frac{p_1}{m/3} = \frac{6v}{\sqrt{7}}$ and $v_2 = \frac{p_2}{2m/3} = \frac{3v}{2\sqrt{7}}$

Increase in kinetic energy of the system due to explosion

$$= \frac{1}{2}\frac{m}{3}\left(\frac{6v}{\sqrt{7}}\right)^2 + \frac{1}{2}\left(\frac{2m}{3}\right)\left(\frac{3v}{2\sqrt{7}}\right)^2 - \frac{1}{2}mv^2 = \frac{13}{28}mv^2$$

128. Magnetic flux in a circular loop of radius r

$$= \int_0^r B(2\pi r)dr = \int_0^r \frac{Kt}{r}2\pi r dr = 2\pi Ktr$$

$\oint \vec{E}.\vec{dr} = \frac{-d\phi}{dt}$

$\Rightarrow \quad E(2\pi r) = -2\pi Kr$

$\Rightarrow \quad E = -K$

Velocity of the bead $V = at = \frac{qE}{m}t = \frac{Kqt}{m}$

Further

$$qvB + N = \frac{mv^2}{R}$$

$$\Rightarrow \quad \frac{K^2q^2t^2}{mR} + N = \frac{m}{R}\frac{K^2q^2t^2}{m^2} \Rightarrow N = 0$$

132.

129. $f_w = \frac{v - v_w}{v} f$

$$f_1 = \frac{v}{v + v_w} f_w = \frac{v}{v + v_w}\left(\frac{v - v_w}{v} f\right)$$

$$f_1 = \frac{v - v_w}{v + v_w} f$$

$$\lambda_1 = \frac{v}{f_1} = \frac{v + v_w}{(v - v_w)}\lambda = \frac{1 + \frac{v_w}{v}}{1 - \frac{v_w}{v}}\lambda = \left(1 + \frac{2v_w}{v}\right)\lambda$$

$$\% \text{ change in wavelength} = \frac{\lambda_1 - \lambda}{\lambda} \times 100 = \frac{2v_w}{v} \times 100 = \frac{200}{1000/3} \simeq 0.6\%$$

Frequency of the echo heard by the observer $= f_1 = \frac{v - v_w}{v + v_w} f \simeq \left(1 - \frac{2v_w}{v}\right) \times 1000$

$$= \left(1 - \frac{2}{1000/3}\right) \times 1000 = 994 \text{ Hz}$$

Beat frequency detected by the receiver $= f - f_1 \simeq 6\,\text{Hz}$

130. When the switch S was open, charge on the capacitors, $q_1 = C_{eq} v = \frac{4}{3} \times 3 = 4\,\mu C$. When the switch S is closed then in steady state charge on capacitor of capacitance $2\mu F$ will be zero and charge on 4 μF capacitor $q_2 = 4 \times 3 = 12\,\mu C$. After the switch S was closed. 133.

Energy supplied by the battery $= \varepsilon \Delta q = 3 \times 8 = 24\ \mu J$

Change in energy stored in capacitors $= \frac{1}{2} 4 \times 9 - \frac{1}{2}\frac{4}{3} \times 9 = 12\ \mu J$

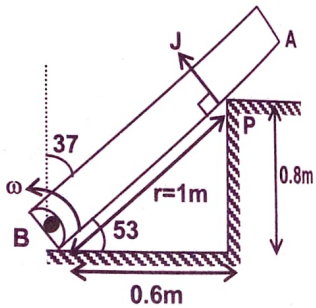
Hence, Heat produced = 24 – 12 = 12 μJ

131. Using conservation of energy

$$mg\frac{\ell}{2}(1 - \cos 37) = \frac{1}{2}\frac{m\ell^2}{3}\omega_0^2$$

$$mg\frac{\ell}{2} \times \frac{1}{5} = \frac{1}{2}\frac{m\ell^2}{3}\omega_0^2$$

$$\omega_0 = \sqrt{\frac{3g}{5\ell}} = \sqrt{\frac{3 \times 10}{5 \times 1.5}} = 2 \text{rad/s}$$



34.

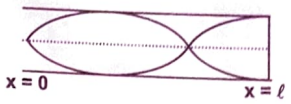
Now, $e = \frac{\omega r}{\omega_0 r} \Rightarrow \omega = e\omega_0 = 0.5 \times 2 = 1$ rad/s

Now, $Jr = \frac{m\ell^2}{3}(\omega + \omega_0)$

$$J \times 1 = \frac{4}{3} \times (1.5)^2 \times 3$$

$\therefore$ $J = 9$ N-S.

132. In first overtone,



$$f = \frac{3v}{4\ell}$$

$$\therefore \ \ell = \frac{3v}{4f}$$

$$= \frac{3 \times 330}{4 \times 220} = \frac{9}{8} m$$

$$\ell = \frac{9}{8} m$$

The amplitude of pressure variation inside the pipe is

$a = |\Delta P_0 \sin kx|$

At $x = \frac{\ell}{2}$

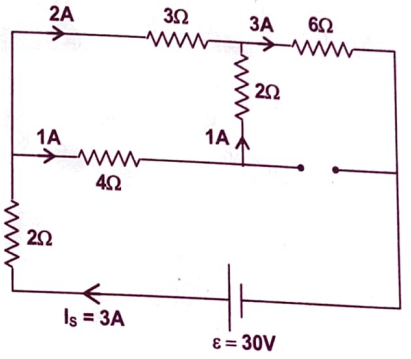
$$a = \left|\Delta P_0 \sin\left(\frac{2\pi}{\lambda} \times \frac{\ell}{2}\right)\right|$$

In first overtone, $\frac{3\lambda}{4} = \ell$

$$a = \left|\Delta P_0 \sin\left(\frac{2\pi}{\lambda} \times \frac{3\lambda}{8}\right)\right| = \left|\Delta P_0 \sin\left(\frac{3\pi}{4}\right)\right| = \frac{\Delta P_0}{\sqrt{2}}$$

$\therefore$ The maximum pressure at the middle of the pipe $= P_0 + \frac{\Delta P_0}{\sqrt{2}}$.

133. The equivalent circuit in steady state is shown



$$\text{Req.} = \left(\frac{6 \times 3}{6+3}\right) + 6 + 2 = 10\Omega.$$

The current through the battery in steady state is

$$I_s = \frac{\varepsilon}{Req} = \frac{30}{10} = 3A$$

The energy stored in the inductor in steady state is

$$U_L = \frac{1}{2} Li^2 = \frac{1}{2} \times 4 \times 10^{-3} \times (2)^2 = 8 \times 10^{-3} J = 8mJ$$

The energy stored in the capacitor in steady state is

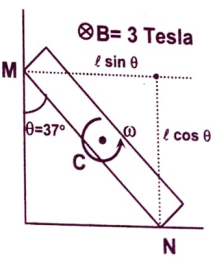
$$U_C = \frac{1}{2} \times CV^2 = \frac{1}{2} \times 5 \times 10^{-6} \times (20)^2 = 1000 \times 10^{-6} J$$

$= 1 \times 10^{-3} J = 1$ mJ

134. Using conservation of energy

$$mg\frac{\ell}{2}(1 - \cos 37) = \frac{1}{2}\frac{m\ell}{3}\omega^2$$

$$\omega = \sqrt{\frac{3g}{5\ell}} = \sqrt{\frac{3 \times 10 \times 3}{5 \times 2}} = 3 \text{ rad/s}$$



The motional emf between the ends M and N of the rod is

$$\varepsilon_{MN} = \frac{1}{2}B\omega\ell^2(\cos^2\theta - \sin^2\theta)$$

$$= \frac{1}{2}\times 3\times 3\times \frac{4}{9}\left(\frac{16}{25}-\frac{9}{25}\right) = 0.56 \text{ volt}$$

The motional emf between the mid-point C and the end N of the rod is

$$\varepsilon_{CN} = \frac{1}{2}B\omega\ell^2\left(\cos^2\theta - \frac{1}{4}\right)$$

$$= \frac{1}{2}\times 3\times 3\times \frac{4}{9}\left(\frac{16}{25}-\frac{1}{4}\right) = 0.78 \text{ volt}$$

135. For central maxima

$$(\mu_s - \mu_m)t + d\sin\theta = (\mu_s' - 1)t'$$

$$d\sin\theta = (\mu_s' - 1)t' - (\mu_s - \mu_m)t$$

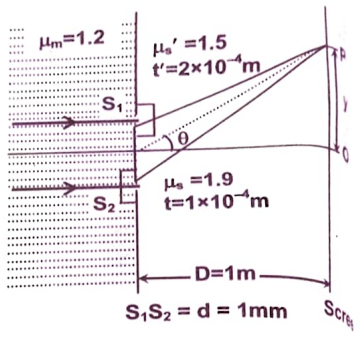
$$1\times 10^{-3}\sin\theta = (1.5-1)\times 2\times 10^{-4} - (1.9-1.2)\times 10^{-4}$$

$$1\times 10^{-3}\sin\theta = (1-0.7)\times 10^{-4}$$

$$\sin\theta = \frac{0.3\times 10^{-4}}{1\times 10^{-3}} = \frac{3}{100}$$

$$\tan\theta = \frac{3}{100}$$

$$\frac{y}{D} = \frac{3}{100} \quad \therefore \ y = 3\text{cm}$$



Phase difference between the light waves for point 'O'

$$\delta = \frac{2\pi}{\lambda}\Delta r, \ = \frac{2\pi}{5\times 10^{-7}}\times 0.3\times 10^{-4} = 2\pi\times 60$$

Maxima is formed at point 'O'.

$$I_{max} = 4I = 4\times 10^{-5}\,w/m^2$$

136. $v_y = 4$ m/s, $\frac{dy}{dt} = 4$

$$v_x = 2y, \ \frac{dx}{dt} = 2y$$

$$\therefore \ \frac{dy}{dx} = \frac{2}{y}$$

$$\Rightarrow \int_0^y y\,dy = 2\int_0^x dx \Rightarrow y^2 = 4x$$

$$\therefore \ \vec{v} = 2y\hat{i} + 4\hat{j}$$

$$\vec{a} = 8\hat{i} \ m/s^2$$

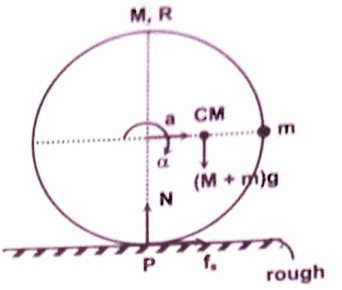
$$\therefore \text{ tangential acceleration } a_t = \vec{a}\cdot\frac{\vec{v}}{v} = \frac{16y}{\sqrt{16+4y^2}} = \frac{8y}{\sqrt{4+y^2}}$$

$$\therefore \text{ normal acceleration } a_n = \left|\vec{a}\times\frac{\vec{v}}{v}\right| = \frac{32}{\sqrt{16+4y^2}} = \frac{16}{\sqrt{4+y^2}}$$

137. Moment of inertia of the system about instantaneous axis of rotation is

$I_P = 2MR^2 + 2mR^2 = 6mR^2$

Distance of centre of mass of the system from centre of the ring is $r = \frac{mR+0}{m+M} = \frac{R}{3}$



Using conservation of energy of the system

$mgR = \frac{1}{2} I_P \omega^2$

$mgR = \frac{1}{2} 6mR^2\omega^2$

$\therefore \omega = \sqrt{\frac{g}{3R}}$

Now, $\tau_P = I_P\alpha$

$(M+m)(g+\omega^2 R)\frac{R}{3} = 6mR^2\alpha$

$3m\left(g+\frac{g}{3}\right)\frac{R}{3} = 6mR^2\alpha$

$\frac{4}{3}mgR = 6mR^2\alpha$

$\therefore \alpha = \frac{2g}{9R}$

For pure rolling motion $a = \alpha R = \frac{2g}{9}$

$\therefore$ acceleration of the centre of the ring, $a = \frac{2g}{9}$

138. $\Delta Q_{AB} = nC_P\Delta T = 2\times\frac{5R}{2}\times 3T_0 = 15RT_0 = 4500\,R$

In the process BC, $PV^{5/2} = \text{constant}$

The molar heat capacity of the gas in the process BC is

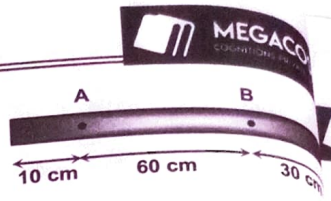
$C = C_V + \frac{R}{1-x}$

$= \frac{3R}{2} + \frac{R}{\left(1-\frac{5}{2}\right)} = \frac{3R}{2} - \frac{2R}{3} - = \frac{5R}{6}$

$C = \frac{5R}{6}$

$\therefore \Delta Q_{BC} = nC\Delta T = 2\times\frac{5R}{6}\times(-3T_0) = -5RT_0 = -1500R$

$\Delta W_{BC} = \frac{nR\Delta T}{1-x} = \frac{2R(-3T_0)}{\left(1-\frac{5}{2}\right)} = 4RT_0 = 1200\,R$



139. The velocity of the longitudinal wave in a solid rod is

$$v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{2\times10^{11}}{8\times10^{3}}} = 5000 \text{ m/s}$$

In fundamental mode of vibration

$n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3$

$= 10 : 60 : 30 = 0.5 : 3 : 1.5$

$n_1 = 0.5, n_2 = 3, n_3 = 1.5$

Total no. of loops formed, $n = 0.5 + 3 + 1.5 = 5$ loops

$$\therefore \frac{5\lambda}{2} = \ell$$

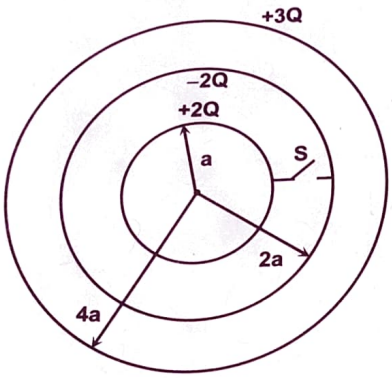
$$\Rightarrow \lambda = \frac{2\ell}{5} = \frac{2\times1}{5} = 0.4 \text{ m}$$

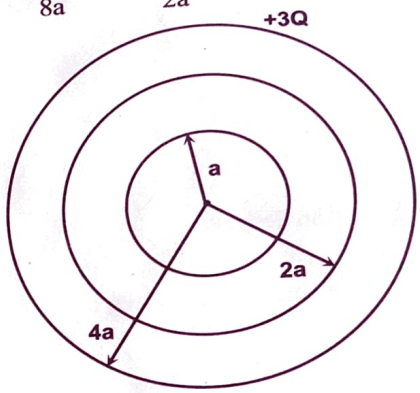
The fundamental frequency of vibration, $f = \frac{v}{\lambda} = \frac{5000}{0.4} = 12.5 \text{ kHz}$

The first overtone frequency of vibration, $f_1 = 3f = 37.5 \text{ kHz}$

140. After closing the switch 'S; the charge flown through the switch 'S' is 2Q.

Before closing the switch 'S' $U_i = \frac{k4Q^2}{2a} + \frac{k4Q^2}{4a} + \frac{k9Q^2}{8a} + \frac{k2Q(-2Q)}{2a}$





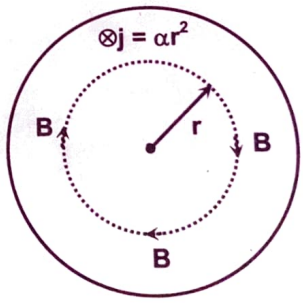
$$U_i = \frac{KQ^2}{a} + \frac{9kQ^2}{8a}$$

After closing the switch 'S', $U_f = \frac{9KQ^2}{8a}$

$\therefore$ The total heat dissipated $= U_i - U_f = \frac{kQ^2}{a} = \frac{Q^2}{4\pi\varepsilon_0 a}$

141. Using Ampere's law

when $r \le R$



$$B2\pi r = \mu_0 \int_0^r j2\pi r dr$$

$$Br = \mu_0 \int_0^r \alpha r^2 r dr$$

$$Br = \frac{\mu_0 \alpha r^4}{4}$$

$B = \frac{\mu_0 \alpha r^3}{4}$, when $r \le R$

$\therefore$ at $r = R/2$, $B = \frac{\mu_0 \alpha R^3}{32}$

When $r > R$

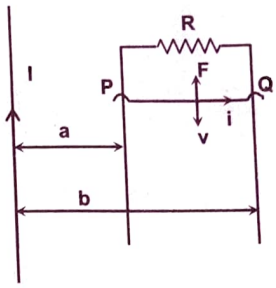
$B2\pi r = \mu_0 \int_0^R j2\pi r dr$

$Br = \mu_0 \int_0^R \alpha r^2 r dr$

$B = \frac{\mu_0 \alpha R^4}{4r}$, when $r > R$

$\therefore$ at $r = 3R$, $B = \frac{\mu_0 \alpha R^3}{12}$

142. The emf induced in the conducting rod 'PQ' is



$\varepsilon = \int_a^b \frac{\mu_0 I}{2\pi r} v dr = \frac{\mu_0 I v}{2\pi} \ell n\left(\frac{b}{a}\right)$

The induced current in the rod is

$i = \frac{\varepsilon}{R} = \frac{\mu_0 I v}{2\pi R} \ell n\left(\frac{b}{a}\right)$

The magnetic force acting on the rod is

$F = \int_a^b \frac{\mu_0 I}{2\pi r} i dr = \left[\frac{\mu_0 I}{2\pi} \ell n\left(\frac{b}{a}\right)\right] i$

$\therefore F = \left[\frac{\mu_0 I}{2\pi} \ell n\left(\frac{b}{a}\right)\right]^2 \frac{v}{R}$

When the terminal velocity is attained by the rod 'PQ'

$F = mg$

$\Rightarrow \left[\frac{\mu_0 I}{2\pi} \ell n\left(\frac{b}{a}\right)\right]^2 \frac{v}{R} = mg$

Terminal velocity attained by the rod is $v_T = \frac{mgR}{\left[\frac{\mu_0 I}{2\pi} \ell n\left(\frac{b}{a}\right)\right]^2}$

The power developed in the resistor after the terminal velocity attained is $P = mgv_T$

$= \frac{m^2 g^2 R}{\left[\frac{\mu_0 I}{2\pi} \ell n\left(\frac{b}{a}\right)\right]^2}$

144. perpendicular to OB, $v' = -v\sin(90^0 - \alpha - \beta) + g\cos\beta t$ ...(i)

Parallel to OB, $0 = v\cos(90^0 - \alpha - \beta) - g\sin\beta t$ ...(ii)

on solving $v' = \frac{v\sin\alpha}{\sin\beta}$

for $\alpha > \beta$, $v' > v$, so by energy conservation $h_2 < h_1$

145. $\vec{\tau}_1 = \vec{P}_1 \times \vec{E} = (P_1)\hat{j} \times \left(\frac{2P_2}{4\pi\varepsilon_0 d^3}\right)\hat{i} = -\frac{2P_1P_2}{4\pi\varepsilon_0 d^3}\hat{k}$

$\vec{\tau}_2 = \vec{P}_2 \times \vec{E} = (P_2)\hat{i} \times \left(-\frac{P_1}{4\pi\varepsilon_0 d^3}\right)\hat{j} = -\frac{P_1P_2}{4\pi\varepsilon_0 d^3}\hat{k}$

147. Let m be the mass of ice and q the rate by which heat is supplied, then

$q \times 1 = m(0.5)(-\theta_i)$ ... (1)

$q \times 4 = m \times 80$ ... (2)

$q \times 2 = m\ 1.\theta_f$ ... (3)

From these equation

$Q_f = 40°C$

$Q_i = -40\ °C$

149. For the process, $P = \frac{R}{aT}$

So, work done by the gas = 200 R Joule

For the process, molar heat capacity $C = \frac{3}{2}R + \frac{RT}{V}2aT = \frac{7R}{2}$

Volume expansion coefficient $\gamma = \frac{1}{V}\frac{dV}{dT} = \frac{2}{T}$

150. Since speed is constant, it means there is no induced emf $\phi = \ell(b+ vt)B$

$e = \ell Bv + \ell(b + vt)\frac{dB}{dt} = 0$

$\Rightarrow (b + vt)\frac{dB}{dt} = -Bv$

$\Rightarrow \int_{B_0}^{B}\frac{dB}{B} = -v\int_0^t \frac{dt}{b+vt}$

$\Rightarrow \ell n\ B/B_0 = -v\left[\frac{1}{v}\ell n\left(\frac{b+vt}{b}\right)\right]$

$= \ell n\left(\frac{b}{b+vt}\right)$

$\Rightarrow B = \frac{B_0 b}{b+vt}$

Second Method:

So flux does not change with time, so

$\phi(t) = \phi(t + dt)$

$\Rightarrow B\ell(b + vt) = B+dB)\ \ell\ [b + v(t + dt)]$

$\Rightarrow (b + vt)dB + Bvdt = 0$

$\Rightarrow \frac{dB}{B} = -v\frac{dt}{b+vt}$

$\Rightarrow B = \frac{B_0 b}{b+vt}$

COMPREHENSION

4. Unit vector along rod $= \hat{\ell} = \frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}$

$\hat{\ell}.\vec{v}_1 = \hat{\ell}.\vec{v}_2 \Rightarrow v_y = (6+\sqrt{3})$ m/s

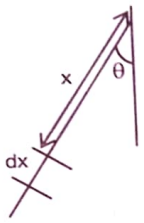
$\omega = .\frac{|\vec{v}_2 - \vec{v}_1|}{\ell} = 1$ rad/sec

13. Consider an element of the rod.

$dL = x \times m \times x \sin\theta = m\omega x^2 \sin\theta$

$\Rightarrow L = \int dL = \frac{m\omega L^2}{3}\sin\theta$

Its horizontal component $= L\cos\theta = \frac{m\omega L^2 \sin 2\theta}{6}$



14. The vertical component of the angular momentum remains constant while its horizontal component keeps changing the direction.

$dL = \frac{m\omega\ell\sin(2\theta)}{6} \times \omega dt \Rightarrow \frac{dL}{dt} = \frac{m\omega^2\ell^2}{6}\sin(2\theta)$

15. $\tau = \frac{dL}{dt}$

$Mg\frac{L}{2}\sin\theta = \frac{\mu\omega^2\ell^2}{6}\sin(2\theta)$

$\Rightarrow \cos\theta = \left(\frac{3g}{2\omega^2\ell}\right)$ or, $\theta = \cos^{-1}\left(\frac{3g}{2\omega^2\ell}\right)$.

16-18: Hint: Use impulse–momentum theorem along the rope and conservation of momentum perpendicular to the rope.

19-21. Hint: First find the point where collision occurs using equation of trajectory. Then use conservation of angular momentum. After collision energy conservation is valid.

22. $mv = \frac{4mv_1}{\sqrt{2}} + mv_2$ ...(1)

$1 = \frac{v_1 = \frac{v_2}{\sqrt{2}}}{v/\sqrt{2}}$ ...(2)

$v_1 = \frac{\sqrt{2}v}{3}$

23. $V - \frac{4v}{3} = V_2$

$V_2 = -V/3$

40. Let v_1 be the velocity of striking mass after collision & v_2 be the velocity of centre of mass of frame after collision by momentum conservation

$$mv = mv_1 + 2mv_2 \quad \ldots(1)$$

By angular momentum conservation

$$mv\frac{L}{2} = mv_1\frac{L}{2} + mL^2\omega \quad \ldots(2)$$

$$e = 1 \Rightarrow v = \left(v_2 + \frac{L}{2}\omega\right) - v_1 \quad \ldots(3)$$

from (1), (2) and (3)

$$\omega = \frac{4v}{7L};\ v_1 = -\frac{v}{7}\ \&\ v_2 = \frac{4v}{7}.$$

41-42. Use conservation of linear and angular momentum along with conservation of kinetic energy. (elastic collision)

43-45. $$PE = 6 \times -\frac{GM^2}{\sqrt{2}a}$$

$$= -3\sqrt{2}\frac{GM^2}{a}$$

$$F = 3 \times \frac{GM^2}{2a^2} \times \frac{\sqrt{2}}{\sqrt{3}} = \sqrt{\frac{3}{2}}\frac{GM^2}{a^2}$$

49. In BC and DE temperature remains constant
⇒ K.E. remains constant
⇒ Heat goes to increasing the P.E.

50. In AB, CD and EF
Temperature increases
⇒ K.E. increases

51. In gases, there is no appreciable molecular interaction.

52. Tension is maximum at the mean position.

53. (C) $2\rho Vg$, $3\pi/2\omega$

Magnitude of component of Buoyant force $= \rho va$

Where a is acceleration of beaker, $a = -\sqrt{3}\ g \sin \omega t$

Net buoyant force $= \sqrt{\rho^2v^2g^2 + \rho^2v^2a^2}$

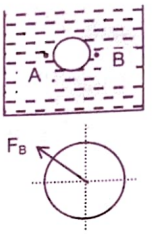
$$= \rho vg\sqrt{1 + 3\sin^2 \omega t}$$

Magnitude of maximum buoyant force $= 2\rho vg$

$\sin^2\omega t = 1$ at $t = \frac{\pi}{2\omega}, \frac{3\pi}{2\omega}$

$\therefore\ 2\rho vg,\ \frac{3\pi}{2\omega}$

54. a = acceleration of beaker $= -\sqrt{3}\, g \sin \omega t$
At $t = \pi/2\omega$, $a = -\sqrt{3}\, g$
Pressure at B > Pressure at A
Horizontal comp. of Buoyant force acts towards left.
$\therefore$ Direction of buoyant force.



58. Time period is not affected by a constant force acting along the line of SHM.

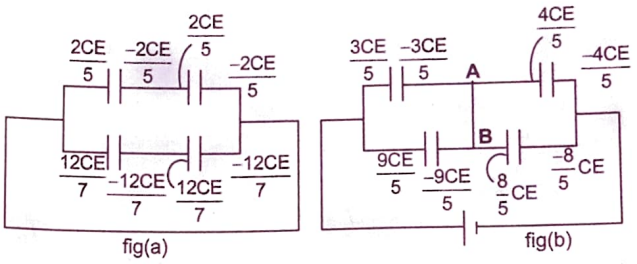
59. mg is balanced by the electrostatic force.

60. Use force balance.

61-63. Initial and final charges on capacitors are shown in figure (a) and (b) respectively charge flown through AB is $\frac{CE}{5}$ work done by battery

$= W_b = (Q_{take} - Q_{take})E$

Heat produced $= W_b - \Delta U = \frac{CE^2}{105}$ J



64. Electric field is parallel to $-\overrightarrow{AB}$.

65. $V_A - V_B = V_A - V_C \Rightarrow W_{A\to B} = W_{A\to C}$

66. $E = -\frac{dV}{dr} \Rightarrow E_p = E_Q = E_R$

67-69. At $t = 0$, capacitor will behave like a short circuit and the inductor as open circuit but as $t \to \infty$, the nature is just opposite.

70. $qv_0B = qE_0 \Rightarrow E_0 = v_0B$

71. (b – a) is the radius of the circle

72. After the fields are switched off, velocity remains constant.

NUMERICAL BASED PROBLEMS

1. $P = kv^2(v)$

$= kv^3 = 0.8 \times \left(18 \times \frac{5}{18}\right)^3 = 100$ watt

2. Required velocity of P for completing the cycle at the instant, when OP is along the direction of motion of platform is

$v_{min} = \sqrt{4 \times 10 \times 0.9} = 6$ m/s

So $T - ma = \frac{mv_{min}^2}{\ell}$

$T = 10 + \frac{(1)36}{0.9} = 10 + 40 = 50\,N$

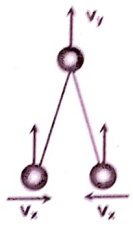
3. Make collision in such a wavy that all balls would be stopped at bottom the bowl.

So, maximum kinetic energy loss = $\frac{1}{2} \times (2 \times 5 \times 10)(3 + 4 + 5) = 600J$

4. from collision

$mv_0 = 3mv_y$

$v_y = \frac{v_0}{3}$



From COE

$\frac{1}{2}mv_0^2 = 3\frac{1}{2}mv_y^2 + 2\frac{1}{2}mv_x^2$

$\frac{1}{2}mv_x^2\left(1 - 2\frac{v_0^2 2}{9}\right) = mv_x$

$v_x = \frac{v_0}{\sqrt{3}}$

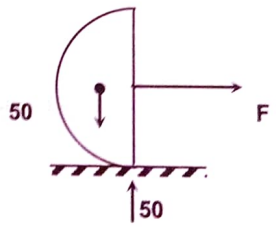
From frame of ball B,

$3T = \frac{mv_x^2}{\ell}$

$\therefore\ T = \frac{mv_x^2}{3\ell} = \frac{(3)(v_0^2)}{3 \times 3 \times 1} = 1$

5. $F = \left(\frac{10}{2}\right)(10)^2 \times \frac{3 \times 0.8}{8}$

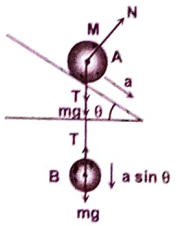
$= 5 \times 100 \times 0.3 = 150\,N$



6. $Mg \sin\theta + T \sin\theta = Ma$ ...(i)

$mg - T = ma \sin\theta$ ...(ii)

solving , we get



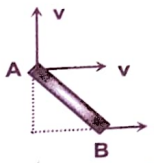
$$a = \frac{(M+m)g\sin^2\theta}{M+m\sin^2\theta}$$

7. Let the velocity of the point A have components 2V along horizontal, v_1 along the vertical.

Since the rod AB is rigid, taking the velocities along the rod AB,

$2v\cos 45° - v_1\cos 45° = v\cos 45°$

or $v_1 = v$



In the frame of B

The angular velocity of the rod AB equals(in the frame in which end B is at rest)

$$\omega = \frac{\sqrt{2}v}{\sqrt{2}R} = \frac{v}{R}$$

The velocity of the C.M. of AB

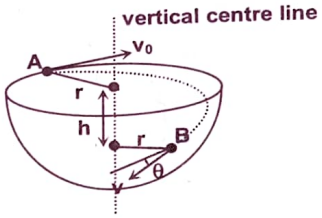
Where $\hat{i}$, $\hat{j}$ are the unit vectors along the horizontal and vertical (upward) direction respectively.

The K.E. of the rod AB

$$= \frac{1}{2}Mv_{CM}^2 + \frac{1}{2}I_{CM}\omega^2 = \frac{1}{2}M\left(\frac{9v^2}{4}+\frac{v^2}{4}\right) + \frac{1}{2}\frac{1}{12}M(\sqrt{2}R)^2\left(\frac{v}{R}\right)^2 = \frac{4}{3}Mv^2$$

8. Angular momentum of the particle is conserved about the vertical centre line

$mv_0r_0 = mvr\cos\theta$



where conservation of mechanical energy gives,

$$\frac{1}{2}mv_0^2 + mgh = \frac{1}{2}mv^2$$

$r^2 = r_0^2 - h^2$

$$\cos\theta = \frac{1}{\sqrt{1+\frac{2gh}{v_0^2}}\sqrt{1-\frac{h^2}{r_0^2}}}$$

9. Apply conservation of angular momentum about O′ (below O lying on ground)

$$\frac{ML^2}{12}\omega_0 = \left(\frac{ML^2}{6}+\frac{ML^2}{12}\right)\omega$$

10. Let the velocity of the end of the falling chain be v, when a length x has already fallen. Taking mass per unit length of the chain as λ, we can write,

$$\lambda xg - \frac{(\lambda vdt)v}{dt} = \lambda x\cdot v\frac{dv}{dx}$$

$$gx - v^2 = x\frac{vdv}{dx}$$

or, $$v\frac{dv}{dx} + \frac{v^2}{x} = g$$

This equation may be solved by using Integrating factor. The solution is:

$v^2 = \frac{2gx}{3} + \frac{C}{x^2}$, where C is an arbitrary constant.

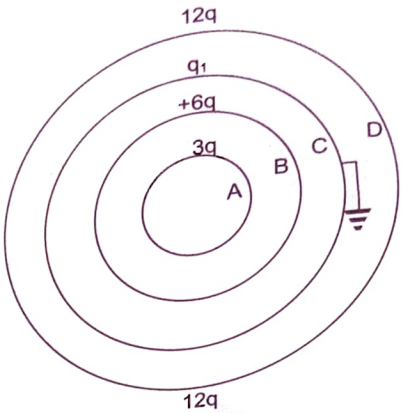
At $t = 0$, $x = 0$ and $v = 0$

$\therefore C = 0$ and $v^2 = \frac{2gx}{3}$

11. $\frac{1}{4\pi\varepsilon_0}\left[\frac{3q}{3a}+\frac{6q}{3a}+\frac{q_1}{3a}+\frac{12q}{4a}\right]=0$

$\Rightarrow \quad q_1 = -18q$

$|\Delta q_1| = 9q$



$\frac{1}{4\pi\varepsilon_0}\left[\frac{3q}{3a}+\frac{q_2}{3a}+\frac{q_3}{3a}+\frac{12q}{4a}\right]=0$

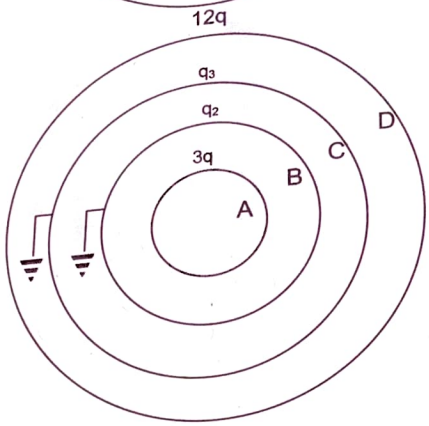
$\frac{1}{4\pi\varepsilon_0}\left[\frac{3q}{2a}+\frac{q_2}{2a}+\frac{q_3}{3a}+\frac{12q}{4a}\right]=0$

Solving

$q_2 = -3q$

$|\Delta q_2| = 9q$

$\frac{|\Delta q_2|}{|\Delta q_1|} = 1$



12. Let u be the velocity of the projectile w.r.t. the trolley.

Case I : The trolley moves forward

horizontal velocity $U_x = u\cos30° + v$

vertical velocity $U_y = u\sin30°$

Time of flight , $T = \frac{2u\sin30°}{g}$

Range $R_1 = U_xT = \frac{2u\sin30°}{g}[u\cos30° + v]$ . . . (1)

Case II : The trolley moves backward

horizontal velocity $U_x = u\cos30° - v$

vertical velocity $U_y = u\sin30°$

Range $R_2 = U_xT = \frac{2u\sin30°}{g}[u\cos30° - v]$ . . . (2)

Adding and subtracting (1) and (2),

$$R_1 + R_2 = \frac{4u^2}{g}(\sin30°\cos30°)$$

$$R_1 - R_2 = \frac{4uv\sin30°}{g}$$

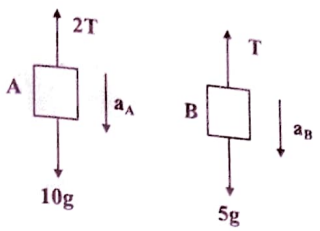
Eliminating u from the two equations

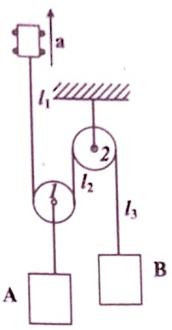
$$v^2 = \frac{g}{4\tan30°}\frac{(R_1-R_2)^2}{(R_1+R_2)}$$

here, $R_1 = 250$ m ; $R_2 = 200$ m ; $g = 10$ m/s^2

$v^2 = 25\ m^2/s^2$ or $v = 5$ m/s.

13. Drawing F.B.D. diagrams.





$10g - 2T = 10a_A$ (1)

$5g - T = 5a_B$ (2)

T from equations (1) & (2) we get

$10a_A - 10a_B = 0$

$\Rightarrow a_A = a_B$

$l_1 + l_2 + l_3 = \text{constant}$

$y_E + y_A + y_A + y_B = \text{constant}$

Differentiating twice w.r.t. time,

$a + 2a_A + a_B = 0$

$2a_A + a_B = -3$

or $3a_0 = -3$

$\Rightarrow a_B = -1 m/s^2$, $a_A = -1\ m/s^2$

$a_A = a_B = 1 m/s^2$ upwards

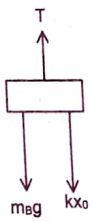
14. Block A and B will not move unless tension in the string $T \geq \mu_A g$ (48 N).

Let blocks A and B do not move and maximum elongation in the spring be Δx.

Applying conservation of mechanical energy for the block C and the spring. We get

$$\Delta x = \frac{2m_c g}{k} = 2\times10^{-2} m$$

In this situation F.B.D. of the block B is



For blocks to be in equilibrium $T = kx_0 + mg =$ 40 N.

$\because T_{max} = 40$ N (< 48 N) Hence blocks A and B would not move.

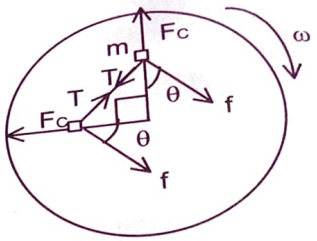
Therefore maximum distance moved by the block C is $\Delta x_0 = \frac{2m_c g}{k} = 2 \times 10^{-2}$ m. = 2 cm

15. For small values of ω friction will be directed radially inwards as the tension in the string is zero. The string will develop tension only if the centrifugal force F_C exceeds the limiting friction f_e i.e. when

$m\omega^2 r > \mu mg$ ($f_e = \mu mg$)

or $\omega = \sqrt{\frac{\mu g}{r}}$

in this case direction of friction will be as shown in the figure.



For equilibrium

$F_c = T\cos 45^0 + f_e \cos\theta$

And $T \sin 45^0 = f_e \sin\theta$

Eliminating T, we get

$F_c = f_e(\sin\theta + \cos\theta)$

i.e. $m\omega^2 r = \mu mg\,[\sqrt{2}\sin(\theta + 45^0)]$

or $\omega^2 = \dfrac{\sqrt{2}\mu g}{r}\sin(\theta + 45^0)$

maximum value of $\sin(\theta + 45^0)$ is 1

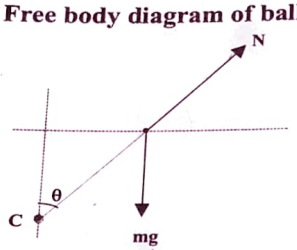
$\therefore$ Maximum value of $\omega = \sqrt{\dfrac{\sqrt{2}\mu g}{r}} = 2$

18.

16. Using conservation of energy principle, if v be the speed of either ball when its radius vector makes angle θ with vertically upward direction.

$$mgR[1 - \cos\theta] = \frac{1}{2}mv^2 \Rightarrow \frac{mv^2}{R} = 2mg[1 - \cos\theta]$$

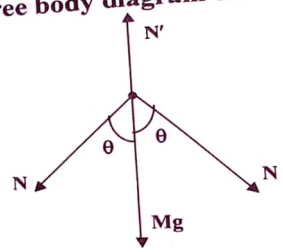
Free body diagram of ball



F.B.D.(i)

N is the normal reaction by tube walls on ball B.

Free body diagram of tube



F.B.D.(ii)

N′ be the normal reaction force by ground on tube.

From F.B.D.(i)

$$N = mg\cos\theta - \frac{mv^2}{R}$$

$$= mg\cos\theta - 2mg\,[1 - \cos\theta]$$

From F.B.D.(ii)

$N' = 2N\cos\theta + Mg$

At the instant tube breakes its contact with ground

$N' = 0$

$\Rightarrow Mg + [mg\cos\theta - 2mg(1 - \cos\theta)]2\cos\theta = 0$

for $\theta = 60°$, we get $m/M = 2$.

17. The minimum force needed to lift the lower block is equal to its weight. During upward motion the spring will get elongated. If elongation in the spring for just lifting the block is x_0 then

$kx_0 = mg$

$$\Rightarrow x_0 = \frac{mg}{k} \qquad \ldots (i)$$

From COE

$$mg(\ell_0 + h) = mg(\ell_0 + x_0) + \frac{1}{2}kx_0^2$$

$$\Rightarrow mgh = mgx_0 + \frac{1}{2}kx_0^2 \Rightarrow mgh = \frac{(mg)^2}{k} + \frac{1}{2}\frac{m^2g^2}{k} \Rightarrow h = \frac{3mg}{2k}$$

During down ward motion, suppose maximum compression in the spring is x

From COE

$$mg(\ell_0 + h) = mg(\ell_0 - x) + \frac{1}{2}kx^2$$

$$\Rightarrow mgh = -mgx + \frac{1}{2}kx^2 \Rightarrow mg\,\frac{3mg}{2k} = -mgx + \frac{1}{2}kx^2$$

$$\Rightarrow 3(mg)^2 = -2mg\,kx + k^2x^2$$

$$\Rightarrow k^2x^2 - 2mgkx - 3(mg)^2 = 0$$

$$\Rightarrow x = \frac{2mgk \pm \sqrt{4(mgk)^2 + 12k^2(mg)^2}}{2k^2} = \frac{2mgk \pm 4mgk}{2k^2} \Rightarrow x = \frac{3mg}{k}$$

18. Let x be the downward displacement of the block from the equilibrium position.
Using constraint relation, the deformation in the springs will be $x' = 2x$
The energy of the system is

$$E = 2\left(\frac{1}{2}kx'^2\right) + \frac{1}{2}mv^2 \qquad \text{or} \quad E = k(2x)^2 + \frac{1}{2}mv^2 = 4kx^2 + \frac{1}{2}mv^2$$

differentiating w.r.t. time, we get

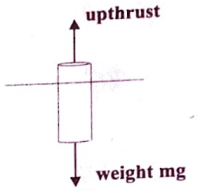
$$\frac{dE}{dt} = 8kx\frac{dx}{dt} + mv\frac{dv}{dt} = 0$$

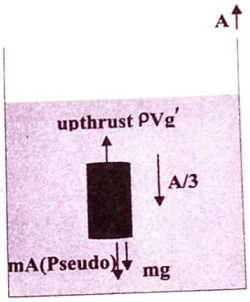
or $$\frac{d^2x}{dt^2} + \frac{8k}{m}x = 0$$

$$\therefore \quad \omega = \sqrt{\frac{8k}{m}} \text{ or } f = \frac{1}{\pi}\sqrt{\frac{2k}{m}}$$

20. Let the volume of the cyliner be V
When the cylinder is floating
upthrust = weight.

$$\rho\left(\frac{3}{4}V\right)g = mg, \quad V = \frac{4m}{3\rho}$$





Let the acceleration of the particle vessel be A(upwards)
In the reference frame of vessel, the acceleration of the cylinder is A/3.

$$\therefore \quad mg + mA - \text{upthrust} = m(A/3)$$

$$mg + mA - \rho Vg' = m(A/3)$$

[Where $g' = g + A$ = effective value of g for upthrust]

$$mg + mA - \rho V(g + A) = m(A/3)$$

$$mg - \frac{4}{3}m(g+A) = -m\frac{2}{3}A$$

$A = (-g/2)$ upwards
The acceleration of the vessel should be g/2 downards.

21. Potential difference across the capacitor is

$$V_B - 0 = \frac{q_0}{C} = \frac{16}{4} = 4 \text{ volt}$$

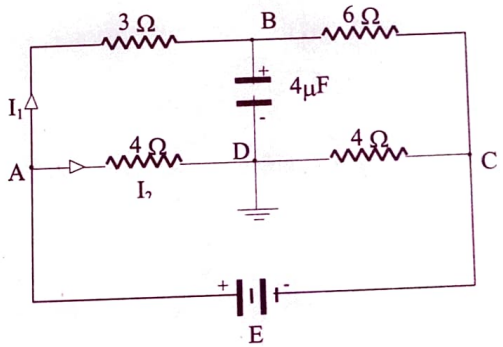
$$9I_1 = 8I_2$$

$$\Rightarrow I_2 = (9/8)I_1$$

Applying Kirchoff's voltage law along ABDA

$$3I_1 + 4 = \left(\frac{9}{8}I_1\right)4$$

or $$I_1 = \frac{8}{3}A$$



$\Rightarrow I_2 = 3A.$

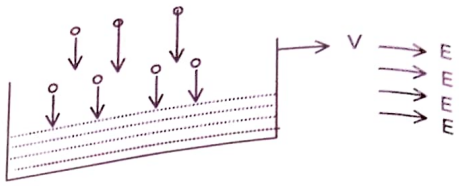
$V_A = 12\ V,\ V_B = 4\ V,$

$V_C = -12\ V$

as $V_D = 0$

E.M.F. of the battery = $V_A - V_C = 24\ V$

22.



Mass of tank at time t

$M = m_0 + nmt$

Let velocity of tank be v

$$M\frac{dv}{dt} = QE - vnm \quad \ldots (i)$$

For v_{max} ; $\frac{dv}{dt} = 0$

$$\Rightarrow v_{max} = \frac{QE}{nm}$$

from (i)

$$(m_0 + nmt)\frac{dv}{dt} = QE - vnm$$

$$\Rightarrow \int_0^v \frac{dv}{QE - vnm} = \int_0^t \frac{dt}{m_0 + nmt}$$

$$\Rightarrow -\frac{1}{nm}\left[\ln(QE - nmv)\right]_0^v = \frac{1}{nm}\left[\ln(m_0 + nmt)\right]_0^t$$

$$\Rightarrow \ln\left(\frac{QE - nmv}{QE}\right) = -\ln\left(\frac{m_0 + nmt}{m_0}\right)$$

$$\Rightarrow \frac{QE - nmv}{QE} = \frac{m_0}{m_0 + nmv}$$

$$\Rightarrow 1 - \frac{nmv}{QE} = \frac{m_0}{m_0 + nmt}$$

$$\Rightarrow \frac{nmv}{QE} = 1 - \frac{m_0}{m_0 + nmt} = \frac{nmt}{m_0 + nmt}$$

$$\Rightarrow t = \frac{m_0}{\frac{QE}{v} - nm}$$

required time when $v = v_{max}/2 = \frac{m_0}{\frac{2QE}{v_{max}} - nm} = \frac{m_0}{nm}$

23. Equivalent emf of the circuit across, AB is given by

$$E_{eq} = \frac{2}{3} \times 6 = 4\ V$$

Equivalent resistance $\Rightarrow \frac{1}{r_{eq}} = \frac{1}{3} + \frac{1}{6} \Rightarrow r_{eq} = 2\Omega$

Now we have,

Current $I = \frac{10-4}{2+R} = \frac{6}{2+R}$

$\therefore$ Power dissipated in R

$P = i^2R = \dfrac{36R}{(2+R)^2}$

for P to be maximum.

$\dfrac{dP}{dR} = 36\left[\dfrac{(2+R)^2 - R(2)(2+R)}{(2+R)^2}\right] = 0$

$R = 2\Omega$.

24. (a) $F_e + mg \geq \mu mg$

$F_e \geq \dfrac{M}{4}g - \dfrac{M}{6}g$

$\dfrac{1}{2}\dfrac{\varepsilon_0 \ell}{d}(k-1)V^2 \geq \dfrac{Mg}{12}$

$\therefore \quad V_{min} = \sqrt{\dfrac{Mg}{12} \times \dfrac{2d}{\varepsilon_0 \ell (k-1)}} = \sqrt{\dfrac{Mgd}{6\varepsilon_0 \ell (k-1)}}$.

25. When the slab is displaced by a distance x, the work done is
= change in stored energy + work done by battery
Initial capacitance of the capacitors

$C = \dfrac{\varepsilon_0 \ell b}{2d} + \dfrac{\varepsilon_0 kb\ell}{2d}$

Final capacitances

$C_{1f} = \dfrac{\varepsilon_0 b}{d}(\ell/2 + x) + \dfrac{\varepsilon_0 bk}{d}(\ell/2 - x)$

$C_{2f} = \dfrac{\varepsilon_0 b}{d}(\ell/2 - x) + \dfrac{\varepsilon_0 bk}{d}(\ell/2 + x)$

Change in energy stored $= - -\dfrac{1}{2}\varepsilon^2 [C_{1f} - C] + \dfrac{1}{2}\varepsilon^2 [C_{2f} - C]$

Work done by battery = εdq
$= \varepsilon \times \varepsilon [(C_{1f} - C) - (C_{2f} - C)]$

$\therefore \quad$ Work done $= \dfrac{\varepsilon^2 \varepsilon_0 b}{d} x(k-1)$

Net acceleration $a = \dfrac{F}{m} = \dfrac{\varepsilon_0 b}{md}(k-1)\varepsilon^2$

Motion is oscillatory but not SHM.

and, $x = \dfrac{1}{2}at^2$

$\therefore \quad t = \sqrt{\dfrac{2x}{a}}$ $\therefore$ Time period $= 4t$

$= \sqrt{\dfrac{32xmd}{\varepsilon_0 b(k-1)\varepsilon^2}}$.

26. Electrostatic energy at t = 0 is

$U(0) = \dfrac{q_0^2}{2C_1}$

Final energy $= U(\infty) = \dfrac{q_0^2}{2(C_1 + C_2)}$

$$\Delta U = U(O) - U(\infty) = \frac{q_0^2 C_2}{2C_1(C_1 + C_2)}$$

27. At 'A' Total energy of particle

$$E_{PA} = \frac{1}{2} m_p V_{PA}^2 + q\varphi + 0$$

And of isotope

$$E_{IA} = \frac{1}{2} m_i V_{IA}^2 + q\varphi + 0 \quad \Rightarrow E_{PA} = E_{iA}$$

At B, $\phi_B = \frac{\sigma a}{2\varepsilon_0}$

Energy of particle is

$$E_{PB} = \frac{1}{2} m_p \times 0 + q\varphi_B + m_p g h$$

Energy of isotope $E_{iB} = \frac{1}{2} m_i V_{IB}^2 + q\varphi_B + m_i g h$

$(m_P - m_i)gh = 10^{-23}$.
$m_P = 1.5 \times 10^{-25}$ kg.
$m_I = 0.5 \times 10^{-25}$ kg

$\therefore$ $E_{PB} = E_{iB}$

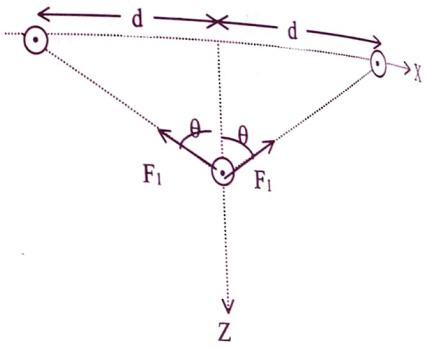
$$\frac{1}{2} m_i V_B^2 = (m_p - m_i) g h = 10^{-23}$$

$$E_{iB} = 10^{-23} + 8 \times 10^{-19} \times 0.5 \times 10^{-4} + \frac{10^{-25}}{2} \times 10 \times 10$$

$$= 5.5 \times 10^{-23} \text{J}.$$

28. Let the central wire is displaced along z axis by a small distance z and released.
Net restoring force on the central wire

$F = -2F_1 \cos\theta$ (F_1 = is the magnitude of force on central wire due to either of the other two wires)



$$= -2 \frac{\mu_0 I^2}{2\pi r} \frac{z}{r} \ell$$

$$= -\frac{\mu_0 I^2 z}{\pi(d^2 + z^2)} \ell$$

$$= -\frac{\mu_0 I^2 z}{\pi d^2} \ell \quad (\text{Since } z << d \Rightarrow z^2 + d^2 \approx d^2)$$

Acceleration of the central wire

$$a = \frac{F}{\lambda \ell} = -\frac{\mu_0 I^2}{\pi \lambda d^2} z$$

comparing this equation with equation of SHM

$a = -\omega^2 x$

$$\Rightarrow \omega = \sqrt{\frac{\mu_0}{\pi\lambda}} \frac{I}{d}$$

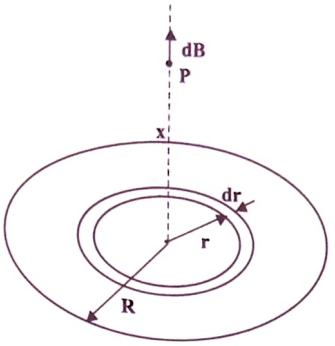
$$\Rightarrow T = \frac{2\pi}{\omega} = \frac{2\pi d}{I} \sqrt{\frac{\pi\lambda}{\mu_0}}$$

29. Consider a ring of radius r and width dr.

$dq = (2\pi r dr)\sigma$

Current due to the ring is

$$dI = \frac{\omega dq}{2\pi} = \sigma\omega r dr$$



Magnetic field due to this ring at point P is,

$$dB = \frac{\mu_0 dI\, r^2}{2\left(r^2 + x^2\right)^{3/2}}$$

or $$\int dB = \frac{\mu_0\sigma\omega}{2}\int_0^R \frac{r^3 dr}{\left(r^2 + x^2\right)^{3/2}}$$

Substituting $r^2 + x^2 = t^2$, and $2rdr = 2tdt$ and integrating we have

$$B = \frac{\mu_0\sigma\omega}{2}\left[\frac{R^2 + 2x^2}{\sqrt{R^2 + x^2}} - 2x\right].$$

30. The equivalent resistance of the circuit is

$$\frac{1}{R_{eq}} = 3 + \frac{1}{2}$$

$$\Rightarrow R_{eq} = \frac{2}{7}\Omega$$

$$\therefore \quad I = \text{total current} = \frac{E}{R_{eq}} = 7 \text{ amp.}$$

The magnetic field at the centre of the loop is

$$B = \frac{\mu_0(2)}{4a} + \frac{\mu_0(2)}{4a} - \frac{\mu_0(2)}{4a} - \frac{\mu_0(1)}{4a}$$

$$\Rightarrow B = \frac{\mu_0}{4a} \text{ (inward)}$$

The magnetic moments of the tiny ring is $M = I_0A = \pi I_0 r^2$

If the ring is rotated through angle θ, then restoring torque is

$$\tau = -MB\sin\theta = -\pi I_0 r^2 \frac{\mu_0}{4a}\theta \qquad \text{(for small angle)}$$

Using IInd law of motion

$$\tau = \frac{mr^2}{2}\frac{d^2\theta}{dt^2} = -\pi I_0 r^2 \frac{\mu_0}{4a}\theta$$

$$\therefore \quad T = 2\pi\sqrt{\frac{2ma}{\pi\mu_0 I_0}}$$

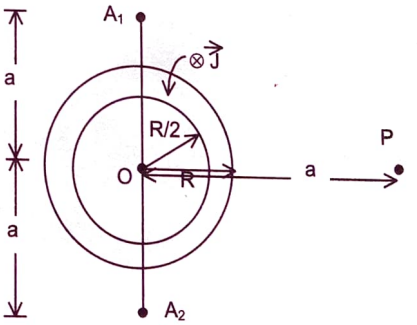
31. Current through conductor

$$I = \int_0^R 2\pi r dr J(r)$$

$$= \int_0^{R/2} 2\pi r J(r) dr + \int_{R/2}^{R} 2\pi r J(r) dr$$

$$= \int_0^{R/2} 2\pi r.0.dr + \int_{R/2}^{R} 2\pi r \rho_0 \frac{r^2}{R^2} dr$$

$$= 0 + \frac{2\pi\rho_0}{R^2}\int_{R/2}^{R} r^3\, dr$$



$$= 0 + \frac{2\pi\rho_0}{R^2}\left[\frac{r^4}{4}\right]_{R/2}^{R} = \frac{2\pi\rho_0}{R^2 4}\left(R^4 - \frac{R^4}{16}\right)$$

$$= \frac{15\pi}{32}\rho_0 R^2 .$$

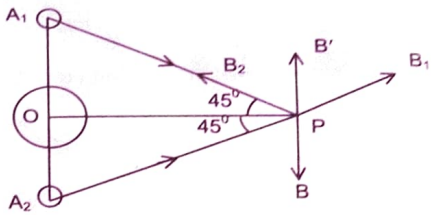
Let us consider a circle with center O and radius OP in plane $\perp$ to conductor, for all point on the circle due to symmetry B is same due to conductor. Applying Amperes law

$$\int_0^{2\pi a} B.d\ell = \mu_0 I \text{ [current inclosed]}$$

$$\Rightarrow \quad B.\,2\pi a = \mu_0 \frac{15}{32}\pi\rho_0 R^2$$

$$\Rightarrow \quad B = \frac{15}{64}\frac{\mu_0\rho_0 R^2}{a}$$

As current is in to the plane B is downward at P. Now field due to wire A_1 and A_2 must cancel B. That is possible when current in the wires is out of the plane.



Field due to A_1 $|\vec{B}_1| = \frac{\mu_0}{4\pi}\frac{2I}{a\sqrt{2}}$

Due to A_2 $|\vec{B}_2| = \frac{\mu_0}{4\pi}\frac{2I}{a\sqrt{2}}$

Resultant of B_1 and B_2

$$\vec{B}' = \vec{B}_1 + \vec{B}_2 = 2\frac{\mu_0}{4\pi}\frac{2I_0}{a\sqrt{2}}\cdot\frac{1}{\sqrt{2}} = \frac{2\mu_0 I_0}{4\pi a} \quad \text{opposite to } \vec{B}$$

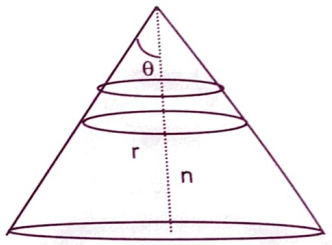
Net field at P

$$\vec{B} + \vec{B}' = 0$$

$$= \frac{\mu_0}{4\pi}\frac{2I_0}{a} - \frac{15}{64}\mu_0\frac{\rho_0 R^2}{a} = 0$$

$$I_0 = \frac{15\pi\rho_0 R^2}{32} .$$

32. A ring of radius r with charge q rotating with angular velocity ω is equivalent to a dipole of dipole moment



$$= NIA = qfA = q\frac{\omega}{2\pi}A$$

dipole moment of differential ring

$$dp = \frac{Q}{\pi R\ell}2\pi r\frac{dx}{\cos\theta}\cdot\frac{\omega}{2\pi}\pi r^2$$

$$\frac{R}{r} = \frac{h}{h-x}$$

$$\Rightarrow \quad dp = \frac{Q\cos\theta}{\pi R.h}\frac{dx}{\cos\theta}\omega\pi\left(\frac{h-x}{n}R\right)^3$$

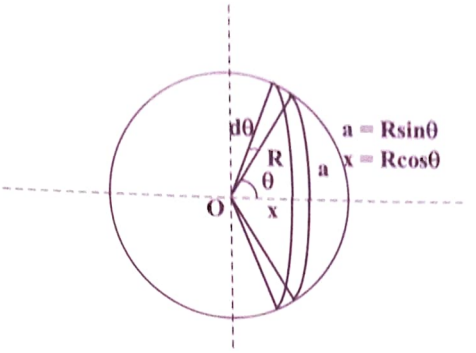
$$\Rightarrow \quad P = \frac{Q\omega.R^2}{h^4}\int_0^h (h-x)^3 dx = \frac{Q\omega R^2}{h^4}\cdot\left|\frac{(h-x)^4}{4}\right|_h^0$$

$$= \frac{\omega Q R^2}{h^4}\frac{h^4}{4} = \frac{Q\omega R^2}{4} = \frac{Q\omega}{4}h^2\tan^2\theta$$

33. Consider a circular ring of radius 'a' at a distance x from the centre O such that

$a = R\sin\theta$

$x = R\cos\theta$



The number of turns dN in the length $Rd\theta$ is

$$dN = \frac{N}{\pi R}(Rd\theta) = \frac{N}{\pi}d\theta$$

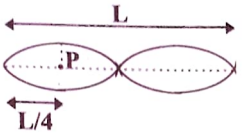
Magnetic induction at the centre of the sphere due to the ring is $dB = \frac{\mu_0 N}{\pi}\frac{d\theta\, I x^2}{2(a^2+x^2)^{3/2}}dx = \frac{\mu_0}{2\pi}\frac{NI}{R}\sin^2\theta d\theta$

On integrating,

$$B = \frac{\mu_0 NI}{2\pi R}\int_0^{\pi}\sin^2\theta d\theta$$

$$B = \frac{\mu_0 NI}{2\pi R}\left[\frac{\pi}{2}\right] \text{ or } B = \frac{\mu_0 NI}{4R}$$

34. The figure rerpesents the mode of vibration for which frequency is 100 Hz .



Here, $L = \lambda$

In the next higher mode with P again an anti-node , for every L/4 length, there will be one extra loop. Thus for the total string, total number of loops = 6

$\Rightarrow$ Desired frequency = (6/2)100 Hz = 300Hz

35. Thermal stress in the wire = $\alpha Y\Delta T$

Tension in the wire = $\alpha Y\Delta T\,\pi r^2$

$$n = \frac{1}{2L}\sqrt{\frac{T}{\pi r^2 D}} = \frac{1}{2L}\sqrt{\frac{\alpha Y\Delta T.\pi r^2}{\pi r^2 D}} = \frac{1}{2L}\sqrt{\frac{\alpha Y\Delta T}{D}}$$

Putting values $n = 600$ Hz.

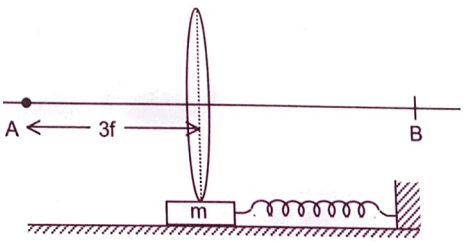
In the fundamental mode as the wire is plucked in the middle.

36. $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ ... (i)

$v-u=x$;

$v = u+x$ putting in (i)

$$\frac{1}{u+x}-\frac{1}{u}=\frac{1}{f}$$



$$\Rightarrow \frac{1}{u+x}=\frac{1}{u}+\frac{1}{f}$$

$$= \frac{u+f}{uf}$$

$$u+x = \frac{uf}{u+f}$$

$$x = \frac{uf}{u+f}-u = u\left[\frac{f}{u+f}-1\right]$$

$$x = \frac{-u^2}{u+f}$$

differentiating w.r.t. x,

$$\therefore \quad \delta x = \left[\frac{-2u}{u+f} + \frac{u^2}{(u+f)^2}\right]\delta u$$

$$= \frac{-2u(u+f)+u^2}{(u+f)^2}\delta u$$

$$= -\frac{u^2+2uf}{(u+f)^2}\delta u$$

$$\delta x = \frac{(u+f)^2+f^2}{(u+f)^2} = \left[\left(\frac{f}{u+f}\right)^2 - 1\right]\delta u$$

$$\delta u = a << |u| = |3f|$$

$$\delta x = \left[\left(\frac{f}{-3f+f}\right)^2 - 1\right]a = -\frac{3}{4}a$$

$$\text{Amplitude} = |\delta x| = \frac{3}{4}a$$

37. Because of the shielding of the nuclear charge by the negative charge of the inner core electrons the energy of an electron in n^{th} level of a multi-electron atom is given by

$$E_n = -\frac{13.6}{n^2}Z_{eff}^2 \text{ eV}$$

For L_α line an electron drops from M shell to L shell

$$\frac{1}{\lambda} = R(Z-7.4)^2\left(\frac{1}{2^2}-\frac{1}{3^2}\right) = R(Z-7.4)^2\left(\frac{5}{36}\right)$$

$$\frac{1}{\lambda} = 109737 \times (47-7.4)^2 \times \frac{5}{36} \text{ cm}^{-1}$$

$$\lambda = \frac{1}{2.39\times10^7}\text{cm}^{-1} = \frac{10^8}{2.39\times10^7} = 4.2\ A^0$$

38. The energy of the He^+ ion in the nth state $= \frac{13.6Z^2}{n^2}$ eV

Here $Z = Z$

$$\therefore \quad E_n = -\frac{54.4}{n^2}\text{eV}$$

The energy in the ground state = -54.4 eV

$$\therefore \quad 54.4\left(1-\frac{1}{n^2}\right) = \frac{hc}{1.6\times10^{-19}}\left(\frac{1}{\lambda_1}+\frac{1}{\lambda_2}\right) = 52.35 \text{ eV}.$$

$$\therefore \quad 1-\frac{1}{n^2} = \frac{52.35}{54.4}$$

$$\Rightarrow \quad n = 5$$

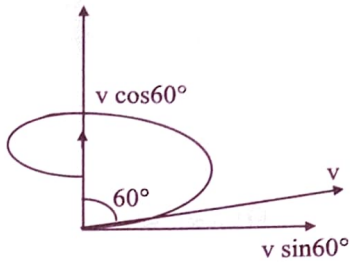
20. Pitch of the helix is given by

$y = (v\cos\theta).T = v(\cos 60)\ T = vT/2$

Where time period of circular motion

$T = \dfrac{2\pi m}{eB}$

Pitch $y = \left(\dfrac{v2\pi m}{Be}\right) v\cos 60^\circ = (\pi mv/Be)$



$\therefore$ KE $= \dfrac{1}{2}mv^2 = \dfrac{1}{2}m\left(\dfrac{Bey}{\pi m}\right)^2$

Using Einstein's equation

$$\frac{1}{2}mv^2 = h\nu - \phi$$

$$\phi = h\nu - \frac{1}{2}m\left(\frac{Bey}{\pi m}\right)^2 = \left[4.9 - \frac{1}{2}\left(\frac{m}{e}\right)\left(\frac{Bey}{\pi m}\right)^2\right] \text{electron} - \text{volts}$$

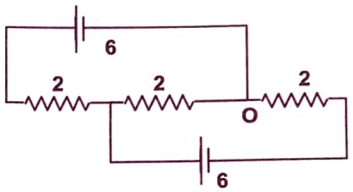
$$= \left[4.9 - \frac{1}{2}\frac{B^2y^2}{4\pi^2}\left(\frac{e}{m}\right)\right] \text{electron} - \text{volt} = (4.9 - 0.4)\text{eV} = 4.5\ \text{eV}.$$

21. $|\varepsilon| = |\vec{v}\cdot(\vec{B}\times\vec{\ell})| = (2\hat{i})\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & 1 & -2\\ 0 & 3/5 & 4/5\end{vmatrix} = 4$

22. $\dfrac{x+6-6}{2} + \dfrac{x+6-0}{2} + \dfrac{x-0}{2} = 0$

$\Rightarrow x = -2$

$\therefore\ i = \dfrac{x+6-0}{2} = \dfrac{4}{2} = 2$

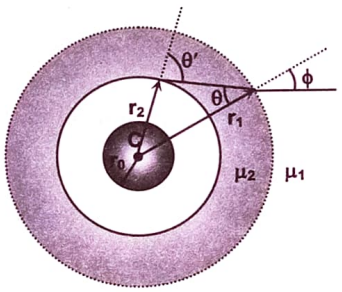


23. $\mu_1 \sin\phi = \mu_2 \sin\theta$

$\dfrac{r_1}{\sin\theta'} = \dfrac{r_2}{\sin\theta}$

$\sin\phi = \dfrac{r_2}{r_1}\sin\theta'$

$\mu_1 \sin\phi = \mu_2 \dfrac{r_2}{r_1}\sin\theta'$



$r_1\mu_1 \sin\phi = r_2\mu_2 \sin\theta'$

$r_1\mu_0 \dfrac{r_1}{r_0}\sin 30^\circ = \mu_0 \dfrac{r_{min}^2}{r_0}\sin 90^\circ$ (At minimum separation)

$r_{min} = r_1\sqrt{\sin 30^\circ} = 8$ cm

24. In fluid frame ball is moving with constant velocity (terminal) and resultant of all forces acting on it zero.

(A) Buoyant force (B) viscous force (C) mg

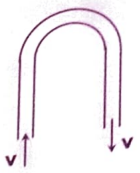
$$v_t = \frac{2}{9}\frac{(\rho - \rho_0)gr^2}{\eta} = \frac{2}{9}\times\frac{900\times 10}{0.1}\times 25\times 10^{-4} = 50\ \text{m/s}$$

25. $2\rho A v^2 = 2F$

$$\sigma = \frac{F}{3\pi r^2} = \frac{\rho A}{3\pi}$$

$$\sigma = \frac{\rho v^2}{3}$$

$v = 30$ m/s

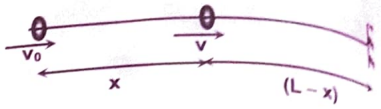


26. $v\frac{dv}{dx} = -k(L-x)$, where k is a constant.

$$\int_{v_0}^{v_0/2} v\,dv = -k\int_0^L (L-x)\,dx$$

$$\Rightarrow \frac{3}{4}v_0^2 = KL^2$$

So initial retardation $= KL = \frac{3}{4}\frac{v_0^2}{L} = \frac{3}{4}\times 25 = 18.75\ \text{m/s}^2$



27. $\Delta\ell_T = \alpha\ell\Delta T$ Extension ...(i)

$$\frac{mg}{S} = \frac{F}{S} = Y\frac{\Delta\ell_e}{\ell}$$

$$\Rightarrow \Delta\ell_e = \frac{mg\ell}{SY} \Rightarrow \text{compression} \quad ...(ii)$$

If there is no change in length, it means extension due to temperature raise must be equal to the elastic compression due to weight. So

$$\alpha\ell\Delta T = \frac{mg\ell}{SY} \Rightarrow m = \frac{\alpha SY\Delta T}{g} = \frac{1.1\times10^{-5}\times10\times10^{-4}\times10\times2.0\times10^{11}}{10} = 2.2\times10^3\,\text{kg}$$

28. $T = \frac{2u\sin\theta}{g} = 4$ s

First collision occur after 5/20 sec

So, $\Delta t = 4 - \frac{5}{20} = 3.75$ s

$S_1 = 3.75 \times 10 = 37.5$ m

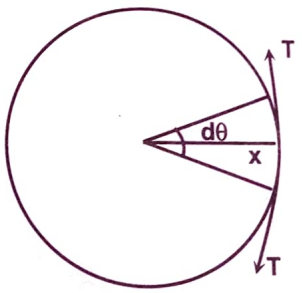
When ball collide with wall B, then total time elapse

$t = 0.25 + 1.5 = 1.75$ s

$\Delta t = 4 - 1.75 = 2.25$ s

$S = 2.25 \times 2.5 = 5.625$ m

29. Let us suppose that mass per unit length of ring initially is λ; In given condition $0 - 2T\sin\frac{d\theta}{2} = \lambda R d\theta\, a$.



$$\Rightarrow a = \frac{-T}{\lambda R} \;\&\; T = \frac{YA}{R}(x-R)$$

$$\Rightarrow a = \frac{-YA}{\lambda R^2}x + \frac{YA}{\lambda R}$$

$$\Rightarrow t = \frac{1}{4}\,2\pi\sqrt{\frac{R^2\lambda}{AY}}$$

$= 1.25$ milisec

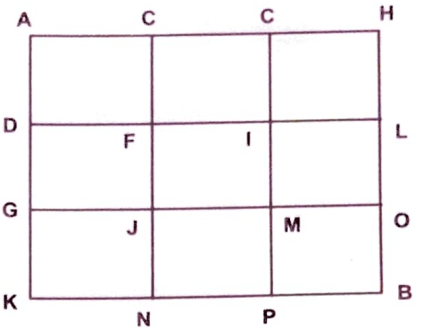
30. $A\omega^2 > g$

$\Rightarrow A > \frac{g}{\omega^2}$

$= \frac{g\lambda^2\mu}{4\pi^2 F}$

$= 62.5\ \mu m;$

31. By symmetry of network H; I; J and K are equipotential points.
Besides it potential of D and C; G and E are also same.



32. $v = v_0 \frac{z^2}{n^3} \Rightarrow \frac{z^2}{n^3} = 2$ ...(i)

$E = E_0 \frac{z^2}{n^2} \Rightarrow \frac{z^2}{n^2} = 4$

Solving (i) and (ii), $n = 2$ and $z = 4$

$L = mvr = \frac{nh}{2\pi}$, $\Delta L = \tau\Delta t = \Delta n \frac{h}{2\pi}$

$\tau = \frac{\Delta n}{\Delta t}\frac{h}{2\pi} = 1.5\times10^{-26}\ Nm$

33. Zero error $= -(50-47)\times0.01$ mm

$= -0.003$ cm

Measured diameter $= 1.10 + 0.001\times12$

$= 1.112$ cm

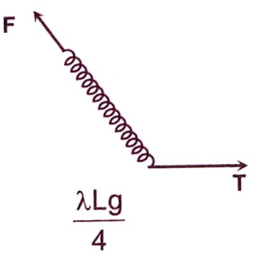
Actual diameter = Measured diameter – Zero Error

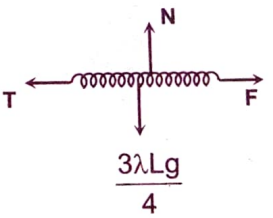
$= 1.112 - (-0.003) = 1.115$ cm $= 11.15$ mm

34. $F\cos 37° = \frac{\lambda L}{4}g$

$F \sin 37° = T = f \le \mu N \qquad \left(N = \frac{3\lambda Lg}{4}\right)$

$\Rightarrow \mu \ge 1/4, \Rightarrow \mu_{min} = 1/4$





35. The frequency of vibration of air column in the pipe is

$f_1 = \frac{3v}{4\ell} = \frac{3\times330}{4\times0.5} = 495$ Hz

The frequency of vibration of string is

$f_2 = f_1 - 5 = 495 - 5 = 490$ Hz

Now,

$$f_2 = \frac{4v}{2\ell}$$

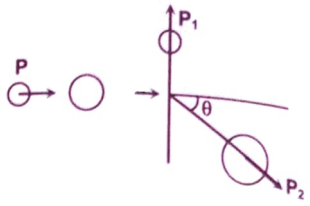
$$490 = \frac{4}{2\ell}\sqrt{\frac{F}{\mu}}$$

$$490 = \frac{4}{2\times0.4}\sqrt{\frac{F\times0.4}{2\times10^{-3}}}$$

$$490 = 50\sqrt{2F}$$

$\therefore$ F = 48.02 Newton

36. $$E_{th} = \left(1+\frac{m}{M}\right)|Q|$$



$$|Q| = \left(\frac{M}{M+m}\right)E_{th} = \frac{7}{11}\times4.4 = 2.8 \text{ Mev}$$

Using conservation of momentum

$P_2\cos\theta = P$

$P_2\sin\theta = P_1$

Squaring and adding, we get

$P_2^2 = P_1^2 + P^2$

$2m_2k_2 = 2m_1k_1 + 2mk$

$10k_2 = k_1 + 4k$

$10k_2 - k_1 = 4 \times 9.8$

$\therefore$ $10k_2 - k_1 = 39.2$ ... (1)

Also, $k_2 + k_1 = k - |Q|$

$K_2 + k_1 = 9.8 - 2.8$

$K_2 + k_1 = 7$ ... (2)

Solving (1) and (2), we get

$k_2 = 4.20$ Mev and $k_1 = 2.80$ Mev

37. The pitch of screw gauge is, P = 1mm

$$\text{Least count} = \frac{P}{N} = \frac{1}{100} = 0.01 \text{ mm}$$

The diameter of the wire is

$d = 2\text{mm} + 45 \times \text{L.C.} + 5 \times \text{L.C.}$

$= 2\text{mm} + 45 \times 0.01 \text{ mm} + 5 \times 0.01 \text{ mm}$

$= 2.50$ mm

$d = 0.250$ cm

The length of the wire is $\ell = 4.48$ cm

$\therefore$ curved surface area of the wire is

$S = \pi d \ell$

$$= \frac{22}{7}\times0.250\times4.48$$

$= 3.52 \text{ cm}^2$

38. Initially V800 = V'1000

$\Rightarrow$ V' = 0.8 V (V' is volume inside water)

$\Rightarrow$ H' = 0.8 H = 0.8 × 50 cm = 40 cm

If we displace the block by dy inside water then

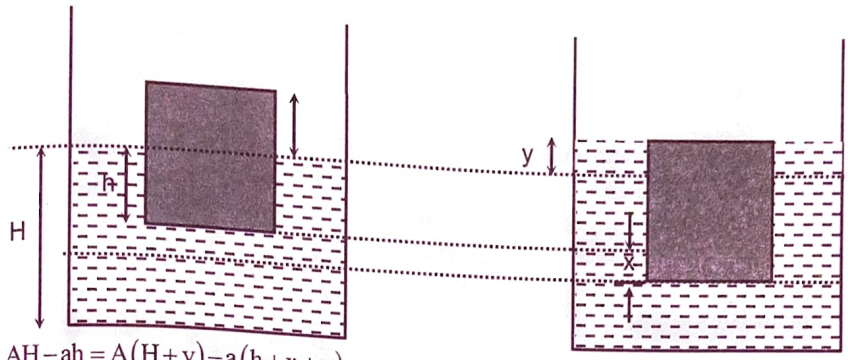
$$dy = dx\left[1-\frac{a}{A}\right]$$

Where dx is the new dipped part of block on displacing the block by dy

$$\Rightarrow dW = \int_{0.4}^{0.5}\left[x(0.5)^2 1000g-(0.5)^3 800g\right]dx\,\frac{3}{4} \Rightarrow W = 9.375\ J$$

Second Method:

a $\Rightarrow$ Area of cube A $\Rightarrow$ Base area of container.



$$AH-ah = A(H+y)-a(h+x+y)$$

$$\Rightarrow \quad y = \frac{a}{A-a}x \Rightarrow x+y = \frac{A}{A-a}x$$

$$F_{ex} = F_B - mg = \rho ga(h+x+y)-a^3\rho_B g$$

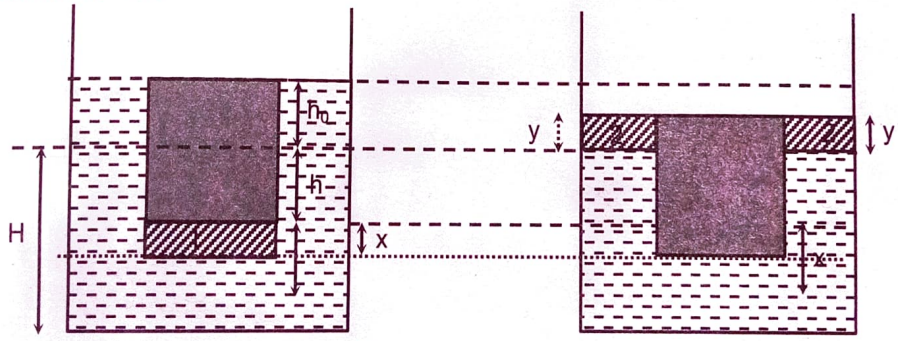
$$= a(x+y)\rho_\ell g = a\left(\frac{A}{A-a}\right)x\rho_\ell g$$

$$W = \int_0^{x_0} F_{ex}dx = \frac{aA\rho_\ell g}{(A-a)}\times\frac{x_0^2}{2} \text{ Here } x_0 = \frac{0.2\sqrt{a}(A-a)}{A}$$

$$\Rightarrow W = \frac{aA\rho_\ell g}{(A-a)}\times\frac{a^2}{2}\times\frac{0.04\times a(A-a)^2}{A^2} = \frac{0.04a^2(A-a)}{2A}\rho_\ell g = \frac{0.04\times\frac{1}{4}\times\frac{3}{4}\times\frac{1}{4}\times1000\times10}{2\times1}$$

$$= \frac{300}{32} = \frac{75}{8} = 9.375 \text{ Joule}$$

Third Method: $a = \ell\times\ell$



$$h_0 = \frac{\ell}{5},\ x+y = \frac{\ell}{5},\ x = 3y \Rightarrow x = \frac{3\ell}{20}$$

m_0 $\Rightarrow$ mass of liquid displace $= ax\rho_\ell = \frac{3a\rho_\ell}{20} = \frac{3V\rho_\ell}{20}$

Vertical displacement of liquid $= h + \frac{x+y}{2} = \frac{4\ell}{5} + \frac{\ell}{10} = \frac{9\ell}{10}$

$W_{FB} \Rightarrow$ work done by force of buoyancy to raise the liquid $= -m_0 g \frac{g\ell}{10} = -\frac{3V}{20}\rho_\ell \times g \times \frac{9\ell}{10}$

$= -\frac{3}{20} \times \frac{1}{8} \times 1000 \times 10 \times \frac{9}{10} \times \frac{1}{2} = -84.375$ J

$W_{mg} \Rightarrow$ Work done by weight to displace the body downward by distance x

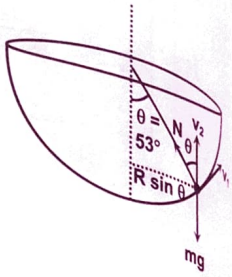
$= mgx = \frac{120}{8} \times 800 \times 10 \times \frac{3}{20} \times \frac{1}{2} = 75$ J

According work energy theorem

$W_{Fex} + W_{FB} + W_{mg} = 0$

$\Rightarrow W_{Fex} = -W_{FB} - W_{mg} = 84.375 - 75 = 9.375$ Joule

39. Let the horizontal component of velocity of the particle at a point where radius makes an angle $\theta = 53°$ with the vertical is v_1.



Using conservation of angular momentum of the particle about the vertical axis of the bowl.

$mv_0 R \sin\theta_0 = mv_1 R \sin\theta$

$\therefore v_1 = \frac{v_0 \sin\theta_0}{\sin\theta}$ ...(i)

Now, $N\sin\theta = \frac{mv_1^2}{R\sin\theta}$

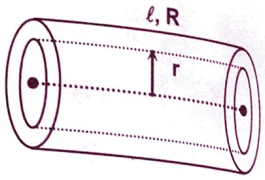
$N = \frac{mv_1^2}{R\sin^2\theta} = \frac{mv_0^2 \sin^2\theta_0}{R\sin^4\theta} = \frac{0.2 \times 64 \times \frac{9}{25}}{0.25 \times \frac{16}{25} \times \frac{16}{25}} = 0.2 \times 225 = 45$ Newton

$N = 45$ Newton

$\therefore n = 9$

40. Given, $v = v_0\left(1 - \frac{r^2}{R^2}\right)$

$\frac{dv}{dr} = v_0\left(\frac{-2r}{R^2}\right)$ ....(i)



The viscous force at a radial distance 'r' is

$F = -\eta 2\pi r\ell \frac{dv}{dr}$

$= -\eta 2\pi r\ell v_0\left(\frac{-2r}{R^2}\right)$ [from equation (i)]

$F = 4\pi\eta\ell v_0\left(\frac{r^2}{R^2}\right)$

In steady flow,

$\Delta P\pi r^2 = F$

$\Delta P\pi r^2 = 4\pi\eta\ell v_0\left(\frac{r^2}{R^2}\right)$

$\Delta P = \frac{4\eta\ell v_0}{R^2}$

$\therefore k = 2$

41. $v\cos\theta = v_0\sin\theta$

$v = v_0\tan\theta$

$a = v_0\sec^2\theta\frac{d\theta}{dt}$ ...(i)

$a = v_0\sec^2\theta\left(\frac{v_0\cos\theta + v\sin\theta}{\ell}\right) = \frac{v_0}{\ell}\sec\theta(v_0 + v\tan\theta)$

$a = \frac{v_0}{\ell}\sec\theta v_0(1+\tan^2\theta)$

$a = = \frac{v_0^2}{\ell}\sec^3\theta$ ...(ii)

$T\sin\theta = mg$ ...(iii)

$T\cos\theta = ma$ ...(iv)

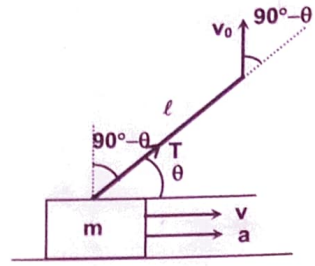
From (iii) and (iv)

$g = a\tan\theta$

$g = \frac{v_0^2}{\ell}\sec^3\theta\tan\theta$

$v_0 = \sqrt{\frac{g\ell}{\sec^3\theta\tan\theta}} = \left(\frac{4}{5}\right)^2\sqrt{\frac{10\times1.5\times5}{3}} = 3.20\text{ m/s}$

$v_0 = 3.20$ m/s



42. $N = mg\cos\theta$

$f_k = \mu N = 2\tan\theta mg\cos\theta$

$f_k = 2mg\sin\theta$

$a = \frac{f_k + mg\sin\theta}{m} = 3g\sin\theta$ ...(i)

$\tau_{CM} = I_{CM}\alpha$

$f_k R = \frac{2}{5}mR^2\alpha$

$\Rightarrow \quad 2mg\sin\theta = \frac{2}{5}m\alpha R$

$\alpha = \frac{5g\sin\theta}{R}$

...(ii)

When pure rolling starts

$v = \omega R$

$v_0 - at = (\omega_0 + \alpha t)R$

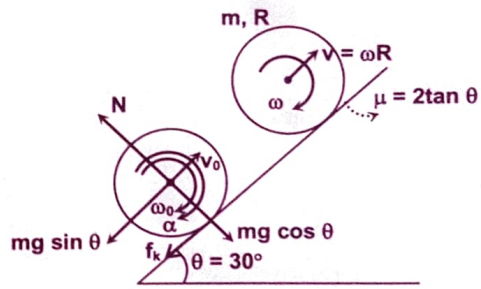
$v_0 - 3g\sin\theta t = \omega_0 R + 5g\sin\theta t$

$v_0 - \omega_0 R = 8g\sin\theta t$

$\therefore\ t = \frac{v_0}{12g\sin\theta}$

$\therefore\ v = v_0 - at = v_0 - 3g\sin\theta\times\frac{v_0}{12g\sin\theta} = v_0 - \frac{v_0}{4} = \frac{3v_0}{4}$

$\therefore\ v = \frac{3v_0}{4} = \frac{3\times15}{4} = 11.25\text{ m/s}$



43. $\tan\theta = \dfrac{a+g\sin\alpha}{g\cos\alpha} = \dfrac{a+g\sin\alpha}{g\cos\alpha}$

$$\frac{5-y}{3} = \frac{4+10\times\frac{3}{5}}{10\times\frac{4}{5}} \Rightarrow \frac{5-y}{3} = \frac{5}{4}$$

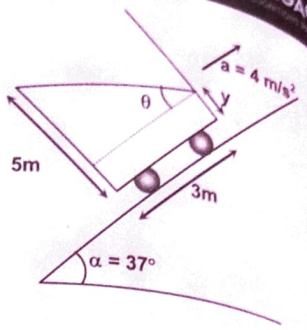
$\Rightarrow y = 1.25$ m

Volume of water remained in the container

$$V = \frac{1}{2}(5+1.25)\times 3\times 4 = 6.25\times 6 = 37.50 \text{ m}^3$$

$\therefore$ volume of water spilt over the tank

$\Delta V = V_0 - V = (3\times 4\times 4) - 37.50 = 10.50 \text{ m}^3$



44. The current through the resistor 'R_1' is

$$i_1 = \frac{\varepsilon}{R_1}\left[1-e^{\frac{R_1 t}{L}}\right]$$

$$i_1 = 4\left[1-e^{\frac{-t}{10^{-3}}}\right] \quad \ldots(i)$$

The current through the resistor 'R_2' is

$$i_2 = \frac{\varepsilon}{R_2}e^{\frac{-t}{R_2 C}}$$

$$i_2 = 2e^{\frac{-t}{2\times 10^{-3}}} \quad \ldots(ii)$$

The current through the battery

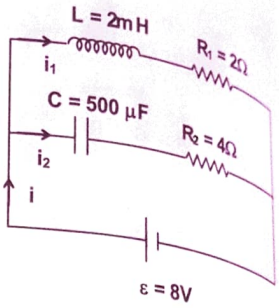
$I = i_1 + i_2$

$$I = 4\left[1-e^{\frac{-t}{10^{-3}}}\right] + \left(2e^{\frac{-t}{2\times 10^{-3}}}\right)$$

At $t = 1\times 10^{-3}$ sec

$I = 4(1-e^{-1}) + 2e^{-0.5}$

$I = 4(1-0.37) + (2\times 0.60) = 3.72$ amp



45. Emf induced in the straight segment AC is $\varepsilon_{AC} = 0$

Emf induced in the circular arc CD is

$$\varepsilon_{CD} = \int_a^{5a/2} B\omega r dr = \frac{1}{2}B\omega\left[r^2\right]_a^{\frac{5a}{2}} = \frac{21}{8}B\omega a^2$$

$$= \frac{21}{8}\times 5\times 10\times (0.2)^2 = 5.25 \text{ volt}$$

$\therefore \varepsilon_{AD} = \varepsilon_{AC} + \varepsilon_{CD} = 0 + 5.25 = 5.25$ volt

46. Least count of screw gauge, $L.C. = \dfrac{1\,\text{mm}}{100} = 0.01\,\text{mm}$

Diameter of the wire, $d = 2$ mm $+ 46\times$ L.C. $+ 4\times$ L.C.

$= 2\text{mm} + 46\times 0.01$ mm $+ 4\times 0.01$ mm $= 2.50$ mm $= 0.250$ cm

The length of wire, $\ell = 4.76$ cm

$\therefore$ The curved surface area of the wire, $S = \pi d\ell$

$$= \frac{22}{7}\times 0.250\times 4.76 = 3.74 \text{ cm}^2$$

47. Thermal resistance of the cylinders are

$$R_1 = \int_a^{2a} \frac{dr}{2\pi K_1 r \ell} = \frac{1}{2\pi K_1 \ell} \ell n 2$$

$$R_2 = \int_{2a}^{4a} \frac{dr}{2\pi K_2 r \ell} = \frac{1}{2\pi K_2 \ell} \ell n 2$$

Heat current flowing between the innermost and outermost cylindrical walls is

$$i = \frac{\theta_1 - \theta_2}{R_1 + R_2}$$

$\therefore \theta_1 - \theta = iR_1$

$$\Rightarrow \theta_1 - \theta = \left(\frac{\theta_1 - \theta_2}{R_1 + R_2}\right) R_1$$

$$\Rightarrow \theta_1 - \theta = \frac{(\theta_1 - \theta_2)\frac{1}{k_1}}{\left(\frac{1}{k_1} + \frac{1}{k_2}\right)}$$

$$\Rightarrow \theta_1 - \theta = \frac{(\theta_1 - \theta_2)k_2}{(k_1 + k_2)}$$

$$\Rightarrow \theta = \left(\frac{k_1\theta_1 + k_2\theta_2}{k_1 + k_2}\right) = \frac{(2k \times 84) + (3k \times 32)}{(2k + 3k)} = \frac{168 + 96}{5} = 52.8°C$$

48. The frequency of vibration of air column in the pipe

$$f_0 = \frac{3v}{4\ell_0} = \frac{3 \times 340}{4 \times 0.85} = 300 \text{ Hz}$$

$$f_s - f_0 = 4$$

$$f_s = f_0 + 4 = 300 + 4$$

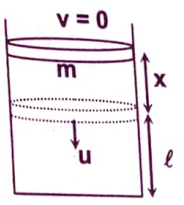
$$\frac{4}{2\ell}\sqrt{\frac{F}{\mu}} = 304$$

$$\frac{4}{2 \times 0.5}\sqrt{\frac{F \times 0.5}{5 \times 10^{-3}}} = 304$$

$$\sqrt{\frac{F}{10^{-2}}} = 76$$

$\therefore F = 57.76$ (Newton)

49. Let the vertical upward displacement of the piston before coming to a complete stop is 'x'.



In initial and final equilibrium of the piston

$$PA = \frac{nRT_0}{\ell} = \frac{nRT}{(\ell + x)} = mg$$

$nRT_0 = mg\ell$ ...(i)

$nRT = mg(\ell + x)$ ...(ii)

Now, $nR\Delta T = mgx$ ...(iii)

$$\frac{1}{2}mu^2 = nC_v\Delta T + mgx$$

$$\frac{1}{2}mu^2 = \frac{n3R}{2}(T - T_0) + mgx$$

$$\frac{1}{2}mu^2 = \frac{3nR\Delta T}{2} + mgx$$

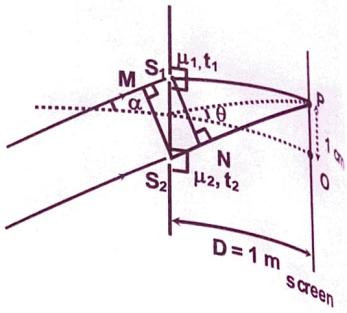
$$\frac{1}{2}mu^2 = \frac{5}{2}mgx$$

$$x = \frac{u^2}{5g} = \frac{8\times 8}{5\times 10} = 1.28 \text{ meter}$$

$\therefore$ vertical upward displacement of the piston = 1.28 meter.

50. Since the central maxima is formed at point 'P'

$d \sin\alpha + (\mu_1 - 1)t_1 = (\mu_2 - 1)t_2 + d \sin\theta$



$$2\times 10^{-3} \times \sin 30° + (1.5-1)t_1 = (1.8-1)3\times 10^{-3} + 2\times 10^{-3} \times \frac{1}{100}$$

$\Rightarrow 1 \times 10^{-3} + 0.5t_1 = 2.4 \times 10^{-3} + 0.02 \times 10^{-3}$

$\Rightarrow t_1 = 2.84 \times 10^{-3}$

$\Rightarrow t_1 = 2.84$ mm

51. Energy of each photon of light incident on the metal surface

$$E = \frac{hC}{\lambda} = \frac{1240}{500} = 2.48 \text{ eV}$$

Kinetic energy of fastest photoelectron emitted

$K_{max} = E - \phi = 2.48 - 1.90 = 0.58$ eV

$$\therefore E_n = -13.6\frac{z^2}{n^2}$$

$$E_4 = -13.6 \times \frac{(2)^2}{4\times 4} = -3.40 \text{ eV}$$

Energy of the photon emitted during combination

$\Delta E = K_{max} - E_4 = 0.58 - (-3.40) = 3.98$ eV

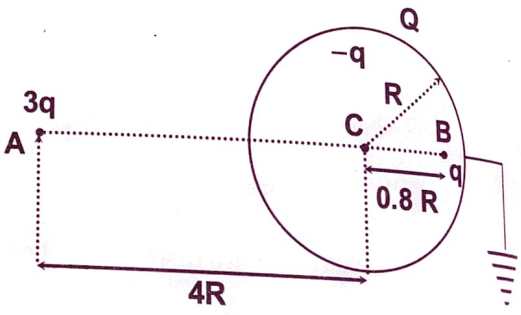
52. Let the charge on the conducting shell after being earthed is Q.

$$\frac{KQ}{R} + \frac{3kq}{4R} = 0$$

$$Q = \frac{-3q}{4}$$

$\therefore$ Total charge that will flow to the earth is

$$\Delta Q = 3q - \left(\frac{-3q}{4}\right) = \frac{15q}{4} = \frac{15}{4} \times 2 = 7.50 \ \mu C$$



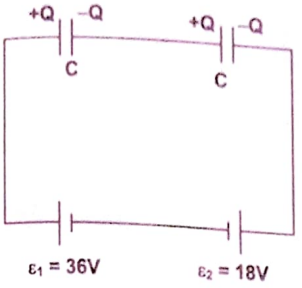
53. Charge on each capacitor, when switch S is opened

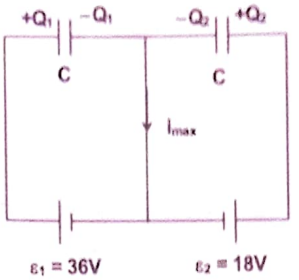
$$Q = \frac{C}{2}(\varepsilon_1 - \varepsilon_2) = \frac{10}{2} \times 18 = 90 \ \mu C$$

Charge on the capacitor when the current through the inductor is maximum after closing the switch 'S'.

$Q_1 = C\varepsilon_1 = 10 \times 36 = 360 \ \mu C$

$Q_2 = C\varepsilon_2 = 10 \times 18 = 180 \ \mu C$





Now, $\Delta Q_1\varepsilon_1 + \Delta Q_2\varepsilon_2 = \left(\frac{1}{2}C\varepsilon_1^2 + \frac{1}{2}C\varepsilon_2^2\right) - \frac{1}{2}\times\frac{C}{2}(\varepsilon_1-\varepsilon_2)^2 + \frac{1}{2}Li_{max}^2$

$10^{-6}(270\times36) + 10^{-6}(270\times18)$

$= 10^{-6}\left\{\frac{1}{2}\times10\times(36)^2 + \frac{1}{2}\times10\times(18)^2 - \frac{1}{2}\times5\times(18)^2\right\} + \frac{1}{2}\times2\times10^{-3}i_{max}^2$

$14580 = (8100-810) + 1\times10^3\times i_{max}^2$

$10^3\times i_{max}^2 = 7290$

$i_{max} = \sqrt{7.29} = 2.70$ ampere

MATRIX-MATCH LIST TYPE

14. (P) $x_1 + 2x_2 + 2x_3 + 2x_4 + 2x_5 = x$

$$\frac{F}{k_1}+\frac{4F}{k_2}+\frac{4F}{k_3}+\frac{4F}{k_4}+\frac{4F}{k_5} = x$$

$$\frac{F}{k}+\frac{4F}{k}+\frac{4F}{k}+\frac{4F}{2k}+\frac{4F}{2k} = x$$

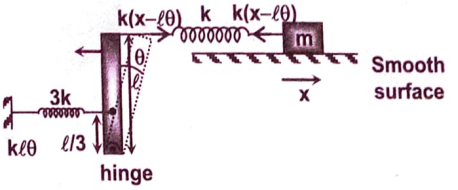
$$\frac{13F}{k} = x \Rightarrow F = \frac{kx}{13} \Rightarrow \frac{d^2x}{dt^2} = -\frac{kx}{13m}$$

$$T = 2\pi\sqrt{\frac{13m}{k}}$$

(Q) $m\frac{d^2x}{dt^2} = -8kx \Rightarrow \frac{d^2x}{dt^2} = -\left(\frac{8k}{m}\right)x$

$$\therefore\ T = 2\pi\sqrt{\frac{m}{8k}} = \pi\sqrt{\frac{m}{2k}}$$

(R) Since rod is light



$$3\frac{k\ell\theta}{3}\frac{\ell}{3} = K(x-\ell\theta)\ell$$

$$\frac{k\ell}{3}\theta = kx - k\ell\theta$$

$$\theta = \frac{3x}{4\ell}$$

$$m\frac{d^2x}{dt^2} = -k(x-\ell\theta) = -k\left(x-\frac{3x}{4}\right)$$

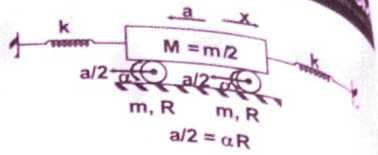
$$\frac{d^2x}{dt^2} = -\frac{kx}{4m} \Rightarrow T = 2\pi\sqrt{\frac{4m}{k}} = 4\pi\sqrt{\frac{m}{k}}$$

(S) $2kx - 2f_1 = Ma = \frac{ma}{2}$ ...(i)

$f_1 + f_2 = \frac{ma}{2}$ ...(ii)

$(f_1 - f_2)R = \frac{mR^2}{2}\alpha$

$(f_1 - f_2) = \frac{m\alpha R}{2} = \frac{ma}{4}$ ...(iii)



Adding (i), (ii) and (iii), we get

$2kx = \frac{5ma}{4} \Rightarrow a = \frac{8kx}{5m}$

$\frac{d^2x}{dt^2} = -\left(\frac{8k}{5m}\right)x \Rightarrow T = 2\pi\sqrt{\frac{5m}{8k}} = \pi\sqrt{\frac{5m}{2k}}$

MATCH THE FOLLOWING

11. For (R) :

$T_1\cos\theta_1 = T_2\cos\theta_2 = T$ (1)

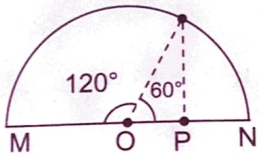
$T_1\sin\theta_2 = m_1 g$ (2)

$T_2\sin\theta_2 = m_2 g$ (3)

$\Rightarrow \frac{m_1}{m_2} = \left(\frac{T_1}{T_2}\right)\frac{\sin\theta_2}{\sin\theta_2} = \left(\frac{\cos\theta_2}{\cos\theta_1}\right)\left(\frac{\sin\theta_1}{\sin\theta_2}\right)$

$\Rightarrow \frac{\ell_1}{\ell_2} = \frac{\tan\theta_1}{\tan\theta_2} = \frac{1}{\sqrt{3}\times\sqrt{3}} = \frac{1}{3}$

For (S):



2.

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